Dynamics

Master Equation

This chapter is the canonical statement of the delayed dynamical law used throughout the dynamics branch. It defines what counts as a causal hit, how the receiver-local force law is assembled from path history, and which exact or regularized structures are firm enough to support later work on binaries, nested shell braids, effective geometry, and quantum closure.

For the primitive-entity ontology, see Architrino. This chapter begins where ontology stops: once continuous transceiver status is turned into a delay-root law, causal-hit branch sum, Jacobian-weighted acceleration, or regularized simulation equation.

The chapter is long because it plays several roles at once: foundational law, theorem spine, analytic benchmark source, and numerical reference. The opening establishes the causal geometry and canonical equation; later sections develop DDE form, self-hit structure, analytic regimes, and the energy-symmetry-conservation interface.

Foundations and Causal Geometry

Purpose and Scope

This document presents the **Master Equation of Motion (EOM)** governing the lawful evolution of all architrinos in the Euclidean void and absolute time. It is the microscopic dynamics input for later closure programs: assemblies, effective continuum descriptions, observer-level geometry, quantum behavior, and gravity must be recovered from this law only after the corresponding assembly, coarse-graining, and validation burdens are met. The chapter keeps c_f explicit in formulas; setting $c_f = 1$ is a nondimensional convention, not a change in the causal law.

Proper time τ does not exist at this layer. The EOM is integrated exclusively over absolute substrate time t. Causal roots, Jacobians, history integrals, and accelerations are all parameterized by t and by emission times $t_0 < t$. Clock readouts, time-dilation language, and effective metric comparisons belong only to later observer-inference chapters after the assembly dynamics have supplied a branch-certified period record.

The Master EOM is:

- **Deterministic**: Given complete initial conditions at t_0, the future is determined, with **deterministic multistability** at threshold regimes.
- **Non-Markovian**: Depends on full path history, not just instantaneous state.
- **Event-local at the receiver**: Only delayed causal intersections at the receiver event contribute to acceleration (no action-at-a-distance).
- **Causal**: All influences propagate at finite field speed c_f.
- **Self-consistent**: Includes self-interaction (self-hit) when same-source causal roots exist; super-field-speed interval history is a necessary warning condition for simple nontrivial self-hit roots, not a sufficient criterion by itself.

The level distinction used throughout the chapter is:

Level	What is asserted here	What is not asserted here
Substrate ontology	Architrinos move in absolute time through the Euclidean void and emit causal wakes.	No fundamental spacetime metric, continuum field substance, or observer reconstruction is assumed.
Dynamics	Acceleration is the receiver-local sum over delayed causal-root hits.	A plotted orbit or numerical residual is not a proof unless its branch chart is certified.
Effective description	Potentials, fields, one-forms, metrics, and wave functions may be reconstructed after coarse-graining.	Effective variables are not promoted to substrate ontology by their predictive usefulness.
Inference and observation	A receiver or observer may infer source configurations from hit records and assembly responses.	Inference does not determine the full ontic history unless the missing path-history data are supplied.

Overview and Key Principle

The Central Idea

Receiver-local relevance principle:

> The only substrate-level contributions to $\mathbf{a}_i(t)$ are causal wake intersections at the receiver event.

At time t, the acceleration of architrino i at position $\mathbf{x}_i(t)$ depends only on causal wake surfaces that intersect its current location.

- **Substrate event**: $\mathbf{x}_i(t)$ coincides with an expanding causal isochron emitted by some source at some past time $t_0 < t$.
- **Path-history input**: the source worldline determines which emission times solve the causal constraint.

- **Effective reconstruction**: a potential or field value away from the receiver is useful only after one has declared a continuum or diagnostic representation.
- **Inference layer**: source identity, distance, and emission velocity may be reconstructed from additional records, but they are not directly supplied by a single hit.

This is an **event-local delayed interaction rule**: the acceleration is evaluated at the receiver event, but depends on path history through the delayed causal roots.

In the absence of any causal hits, an architrino follows inertial motion: straight-line, constant-velocity trajectories in the fixed Euclidean background.

Operationally, the expanding causal wake is also the theory's minimal bridge between time and space. Absolute time orders emissions, Euclidean distance sets the propagation delay, and the receiver event is where those two inputs are rejoined into one physical interaction. The wake law is therefore not just a force prescription; it is the mechanism that turns temporal ordering plus spatial separation into concrete dynamics.

Abstract Form

The Master Equation of Motion (abstract level):

$$\boxed{\frac{d^2\mathbf{x}_i}{dt^2} = \sum_{j\neq i} \mathbf{a}_{ij}(\text{causal history}) + \mathbf{a}_{ii}(\text{self-hit})}$$

where:

- $\mathbf{a}_{ij}(\text{causal history})$: Sum of all per-hit accelerations from source $j \neq i$ arriving at receiver i at time t
- $\mathbf{a}_{ii}(\text{self-hit})$: Sum of all self-hit acceleration contributions (architrino i intersecting its own past emissions)

(The per-hit acceleration $\mathbf{a}_{ij}(t;t_0)$ is defined below in canonical form. The substrate law is acceleration-first. If a force-like bookkeeping symbol is desired, introduce one universal conversion constant μ_{arch} and define $\mathbf{F}_{ij} \equiv \mu_{\text{arch}}\mathbf{a}_{ij}$.)

Key insight: Both terms have the same functional form: a radial inverse-square law modulated by the causal Jacobian $J_{ij}(t;t_0)$. They differ only in source identity ($j = i$ vs $j \neq i$).

Path-History Sum and Integral Representation

The Master EOM is most naturally understood as a **path-history branch sum**: all of the physical content resides in the past worldlines of the sources, and the causal constraint selects the emission points on those worldlines whose influence reaches the receiver event "now."

In integral form, the same branch-sum law can be written as

$$\frac{d^2\mathbf{x}_i}{dt^2} = \sum_j \kappa\,\sigma_{ij}\,|q_i q_j| \int_{-\infty}^{t} \mathrm{d}t_0\, \frac{\hat{\mathbf{r}}_{ij}(t;t_0)}{r_{ij}^2(t;t_0)}\,\delta\Big(g_{ij}(t;t_0)\Big)$$

where

- $r_{ij}(t;t_0) = \|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|$,
- $\hat{\mathbf{r}}_{ij} = (\mathbf{x}_i(t) - \mathbf{x}_j(t_0))/r_{ij}$,
- $g_{ij}(t;t_0) = r_{ij}(t;t_0) - c_f(t - t_0)$,
- $\partial_{t_0} g_{ij}(t;t_0) = c_f - \hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)$,
- $\delta(\cdot)$ enforces the causal constraint $g_{ij} = 0$, and
- $\sigma_{ij} = \text{sign}(q_i q_j)$ encodes attraction/repulsion.

The causal delta collapses with the standard root Jacobian, so the integral evaluates to

$$\int_{-\infty}^{t} \mathrm{d}t_0\, f(t_0)\,\delta\big(g_{ij}(t;t_0)\big) = \sum_{t_0\in\mathcal{C}_{ij}(t)} \frac{f(t_0)}{|\partial_{t_0} g_{ij}(t;t_0)|}$$

provided the active roots are simple. This is the path-history integral representation of the exact branch law: acceleration at t depends on the causal contributions selected by the source worldline, with no contribution from noncausal points on that worldline.

For a certified branch chart, simplicity is recorded as a transversality floor:

$$|\partial_{t_0} g_{ij}(t;t_0)| = |c_f - \hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)| \geq \kappa_{\text{hit}} > 0$$

When this floor fails, the active root is caustic-like or degenerate and must be routed to a different branch chart or regularization regime. In special geometries the floor can be computed rather than declared; the principal circular partner branch derives $\kappa_{\rm hit}^{\rm bin} = c_f(1 + \beta \sin(\phi/2)) > c_f$ in Binary Dynamics.

Caustic Transit and Finite Impulse

The branch expression with $|J_{ij}|^{-1}$ should not be interpreted as a permission to pin an architrino at an infinite pointwise force. At a simple delay-map caustic, the branch chart fails, but the time-integrated velocity change can remain finite.

Let s denote the source emission-time variable near a degenerate root (t_*, s_*), and assume the local delay map has the nondegenerate fold form

$$g(t,s) = \alpha(s-s_*)^2 - \lambda(t-t_*) + O\big(|s-s_*|^3 + |t-t_*|\,|s-s_*| + |t-t_*|^2\big)$$

with $\alpha > 0$, $\lambda > 0$, and $r_{ij} \geq r_{\rm min} > 0$ on the local support. For $t < t_*$ the two simple roots satisfy

$$s_\pm(t) = s_* \pm \sqrt{\frac{\lambda}{\alpha}(t_* - t)} + O(t_* - t)$$

and the Jacobian factor scales as

$$|\partial_s g(t, s_\pm(t))| = 2\sqrt{\alpha\lambda}\sqrt{t_* - t} + O(t_* - t)$$

Thus each branch contribution has at worst the local bound

$$\|\mathbf{a}_{ij,\pm}(t)\| \leq \frac{C}{\sqrt{t_* - t}}$$

where C absorbs the bounded numerator, $r_{\rm min}^{-2}$, polarity factor, and coupling. The mechanical impulse through the caustic window is finite:

$$\int_{t_*-\varepsilon}^{t_*} \|\mathbf{a}_{ij,+}(t) + \mathbf{a}_{ij,-}(t)\|\, dt \leq 4C\sqrt{\varepsilon}$$

The same conclusion holds for a finite-order algebraic caustic $g \sim (s-s_*)^m - \lambda(t-t_*)$ with finite $m > 1$: the branch weight scales like $|t - t_*|^{-(m-1)/m}$, which is locally integrable in receiver time. A persistent interval with $J = 0$, a cusp with no finite-order normal form, or a simultaneous collision-floor failure is not covered by this impulse lemma and must remain in the regularized chart. The simulation rule is therefore: integrate the regularized acceleration through a caustic transit and record the finite $\Delta\mathbf{v}$; do not hold the state exactly on $J = 0$ as an infinite-force constraint.

The singular set should be routed by stratum, not by the single phrase “small denominator.” Let

$$\Sigma_{ij} = \{(t,s,\lambda) : g(t,s;\lambda) = 0,\ \partial_s g(t,s;\lambda) = 0\}$$

inside a declared one- or multi-parameter branch family. The finite-impulse lemma covers the fold stratum Σ^1, where $\partial_{ss} g \neq 0$ and the control parameter crosses the fold transversely. Cusp and higher strata, such as $\Sigma^{1,1}$, require extra degeneracy conditions and are not ordinary force rows. They may merge or split more than one opposite-sign root pair at once, so their ledger transition is not certified by the generic $\Delta N = \pm 2,\ \Delta D = 0$ fold law until a separate regularized normal form is supplied. Thus Σ^1 routes to finite impulse plus transition metadata, while $\Sigma^{1,1}$ or deeper routes to a singular-stratum chart before promotion.

The word “set” in $\mathcal{C}_{ij}(t)$ should therefore be read as a root set extracted from a continuous path-history integral, not as a replacement for that history. The source worldline is continuous data. In the sharp causal-wake limit the delta constraint collapses the received contribution to the emission times in $\mathcal{C}_{ij}(t)$; with $\eta > 0$ mollification, the received contribution comes from finite-width neighborhoods of those roots. A single source can contribute more than one root at the same receiver event when its worldline crosses the receiver’s backward causal surface more than once, especially in curved or super-field-speed history. The same bookkeeping applies to nontrivial self-history roots when $j = i$.

Writing

$$J_{ij}(t;t_0) \equiv 1 - \frac{\mathbf{v}_j(t_0)\cdot \hat{\mathbf{r}}_{ij}(t;t_0)}{c_f}$$

one obtains the exact branch-resolved form

$$\frac{d^2\mathbf{x}_i}{dt^2} = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij}\, \frac{|q_i q_j|}{r_{ij}^2(t;t_0)\ |J_{ij}(t;t_0)|}\, \hat{\mathbf{r}}_{ij}(t;t_0)$$

Since $\partial_{t_0} g_{ij}(t;t_0) = c_f J_{ij}(t;t_0)$, the collapse produces an overall factor $1/c_f$ together with $|J_{ij}|^{-1}$; by convention that constant factor is absorbed into κ.

An architrino emits potential at a constant rate per unit absolute time, but a moving source lays that steady output down on a moving family of causal surfaces. The **received causal flux** is therefore velocity dependent through the delay-map Jacobian: source motion geometrically compresses or dilates successive wake arrivals at the receiver. That J_{ij}^{-1} factor is therefore part of the fundamental law, not an optional correction.

Numerical implementations discretize this representation by sampling candidate emission times and solving for the active roots. The familiar "sum over spherical wake surfaces" is therefore a numerical realization of the same branch-selection rule, not a separate physical mechanism.

Simple-root transport lemma. Let

$$F_{ij}(t,s) = \|\mathbf{x}_i(t) - \mathbf{x}_j(s)\| - c_f(t-s)$$

and suppose $F_{ij}(t,s(t)) = 0$ on an interval where the active root is simple. Then $s(t)$ is differentiable and

$$\frac{ds}{dt} = \frac{c_f - \hat{\mathbf{r}}_{ij}(t;s)\cdot\mathbf{v}_i(t)}{c_f - \hat{\mathbf{r}}_{ij}(t;s)\cdot\mathbf{v}_j(s)} = \frac{1 - \hat{\mathbf{r}}_{ij}\cdot\mathbf{v}_i/c_f}{J_{ij}(t;s)}$$

Thus a simple causal root moves continuously with receiver time as long as the denominator stays away from zero. Simulations should track this root-transport residual alongside the root residual and the J floor; failure of the transport equation is a branch-chart failure, not an ordinary force fluctuation.

Branch-Chart Closure Object

A local master-equation closure claim should be attached to an explicit branch-chart object, not just to a plotted orbit or a small force residual. For a branch chart on a section $\mathcal{S}$, define

$$\mathfrak{B}(\Gamma,\mathcal{S};h,\eta,\epsilon_c) = \left(\mathcal{R}^{\text{act}},\mathcal{G}^{\text{inact}},\nu_J,h_{\text{mem}},\mathcal{R}_{\text{return}},\lambda_{\text{sec}}\right)$$

Here $\mathcal{R}^{\text{act}}$ is the active causal-root set retained by the chart, $\mathcal{G}^{\text{inact}}$ is the collection of inactive branch-gap functions, ν_J is the active-root Jacobian floor, h_{mem} is the required memory depth, $\mathcal{R}_{\text{return}}$ is the return residual on the section, and λ_{sec} is the transverse section-stability margin.

The object is acceptable only when

$$\nu_J > 0, \qquad \inf_{\mathcal{G}^{\text{inact}}} g_a^{ij} > 0, \qquad 0 < h_{\text{mem}} < h < \infty, \qquad \|\mathcal{R}_{\text{return}}\| \le \epsilon_{\text{return}}$$

and the section return is stable, for example

$$\rho(M_{\mathcal{S}}|_{E_\perp}) \le 1 - \lambda_{\text{sec}}, \qquad \lambda_{\text{sec}} > 0$$

The inactive-gap condition means that nearby discarded causal roots remain separated from the active chart; the stability condition means that a small transverse section error is trapped rather than amplified.

Local promotion lemma. If a candidate history supplies $\mathfrak{B}(\Gamma,\mathcal{S};h,\eta,\epsilon_c)$ with positive active-root floors, positive inactive gaps, finite memory, bounded return residual, and stable section monodromy, then the history may support a local master-equation closure claim on that section. The lemma does not prove global closure, eliminate all folds, control the $\eta \to 0$ limit, or certify unrelated histories. It only promotes the branch chart from a numerical trace to a locally replayable causal-root closure record.

State-Dependent Delay Compatibility

A branch-chart closure object must also be compatible with the delayed history space that generates the active roots. Fix a retained history tube

$$\mathcal{U}_{\mathfrak{B}} \subset C^1\left([-h,0],(\mathbb{R}^3)^N\right)$$

around the returned history segment. For each active branch row ℓ, write its emission offset as $s_\ell(\phi) \in [-h,0)$ for a history $\phi \in \mathcal{U}_{\mathfrak{B}}$, and define

$$F_\ell(\phi,s) = \|\phi_i(0) - \phi_j(s)\| - c_f(0-s)$$

The branch chart is history-compatible on $\mathcal{U}_{\mathfrak{B}}$ only if

$$F_\ell(\phi,s_\ell(\phi)) = 0, \qquad |\partial_s F_\ell(\phi,s_\ell(\phi))| \ge c_f\nu_J > 0$$

and if every inactive complement remains separated by the declared positive gap. Under these conditions the implicit-function theorem gives C^1 dependence of s_ℓ on the retained history, so the branch acceleration, root-transport residual, and wake-history Noether increments are functionals on one local history chart rather than pointwise rows that only happen to close at one evaluation time.

This compatibility condition is a theorem-target requirement, not a new force law. It says that a promoted branch chart must define a locally replayable delayed functional system: nearby retained histories must keep the same root identities, positive Jacobian floor, inactive gaps, and finite memory depth until a declared fold, branch transition, or chart boundary is reached.

Local-To-Global Branch-Chart Gluing Target

The branch-chart object above also defines a candidate causal-root sheaf for global closure. For an open history or parameter neighborhood U, let

$$\mathcal{F}_{\mathrm{root}}(U) = \{\text{admissible branch charts on } U \text{ with the declared regularization data}\}$$

and restrict a chart from U to $V \subset U$ by restricting its active-root rows, inactive gaps, memory tube, and endpoint convention. The implicit-function theorem supplies the local restriction maps while the root identities remain simple.

Global closure is the additional statement that local sections of $\mathcal{F}_{\mathrm{root}}$ glue. On an overlap $U_\alpha \cap U_\beta$, two local charts must agree not merely on the plotted trajectory but on the signed causal-root ledger, branch labels, endpoint convention, wake-history charges, and transition metadata. A mismatch on triple overlaps is a Cech-style gluing obstruction: locally replayable charts may exist while no single global branch chart exists. This is a theorem target, not a new postulate. It gives proof programs an explicit failure mode between “local residuals are small” and “the Master Equation branch is globally closed.”

Dual-Mollified Absolute-Time Evolution Law

For proof work, branch sums should be derived from one regularized absolute-time law rather than treated as the primary definition through every causal fold. Fix a memory horizon

$$h > 0$$

a causal-wake-surface width

$$\eta > 0$$

and a short-distance core scale

$$\epsilon_c > 0$$

Define

$$\mathbf{r}_{ij}(t,s) \equiv \mathbf{x}_i(t) - \mathbf{x}_j(s), \qquad r_{ij}(t,s) \equiv \|\mathbf{r}_{ij}(t,s)\|$$

and, away from the zero vector,

$$\hat{\mathbf{r}}_{ij}(t,s) \equiv \frac{\mathbf{r}_{ij}(t,s)}{r_{ij}(t,s)}$$

The dual-mollified finite-memory evolution law is

$$\boxed{\ddot{\mathbf{x}}_i(t) = \kappa \sum_j \sigma_{ij} |q_i q_j| \int_{t-h}^{t} \frac{\hat{\mathbf{r}}_{ij}(t,s)}{r_{ij}^2(t,s) + \epsilon_c^2}\, \delta_\eta\big(r_{ij}(t,s) - c_f(t-s)\big)\, ds}$$

with the same sign convention

$$\sigma_{ij} = \mathrm{sign}(q_i q_j)$$

used in the exact branch law. For equal-magnitude charges

$$|q_i| = \epsilon$$

the factor

$$|q_i q_j|$$

reduces to

$$\epsilon^2$$

This equation is the certification-level law for the dual-mollified problem. The causal-surface mollifier

$$\delta_\eta$$

selects causal surfaces with finite width, while

$$\epsilon_c$$

caps the near-collision inverse-square amplitude. Branch-resolved formulas with Jacobian factors are local reductions of this equation on finite simple-root charts. They should not be used as the global definition across causal folds, caustic transit, or chart-boundary verification.

Finite-η Pathology Quarantine Theorem Target

The classical point-source pathologies are not closed by the sharp branch formula alone. The theorem target is finite-η, branch-chart local, and conditional on one regularized action and energy convention. Fix a finite architrino set, a finite window $W = [t_a, t_b]$, a memory depth $h < \infty$, a causal-surface width $\eta > 0$, a core scale $\epsilon_c > 0$, and a branch chart

$$\mathfrak{B}(\Gamma, \mathcal{S}; h, \eta, \epsilon_c)$$

whose retained rows are generated by the dual-mollified evolution law above.

The admissibility assumptions are:

1. The evolution is causal in absolute time: every force row is generated from $s < t$, and the self-coincident endpoint is excluded by the $H(0) = 0$ convention or by the declared core regularization.
2. The retained chart has finite memory, positive inactive-root gaps, and either a positive active-root Jacobian floor $\nu_J > 0$ or an explicitly declared finite-order caustic transit integrated in the dual-mollified law.
3. The active support stays away from an unregularized collision: either $r_{ij,\ell} \geq d > 0$ on the retained rows or the same ϵ_c cutoff is used in the force, action, and energy rows.
4. The regularized right-hand side is locally Lipschitz on the retained history tube, so the finite-η state-dependent delay problem has existence, uniqueness, and continuation until a declared boundary of the admissible class is reached.
5. The same regularized action or compatible realized-trajectory reconstruction supplies the force row, wake-history energy, momentum, and angular-momentum rows. Endpoint leakage, omitted branch rows, and period-cut terms must appear as residuals rather than hidden corrections.

Under these assumptions, the finite-η theorem packet should prove the following local conclusions.

- **Divergent self-energy is quarantined.** There is no accepted instantaneous self-kick, no unregularized $r = 0$ self-root inside the chart, and every retained self-hit contribution is bounded by the declared d, ϵ_c, η, and ν_J data. Any remaining divergence is therefore a failure of the branch floor, core convention, memory window, or $\eta \to 0^+$ convergence claim, not an accepted finite-η state.
- **Runaway solution branches are quarantined.** If the same action-level bookkeeping gives

$$E_{\text{tot}}^{(\eta)}(t) = K_\mu(t) + E_{\text{wake}}^{(\eta)}(t), \qquad E_{\text{wake}}^{(\eta)}(t) \geq U_{\min}^{(\eta)} > -\infty,$$

then $K_\mu(t)$ remains bounded on W. A branch with $v_{\max} \to \infty$ must therefore leave the admissible class by driving the interaction charge downward without bound, losing a floor, or breaking the declared action-energy residual.

- **Pre-acceleration is excluded at finite η.** The acceleration at t is a functional of the retained history segment $[t - h, t)$, together with the current receiver event, and contains no future state. A proposed action repair is admissible only when its endpoint convention contributes a wake-history boundary term or a vanishing residual, not a future-boundary force selection rule.
- **Caustic and Jacobian blow-up are quarantined.** Simple-root charts require $\nu_J > 0$. A finite-order caustic may be crossed only through the dual-mollified equation with finite integrated impulse and stable transition metadata. Persistent $J = 0$, cusp behavior outside a finite-order normal form, simultaneous collision-floor failure, or regulator-dependent transition observables are chart failures rather than ordinary force rows.
- **Finite deterministic multistability is routed, not quarantined.** At a fold boundary, if the regularized post-transit data supply exactly one admissible continuation chart, the event is an ordinary branch transition. If they supply two or more inequivalent admissible charts with positive floors and finite memory, the complete microstate still selects one deterministic continuation, but a record-limited comparison must route the event to a finite basin-weight or multistability record. Multistability is a failure only when the finite continuation family is empty, infinite, unlabeled, or lacks the common branch data needed for comparison.

The failure boundary for the theorem packet is the union of the following conditions:

$$\partial\mathcal{A}_\eta = \{\nu_J = 0\} \cup \{r_{ij,\ell} = d\} \cup \{\inf \mathcal{G}^{\text{inact}} = 0\} \cup \{h_{\text{mem}} \geq h\} \cup \{E_{\text{wake}}^{(\eta)} \downarrow -\infty\} \cup \{\text{non-vanishing action or boundary residual}\}$$

The first component is refined as

$$\{\nu_J = 0\} = \Sigma_{\text{transit}} \sqcup \Sigma_{\text{bif}}^{\text{multi}} \sqcup \Sigma_{\text{sing}}^{\text{fail}}$$

where $\Sigma_{\rm transit}$ has a unique finite post-transit chart, $\Sigma_{\rm bif}^{\rm multi}$ has a finite labeled family of admissible continuations, and $\Sigma_{\rm sing}^{\rm fail}$ lacks a promoted finite chart. A trajectory crossing this boundary is not promoted as a closed Master Equation solution until the appropriate route is certified. It is routed to branch transition, finite multistability, caustic transit, core-regularization repair, finite-window leakage, or η-ladder failure according to which boundary component is reached.

The validation residuals consumed by this theorem target are the root residual, root-transport residual, active Jacobian floor, inactive-gap residual, finite-memory residual, return residual, finite-window energy residual $\mathcal{R}_E$, momentum residual $\mathcal{R}_P$, angular-momentum residual $\mathcal{R}_J$, Euler residual of the same action, endpoint or period-cut leakage, transition-observable residuals across η refinement, and the symplectic residual $\mathcal{R}_\Omega$ when the branch is promoted to a reduced Hamiltonian chart. The theorem is finite-η only; any zero-width or infinite-system statement requires the separate convergence boundary stated in the regularization package.

Regularized Energy Diagnostic for the Exact Charge

For computation with finite causal-wake-surface width $\eta > 0$, it is useful to introduce an η-regularized diagnostic for the same history-aware energy charge tracked by the exact nonlocal action. When one wants a quadratic kinetic bookkeeping proxy, use a single universal conversion constant $\mu_{\rm arch}$ rather than particle-specific substrate masses. This smooth expression is used for numerical evaluation and convergence testing of the conserved quantity:

$$E_{\rm tot}^{(\eta)}(t) = \sum_i \frac{1}{2}\mu_{\rm arch}\,\|\dot{\mathbf{x}}_i(t)\|^2 + E_{\rm wake}^{(\eta)}(t)$$

For the regularized interaction diagnostic, a convenient working expression is:

$$E_{\rm wake}^{(\eta)}(t) = \frac{1}{2}\sum_{i,j}\kappa\,\sigma_{ij}\,|q_i q_j| \int_{t-\tau_{\max}}^{t} dt_0\, \frac{1}{r_{ij}^2(t;t_0)\,\,|J_{ij}(t;t_0)|}\,\delta_\eta\big(r_{ij}(t;t_0) - c_f(t-t_0)\big)$$

where $\tau_{\max}$ bounds the causal memory depth used in analysis and simulation. Because this expression is written with the branch-level inverse-square force density, it should be treated as a diagnostic candidate unless it is derived from the same time-translation-invariant action-level regularization as the action charge below. If the dual-mollified law with a core cutoff ϵ_c is used, the energy diagnostic must carry the same cutoff convention. The nonlocal Noether charge used for theorem-level conservation is the boundary functional in Action-level wake-energy functional at time boundary t.

Causal Interaction Set (The Geometry of Delay)

Definition of Causal Emission Times

For a receiver at position $\mathbf{x}_i(t)$ and a source with worldline $\mathbf{x}_j(t')$, the **causal emission times** $\mathcal{C}_{ij}(t)$ are all past times $t_0 < t$ such that a causal wake surface emitted by source j at t_0 arrives at receiver i at time t.

Causal constraint:

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t-t_0)$$

where c_f is the field speed (set to 1 in natural units).

Notation:

$$\mathcal{C}_{ij}(t) = \Big\{t_0 < t \;\Big|\; \|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t-t_0)\Big\}$$

Causal-Time Map and Root Topology

For fixed receiver time t, define the causal-time map

$$f_t^{(ij)}(t_0) \equiv t_0 + \frac{1}{c_f}\,\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|, \qquad F_t^{(ij)}(t_0) \equiv f_t^{(ij)}(t_0) - t$$

Then causal emission times are exactly the roots:

$$t_0 \in \mathcal{C}_{ij}(t) \quad\Longleftrightarrow\quad F_t^{(ij)}(t_0) = 0$$

Operationally, this is the branchwise source-to-receiver reading of the dynamics. The source worldline supplies a path-history map $t_0 \mapsto (\mathbf{x}_j(t_0), \mathbf{v}_j(t_0))$, while the receiver supplies the event data $(\mathbf{x}_i(t), \mathbf{v}_i(t), t)$. Solving $F_t^{(ij)}(t_0) = 0$ selects exactly those source-history points whose causal isochrons are received at that event. Each selected root therefore maps one source-history branch into one receiver-local line of action; the delay-map Jacobian below records how constant source emission cadence is compressed or dilated when read at the receiver. When multiple roots exist, the causal-root ledger is the bookkeeping of these simultaneous source-to-receiver branch matches.

The one-dimensional delay-map Jacobian is

$$\frac{dF_t^{(ij)}}{dt_0} = 1 - \frac{\hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)}{c_f}$$

On a bounded history interval I_t (e.g., simulation memory window), define:

- Unsigned root count: $N_{ij}(t) \equiv \#\mathcal{C}_{ij}(t)$,
- Signed Brouwer degree:

$$D_{ij}(t) \equiv \deg(F_t^{(ij)}, I_t, 0) = \sum_{t_0\in\mathcal{C}_{ij}(t)} \operatorname{sign}\left(\frac{dF_t^{(ij)}}{dt_0}\Big|_{t_0}\right)$$

Delay-Map Theorem Pack (Formalized)

Fix a bounded history interval $I_t = [a,b] \subset (-\infty, t)$ and define regularity conditions:

- **(R1) Boundary regularity:** $0 \notin F_t^{(ij)}(\partial I_t)$ (no root crossing at a or b).
- **(R2) Simple roots:** if $F_t^{(ij)}(t_0) = 0$, then $\frac{dF_t^{(ij)}}{dt_0}(t_0) \neq 0$.

Theorem 1 (Degree invariance on regular families).
For any continuous deformation of worldlines/parameters that preserves (R1)-(R2), the signed degree $D_{ij}(t) = \deg(F_t^{(ij)}, I_t, 0)$ is invariant.

Proof sketch: In 1D, D_{ij} is the oriented count of simple roots. Under a regular homotopy, roots move continuously and cannot appear/disappear in the interior without becoming critical, and cannot enter/leave through the boundary by (R1). Hence the oriented count is constant.

Proposition 2 (Sub-c_f monotonic single-hit regime).
If there exists $v_* < c_f$ such that $\|\mathbf{v}_j(t_0)\| \le v_*$ for all $t_0 \in I_t$, then

$$\frac{dF_t^{(ij)}}{dt_0} \ge 1 - \frac{v_*}{c_f} > 0$$

so $F_t^{(ij)}$ is strictly increasing on I_t. Therefore it has at most one root. If additionally $F_t^{(ij)}(a) < 0 < F_t^{(ij)}(b)$ (or the opposite sign ordering), then exactly one root exists and

$$N_{ij}(t) = 1, \qquad D_{ij}(t) = +1$$

Proof sketch: Strict positivity of the Jacobian gives monotonicity, hence injectivity. Existence under endpoint sign change follows by the intermediate value theorem.

This proposition is retained-interval local. It controls roots whose emission times lie inside the declared interval I_t; it does not remove older path-history roots emitted outside I_t, including self-hit candidates from an earlier super-field-speed interval that remain inside a longer memory window. If the model retains such persistent-memory roots, the speed bound must be checked on the enlarged interval that contains their emission times.

Proposition 3 (Fold criterion and even-jump law).
In a one-parameter family $F^{(ij)}(t_0;\lambda)$ (with λ a control parameter, e.g. receiver time or orbit parameter), interior root-count changes occur only at fold points:

$$F^{(ij)}(t_0;\lambda) = 0, \qquad \partial_{t_0}F^{(ij)}(t_0;\lambda) = 0$$

For generic folds ($\partial_{t_0t_0}F \neq 0$, $\partial_\lambda F \neq 0$), one root pair is created/annihilated, so

$$\Delta N_{ij} = \pm 2, \qquad \Delta D_{ij} = 0$$

between regular intervals.

Proof sketch: Local normal form near a generic fold is equivalent to $u^2 \pm \mu = 0$, yielding either 0 or 2 simple roots. The two roots carry opposite Jacobian signs, so the degree is unchanged.

This delay-map theorem pack is foundational rather than merely model-specific. Within this chapter it serves as the fold-geometry reference for delayed-root constructions: regular charts preserve signed degree, while branch creation or annihilation requires a Jacobian-degenerate fold.

Signed Causal-Root Complex

For a receiver-source pair (i,j) on a regular interval, the root ledger can be sharpened from an unsigned set to a two-term signed complex. Split the simple active roots by Jacobian sign:

$$C_+^{ij}(t) = \operatorname{span}\{s_\ell \in \mathcal{C}_{ij}(t) : \partial_s F_t^{(ij)}(s_\ell) > 0\}, \qquad C_-^{ij}(t) = \operatorname{span}\{s_\ell \in \mathcal{C}_{ij}(t) : \partial_s F_t^{(ij)}(s_\ell) < 0\}.$$

Then

$$N_{ij} = \dim C_+^{ij} + \dim C_-^{ij}, \qquad D_{ij} = \dim C_+^{ij} - \dim C_-^{ij}.$$

At a generic fold, the local boundary pairing creates or removes one positive and one negative generator, preserving D_{ij} while changing N_{ij} by two. In this reading, Theorem 1 is invariance of the Euler-characteristic-like signed count, and Proposition 3 is the elementary opposite-sign pair surgery.

For assembly work, solver artifacts should therefore report the signed grading (C_+^{ij}, C_-^{ij}), not only raw hit counts. The binary and tri-binary ledgers N_s and M_p are admissible topological labels only after their self-hit and partner-hit rows inherit this signed-root-complex data, together with the phase-bundle integer used by the assembly topological charge and resonance-lock chapters.

Proposition 4 (forward partner-root starvation under field-speed drift).
Let a candidate translating branch have center drift $u\hat{\mathbf{e}}$ on the retained interval, with $u \ge 0$ and $\|\hat{\mathbf{e}}\| = 1$. Write two partner constituents as

$$\mathbf{x}_i(t) = ut\,\hat{\mathbf{e}} + \boldsymbol{\rho}_i(t), \qquad \mathbf{x}_j(t_0) = ut_0\,\hat{\mathbf{e}} + \boldsymbol{\rho}_j(t_0)$$

where i is the receiver and $j \ne i$ is the source. Suppose the retained partner row is forward-directed in the co-moving branch chart:

$$d_\parallel(t, t_0) \equiv \hat{\mathbf{e}} \cdot \big(\boldsymbol{\rho}_i(t) - \boldsymbol{\rho}_j(t_0)\big) \ge d_{\min} > 0$$

For any positive-delay candidate root with $\tau = t - t_0 > 0$,

$$c_f \tau = \big\|u\tau\,\hat{\mathbf{e}} + \boldsymbol{\rho}_i(t) - \boldsymbol{\rho}_j(t_0)\big\| \ge u\tau + d_{\min}$$

Hence

$$(c_f - u)\,\tau \ge d_{\min}$$

If $u \ge c_f$, no such forward partner root exists. If $u < c_f$, any such row has the lower delay bound

$$\tau \ge \frac{d_{\min}}{c_f - u}$$

so the required memory depth diverges as $u \to c_f^-$.

This is a kinematic starvation result, not a force-balance approximation. It says that a forward structural partner row cannot be retained at or above field-speed center drift because the causal wake cannot catch the leading receiver. A bound assembly branch that requires at least one such forward partner row for structural closure therefore cannot preserve the same causal-root ledger for sustained drift $u \ge c_f$. The proposition does not impose a speed cap on a single architrino, on internal curved self-hit motion, or on history-supported super-field-speed components; it applies to center translation of an internally bound branch whose leading-side partner closure is part of the retained ledger.

With finite retained memory h and internal branch period T_{int}, the same obstruction has a graded scale:

$$\Lambda_{\text{starv}} = \frac{\tau_{\text{fwd}}}{T_{\text{int}}} \ge \frac{d_{\min}}{(c_f - u)T_{\text{int}}}.$$

The forward row remains available to the retained chart only while $\tau_{\text{fwd}} < h$, equivalently

$$u < u_{\text{crit}} \equiv c_f - \frac{d_{\min}}{h}.$$

Thus starvation is a root-complex obstruction before it is a speed slogan: if the assembly requires that forward generator, the bare causal kernel cannot carry the same branch chart through u_{crit}. Any promoted supra-u_{crit} branch must show a Noether-sea or assembly reorganization that removes or replaces the forward row without hiding a memory-window failure.

Single-Hit Regime (Unique t_0)

In the **sub-field-speed regime** ($\|\mathbf{v}_j(t_0)\| < c_f$ locally), Proposition 2 applies, and the map is strictly monotone:

$$\frac{dF_t^{(ij)}}{dt_0} \ge 1 - \frac{\|\mathbf{v}_j(t_0)\|}{c_f} > 0$$

so $f_t^{(ij)}$ is a diffeomorphic time map on I_t, and the causal set is generically a singleton:

$$N_{ij}(t) = 1, \qquad D_{ij}(t) = +1$$

Intuition: If the source is moving slower than the field speed, its past emissions form a non-overlapping family of concentric (or nearly concentric) isochrons. Any given receiver location lies on exactly one of those causal surfaces.

Multi-Hit Regime (Multiple t_0)

In the **super-field-speed regime** ($\|\mathbf{v}_j\| > c_f$ at some past times), the delay map can fold when $\hat{\mathbf{r}}_{ij} \cdot \mathbf{v}_j > c_f$, i.e. when $dF_t^{(ij)}/dt_0$ changes sign. Then $\mathcal{C}_{ij}(t)$ can contain multiple solutions:

$$\mathcal{C}_{ij}(t) = \{t_{0,1}, t_{0,2}, \ldots, t_{0,m}\}$$

Fold bifurcations create/annihilate roots in pairs. The signed degree D_{ij} stays topologically fixed between folds, while the unsigned branch count N_{ij} jumps by even integers.

For a suggestive first folded branch used as a nested shell braid closure target, the reduced root-count analogy is

$$N_O = 1 \;\longrightarrow\; N_I = 2$$

This is not yet a nested shell braid closure result. It is the root-count counterpart one would need to justify before using the action-partition doubling target ($w_I = 2w_O$) or the associated $1:2:4$ frequency-lock discussion as derived structure.

Intuition: If the source outruns its own emissions, it can emit multiple wake surfaces that later converge and intersect the same receiver location simultaneously (or nearly so, within regularization width η).

Example: In uniform circular motion at $v > c_f$, a receiver can be hit by wake surfaces from multiple points on the source's orbit (different "winding numbers" m due to self-hit dynamics).

Self-Hit (Source = Receiver, $j = i$)

When $j = i$ (source and receiver are the same architrino), the causal set $\mathcal{C}_{ii}(t)$ represents **self-hits**: times when architrino i intersects its own past emissions.

Self-hit condition:

$$\|\mathbf{x}_i(t) - \mathbf{x}_i(t_0)\| = c_f(t - t_0), \quad t_0 < t$$

Interval-speed lemma. Let $\Delta = t - t_0 > 0$ and suppose $\mathbf{x}_i$ is absolutely continuous on $[t_0, t]$. If

$$\|\mathbf{x}_i(t) - \mathbf{x}_i(t_0)\| = c_f \Delta$$

then

$$\frac{1}{\Delta}\int_{t_0}^{t} \|\dot{\mathbf{x}}_i(s)\|\, ds \geq c_f$$

This follows immediately from the triangle inequality. Therefore strict sub-field-speed motion on the whole interval forbids a nontrivial self-hit. A simple noncoincident self-hit requires super-field-speed motion somewhere along the interval, except for the degenerate straight field-speed case where the causal branch is tangent and the simple-root Jacobian condition fails.

Critical requirements for self-hit:

1. **Curvature**: Straight-line motion admits no self-hits (the worldline never intersects its own past causal isochrons).
2. **Super-field-speed interval history**: along the interval from emission to reception, the architrino must have exceeded c_f somewhere, unless the branch is the degenerate straight field-speed case excluded by the simple-root assumptions.

Key clarification:

- **Self-hits can be plural**: $\mathcal{C}_{ii}(t)$ can contain multiple emission times (e.g., multiple winding numbers in circular motion).
- **Persistent memory**: Once an architrino has exceeded $v > c_f$ in its past, it can **later slow down** to $v < c_f$ and **still receive self-hits** from wake surfaces emitted during the super-field-speed phase. The self-hit regime is **not** instantaneously tied to current velocity; it depends on **path history**.

Implication: Self-hit is a **non-Markovian memory effect**. The architrino's current acceleration depends on whether it **ever** exceeded c_f in the past and curved, not just on its current state.

Geometric Interpretation

Visualize the causal constraint as:

- Receiver at $\mathbf{x}_i(t)$ "now"

- Source worldline $\{\mathbf{x}_j(t') : t' < t\}$ in the past
- Field-speed causal wake surface: the expanding isochron at radius $c_f(t - t_0)$ centered at $\mathbf{x}_j(t_0)$
- **Causal emission times**: where this wake surface **intersects** the receiver's current location

For each $t_0 \in \mathcal{C}_{ij}(t)$, draw a line from $\mathbf{x}_j(t_0)$ to $\mathbf{x}_i(t)$; this is the **line of action** $\hat{\mathbf{r}}_{ij}$ for the force.

This geometry should be read in terms of the source worldline, the expanding causal isochrons centered on past emission points, and the receiver event at which one or more of those isochrons are intersected.

Reduced Translating-Loop Delay Checkpoint

To obtain a nontrivial analytic checkpoint from the same causal constraint, consider a translating phase-locked two-leg internal loop, with one leg parallel to motion and one transverse. Let the loop center translate with speed v through the Euclidean void while every wake still propagates at the primitive field speed c_f. Define

$$\beta \equiv \frac{v}{c_f}, \qquad C(v) \equiv \frac{L_\parallel(v)}{L_0}$$

with rest bond length L_0.

Parallel round-trip delay:

$$T_\parallel(v) = \frac{L_\parallel}{c_f - v} + \frac{L_\parallel}{c_f + v} = \frac{2L_0}{c_f}\frac{C(v)}{1-\beta^2}$$

Transverse one-way delay satisfies

$$c_f^2\tau^2 = L_0^2 + v^2\tau^2 \;\Rightarrow\; \tau = \frac{L_0}{c_f\sqrt{1-\beta^2}}$$

so

$$T_\perp(v) = 2\tau = \frac{2L_0}{c_f}\frac{1}{\sqrt{1-\beta^2}}$$

If internal phase locking is operationally isotropic (no orientation-dependent clock leakage),

$$T_\parallel(v) = T_\perp(v)$$

then necessarily

$$C(v) = \sqrt{1-\beta^2}, \qquad T(v) = \frac{T_0}{\sqrt{1-\beta^2}}, \quad T_0 = \frac{2L_0}{c_f}$$

This gives a purely substrate-level period-stretch checkpoint. It says only that preserving the same internal phase closure while the receiver translates forces the physical period T to increase in absolute time unless the longitudinal leg shortens. The full unresolved step is proving the same absolute-period scaling for the complete multi-hit NFDE nested shell braid dynamics without reducing to a two-leg closure model.

The forward parallel leg carries the same starvation constraint as Proposition 4 with $d_{\min}$ replaced by the leading longitudinal leg length $L_\parallel$. Therefore the two-leg checkpoint is admissible as a retained-row model only in the starvation-free regime

$$v < c_f - \frac{L_\parallel}{h}.$$

Above that scale, the checkpoint no longer tracks the same causal-root ledger: the full nested shell braid proof must show a ledger reorganization, a sea-mediated replacement row, or a declared failure of the translating-loop reduction.

The two-leg loop is only a checkpoint. It has two phase points and one chosen orientation relative to the absolute motion. A real assembly has an effective internal phase distribution over a finite three-dimensional volume, and operational isotropy has to hold for all loop orientations at once. The closure target is therefore a full oblate-envelope-to-sphere reduction in the internal nested shell braid phase space, not just the equality

$$T_\parallel = T_\perp$$

for one leg pair.

Accelerated motion adds a second burden. Even if the inertial translating-loop scaling is recovered, acceleration requires a transport law for the internal phase ledger through the Noether sea. For a stable branch with rest size L_0, center speed $v(t)$, and small acceleration scale $a(t)$, the dynamics target is a branch-period transport law of the schematic form

$$T_q[v(t), a(t)] = T_q[v(t), 0]\left(1 + O\left(\frac{a^2 L_0^2}{c_f^2}\right)\right)$$

with every term evaluated in absolute time. The residual is the finite-loop-size, non-Markovian correction caused by acceleration during one internal phase cycle. Observer-inference chapters may later translate a branch-certified period record into clock and metric language, but no such translation is part of the Master EOM.

Master Equation and DDE Formulation

The Master Equation (Canonical Form)

Per-Hit Acceleration

For each causal emission time $t_0 \in \mathcal{C}_{ij}(t)$, define:

Separation vector and distance:

$$\mathbf{r}_{ij}(t; t_0) = \mathbf{x}_i(t) - \mathbf{x}_j(t_0), \quad r_{ij} = \|\mathbf{r}_{ij}\|$$

Unit direction (line of action):

$$\hat{\mathbf{r}}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}} = \frac{\mathbf{x}_i(t) - \mathbf{x}_j(t_0)}{\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|}$$

Polarity sign factor:

$$\sigma_{ij} = \mathrm{sign}(q_i q_j) = \begin{cases} +1 & \text{like polarities (repel)} \\ -1 & \text{unlike polarities (attract)} \end{cases}$$

Delay-map Jacobian:

$$J_{ij}(t; t_0) \equiv 1 - \frac{\mathbf{v}_j(t_0) \cdot \hat{\mathbf{r}}_{ij}(t; t_0)}{c_f}$$

Per-hit acceleration contribution:

$$\mathbf{a}_{ij}(t; t_0) = \kappa\, \sigma_{ij} \frac{|q_i q_j|}{r_{ij}^2\, |J_{ij}(t; t_0)|} \hat{\mathbf{r}}_{ij}$$

If a force-like bookkeeping symbol is desired, define

$$\mathbf{F}_{ij}(t; t_0) \equiv \mu_{\text{arch}}\, \mathbf{a}_{ij}(t; t_0)$$

where μ_{arch} is a universal conversion constant used only for force/energy bookkeeping. It is not a particle-specific inertial mass.

where:

- κ: universal coupling constant
- q_i, q_j: intrinsic polarities of receiver and source ($\pm\epsilon$ for electrinos/positrinos)
- r_{ij}: distance from emission point to reception point
- $\hat{\mathbf{r}}_{ij}$: radial direction from emission to reception
- J_{ij}: causal Jacobian controlling geometric bunching or dilation of the received wake flux

Note on interaction structure: The per-hit acceleration $\mathbf{a}_{ij}(t; t_0)$ is **radial in direction**: it points along the line of action $\hat{\mathbf{r}}_{ij}$ from the source's past position to the receiver's current position. There are **no velocity-dependent cross-product terms** (no $\mathbf{v}_i \times \mathbf{B}$-like contributions) in the fundamental interaction kernel. However, the force magnitude is not purely $1/r^2$; it is modulated by $|J_{ij}|^{-1}$. Constant emission per unit absolute time at the source is therefore received as a Jacobian-weighted causal flux at the receiver, with the spatial deposition pattern itself changing as the source moves.

Implication for emergent forces: All "magnetic" or velocity-dependent forces (e.g., Lorentz force $\mathbf{v} \times \mathbf{B}$) must arise from **delay geometry**, **Jacobian-modulated flux**, and **superposition of radial hits**, not from intrinsic cross-product terms in the fundamental law. This places the burden of magnetic-field emergence on the assembly structure, Noether sea dynamics, and the finite-speed causal geometry itself.

Total Acceleration (Sum Over All Causal Hits)

The total acceleration on architrino i at time t is the **vector sum** over:

1. All sources $j \neq i$ (partner hits)

2. All causal emission times $t_0 \in \mathcal{C}_{ij}(t)$ for each source
3. Self-hits ($j = i$), if any exist

Master Equation of Motion (Canonical Form):

$$\boxed{\frac{d^2\mathbf{x}_i}{dt^2} = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij}\, \frac{|q_i q_j|}{r_{ij}^2\, |J_{ij}(t;t_0)|}\, \hat{\mathbf{r}}_{ij}}$$

where:

- Outer sum: over all sources j (including $j = i$ for self-hits)
- Inner sum: over all causal emission times $t_0 \in \mathcal{C}_{ij}(t)$
- Each term: radial inverse-square acceleration with sign σ_{ij} and Jacobian weight $|J_{ij}|^{-1}$

Explicit separation of partner and self-hit terms:

$$\frac{d^2\mathbf{x}_i}{dt^2} = \underbrace{\sum_{j\neq i} \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij}\, \frac{|q_i q_j|}{r_{ij}^2\, |J_{ij}(t;t_0)|}\, \hat{\mathbf{r}}_{ij}}_{\text{Partner hits}} + \underbrace{\sum_{t_0 \in \mathcal{C}_{ii}(t)} \kappa\, \sigma_{ii}\, \frac{|q_i q_i|}{r_{ii}^2\, |J_{ii}(t;t_0)|}\, \hat{\mathbf{r}}_{ii}}_{\text{Self-hits}}$$

Note: $\sigma_{ii} = +1$ (like polarities repel), so self-hits are always **repulsive**.

This sum can be viewed as a **path-history branch sum**: each emission time in $\mathcal{C}_{ij}(t)$ marks where the receiver's worldline crosses the causal wake surface emitted at t_0. The integral representation above is simply the distributional encoding of this branch-selection rule.

Conventions and Exclusions

Heaviside Convention ($H(0) = 0$):

The emission at $t_0 = t$ (instantaneous self-force) is **excluded**. Formally, this is enforced by writing:

$$\mathcal{C}_{ij}(t) = \left\{ t_0 < t \;\middle|\; \|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t - t_0) \right\}$$

(Strict inequality $t_0 < t$; no $t_0 = t$ allowed.)

Physical justification: The causal wake surface at the instant of emission ($r = 0$, $\tau = 0$) has not yet expanded; it cannot exert a force on the emitter "now." Under symmetric regularization (mollification), the $r \to 0$ limit yields zero net push.

No $r = 0$ causal roots beyond $\tau = 0$:

Because $r = c_f(t - t_0)$, $r = 0$ implies $\tau = t - t_0 = 0$. This case is excluded by $H(0) = 0$. There are no "collision singularities" in the causal set (architrinos can pass through each other; forces are mediated by expanding wake surfaces, not by contact).

Superposition Principle

The Master EOM is **linear in source contributions** on a declared branch chart:

$$\mathbf{a}_{\text{total}}(t) = \sum_j \mathbf{a}_j(t)$$

The causal-wake distributions from distinct sources superpose without mutual interference, and the receiver sums the branch accelerations that actually intersect it. Effective potentials reconstructed from those wakes also superpose in the corresponding linear diagnostic or continuum limit, but the substrate law remains the receiver-local branch sum.

Consequence: The problem of N interacting architrinos reduces to solving N coupled delay differential equations (DDEs), one per architrino, with each depending on the retained history of all sources and on the certified active causal-root rows.

Terms and Conventions (Detailed Breakdown)

Direction and Sign

Direction of $\hat{\mathbf{r}}_{ij}$:

$\hat{\mathbf{r}}_{ij}$ points **from the source's historical position $\mathbf{x}_j(t_0)$ to the receiver's current position $\mathbf{x}_i(t)$.**

Sign of the acceleration:

- **Like polarities** ($\sigma_{ij} = +1$): acceleration along $+\hat{\mathbf{r}}_{ij}$ (repulsion; pushes receiver away from emission point)

- **Unlike polarities** ($\sigma_{ij} = -1$): acceleration along $-\hat{\mathbf{r}}_{ij}$ (attraction; pulls receiver toward emission point)

Two-body checks (stationary sources):

- Electrino + Electrino (like polarities): repulsion
- Positrino + Positrino (like polarities): repulsion
- Electrino + Positrino (unlike polarities): attraction
- All symmetric: if source and receiver swap roles, the force direction reverses (Newton's third law in the instantaneous-interaction limit)

Scaling and Normalization

The $1/r^2$ factor:

Reflects the **surface density** of potential on the causal isochron. As that surface grows, the potential spreads over area $4\pi r^2$, so the density at any point scales as $1/r^2$.

The Jacobian factor $|J_{ij}|^{-1}$:

Under the constant-time emission rule stated above, source motion between emission instants deposits the output onto a history-dependent family of expanding causal surfaces. Motion of the source toward the active branch compresses the spacing of successive wake arrivals and increases the received flux; motion away from the branch dilates the spacing and decreases it. The geometric compression/dilation factor is exactly $|J_{ij}|^{-1}$.

Absorption of geometric constants into κ:

All geometric normalization factors (e.g., $1/(4\pi)$ from spherical surface area and overall $1/c_f$ factors from δ-function change-of-variables) are absorbed into the coupling constant κ by convention. The canonical per-hit law is therefore written with an explicit inverse-square factor together with the dimensionless Jacobian weight $|J_{ij}|^{-1}$.

Dimensional analysis:

$$[\kappa] = \frac{[\text{Length}]^3}{[\text{Time}]^2[\text{Polarity}]^2}, \quad [\mathbf{a}] = \frac{[\text{Length}]}{[\text{Time}]^2}$$

If the force-like bookkeeping variable $\mathbf{F} = \mu_{\text{arch}}\mathbf{a}$ is introduced, then $[\mathbf{F}] = [\mu_{\text{arch}}][\text{Length}]/[\text{Time}]^2$. In natural units with $c_f = 1$, $[\text{Length}] = [\text{Time}]$, and κ has dimensions of $[\text{Length}]/[\text{Polarity}]^2$.

Receiver Kinematics (Radial vs Orthogonal Components)

At a given hit $(t; t_0)$, decompose the receiver's velocity into components parallel and orthogonal to the line of action $\hat{\mathbf{r}}_{ij}$:

$$\mathbf{v}_i(t) = v_r\hat{\mathbf{r}}_{ij} + \mathbf{v}_\perp$$

where:

- $v_r = \mathbf{v}_i(t) \cdot \hat{\mathbf{r}}_{ij}$ (radial component; positive = moving away from emission point)
- $\mathbf{v}_\perp = \mathbf{v}_i(t) - v_r\hat{\mathbf{r}}_{ij}$ (orthogonal component)

Instantaneous effect of the hit:

Because $\mathbf{a}_{ij}(t; t_0) \parallel \hat{\mathbf{r}}_{ij}$, its instantaneous effect satisfies:

$$\frac{d}{dt}\mathbf{v}_\perp\Big|_{\text{hit}} = \mathbf{0}, \quad \frac{d}{dt}v_r\Big|_{\text{hit}} = \mathbf{a}_{ij} \cdot \hat{\mathbf{r}}_{ij} = \kappa\,\sigma_{ij}\,\frac{|q_i q_j|}{r_{ij}^2\;|J_{ij}(t; t_0)|}$$

Plain language: A hit only changes the along-the-line velocity component right now; sideways motion continues unaffected at the instant of the hit. Over time, the changing radial motion alters the trajectory and thus the subsequent orthogonal component.

Translating-assembly deformation requirement: The receiver kinematics described here must mechanically produce the moving-assembly deformation, branch-period stretch, and two-way signal-synchronization records that later observer-inference chapters consume. If nested shell braids do not squash along the direction of motion and do not preserve one retained causal-root ledger while translating through the Noether sea, the downstream recovery program fails at the dynamics layer.

Work and Power

The **instantaneous power** (rate of kinetic energy change) from a single hit is:

$$\frac{dE_k}{dt}\Big|_{\text{hit}} = \mathbf{F}_{ij} \cdot \mathbf{v}_i = \big(\mu_{\text{arch}}\mathbf{a}_{ij} \cdot \hat{\mathbf{r}}_{ij}\big)v_r = \mu_{\text{arch}}\,\kappa\,\sigma_{ij}\,\frac{|q_i q_j|}{r_{ij}^2\;|J_{ij}(t; t_0)|}\,v_r$$

Key insight: There is **no instantaneous work** on the orthogonal component. Power depends only on the radial velocity v_r.

Radial motion and the $1/r^2$ factor (local trend):

- **Inward motion** ($v_r < 0$, receiver moving toward the emission point): decreases r_{ij} between close successive hits, tending to **increase** subsequent per-hit strengths via $1/r^2$ (all else equal).
- **Outward motion** ($v_r > 0$): increases r_{ij}, tending to **decrease** subsequent per-hit strengths.

Important caveat: Path-history delay shifts both the causal root t_0 and $\hat{\mathbf{r}}_{ij}$ over finite intervals, so these are strictly **local** statements about infinitesimal time evolution. The global trajectory depends on the full history of all sources.

Moving-Source Geometry and Received Flux

Critical modeling note:

- **Emission rule**: fixed by the constant-time law stated above
- **Spatial deposition**: velocity dependent because the source changes position between emission instants

The **emitted potential pattern in space** is not speed independent: a moving source lays down successive wake surfaces from different points on its worldline. The **received** force magnitude is therefore not purely a function of r_{ij}. It is modulated by the causal Jacobian $|J_{ij}|^{-1}$, which measures how the source motion compresses or dilates the spacing of wake surfaces along the active branch.

The receiver's velocity $\mathbf{v}_i(t)$ does **not** appear as a separate source-strength factor in $\|\mathbf{F}_{ij}\|$ itself (at fixed r_{ij}, $\hat{\mathbf{r}}_{ij}$, and J_{ij}). It influences:

1. The **instantaneous power** through $\mathbf{F} \cdot \mathbf{v} = \|\mathbf{F}\| v_r$.
2. The **subsequent evolution of** r_{ij} (and thus future force magnitudes).
3. Which delayed branches are actually sampled along the receiver worldline over time.

For a uniformly moving source on a simple branch, this flux modulation is closed form. Let

$$\mathbf{x}_j(t_0) = \mathbf{x}_{j,0} + \mathbf{u}t_0, \qquad \beta = \frac{\|\mathbf{u}\|}{c_f}, \qquad \cos\theta = \frac{\mathbf{u} \cdot \hat{\mathbf{r}}_{ij}}{\|\mathbf{u}\|}$$

at the emission event. Then

$$J_{ij} = 1 - \beta\cos\theta$$

and the received wake density attached to the simple root is proportional to

$$\frac{1}{r_{ij}^2\,|1 - \beta\cos\theta|}$$

For $0 \le \beta < 1$ this gives a headlight-style causal-wake anisotropy:

$$\mathcal{D}_{\text{wake}}(\theta;\beta) = \frac{1}{1 - \beta\cos\theta}$$

up to the common inverse-square dilution and coupling normalization. The formula is not a Lorentz transformation and does not add electrodynamic velocity-field or acceleration-field terms. It is the microscopic source of leading/trailing wake-density asymmetry: forward directions with $\cos\theta > 0$ receive compressed isochron spacing, while trailing directions receive diluted spacing. Doppler shift, aberration, magnetic-like response, preferred-frame leakage estimates, and translating-binary asymmetry must be derived from this branch geometry plus assembly and observer-channel closure rather than inserted as independent laws.

For $\beta \ll 1$,

$$\mathcal{D}_{\text{wake}}(\theta;\beta) = 1 + \beta\cos\theta + O(\beta^2).$$

The even scalar part feeds the coarse scalar wake potential, while the odd velocity-directed part is the single-hit seed of the recoil and vector-transport channel described in Effective Lagrangian. Under coarse-graining this is the same source of $\mathbf{A}_{\text{wake}}$ and antisymmetric wake-stress diagnostics; it is not an independent magnetic law added on top of the Master Equation.

Causal-flux modulation: Unlike models that make source strength itself a function of speed, the velocity dependence here enters through the **moving-source geometry** of emission, the **geometry of causal intersections**, and the **bunching or dilation of received wake flux** in the Euclidean void. This is the origin of the Jacobian denominator and the seed of relativistic and magnetic behavior in the emergent theory.

Delay Differential Equation (DDE) Formulation

State Vector and Evolution

Define the **state vector** for architrino i:

$$\mathbf{X}_i(t) = \begin{pmatrix} \mathbf{x}_i(t) \\ \mathbf{v}_i(t) \end{pmatrix} \in \mathbb{R}^6$$

The Master EOM is a **second-order ODE** in $\mathbf{x}_i$, or equivalently a **first-order system** in $\mathbf{X}_i$:

$$\frac{d\mathbf{X}_i}{dt} = \begin{pmatrix} \mathbf{v}_i(t) \\ \mathbf{a}_i(t) \end{pmatrix}$$

where:

$$\mathbf{a}_i(t) = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij}\, \frac{|q_i q_j|}{r_{ij}^2\; |J_{ij}(t;t_0)|}\, \hat{\mathbf{r}}_{ij}$$

Causal Functional Form

The acceleration $\mathbf{a}_i(t)$ depends on the **history** of all worldlines $\{\mathbf{X}_j(t') : t' < t\}$ through the implicit causal constraint:

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t - t_0)$$

This makes the system a **delay differential equation (DDE)** with **state-dependent delays** (the delay $\tau_j = t - t_0$ is not constant; it depends on the solution itself).

Functional notation:

$$\frac{d\mathbf{X}_i}{dt} = \mathcal{F}\Big[\mathbf{X}_i(t), \{\mathbf{X}_j(\cdot)\}_j, t\Big]$$

where $\mathcal{F}$ is a **causal functional**: it depends on the current state $\mathbf{X}_i(t)$ and the past states $\{\mathbf{X}_j(t') : t' < t\}$ of all architrinos (including i itself for self-hits).

Regularization (Mollified Causal Wake Surfaces, Finite η)

The ideal model uses **surface-delta causal isochrons** in the emission-time integral. On a simple branch with a distance floor and a Jacobian floor, the delta collapses to a continuous receiver-time branch contribution; singular or impulse-like behavior arises only when branches hit collision support, lose transversality, accumulate, or are sampled as unresolved numerical events. One may treat the singular limit as a measure-valued branch law, or regularize by replacing the surface delta with a narrow wake surface of thickness $\eta > 0$:

$$\delta(r - c_f\tau) \longrightarrow \delta_\eta(r - c_f\tau) = \frac{1}{\sqrt{2\pi}\,\eta}\exp\Big(-\frac{(r - c_f\tau)^2}{2\eta^2}\Big)$$

while preserving total emission q.

Effect: Under the finite-branch, distance-floor, and transversality assumptions stated below, this supports **continuous-in-time force diagnostics** and classical C^1 solutions for $\mathbf{x}_i(t)$ given C^1 initial data.

In the super-field-speed regime ($\|\mathbf{v}_a\| > c_f$), multiple self-roots can occur; summing over all causal times with an integrable regularization gives a finite contribution only while the active-root count, separation floor, and Jacobian floor remain controlled.

Convergence requirement: As $\eta \to 0$, numerical solutions must converge to a well-defined limit.

Conditional Well-Posedness for the Regularized Exact Model

To make the existence/uniqueness claim precise for the finite-η regularization used in this chapter, we formalize the dynamics as a state-dependent delay system in first-order form:

$$\dot{\mathbf{Y}}(t) = \mathcal{G}(\mathbf{Y}_t), \qquad \mathbf{Y}_t(\theta) = \mathbf{Y}(t+\theta),\ \theta \in [-h, 0]$$

with phase space $\mathcal{H} = C^1([-h,0], \mathbb{R}^{6N})$.
This is the convenient proof scaffold used here because the active-root extraction uses the implicit-function theorem on

$$C^1$$

histories. For sharper state-dependent delay work, especially when acceleration bounds rather than classical second derivatives are the natural control, the phase space may need to be

$$W^{1,\infty}([-h,0],\mathbb{R}^{6N})$$

or an absolutely continuous history class. The exact choice is a regularity burden of the theorem being proved, not a change in the causal law.

Assumptions (regularized regime):

- **(W1) Kernel regularity:** δ_η is C^1, bounded, and integrable.
- **(W2) Uniform branch finiteness:** on the considered history neighborhood, each pair (i,j) has at most $B_{ij} < \infty$ active causal branches.
- **(W3) Root transversality:** for every active branch $\tau_{ij,\ell}$,

$$|\partial_\tau g_{ij}(\tau,\phi)| \geq \nu > 0, \qquad g_{ij}(\tau,\phi) = \|\phi_i(0) - \phi_j(-\tau)\| - c_f\tau$$

- **(W4) Distance floor on the branch support:** $\|\phi_i(0) - \phi_j(-\tau_{ij,\ell}(\phi))\| \geq d_{\min} > 0$.
- **(W5) Bounded charges/couplings:** κ, $|q_i|$ finite.

Conditional theorem (local well-posedness and continuation).
Under (W1)-(W5), for any initial history $\phi^0 \in \mathcal{H}$ there exists $T > 0$ and a unique solution

$$\mathbf{Y} \in C^1([t_0 - h, t_0 + T), \mathbb{R}^{6N}), \qquad \mathbf{Y}_{t_0} = \phi^0$$

The solution extends uniquely to a maximal interval $[t_0 - h, t_{\max})$. If on every finite interval

$$\sup_{t<t^*} \|\mathbf{v}(t)\| < \infty, \qquad \inf_{t<t^*,\, i,j,\ell} r_{ij,\ell}(t) > 0, \qquad \inf_{t<t^*,\, i,j,\ell} |\partial_\tau g_{ij,\ell}(t)| > 0$$

and

$$\sup_{t<t^*,\, i,j} B_{ij}^{\text{active}}(t) < \infty$$

then $t_{\max} = \infty$.
Here $r_{ij,\ell}(t)$ denotes the source-receiver distance on branch ℓ, and

$$B_{ij}^{\text{active}}(t)$$

denotes the number of active causal branches of pair

$$(i,j)$$

inside the chosen memory horizon at receiver time

$$t$$

Proof.

1. By (W3), each active delay branch is simple; the Implicit Function Theorem gives $\tau_{ij,\ell}(\phi) \in C^1$ on a neighborhood of ϕ^0.
2. Each per-branch acceleration term is a composition of C^1 maps (evaluation, subtraction, norm, mollifier, and unit-direction projection). By (W4), denominators stay away from zero; by (W5), coefficients are bounded. Hence each branch term is locally Lipschitz in ϕ.
3. By (W2), only finitely many branches contribute, so their sum $\mathcal{G}$ is locally Lipschitz on an open subset of $\mathcal{H}$ where (W3)-(W4) hold.
4. Standard state-dependent DDE existence/uniqueness theory on Banach spaces applies, yielding a unique local C^1 solution and a maximal extension.
5. Continuation follows from the same theorem: finite-time breakdown can occur only by leaving every bounded subset of the admissible set, i.e. via unbounded speed, vanishing separation on active support, transversality loss/root accumulation, or unbounded active branch-count growth.

Therefore the regularized delayed dynamics are locally well-posed, with global existence whenever those failure modes are excluded. This conditional statement applies to the finite-η regularized model; the ideal $\eta \to 0$ surface-delta limit still requires separate control of root accumulation and Jacobian-degenerate branches. □

Finite-Continuation Criterion for Global Comparisons

The well-posedness theorem is the dynamics-side home for global-continuation comparisons used later in General Relativity and Singularity Resolution. It should not be read as a claim that observer records determine a unique global spacetime. Its native claim is narrower: a declared finite history, boundary wake record, and branch chart either determine a finite continuation family or they do not.

For a compact subsystem Ω and window $W = [t_i, t_f]$, let $\mathcal{A}^{(\eta)}_{\Omega,W}$ be the set of branch charts that satisfy the regularized assumptions (W1)-(W5), the bounded active-branch condition, the distance floor, and the root-transversality floor on W using the same finite boundary data $\mathcal{B}_{\partial\Omega}|_W$. The dynamics-side continuation family is

$$\mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} = \left\{ \mathbf{Y}^a_{t_f} : a \in \mathcal{A}^{(\eta)}_{\Omega,W},\ \mathbf{Y}^a_{t_f} \text{ is generated by the regularized master equation from } (\mathbf{Y}_{t_i}, \mathcal{B}_{\partial\Omega}|_W) \right\}$$

The comparison passes only if

$$0 < \left| \mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} \right| < \infty$$

with every element carrying the causal-root ledger, energy diagnostic or exact charge used for the run, and the boundary wake data that selected it. Empty, infinite, or unlabeled families are not global closure; they mark an unresolved continuation ambiguity. A later strong-field or cosmology chapter may quotient this family by observer-accessible records, but the quotient must be derived from the same master-equation data rather than imposed as a global-hyperbolicity assumption.

The cardinality is itself a structural diagnostic:

- $\left| \mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} \right| = 0$ means no compatible global section of the declared branch-chart data exists.
- $\left| \mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} \right| = 1$ means the finite data select a unique deterministic continuation.
- $2 \leq \left| \mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} \right| < \infty$ means deterministic multistability: the complete state occupies one branch, while record-limited comparisons must use the basin weights induced on the finite labeled family.
- $\left| \mathfrak{S}^{\mathrm{ME},\eta}_{\Omega,W} \right| = \infty$ means root accumulation, noncompact branch freedom, or unresolved chart gluing, not a mature global comparison.

Equivalently, this is the section count of the causal-root gluing target over the finite window. Finite labeled multistability is an admissible branch-statistics object; empty, infinite, or unlabeled continuation families remain closure failures.

Operational Principles, Self-Interaction, and Examples

Core Principles (Operational Summary)

Superposition

Statement: The potential wake contributions from all sources **superpose linearly**. The net potential at any point is the sum of the individual wake potentials:

$$\Phi_{\text{net}}(\mathbf{x}, t) = \sum_i \Phi_i(\mathbf{x}, t)$$

The total acceleration on a particle at any instant is the **vector sum** of the contributions from every causal entry in its path history.

Operational implication: Every architrino is continuously immersed in the superposed wakes of all others (and, when the same-source root condition permits, its own). Tractability comes from treating each causal emission independently with $1/r^2$ distance weighting, branch gaps, and screening or cancellation assumptions that make the retained sum finite.

Inverse-square dilution alone is not a global convergence theorem. For an infinite source family, a branch chart must declare a summation or continuum prescription under which

$$\lim_{R\to\infty} \sum_{j:\|\mathbf{x}_j\|<R} \sum_{t_0\in\mathcal{C}_{ij}(t)} \mathbf{a}_{ij}(t; t_0)$$

exists, or it must supply local neutrality, angular cancellation, shielding, a screened kernel, finite active horizon, or a mean-field/principal-value subtraction. Without this condition, the many-source wake sum is not a well-defined acceleration law even though each individual hit has the correct surface-density falloff.

Velocity Dependence

Statement: The dynamics are **delayed** and **radial in direction**. Because the source moves while emitting, both the emitted wake pattern and the received force are velocity dependent through causal geometry. The received force magnitude is modulated by the causal Jacobian $|J_{ij}|^{-1}$, while the receiver's speed affects the **work rate** and branch sampling via $\mathbf{F}\cdot\mathbf{v} = \|\mathbf{F}\| v_r$.

Self-interaction requirement: Self-hit requires super-field-speed interval history: the worldline must exceed c_f somewhere along a nontrivial emission-to-reception interval, except for the degenerate field-speed tangent case excluded by the simple-root branch condition. Curvature alone is insufficient if $\|\mathbf{v}_a\| < c_f$ everywhere on the relevant interval.

Persistent memory: Once an architrino has exceeded $\|\mathbf{v}\| > c_f$ in its past and emitted wake surfaces, it can **later slow down** to $\|\mathbf{v}\| < c_f$ and **still receive self-hits** from those earlier emissions. The self-hit regime is **not instantaneously tied to current velocity**; it is a **path-history memory effect**.

Causality and Locality

Causal structure: Event A at $(t_A, \mathbf{x}_A)$ can influence event B at $(t_B, \mathbf{x}_B)$ only if:

$$t_B > t_A \quad \text{and} \quad \|\mathbf{x}_B - \mathbf{x}_A\| \leq c_f(t_B - t_A)$$

This defines a **field-speed causal cone** centered at each event. The filled inequality is the reachability condition; exact hits still occur only on causal wake surfaces satisfying the equality root.

No action-at-a-distance: All influences propagate at finite speed c_f. There are no instantaneous interactions across spatial separation.

Event-locality at the receiver: The Master EOM is evaluated **at the receiver event**: only the causal wake surfaces intersecting $\mathbf{x}_i(t)$ contribute to the acceleration there and then. However, it is **path-history dependent**: the active branches depend on the **entire past worldline** of all sources.

Self-Interaction (Self-Hit Dynamics)

Self-Hit Condition

An architrino i experiences self-hit at time t if there exists $t_0 < t$ such that:

$$\|\mathbf{x}_i(t) - \mathbf{x}_i(t_0)\| = c_f(t - t_0)$$

Geometric interpretation: The architrino's current position $\mathbf{x}_i(t)$ lies on the causal isochron emitted from its past position $\mathbf{x}_i(t_0)$.

Requirements:

1. **Curvature**: The worldline must curve (straight-line motion admits no self-hits).
2. **Super-field-speed interval history**: the speed must exceed c_f somewhere on the interval from emission to reception, except for the degenerate straight field-speed riding case excluded by the branch Jacobian condition.

Multiple Self-Hits (Plural)

Key insight: An architrino can experience **multiple self-hits simultaneously** (or within a regularization window η).

Mechanism: In curved motion at super-field-speed, the worldline may intersect **multiple past isochrons** at the same observation time t. Each intersection corresponds to a distinct emission time $t_{0,k} \in \mathcal{C}_{ii}(t)$.

Example: In uniform circular motion at speed $v > c_f$, an architrino can be hit by wake surfaces from multiple points on its own orbit, corresponding to different "winding numbers" $m = 0, 1, 2, \ldots$ (see Maximum-Curvature Orbit).

Sum over all self-hit roots:

$$\mathbf{F}_{ii}(\text{self-hit}) = \sum_{t_0 \in \mathcal{C}_{ii}(t)} \kappa\, \sigma_{ii} \frac{|q_i q_i|}{r_{ii}^2\, |J_{ii}(t; t_0)|} \hat{\mathbf{r}}_{ii}$$

where $\sigma_{ii} = +1$ (like polarities repel), so each self-hit contributes an **outward** (repulsive) force.

Persistent Memory (Self-Hit After Slowing Down)

Critical clarification:

Self-hit is **not** instantaneously tied to current velocity. An architrino that has **previously** exceeded $\|\mathbf{v}\| > c_f$ and emitted wake surfaces can **later slow down** to $\|\mathbf{v}\| < c_f$ and **still receive self-hits** from those earlier emissions.

Scenario:

1. At time t_1: Architrino accelerates to $\|\mathbf{v}\| > c_f$ and emits wake surfaces while in super-field-speed regime.
2. At time $t_2 > t_1$: Architrino slows down to $\|\mathbf{v}\| < c_f$ (e.g., due to partner attraction or external forces).
3. At time $t_3 > t_2$: The architrino's trajectory curves such that it intersects one of the wake surfaces emitted at t_1 (when $\|\mathbf{v}\| > c_f$).

Result: Self-hit occurs at t_3 even though current velocity $\|\mathbf{v}(t_3)\| < c_f$.

Implication: Self-hit is a **path-history memory effect**. The architrino's current acceleration depends on **whether it ever exceeded c_f in the past and curved**, not just on its instantaneous state.

Non-Markovian nature: Knowing $\mathbf{x}_i(t)$ and $\mathbf{v}_i(t)$ is insufficient to determine $\mathbf{a}_i(t)$. The **full past worldline** $\{\mathbf{x}_i(t') : t' < t\}$ is needed to identify all causal self-hit times $t_0 \in \mathcal{C}_{ii}(t)$.

Self-Hit as Stabilization Mechanism

Role in binary formation: Self-hit provides a **repulsive radial contribution** that opposes the attractive pull of opposite-polarity partners. This competition produces:

- **Maximum-curvature candidates**: the circular toy model identifies where a minimum-radius barrier must be analyzed.
- **Null-separatrix protection**: the Jacobian-degenerate boundary $J = 0$ acts as a geometric wall against collapse in the exact kernel.
- **A closure test, not a closure proof**: the same $1/|J|$ amplification multiplies tangential as well as radial projections, so a Jacobian-null branch does not by itself prove vanishing tangential power or an exact locked orbit.

Connection to quantum behavior: At this chapter's claim level, non-Markovian memory and deterministic-but-complex self-hit dynamics are a candidate substrate mechanism for effective quantum-like behavior, not yet a derivation of the quantum formalism:

- effective guidance by self-interference and causal-wake history,
- discrete stable states as attractors in phase space,
- measurement uncertainty as receiver-level informational ambiguity.

An important open problem is to map the phase-space attractor landscape for self-hit binaries, including basin size for maximum-curvature orbits, escape conditions, and the existence of secondary attractors such as long-lived elliptical families.

Worked Examples (Analytic Baselines)

Stationary Opposite Charges (Radial Fall)

Setup:

- Two architrinos: Electrino at $\mathbf{x}_1(t)$, Positrino at $\mathbf{x}_2(t)$
- Initial conditions: Both at rest, separated by distance d_0
- No self-hits (speeds remain $< c_f$ if d_0 is not too small)

Symmetry: By polarity symmetry, both fall toward their common center of mass.

Equations: Radial coordinate $r(t) = \|\mathbf{x}_2(t) - \mathbf{x}_1(t)\|$ satisfies:

$$\frac{d^2 r}{dt^2} = -\frac{2\kappa\epsilon^2}{r^2}$$

where the factor of 2 comes from the symmetry (each feels the same magnitude force).

Solution structure: This has the same quadrature structure as Keplerian radial fall at leading order in the slow, single-branch regime.

Key insight: Partner attraction dominates; no self-hit (speeds remain sub-field-speed for moderate d_0).

Sub-Field-Speed Circular Orbit (Instability)

Setup:

- Two opposite polarities in symmetric circular orbit at radius R, speed $v < c_f$
- No self-hits (sub-field-speed regime)

Partner contribution:

- Provides inward radial force (centripetal)
- Also provides **tangential force** (always positive, i.e., in direction of motion)

Result: Net tangential power $T > 0$ means the partner-only circular branch is anti-damped. It accelerates along the orbit and cannot remain a constant-speed circle. The sign of this tangential work does not prove inward tightening; any contraction must come from a separate non-circular branch, capture basin, wake-flux/recoil channel, or multi-root ledger.

Conclusion within this circular benchmark: No stable circular orbit appears in the sub-field-speed regime for isolated opposite-polarity binaries.

Maximum-Curvature Orbit (Self-Hit Stabilization)

Setup:

- A candidate opposite-polarity branch reaches super-field-speed curved history after a non-circular contraction, capture, or forced branch transition
- Self-hits activate → repulsive outward force

Geometric definition (Null Separatrix):
For an active causal root $t_0 \in \mathcal{C}_{ii}(t)$ on the self-hit branch, define

$$J_{ii}(t;t_0) \equiv 1 - \frac{\mathbf{v}_i(t_0) \cdot \hat{\mathbf{r}}_{ii}(t;t_0)}{c_f}$$

The maximum-curvature binary (MCB) boundary is the Jacobian-degenerate set

$$J_{ii}(t;t_0) = 0$$

with approach from the admissible side $J_{ii} > 0$. Geometrically, this is the state where the receiver trajectory is tangent to the causal wake surface of its own past emission (the “riding-the-shock” limit).

Why this is a hard wall in the exact theory:
In the exact branch-resolved force, the self-hit contribution carries the factor

$$\frac{1}{r_{ii}^2(t;t_0)\ |J_{ii}(t;t_0)|}$$

Hence as $J_{ii} \to 0^+$ the ideal branch-resolved pointwise response diverges, producing a restoring barrier that blocks naive continuation into a collapsing branch. This divergence should not be treated as a literal state pinned at infinite force. Across a simple caustic transit, Caustic Transit and Finite Impulse shows that the integrated velocity change can remain finite; with finite numerical regularization $\eta > 0$, the event appears as a large but finite impulse that sharpens as $\eta \to 0$.

This null-separatrix is therefore an **amplitude wall** for the self branch. It is not, by itself, a theorem of circular closure. The same branch weight multiplies every projection of the self-hit force, including the tangential component, so contact with $J_{ii} = 0$ obstructs collapse but does not by itself establish a periodic orbit or zero net cycle-averaged power.

Operational characterization of MCB:

- The inner branch evolves by caustic grazing near $J_{ii} = 0$, with finite impulses across the regularized boundary rather than exact pinning on an infinite-force surface.
- The minimum radius $R_{\min}$ is the smallest orbit radius compatible with $J_{ii} \geq 0$ on active roots.
- Tangential power must be controlled separately; near-zero cycle-average power is an additional closure condition, not a consequence of $J_{ii} = 0$ alone.

Significance:

- Defines a **fundamental length scale** $R_{\min}$ that sets the tightest stable orbit radius
- In the exact geometric model, excludes classical $r \to 0$ collapse by a null-separatrix barrier
- Supplies one geometric ingredient in candidate stable particle assemblies such as nested shell braids

Status split (analytic vs numeric):

- **Analytic:** Existence of the Jacobian-null boundary and its singular restoring scaling in the exact kernel.
- **Numeric still required:** Basin size, global attractivity, and long-time capture probability for realistic multi-body assemblies.

Informational Ambiguity at the Receiver

Limited Information Per Hit

From the perspective of the receiving architrino, the information carried by an intersecting causal isochron is **limited**. The receiver only knows:

1. The **net strength** of the potential at the point of intersection (through the acceleration magnitude $\|\mathbf{F}\|$ when force bookkeeping is used).
2. The **unoriented line of action** through its current position (the line along which the acceleration points).

The receiver does **not** have direct knowledge of:

- The source’s identity (which architrino j?)
- The source’s precise distance r_{ij} (without additional assumptions)
- The source’s velocity at emission $\mathbf{v}_j(t_0)$

Ambiguity: Electrino vs Positrino on Opposite Sides

A particularly important ambiguity: the receiver cannot distinguish between:

- A **negative potential** due to an Electrino (polarity $-\epsilon$) on one side of the line of action, and
- A **positive potential** due to a Positrino (polarity $+\epsilon$) on the **opposite side** of the same line,

if the resulting radial acceleration is the same.

Example: An acceleration **towards** a point along the line of action could be interpreted as:

- Attraction to a Positrino at that point, **or**
- Repulsion from an Electrino located at the diametrically opposite point on the same line.

Rest-Frame Recast (Useful Inference Device)

Any single hit can be **equivalently described** with a **stationary emitter** ($\|\mathbf{v}\| = 0$) placed somewhere along the same unoriented line of action, with the emitter's actual speed at emission accounted for by an adjusted emission time and, if desired, a surrogate location along that line.

Key property: The same emission law is preserved in this recast; the velocity dependence is transferred into the adjusted emission geometry and the matched Jacobian-weighted flux.

Utility: This recast simplifies some analytic calculations and provides intuition for the receiver's "inference problem" (what source configurations are consistent with a given hit?).

Superposition Complicates Inference

The ambiguity is compounded by **superposition**: The net potential at any instant is the sum of all intersecting expanding causal wake surfaces. A measured potential along a single radial can be the consequence of a **complex confluence of wakes** from many different emitters located along that line of action, arriving from both directions.

Consequence: The receiver experiences a **deterministic acceleration** (given full microstate knowledge, as known to the $\mathbb{U}_{\text{now}}$ universe-state perspective), but has **incomplete local information** about the source configuration.

Connection to Quantum Measurement Uncertainty

This limited, unoriented, and source-ambiguous information at the hit level is a candidate bridge to effective quantum-like behavior and measurement uncertainty from deterministic micro-dynamics. The bridge remains a closure target until the coarse-grained state map and record-formation dynamics are derived:

- **Wave function transition**: ψ may be interpreted as a coarse-grained representation of the wake-defined potential landscape only after a density/phase map has been supplied.
- **Measurement interaction**: an outcome is a record formed by assembly interactions and causal-hit history, not by adding a fundamental collapse postulate at this level.
- **Uncertainty**: the native candidate mechanism is informational ambiguity at the receiver plus unresolved microstate sensitivity, not ontic randomness.

Parameters and Numerical Implementation

Parameter Definitions

The core parameters entering the Master Equation are:

Parameter	Symbol	Working convention	Dimensional	Comment
Wake speed	c_f	Set to 1 in natural units unless otherwise stated	$\mathrm{L\,T^{-1}}$	Propagation speed in the causal constraint
Coupling constant	κ	Universal coupling parameter	$\mathrm{L^3\,T^{-2}\,Q^{-2}}$	Controls the strength of the inverse-square interaction
Architrino polarity unit	ϵ	$\lvert e\rvert/6$	Q	Fundamental polarity magnitude
Causal-wake-surface thickness (regularization)	η	Positive regularization width used in analysis and simulation	L	Mollifies delta singularities

In this document, c_f is treated primarily as a unit-setting convention, κ as the universal coupling scale of the delayed interaction law, ϵ as the fundamental polarity unit, and η as a regularization parameter used only when a smooth surrogate of the exact causal-wake dynamics is required.

Numerical Implementation Notes

Delay Root-Finding Algorithms

At each time step t, the numerical integrator must solve the **implicit causal constraint** for each source j:

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t - t_0), \quad t_0 < t$$

Algorithm (schematic):

1. For each source j, search the history buffer $\{\mathbf{x}_j(t') : t' < t\}$ for all t_0 satisfying the constraint.

2. Use **bisection** or **Newton-Raphson** to refine roots to tolerance ϵ_{root}.
3. If multiple roots exist (multi-hit regime), enumerate all; sum their contributions.
4. If no roots exist (source too far away or not yet causal), skip source j at this time step.

Efficiency: Use **history binning** or **spatial hashing** to avoid exhaustive search over all past times.

Spatial Hashing for History Buffers

Efficiency requirement: Naïve all-pairs history search scales as $O(N^2 T_{\text{history}})$, intractable for $N > 100$ particles.

Required optimization: Implement spatial hash grid with cell size $\sim c_f \Delta t_{\max}$; only search cells within causal range of receiver. Expected scaling: $O(N \log N)$.

Implementation notes:

- Partition spatial domain into cubic cells of side length $\Delta_{\text{cell}} \approx c_f T_{\text{history,max}}$
- At each time step, bin all architrino positions into cells
- For receiver at $\mathbf{x}_i(t)$, only search cells within causal radius $r_{\max} = c_f T_{\text{history}}$
- Update hash grid incrementally (not from scratch each step)

Time-Stepping Schemes for DDEs

The Master EOM is a **state-dependent DDE** (delay depends on the solution itself). Standard ODE integrators (e.g., RK4) must be adapted:

Recommended methods:

- **Fixed-point iteration** with predictor-corrector (for implicit delays)
- **Adaptive time-stepping** (small Δt when roots are close or numerous)
- **Event detection** for exact root crossings (optional; improves accuracy in sharp-hit regime)

Stability: Ensure $\Delta t < \eta / c_f$ (resolve mollified wake surface width); adjust η and Δt together in convergence tests.

Provenance Tracking (Emission Event → Receiver → Response)

For debugging and interpretation:

At each hit, log:

- Source ID j
- Emission time t_0
- Emission position $\mathbf{x}_j(t_0)$
- Reception time t
- Reception position $\mathbf{x}_i(t)$
- Force contribution $\mathbf{F}_{ij}(t; t_0)$

Use cases:

- Visualize causal cones and causal isochrons
- Identify self-hit events and winding numbers
- Trace energy transfer pathways
- Validate superposition (sum of logged forces = total acceleration?)

Analytic Regimes and Research Roadmap

Summary and Key Takeaways

What This Document Establishes

The **Master Equation of Motion** is the deterministic law governing the evolution of all architrinos:

$$\frac{d^2\mathbf{x}_i}{dt^2} = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij} \, \frac{|q_i q_j|}{r_{ij}^2 \, |J_{ij}(t; t_0)|} \, \hat{\mathbf{r}}_{ij}$$

Key features:

1. **Event-local at the receiver**: Only intersecting delayed causal wake surfaces contribute (no action-at-a-distance).
2. **Non-Markovian**: Depends on full path history (self-hit memory).
3. **Superposition**: Linear sum over all sources and causal roots.

4. **Self-hit**: Repulsive same-source interaction when $\mathcal{C}_{ii}(t)$ is nonempty with a valid transversality floor; super-field-speed interval history is a necessary warning condition for simple nontrivial roots and can persist as memory after slowing down.
5. **Radial line of action with Jacobian flux weighting**: No magnetic or velocity-cross-product terms; all per-hit accelerations point along $\hat{\mathbf{r}}_{ij}$, with magnitude modulated by $|J_{ij}|^{-1}$.

Implications for Emergent Phenomena

The equation supplies the microscopic input for later emergence claims, but it does not by itself prove those claims. The status split is:

- **Binary stabilization**: supported by self-hit barriers and circular/spiral benchmarks; exact stable branches still require certified branch charts and tangential-power closure.
- **Nested shell braids and particle assemblies**: downstream assembly claims that must be derived from multi-body causal-root locking and hierarchy averaging.
- **Quantum behavior**: an effective closure target based on non-Markovian memory, attractor basins, and receiver-level informational ambiguity.
- **Observer-level geometry and gravity**: effective descriptions that must be recovered from Noether sea constitutive response and clock/ruler closure, not inserted into the substrate law.
- **Cosmology**: an effective observer-side program tied to Noether sea evolution, transport, and clock-rate comparison; the Euclidean void itself is not claimed to expand.

Fully general case (arbitrary N, arbitrary trajectories)

The master EOM is a coupled system of **state-dependent delay differential equations** with:

- non-linear dependence on all worldlines,
- implicit causal roots defined by
  $\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t - t_0)$,
- potentially **multiple roots** per pair (multi-hit, self-hit),
- non-smooth behavior in the $\eta \to 0$ limit.

In PDE/DDE theory, systems of this type almost never admit closed-form analytic solutions except in toy limits.

So:

- **Claim:** There is no expectation of general analytic solutions for arbitrary N and trajectories.
- What we can aim for instead:
    - Existence/uniqueness theorems in broad classes,
    - qualitative theory (invariants, attractors, bifurcations),
    - asymptotic approximations (multipole / far-field, continuum limits),
    - special highly symmetric exact solutions.

That's standard: even Newtonian N-body gravity is analytically intractable generically; we're strictly more complex than that.

Ideal / symmetric cases where analytic work is realistic

The most tractable cases are the highly symmetric regimes in which closed forms or controlled approximations remain plausible.

Static / quasi-static limit (Coulomb analogue)

Assumptions:

- All particles move slowly: $\|\mathbf{v}_j\| \ll c_f$,
- Configuration changes on timescales long compared to light-crossing time across the system,
- No self-hits (sub-c_f everywhere, weak curvature).

Then:

- For each pair (i, j), the causal root is essentially unique and very close to the instantaneous causal-delay emission time.
- We can neglect acceleration and velocity corrections in the past-emission position.

To leading order, we should recover:

- A **Coulomb-like 1/r^2 law** between quasi-static sources,
- The usual Kepler-like two-body dynamics.

Analytic status:

- Two-body problem in this limit: solvable exactly (ellipses, etc.).
- N-body: same qualitative status as Newtonian gravity/electrostatics—no closed form in general, but standard perturbation methods apply.

This is the basic consistency-check regime of the theory.

Two-body, 1D radial motion (head-on, no angular momentum)

Setup:

- Two opposite polarities on a line, starting at rest, moving directly toward each other,
- Symmetry: center-of-mass at rest, only radial variable $r(t)$,
- Speeds sub-c_f so no self-hit.

Then:

- Causal delay gives a small correction; in the slow regime we can treat it perturbatively.
- To zeroth order, the reduced equation is:

$$\frac{d^2r}{dt^2} = -\frac{2\kappa\epsilon^2}{r^2}$$

which has an exact analytic solution for $r(t)$ (same mathematics as Kepler fall-to-center).

We can:

- Write the exact integral for $t(r)$, and invert in special cases.
- Then treat causal delay as a small parameter $\epsilon_{\mathrm{ret}} \sim r/c_f T$ and develop a systematic expansion.

So: **analytic yes** (up to standard quadratures), and corrections doable.

For the self-hit-capable reduced problem that goes beyond the sub-c_f perturbative regime and sets up a return-map breather question, see collinear-breather.md.

For the local origin-crossing theorem program in that reduced note, the working 1D model is dual-mollified rather than merely causal-surface-regularized: the causal-surface mollifier δ_η still selects delayed roots, while a separate core mollifier ϵ_c is imposed on the inverse-square amplitude so the post-crossing local vector field remains finite. That dual-mollified local model is the one used for the first recapture lemmas there.

Two-body uniform circular orbit, sub-c_f (no self-hit)

Consider the symmetric opposite-polarity circular ansatz

$$\mathbf{x}_1(t) = R(\cos\omega t, \sin\omega t, 0), \qquad \mathbf{x}_2(t) = -\mathbf{x}_1(t), \qquad \beta \equiv \frac{v}{c_f} = \frac{\omega R}{c_f} \in (0,1)$$

Fix receiver 1 at time t and let the unique partner emission time be $t_0 = t - \Delta$, with

$$\xi \equiv \frac{\omega\Delta}{2} \in \left(0, \frac{\pi}{2}\right)$$

Write $\mathbf{e}_r(t) = (\cos\omega t, \sin\omega t, 0)$ and
$\mathbf{e}_\theta(t) = (-\sin\omega t, \cos\omega t, 0)$ for the receiver polar frame.

Proposition (Unique partner branch and exact delay equation)

In the symmetric sub-c_f circular ansatz, the partner branch is unique and its delay angle ξ is the unique solution of

$$\cos\xi = \frac{\xi}{\beta}, \qquad 0 < \xi < \frac{\pi}{2}$$

Proof.
The partner separation is

$$\mathbf{r}_{12}(t;t_0) = \mathbf{x}_1(t) - \mathbf{x}_2(t_0) = R\big(\mathbf{e}_r(t) + \mathbf{e}_r(t-\Delta)\big)$$

so

$$r_{12}(t;t_0) = 2R\cos\frac{\omega\Delta}{2} = 2R\cos\xi$$

The causal condition $r_{12} = c_f \Delta$ therefore becomes

$$2R\cos\xi = c_f \frac{2\xi}{\omega}$$

hence $\cos\xi = \xi/\beta$.
Define $h_\beta(\xi) = \cos\xi - \xi/\beta$ on $[0, \pi/2]$. Then

$$h_\beta(0) = 1 > 0, \qquad h_\beta\left(\frac{\pi}{2}\right) = -\frac{\pi}{2\beta} < 0, \qquad h'_\beta(\xi) = -\sin\xi - \frac{1}{\beta} < 0$$

So h_β is strictly decreasing and has exactly one root on $(0, \pi/2)$. □

Proposition (Exact partner-only circular force decomposition)

For the unique partner branch above,

$$\hat{\mathbf{r}}_{12} = \cos\xi\,\mathbf{e}_r(t) - \sin\xi\,\mathbf{e}_\theta(t), \qquad r_{12} = 2R\cos\xi, \qquad J_{12} = 1 + \beta\sin\xi$$

Since the charges are opposite, the partner acceleration on receiver 1 is

$$\mathbf{a}_{12} = -\frac{\kappa|q_1 q_2|}{4R^2\cos^2\xi\,(1+\beta\sin\xi)}\,(\cos\xi\,\mathbf{e}_r(t) - \sin\xi\,\mathbf{e}_\theta(t))$$

Therefore the exact radial and tangential components are

$$a_r^{(\mathrm{part})} = -\frac{\kappa|q_1 q_2|}{4R^2\cos\xi\,(1+\beta\sin\xi)} < 0$$

$$a_\theta^{(\mathrm{part})} = \frac{\kappa|q_1 q_2|\,\sin\xi}{4R^2\cos^2\xi\,(1+\beta\sin\xi)} = \frac{\kappa|q_1 q_2|\,\tan\xi}{4R^2\cos\xi\,(1+\beta\sin\xi)} > 0$$

Proof.
Using

$$\mathbf{e}_r(t-\Delta) = \cos(2\xi)\,\mathbf{e}_r(t) - \sin(2\xi)\,\mathbf{e}_\theta(t)$$

one finds

$$\mathbf{r}_{12} = R\big(\mathbf{e}_r(t) + \mathbf{e}_r(t-\Delta)\big) = 2R\cos\xi\,(\cos\xi\,\mathbf{e}_r(t) - \sin\xi\,\mathbf{e}_\theta(t))$$

which gives the stated r_{12} and $\hat{\mathbf{r}}_{12}$.
The source velocity at emission is

$$\mathbf{v}_2(t_0) = -v\,\mathbf{e}_\theta(t-\Delta)$$

and

$$\mathbf{e}_\theta(t-\Delta)\cdot\hat{\mathbf{r}}_{12} = \sin\xi$$

so

$$\mathbf{v}_2(t_0)\cdot\hat{\mathbf{r}}_{12} = -v\sin\xi, \qquad J_{12} = 1 - \frac{\mathbf{v}_2(t_0)\cdot\hat{\mathbf{r}}_{12}}{c_f} = 1 + \beta\sin\xi$$

Because $\sigma_{12} = -1$ for opposite polarities, the branch acceleration is $-\kappa|q_1 q_2|\hat{\mathbf{r}}_{12}/(r_{12}^2 J_{12})$, and projecting onto $\mathbf{e}_r(t)$ and $\mathbf{e}_\theta(t)$ yields the stated components. Since $\xi \in (0, \pi/2)$, every factor in the denominators is positive and $\sin\xi > 0$, proving the sign claims. □

Corollary (Tangential positivity and circular instability)

Within the isolated partner-only circular ansatz, the tangential power is strictly positive:

$$\mathbf{a}_{12}\cdot\mathbf{v}_1(t) = v\,a_\theta^{(\mathrm{part})} > 0$$

Therefore an isolated opposite-polarity binary cannot realize an exact constant-speed circular orbit from partner delay alone.

Interpretation.

These are the exact partner-only circular formulas needed elsewhere in the chapter. They show that the delayed partner branch supplies inward radial pull, but it also drives the motion forward along $\mathbf{e}_\theta$. The circular ansatz therefore fails by anti-damping rather than by proving an inward spiral. Any tightening history must be certified on a non-circular branch or by an explicit finite-window energy ledger.

Self-hit for uniform circular motion, $v > c_f$ (single particle)

This is the key toy model for self-hit/maximum curvature.

Take:

- One architrino on a circle of radius R, angular speed ω, velocity $v = \omega R > c_f$.
- We ignore partner forces; pure self-hit geometry.

Then causal condition:

$$\|\mathbf{x}(t) - \mathbf{x}(t_0)\| = 2R\left|\sin\frac{\omega(t-t_0)}{2}\right| = c_f(t-t_0)$$

Let $\Delta = t - t_0 > 0$. Then:

$$2R\left|\sin\frac{\omega\Delta}{2}\right| = c_f\Delta$$

Introduce the dimensionless variables

$$\beta = \frac{v}{c_f} = \frac{\omega R}{c_f}, \qquad \xi = \frac{\omega\Delta}{2}$$

Then the circular self-hit condition becomes

$$\sin\xi = \frac{\xi}{\beta}, \qquad 0 < \xi < \beta$$

For fixed $\beta > 1$, the admissible self-hit set is therefore **finite**, not infinite: roots are exactly the intersections of $\sin\xi$ with the line ξ/β inside the compact interval $(0, \beta)$.

The principal branch turns on at $\beta = 1$. Writing $\beta = 1 + \mu$ with $\mu > 0$ small, the smallest root obeys

$$\xi_0 \sim \sqrt{6\mu}, \qquad \Delta_0 \sim \frac{2\sqrt{6\mu}}{\omega}, \qquad r_0 = c_f\Delta_0 \sim 2R\sqrt{6\mu}$$

The associated circular branch Jacobian is

$$J_n = 1 - \frac{\mathbf{v}(t-\Delta_n)\cdot\hat{\mathbf{r}}_n}{c_f} = 1 - \beta\cos\xi_n = 1 - \xi_n\cot\xi_n$$

On the principal branch,

$$J_0 \sim 2\mu$$

Hence the near-threshold self-hit weight scales like

$$\frac{1}{r_0^2|J_0|} \sim \frac{1}{48R^2\,\mu^2}$$

This is the circular caustic behind the null-separatrix wall: the first self branch does not merely appear at $\beta = 1$, it appears with a Jacobian-amplified weight that is already singular in the excess speed.

Higher branches are also tractable. For the circular root function

$$g_\beta(\xi) \equiv \sin\xi - \frac{\xi}{\beta}$$

new admissible roots can appear only at interior tangencies satisfying

$$g_\beta(\xi) = 0, \qquad g'_\beta(\xi) = 0$$

Eliminating β gives the tangency equation

$$\tan\xi = \xi$$

and the corresponding threshold speed is

$$\beta^\star = \sec\xi^\star$$

At every such tangency,

$$J^\star = 1 - \beta^\star \cos \xi^\star = 0$$

So each new circular self branch is born directly on a Jacobian-null boundary: branch creation and null-separatrix contact are the same event in the uniform circular toy model.

> **Proposition (Signed higher-winding circular branch birth).**
> The circular distance equation should be read branchwise as
>
> $$> g_{\beta,s}(\xi) \equiv s \sin \xi - \frac{\xi}{\beta} = 0, > \qquad > s = \operatorname{sign}(\sin \xi) \in \{+1, -1\}. >$$
>
> For each higher half-winding $n \geq 1$, set
>
> $$> I_n = \left(n\pi, \left(n + \frac{1}{2}\right)\pi\right), > \qquad > s_n = (-1)^n, >$$
>
> and let $\xi_n^\star \in I_n$ be the unique positive solution of
>
> $$> \tan \xi_n^\star = \xi_n^\star. >$$
>
> The signed branch-birth speed is
>
> $$> \beta_n^\star >=> s_n \sec \xi_n^\star >=> |\sec \xi_n^\star| >=> \sqrt{1 + (\xi_n^\star)^2}. >$$
>
> If
>
> $$> a_n = \left(n + \frac{1}{2}\right)\pi, >$$
>
> then
>
> $$> \xi_n^\star = a_n - \frac{1}{a_n} + O(a_n^{-3}), > \qquad > \beta_n^\star = a_n - \frac{1}{2a_n} + O(a_n^{-3}). >$$
>
> For $\beta = \beta_n^\star + \mu$ with $0 < \mu \ll 1$, the two newly active roots satisfy
>
> $$> \xi_{n,\pm}(\beta) >=> \xi_n^\star > \pm > \sqrt{\frac{2\mu}{\beta_n^\star}} > +O(\mu), >$$
>
> and their force-law Jacobians have opposite signs:
>
> $$> J_{n,\pm} >=> 1 - \beta s_n \cos \xi_{n,\pm} >=> \pm \xi_n^\star > \sqrt{\frac{2\mu}{\beta_n^\star}} > +O(\mu). >$$
>
> Thus a higher-winding fold creates a signed root pair on a Jacobian-null boundary. Since $r_{n,\pm} \to 2R\xi_n^\star/\beta_n^\star \neq 0$, the master-equation force-law weight scales as
>
> $$> \frac{1}{r_{n,\pm}^2 |J_{n,\pm}|} >=> O(\mu^{-1/2}). >$$
>
> The causal-action coarea weight is a separate collapse factor:
>
> $$> g'_{\beta,s_n}(\xi_{n,\pm}) >=> s_n \cos \xi_{n,\pm} - \frac{1}{\beta} >=> -\frac{J_{n,\pm}}{\beta}, >$$
>
> so the action-counting density carries an additional $|g'_{\beta,s_n}|^{-1}$ and scales as $O(\mu^{-1})$ at fixed nonzero $r_n^\star$. The coarea factor does not replace the force-law Jacobian weight; it is an additional measure factor from collapsing the causal-action integral onto the circular causal roots.
>
> Consequently the circular self-hit combinatorics remain linearly bounded in β. A one-sign subchart has
>
> $$> N_{\text{self}}^{(+)}(\beta) = \frac{\beta}{\pi} + O(1), >$$
>
> while the full signed $|\sin \xi|$ chart has the same no-proliferation form with the convention-dependent leading constant.

> **Benchmark Proposition (Circular branch-count bound).**
> In the symmetric circular benchmark, if the speed ratio obeys

$$> |\beta(t)| \le \beta_* < \infty >$$

> uniformly, then the active circular self-hit count is uniformly bounded:

$$> N_{\text{self}}(t) > \le > \frac{\beta_*}{\pi} + C_{\text{circ}}, >$$

> where

$$> C_{\text{circ}} >$$

> is an absolute endpoint-count constant for the circular root equation. This supplies the missing branch-count input in the continuation criterion for that benchmark. A general super-field-speed trajectory still needs its own no-proliferation theorem; tight spirals or repeatedly folded histories can otherwise leave the finite-branch chart even without speed blowup, collision, or a single Jacobian floor loss.

That already:

- Gives us analytic control of the causal roots (as solutions of a simple scalar transcendental),
- Lets us write the self-force as

$$\mathbf{a}_{\text{self}}(t) = \sum_n \kappa \frac{q^2}{r_n^2\,|J_n|} \hat{\mathbf{r}}_n$$

with $r_n = c_f \Delta_n$, $J_n = 1 - \mathbf{v}(t - \Delta_n) \cdot \hat{\mathbf{r}}_n / c_f$, and directions that can be written explicitly in terms of the phase difference.

We will not get a *closed-form sum*, but:

- The geometry is 100% analyzable,
- At high speed the number of admissible roots grows only linearly with β because all roots lie in $(0, \beta)$,
- Large-n roots admit asymptotic expansions,
- We can show convergence of the self-force series away from Jacobian-degenerate roots ($J = 0$),
- And derive asymptotic radial/tangential components as functions of v/c_f.

Near the null-separatrix condition $J \to 0$, the exact branch weight carries a $1/|J|$ singularity (see Maximum-Curvature Orbit above), so this toy model also captures the geometric-wall limit.

This is again an amplitude statement, not a closure theorem. A circular self branch born on $J = 0$ is born with singular weight, but the same singular factor multiplies tangential as well as radial projections. The null wall therefore obstructs continuation through the branch boundary without, by itself, proving an exactly locked circular orbit.

So: **strong analytic handle**, though not “closed form in elementary functions.”

This is the right playground to:

- Derive a condition for equilibrium between self-repulsion and an imposed centripetal requirement,
- Define $R_{\min}(v)$ and in particular the extremal radius / speed.

The same circular chart also gives a branchwise force decomposition. On the positive-sine self-hit subchart, every active root satisfies

$$\sin\xi = \frac{\xi}{\beta}, \qquad r = 2R\sin\xi = 2R\frac{\xi}{\beta}, \qquad J = 1 - \beta\cos\xi = 1 - \xi\cot\xi$$

Resolving the line-of-action direction into the instantaneous circular frame gives

$$\hat{\mathbf{r}}(\xi) = \sin\xi\,\mathbf{e}_r + \cos\xi\,\mathbf{e}_\theta$$

With

$$C = \frac{\kappa q^2}{4R^2}$$

the branchwise self-hit projections are therefore

$$a_r(\xi) = C\,\frac{\beta}{\xi|J|}, \qquad a_\theta(\xi) = C\,\frac{\beta^2\cos\xi}{\xi^2|J|}$$

Thus the radial projection is outward on every active self root, while the tangential projection is controlled entirely by the sign of $\cos\xi$.

The branch sheets have the following one-sign structure:

Sheet	Root status for $\sin\xi = \xi/\beta$	Radial projection	Tangential projection
Negative sine lobes	No roots, because $\xi/\beta > 0$	Inactive	Inactive
First positive lobe	One nonzero root for $\beta > 1$	Outward	Forward for $\cos\xi > 0$, backward for $\cos\xi < 0$
Higher positive left sheets	One root after birth from a Jacobian-null fold	Outward	Forward
Higher positive right sheets	Paired root after the same birth	Outward	Backward

For large β, away from arbitrarily small Jacobian-null birth windows, the signed circular self-hit sums obey

$$A_r(\beta) = \sum_{\xi_n} a_r(\xi_n) = \frac{C}{\pi}\log\beta + O(C)$$

and

$$A_\theta(\beta) = \sum_{\xi_n} a_\theta(\xi_n) = -\frac{C\beta}{12} + O(C\log\beta)$$

The corresponding absolute tangential activity is

$$\sum_{\xi_n} |a_\theta(\xi_n)| = \frac{C\beta}{6} + O(C\log\beta)$$

The full signed $|\sin\xi|$ circular chart uses $s = \operatorname{sign}(\sin\xi)$ and

$$s\sin\xi = \frac{\xi}{\beta}, \qquad J = 1 - \beta s\cos\xi, \qquad \hat{\mathbf{r}}(\xi) = |\sin\xi|\,\mathbf{e}_r + s\cos\xi\,\mathbf{e}_\theta$$

Thus the full signed-chart projections are

$$a_r^{|\sin|}(\xi) = C\,\frac{\beta}{\xi|J|}, \qquad a_\theta^{|\sin|}(\xi) = C\,\frac{\beta^2 s\cos\xi}{\xi^2|J|}$$

The radial contribution is still outward on every active self root. The tangential contribution is forward on each left sheet and backward on each right sheet, independent of the sine-lobe sign. In the full signed chart, the order-β signed tangential terms cancel pairwise between adjacent sheets, giving the bound

$$A_r^{|\sin|}(\beta) = \frac{2C}{\pi}\log\beta + O(C), \qquad A_\theta^{|\sin|}(\beta) = O(C)$$

again away from arbitrarily small Jacobian-null birth windows. The absolute tangential activity remains large:

$$\sum_{\xi_n} |a_\theta^{|\sin|}(\xi_n)| = \frac{C\beta}{3} + O(C\log\beta)$$

Pure circular self-hit is therefore not tangentially neutral branchwise. It supplies outward radial support and large cancelling forward/backward tangential activity; on the positive-sine subchart alone the signed large-β residue is backward and order β, while on the full signed chart the linear signed terms cancel to a bounded remainder. This corrects the stronger blanket statement that self branches are always positive-tangential, without by itself proving or disproving full binary closure.

Large-β partner/self circular residual

The exact partner branch can now be combined with the self-hit sums to get a high-speed obstruction for the equal-magnitude bare circular binary. Let $\xi_p(\beta)$ solve

$$\cos\xi_p = \frac{\xi_p}{\beta}, \qquad 0 < \xi_p < \frac{\pi}{2}$$

and set

$$C = \frac{\kappa q^2}{4R^2}$$

Then

$$\xi_p = \frac{\pi}{2} - \frac{\pi}{2\beta} + O(\beta^{-2})$$

so the partner projections satisfy

$$a_\theta^{(\mathrm{part})} = \frac{4C}{\pi^2}\beta + O(C), \qquad a_r^{(\mathrm{part})} = -\frac{2C}{\pi} + O(C\beta^{-1})$$

On the positive-sine self chart,

$$a_\theta^{(\mathrm{part})} + A_\theta(\beta) = C\left(\frac{4}{\pi^2} - \frac{1}{12}\right)\beta + O(C\log\beta) > 0$$

for sufficiently large β, and

$$a_r^{(\mathrm{part})} + A_r(\beta) = \frac{C}{\pi}\log\beta - \frac{2C}{\pi} + O(C)$$

is outward for sufficiently large β outside Jacobian-null birth windows.

On the full signed $|\sin\xi|$ chart,

$$a_\theta^{(\mathrm{part})} + A_\theta^{|\sin|}(\beta) = \frac{4C}{\pi^2}\beta + O(C) > 0$$

while

$$a_r^{(\mathrm{part})} + A_r^{|\sin|}(\beta) = \frac{2C}{\pi}\log\beta - \frac{2C}{\pi} + O(C)$$

is again outward for sufficiently large β. Thus, on this equal-magnitude bare circular chart and away from Jacobian-null birth windows, an exact constant-radius orbit is excluded for sufficiently large β: the tangential residual remains forward, and the radial branch sum does not provide the required inward acceleration $-\omega^2 R$. This is a conditional high-speed chart result, not a global MCB no-go theorem and not a finite-β exclusion across all ledgers; any surviving finite-speed window still requires a certified branch chart with positive Jacobian floor, inactive gaps, finite memory depth, and signed residual closure.

The circular self-hit and partner-hit formulas are kernel benchmarks. They are not the Noether braid model. The Noether braid model is the six-body branch chart containing self, partner, and inter-layer causal roots, with hierarchy averaging only where justified by separated scales and certified branch data.

Maximum-curvature binary (inner binary idealization)

For the full **two-body** maximum-curvature orbit (inner binary), we have:

- Two charges on roughly circular orbits about their COM,
- Both potentially with self-hit,
- Plus partner forces with causal delay.

Analytic expectations:

- An *exact closed form* is very unlikely.
- But:
  - We can construct a controlled circular ansatz:
    - Assume perfectly circular orbits with fixed R, ω,
    - Compute partner force including causal delay (as in Sub-Field-Speed Circular Orbit (Instability)),
    - Compute self-force (as in Self-Interaction (Self-Hit Dynamics)),
    - Demand that time-averaged radial force gives exactly $\omega^2 R$,
    - Demand that time-averaged tangential force vanish.
  - That gives us a **pair of algebraic conditions** in R and ω (or equivalently R and v).
  - Solving those algebraic conditions (perhaps numerically) defines a maximum-curvature solution family.

However, the circular benchmark still exposes a serious obstruction in the bare two-body kernel. In the symmetric isolated binary, every active partner branch contributes a positive tangential component. The self sector is different: the circular self-hit branch table above gives outward radial support, while the tangential projection changes sign by sheet; on the full signed chart its order-β signed tangential terms cancel to a bounded remainder, with order-β absolute tangential activity still present. Therefore the former blanket self-branch positive-tangential claim is too strong. Exact constant-speed closure of the bare circular two-body ansatz is not ruled out by a branchwise sign argument alone; it requires the actual signed partner-plus-self tangential sum to vanish on a certified branch chart.

This sharpens the maximum-curvature program into a concrete fork:

- either the certified signed branch sum fails to cancel the partner drive, so the isolated two-body MCB does **not** exist as an exact constant-speed circular orbit of the bare kernel, or
- an algebraic cancellation exists, after which stability still requires a separate delay-operator proof and may require additional structure beyond the bare circular two-body ansatz, such as medium coupling, genuine nested shell braid multi-body locking, or a more subtle non-circular periodic balance.

So:

- Analytically: we can reduce the existence question to algebraic conditions and asymptotic expansions, and in the bare circular ansatz we can identify the partner-positive/self-signed tangential balance that any closure certificate must satisfy.
- Dynamically: the stability question remains separate from the algebraic construction and requires numerical analysis of attractivity versus fine-tuned orbit families.

The expected answer for the bare two-body kernel is instability, not robust attraction. Existence of a circular or maximum-curvature solution would only solve the algebraic balance conditions

$$\overline{F}_{\mathrm{rad}}(R,v)=\omega^2 R,\qquad \overline{F}_{\tan}(R,v)=0$$

Stability is a different question: linearizing the delayed dynamics about the candidate orbit gives a delay operator

$$L(\lambda)$$

and the characteristic equation

$$\det(\lambda I-L(\lambda))=0$$

Any root with

$$\operatorname{Re}\lambda>0$$

is an unstable mode.

> **Target Proposition (MCB transverse stability diagnostic).**
> For any candidate bare two-body maximum-curvature binary, compute the linearized delay operator on radial and tangential perturbations. The null-separatrix self-hit wall may stabilize or block the radial collapse channel; partner branches supply sign-definite forward drive, while circular self branches supply outward radial support and a signed tangential channel whose full signed-chart linear terms cancel only after summing adjacent sheets. Thus a bare MCB should be treated as an uncertified organizing orbit in

$$>(R,v)>$$

> space until the net signed tangential balance and transverse eigenvalues are certified.

This is the intended dynamical interpretation. Stable particles in the present architecture are Noether braid assemblies; a bare MCB, if it exists, is a high-curvature component or limiting scaffold whose instability explains why additional locking structure is needed.

This would be an “analytic scaffold + numerical check” situation, not full closed forms.

Symmetric delayed spiral (advanced non-circular benchmark)

The circular obstruction makes a non-circular benchmark worthwhile. A workable first ansatz is the symmetric logarithmic spiral

$$r(\theta)=R_0e^{-a\theta},\qquad t(\theta)=\frac{\theta}{\Omega},\qquad \mathbf{x}_1(\theta)=r(\theta)\,\mathbf{e}_r(\theta),\qquad \mathbf{x}_2(\theta)=-r(\theta)\,\mathbf{e}_r(\theta)$$

with fixed pitch $a>0$ and constant angular rate $\Omega>0$.

The variable-pitch extension replaces the constant pitch by

$$p(\theta)\equiv-\frac{r'(\theta)}{r(\theta)}$$

At a source angle $\theta_0=\theta-\Delta$, write

$$p_0\equiv p(\theta-\Delta),\qquad \omega_0\equiv\dot{\theta}(\theta-\Delta),\qquad \rho\equiv\frac{r(\theta-\Delta)}{r(\theta)}$$

The logarithmic benchmark is the special case $p(\theta)=a$, $\omega_0=\Omega$, and $\rho=e^{a\Delta}$. This extension is useful because a true minimum-radius event requires

$$\dot{r} = 0, \qquad \ddot{r} \geq 0$$

which in the pitch variable means

$$p(\theta_*) = 0, \qquad p'(\theta_*) \leq 0$$

when $\dot{\theta}(\theta_*) \neq 0$.

For a receiver event at angle θ and a partner emission at $\theta_0 = \theta - \Delta$ with $\Delta > 0$, define

$$\Lambda_p(\theta, \Delta) \equiv \sqrt{1 + \rho^2 + 2\rho \cos \Delta}$$

Then

$$\mathbf{r}_{12}(\theta; \theta_0) = r(\theta) \Big[(1 + \rho \cos \Delta) \mathbf{e}_r(\theta) - \rho \sin \Delta \, \mathbf{e}_\theta(\theta) \Big]$$

so the exact delayed-hit condition is

$$r(\theta) \, \Lambda_p(\theta, \Delta) = c_f \, (t(\theta) - t(\theta - \Delta))$$

For constant angular rate this reduces to

$$\Lambda_p(\theta, \Delta) = \frac{\Delta}{b(\theta)}, \qquad b(\theta) \equiv \frac{\Omega r(\theta)}{c_f}$$

which is the non-circular analogue of the circular partner equation $\cos \xi = \xi / \beta$.

The receiver Frenet frame for the variable-pitch spiral is

$$\hat{\mathbf{T}} = \frac{-p \, \mathbf{e}_r(\theta) + \mathbf{e}_\theta(\theta)}{\sqrt{1 + p^2}}, \qquad \hat{\mathbf{N}} = \frac{-\mathbf{e}_r(\theta) - p \, \mathbf{e}_\theta(\theta)}{\sqrt{1 + p^2}}$$

where $p = p(\theta)$ and $\hat{\mathbf{N}}$ points inward in the circular limit.
Using the branch unit vector

$$\hat{\mathbf{r}}_{12} = \frac{(1 + \rho \cos \Delta) \mathbf{e}_r(\theta) - \rho \sin \Delta \, \mathbf{e}_\theta(\theta)}{\Lambda_p}$$

the partner source-velocity projection entering the Jacobian is

$$\mathbf{v}_2(\theta - \Delta) \cdot \hat{\mathbf{r}}_{12} = \frac{r(\theta) \rho \, \omega_0}{\Lambda_p} \Big[p_0 (\cos \Delta + \rho) - \sin \Delta \Big]$$

Hence

$$J_{12} = 1 + \frac{r(\theta) \rho \, \omega_0}{c_f \, \Lambda_p} \Big[\sin \Delta - p_0 (\cos \Delta + \rho) \Big]$$

The sign is fixed by the circular limit: when $p_0 = 0$ and $\rho = 1$, this gives $J_{12} = 1 + \beta \sin(\Delta/2)$.

For opposite polarities, the branch acceleration is

$$\mathbf{a}_{12} = \frac{-\kappa |q_1 q_2|}{r(\theta)^2 \Lambda_p^2 |J_{12}|} \, \hat{\mathbf{r}}_{12}$$

Projecting onto the variable-pitch Frenet frame gives

$$a_T^p = \frac{\kappa |q_1 q_2|}{r(\theta)^2 \Lambda_p^3 |J_{12}| \, \sqrt{1 + p^2}} \Big[p (1 + \rho \cos \Delta) + \rho \sin \Delta \Big]$$

$$a_N^p = \frac{\kappa |q_1 q_2|}{r(\theta)^2 \Lambda_p^3 |J_{12}| \, \sqrt{1 + p^2}} \Big[1 + \rho \cos \Delta - p \rho \sin \Delta \Big]$$

The partner tangential numerator is therefore

$$S_T^p(\theta, \Delta) \equiv p (1 + \rho \cos \Delta) + \rho \sin \Delta$$

The missing self-branch analogue uses

$$\Lambda_s(\theta,\Delta) \equiv \sqrt{1+\rho^2-2\rho\cos\Delta}$$

$$\hat{\mathbf{r}}_{11} = \frac{(1-\rho\cos\Delta)\mathbf{e}_r(\theta) + \rho\sin\Delta\,\mathbf{e}_\theta(\theta)}{\Lambda_s}$$

The self-hit delay equation is

$$r(\theta)\,\Lambda_s(\theta,\Delta) = c_f\,(t(\theta)-t(\theta-\Delta))$$

and the self-branch Jacobian is

$$J_{11} = 1 - \frac{r(\theta)\rho\,\omega_0}{c_f\,\Lambda_s}\Big[\sin\Delta + p_0(\rho-\cos\Delta)\Big]$$

Again the circular limit agrees with the uniform circular self-hit formula, $J_{11} = 1-\beta\cos(\Delta/2)$.

For self-hit, $\sigma_{11} = +1$, so

$$\mathbf{a}_{11} = \frac{\kappa q_1^2}{r(\theta)^2\Lambda_s^2|J_{11}|}\,\hat{\mathbf{r}}_{11}$$

The self-branch tangential projection is

$$a_T^s = \frac{\kappa q_1^2}{r(\theta)^2\Lambda_s^3|J_{11}|\,\sqrt{1+p^2}}\Big[-p(1-\rho\cos\Delta)+\rho\sin\Delta\Big]$$

so

$$S_T^s(\theta,\Delta) \equiv -p(1-\rho\cos\Delta)+\rho\sin\Delta$$

The circular obstruction is now converted into a branch-chart test. A non-circular spiral can beat the isolated circular tangential obstruction only if the certified active roots satisfy a negative weighted tangential sum on enough of the controlled cycle:

$$\sum_{\text{part}} \frac{|q_1q_2|\,S_T^p}{\Lambda_p^3|J_{12}|} + \sum_{\text{self}} \frac{q_1^2\,S_T^s}{\Lambda_s^3|J_{11}|} < 0$$

after the common positive factors are removed. Algebraic sign allowance is not enough; the delayed-root equations must actually admit those roots with positive Jacobian floors and finite memory depth.

At a minimum-radius event θ_*, the pitch condition gives $p(\theta_*) = 0$. Therefore both tangential numerators reduce locally to

$$S_T^p(\theta_*,\Delta) = S_T^s(\theta_*,\Delta) = \rho\sin\Delta$$

Principal roots with $0<\Delta<\pi$ still carry the same positive tangential sign as the circular benchmark. The only bare-kernel escape routes are therefore:

1. admissible older or wrapped roots with $\sin\Delta < 0$ and enough Jacobian weight;
2. off-turn variable-pitch intervals where the p-terms dominate the positive principal branches;
3. additional medium, nested shell braid, or multi-body structure outside the isolated two-body spiral ansatz.

The radial turn condition is equally explicit. Since

$$\ddot{r} = a_r + r\dot{\theta}^2$$

at a point with $\dot{r}=0$, a minimum-radius turn requires

$$r_*\dot{\theta}_*^2 - \sum_{\text{part}} \frac{\kappa|q_1q_2|\,(1+\rho_p\cos\Delta_p)}{r_*^2\Lambda_p^3|J_{12,p}|} + \sum_{\text{self}} \frac{\kappa q_1^2\,(1-\rho_s\cos\Delta_s)}{r_*^2\Lambda_s^3|J_{11,s}|} > 0$$

This is a theorem target, not a closure proof. It supplies the concrete falsification gate: enumerate the admissible partner and self roots on a variable-pitch candidate, certify their Jacobian floors, and test both the radial turn inequality and the weighted tangential sum. If all admissible roots keep the weighted tangential sum nonnegative on every candidate turn corridor, the bare isolated spiral does not beat the circular obstruction.

For a retained chart at a turn center, the radial row can be normalized by the common force factor, but that normalization separates the branch sum from the independent force ratio. In the equal-magnitude opposite-polarity case, one may write

$$\Gamma \equiv \frac{r_*^3\Omega^2}{\kappa q_1^2}, \qquad B_r(\theta_*) = -\sum_{\text{part}} \frac{1+\rho_p\cos\Delta_p}{\Lambda_p^3|J_{12,p}|} + \sum_{\text{self}} \frac{1-\rho_s\cos\Delta_s}{\Lambda_s^3|J_{11,s}|}$$

so the normalized turn row is

$$\Gamma + B_r(\theta_*) > 0$$

The retained branch chart fixes B_r only. It does not determine Γ from $b_* = \Omega r_*/c_f$, from the delayed-root offsets, or from a branch-sum threshold. A branch certificate must therefore either supply an independently derived force-ratio interval or report the radial row as blocked.

A fixed retained-chart benchmark illustrates a sharper prescribed-history failure. For the $a_{\mathrm{A1}} = 0.204$, $b_* = 7/2$ constant-Ω variable-pitch spiral on $I_* = [-\pi/6, \pi/6]$, the retained $3+1$ chart has certified active-root, inactive-gap, Jacobian-floor, finite-memory, and root-transport rows. Its exact radial kinematics at $\theta_* = 0$ fix the force-ratio row by

$$B_r(C_{\mathrm{A1}}; 0) = (a_{\mathrm{A1}} - 1)\Gamma, \qquad \Gamma \in [0.007531050241046427,\ 0.007531144882881889]$$

which strictly passes the minimum-turn inequality. The same prescribed history fails exact tangential compatibility at the turn center: constant Ω and $p(0) = 0$ require the normalized pointwise tangential force sum $T_0(C_{\mathrm{A1}})$ to vanish, while the retained chart gives

$$T_0(C_{\mathrm{A1}}) \in [-0.007585901776635041,\ -0.007585740886803276]$$

Thus A1 is a constant-Ω kinematic-balance no-go for this prescribed isolated two-body history. It remains a replayable retained-chart benchmark, not a closed isolated spiral certificate and not a rejection of variable-angular-rate, medium-supplemented, nested shell braid, or other non-circular histories.

The no-go is also constructive. If the same turn-center radial curve is allowed a variable angular rate, with $\omega_* = \dot\theta(0) > 0$ and $\alpha_* = \ddot\theta(0)$, then $r'(0) = 0$ and the exact local balance equations become

$$B_r(C_{\mathrm{A1}}; 0) = (a_{\mathrm{A1}} - 1)\Gamma_*, \qquad T_0(C_{\mathrm{A1}}) = \Gamma_* \frac{\alpha_*}{\omega_*^2}$$

where $\Gamma_* = r_*^3\omega_*^2/(\kappa q_1^2)$. Combining the retained A1 intervals gives

$$\frac{\alpha_*}{\omega_*^2} \in [-1.0072833846320208,\ -1.007249363114164]$$

Thus the constant-Ω failure supplies a precise local angular-deceleration target for a variable-angular-rate continuation. It does not by itself close such a continuation, because the delayed roots and Jacobian weights must be recomputed for the nonconstant time law.

The stronger invariant form of the target is the angular slope of the time law,

$$\left.\frac{d}{d\theta}\log\dot\theta\right|_{\theta=0} = \frac{\ddot\theta(0)}{\dot\theta(0)^2} = \frac{T_0(C_{\mathrm{A1}})}{\Gamma_*}$$

However, the delayed roots are controlled by a finite-memory integral, not by this local slope alone. If

$$H(\Delta) = \omega_* \int_{-\Delta}^{0} \frac{d\phi}{\dot\theta(\phi)}$$

then the turn-center root equation is $\Lambda_{P/S}(0, \Delta) = H(\Delta)/b_*$. Retaining an old constant-rate root at the same offset would require $H(\Delta_\alpha) = \Delta_\alpha$, or

$$\int_{-\Delta_\alpha}^{0} \left(\frac{\omega_*}{\dot\theta(\phi)} - 1\right) d\phi = 0$$

Thus the variable-rate A1 continuation is a finite-memory time-law problem: the local angular-deceleration target must be reconciled with inverse-rate averages over the delayed branch intervals. Simple one-parameter extensions of the local slope do not preserve A1; they either lose the retained roots or move to a branch ledger with the wrong radial sign for positive Γ.

This finite-memory condition is nevertheless not an algebraic no-go at the turn center. In past-lag coordinates $x = -\phi$, define

$$q(x) = \frac{\omega_*}{\dot\theta(-x)}$$

A retained-root profile must satisfy both moment and endpoint constraints,

$$\int_0^{\Delta_\alpha} \big(q(x) - 1\big)\, dx = 0, \qquad q(\Delta_\alpha) = 1$$

for each retained A1 delay. Because the local target gives $q'(0) = T_0(C_{\mathrm{A1}})/\Gamma_* < 0$, the inverse-rate profile dips below 1 just behind the turn and must compensate by rising above 1 before the first retained delay. A positive retained-root inverse-rate profile can satisfy these constraints, keep the active source-speed factors at their constant-rate values at the retained offsets, and make the same branch sums give the required local angular-rate slope.

The first off-center transport row is also fixed at the turn center. If $q_\theta(u) = \dot\theta(\theta)/\dot\theta(\theta-u)$ and $H(\theta,\Delta) = \int_0^\Delta q_\theta(u)\,du$, then the retained endpoint constraints imply

$$\partial_\theta H(\theta,\Delta_\alpha)|_{\theta=0} = k_*\Delta_\alpha, \qquad k_* = \frac{T_0(C_{\mathrm{A1}})}{\Gamma_*}$$

Since $b'(\theta)/b(\theta) = k_*$ at the turn center, the first θ-derivative of H/b cancels at the retained endpoints. Thus the retained-memory witness inherits the constant-chart first-order root-transport identity at $\theta = 0$. This is still only a branch-chart search target, not an orbit certificate: the active roots, inactive gaps, source-speed Jacobians, finite-memory depth, generalized root-transport residuals, and force-balance rows still have to be recomputed on a finite θ interval for the chosen nonconstant time law.

The finite-collar target can be stated without adding a new law. Let

$$Q(\theta) = \frac{\omega_*}{\dot\theta(\theta)}, \qquad \sigma(\theta) = \frac{r(\theta)}{r_*}, \qquad K_Q(\theta,\Delta) = \int_{\theta-\Delta}^{\theta} Q(\phi)\,d\phi$$

Then the transported retained-root equation is

$$F_{\alpha,Q}(\theta,\Delta) = \Lambda_\alpha(\theta,\Delta) - \frac{K_Q(\theta,\Delta)}{b_*\sigma(\theta)} = 0$$

At each retained endpoint, $K_Q(0,\Delta_\alpha) = \Delta_\alpha$ and $\partial_\theta K_Q(0,\Delta_\alpha) = 0$, so the first memory drift begins at second order in θ. The branch-chart certificate must bound that drift while satisfying the tangential transport equation for Q and the radial residual on the same active root ledger.

A local convergence diagnostic can sharpen this finite-collar target, but it does not by itself fix the full continuation class. After the tangential row is imposed on the retained ledger, the transported radial row should be tested through the leading one-sided jet of $\mathcal{R}_R^{\mathrm{tr}}(\theta)$ near $\theta = 0$. For a specified tangential-transport profile, the jet coefficient is

$$\left(\mathcal{R}_R^{\mathrm{tr}}\right)'_+(0) = B'_+(0) - (3a-2)T_0$$

The retained endpoint and moment constraints do not yet fix all source-side endpoint-slope data entering $B'_+(0)$. A nonzero sampled coefficient is therefore a local obstruction candidate for that profile, not a theorem that every positive C^2 variable-rate continuation fails.

A sampled endpoint-slope construction sharpens the same caution. By perturbing the retained past inverse-rate profile while preserving the retained endpoint values, moment rows, compact C^2 tail, and center slope, one can cancel the leading affine radial jet at sampled level and still keep a positive retained past profile with the expected $3+1$ active-root ledger after tangential transport. This does not certify A1 closure. It moves the theorem-grade burden to finite-collar control after endpoint-slope cancellation: positivity, inactive gaps, Jacobian floors, finite memory, tangential transport, and the full radial residual must all be bounded on the same branch chart.

Effective Continuum Limits

Another class of analytic work appears only after coarse-graining the microscopic DDE:

Homogeneous, isotropic Noether Sea

Assume:

- Very large number of architrinos,
- Statistically homogeneous and isotropic distribution,
- Global neutrality.

Then, at coarse-grained level:

- Symmetry dictates the net acceleration on a test architrino at rest is zero.

- Small perturbations can be analyzed by linearizing around the homogeneous background.

We can:

- derive an effective wave equation for small perturbations in density or potential diagnostics,
- show that disturbances propagate at an emergent channel speed tied to c_f and medium response,
- recover Maxwell-like or acoustic-like behavior as effective continuum behavior.

These are field-theory-style analytic solutions (plane waves, Green's functions) of the **coarse-grained** equations, not of the micro DDEs. They are useful only when the continuum variables are explicitly derived from the master equation by a declared coarse-graining limit.

This regime is analytically tractable and important for:

- Emergent electromagnetism,
- Emergent metric propagation (gravitational-wave analogues),
- Stability of the Noether sea itself.

Analytic footholds and remaining targets

Several formerly open checks are now footholds rather than blank targets:

1. **Partner-only circular orbit with causal delay ($v < c_f$)** now has explicit radial and tangential components, including the positive tangential-drive obstruction for a bare constant-speed circle.
2. **Uniform circular self-hit ($v > c_f$)** now has principal-root onset asymptotics, signed higher-winding branch birth, branchwise radial/tangential projections, and large-β self-hit estimates.
3. **Variable-pitch spiral retained-chart benchmarks** now expose both branch-chart rows and prescribed-history compatibility rows. The fixed A1 constant-Ω history has certified active-root, inactive-gap, Jacobian-floor, finite-memory, and root-transport rows; its exact radial kinematics fix Γ in the accepted normalization and pass the minimum-turn inequality, while the exact turn-center tangential residual excludes zero. A1 is therefore a replayable constant-Ω kinematic-balance no-go for that prescribed isolated two-body history, not a closure result and not a global no-go for non-circular histories. The same calculation turns the failure into a local continuation equation: a variable-angular-rate A1 turn would need $\ddot{\theta}(0)/\dot{\theta}(0)^2 \in [-1.0072833846320208,\ -1.007249363114164]$ before the delayed-root chart is recomputed for the new time law. The recomputation is now sharpened as a finite-memory problem: the root equation uses $H(\Delta) = \omega_* \int_{-\Delta}^{0} d\phi/\dot{\theta}(\phi)$, so any viable nonconstant A1 history must match branch-memory averages as well as the local turn slope. A retained-root inverse-rate profile can satisfy those turn-center memory equations, its endpoint constraints cancel the first off-center derivative of H/b at $\theta = 0$, and sampled endpoint-slope freedom can cancel the leading transported radial jet. The remaining burden is finite-interval transport and radial control of that profile class inside a certified branch chart rather than a pointwise or first-order algebraic obstruction.

The remaining analytic targets are sharper:

1. build the maximum-curvature branch certificate from active roots, inactive gaps, Jacobian floors, finite memory, root transport, returned-section residuals, radial/tangential balance, and the independent force-ratio row;
2. coarse-grain the master equation around a homogeneous Noether sea and extract the linear response and dispersion relation $\omega(k)$;
3. prove which regularized energy diagnostic is actually induced by a symmetry-preserving action-level regularization.

These targets keep the bridge between the formal law and the broader closure program mathematical: a branch chart, a conserved charge, or a response equation must be supplied before a stability or mass claim is promoted.

Bottom line

- **General N-body analytic solution:** No; the structure is too complex (DDE with state-dependent delays and self-hit multiplicity).
- **Idealized / symmetric cases:** Yes, in several important classes:
  - 1D radial two-body,
  - sub-c_f circular orbit,
  - uniform circular self-hit,
  - algebraic maximum-curvature conditions,
  - continuum/wave limits of the Noether sea.

Energy, Symmetry, and Conservation

Energy, Lagrangian, and Hamiltonian Structure of the Architrino Dynamics

In this section we outline how **energy** and **variational structure** are handled in $\mathbb{AAA}$, given the Master Equation of Motion:

$$\frac{d^2\mathbf{x}_i}{dt^2} = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij}\, \frac{|q_i q_j|}{r_{ij}^2(t;t_0)\ |J_{ij}(t;t_0)|}\, \hat{\mathbf{r}}_{ij}(t;t_0)$$

where each contribution comes from a **causal wake intersection** at time t between architrino i and a wake emitted by architrino j at earlier time t_0. The set $\mathcal{C}_{ij}(t)$ encodes all such emission times selected by the causal constraint

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\| = c_f(t - t_0), \quad t_0 < t$$

Once any internal binary reaches the $v > c_f$ regime at some stage in its curved history, **self-hit** becomes a live branch candidate and must be checked explicitly in realistic energy accounting. Completed assemblies cannot be assigned a “no self-hit” energy row merely from current sub-field-speed motion; the retained path history must show that same-source roots are absent or inactive with a certified branch gap.

We organize the discussion into four pieces:

1. Aggregate kinetic energy for a finite, isolated set of architrinos,
2. An action-level nonlocal Noether energy charge compatible with path-history dynamics,
3. A nonlocal Lagrangian scaffold whose variations reproduce the Master Equation only when the constraint residual closes,
4. A corresponding Hamiltonian / total energy functional, with energy exchange only at $t = \text{now}$ between architrinos.

Aggregate Kinetic Energy

We work with **absolute time** t and Euclidean 3-space. For each architrino i, define:

- Position $\mathbf{x}_i(t)$,
- Velocity $\mathbf{v}_i(t) = d\mathbf{x}_i/dt$,
- Optional universal bookkeeping constant μ_{arch} when a quadratic kinetic proxy is desired.

We do **not** a priori assign energy to any continuous field; energy is carried by architrinos and their assemblies and is updated only at the instants where wake surfaces intersect receivers.

Definition (Quadratic kinetic bookkeeping proxy). For a finite isolated set of architrinos $\{i = 1, \ldots, N\}$,

$$K_\mu(t) \equiv \sum_{i=1}^{N} \frac{1}{2} \mu_{\text{arch}} \|\mathbf{v}_i(t)\|^2$$

Remarks:

- This is a bookkeeping choice for analysis, numerics, and Noether-style energy accounting. The substrate law itself remains acceleration-first.
- Because μ_{arch} is universal, it can be absorbed into units or into an overall normalization of force-like quantities if desired.
- For assemblies (binaries, tri-binaries), one defines an effective assembly mass M_{assembly} as

$$M_{\text{assembly}} = \frac{1}{V_{\text{CM}}} \frac{d}{dV_{\text{CM}}} (\text{total kinetic} + \text{interaction energy of internal motion})$$

where V_{CM} is the center-of-mass speed. In practice, this is computed from the internal architrino motions (e.g., the tight inner binary self-hit orbit plus its interaction with partner binaries).

Thus kinetic energy splits naturally into:

- **Internal kinetic energy** of bound assemblies (setting rest mass),
- **Center-of-mass kinetic energy** of assemblies relative to the Noether sea.

Action-Level Nonlocal Noether Energy

With finite-speed causal wakes and path-history dependence, an instantaneous position-only potential is not fundamental. Time-translation symmetry of a symmetry-preserving nonlocal action model supplies the corresponding nonlocal Noether charge. The formulas in this subsection therefore belong to the action-derived delayed model, not to every regularized implementation of the Master Equation.

For the dual-mollified local 1D model used later in collinear-breather.md, the same conservation language should be read more carefully: the causal-surface mollifier δ_η and core mollifier ϵ_c support a finite local vector field and a tractable return-map theorem program, but exact Noether-charge statements transfer automatically only if that dual mollification is itself derived from a time-translation-invariant action-level regularization of the causal kernel.

Energy exchange per causal hit

Consider a single contribution to the acceleration of architrino i at time t from a causal hit emitted by j at time $t_0 \in \mathcal{C}_{ij}(t)$. The acceleration contribution is:

$$\mathbf{a}_{ij}(t;t_0) = \kappa\,\sigma_{ij}\,\frac{|q_i q_j|}{r_{ij}^2\;|J_{ij}(t;t_0)|}\,\hat{\mathbf{r}}_{ij}$$

The instantaneous power delivered to architrino i by this hit is:

$$P_{ij}(t;t_0) = \mu_{\text{arch}}\,\mathbf{a}_{ij}\cdot\mathbf{v}_i = \mu_{\text{arch}}\,\kappa\,\sigma_{ij}\,\frac{|q_i q_j|}{r_{ij}^2\;|J_{ij}(t;t_0)|}\,v_{r,ij}$$

where $v_{r,ij} = \mathbf{v}_i(t)\cdot\hat{\mathbf{r}}_{ij}$ is the radial component of the receiver's velocity along the line of action. This is the **only instant** when the interaction can change the kinetic energy of i. Between hits, $\mathbf{a}_{ij}$ from this specific emission is zero.

Summing over all contributing sources and all causal emission times at a given t,

$$\frac{dK_\mu}{dt}(t) = \sum_i\sum_j\sum_{t_0\in\mathcal{C}_{ij}(t)} P_{ij}(t;t_0)$$

with the understanding that for self-hit we include $j=i$ as well.

Action-level wake-energy functional at time boundary t

Let $\mathcal{K}_{ij}(t_1,t_0)$ denote the causal-delay interaction kernel appearing in the nonlocal action scaffold below:

$$\mathcal{K}_{ij}(t_1,t_0) = \frac{\kappa\,\sigma_{ij}\,|q_i q_j|}{c_f}\,\Theta(t_1-t_0)\,\frac{\delta\big(g_{ij}(t_1,t_0)\big)}{r_{ij}(t_1,t_0)}, \qquad g_{ij}(t_1,t_0) = t_1 - t_0 - \frac{r_{ij}(t_1,t_0)}{c_f}$$

For an isolated system, the nonlocal Noether charge associated with $t\mapsto t+\tau$ is

$$E_{\text{tot}}(t) = K_\mu(t) + E_{\text{wake}}(t)$$

with

$$E_{\text{wake}}(t) = \frac{1}{2}\sum_{i,j}\int_{-\infty}^{t} dt_0 \int_t^{\infty} dt_1\,\partial_{t_1}\mathcal{K}_{ij}(t_1,t_0)$$

For $i=j$, the same rule applies with the trivial coincidence branch ($t_1=t_0$) excluded, matching the self-hit convention used throughout this chapter.

Interpretation: the double integral measures interaction links that cross the time boundary t (past emission side $t_0\le t$ and future reception side $t_1\ge t$). This is the exact "in-flight" interaction contribution in the nonlocal theory.

For exact solutions of the causal action, nonlocal Noether's theorem gives

$$\frac{d}{dt}\Big(K_\mu(t) + E_{\text{wake}}(t)\Big) = 0$$

No separate spatial field-energy ontology is required; conservation is encoded directly in worldline geometry and the causal kernel.

For proof and simulation, the same statement can be written as a residual balance. Let

$$\mathbf{R}_i^{(\eta)}(t) = \mu_{\text{arch}}\mathbf{a}_i(t) - \mathbf{F}_{i,\text{act}}^{(\eta)}(t)$$

be the Euler-Lagrange residual of the symmetry-preserving regularized action, where $\mathbf{F}_{i,\text{act}}^{(\eta)}$ includes the scale term and any nonzero constraint-variation residual from the action. Let $\mathcal{B}_E^{(\eta)}(t)$ collect energy flux through finite history-window endpoints, period cuts, and excluded self-coincidence boundaries. Then the action-level energy balance is

$$\frac{d}{dt}\left(K_\mu(t) + E_{\text{wake}}^{(\eta)}(t)\right) = \sum_i \mathbf{v}_i(t)\cdot\mathbf{R}_i^{(\eta)}(t) + \mathcal{B}_E^{(\eta)}(t)$$

For isolated compactly supported or period-matched histories, $\mathbf{R}_i^{(\eta)} = \mathbf{0}$ and $\mathcal{B}_E^{(\eta)} = 0$ give the exact conserved charge. A nonzero residual identifies a real failure mode: branch-chart loss, nonsymmetric regularization, leakage through the finite memory window, or an unaccounted derivative-of-delta counterterm.

Equivalent work-integral form

For direct trajectory evaluation, one may reconstruct a compatible interaction contribution through the accumulated power exchange along the realized trajectory:

$$U(t) = U_* - \int_{t_*}^{t} \sum_i \mu_{\text{arch}}\, \mathbf{a}_i(t') \cdot \mathbf{v}_i(t')\, dt'$$

This work-integral form is a practical trajectory-level reconstruction when the same action-derived force law and boundary convention are used. It should not be treated as an independent off-shell Noether functional; outside the symmetry-preserving action model it is a diagnostic bookkeeping quantity rather than a proved conserved charge.

In short-delay effective limits, E_{wake} reduces to an approximate instantaneous pair form

$$E_{\text{wake}}(t) \approx \sum_{i<j} U_{ij}\big(\mathbf{x}_i(t), \mathbf{x}_j(t)\big)$$

with leading $1/r_{ij}$ behavior plus geometry-dependent self-hit corrections.

Exact Nonlocal Lagrangian

To connect with variational methods and with later continuum approximations, it is useful to exhibit the **action principle** for the delayed dynamics. Because the interactions depend on path history via causal wakes, the action is necessarily nonlocal in time.

Exact causal-delay Fokker-type interaction term

For the focused scalar causal-locus statistic (definitions, theorem spine, and circular branch-count benchmark), see Causal Action Functional. That chapter's $1/(r^2 J)$ functional is an action-counting diagnostic built from the received branch density; it is not automatically identical to the exact Fokker-type variational action below, whose $1/r$ causal kernel yields the inverse-square branch law after variation.

Let the worldline of architrino i be $\mathbf{x}_i(t)$. For the action-scaffold discussion, the same universal bookkeeping constant may be inserted in the quadratic kinetic term:

$$S[\{\mathbf{x}_i\}] = \sum_i \int dt\, \frac{1}{2}\mu_{\text{arch}} \|\mathbf{v}_i(t)\|^2 \;-\; \frac{1}{2}\sum_{i \neq j} S_{ij}$$

with interaction contributions

$$S_{ij} = \frac{\kappa\, \sigma_{ij}\, |q_i q_j|}{c_f} \int dt \int dt'\, \Theta(t-t')\, \frac{\delta\big(g_{ij}(t,t')\big)}{r_{ij}(t,t')}$$

where

$$g_{ij}(t,t') \equiv t - t' - \frac{r_{ij}(t,t')}{c_f}, \qquad r_{ij}(t,t') = \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\|$$

Key points:

- $\Theta(t-t')$ enforces the purely past-causal branch ($t' \le t$).
- $\delta(g_{ij})$ restricts support to the characteristic causal surface $r_{ij} = c_f(t-t')$.
- The sharp action scaffold contains no fundamental mollifier: η is a regularization parameter used to test branch limits.

Integrating out the delta via the delay-map Jacobian gives the branch-resolved form:

$$\delta\big(g_{ij}(t,t')\big) = \sum_{t_0 \in \mathcal{C}_{ij}(t)} \frac{\delta(t'-t_0)}{|\partial_{t'} g_{ij}(t,t_0)|}, \qquad \partial_{t'} g_{ij} = -1 + \frac{\hat{\mathbf{r}}_{ij}(t;t_0)\cdot \mathbf{v}_j(t_0)}{c_f}$$

Hence

$$S_{ij} = \frac{\kappa\, \sigma_{ij}\, |q_i q_j|}{c_f} \int dt \sum_{t_0 \in \mathcal{C}_{ij}(t)} \frac{1}{r_{ij}(t;t_0)\, |1 - \hat{\mathbf{r}}_{ij}(t;t_0)\cdot \mathbf{v}_j(t_0)/c_f|}$$

Variation and line-of-action forces

The branch law targeted by the action-level variation is:

$$\frac{d}{dt}\left(\mu_{\text{arch}} \mathbf{v}_i(t)\right) = \sum_j \mathbf{F}_{ij}(t)$$

and the branch-resolved force is

$$\mathbf{F}_{ij}(t) = \mu_{\text{arch}}\, \kappa\, \sigma_{ij}\, |q_i q_j| \sum_{t_0 \in \mathcal{C}_{ij}(t)} \frac{\hat{\mathbf{r}}_{ij}(t;t_0)}{r_{ij}^2(t;t_0)\ |1 - \hat{\mathbf{r}}_{ij}(t;t_0) \cdot \mathbf{v}_j(t_0)/c_f|}$$

The inverse-square factor follows in the theorem sketch from the variation of the scale-invariant kernel. On a simple-root chart, the interaction density is

$$\frac{1}{r_{ij}} \delta(g_{ij}), \qquad g_{ij} = t - t' - \frac{r_{ij}}{c_f}$$

Varying the receiver position gives

$$\delta r_{ij} = \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_i, \qquad \delta\left(\frac{1}{r_{ij}}\right) = -\frac{\hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_i}{r_{ij}^2}$$

The variation of

$$\delta(g_{ij})$$

is the remaining distributional part of the proof. After the source-side variation, integration by parts on the root-selected chart, and boundary terms are accounted for, the target branch-resolved Euler-Lagrange term is proportional to

$$\frac{\hat{\mathbf{r}}_{ij}}{r_{ij}^2 |J_{ij}|}$$

This $1/r^2$ scaling is not an added ansatz in the accepted proof route: it is the pull-back expected from a scale-invariant causal-cone constraint in 3D when varying a $1/r$ Fokker kernel. The full proof requires controlling the derivative-of-delta term under the same symmetry-preserving regularization.

The derivative-of-delta term has a useful exact reduction on any transversal branch. Since

$$\partial_{t'} g_{ij}(t, t') = -J_{ij}(t; t')$$

one has

$$\delta'_\eta(g_{ij}) = -\frac{1}{J_{ij}} \partial_{t'} \delta_\eta(g_{ij})$$

Thus the root-constraint variation can be integrated by parts in the source time t':

$$\int dt'\, \Theta(t - t') \frac{\delta'_\eta(g_{ij})}{c_f r_{ij}} \hat{\mathbf{r}}_{ij} = \mathcal{B}_{ij}^{(\eta)}(t) + \int dt'\, \delta_\eta(g_{ij}) \partial_{t'} \left[\Theta(t - t') \frac{\hat{\mathbf{r}}_{ij}}{c_f r_{ij} J_{ij}}\right]$$

The first term is an endpoint or excluded-coincidence contribution; the second is the root-chart interior derivative that must be accounted for before the pure scalar kernel can be claimed to derive the branch-resolved force law. Therefore the action proof does not license dropping $\delta'_\eta(g_{ij})$ by fiat. It requires the symmetry-preserving regularization to make this interior derivative vanish, become a boundary/source-side contribution under the allowed variations, or be cancelled by an explicit counterterm. If that cancellation fails, the branch-resolved law requires an additional regularized counterterm beyond $\hat{\mathbf{r}}_{ij}/(r_{ij}^2 |J_{ij}|)$.

The source-side variation narrows the issue further. Holding the receiver point fixed and varying the emission point gives

$$\delta r_{ij} = -\hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_j(t'), \qquad \delta g_{ij} = \frac{1}{c_f} \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_j(t')$$

so

$$\delta_{\text{src}}\left(\frac{\delta_\eta(g_{ij})}{r_{ij}}\right) = \left[\frac{\delta_\eta(g_{ij})}{r_{ij}^2} + \frac{\delta'_\eta(g_{ij})}{c_f r_{ij}}\right] \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_j(t')$$

Selecting the future reception time as the root gives

$$\partial_t g_{ij}(t, t') = 1 - \frac{\hat{\mathbf{r}}_{ij}(t, t') \cdot \mathbf{v}_i(t)}{c_f}$$

and hence the source-side derivative-of-delta term integrates by parts as

$$\int dt\,\Theta(t-t')\frac{\delta'_\eta(g_{ij})}{c_f r_{ij}}\hat{\mathbf{r}}_{ij} = \widetilde{\mathcal{B}}^{(\eta)}_{ij}(t') - \int dt\,\delta_\eta(g_{ij})\partial_t\left[\Theta(t-t')\frac{\hat{\mathbf{r}}_{ij}}{c_f r_{ij}\,(1-\hat{\mathbf{r}}_{ij}\cdot\mathbf{v}_i/c_f)}\right]$$

This is a coefficient of $\delta\mathbf{x}_j(t')$, not of $\delta\mathbf{x}_i(t)$. For arbitrary compactly supported interior variations, the receiver and source variations are independent. The source-side term therefore does not generically cancel the receiver-side root-chart derivative in the Euler-Lagrange equation for $\mathbf{x}_i(t)$. Noether boundary terms control endpoint contributions and global time-translation, spatial-translation, and rotation charges; they do not remove an interior coefficient under compact variations.

In the sharp positive-delay, transversal limit,

$$\int dt'\,\delta_\eta(g_{ij})\partial_{t'}\left[\Theta(t-t')\frac{\hat{\mathbf{r}}_{ij}}{c_f r_{ij}J_{ij}}\right] \longrightarrow \frac{1}{|J_{ij}(t;t_0)|}\,\partial_{t'}\left[\frac{\hat{\mathbf{r}}_{ij}(t,t')}{c_f r_{ij}(t,t')J_{ij}(t;t')}\right]\Bigg|_{t'=t_0}$$

Thus the exact $1/r$ causal kernel proves the target inverse-square branch term only if the admitted branch also satisfies the local stationarity condition

$$\partial_{t'}\left[\frac{\hat{\mathbf{r}}_{ij}(t,t')}{r_{ij}(t,t')J_{ij}(t;t')}\right]\Bigg|_{t'=t_0} = \mathbf{0}$$

as a sufficient special case, or if the action is supplemented by an explicit regularized counterterm whose receiver Euler derivative cancels this residual interior vector. Such a counterterm must come from an invariant action-level mechanism, not from fitting the already accepted force law. Without one of those conditions, the displayed Master EOM remains the accepted causal law and branch diagnostic, while the pure scalar $1/r$ Fokker-type action is only a partial variational scaffold for it.

Equivalently, define the direct scale term

$$\mathbf{F}^{(\eta)}_{ij,\mathrm{scale}}(t) = \int_{-\infty}^{t} dt'\,\frac{\hat{\mathbf{r}}_{ij}(t,t')}{r_{ij}^2(t,t')}\,\delta_\eta(g_{ij}(t,t'))$$

and the constraint residual

$$\mathbf{C}^{(\eta)}_{ij}(t) = \mathbf{C}^{(\eta)}_{ij,\mathrm{recv}}(t) + \mathbf{C}^{(\eta)}_{ij,\mathrm{src}}(t) + \mathbf{C}^{(\eta)}_{ij,\mathrm{bdry}}(t)$$

where $\mathbf{C}^{(\eta)}_{ij,\mathrm{recv}}$ is the receiver-side interior derivative displayed above, $\mathbf{C}^{(\eta)}_{ij,\mathrm{src}}$ is the source-side coefficient on $\delta\mathbf{x}_j(t')$, and $\mathbf{C}^{(\eta)}_{ij,\mathrm{bdry}}$ is the boundary contribution. On a regularized chart the action-derived equation has the diagnostic form

$$\mu_{\mathrm{arch}}\mathbf{a}_i(t) = \sum_j \kappa\,\sigma_{ij}|q_i q_j|\left(\mathbf{F}^{(\eta)}_{ij,\mathrm{scale}}(t) + \mathbf{C}^{(\eta)}_{ij}(t)\right)$$

The canonical branch law is recovered on a tested window W in the weak simple-root limit only if

$$\lim_{\eta\to 0^+}\int_W \left\|\sum_j \kappa\,\sigma_{ij}|q_i q_j|\mathbf{C}^{(\eta)}_{ij}(t)\right\| dt = 0$$

with the same branch floors and boundary convention used to define the action. This windowed residual condition is the minimal proof obligation for upgrading the variational scaffold to an exact action derivation of the Master EOM.

Decision (pure scalar action). The pure scalar $1/r$ Fokker-type scaffold does not generically derive the canonical branch law. The obstruction is local, not merely a boundary convention: for compactly supported receiver variations, the source-side coefficient and Noether endpoint terms cannot cancel the receiver-side interior derivative unless the branch satisfies the stationarity condition above or an invariant action-level counterterm supplies the missing Euler derivative.

Equivalently, on an admissible branch with $r_{ij} > 0$ and $|J_{ij}| > J_{\min} > 0$,

$$\partial_{t'}\left[\frac{\hat{\mathbf{r}}_{ij}(t,t')}{r_{ij}(t,t')J_{ij}(t;t')}\right]\Bigg|_{t'=t_0} \neq \mathbf{0}$$

is a certificate that the pure scalar scaffold leaves a nonzero receiver-force residual on that branch. This falsifies the universal claim “the scalar $1/r$ action by itself is the exact action for the Master EOM.” It does not falsify the Master EOM, the action-level Noether bookkeeping on closed charts, or the possibility of a later invariant counterterm derived from a richer regularized action.

No-go scaffold (same-support local scalar counterterm). The clean local scalar counterterm route is closed under the following restricted assumptions: the added term has the same causal-surface support as the $1/r$ kernel, uses only g_{ij}, r_{ij}, and J_{ij} on the existing branch chart, introduces no new variables, adds no off-surface support, and is not fitted after the force law is already known. Suppressing the common coupling and sign factors, the allowed branch-pair form is

$$S^{(\eta)}_{\mathrm{ct},ij} = \int dt\, dt'\, \Theta(t-t')\, a(r_{ij}, J_{ij})\, \delta_\eta(g_{ij})$$

For receiver variation,

$$\delta r_{ij} = \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_i, \qquad \delta g_{ij} = -\frac{1}{c_f} \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_i$$

Before any optional J_{ij}-variation is included, the radial part of the counterterm variation contains

$$\delta_{\mathbf{x}_i} S^{(\eta)}_{\mathrm{ct},ij} \supset \left[\partial_{r_{ij}} a\, \delta_\eta(g_{ij}) - \frac{a}{c_f} \delta'_\eta(g_{ij}) \right] \hat{\mathbf{r}}_{ij} \cdot \delta \mathbf{x}_i$$

The optional J_{ij}-dependence can add transverse and source-velocity terms, but it does not remove the scalar radial coefficient that must cancel the original derivative-of-delta residual. Cancelling that coefficient for all admitted receiver variations requires

$$a(r_{ij}, J_{ij}) = -\frac{1}{r_{ij}}$$

This choice necessarily adds

$$\partial_{r_{ij}} a\, \delta_\eta(g_{ij}) = \frac{\delta_\eta(g_{ij})}{r_{ij}^2}$$

which changes the accepted inverse-square scale term. Any further same-support scalar correction that removes this scale change reintroduces a derivative-of-delta coefficient. A g_{ij}-antiderivative of $\delta_\eta(g_{ij})$ would move support away from the causal wake surface and is outside the assumptions. Therefore no same-support local scalar counterterm built only from g_{ij}, r_{ij}, and J_{ij} is admissible under this restricted route.

The obstruction also survives a finite local delta-jet extension. Let

$$K^{(\eta)}_{\mathrm{ct}}(r,g) = \sum_{n=0}^{N} a_n(r) \delta^{(n)}_\eta(g), \qquad D_{ij} \equiv \partial_r - \frac{1}{c_f} \partial_g$$

The direct kernel $K^{(\eta)}_0 = \delta_\eta(g)/r$ has

$$D_{ij} K^{(\eta)}_0 = -\frac{\delta_\eta(g)}{r^2} - \frac{\delta'_\eta(g)}{c_f r}$$

Cancelling only the derivative-of-constraint residual would require

$$D_{ij} K^{(\eta)}_{\mathrm{ct}} = \frac{\delta'_\eta(g)}{c_f r}$$

without adding another $\delta_\eta(g)/r^2$ scale term. For $N \geq 1$, the highest derivative coefficient is $-a_N(r)\delta^{(N+1)}_\eta(g)/c_f$, so $a_N = 0$; descending through the jet order forces $a_n = 0$ for every $n \geq 1$. The remaining $N = 0$ case requires $a_0(r) = -1/r$, but then $\partial_r a_0 = 1/r^2$, so the counterterm again changes the inverse-square scale term it was supposed to preserve.

The conclusion is narrow but decisive for local repairs: no finite same-support local scalar or delta-jet counterterm cancels the scalar-kernel residual while leaving the canonical branch strength intact. A viable action-level repair must instead be nonlocal along the (r, g) characteristic, or must use a richer velocity/history-dependent invariant action. Either route changes the action ontology enough that it should be discussed explicitly before canonization.

The terminal common-center inter-layer chart gives a concrete obstruction to the remaining per-branch stationarity route. In that specialization, stationarity of $\hat{\mathbf{r}}/(rJ)$ forces the source tangent to be parallel to the source-receiver separation. The scalar part then reduces to $\rho_\delta(1 - \rho_\delta) = 0$: the first factor collapses a positive-delay branch when the source speed is nonzero, and the second factor is $J = 0$, a grazing branch excluded by the Jacobian floor. Thus terminal inter-layer charts should not expect the scalar scaffold to close by per-branch stationarity. The remaining local target is either branch-summed residual closure for a scale-only scaffold, or a recoil-inclusive action ledger that retains the residual as wake-emission resistance.

For the scale-only Master EOM, the branch-summed residual target is the vanishing of the signed receiver-side interior Euler derivative after the direct inverse-square term is removed:

$$\sum_{b:\, o_b = i} \kappa\, \mathrm{sign}(q_{j_b} q_i)\, |q_{j_b} q_i|\, \mathbf{C}^{(0)}_b(t) = \mathbf{0}$$

with the same positive-delay, Jacobian-floor, and boundary convention used by the branch chart. This is not the Master EOM force residual and not the Noether conservation ledger. It is the additional condition needed for the scalar action scaffold to have no leftover interior Euler derivative on that receiver. If the same signed sum is nonzero and is retained by the action rather than cancelled, it is the local recoil term that must appear in the finite-window force and energy ledger.

Nonlocal characteristic repair target. The least invasive remaining action-level route is to solve the counterterm equation before imposing causal-surface support. In the reduced scalar variables, the required receiver-gradient correction has the form

$$D_{ij}K_{\mathrm{ct}}^{(\eta)}(r,g)=\frac{\delta_\eta'(g)}{c_f r},\qquad D_{ij}=\partial_r-\frac{1}{c_f}\partial_g$$

The characteristics of D_{ij} preserve

$$u=g+\frac{r}{c_f}$$

Thus a formal characteristic solution is

$$K_{\mathrm{ct}}^{(\eta)}(r,g)=H_{\mathrm{ct}}^{(\eta)}\left(g+\frac{r}{c_f}\right)+\int_{r_*}^{r}\frac{1}{c_f\rho}\,\delta_\eta'\left(g+\frac{r-\rho}{c_f}\right)d\rho$$

with $H_{\mathrm{ct}}^{(\eta)}$ and the lower characteristic endpoint r_* fixed by the history-window, core-regularization, or boundary convention. This expression is invariant under time translation, spatial translation, and spatial rotation because it depends only on the causal scalar g, the Euclidean separation r, and declared scalar endpoints. It is not a same-support wake-surface term: it carries a characteristic tail in (r,g) and therefore changes the action scaffold.

This gives a concrete proof target rather than a completed replacement action. A candidate nonlocal action may be promoted only if its endpoint convention preserves $H(0)=0$, its Euler derivative cancels the residual above without changing the accepted inverse-square branch term, and its Noether boundary terms close the same energy, momentum, and angular-momentum ledger used by the Master EOM.

The endpoint calculation sharpens that target. The lower-endpoint form above is only the formal characteristic integral. A delayed-interior tail should instead be oriented toward an outgoing endpoint:

$$K_{\mathrm{ct},+}^{(\eta)}(r,g)=H_{+}^{(\eta)}\left(g+\frac{r}{c_f}\right)-\int_{r}^{R_+}\frac{1}{c_f\rho}\,\delta_\eta'\left(u-\frac{\rho}{c_f}\right)d\rho,\qquad u=g+\frac{r}{c_f}$$

with $R_+\geq r$ on the retained finite history chart, or with an explicitly controlled $R_+=\infty$ limit. A characteristic endpoint is the special case $R_+=R_+(u)$. Since $D_{ij}u=0$, differentiation gives

$$D_{ij}K_{\mathrm{ct},+}^{(\eta)}=\frac{\delta_\eta'(g)}{c_f r}-\frac{D_{ij}R_+}{c_f R_+}\,\delta_\eta'\left(u-\frac{R_+}{c_f}\right)$$

Thus the desired interior cancellation holds without an extra endpoint source only when R_+ is itself a characteristic endpoint, $D_{ij}R_+=0$.

Endpoint-ledger decision. The displayed endpoint term is not automatically a Noether boundary term. Its support is

$$u=\frac{R_+}{c_f},\qquad g=\frac{R_+-r}{c_f}$$

which is generally an interior tail surface of the retained delayed chart, not the primary arrival surface $g=0$. If $D_{ij}R_+\neq 0$, compact receiver variations inside the retained chart see

$$-\frac{D_{ij}R_+}{c_f R_+}\,\delta_\eta'\left(u-\frac{R_+}{c_f}\right)$$

as an Euler coefficient. Moving that term into the Noether wake-history ledger would hide a force-law change unless the endpoint is a declared fixed history boundary whose variation is held fixed. Therefore the characteristic-tail repair preserves the accepted Master EOM branch force only under one of the following conditions:

$$D_{ij}R_+=0,\qquad \text{or}\qquad \lim_{\eta\to 0^+}\int_W\left\|\frac{D_{ij}R_+}{c_f R_+}\,\delta_\eta'\left(u-\frac{R_+}{c_f}\right)\right\|dt=0$$

for the declared branch chart and fixed endpoint convention. In that admissible case the endpoint contributes only a boundary wake-history flux, not a new receiver force. In the generic non-characteristic case, the repair is a no-go for the current Master EOM because it adds an extra interior action force.

In the sharp-support limit, the outgoing form is supported on

$$0 \le g \le \frac{R_+ - r}{c_f}$$

so it is a causal interior tail behind the arriving wake surface, not a same-support surface density. Conversely, a lower endpoint $r_* < r$ supports a tail with $g \le 0$ and is not delayed-interior causal support unless the boundary convention supplies a separate interpretation. This proves a useful but limited result: the characteristic-tail equation can cancel the scalar scaffold's interior derivative residual at the level of the Euler derivative, but it does so by adding a nonlocal tail and endpoint ledger obligation. It is not yet an exact action for the Master EOM.

A nondegenerate characteristic endpoint gives the cleanest current candidate. On a retained chart choose

$$R_+(u) = c_f(u + h_+), \qquad h_+ > 0$$

or take the controlled $R_+ = \infty$ limit. Since $R_+ = R_+(u)$ and $D_{ij}u = 0$, the endpoint is characteristic and the Euler leakage term proportional to $D_{ij}R_+$ vanishes. The outgoing counterterm can then be written, after the change of variable $s = u - \rho/c_f$, as

$$K^{(\eta)}_{\mathrm{ct},+}(r,g) = H^{(\eta)}_+(u) - \int_{-h_+}^{g} \frac{\delta'_\eta(s)}{c_f(u-s)} ds$$

Integrating by parts gives

$$\frac{\delta_\eta(g)}{r} + K^{(\eta)}_{\mathrm{ct},+}(r,g) = H^{(\eta)}_+(u) + \frac{\delta_\eta(-h_+)}{c_f(u+h_+)} + \int_{-h_+}^{g} \frac{\delta_\eta(s)}{c_f(u-s)^2} ds$$

The finite-endpoint clearance condition is therefore

$$\mathcal{B}^{(\eta)}_+(u,h_+) \equiv \frac{\delta_\eta(-h_+)}{c_f(u+h_+)} = 0$$

for a compactly supported mollifier with h_+ outside the support, or $\mathcal{B}^{(\eta)}_+ \to 0$ in the declared weak limit for a Gaussian mollifier. If finite-η endpoint clearance is not exact, the characteristic gauge must be fixed by

$$H^{(\eta)}_+(u) = -\mathcal{B}^{(\eta)}_+(u,h_+)$$

before the kernel is treated as a normalized action object. This condition is invisible to the receiver Euler derivative because it depends only on u , but it is visible to the Noether wake-history charge.

With the endpoint-clear normalization imposed, the delayed-interior effective kernel is

$$K^{(\eta)}_{\mathrm{eff},h_+}(r,g) = \int_{-h_+}^{g} \frac{\delta_\eta(s)}{c_f(u-s)^2} ds, \qquad u = g + \frac{r}{c_f}$$

In the infinite-endpoint form,

$$K^{(\eta)}_{\mathrm{eff}}(r,g) = \int_{-\infty}^{g} \frac{\delta_\eta(s)}{c_f(u-s)^2} ds$$

Both forms satisfy the receiver-gradient identity

$$D_{ij}K^{(\eta)}_{\mathrm{eff}} = -\frac{\delta_\eta(g)}{r^2}$$

with the finite form using the same identity after endpoint clearance. Thus the delayed-interior characteristic-tail kernel cancels the derivative-of-constraint residual without adding a second inverse-square scale term. The accompanying Noether boundary terms for energy, momentum, and angular momentum must be taken from the same normalized kernel, as below; replacement of a diagnostic inverse-square adapter on any concrete branch still requires the branch-chart residual and conservation checks.

Noether boundary increments for the normalized tail. With the endpoint-clear normalization imposed, there is no remaining free $H^{(\eta)}_+(u)$ gauge term that can shift the wake-history charge. Define the weighted effective action kernel

$$\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0) = \frac{\kappa\,\sigma_{ij}|q_i q_j|}{c_f}\Theta(t_1 - t_0)K^{(\eta)}_{\mathrm{eff}}(r_{ij}(t_1;t_0), g_{ij}(t_1,t_0))$$

with the same finite-endpoint version when the chart uses $h_+ < \infty$. For a time cut t_*, let

$$X_{ij}(t_*) = \{(t_1,t_0) : t_0 \le t_* < t_1,\ t_1 > t_0\}$$

with the trivial self-coincidence branch excluded when $i = j$. The normalized characteristic-tail wake increments are

$$E^{(\eta)}_{\mathrm{wake,eff}}(t_*) = \frac{1}{2}\sum_{i,j}\int_{X_{ij}(t_*)} \partial_{t_1}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

$$\mathbf{P}^{(\eta)}_{\mathrm{wake,eff}}(t_*) = -\frac{1}{2}\sum_{i,j}\int_{X_{ij}(t_*)} \nabla_{\mathbf{x}_i(t_1)}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

and

$$\mathbf{J}^{(\eta)}_{\mathrm{wake,eff}}(t_*) = -\frac{1}{2}\sum_{i,j}\int_{X_{ij}(t_*)} \mathbf{x}_i(t_1)\times\nabla_{\mathbf{x}_i(t_1)}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

The minus signs in the spatial charges follow the sign convention that the interaction contribution appears with a minus sign in the action. The receiver-gradient identity gives

$$\nabla_{\mathbf{x}_i(t_1)}K^{(\eta)}_{\mathrm{eff}} = \hat{\mathbf{r}}_{ij}D_{ij}K^{(\eta)}_{\mathrm{eff}} = -\frac{\delta_\eta(g_{ij})}{r_{ij}^2}\hat{\mathbf{r}}_{ij}$$

while the source-end gradient is the opposite. Therefore a global spatial translation or rotation of both endpoints changes no interior action density, and a step translation or step rotation across t_* exposes exactly the boundary increments above. The characteristic endpoint condition $D_{ij}R_+ = 0$, together with endpoint clearance, is the local reason these increments are wake-history boundary terms rather than a hidden extra receiver force.

This closes the local kernel-normalization and Noether-increment definition for the delayed-interior characteristic-tail repair. It does not by itself certify any proposed branch, terminal label, or nested shell braid attractor: a branch chart must still show vanishing Euler residual, finite memory depth, positive Jacobian floors, and closure of $K_\mu + E^{(\eta)}_{\mathrm{wake,eff}}$, $\mathbf{P}_{\mathrm{mech}} + \mathbf{P}^{(\eta)}_{\mathrm{wake,eff}}$, and $\mathbf{J}_{\mathrm{mech}} + \mathbf{J}^{(\eta)}_{\mathrm{wake,eff}}$ over the same retained branch set.

Branch-chart conservation pullback. Let $\mathfrak{B}(\Gamma,\mathcal{S};h,\eta,\epsilon_c)$ be a retained branch chart with active causal-root rows $\mathcal{R}^{\mathrm{act}}$, positive inactive-root gaps, positive Jacobian floor, finite memory depth, and declared endpoint convention. For a time cut t_*, define the chart-restricted crossing domain

$$X^{\mathfrak{B}}_{ij}(t_*) \equiv X_{ij}(t_*) \cap \{(t_1,t_0) : (i,j,t_1,t_0)\ \text{lies on a retained row of}\ \mathcal{R}^{\mathrm{act}}\}$$

with trivial self-coincidence excluded when $i=j$. The pulled-back wake-history charges are the same Noether boundary terms above, restricted to $X^{\mathfrak{B}}_{ij}(t_*)$:

$$E^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}(t_*) = \frac{1}{2}\sum_{i,j}\int_{X^{\mathfrak{B}}_{ij}(t_*)} \partial_{t_1}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

$$\mathbf{P}^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}(t_*) = -\frac{1}{2}\sum_{i,j}\int_{X^{\mathfrak{B}}_{ij}(t_*)} \nabla_{\mathbf{x}_i(t_1)}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

and

$$\mathbf{J}^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}(t_*) = -\frac{1}{2}\sum_{i,j}\int_{X^{\mathfrak{B}}_{ij}(t_*)} \mathbf{x}_i(t_1)\times\nabla_{\mathbf{x}_i(t_1)}\mathcal{K}^{(\eta)}_{ij,\mathrm{eff}}(t_1,t_0)\,dt_0\,dt_1$$

The matching mechanical charges on the same chart are

$$K_{\mu,\mathfrak{B}}(t) = \sum_{i\in\mathfrak{B}}\frac{1}{2}\mu_{\mathrm{arch}}\|\mathbf{v}_i(t)\|^2, \qquad \mathbf{P}_{\mathrm{mech},\mathfrak{B}}(t) = \sum_{i\in\mathfrak{B}}\mu_{\mathrm{arch}}\mathbf{v}_i(t)$$

$$\mathbf{J}_{\mathrm{mech},\mathfrak{B}}(t) = \sum_{i\in\mathfrak{B}}\mathbf{x}_i(t)\times\mu_{\mathrm{arch}}\mathbf{v}_i(t)$$

For a retained window $W=[t_a,t_b]$, the branch-chart conservation test is

$$\Delta_W\left(K_{\mu,\mathfrak{B}} + E^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}\right) = \int_W\sum_i \mathbf{v}_i(t)\cdot\mathbf{R}^{(\eta)}_{i,\mathrm{eff},\mathfrak{B}}(t)\,dt + \int_W \mathcal{B}^{(\eta)}_{E,\mathfrak{B}}(t)\,dt$$

$$\Delta_W\left(\mathbf{P}_{\mathrm{mech},\mathfrak{B}} + \mathbf{P}^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}\right) = \int_W\sum_i \mathbf{R}^{(\eta)}_{i,\mathrm{eff},\mathfrak{B}}(t)\,dt + \int_W \mathcal{B}^{(\eta)}_{P,\mathfrak{B}}(t)\,dt$$

$$\Delta_W\left(\mathbf{J}_{\mathrm{mech},\mathfrak{B}} + \mathbf{J}^{(\eta)}_{\mathrm{wake,eff},\mathfrak{B}}\right) = \int_W\sum_i \mathbf{x}_i(t)\times\mathbf{R}^{(\eta)}_{i,\mathrm{eff},\mathfrak{B}}(t)\,dt + \int_W \mathcal{B}^{(\eta)}_{J,\mathfrak{B}}(t)\,dt$$

The theorem-level branch claim requires the three residual balances to converge to zero with $\epsilon_{\rm var}^{(\eta)}(W) \to 0$, vanishing declared endpoint or period-cut leakage, stable branch floors, and the same retained row set in the force residuals and in the three wake-history charges. A work-integral reconstruction $U(t)$ or a projected torque increment is only a numerical diagnostic unless it is derived from this same action kernel, endpoint convention, and retained branch chart.

For a finite spatial window, the energy boundary row can be read as the wake-escapement flux defined in Energy. In that notation, a theorem-level conservation packet should be able to rewrite the energy balance in the form

$$\frac{dE_W}{dt} = -\Phi_{\text{wake},\partial W}(t) + P_{\text{ext},W}(t) + \mathcal{R}_{E,W}(t), \qquad \Phi_{\text{wake},\partial W}(t) = \frac{d}{dt} \sum_{\alpha \in \mathcal{E}_{\text{esc}}(W)} E_\alpha^{\text{emit}}.$$

This does not add a second energy channel. It identifies the endpoint and boundary term of the delayed action with the causal-wake history that exits the retained window without a retained receiver. The same boundary object is the entropy-arrow theorem target in Entropy: energy flux, wake escapement, and observer-window entropy production are three projections of one finite-window path-history defect.

Self-interaction ($i = j$) is included by adding S_{ii} with the same kernel, but explicitly excluding the trivial coincidence $t' = t$ (no instantaneous self-push at the moment of emission). Self-hit corresponds to nontrivial roots $t_0 < t$ where the worldline re-intersects its own causal isochrons, which are captured naturally by the same double-integral structure.

Thus:

- The scalar $1/r$ action above is a nonlocal variational scaffold for the delayed dynamics under the stated branch and regularization assumptions,
- It becomes an exact action derivation of the Master EOM only on branch charts where the constraint residual vanishes or is cancelled by an invariant action-level counterterm,
- A finite same-support local scalar or delta-jet counterterm has been ruled out because it cancels the derivative residual only by disturbing the inverse-square scale term,
- The remaining minimal action repair is the delayed-interior characteristic-tail kernel above; its receiver Euler derivative has the desired inverse-square identity, and its normalized wake-history boundary increments are now explicit,
- Without such closure, the pure scalar action is falsified as the universal exact action for the Master EOM and should be treated as a diagnostic scaffold,
- Any δ_η replacement must preserve the symmetries that supply the Noether charges if conservation claims are to remain exact.

Total Energy for an Isolated Set

Given the kinetic energy definition, we now address the most useful history-aware total energy for an isolated architrino set.

General structure

We define a functional H_{tot} such that:

- H_{tot} is constant in t for isolated exact trajectories,
- H_{tot} reduces to $K_\mu + U$ in regimes where an effective potential description is adequate.

For an isolated action-derived system,

$$H_{\text{tot}}[\{\mathbf{x}_i(\cdot)\}, \{\mathbf{v}_i(\cdot)\}; t] \equiv K_\mu(t) + U(t)$$

with $U(t)$ reconstructed along the realized trajectory by:

$$U(t) = U_* - \int_{t_*}^{t} \sum_i \mu_{\text{arch}}\, \mathbf{a}_i(t') \cdot \mathbf{v}_i(t')\, dt'$$

where U_* is a fixed reference and $\mathbf{a}_i$ is the actual acceleration given by the Master Equation (including self-hit and partner contributions). Then:

$$\frac{dH_{\text{tot}}}{dt} = \frac{dK_\mu}{dt} + \frac{dU}{dt} = \sum_i \mu_{\text{arch}}\, \mathbf{a}_i \cdot \mathbf{v}_i - \sum_i \mu_{\text{arch}}\, \mathbf{a}_i \cdot \mathbf{v}_i = 0$$

In this sense, the total energy functional is:

- Not a local function only of $(\mathbf{x}_i(t), \mathbf{p}_i(t))$,
- But a **history-aware conserved quantity** that accounts for all past wake emission and all energy exchange via causal intersections.

This matches the ontology:

- **Energy resides in the architrinos and their assemblies**, not in a separate field substance,

- It is only updated at the times t when wake surfaces intersect receivers,
- Yet there exists a global invariant H_{tot} for isolated trajectories, defined entirely from the particle worldlines and their induced accelerations.

Local canonical form in effective limits

In regimes where:

- Internal binaries are in tight, quasi-stationary maximum-curvature orbits (giving approximately fixed internal energies),
- Wake travel times across the system are short compared to dynamical timescales of the center-of-mass motion of assemblies,
- Self-hit contributions primarily renormalize the internal energies (rest masses),

we can introduce **effective assemblies** with:

- Effective masses M_A,
- Positions $\mathbf{X}_A$,
- Momenta $\mathbf{P}_A = M_A \dot{\mathbf{X}}_A$,
- An approximate instantaneous interaction potential $U_{\text{eff}}(\{\mathbf{X}_A\})$,

and write a **local canonical Hamiltonian**:

$$H_{\text{eff}}(\{\mathbf{X}_A\}, \{\mathbf{P}_A\}) = \sum_A \frac{\|\mathbf{P}_A\|^2}{2M_A} + U_{\text{eff}}(\{\mathbf{X}_A\})$$

which:

- Approximates the full history-aware energy functional when assemblies are well separated and slowly varying,
- Recovers familiar particle-mechanics structure for many emergent phenomena (orbital motion, scattering, bound states) without ever attributing energy to a continuous field.

Summary

- **Kinetic energy** is defined in the usual way at the architrino level, with internal kinetic energy of tightly bound self-hit binaries contributing to assembly rest masses.
- **Interaction energy** is not primitive as an instantaneous position function; it is encoded in the nonlocal causal charge E_{wake} and may be reconstructed from the work-integral form U.
- A **nonlocal variational scaffold** is available under the regularity and boundary assumptions stated above: a multi-time Lagrangian whose kernel enforces the causal isochron geometry and targets the Master EOM with its Jacobian-weighted inverse-square law, becoming an exact action derivation only when the constraint residual vanishes or is explicitly cancelled.
- The **total energy** for an isolated trajectory is history-aware; in suitable limits it reduces to a canonical $H_{\text{eff}} = \sum \mathbf{P}^2/2M + U_{\text{eff}}$ for effective assemblies, with no separate "field energy" ontology.

All energy accounting remains localized to **architrinos and their assemblies** and is only updated at the instants when **causal wake surfaces intersect receivers** at $t = \text{now}$. The action-derived conserved charge is written as $K_\mu(t) + E_{\text{wake}}(t)$; a work-integral reconstruction $K_\mu(t) + U(t)$ is compatible only along realized trajectories after the same boundary convention and force law have been declared.

Symmetry, Conservation, and Lyapunov Functionals

Introduction

The Master Equation is a state-dependent delay system: acceleration at time t depends on the path-history segment over $[t-h, t]$. In this setting, conservation laws are not functions of the instantaneous state $(\mathbf{x}, \mathbf{v})$ alone. Instead, they are **functionals on path history** that track "in-flight" wake contributions.

This section makes the symmetry group explicit and states the corresponding conserved functionals for isolated systems with $\eta > 0$.

Fundamental Symmetry Group

Definition (Fundamental symmetry group). The substrate and interaction kernel are invariant under

$$G_{\text{fund}} = E(3) \times \mathbb{R}_{\text{time}}$$

where $E(3) = \mathbb{R}^3 \rtimes O(3)$ acts by spatial translations and rotations, and $\mathbb{R}_{\text{time}}$ acts by time translation.

Theorem (Invariance of the Master Equation). If $\mathbf{x}(t)$ is a solution, then:

1. **Time translation:** $\mathbf{y}(t) = \mathbf{x}(t+\tau)$ is a solution for any $\tau \in \mathbb{R}$.
2. **Spatial isometry:** $\mathbf{y}(t) = R\mathbf{x}(t) + \mathbf{b}$ is a solution for any $R \in O(3)$ and $\mathbf{b} \in \mathbb{R}^3$.

Proof sketch. The causal constraint depends only on Euclidean distances and time differences. Both are invariant under G_{fund}. The line-of-action vector $\hat{\mathbf{r}}_{ij}$ transforms covariantly under rotations, so the per-hit acceleration retains the same form.

This symmetry statement applies to the background, the interaction kernel, and the transformed full histories. It does not license arbitrary architrino flips or permutations after provenance has been assigned. Let H_i^t denote the path-history/provenance record of architrino i up to time t. A label permutation P is an exact symmetry only on the restricted histories for which

$$H_{P(i)}^t = P(H_i^t)$$

for every i, with the same transformation also preserving all causal-root relations $\mathcal{C}_{ij}(t)$. For generic states this condition fails, so effective indistinguishability must be treated as coarse-grained observer bookkeeping rather than substrate identity.

Generalized Momentum and Angular Momentum

The delayed theory separates ordinary mechanical motion from the causal-wake history that is still in flight. For an action-derived delayed model with translation and rotation symmetry, the full Noether charges are history functionals: the particle-only quantities need not be conserved by themselves, but the particle-plus-wake totals are. In regularized or numerical variants, the same expressions should be treated as conserved diagnostics only when the chosen regularization preserves those symmetries.

Definition (Mechanical momentum).

$$\mathbf{P}_{\text{mech}}(t) = \sum_i \mu_{\text{arch}} \mathbf{v}_i(t)$$

Because the forces are delayed, $\dot{\mathbf{P}}_{\text{mech}}(t)$ is generally nonzero.

Definition (Wake momentum functional). For an isolated system, define

$$\mathbf{P}_{\text{wake}}(t) = \mathbf{P}_{\text{wake}}(t_*) - \int_{t_*}^{t} \sum_i \mathbf{F}_i(s)\, ds$$

with $\mathbf{F}_i = \mu_{\text{arch}} \mathbf{a}_i$ from the Master Equation.

Conservation law (total momentum).

$$\mathbf{P}_{\text{tot}}(t) \equiv \mathbf{P}_{\text{mech}}(t) + \mathbf{P}_{\text{wake}}(t)$$

is constant in time for isolated solutions. It is the momentum decomposition associated with spatial-translation invariance of the nonlocal causal action.

Definition (Mechanical angular momentum).

$$\mathbf{L}_{\text{mech}}(t) = \sum_i \mathbf{x}_i(t) \times \mu_{\text{arch}} \mathbf{v}_i(t)$$

Definition (Wake angular momentum functional).

$$\mathbf{L}_{\text{wake}}(t) = \mathbf{L}_{\text{wake}}(t_*) - \int_{t_*}^{t} \sum_i \mathbf{x}_i(s) \times \mathbf{F}_i(s)\, ds$$

Conservation target (total angular momentum).

$$\mathbf{L}_{\text{tot}}(t) \equiv \mathbf{L}_{\text{mech}}(t) + \mathbf{L}_{\text{wake}}(t)$$

is the angular-momentum decomposition associated with rotational invariance of the nonlocal causal action. For isolated solutions of the symmetry-preserving action model it is conserved. For working regularized models, conservation of $\mathbf{L}_{\text{tot}}$ is a validation condition rather than an automatic consequence.

Remark. These definitions mirror the energy decomposition used earlier: the apparent "missing" momentum and angular momentum are assigned to in-flight causal-wake geometry. The total quantities are therefore functionals of the path history, not functions of the instantaneous particle state alone.

Energy Functional and No-Runaway Criterion

Time-translation invariance implies a conserved history functional, which we write as

$$E_{\text{tot}}(t) = K_\mu(t) + E_{\text{wake}}(t)$$

where K_μ is the quadratic kinetic bookkeeping proxy and E_{wake} denotes the exact nonlocal interaction charge. In direct trajectory evaluation, U may be used as a compatible reconstruction up to a constant offset when it is derived from the same action-level force and boundary convention.

This statement is exact for the action-based delayed theory discussed in this section. For regularized working models, especially the dual-mollified local recapture model of collinear-breather.md, it should be interpreted as exact only when the regularization preserves the same symmetry structure; otherwise it is the natural history-aware bookkeeping candidate rather than a proved invariant.

Lemma (Bounded work rate under regularization). If $\eta > 0$ and the mollified kernel bounds the per-hit force, then there exists $F_{\max}(\eta)$ such that

$$\left|\frac{dK_\mu}{dt}\right| \leq \sum_i \|\mathbf{F}_i\| \, \|\mathbf{v}_i\| \leq N \, F_{\max}(\eta) \, v_{\max}(t)$$

Theorem target (No-runaway criterion). For an isolated system with fixed $\eta > 0$, if the action-derived interaction charge $E_{\text{wake}}(t)$, or a compatible realized-trajectory reconstruction $U(t)$, is bounded below on the admissible history class (for example, by enforcing a minimum separation within the regularized kernel support), then $K_\mu(t)$ is bounded for all times where the solution exists. In particular, a runaway $v_{\max}(t) \to \infty$ is only possible if the corresponding interaction term tends to $-\infty$, which requires a collapse toward the singular regime or a breakdown of the regularized assumptions.

Interpretation. Self-hit repulsion can transfer energy between U and K, but it cannot generate unbounded kinetic energy without a corresponding unbounded decrease in U. This is the core conservation argument for excluding unphysical runaway acceleration in the regularized model.

Simulation Diagnostics (Symmetry and Conservation)

In addition to the convergence checks in Numerical Implementation Notes, track these conserved functionals in any isolated run:

- **Total energy**: $H_{\text{tot}}(t) = K_\mu(t) + E_{\text{wake}}(t)$, or a declared compatible reconstruction $K_\mu + U$, should remain constant within the chosen numerical tolerance.
- **Total momentum**: $\mathbf{P}_{\text{tot}}(t)$ should be constant; monitor $\|\mathbf{P}_{\text{tot}}(t) - \mathbf{P}_{\text{tot}}(0)\|$.
- **Total angular momentum**: $\mathbf{L}_{\text{tot}}(t)$ should be constant; in planar runs, the unit axis $\hat{\mathbf{n}} = \mathbf{L}_{\text{tot}}/\|\mathbf{L}_{\text{tot}}\|$ should remain fixed.
- **Binary symmetry defect** (for symmetric initial data):

$$\Delta_{\text{sym}}(t) = \|\mathbf{x}_1(t) + \mathbf{x}_2(t)\|$$

A secular drift indicates numerical asymmetry or a symmetry-breaking perturbation.

These diagnostics operationalize the symmetry constraints and provide early warning of numerical artifacts or model inconsistencies.

Closure Interface: Coarse-Graining Gate to Effective Quantum Envelope

For integration with the quantum closure program, the master equation provides the microscopic gate:

$$\ddot{\mathbf{x}}_i(t) = \text{delayed causal-hit sum over } \mathcal{C}_{ij}(t)$$

The required next reduction is a controlled map to mesoscopic density dynamics:

$$f(t, \mathbf{x}, \mathbf{v}) \Longrightarrow (\rho, \mathbf{u}, S) \Longrightarrow \psi = \sqrt{\rho}\, e^{iS/\hbar_{\text{eff}}}$$

Closure condition for this interface:

- the same coarse-graining window that preserves validated dynamical invariants must recover the effective Schrödinger limit in the non-relativistic, weak-field, fixed-particle-number regime;
- residual non-Markovian terms must be explicitly retained as correction operators, not absorbed into uncontrolled fitting.

Return-map symplectic residual for action-derived branch promotion. When a replayable branch chart is promoted to an action-derived reduced Hamiltonian chart, the section return map must preserve the reduced symplectic structure. Let $z = (Q^a, \Pi_a)$ be local reduced coordinates after the retained root constraints and section condition have been solved, let

$$\mathcal{P}_{\mathcal{S}} : z_n \mapsto z_{n+1}, \qquad M_{\mathcal{S}} = D\mathcal{P}_{\mathcal{S}}$$

and let $\Omega_{\mathcal{S}}$ be the pulled-back symplectic matrix on the reduced section. Define

$$\mathcal{R}_\Omega \equiv \left\|M_{\mathcal{S}}^T \Omega_{\mathcal{S}} M_{\mathcal{S}} - \Omega_{\mathcal{S}}\right\|, \qquad \mathcal{R}_{\text{vol}} \equiv |\det M_{\mathcal{S}} - 1|$$

For an exact finite-dimensional Hamiltonian reduction, $\mathcal{R}_\Omega = 0$ and therefore $\mathcal{R}_{\text{vol}} = 0$. For the delayed Master EOM these are not automatic consequences of a small orbit residual: they are closure diagnostics for the claim that the retained branch chart has captured the missing path-history degrees of freedom well enough to behave like a canonical return map. A nonzero $\mathcal{R}_\Omega$ means at least one of the following remains unresolved: omitted causal-root rows, window-boundary wake flux, an action-level residual, or a reduction that is not actually Hamiltonian. Thus a local master-equation closure claim still uses $\mathfrak{B}(\Gamma, \mathcal{S}; h, \eta, \epsilon_c)$ as defined above, while the stronger Hamiltonian claim must additionally report $\mathcal{R}_\Omega \le \epsilon_\Omega$.

Standard charged-particle comparison target. In ordinary electromagnetic mechanics, a charged particle can be described by

$$L_{\mathrm{EM}} = \frac{1}{2}m\|\dot{\mathbf{r}}\|^2 - e\phi(\mathbf{r},t) + e\,\dot{\mathbf{r}}\cdot\mathbf{A}(\mathbf{r},t), \qquad \mathbf{p}_{\mathrm{can}} = m\dot{\mathbf{r}} + e\mathbf{A}$$

The velocity-coupled one-form shifts canonical momentum and yields the effective Lorentz-force law. Under

$$\phi \mapsto \phi - \partial_t\chi, \qquad \mathbf{A} \mapsto \mathbf{A} + \nabla\chi$$

the Lagrangian changes only by $e\,d\chi/dt$, so the effective equations are unchanged. In $\mathbb{AAA}$ this is a comparison structure, not substrate ontology: the primitive kernel still contains only radial causal hits. The corresponding closure target is to extract an assembly-level effective one-form

$$\mathcal{A}_{\mathrm{eff}} = A_a^{\mathrm{eff}}(Q,t)\,dQ^a - \phi_{\mathrm{eff}}(Q,t)\,dt$$

from coarse-grained causal-root geometry, then show that the observer-level residual

$$\mathcal{R}_{\mathrm{EM}}(W) = \int_W \|\mu_A\mathbf{a}_A(t) - e_A\left(\mathbf{E}_{\mathrm{eff}}(\mathbf{X}_A,t) + \mathbf{V}_A(t)\times\mathbf{B}_{\mathrm{eff}}(\mathbf{X}_A,t)\right)\|\,dt$$

vanishes in the stated approximation while the underlying branch ledger remains a sum of line-of-action contributions. Failure of this residual is a magnetic-emergence failure, not evidence for inserting an intrinsic cross-product term into the Master EOM.

Constrained branch-multiplier formulation. On a fixed retained branch chart, the causal roots may be represented as constrained variables rather than solved away immediately. Let $s_{ij,\ell}(t)$ be the emission time assigned to retained row ℓ and define

$$G_{ij,\ell}(t) \equiv r_{ij,\ell}(t) - c_f\big(t - s_{ij,\ell}(t)\big), \qquad r_{ij,\ell}(t) = \|\mathbf{x}_i(t) - \mathbf{x}_j(s_{ij,\ell}(t))\|$$

A branch-reduced constrained scaffold on a window W has the form

$$S_{\mathfrak{B}}^{(\eta)} = \int_W \left[\sum_i \frac{1}{2}\mu_{\mathrm{arch}}\|\mathbf{v}_i(t)\|^2 - \sum_{(i,j,\ell)\in\mathcal{R}^{\mathrm{act}}} \alpha_{ij}\frac{w_{ij,\ell}^{(\eta)}(t)}{r_{ij,\ell}(t)\,|J_{ij,\ell}(t)|} + \sum_{(i,j,\ell)\in\mathcal{R}^{\mathrm{act}}} \lambda_{ij,\ell}(t)G_{ij,\ell}(t)\right]dt$$

where $\alpha_{ij} = \kappa\,\sigma_{ij}|q_iq_j|/c_f$, $w_{ij,\ell}^{(\eta)}$ carries the retained mollified branch weight and cutoff convention, and $\lambda_{ij,\ell}$ is a Lagrange multiplier for the causal-root constraint. Variation with respect to $\lambda_{ij,\ell}$ enforces $G_{ij,\ell} = 0$. Variation with respect to the root variable gives the row equation

$$0 = \partial_{s_{ij,\ell}}\left[-\alpha_{ij}\frac{w_{ij,\ell}^{(\eta)}}{r_{ij,\ell}|J_{ij,\ell}|} + \lambda_{ij,\ell}G_{ij,\ell}\right]$$

provided the row has no explicit $\dot{s}_{ij,\ell}$ dependence after the chosen reduction. Variation with respect to the receiver position exposes the constraint contribution

$$\delta_{\mathbf{x}_i}\int_W \lambda_{ij,\ell}G_{ij,\ell}\,dt = \int_W \lambda_{ij,\ell}(t)\,\hat{\mathbf{r}}_{ij,\ell}(t)\cdot\delta\mathbf{x}_i(t)\,dt$$

Thus the multiplier term is not a new substrate force. It is the finite-dimensional record of the work required to keep the retained branch row on the causal-root surface while the surrounding path history is varied. The unconstrained branch action is recovered only when these multiplier contributions are either solved into the same invariant action-level counterterm used above, converted into legitimate boundary wake-history terms, or shown to vanish in the branch-summed residual:

$$\mathcal{R}_{\lambda,i}(W) \equiv \int_W \left\|\sum_{\ell:\,o_\ell=i}\lambda_\ell(t)\hat{\mathbf{r}}_\ell(t) - \sum_j \kappa\,\sigma_{ij}|q_iq_j|\mathbf{C}_{ij}^{(\eta)}(t)\right\|dt \longrightarrow 0$$

Here o_ℓ denotes the receiver index of branch row ℓ. This is the delayed-action analogue of ordinary holonomic constraint handling: one may solve constraints into generalized coordinates, or retain them with multipliers, but the multiplier ledger must not be hidden inside a claimed exact force law.

Noether history-functional balance target. Let $S_{\mathfrak{B}}^{(\eta)}$ be a symmetry-preserving regularized action on a retained branch chart, and let a one-parameter transformation have infinitesimal generator $\boldsymbol{\xi}_i(t)$ on each worldline. If the action changes only by endpoint terms,

$$\delta_\xi S_{\mathfrak{B}}^{(\eta)} = \left[B_\xi^{(\eta)}(t)\right]_{t_a}^{t_b}$$

after the retained causal-root constraints, endpoint convention, and excluded self-coincidence convention are applied, then the corresponding history charge at a cut t_* has the form

$$Q_\xi^{(\eta)}(t_*) = \sum_i \mu_{\rm arch}\mathbf{v}_i(t_*)\cdot\boldsymbol{\xi}_i(t_*) + Q_{\xi,\rm wake}^{(\eta)}(t_*) - B_\xi^{(\eta)}(t_*)$$

Its finite-window balance is

$$\frac{dQ_\xi^{(\eta)}}{dt} = \sum_i \boldsymbol{\xi}_i(t)\cdot\mathbf{R}_i^{(\eta)}(t) + \mathcal{B}_\xi^{(\eta)}(t)$$

where $\mathbf{R}_i^{(\eta)}$ is the Euler residual of the same action and $\mathcal{B}_\xi^{(\eta)}$ collects leakage through finite memory endpoints, period cuts, omitted branch rows, and non-characteristic tail endpoints. Exact conservation follows only when both terms vanish. Time translation, spatial translation, and rotation are the special cases that produce energy, momentum, and angular momentum above. This is the delayed version of the standard symmetry-to-conservation statement, with the crucial difference that the conserved object is a particle-plus-wake history functional rather than an equal-time particle function.

End of Master Equation of Motion Document

Energy

In $\mathbb{AAA}$, energy accounting begins with architrinos and the causal wakes they generate. Architrinos carry primitive kinetic energy through motion and supply potential-energy bookkeeping through delayed interactions; the wake itself is not a standalone substance or vacuum reservoir. A **wake** is the source-dependent causal-isochron record of an architrino's emissions: motion changes its geometry, branch timing, and received potential, not the fact that an emission record exists. The term `wake` is the architrino-native description of what appears as a field at the effective level.

This chapter underwrites Particle Masses, Nested Shell Braid Dynamics, Noether Braid, Noether Sea Pro/Anti Coupling, Emergent Metric, and the constructive delay-energy standard in Delay Dynamics Energy.

All such dynamics unfold on a fixed ontological background: absolute time plus the Euclidean void. Forces and motion arise from **delayed causal hits from causal isochrons**, with line-of-action direction and Jacobian-weighted magnitude, on this fixed background. We work in units with causal-wake propagation speed $c_f = 1$.

The chapter keeps four levels separate. At the substrate level, kinetic and potential terms are architrino and causal-wake records on absolute time and the Euclidean void. At the dynamical level, energy changes through Jacobian-weighted causal hits and radial power. At the effective level, assemblies acquire inertia, apparent energy, and effective metric response through Noether sea coupling. At the inference level, scalar masses, thermodynamic records, and cosmological inventories are accepted only after a window, boundary record, and residual are declared.

Spacetime in this framework belongs to the effective level, not the ontological one. The ambient Noether sea is a **dense sea of scalable high-energy Noether braid assemblies** occupying the Euclidean void. These Noether braids are extremely small compared to ordinary Standard Model particles and constitute the Noether sea through which all other assemblies move and interact. The energetic state and configuration of this Noether braid sea control how energy, inertia, and effective geometry appear at larger scales.

Kinetic Energy and Momentum of a Single Architrino

An architrino in motion possesses kinetic energy and momentum.

- **Kinetic Energy E_k**

  A scalar quantity representing the energy of motion. For a single architrino a with velocity $\mathbf{v}_a(t)$, we write

$$E_{k,a}(t) = K\big(\|\mathbf{v}_a(t)\|\big),$$

where s denotes the speed argument and K is a strictly convex, monotonically increasing function with $K(0) = 0$. If an effective saturation proxy is being used, $K'(s) \to \infty$ at the saturation scale; in the primitive limit, K grows unboundedly. K is left unspecified because mass is emergent from interactions between assemblies, especially the Noether braids in the Noether sea. Strict convexity ensures a one-to-one mapping between kinetic energy and speed magnitude. Because a free architrino has no intrinsic speed limit in the micro-model, E_k is, in principle, unbounded as $\|\mathbf{v}_a\| \to \infty$.

- **Momentum $\mathbf{p}_a$**

The vector counterpart of kinetic energy:

$$\mathbf{p}_a(t) = P\big(\|\mathbf{v}_a(t)\|\big)\,\hat{\mathbf{v}}_a(t), \quad \hat{\mathbf{v}}_a = \frac{\mathbf{v}_a}{\|\mathbf{v}_a\|},$$

where P is a speed-dependent magnitude. Its detailed form is not postulated at the architrino level; it emerges from matching to assembly behavior.

If this momentum is treated as the conjugate momentum for the primitive kinetic scalar and $\mathbf{F} = d\mathbf{p}/dt$ is used in the work-energy relation, then P and K are not independent. For arbitrary nonzero velocity and acceleration, consistency requires

$$P'(s) = \frac{K'(s)}{s} = \mu_K(s), \qquad P(s) = \int_0^s \frac{K'(u)}{u}\,du$$

after choosing $P(0) = 0$. If this condition is not imposed, $\mathbf{p}$ should be read as a momentum-like bookkeeping vector rather than the canonical momentum of K.

Kinetic-scalar / closure compatibility. The conjugacy relation above also prevents a hidden second speed scale. If the primitive kinetic scalar is modeled with a finite saturation scale c_K, meaning $K'(s) \to \infty$ as $s \to c_K^-$, then any effective assembly closure using a signal speed $c_{\rm eff}$ is admissible on the declared comparison window only when

$$\left|\frac{c_{\rm eff}}{c_K} - 1\right| \le \epsilon_{cK}$$

with ϵ_{cK} declared before the comparison is promoted. If K is instead kept in the primitive unbounded-speed limit, then $c_{\rm eff}$ is wholly a Noether sea response quantity and no substrate-level particle speed cap may be invoked in the energy or mass-shell argument.

No fundamental mass:

In this model, there is no **particle-specific substrate mass** assigned to individual architrinos. We do **not** assume $E_k = \frac{1}{2}m\|\mathbf{v}\|^2$ or $\mathbf{p} = m\mathbf{v}$ at the substrate level for distinct architrino species. Instead:

- Kinetic energy and momentum are **primitive kinematic quantities** of architrinos.
- The substrate law is written in **acceleration-first** form.
- If force-like or quadratic-kinetic bookkeeping is needed, one may introduce a single universal conversion constant $\mu_{\rm arch}$, but this is not a particle-specific inertial mass.
- “Mass” in the usual observer sense appears **only at the assembly level** as a derived property of how a large internal energy distribution responds to external forcing in the Noether sea.

Work–Energy Relation and Per-Hit Power

Kinetic-energy accounting is controlled by the acceleration-first master law, but the familiar quadratic work-energy form applies only after a kinetic proxy has been chosen. For a general primitive kinetic scalar with $s_a = \|\mathbf{v}_a\|$,

$$\frac{dE_{k,a}}{dt} = K'(s_a)\frac{\mathbf{v}_a \cdot \mathbf{a}_a}{s_a} = \mu_K(s_a)\,\mathbf{a}_a \cdot \mathbf{v}_a, \qquad \mu_K(s) \equiv \frac{K'(s)}{s}$$

If one introduces the optional universal bookkeeping constant $\mu_{\rm arch}$ and defines $\mathbf{F}_a \equiv \mu_{\rm arch}\mathbf{a}_a$, then the quadratic bookkeeping proxy $K_{\mu,a} = \frac{1}{2}\mu_{\rm arch}\|\mathbf{v}_a\|^2$ satisfies

$$\frac{dK_{\mu,a}}{dt} = \mathbf{F}_a(t) \cdot \mathbf{v}_a(t)$$

Here $\mathbf{F}_a$ is the optional force-like bookkeeping quantity associated with the net acceleration from all causal hits; it is not a particle-specific substrate mass law.

From the canonical per-hit law

$$\mathbf{a}_{o'\leftarrow o}(t;t_0) = \kappa\,\sigma_{q_o q_{o'}}\,\frac{|q_o q_{o'}|}{r^2\,|J_{o'\leftarrow o}(t;t_0)|}\,\hat{\mathbf{r}}$$

where

$$J_{o'\leftarrow o}(t;t_0) \equiv 1 - \frac{\mathbf{v}_o(t_0)\cdot\hat{\mathbf{r}}}{c_f}$$

is the causal Jacobian encoding geometric bunching or dilation of the received wake flux.

Decompose the receiver's velocity into radial and transverse components:

$$\mathbf{v}_{o'} = v_r\hat{\mathbf{r}} + \mathbf{v}_\perp, \quad v_r = \mathbf{v}_{o'} \cdot \hat{\mathbf{r}}.$$

Because $\mathbf{a}_{o'\leftarrow o} \parallel \hat{\mathbf{r}}$:

- The **instantaneous work rate** from this hit is

$$\left.\frac{dK_\mu}{dt}\right|_{\text{hit}} = \mu_{\text{arch}}\mathbf{a}_{o'\leftarrow o} \cdot \mathbf{v}_{o'} = \mu_{\text{arch}} \frac{\kappa\,\sigma_{q_o q_{o'}}\,|q_o q_{o'}|}{r^2\,|J_{o'\leftarrow o}(t;t_0)|}\,v_r$$

Only v_r contributes to instantaneous quadratic-proxy power. For the primitive scalar K, replace μ_{arch} by $\mu_K(\|\mathbf{v}_{o'}\|)$.

- A hit only changes the **along-the-line** component of velocity; sideways motion $\mathbf{v}_\perp$ is unchanged instantaneously.

Potential Energy and Causal-Wake Potential

Potential energy arises from the interaction of an architrino with the **net causal-wake potential** generated by all architrinos, including in some regimes its own past emissions.

Net Causal-Wake Potential

At a point $\mathbf{s}$ and time t, the net potential is the **superposition** of contributions from all sources:

$$\Phi_{\text{net}}(\mathbf{s},t) = \sum_o \Phi_o(\mathbf{s},t).$$

Each Φ_o is built from the expanding causal isochrons emitted by source o, using the measure-valued or mollified emission density described in the architrino section. In the mollified representation with causal-surface width $\eta > 0$, Φ_{net} is a smooth function of $(\mathbf{s},t)$; in the ideal limit $\eta \to 0$ it becomes a measure-valued distribution supported on causal isochrons.

Potential Availability Is Geometric

The phrase "an architrino emits potential" should not be read as a source continually spending an internal fuel. The emission is the causal-wake geometry of the architrino itself: at each emission time, an expanding causal isochron is added to the source's path history. That causal structure can later participate in work, but it is not a material energy substance stored inside the Euclidean void.

Potential energy is therefore relational. It is assigned when a receiver is placed in a source's path-history causal-wake record and its trajectory intersects the relevant causal wake surfaces. The receiver's energy accounting depends on the active causal roots, their inverse-square distance factors, their polarity signs, the branch Jacobians, and the receiver's radial motion through the line of action. In the general per-hit law the source-side branch factor is

$$J_{o'\leftarrow o}(t;t_0) = 1 - \frac{\mathbf{v}_o(t_0)\cdot\hat{\mathbf{r}}}{c_f}$$

while the instantaneous power delivered to the receiver is controlled by

$$\mathbf{a}_{o'\leftarrow o} \cdot \mathbf{v}_{o'} = \|\mathbf{a}_{o'\leftarrow o}\|\,v_r$$

On the affine partner chart used in the closed-form collinear breather ansatz, the same causal bunching appears in the simple branch factor $J_p = 1 + \dot{x}/c_f$. That formula is not a new global definition of energy; it is the one-dimensional branch expression for how emission cadence is received on that chart.

Thus the potential to do work is broadly available wherever causal wakes pass, but work is realized only through an actual receiver trajectory. A quiet region is not a region with no causal activity; it is a region where the active wake contributions sum to negligible net acceleration and negligible net power for the assemblies present there.

Potential Energy

For a receiver architrino o' with polarity $q_{o'}$ at position $\mathbf{s}_{o'}(t)$, the potential energy $U_{o'}(t)$ is the fixed-history bookkeeping value assigned to the current configuration against the causal path-history wake record:

$$U_{o'}(t) = q_{o'}\,\Phi_{\text{net}}[\text{history}]\big(\mathbf{s}_{o'}(t),t\big).$$

Unlike electrostatics, Φ_{net} is not a function of instantaneous source positions but a functional of their past worldlines intercepted by the backward causal-wake record of $\mathbf{s}_{o'}(t)$. The gradient $\nabla\Phi_{\text{net}}$ is taken with respect to the receiver's spatial coordinates on the fixed background, holding the causal history fixed. In the idealized picture, Φ is a distribution supported on causal isochrons, not a smooth continuum field.

When we work with the mollified effective potential Φ_η, we can also write the fixed-history, force-like relation:

$$\mathbf{F}_{o'}(t) = -\nabla_{\mathbf{s}_{o'}} U_{o'}(t) = -q_{o'} \nabla_{\mathbf{s}_{o'}} \Phi_\eta[\text{history}]\big(\mathbf{s}_{o'}(t), t\big),$$

and this is equivalent to the Master Equation in the quasi-static, resolved-in-time limit after the same force normalization, such as $\mathbf{F}_{o'} = \mu_{\text{arch}} \mathbf{a}_{o'}$ or the appropriate $\mu_K \mathbf{a}_{o'}$, has been declared.
The force-as-gradient identity is valid only when taking the gradient at fixed causal history; the fundamental force law remains the per-hit sum of the Master EOM.

Macroscopic Cancellation and Localized Resonance

Constant causal emission by many architrinos does not imply a large random macroscopic force. The net causal-wake potential is a superposition, and in a large, incoherent population the leading gradients arrive with many signs, distances, phases, and line-of-action directions. For a receiver sampling such a population, macroscopic quietness is a two-moment condition, not only a mean-zero statement:

$$\|\langle \nabla\Phi_{\text{net}}\rangle_W\| \le \epsilon_{\text{mean}}, \qquad \frac{\text{Var}_W(\nabla\Phi_{\text{net}})}{\left\|\nabla\Phi_{\text{coh}}^{\text{bound}}\right\|^2} \le \epsilon_{\text{var}}$$

Both bounds must use the same horizon, screening, and summation prescription that makes the many-source wake sum converge. The variance bound is the load-bearing part: incoherent fluctuation must remain small compared with the coherent bound-state gradient that phase-locked assemblies preserve. This cancellation is one reason the Noether sea can be densely active while remaining macroscopically quiet. What standard prose may call a vacuum state is not empty Euclidean void; it is the effective limit in which the local Noether sea assemblies and their causal wakes balance so well that only small residual gradients remain available to ordinary probes.

Phase-locked bound states are the important exception. In a localized assembly, nearby constituents do not sample random phases; their active causal roots are correlated, and the $1/r^2$ distance factor lets the nearest coherent branches dominate over the far incoherent background. A collinear breather, for example, is precisely a reduced setting in which two opposite-polarity architrinos can form a localized, non-canceling causal resonance. The breather ansatz isolates this effect: instead of averaging away, the partner-hit and self-hit branches stay phase organized enough to exchange kinetic and potential energy across a bounded cycle.

Energy Conservation and Exchange

In the exact causal theory, energy conservation is enforced through exchange between kinetic motion and the causal-history interaction content encoded by wakes. This wake term should not be read as an independent material reservoir that drains from the emitter with every unreceived isochron; it is the nonlocal bookkeeping required by the same delayed causal action that generates the hits. For mollified working models, the strongest exact conservation claims remain conditional on the regularization being derived from the same time-translation-invariant causal action rather than inserted only at the equation-of-motion level.

For a single architrino:

$$\Delta E_k = \int \mathbf{F} \cdot d\mathbf{s} = -\Delta U$$

(when we restrict attention to its interactions with a fixed set of sources). For an **isolated system** of architrinos and their wakes, the total energy is:

$$E_{\text{total}} = \sum_a E_{k,a} + U_{\text{int}} + E_{\text{wake}},$$

and is constant in time for exact isolated solutions of the causal action. In mollified working models, this same bookkeeping should be treated as exact only when the mollified kernel inherits that action-level time-translation symmetry; otherwise it remains the natural candidate history functional to monitor, but not yet an established exact invariant.

- U_{int} is an optional effective decomposition of near-field interaction energy.
- E_{wake} accounts for the exact nonlocal interaction content carried by wake structures and any radiation-like transport through the Noether sea.

Consistency rule: either use E_{wake} alone for all interaction energy, or, if a U_{int} pairwise term is retained as an effective decomposition inside assemblies, then E_{wake} must explicitly omit the corresponding near-field content to prevent double counting.

For a hybrid decomposition, the omission must be checkable on the same finite window. Let the near/far split be made with the same coarse-graining window W_ℓ used in the matter-to-sea source $S^{(\ell)}_{\text{mat}\to\text{sea}}$. Define the partition-overlap residual

$$\mathcal{R}_{\text{dbl},W} = \frac{\left|E^{(\eta)}_{\text{wake},W} - \left(E^{\text{far}}_{\text{wake},W} + E^{\text{near}}_{\text{wake},W}\right)\right| + \left|U_{\text{int},W} - E^{\text{near}}_{\text{wake},W}\right|}{\left|E^{(\eta)}_{\text{wake},W}\right| + |U_{\text{int},W}| + \varepsilon_E}$$

with $\varepsilon_E > 0$ a declared denominator floor. A retained $U_{\text{int}} + E_{\text{wake}}$ decomposition is admissible only when $\mathcal{R}_{\text{dbl},W} \to 0$ under refinement of the same window, boundary record, and regularized causal action.

Conservation Status

The conservation claim is a level-specific statement. For an isolated branch whose force law comes from a time-translation-invariant causal action,

$$\frac{d}{dt}E_{\text{total}}(t) = 0, \qquad E_{\text{total}}(t) = \sum_a E_{k,a}(t) + U_{\text{int}}(t) + E_{\text{wake}}(t)$$

This is not a claim that $\sum_a E_{k,a}$ is constant on Σ_t, nor that a finite simulation window conserves its particle-only ledger. Delayed hits move energy between mechanical motion and causal-wake history, and finite windows must also name boundary flux, external work, and residuals. A calculation that omits one of those terms has not established energy nonconservation; it has exposed an incomplete retained record.

In working models the exact claim is conditional. If the mollifier, history window, self-branch cutoff, or characteristic-tail repair is inserted only at the equation-of-motion level, then the same expression is a diagnostic to monitor, not a proved Noether charge. Exact conservation is promoted only when the same regularized action supplies both the force row and the energy row, and when the energy residual in this section vanishes under refinement. The formal construction routes, crosswalk residual, and promotion conditions for E_{wake} are isolated in Delay Dynamics Energy.

The finite-η pathology theorem target in Master Equation uses this conservation status in a restricted way. The no-runaway conclusion is available only when the action-derived $E^{(\eta)}_{\text{wake}}$, or a compatible realized-trajectory reconstruction, has a declared lower bound on the same admissible branch chart. If the lower bound is absent, the run is not promoted as a closed solution; it is routed to the continuation boundary where collapse, missing wake-history bookkeeping, regulator dependence, or endpoint leakage must be resolved.

For reaction or radiation events, energy can leave the source assembly as photon output, recoil, medium excitation, remnant excitation, wake-carried exchange, or handoff terms, but those are named outputs rather than hidden losses. The event-level version is the componentwise ledger closure in Reaction Ledger.

Wake Escapement

For a finite local window $W \subset \Sigma_t$, **wake escapement** is the subset of emitted causal isochrons that exit the retained window without intersecting any retained receiver inside that window. More explicitly, if architrino a emits at t_0, define the causal isochron at later time t by

$$C_a(t; t_0) = \{\mathbf{y} \in \Sigma_t : \|\mathbf{y} - \mathbf{x}_a(t_0)\| = c_f(t - t_0)\}$$

The emitted isochron belongs to the escapement set $\mathcal{E}_{\text{esc}}(W)$ when it has a first retained boundary crossing

$$C_a(t_{\partial W}; t_0) \cap \partial W \neq \varnothing$$

and there is no retained receiver hit before that crossing:

$$\nexists\, b, t_r \quad \text{with} \quad t_0 < t_r < t_{\partial W}, \quad \mathbf{x}_b(t_r) \in W, \quad \mathbf{x}_b(t_r) \in C_a(t_r; t_0)$$

Wake escapement is therefore a finite-window boundary classification, not a new substance in the Euclidean void. It names the portion of causal-wake history that cannot be balanced by local receiver work because no local receiver intercepted it. In a contracting binary, the persistent positive tangential drive identified in Binary Dynamics should be read against this boundary ledger: particle kinetic gain, local interaction-energy change, recoil, and escaped wake flux are parts of one balance law.

For a finite spatial window $W \subset \Sigma_t$, conservation is a balance law rather than a claim that the window is isolated. This is the conservation-law upgrade relative to instantaneous mechanics: energy, momentum, and angular momentum are not generally conserved equal-time particle snapshots, but finite-window history functionals whose apparent deficits must be carried by causal-wake fluxes or by an explicit residual. Write

$$E_W(t) = \sum_{a:\mathbf{s}_a(t)\in W} K_a(t) + U_{\text{int},W}(t) + E_{\text{wake},W}(t)$$

where the terms include only the kinetic, interaction, and wake-history content retained by the declared window record. The finite-window energy balance should take the residual form

$$\frac{dE_W}{dt} + \int_{\partial W} \mathbf{J}_E \cdot \hat{\mathbf{n}}\, dA = P_{\text{ext},W} + \mathcal{R}_E(\eta, \Delta t, W)$$

Here $\mathbf{J}_E$ is the boundary flux of causal-wake energy bookkeeping, including any wake escapement through ∂W; $P_{\text{ext},W}$ is declared external work through sources or controls not included in W; and $\mathcal{R}_E$ records mollifier, timestep, and omitted-boundary-history error. A finite-window conservation claim is mature only when $\mathcal{R}_E \to 0$ under the same regularized causal action used for the local equation of motion.

The characteristic-tail repair target in Master Equation inherits this same rule. If the outgoing tail kernel $K^{(\eta)}_{\text{ct},+}$ is used, its endpoint contribution may be counted as a Noether wake-history boundary flux only when the endpoint is characteristic, or when it is a declared fixed history boundary whose leakage residual vanishes:

$$\mathcal{B}_{E,+}^{(\eta)} \sim \frac{D_{ij}R_+}{c_f R_+}\,\delta'_\eta\left(u-\frac{R_+}{c_f}\right) \longrightarrow 0$$

on the retained branch chart. If $D_{ij}R_+ \neq 0$ and this residual does not vanish, the endpoint is an interior Euler source rather than a conservation-boundary term. In that case the characteristic-tail action changes the accepted branch force and cannot be used to close exact energy conservation for the current Master EOM.

The analogous momentum and angular-momentum closures must also remain tied to the same window and boundary data. The finite-window momentum functional P_W^i contains the mechanical momentum retained in W plus the retained wake-history momentum record:

$$\frac{dP_W^i}{dt} + \int_{\partial W} \Pi^{ij}\hat{n}_j\, dA = F^i_{\text{ext},W} + \mathcal{R}_P^i(\eta, \Delta t, W)$$

For a declared origin $\mathbf{x}_0$, the corresponding angular-momentum history functional has the schematic form

$$\mathbf{L}_W(t) = \sum_{a:\mathbf{s}_a(t)\in W} \big(\mathbf{s}_a(t)-\mathbf{x}_0\big)\times \mathbf{p}_a(t) + \mathbf{L}_{\text{wake},W}(t)$$

where $\mathbf{p}_a$ is the declared mechanical momentum proxy for the chosen kinetic bookkeeping. Its finite-window balance target is

$$\frac{dL_W^i}{dt} + \int_{\partial W} \Lambda^{ij}\hat{n}_j\, dA = \tau^i_{\text{ext},W} + \mathcal{R}_L^i(\eta, \Delta t, W)$$

Here Π^{ij} and Λ^{ij} are finite-window flux diagnostics for retained causal wakes and assembly crossings, not new substrate fields. $\tau^i_{\text{ext},W}$ is the external torque about the same origin $\mathbf{x}_0$. If the energy, momentum, and angular-momentum residuals can be made small only by changing the window measure, boundary wake record, or regularization separately for each observable, the calculation has fitted separate summaries rather than demonstrated one causal-history conservation law.

Cosmological inventory comparisons add one more finite-window caution. A gravitational binding contribution is negative relative to dispersed matter in the declared window, but the sign is meaningful only after the boundary and coarse-graining are fixed. In this chapter, G_{eff} in the binding line is a provisional external comparison input until the mass map and Noether sea response tensor independently derive it. For a component inventory over W,

$$E^{\text{grav}}_{\text{bind},W} = -\frac{1}{2}\int_W\int_W \frac{G_{\text{eff}}(\theta;\mathbf{x},\mathbf{y})\,\rho_{\text{eff}}(\mathbf{x})\rho_{\text{eff}}(\mathbf{y})}{\|\mathbf{x}-\mathbf{y}\|}\, dV_{\mathbf{x}}\, dV_{\mathbf{y}} + \mathcal{B}_{\partial W}$$

where $\mathcal{B}_{\partial W}$ records boundary and embedding terms. The corresponding inventory residual is

$$\mathcal{R}_{\text{grav bind},W} = \frac{\left|E^{\text{grav}}_{\text{bind},W} - E^{\text{obs}}_{\text{bind},W}\right|}{\epsilon_{\text{bind}}} + \frac{|\mathcal{B}_{\partial W}|}{\epsilon_{\partial W}}$$

The circularity check is the post-handoff residual

$$\mathcal{R}_{G\text{-consist},W} = \frac{\left|G^{\text{bind}}_{\text{eff}} - G^{(\zeta,\mathcal{M})}_{\text{eff}}\right|}{\left|G^{(\zeta,\mathcal{M})}_{\text{eff}}\right| + \varepsilon_G} \le \epsilon_G$$

where $G^{\text{bind}}_{\text{eff}}$ is the value used in the inventory comparison and $G^{(\zeta,\mathcal{M})}_{\text{eff}}$ is the value derived from shielding, exposed response, and the Noether sea response tensor. Until $\mathcal{R}_{G\text{-consist},W}$ is reported on the same window, the cosmological binding line is comparison bookkeeping only, not a derived inventory contribution. This keeps gravitational binding from being used as an adjustable bookkeeping sign that can repair the cosmic energy inventory without specifying the same window, boundary wake history, and effective G_{eff} used by the rest of the cosmology branch.

Theorem target (center of response). The standard center-of-mass theorem depends on equal-time internal force cancellation. In delayed causal dynamics that cancellation is not available as a particle-only statement on Σ_t: the reciprocal hit generally belongs to a different emission time, a different causal-root branch, or a boundary wake record not retained by the finite window. For an assembly window $W_A(t)$, the replacement target is to prove that there is a response center $\mathbf{X}_{\text{resp}}(t)$ and an assembly response tensor M_A^{ij} such that the finite-window momentum balance reduces, over resolved windows, to

$$\frac{d}{dt}\left(M_A^{ij}\dot{X}_{\text{resp},j}\right) = F^i_{\text{ext},W_A} - \int_{\partial W_A}\Pi^{ij}\hat{n}_j\, dA + \mathcal{R}^i_{\text{resp}}(\eta, \Delta t, W_A)$$

The pair $(\mathbf{X}_{\text{resp}}, M_A^{ij})$ is not free to be chosen after the balance is fitted. The response center must be pinned independently by the exposed internal-energy ledger,

$$\mathbf{X}_{\rm resp}(t) \equiv \frac{\displaystyle\int_{W_A} \mathbf{x}\,\zeta_{\rm loc}(\mathbf{x},t)\,e_{\rm internal}(\mathbf{x},t)\,dV}{\displaystyle\int_{W_A} \zeta_{\rm loc}(\mathbf{x},t)\,e_{\rm internal}(\mathbf{x},t)\,dV}$$

whenever the denominator is positive and the window contains the exposed assembly record. The tensor M_A^{ij} must then reduce to the independently extracted response tensor I_A^{ij} on the same branch chart. Only when these independently defined objects satisfy the balance with $\mathcal{R}^i_{\rm resp} \to 0$ does it reduce to the familiar center-of-mass form. A non-vanishing irreducible residual means the exposed-energy center is not the inertial response center for that branch, rather than a license to redefine the center. Until that theorem is closed, a center-of-mass trajectory is an effective readout of the assembly response, not a substrate-level proof that internal delayed forces cancel instantaneously.

In practice, finite systems or simulation domains should monitor $E_W(t)$, $P^i_W(t)$, and $L^i_W(t)$ together with their boundary fluxes and residuals. $E_{\rm total}(t)$ is the isolated-system limit when the declared window contains the full wake-history record and the boundary terms vanish.

Entropy, Free Energy, and Coarse Residuals

Entropy and free-energy language belongs to coarse-grained records, not to empty Euclidean void. It is useful when a simulation or continuum reduction groups many microhistories into the same retained macrostate. For a declared coarse map $\mathcal{Q} : S(t) \mapsto z$ with cell probabilities p_α over the retained histories, the entropy diagnostic is

$$S_{\mathcal{Q}} = -k_B \sum_\alpha p_\alpha \log p_\alpha$$

When a temperature-like channel $T_{\mathcal{Q}}$ is declared by the same record, the Helmholtz-style free-energy diagnostic is

$$F_{\mathcal{Q}} = E_{\mathcal{Q}} - T_{\mathcal{Q}} S_{\mathcal{Q}}$$

This is not an added thermodynamic postulate. It is a test that the chosen coarse variables have retained enough state counting to make relaxation and response claims reproducible.

The distinction matters because energy conservation does not by itself measure work availability. Two records with the same total energy can have different free-energy diagnostics when one retains a concentrated heat, chemical, photon-channel, or potential-gradient channel and the other has dispersed the same energy into unresolved thermal, boundary, or wake-history records. A finite-window calculation must therefore close the energy ledger and the entropy ledger on the same retained record before claiming that energy remained useful, became waste heat, or crossed the boundary as low-grade radiation.

For an isolated finite window, the minimum coarse thermodynamic gate is the same-record entropy-production residual

$$\mathcal{R}_{S,W} = \frac{\left[-\Delta_W S_{\mathcal{Q}} + \int_W \frac{\mathcal{D}_{\mathcal{Q}}}{T_{\mathcal{Q}}+\varepsilon_T}\,dt\right]_+}{|\Delta_W S_{\mathcal{Q}}| + \int_W \left|\frac{\mathcal{D}_{\mathcal{Q}}}{T_{\mathcal{Q}}+\varepsilon_T}\right| dt + \varepsilon}$$

where $[x]_+ = \max(x, 0)$ and $\mathcal{D}_{\mathcal{Q}}$ is the declared coherent-to-incoherent transfer rate, including viscous, thermal, wake-boundary, or Noether sea response channels retained by the packet. Passing this gate means only that the selected coarse record has not made entropy decrease after unresolved boundary leakage is accounted for. It does not prove a fundamental stochastic substrate.

For the consolidated mapping from legacy entropy formulas into $\mathbb{AAA}$ record projections, see Entropy.

In near-equilibrium comparison runs, response and fluctuation must also come from one record. The fluctuation-dissipation map may be invoked only after that same retained record supplies an admissible temperature channel. Concretely, the record θ_W that supplies χ''_{AB} and $S^{\rm meas}_{AB}$ must pass $\mathcal{R}_{S,W}$ and must yield consistent temperatures from at least two independent observable pairs:

$$\mathcal{R}_{T,W} = \frac{\left|T_{\mathcal{Q}}^{(AB)} - T_{\mathcal{Q}}^{(A'B')}\right|}{\left|T_{\mathcal{Q}}^{(AB)}\right| + \left|T_{\mathcal{Q}}^{(A'B')}\right| + \varepsilon_T} \le \epsilon_T$$

If this sea-temperature admissibility check fails, the packet may report the dissipative response χ''_{AB} alone, but it may not use an equilibrium fluctuation-dissipation map as closure evidence. If an observable O_A has response kernel $\chi_{AB}(\omega)$ to a controlled source coupled to O_B, the causal-response check is that the dissipative part and the equilibrium fluctuation spectrum $S_{AB}(\omega)$ obey a declared classical or quantum fluctuation-dissipation row. A dimensionless packet residual can be written as

$$\mathcal{R}_{\rm FD}(A,B) = \frac{\|S^{\rm meas}_{AB}(\omega) - \mathcal{F}_T(\chi''_{AB}(\omega))\|_\omega}{\|S^{\rm meas}_{AB}(\omega)\|_\omega + \|\mathcal{F}_T(\chi''_{AB}(\omega))\|_\omega + \varepsilon}$$

Here $\mathcal{F}_T$ is the packet's chosen fluctuation-dissipation map, and χ''_{AB} is the imaginary, dissipative response. A passing value supports the coarse response chart; a failing value means the noise, dissipation, and energy ledger have been fitted separately.

Noether Sea, Effective Spacetime, and Energy Storage

At the fundamental level, the Euclidean void is an empty container. **Effective spacetime** is the observer-level summary of a **sea of high-energy Noether braid assemblies**:

- These Noether braids are extremely small compared to ordinary particles (electrons, protons, etc.).
- Each nested shell braid is itself a tightly bound architrino assembly with very high internal kinetic and potential energy.
- As a sea, they form a **dense population of coupled assemblies** occupying the Euclidean void. This ambient Noether sea content carries non-zero assembly density and internal stress. It provides the constitutive relations (permittivity, permeability, and medium-dressed inertial response) that deform the primitive architrino dynamics into effective relativistic kinematics, providing the bridge-level spacetime medium for:
  - Emergent inertia and mass,
  - Effective causal-cone behavior and Lorentz-like behavior,
  - Effective gravitational coupling (emergent geometry at large scales).

Energy in this picture is distributed across:

1. **Unbound Architrinos** (rare at low energies),
2. **Standard Model assemblies** (electrons, nucleons, etc.),
3. The **Noether sea** and, in bridge prose, the spacetime medium.

Assemblies: Internal vs Apparent Energy

For composite systems such as Standard Model particles, nuclei, and composite bound states formed from architrinos and embedded in the Noether sea, we distinguish:

- **Total internal energy**: energy retained by the assembly and by its immediate nested shell braid environment,
- **Apparent energy**: what leaks out as a long-range wake signature and governs how the assembly interacts with the outside world.

Internal Energy of an Assembly

For an assembly A (e.g., nested shell braid or higher structure), let $i \in A$ run over its constituent architrinos. Then:

$$E_{\text{internal}}(A) = \sum_{i \in A} E_{k,i} + \frac{1}{2} \sum_{\substack{i,j \in A \\ i \neq j}} U_{ij} + E_{\text{coupling to sea}}(A),$$

where:

- $E_{k,i}$ is the kinetic energy of architrino i,
- U_{ij} is mutual potential energy of pair (i, j),
- $E_{\text{coupling to sea}}$ accounts for how the assembly deforms and polarizes the surrounding Noether sea, that is, the local Noether sea environment (or in bridge prose, the local spacetime medium).

This internal energy can be **very large**: accepted high-energy branches may retain Planck-scale or higher internal energy, even when the assembly appears externally as a low-mass effective particle.

Apparent Energy and Shielding

The surrounding Noether sea, and the arrangement of positive- and negative-polarity architrinos inside an assembly, can **shield** internal energy from the external world through:

- **Polarity cancellation**: positive- and negative-polarity architrinos within the assembly (and in surrounding Noether braids) emit wakes that interfere destructively at larger distances.
- **Phase-structured far-field cancellation**: the geometry of internal orbits and Noether braid polarization patterns generates cancellation of most multipoles at scales $r \gg$ assembly size.
- **Nested shielding**: in multi-binary fermion cores, outer binaries partially screen the deeper binaries from the surrounding sea. Generation shifts can therefore be read as loss of shielding tiers, not only as loss of constituent count.

At the reference-attractor level, define the **shielding (leakage) factor** as the leading isotropic projection of a larger far-field wake ledger:

$$\zeta(A_0) \equiv \frac{\|\Pi_0 \mathcal{L}_{\text{wake}}(A_0)\|}{\|\mathcal{L}_{\text{naive}}(A_0)\|}, \qquad \mathcal{L}_{\text{aniso}}(A_0) \equiv \mathcal{L}_{\text{wake}}(A_0) - \Pi_0 \mathcal{L}_{\text{wake}}(A_0)$$

evaluated in a regime where the assembly appears as an effective point source. Here Π_0 extracts the monopole/isotropic component of the far-field wake ledger and $\mathcal{L}_{\text{aniso}}$ retains anisotropic leakage instead of hiding it inside a scalar error term. For a strongly shielded, neutral Noether

braid in the Noether sea, we expect $\zeta \ll 1$.
Operationally, extract $\zeta(A)$ from a far-field fit of $\Phi_{\rm net}$ (or hit amplitude) at $r \gg \mathrm{size}(A)$: $\zeta \equiv A_{\rm measured}/A_{\rm naive}$, the ratio of the leading $1/r^2$ (or multipole) coefficient to the naive constituent sum, with anisotropic residuals reported separately.
The scalar shielding summary is admissible only when anisotropic leakage is small enough for the comparison being made, for example

$$\frac{\|\mathcal{L}_{\rm aniso}(A_0)\|}{\|\mathcal{L}_{\rm naive}(A_0)\|} \le \epsilon_{\rm aniso}$$

with $\epsilon_{\rm aniso}$ declared before the branch is promoted to a scalar mass-facing result.

The scalar apparent-energy proxy that influences other assemblies at large distances is then:

$$E_{\rm apparent}(A) \sim \zeta(A)\, E_{\rm internal}(A),$$

This is a roadmap relation, not a substrate identity; proportionality constants must be fixed by matching to effective low-energy theory (e.g. mapping to mc^2).

The exposed energy cannot be counted twice as both the direct probe readout and the sea-retuning source. On a declared comparison window, split the exposed ledger into a probe channel and a sea-coupled channel:

$$\zeta(A)E_{\rm internal}(A) = E_{\rm probe}(A) + E_{\text{sea-coupled}}(A) + E_{\rm unresolved}(A), \qquad E_{\rm probe}(A) + E_{\text{sea-coupled}}(A) \le \zeta(A)E_{\rm internal}(A)$$

with partition residual

$$\mathcal{R}_{\rm part}(A) = \frac{|\zeta(A)E_{\rm internal}(A) - E_{\rm probe}(A) - E_{\text{sea-coupled}}(A)|}{|\zeta(A)E_{\rm internal}(A)| + \varepsilon_{\rm part}}$$

The mass map couples distant probes to $E_{\rm probe}$ through the retuned Noether sea; the matter-to-sea source uses $E_{\text{sea-coupled}}$. A calculation that uses the raw $\zeta E_{\rm internal}$ in both roles must report $\mathcal{R}_{\rm part}$ as unresolved rather than treating the two uses as independent evidence.

Emergent Inertia (Mass) from Shielded Energy

Inertia is not fundamental; it is the externally exposed response of an assembly's closed internal causal-history ledger, shielding factor, and Noether sea coupling to changes in bulk motion.

Operational Definition of Inertial Mass

For an assembly A, define its inertial mass $m_{\rm inertial}(A)$ operationally via:

- Apply a small external wake potential (from a distant test source) that exerts a known net force $\mathbf{F}_{\rm ext}$ on A,
- Measure the resulting acceleration of the response center; in regimes where the effective center-of-mass readout has been justified, denote this acceleration by $\mathbf{a}_{\rm cm}$,
- Define:

$$m_{\rm inertial}(A) \equiv \frac{\|\mathbf{F}_{\rm ext}\|}{\|\mathbf{a}_{\rm cm}\|}.$$

Because the external wake couples mainly to the probe-facing exposed energy, not the full internal circulation, the scalar roadmap limit is:

$$m_{\rm inertial}(A) \approx \alpha_{\rm m}\, \frac{E_{\rm probe}(A)}{c_{\rm eff}^2}.$$

The tensor handoff is more precise. In the formulas below, $\mathcal{Z}_A^{ab}$ is the probe-channel exposure tensor after the exposed-energy partition has been declared; the sea-coupled channel enters through $S^{(\ell)}_{\rm mat\to sea}$ and the resulting Noether sea response, not as a second direct inertial source. For a small center-of-mass velocity $V_{{\rm cm},b}$ through a declared Noether sea response record,

$$p^a_{\rm int} \approx \alpha_{\rm m}\, \zeta(A)E_{\rm internal}(A)\, \mathcal{M}^{ab}_{\rm sea}\, V_{{\rm cm},b}$$

with homogeneous isotropic limit

$$\mathcal{M}^{ab}_{\rm sea} \to \frac{h^{ab}}{c_{\rm eff}^2}$$

A more complete first-order handoff keeps the scalar and trace-free exposure pieces visible. Write

$$\mathcal{Z}_A^{ab} = \zeta(A)h^{ab} + \mathcal{Z}^{ab}_{\rm tf}(A), \qquad h_{ab}\mathcal{Z}^{ab}_{\rm tf}(A) = 0$$

and split the local Noether sea response as

$$\mathcal{M}_{\text{sea}}^{ab} = \frac{1}{c_{\text{eff},0}^2}\left[(1+\delta\mathcal{M}_0)h^{ab} + \delta\mathcal{M}_{\text{tf}}^{ab}\right]$$

Then the exposed inertial-response tensor is

$$\mathsf{I}_A^{ab} = \frac{\alpha_{\text{m}}E_{\text{internal}}(A)}{2}\left(\mathcal{Z}^a_{A\,c}\mathcal{M}_{\text{sea}}^{cb} + \mathcal{Z}^b_{A\,c}\mathcal{M}_{\text{sea}}^{ca}\right), \qquad p_{\text{int}}^a \approx \mathsf{I}_A^{ab}V_{\text{cm},b}$$

Its rotational scalar trace is

$$m_{\text{tr}}(A) \equiv \frac{1}{3}h_{ab}\mathsf{I}_A^{ab} = \alpha_{\text{m}}\frac{E_{\text{internal}}(A)}{c_{\text{eff},0}^2}\left[\zeta(A)(1+\delta\mathcal{M}_0) + \frac{1}{3}\mathcal{Z}_{\text{tf},ab}(A)\delta\mathcal{M}_{\text{tf}}^{ab}\right]$$

Only in the homogeneous isotropic limit does the scalar mass formula above follow. The trace formula gives a stricter diagnostic: pure exposure anisotropy does not shift scalar mass in an isotropic medium, and pure trace-free medium response does not shift scalar mass for scalar exposure. A scalar mass shift from anisotropy appears only through the contraction $\mathcal{Z}_{\text{tf},ab}\delta\mathcal{M}_{\text{tf}}^{ab}$; otherwise the residue remains directional inertia in I_A^{ab}. Here E_{internal} names the large internal energy circulation, while $\zeta(A)$ names the small external leakage that survives cancellation and Noether sea shielding. The formula therefore explains weak long-range gravitational and inertial footprints without making the internal energy small: ordinary probes couple to the leaked pattern, not to every internal exchange branch.

This scalar trace is admissible as a positive inertial mass only inside the shielding window

$$\zeta(A)(1+\delta\mathcal{M}_0) > \frac{1}{3}\left|\mathcal{Z}_{\text{tf},ab}(A)\delta\mathcal{M}_{\text{tf}}^{ab}\right|$$

with the comparison sea state declared. If a certified A_0 branch reports ζ so small that this inequality fails for plausible $\delta\mathcal{M}_{\text{tf}}^{ab}$ in the accepted environment, the shielded-energy mass map is falsified for that branch. Thus $\zeta \ll 1$ is a constrained exposure window, not an unconstrained way to suppress all long-range response.

At the matter-to-medium interface, a Standard Model fermion assembly should therefore be treated as a localized source of exposed response, not as an unshielded transfer of all internal energy into the surrounding Noether sea. For a coarse cell Ω_ℓ, the source supplied by stable matter assemblies can be written schematically as

$$S_{\text{mat}\to\text{sea}}^{(\ell)}(\mathbf{x},t) = \sum_{A\subset\Omega_\ell} W_\ell(\mathbf{x}-\mathbf{X}_A(t))\,E_{\text{sea-coupled}}(A) + S_{\text{aniso}}^{(\ell)}(\mathbf{x},t)$$

where W_ℓ is the coarse-graining window, $\mathbf{X}_A$ is the assembly center, and $S_{\text{aniso}}^{(\ell)}$ records exposed tensor, orientation, spin, or wake-history residue that cannot be collapsed into the scalar shielding factor. This source then perturbs the local Noether sea state through a constitutive response map,

$$\delta\theta_{\text{sea}}^{(\ell)} = \mathcal{C}_{\text{mat}\to\text{sea}}\left(S_{\text{mat}\to\text{sea}}^{(\ell)}, \lambda_A, \xi_A, \mathcal{H}_A, \theta_{\text{sea},0}^{(\ell)}\right)$$

with $\delta\theta_{\text{sea}}^{(\ell)}$ projecting into n, χ_{sea}, Γ_N, strain, orientation, cadence, and envelope-scale variables. In this language, saying that neighboring Noether braids absorb the exposed potential means that they retune their branch state. Depending on the accepted branch, that retuning may appear as higher cadence, changed strain, stronger alignment, envelope-scale shift, or altered coupling to nearby Noether braids; it should not be compressed into a generic statement that the cores simply gain energy and expand.

This is the same shielding-based logic developed more directly in Particle Masses. The matching factor α_{m} should be fixed only after a calibration-free reference attractor has supplied E_{internal}, ζ, and the medium-response map; it should not be fitted separately to each particle species. Universality is a cross-species invariant, not a notation choice. For any certified assembly A, define the back-solved value

$$\alpha_{\text{m}}(A) \equiv \frac{m_{\text{tr}}(A)c_{\text{eff},0}^2}{E_{\text{internal}}(A)\left[\zeta(A)(1+\delta\mathcal{M}_0) + \frac{1}{3}\mathcal{Z}_{\text{tf},ab}(A)\delta\mathcal{M}_{\text{tf}}^{ab}\right]}$$

on branches that pass the positivity gate above. For any pair A, A' in the mass-map test set, require

$$\mathcal{R}_\alpha(A,A') \equiv \frac{|\alpha_{\text{m}}(A)-\alpha_{\text{m}}(A')|}{|\alpha_{\text{m}}(A)|+|\alpha_{\text{m}}(A')|} \le \epsilon_\alpha$$

with ϵ_α declared before promotion. If this residual cannot be held small without per-species tuning, the universality claim fails and the parameter count must be raised explicitly.

Thermodynamic or entropic derivations of gravitational force are therefore comparison benchmarks for this chapter, not replacements for the mass mechanism. They may sharpen the observer-level equation-of-state target for gravity, but $m_{\text{inertial}}(A)$ is not closed until the same assembly ledger supplies its closed internal causal-history record, shielding extraction, Noether sea response tensor, and acceleration response.

The immediate hand-off is the A_0 reference attractor gate. The energy chapter owns the internal-energy and apparent-energy definitions that A_0 must report: layer energies, interaction and wake terms, total $E_{\text{internal}}(A_0)$, far-field wake coefficients, $E_{\text{probe}}(A_0)$, $E_{\text{sea-coupled}}(A_0)$, and $\mathcal{R}_{\text{part}}(A_0)$. Those outputs are still closure targets until a stable branch, shielding extraction, and response tensor are computed. Compact finite-coordinate no-go records and branch-chart checker results cannot be consumed as energy-accounting inputs: a rejection blocks the chart path, and a clearance authorizes only a rerun candidate until Tier 2 shielding exists on an accepted branch.

The multi-scale status of A_0 matters for this accounting. Fast internal corrections should not be removed until they are classified. Nonresonant inner-layer motion may average out of the leading apparent-energy fit, but corrections that change self-hit counts, the branch Jacobian near c_f, or the leakage tensor can change $E_{\text{internal}}(A_0)$, $\zeta(A_0)$, or both. Apparent energy is therefore downstream of closure and stability, not an input used to force a convenient branch.

Noether Sea and Effective Relativistic Behavior

The nested shell braid Noether sea adds an additional layer:

- Moving assemblies must retune their internal causal ledger and reorganize local Noether sea coupling.
- The effective resistance to high center-of-mass speed (near the internal nested shell braid causal-wake propagation scale) increases steeply, producing an emergent “speed of light” scale c_{eff} at which assemblies effectively saturate.

Thus:

- At low center-of-mass speeds $v_{\text{CM}} \ll c_{\text{eff}}$, the effective readout recovers $E_k \approx \frac{1}{2} m_{\text{inertial}} v_{\text{CM}}^2$ for assemblies.
- At high center-of-mass speeds approaching c_{eff}, internal coupling to the Noether sea and self-hit effects yield a relativistic-like $E_k \sim m_{\text{inertial}} c_{\text{eff}}^2 (\gamma_{\text{eff}} - 1)$, with $\gamma_{\text{eff}} = 1/\sqrt{1 - v_{\text{CM}}^2 / c_{\text{eff}}^2}$, as an **effective law**.
- Near c_{eff}, axial architrino stripping and oblation are failure channels or branch-transition hypotheses to test, not assumed parts of the mass mechanism.

The details of this emergent relativistic law arise from the combined dynamics of the assembly and the Noether sea; they are not postulated but must be confirmed by coefficient extraction, simulation, and matching to known particle kinematics. Ordinary dissipative drag is a failure channel for this program, not the mass mechanism. The mass-side integration and quantitative derivation path is tracked in Particle Masses.

Effective Energy-Momentum Closure

For assembly center-of-mass motion in the Lorentz-suppressed regime, impose the relativistic mass-shell relation as an **effective closure test** (not a substrate postulate):

$$E_{\text{CM}}^2 = p_{\text{CM}}^2 c_{\text{eff}}^2 + M_0^2 c_{\text{eff}}^4$$

Here:

- M_0 is the assembly rest/internal invariant extracted at $v_{\text{CM}} = 0$ in a locally homogeneous sea.
- E_{CM} and p_{CM} are the total center-of-mass energy and momentum measured from trajectory dynamics.
- c_{eff} is the isotropic projection of the local Noether sea response-speed record; in weak-field homogeneous and neutral conditions that also pass the two-moment quietness condition above, $c_{\text{eff}} \to c_f$.

More precisely, the response-speed tensor may be written schematically as

$$\left(c_{\text{eff}}^2\right)^{ab} = c_f^2 \left[(1 + \delta c_0) h^{ab} + \delta c_{\text{tf}}^{ab}\right], \qquad c_{\text{eff}}^2 \equiv \frac{1}{3} h_{ab} \left(c_{\text{eff}}^2\right)^{ab}$$

The scalar mass-shell closure is admissible only when the anisotropic propagation correction is bounded,

$$\|\delta c_{\text{tf}}\| \le \epsilon_{c,\text{tf}}$$

on the same comparison window. The scalar offset $\delta c_0 \to 0$ is not assumed by isotropy language alone; it must follow from the same homogeneous neutral summation and screening conditions that make the Noether sea macroscopically quiet.

Equivalent parameterization:

$$E_{\text{CM}} = \gamma_{\text{eff}} M_0 c_{\text{eff}}^2, \qquad p_{\text{CM}} = \gamma_{\text{eff}} M_0 v_{\text{CM}}, \qquad \gamma_{\text{eff}} = \frac{1}{\sqrt{1 - v_{\text{CM}}^2 / c_{\text{eff}}^2}}$$

Consistency requirement: if this closure fails in regimes where emergent Lorentz behavior is claimed, the current mass-loading and medium-response model is incomplete.

Cross-links:

- Proper-time closure test
- SR mapping entry

Energy and Self-Hit in the Noether Sea

In the **super-field-speed** regime ($\|\mathbf{v}_a\| > 1$ somewhere along the relevant path-history interval), architrinos and assemblies can intersect their own past isochrons (self-hit). In the presence of the Noether sea:

- Self-hit repulsion acts as an internal **stiffening mechanism** for nested shell braids and more complex assemblies, contributing to their stability.
- Energy represented in an architrino's causal wake and local Noether sea response can be partially routed back through delayed self-interaction. At the bookkeeping level, this is an exchange between internal kinetic energy and wake/medium energy associated with the local nested shell braid configuration.

At the exact causal-action level, global energy is conserved: self-hit just routes energy along more complex paths (architrino → causal isochron → local Noether sea → back to architrino/assembly). In dual-mollified local theorem models, the same statement should be read conditionally unless the mollified kernel is explicitly tied to an action-level regularization.

Intuition (Plain Language)

Inside an assembly, large internal causal-history energy can circulate through many branch channels. Outside the assembly, distant probes couple only to the portion of that ledger that survives phase cancellation, shielding, and Noether sea response.

Architrinos and their assemblies are where the energy bookkeeping lives. The Noether sea is a dense population of high-energy Noether braid assemblies whose net long-range wake response is usually quiet because incoherent contributions cancel and shielded internal layers leak only weakly. In nested fermion cores, outer binaries screen deeper layers from the ambient Noether sea. The small residual exposure is what observer-level mass and gravitational response measure.

Summary and Role in the Larger Theory

- **At the architrino level:**

  Kinetic energy and potential energy are defined via the Master EOM. Exact global conservation belongs to the exact causal-action theory; in mollified working models it is the target bookkeeping structure and is exact only when the regularization preserves the underlying time-translation symmetry. The substrate law is acceleration-first; no particle-specific fundamental mass is assigned to architrinos, and speeds are unbounded in principle.
  Potential availability is geometric rather than fuel-like: causal wakes are emitted as path-history structure, while work appears only when a receiver intersects active wake branches with nonzero radial power.

- **At the assembly level:**

  Large internal energies, plus coupling to the Noether sea, generate:

  - Effective inertia (mass),
  - Shielded external wake signatures (tiny apparent energy compared to internal),
  - Generation dependence through how many outer screening layers still surround the deepest core,
  - An emergent speed scale c_{eff} and relativistic-like behavior.
    Macroscopic quietness follows from superposition and shielding: incoherent populations cancel statistically, while phase-locked assemblies such as collinear breathers preserve localized, non-canceling wake structure.

- **For spacetime and gravity:**

  The sea of small, high-energy Noether braids forms the Noether sea and, at coarse-grained level, the effective spacetime medium whose energy density and stress give rise to an emergent metric. The shielding factors and internal energies of both Noether braids in the Noether sea and “matter” assemblies contribute to:

  - The effective Newton constant G,
  - The cosmological Noether sea energy density,
  - How strongly observer-level effective metric response is reconstructed from different kinds of energy.

  Density-driven oblation is a candidate contribution to the effective gravitational-coupling closure: as the Noether sea encounters denser matter, local Noether braids may scale down and oblate, creating a compliance gradient that must be mapped through the Noether sea response tensor before it can be read as part of G.

Appendix A: Energy Zero and Bookkeeping

$\mathbb{AAA}$ uses a **binding-energy convention** that fixes the zero of potential energy at the **inner turning point** of an accepted bound branch (the self-hit / max-curvature radius when that branch has been certified). This choice is operational: on a branch with a self-hit lower boundary, the deepest accessible state supplies the reference. It should not be read as a proof that every isolated two-body candidate already has a unique, history-independent cutoff.

Cosmology inventory prose uses the same convention only after declaring the comparison window. Positive component entries such as matter, radiation, dark-sector bookkeeping, and thermal reservoirs are mass-equivalent or energy-density terms measured relative to that window, while gravitational binding is a negative finite-window contribution. Mixing a local branch convention with a cosmological inventory convention without naming the window and boundary term risks double counting the same retained wake-history energy.

Physical Setup and Why a New Zero is Needed

For an accepted attractive bound branch (opposite polarities), the inward motion accelerates until it reaches a **minimum radius** $r_{\min}$ where the certified self-hit and curvature records prevent further collapse. The motion then rebounds or orbits. Unlike a pure Coulomb potential, this branch has a lower bound on radius (and hence on accessible energy states).

Because a lower bound exists, the natural reference is **not** “infinite separation” but the **ground configuration** at $r_{\min}$.

The Bookkeeping Convention

We adopt a **singular-boundary gauge**: on a certified branch chart with a declared self-hit lower boundary $r_{\min}$ (the maximum curvature attractor), we fix the potential gauge at this wall.

$$U(r_{\min}) \equiv 0.$$

In this gauge, $U(r)$ represents the **accumulated work** performed to separate the binary from its ground state to radius r. Total energy is thus partitioned into *kinetic* (motion) and *deformation* (separation) components, with fully separated (unbound) pairs carrying maximal deformation energy $U_{\max} \equiv B_{\max}$.

When the active causal-root ledger changes, this gauge must be indexed by the branch ledger. For ledger cell b,

$$U^{(b)}(r) \equiv B^{(b)}_{\max} - B^{(b)}(r), \qquad B^{(b)}_{\max} = B^{(b)}\left(r^{(b)}_{\min}\right)$$

so a separator crossing that changes the effective inner wall cannot be counted once as a gauge-origin jump and again as an independent h-like energy quantum. At a crossing radius r_* between ledger cells b and b', the physical bookkeeping must satisfy

$$[E_{\text{total}}]_{b \to b'} = K^{(b)}(r_*) + U^{(b)}(r_*) = K^{(b')}(r_*) + U^{(b')}(r_*) + \Delta_{\text{ledger}}$$

where Δ_{ledger} is the declared root-change energy routed through the table entries such as ε_o, ε_w, and the middle-channel adjustment. The visible step is the ledger/gauge matching term; it is not additional to that matching.

Binding Energy and Total Energy

Let $B(r)$ denote the **binding energy** at radius r, with

$$B(r_{\min}) = B_{\max}.$$

Define

$$U(r) = B_{\max} - B(r).$$

Then total energy bookkeeping is:

$$E_{\text{total}} = K(r) + U(r), \qquad U(r) \geq 0.$$

At the minimum radius:

$$E_{\text{total}} = K_{\max}, \quad U(r_{\min}) = 0.$$

All available mechanical energy is kinetic at the inner turning point. Moving outward converts kinetic energy into potential energy (the rebound / climb-out phase).

Effective Potential Language

If an effective potential is used, the centrifugal term and the self-hit barrier both contribute:

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2m_{\text{eff}}r^2} + V_{\text{self-hit}}(r).$$

Here m_{eff} is an **effective inertial scale** (a bookkeeping proxy for mass in the coarse-grained description), not a primitive architrino mass.

The convention above fixes:

$$V_{\text{eff}}(r_{\min}) = 0.$$

This does **not** change dynamics; it sets a physically meaningful reference.

Self-Hit Echo and Discrete Steps (Working Note)

In the current picture, the self-hit region is **not** assumed to change the local force law. The radial slope remains smooth:

$$\frac{dU}{dr} \text{ remains finite and continuous across the retained regularized branch chart.}$$

So the transition between the $v = c_f$ regime and the self-hit regime is a **regularized branch transition**, not a kink in the potential. The distinction shows up in **how action and energy bookkeeping are routed** between binaries, not in a new macroscopic slope.

The discrete step is a causal-root ledger effect, not an assumption that energy itself is made of independent chunks. On a fixed branch chart, the active causal intersections have an integer multiplicity: a self-hit count N and an analogous partner-hit or channel count M in the root-ledger language developed in the closed-form collinear breather ansatz. In the circular binary notation this same idea appears as the pair (N_s, M_p) in Super-Field-Speed Root Ledgers and Resonance Lock. Within one ledger cell the underlying trajectory and $U(r)$ remain continuous. A visible h-like transaction occurs when a separator crossing changes the admissible integer ledger, for example by adding one grouped channel or, in the raw simple-root table, by a fold-pair jump satisfying $\Delta N \in 2\mathbb{Z}$ with $\Delta D = 0$.

The mechanical event behind such a ledger change can be a caustic-grazing impulse. When a regularized branch crosses a $J = 0$ caustic, the pointwise branch expression may become large while the integrated velocity change remains finite, as in Caustic Transit and Finite Impulse:

$$\Delta \mathbf{v}_{a,n} = \int_{t_n^-}^{t_n^+} \mathbf{a}_a^{(\eta)}(t)\, dt$$

This finite impulse is a candidate substrate mechanism for changing the active causal-root ledger by a discrete amount without making primitive energy granular.

Thus the candidate quantum of action is geometric bookkeeping: it is the action scale assigned to a threshold crossing of the causal-root ledger. The energy shift appears in steps because the allowed causal intersections have changed discretely, even though the path-history geometry and the local potential slope remain continuous through the regularized fold layer. A closed branch chart must still expose the root-change energy, wake exchange, middle-channel adjustment, and any mismatch routed into unresolved modes.

Working bookkeeping hypothesis:

- Outer binary registers a single-step transaction (h-like unit), meaning one minimal admissible update of its active partner and self channel ledger.
- Middle binary adjusts to conserve total energy.
- Inner binary executes a two-step shift ($2h$-like unit), i.e., two discrete ledger updates rather than one. The “step” corresponds to the system crossing a separatrix between basins of attraction in the nonlinear delay dynamics. While the underlying trajectory is continuous, the energy redistribution stabilizes only at discrete resonances (winding numbers and causal-root multiplicities), making the effective energy transfer appear quantized.

This can read as an “amplified” response, but only because the inner binary is **releasing or reconfiguring retained internal energy** when the self-hit echo is engaged. It is **not** net energy creation; it is a redistribution between internal stores under a smooth $U(r)$.

Nested Shell Braid as Routing/Locking Circuit (Analogy)

It is useful (as a **bookkeeping analogy**) to think of the nested shell braid as a **routing/locking circuit** rather than a simple reservoir. An incoming single-step transaction (h-like) couples most strongly to the **outer binary**, the **middle binary** acts as a buffer/fulcrum that maintains overall consistency, and the **inner binary** can respond with a two-step reconfiguration when the self-hit echo is engaged. The effective response can resemble a geared or ratcheted redistribution, but the mechanism is still deterministic energy routing, not creation.

In this language, a discrete input can **lock in** a new nested shell braid configuration: a threshold-triggered, history-dependent update that selects one stable branch over another. This is a **collapse-like** event in the phenomenological sense (a sudden, discrete state update), but in $\mathbb{AAA}$ it is treated as a **deterministic, microstate-sensitive bifurcation**, not an intrinsically stochastic collapse.

Bookkeeping Table: One h of Closed-Cycle Action (Outer $v < c_f$)

For the h versus $\hbar$ convention used here, see Angular Momentum and Spin.

Assumptions for this bookkeeping pass:

- f labels a discrete outer-binary orbital state (frequency index). The three rows are **pre-hit** ($f-1$), **action/transition** (f_ψ), and **post-redistribution** (f). There is **one** step in frequency. The f_ψ label is a transient bookkeeping state, not a new frequency index or literal wave function.
- The transaction is a single closed-cycle action unit, $\Delta A_{\text{cycle}} = +h$, coupled first to the **outer** binary while $v_{\text{out}} < c_f$.
- The symbol h labels action per full causal phase cycle. The associated radian-normalized rotational-action increment is $\hbar = h/(2\pi)$; in this local bookkeeping pass ΔI denotes that angular-momentum/action variable.
- Energy bookkeeping uses action-angle language: for a small discrete step, $\Delta E \approx \omega\,\Delta I = f\,\Delta A_{\text{cycle}}$. This is a **notation choice**, not a claim about the exact micro-law.
- The **inner binary** responds with a two-step reconfiguration. The **middle binary** adjusts to satisfy conservation of total energy and total angular momentum (including any causal-wake exchange).

Notation in the table:

- K_o, U_o = outer-binary kinetic and potential energies.
- K_m, U_m = middle-binary kinetic and potential energies.
- K_i, U_i = inner-binary kinetic and potential energies.
- Superscripts $(f-1)$, (f_ψ), and (f) denote the state index (one-step update).

Per-step increments (explicit, no deltas):

- Outer step energy: $\varepsilon_o \equiv \omega_o \hbar$ with

$$k_o \equiv \chi_o\,\varepsilon_o, \quad u_o \equiv (1-\chi_o)\,\varepsilon_o,$$

so $k_o + u_o = \varepsilon_o$.

- Inner step energy: $\varepsilon_i \equiv \omega_i \hbar$ with

$$k_i \equiv \chi_i\,\varepsilon_i, \quad u_i \equiv (1-\chi_i)\,\varepsilon_i,$$

so $k_i + u_i = \varepsilon_i$. Because the inner binary takes **two steps**, it adds $2k_i$ and $2u_i$.

- Middle adjustment energy: ε_m is whatever is needed to close the ledger. Here ε_w denotes the **causal-wake exchange energy** during the step:

$$\varepsilon_m \equiv \varepsilon_w - 2\varepsilon_i,$$

and we split it as

$$k_m \equiv \chi_m\,\varepsilon_m, \quad u_m \equiv (1-\chi_m)\,\varepsilon_m.$$

State	Outer (o)	Middle (m)	Inner (i)	Notes
$f-1$	K_o^{f-1}, U_o^{f-1}	K_m^{f-1}, U_m^{f-1}	K_i^{f-1}, U_i^{f-1}	Baseline. No pending transaction.
f_ψ	$K_o^{f_\psi} = K_o^{f-1} + k_o$ $U_o^{f_\psi} = U_o^{f-1} + u_o$	$K_m^{f_\psi} = K_m^{f-1}$ $U_m^{f_\psi} = U_m^{f-1}$	$K_i^{f_\psi} = K_i^{f-1}$ $U_i^{f_\psi} = U_i^{f-1}$	Immediate post-hit. Outer receives $\Delta I_o = +\hbar$ in the initial bookkeeping gauge. Outer records a (k_o, u_o) increment.
f	$K_o^f = K_o^{f-1} + k_o$ $U_o^f = U_o^{f-1} + u_o$	$K_m^f = K_m^{f-1} + k_m$ $U_m^f = U_m^{f-1} + u_m$	$K_i^f = K_i^{f-1} + 2k_i$ $U_i^f = U_i^{f-1} + 2u_i$	Post-redistribution. Outer update is complete at f_ψ; only middle/inner continue to settle.

Constraints to apply across the $f-1 \to f$ transition (bookkeeping level):

- **Angular momentum / rotational action**: the sign rule is gauge-invariant only after declaring the allowed wake share:

$$\Delta I_{\text{out}} + \Delta I_{\text{mid}} + \Delta I_{\text{in}} + \Delta I_{\text{wake}} = +\hbar, \qquad |\Delta I_{\text{wake}}| \le \epsilon_w \hbar$$

For a **net positive** transaction, the layer increments must satisfy $\Delta I_k \ge -\epsilon_w \hbar$ for $k \in \{\text{out}, \text{mid}, \text{in}\}$. For a **net negative** transaction, the same bound applies with signs reversed. The nonnegative-increment claim is therefore an up-to-wake-tolerance statement, not a gauge-free statement that the wake channel carries exactly zero rotational action.

- **Energy**: $(k_o + u_o) + (k_m + u_m) + 2(k_i + u_i) = \varepsilon_o + \varepsilon_w$. This is the explicit version of conservation using the per-step increments defined above.
- **Root-ledger closure**: the transition must move from one admissible integer causal-root ledger to another and then close consistently over the full cycle. In a raw self-root table, separator crossings obey the parity rule $\Delta N \in 2\mathbb{Z}$ and $\Delta D = 0$; in a grouped channel ledger, the same event may be recorded as one newly active channel.

- **Cross-ledger gauge matching**: any jump in $r_{\min}^{(b)}$ and $B_{\max}^{(b)}$ is part of the declared Δ_{ledger} budget above. A table row may not count the same gauge-origin shift once in $U^{(b)}$ and again as an extra wake or oscillator energy.
- **Smooth slope**: dU/dr remains continuous across the graft; the discrete behavior comes from **state updates**, not a kink in $U(r)$.

This table is intentionally explicit: every h closed-cycle action transaction is represented by a radian-normalized $\hbar$ rotational-action increment, split into a kinetic part (k) and a potential part (u). The remaining freedom is **how** each binary partitions its step (the χ fractions) and how the middle, inner, and causal-wake channels redistribute the initial outer-binary coupling.

Comparison to Coulomb and Standard Conventions

In pure Coulomb,

$$V(r) = -\frac{kq^2}{r},$$

so there is no inner bound and no natural finite zero. Classical mechanics therefore chooses $V(\infty) = 0$.

In $\mathbb{AAA}$, a certified hard inner bound **supplies** a natural zero at $r_{\min}$, which is the lowest accessible state. The bookkeeping therefore switches from “energy relative to infinity” to “energy relative to the ground state.”

Summary Table (Operational Meaning)

Region	K	U	Meaning
$r = r_{\min}$	max	0	Fully bound (ground)
$r > r_{\min}$	$\downarrow$	$\uparrow$	Climbing out / rebound
escape limit	0	$B_{\max}$	Free (unbound)

One-Line Rule

If the model has a hard inner bound, **set the potential zero at that bound** and measure all energies outward from it.

Adiabatic branch invariant target. On a certified branch chart for binary layer a, suppose the reduced cycle admits a canonical pair (Q_a, Π_a) and a slowly varying branch parameter $\lambda(t)$, such as a local Noether sea response variable, shielding parameter, or neighboring-layer phase parameter. Define the rotational action

$$I_a(\lambda) \equiv \frac{1}{2\pi} \oint_{\gamma_a(\lambda)} \Pi_a \, dQ_a$$

If the parameter changes slowly compared with the cycle period $T_a(\lambda)$,

$$\epsilon_{\text{ad},a} \equiv \max_{t \in W} \left(T_a(\lambda(t)) \left\| \frac{d\lambda}{dt} \right\| \ell_\lambda^{-1} \right) \ll 1$$

and the path remains a positive distance from the causal-root ledger-cell boundary,

$$\operatorname{dist}(\gamma_a(\lambda), \partial \mathcal{G}_a) \geq \delta_{\text{cell}} > 0$$

the interior adiabatic theorem target is

$$\frac{dI_a}{dt} = O(\epsilon_{\text{ad},a}) + \mathcal{R}_{\text{int},a}(t)$$

Here ℓ_λ is the declared scale over which the reduced Hamiltonian changes appreciably, and $\mathcal{R}_{\text{int},a}$ records omitted wake-history exchange, non-characteristic boundary leakage, or small chart error while the branch stays inside one ledger cell. At a separator crossing or root-fold boundary, the interior estimate is void. The crossing rule is instead

$$\Delta I_a \big|_{\text{fold}} = \frac{1}{2\pi} \left(\oint_{\gamma_a'} \Pi_a \, dQ_a - \oint_{\gamma_a} \Pi_a \, dQ_a \right) = \Delta I_{\text{ledger},a}$$

where $\Delta I_{\text{ledger},a}$ is the declared quantized ledger increment associated with the change in active causal-root multiplicity or branch chart. Thus the action variable is expected to drift only adiabatically inside a ledger cell, while a root-ledger transition may produce the discrete ΔI recorded above. This turns the h-like bookkeeping into a branch-boundary invariant target rather than an assumption that energy itself is quantized at the primitive level.

Action-Energy

Action Model

This note compares three modeling options for the emission-propagation-interaction pipeline and recommends a primary approach, with supporting roles for the others. We work in units with field speed $v = 1$ unless stated otherwise; emission cadence and per-wavefront amplitude are constant at the source; per-hit actions are directed along $\hat{\mathbf{r}}$ with inverse-square geometric decay and Jacobian-weighted magnitude; $H(0) = 0$ excludes the coincident-time self-kick; no cross products or right-hand-rule terms appear.

Setup / assumptions

- The emitter is at position $\mathbf{x}_s(t)$ in 3-D space (it can move).
- The emitter emits **thin causal wake surfaces**. Each wake surface is created at a single instant τ and then expands outward **spherically** from the creation point.
- The wake surface radius after emission time τ is

$$r(t, \tau) = c\,(t - \tau) \quad \text{for } t \geq \tau$$

where c is the constant **field speed**.

- Each emitted wake surface carries a **strength** Q, interpreted here as wake-surface amplitude. Its physical bookkeeping role depends on the comparison target: polarity, potential impulse, energy, or another declared quantity.
- Continuous source (preferred): model the emitter as a moving point injection with time-density $q(t)$ (amplitude per unit time) at its instantaneous position, i.e., $S(\mathbf{x}, t) = q(t)\,\delta\big(\mathbf{x} - \mathbf{x}_s(t)\big)$. Each instant t_0 contributes a causal wake surface; we do not count "wake surfaces per second" (pulse trains are merely numerical surrogates).
- The diagnostic target is the effective scalar potential $\phi(\mathbf{x}, t)$ produced at any point $\mathbf{x}$ and time t.
- Global neutrality (working hypothesis): on large scales the total architrino polarity inventory sums to zero (equal counts of $\pm\epsilon$); use this as the default boundary condition in PDE/Green's-function comparisons.

We compare three frameworks: (1) a time-domain PDE/source, (2) an integral/Green's-function (path history) solution, and (3) an event-driven radial-transport plus per-hit EOM. For each, we define symbols, show how the expanding causal wake surfaces appear, discuss how slowing or stopping the emitter is handled, and weigh trade-offs to inform a recommendation.

Time-based PDE (wave equation with a moving point source)

Physical idea: keep the source as "something injected per unit time at the emitter location," put that into a PDE surrogate for causal wake propagation at speed c, and let the PDE produce expanding spherical causal wake surfaces automatically. Numerically this is usually the easiest and most robust approach.

PDE model

Use the scalar wave equation as a continuum comparison surrogate for finite-speed causal-wake reconstruction:

$$\boxed{\frac{\partial^2 \phi}{\partial t^2}(\mathbf{x}, t) - c^2\,\nabla^2 \phi(\mathbf{x}, t) \;=\; S(\mathbf{x}, t)}$$

Symbols

- $\phi(\mathbf{x}, t)$: scalar potential surrogate at position $\mathbf{x} \in \mathbb{R}^3$ and time t.
- c: field propagation speed (units length/time).
- ∇^2: Laplacian operator in space (sums second spatial derivatives).
- $S(\mathbf{x}, t)$: source term (right-hand side) — this is how the emitter injects wake surfaces into the field.

Point (moving) source form

Use a continuous time-density of emission at the moving point:

$$S(\mathbf{x}, t) \;=\; q(t)\,\delta\big(\mathbf{x} - \mathbf{x}_s(t)\big)$$

Here $q(t)$ has units "amplitude per unit time." The finite-speed wave operator then generates outgoing spherical causal wake surfaces automatically; no discrete wake surface count is assumed.

How expanding causal wake surfaces appear

- The source term does not explicitly insert a radius into the right-hand side. Instead, the PDE and the finite speed c cause any instantaneous injection at the point $\mathbf{x}_s(\tau)$ to produce an outgoing spherical causal wake surface whose front moves outward at speed c.

That is the built-in behavior of the wave-equation surrogate.

- The Green's function ensures that, at $(\mathbf{x}, t)$, only the path history emission $q(\tau)$ with $\tau = t - r/c$ contributes, producing an outgoing spherical wave with amplitude $q(\tau)/(4\pi r)$ supported on $r = c(t - \tau)$. Thus Method 1 with $S(\mathbf{x}, t) = q(t)\delta(\mathbf{x} - \mathbf{x}_s(t))$ naturally yields expanding causal wake surfaces at speed c.

Why $\|\mathbf{v}\|$ (emitter speed) does not cause blow-ups

- If the emitter slows or stops, $S(\mathbf{x}, t)$ remains nonzero at the same spatial location; the wave equation spreads each injection outward at speed c. No $1/\|\mathbf{v}\|$ singularity appears because the formulation does not convert from per-time emission to per-distance emission.
- Numerically, represent the point delta by a small, smooth kernel when avoiding grid artifacts. For example, instead of $\delta(\mathbf{x} - \mathbf{x}_s)$ use a small Gaussian of width σ comparable to grid spacing.

Numerical recipe (simple)

- Choose spatial grid $\mathbf{x}_i$ and time step Δt satisfying CFL stability (roughly $c\Delta t/\Delta x \leq \text{const}$).
- Use a standard finite-difference time stepping for the wave equation (centered difference in time and space).
- At each time step t_n add the source contribution $S(\cdot, t_n)$ to the RHS at the grid cells nearest $\mathbf{x}_s(t_n)$. If the emitter stops, it remains injecting at that grid location — the solver propagates outgoing wake surfaces.
- To avoid a numerical spike, spread the delta over a few cells with a mollifier, representing a thin wake surface of finite thickness.

Summary for Method 1

- Model is explicit, straightforward, numerically robust.
- Emission is naturally time-based; wake surfaces expand automatically at speed c.
- No division by the emitter speed appears; stopping the emitter is handled simply by keeping the source at the same location.

Integral (Green's function / path-history potential) approach

Physical idea: instead of evolving a PDE in time, write the solution as the sum of contributions from every past emission. For the wave equation the contribution from an impulse emitted at time τ and place $\mathbf{x}_s(\tau)$ arrives at a field point $\mathbf{x}$ only at the **path-history time** when the causal wake surface reaches $\mathbf{x}$. The Green's function neatly encodes the expanding causal wake surface.

Fundamental formula (general)

If the wave equation is

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \nabla^2 \phi = S(\mathbf{x}, t)$$

then the solution may be written as the space–time convolution with the Green's function G:

$$\boxed{\phi(\mathbf{x}, t) \;=\; \iint G\big(\mathbf{x}, t; \mathbf{y}, \tau\big)\; S(\mathbf{y}, \tau)\; d\tau\, d^3y}$$

- $G(\mathbf{x}, t; \mathbf{y}, \tau)$ is the response at $(\mathbf{x}, t)$ to an instantaneous unit impulse at $(\mathbf{y}, \tau)$.

The 3-D free-space wave Green's function

For three spatial dimensions (the usual case for causal wake surfaces), the causal Green's function is

$$G(\mathbf{x}, t; \mathbf{y}, \tau) \;=\; \frac{\delta\big(t - \tau - \frac{\|\mathbf{x}-\mathbf{y}\|}{c}\big)}{4\pi\,\|\mathbf{x} - \mathbf{y}\|}, \qquad t > \tau$$

Interpretation: a unit impulse at location $\mathbf{y}$ and time τ influences $\mathbf{x}$ at time t only when the travel time $\|\mathbf{x} - \mathbf{y}\|/c$ has elapsed; the $1/(4\pi r)$ factor is the usual geometric decay of an outgoing spherical wave in 3D.

Plugging in a moving point source

If the emitter is a moving point source with a time-dependent source amplitude $q(\tau)$ at location $\mathbf{x}_s(\tau)$, then $S(\mathbf{y}, \tau) = q(\tau)\,\delta(\mathbf{y} - \mathbf{x}_s(\tau))$. Plugging this into the convolution gives (integral over τ only):

$$\boxed{\phi(\mathbf{x}, t) \;=\; \int_{-\infty}^{t} \frac{q(\tau)\;\delta\big(t - \tau - \frac{\|\mathbf{x}-\mathbf{x}_s(\tau)\|}{c}\big)}{4\pi\,\|\mathbf{x} - \mathbf{x}_s(\tau)\|}\; d\tau}$$

- $q(\tau)$ is the continuous emission density per unit time at the emission instant τ. For a steady source, $q(\tau) = q_0$ (constant); more generally, q may vary smoothly with τ.

Evaluating the integral — the path-history time

The δ-function in the integrand enforces the *path-history-time condition*:

$$t-\tau=\frac{r(\tau)}{c}, \qquad r(\tau)\equiv\|\mathbf{x}-\mathbf{x}_s(\tau)\|$$

So the contribution to $\phi(\mathbf{x},t)$ comes only from times τ such that the expanding causal wake surface emitted at τ has just reached $\mathbf{x}$ at time t.

Mathematically, use the identity $\delta(g(\tau))=\sum_i \delta(\tau-\tau_i)/|g'(\tau_i)|$ where τ_i are simple roots of g. With $g(\tau)=t-\tau-r(\tau)/c$ we find (after algebra) the standard path-history solution:

$$\boxed{\phi(\mathbf{x},t) \;=\; \sum_{\tau_i}\frac{q(\tau_i)}{4\pi\, r(\tau_i)\,\left|1+\frac{1}{c}\,r'(\tau_i)\right|} \;=\; \sum_{\tau_i}\frac{q(\tau_i)}{4\pi\, r(\tau_i)\,\left|1-\frac{\mathbf{n}(\tau_i)\cdot\mathbf{v}_s(\tau_i)}{c}\right|}}$$

where:

- the sum runs over **path-history times** τ_i solving $t-\tau_i=r(\tau_i)/c$ (usually there is a single relevant root).
- $r(\tau_i)=\|\mathbf{x}-\mathbf{x}_s(\tau_i)\|$.
- $r'(\tau)=\dfrac{d}{d\tau}\|\mathbf{x}-\mathbf{x}_s(\tau)\|=-\mathbf{n}(\tau)\cdot\mathbf{v}_s(\tau)$.
- $\mathbf{v}_s(\tau)=\dfrac{d\mathbf{x}_s}{d\tau}$ is the source velocity at emission time τ.
- $\mathbf{n}(\tau)=\dfrac{\mathbf{x}-\mathbf{x}_s(\tau)}{r(\tau)}$ is the unit vector pointing from source (at emission) to the field point.

In standard wave-equation solutions, a Jacobian factor $|1-\mathbf{n}\cdot\mathbf{v}_s/c|$ arises from the change of variables used to evaluate the path history time delta. In this project's canonical per-hit law, emission cadence and per-wavefront amplitude are constant and do not depend on emitter speed; the corresponding branch Jacobian enters as received causal-flux weighting, not as an extra source-amplitude modulation.

Special simple case — stationary emitter

If $\mathbf{x}_s(\tau)=\mathbf{x}_0$ (emitter fixed) and $q(\tau)=Q\,\delta(\tau-\tau_0)$ (single wake surface at τ_0), then the formula reduces to the intuitive result:

- The field at $\mathbf{x},t$ is nonzero only when $t-\tau_0=\|\mathbf{x}-\mathbf{x}_0\|/c$, i.e., when the causal wake surface of radius $r=c(t-\tau_0)$ reaches $\mathbf{x}$.
- The amplitude is $\phi(\mathbf{x},t)=\dfrac{Q}{4\pi\, r}$ (no extra Jacobian factor because $v_s=0$).

How wake surfaces show up here

- Each emitted wake surface corresponds to one emission time τ. The delta in the Green's function selects the observation times t at which the wake surface reaches $\mathbf{x}$.
- The shape of the contribution is the $1/(4\pi r)$ geometric factor for wave amplitude; the wake surface is thin in time if $q(\tau)$ is a delta in τ, so the receiver gets a short impulse when the wavefront passes.

Handling an emitter that stops / $\|\mathbf{v}_s\|\to 0$

- If the emitter slows or stops, the Jacobian factor $1-\mathbf{n}\cdot\mathbf{v}_s/c$ tends to 1 and nothing singular happens. The path-history equation still has a solution and each wake surface arrives at the predicted time.
- If the emitter sits still and emits many wake surfaces (continuous $q(\tau)$), the field is the time integral (or sum) of all wake surface contributions evaluated at their respective causal times. No $1/\|\mathbf{v}_s\|$ blowup occurs.

Event-driven radial-transport + per-hit EOM (current canonical method)

Physical idea: represent emission as a conserved, razor-thin causal wake surface (a measure on the causal isochron), then drive particle motion by summing line-of-action per-hit accelerations with Jacobian-weighted magnitude at causal intersection times. We work in units with field speed $v=1$ unless noted; replace v by c otherwise.

Field representation (transport/continuity form)

- Source impulse at $(t_0,\mathbf{s}_0)$ creates a wake surface supported on $r=v(t-t_0)$ with surface density that conserves a constant per-wake surface amplitude q:

$$\rho(t,\mathbf{s}) \;=\; \frac{q}{4\pi r^2}\,\delta\big(r-v(t-t_0)\big)\,H(t-t_0), \quad r=\|\mathbf{s}-\mathbf{s}_0\|$$

- This solves the radial continuity (transport) equation

$$\partial_t\rho+\nabla\cdot\big(v\,\hat{\mathbf{r}}\,\rho\big) \;=\; q\,\delta(t-t_0)\,\delta^{(3)}(\mathbf{s}-\mathbf{s}_0)$$

- Emission is continuous with constant time-density $q(t) \equiv q_0$.

Per-hit equation of motion (EOM)

- For a receiver o' at time t and a source j, causal emission times satisfy

$$\|\mathbf{s}_{o'}(t) - \mathbf{s}_j(t_0)\| = v\,(t - t_0), \qquad t_0 < t$$

- Each root contributes a line-of-action acceleration

$$\mathbf{a}_{o' \leftarrow j}(t; t_0) \;=\; \kappa\, \sigma_{q_j q_{o'}} \frac{|q_j q_{o'}|}{r^2\, |J_{o'j}(t; t_0)|}\, \hat{\mathbf{r}}, \quad \hat{\mathbf{r}} = \frac{\mathbf{s}_{o'}(t) - \mathbf{s}_j(t_0)}{r},\ r > 0$$

with total acceleration the sum over sources and roots. Convention $H(0) = 0$ removes the instantaneous self-kick at $\tau = 0$. Optional mollification replaces $\delta(\cdot)$ by $\delta_\eta(\cdot)$ to produce smooth pushes.

Implementation checklist

- Root finding: solve $F(t_0; t) = \|\mathbf{s}_{o'}(t) - \mathbf{s}_j(t_0)\| - v(t - t_0) = 0$ for all j (including $j = o'$ for self-hits when kinematics permit).
- Accumulation: compute $r, \hat{\mathbf{r}}$, apply $1/r^2$, then superpose.
- Time stepping: impulsive mode (events) or mollified mode ($\eta > 0$) with standard ODE integrators.
- Self-interaction: appears when the worldline outruns recent wake surfaces ($\|\mathbf{v}\| > v$ for some emissions); self-hits are repulsive (like-on-like).

Relation to Methods 1 and 2

- This is a transport/continuity model, not the scalar wave equation. The $1/r^2$ factor is a surface-density normalization (Gauss-like on the spherically expanding causal wake surfaces); it is compatible with conserving total emission per wake surface. In Method 2 the $1/(4\pi r)$ factor appears for a wave amplitude; taking gradients connects these scalings when mapping to forces.
- The Doppler-type Jacobian $1 - \mathbf{n}\cdot\mathbf{v}_s/c$ from Method 2 is the same branch-transversality factor that appears as $|J|^{-1}$ in the canonical per-hit law. Geometric constants are absorbed into κ by convention, but the branch Jacobian is retained as received causal-flux weighting; no additional source-speed amplitude factor is introduced.
- Numerically, this method targets particle dynamics directly (per-hit ODEs) rather than evolving a full field (Method 1) or evaluating fields at sparse probes (Method 2).

Operator diagnostics (finite-window checks)

- Use vector-calculus identities only on declared, reconstructed diagnostic channels such as $\nabla\Phi_\eta$ or the mollified transport current $\mathbf{J}_\eta = v\,\hat{\mathbf{r}}\,\rho_\eta$. These channels are validation objects, not new substrate ontology.
- For any finite control volume $V \subset \Sigma_t$ with outward unit normal $\hat{\mathbf{n}}$, define the Gauss residual

$$R_G[V, t; \mathbf{Y}_\eta] \equiv \frac{\left|\int_{\partial V} \mathbf{Y}_\eta\cdot\hat{\mathbf{n}}\, dS - \int_V \nabla\cdot\mathbf{Y}_\eta\, dV\right|}{\int_{\partial V} |\mathbf{Y}_\eta\cdot\hat{\mathbf{n}}|\ dS + \int_V |\nabla\cdot\mathbf{Y}_\eta|\ dV + \varepsilon_G}$$

- For any oriented smooth surface $S \subset \Sigma_t$ with boundary ∂S, define the Stokes residual

$$R_S[S, t; \mathbf{Y}_\eta] \equiv \frac{\left|\oint_{\partial S} \mathbf{Y}_\eta\cdot d\mathbf{x} - \int_S (\nabla\times\mathbf{Y}_\eta)\cdot\hat{\mathbf{n}}\, dS\right|}{\oint_{\partial S} |\mathbf{Y}_\eta\cdot d\mathbf{x}| + \int_S |(\nabla\times\mathbf{Y}_\eta)\cdot\hat{\mathbf{n}}|\ dS + \varepsilon_S}$$

- PDE and event-root simulations should agree not only pointwise after resampling, but also as operators on finite windows. If $\Delta\mathbf{Y}_\eta = \mathbf{Y}_\eta^{\mathrm{PDE}} - R(\mathbf{Y}_\eta^{\mathrm{root}})$, use

$$E_{\mathrm{op}}(V, S, t) \equiv \max\{R_G[V, t; \Delta\mathbf{Y}_\eta],\ R_S[S, t; \Delta\mathbf{Y}_\eta]\}$$

For the conservative potential channel $\mathbf{Y}_\eta = \nabla\Phi_\eta$, nonzero circulation is a numerical, boundary, or coordinate-operator error unless a non-gradient effective channel has been explicitly declared.

Plain language: treat the potential contribution as a conserved amount spread over a growing causal wake surface. When a wake surface reaches a receiver, the receiver gets a straight-line push that falls off like $1/r^2$; the calculation may treat it as a sharp kick or as a short, smooth nudge.

Cross-Method Guidance

Cross-Method Selection

- Method 1 (PDE): whole-field grid simulations, visualization, and complex media/boundaries. Deposit a smeared source each step; robust when an emitter slows or stops. Aggregate particle data to coarse-grained densities n(x,t), ρ(x,t), and $\mathcal{E}$(x,t) as inputs/targets for PDE runs and validation.

- Method 2 (Green's function / path-history integral): closed forms and sparse probe evaluation. Enforce the path-history condition $t - \tau = \|\mathbf{x} - \mathbf{x}_s(\tau)\|/c$ and handle the geometric factor $1 - \mathbf{n} \cdot \mathbf{v}_s/c$ during evaluation; root-solve one (or more) τ per (observer, time) pair.
- Method 3 (Event-driven canonical): production many-body dynamics. Find causal roots and sum per-hit $1/(r^2|J|)$ pushes; prefer η-mollified mode for smooth ODEs when needed.

Short worked example — stationary emitter, continuous source (consistent across methods)

- Setup: emitter at origin $\mathbf{x}_s = 0$ with $q(t) \equiv q_0$ (constant).
- Method 1: solving the wave PDE with $S(\mathbf{x}, t) = q_0\, \delta(\mathbf{x})$ reproduces the same spherical profile $\phi(r, t) = q_0/(4\pi r)$ on the outgoing wavefront.
- Method 2: the path-history formula gives $\phi(r, t) = \dfrac{q_0}{4\pi r}$ with the path-history time $\tau = t - r/c$.
- Method 3: the path-history condition selects the single causal time $t_0 = t - r/c$; the per-hit EOM yields one radial push along $\hat{\mathbf{r}}$ with $1/r^2$ scaling, consistent with taking spatial gradients of the $1/r$ potential to connect amplitude to force.

Practical implementation notes (concise)

- PDE: smear $\delta(\mathbf{x} - \mathbf{x}_s)$ to grid scale; enforce CFL ($c\,\Delta t/\Delta x$ within the scheme's bound).
- Path-history: robust root-finding for τ from $t - \tau = r(\tau)/c$; take care near grazing geometries where $1 - \mathbf{n} \cdot \mathbf{v}_s/c$ is small.
- Event-driven: bracket causal roots for continuity, optionally use δ_η for smooth pushes, and limit step sizes so only a controlled number of mollified wake surfaces overlap.

Operational Summary

- Model sources as $S(\mathbf{x}, t) = q(t)\, \delta\big(\mathbf{x} - \mathbf{x}_s(t)\big)$ (time-based emission density).
- Use Method 3 as the primary dynamics engine; use Method 2 for calibration/spot checks; use Method 1 for whole-field/media studies.
- All three agree on simple stationary cases; they differ mainly in computational scope: grids (1), closed-form probes (2), and event-driven ODEs (3).

Differential analysis (criteria-by-criteria)

Axiomatic fidelity (delayed-only, line-of-action, constant source emission)

- Method 1: Partially aligned. The PDE yields $1/(4\pi r)$ wave amplitudes; mapping to $1/r^2$ per-hit accelerations requires gradients and conventions. Radial-only action is not built-in.
- Method 2: Causality and superposition are exact; amplitudes are $1/(4\pi r)$ with a Jacobian $|1 - \mathbf{n} \cdot \mathbf{v}_s/c|^{-1}$ when evaluating the path-history time delta. The canonical law keeps that Jacobian weighting explicitly, while overall geometric normalizations are absorbed into κ when comparing accelerations.
- Method 3: Exact match. Delayed-only, line-of-action per-hit with constant source emission is native, and the branch Jacobian appears explicitly in the received force magnitude. Geometric normalizations are conventionally absorbed into κ.

Causal root structure, self-interaction, multiplicity

- Method 1: Self-hits and multiple roots are implicit in the evolving field; they are not directly enumerated as discrete events.
- Method 2: Causal roots arise via solving $t - \tau = r(\tau)/c$; multiple roots and tangencies are explicit but require robust root-finding.
- Method 3: Roots are primitive; multi-hit and self-hit regimes are treated natively. Conventions H(0)=0 and exclusion of $r = 0$ beyond $\tau = 0$ are explicit.

Energetics and work

- Method 1: Continuum energy bookkeeping is natural (ϕ, $\partial_t\phi$, $\nabla\phi$). Mapping to radial per-hit work needs careful averaging and alignment with the EOM.
- Method 2: Exact potentials in free space; gradients give forces; care is needed near $|1 - \mathbf{n} \cdot \mathbf{v}_s/c| \to 0$ geometries.
- Method 3: Energetics are validated via η-mollified potentials Φ_η and work–energy on resolved windows; impulses are recovered as $\eta \to 0$ in the weak sense.

Numerical stability and well-posedness

- Method 1: CFL constraints; dispersion/reflection control needed; robust under regularized sources; well posed on grids.
- Method 2: Stable as an evaluation formula; computational issues concentrate in robust, multi-root solving and handling near-tangency Jacobians.
- Method 3: Well posed with event handling or η-regularization; stability governed by root-tracking and step control; lightweight for many-body ODEs.

Computational cost and scalability

- Method 1: Heavy (3D grid + CFL time stepping). Cost grows with volume, resolution, and duration—independent of number of receivers.
- Method 2: Moderate to heavy depending on receivers × times × sources × roots; efficient for few probes, costly for dense sampling.
- Method 3: Light for particle dynamics. Cost scales with sources × average roots per step; independent of any spatial grid.

Boundaries, media, and heterogeneity

- Method 1: Natural—modify PDE coefficients (inhomogeneous c, damping, boundaries).
- Method 2: Natural only in homogeneous free space; complex media/boundaries require bespoke Green's functions.
- Method 3: Natural in free space. Media/boundaries need additional modeling (e.g., corridor-level effective rules); not PDE-native.

Observables and Inference

- Method 1: Full-field pictures aid intuition and corridor studies but obscure per-hit ambiguity without extra processing.
- Method 2: Clarifies causal timing and geometry at probes; good for inference templates and surrogate-location recasts.
- Method 3: Directly aligned with hit histories {A(t_k), L(t_k)}; best substrate for event-driven inference and assembly dynamics.

Summary (one line each)

- Method 1: Best for whole-field, media, and visualization; poorest fit to per-hit radial-only axioms without translation layers.
- Method 2: Best for exact, pointwise, causal analysis in free space; good for calibration and sparsely sampled validation.
- Method 3: Best for dynamics of many particles/assemblies under the canonical law; scales and matches axioms directly.

Operational guidance — when to use which method

- Method 1 (PDE): use this for whole-field grid simulations, visualization, and complex media or boundaries; step the wave PDE forward with a smeared source. Robust when an emitter slows or stops.
- Method 2 (Path history integral): use this for closed forms, analytic insight, or sparse probe evaluation; enforce the path-history condition $t-\tau=\|\mathbf{x}-\mathbf{x}_s(\tau)\|/c$ and handle the geometric factor $1-\mathbf{n}\cdot\mathbf{v}_s/c$ in evaluation; solve one root per (observer, time) pair in slow-motion, more if sources move fast.
- Method 3 (Event-driven canonical): use this for production many-body dynamics; find causal roots and sum per-hit $1/(r^2|J|)$ pushes; prefer η-mollified mode for smooth ODEs when needed.

Pros and cons (comparative)

Method 1 — Time-based PDE (wave equation)

- Pros
- Physically standard propagation at fixed speed c; expanding causal wake surfaces emerge automatically.
  - Robust on grids; handles inhomogeneous media, damping, and boundaries.
  - Good for full-field visualization and energy bookkeeping in continuum form.
- Cons
  - Computationally heavy for many-particle dynamics (3D grids, CFL constraints).
  - Requires careful numerics to avoid dispersion/reflection; mesh choices can bias results.
  - Mapping grid fields to the radial-only per-hit ODE can add another modeling layer.

Method 2 — Green's function (path-history integral)

- Pros
  - Exact in homogeneous free space; no grid or time stepping for the field.
  - Makes causality explicit via path-history times; captures Doppler/Jacobian $1-\mathbf{n}\cdot\mathbf{v}_s/c$ automatically.
  - Efficient for field evaluation at a few observation points; excellent for analysis and cross-checks.
- Cons
  - Requires root-finding for each (observer, time) pair; multiple roots possible when sources outrun wake surfaces.
  - Costly when many receivers/sources are present; bookkeeping grows quickly.
  - Needs careful handling near tangencies (small Jacobians) and in multi-hit/self-hit regimes.

Method 3 — Event-driven radial-transport + per-hit EOM (current canonical)

- Pros
  - Directly implements the project's delayed, radial-only interaction law with constant emission cadence.
  - Natural support for self-hits and superposition; local $1/r^2$ weighting makes nearby coherent roots dominate once the far-field cutoff, screening, cancellation, or summation prescription is declared.

- Numerically lightweight for particle dynamics; works cleanly with impulsive or mollified ODE integration.

- Cons
  - Not derived from the scalar wave equation; global field-energy accounting is indirect (via mollified potentials).
  - Must retain the causal-root Jacobian factor from the master equation; a reduced test harness that omits it is a noncanonical approximation rather than a calibration of κ.
  - Accuracy depends on robust causal-root finding and regularization choices in complex multi-hit scenarios.

Recommendation

- Use Method 3 as the primary engine for particle dynamics and assemblies. It matches the model's axioms (radial-only action, constant emission cadence) and scales well.
- Adopt Method 2 as the analytic reference for calibration and validation. Calibrate κ so simple benchmarks (stationary/slow sources, symmetric binaries) agree between Methods 2 and 3 at the per-hit level; do not introduce any per-hit emitter-speed weighting.
- Baseline formula (stationary emitter at origin): with $q(t) \equiv q_0$, $\phi(r,t) = \dfrac{q_0}{4\pi r}$ since the path history condition selects $\tau = t - r/c$; if q varies, $\phi(r,t) = \dfrac{q(t - r/c)}{4\pi r}$.
- Reserve Method 1 for full-field studies (visualization, media, boundary effects) and for end-to-end tests of numerical stability; it is valuable but unnecessary for routine ODE-based assembly simulations.
- Documentation/actionables: keep the continuity-form field definition and per-hit EOM as the canonical statement; add a brief appendix mapping densities (Method 3) to potentials (Method 2) to clarify when $1/r$ vs $1/r^2$ factors appear and how calibration preserves totals.
- Numerical cautions:
  - Always smear $\delta(\mathbf{x} - \mathbf{x}_s)$ to a normalized kernel of width σ comparable to the grid spacing in PDE runs to avoid grid-scale artifacts.
  - Enforce CFL: choose Δt so that $c\,\Delta t/\Delta x$ meets the stability bound for the chosen stencil to prevent instability.
  - Path history solving: solve $t - \tau = r(\tau)/c$ carefully; near $\|\mathbf{v}_s\| \approx c$, root finding and the factor $1 - \mathbf{n}\cdot\mathbf{v}_s/c$ require extra care.
  - Finite temporal thickness: if wake surfaces have duration, replace $\delta(t-\tau)$ with a smooth profile to model finite-width wavefronts.

Plain language: use the event-driven, radial-only method for dynamics, check it against the path-history integral to calibrate parameters, and use the PDE only when the calculation needs whole-field pictures or complex media.

Recap (in three lines)

- Model sources as $S(\mathbf{x},t) = q(t)\,\delta\big(\mathbf{x} - \mathbf{x}_s(t)\big)$ (time-based emission density).
- Method 1: easiest for grid-based whole-field runs; wake surfaces emerge at speed c.
- Method 2: exact path-history formula; contributions occur only when $t - \tau = \|\mathbf{x} - \mathbf{x}_s(\tau)\|/c$, with amplitude decaying as $1/(4\pi r)$ and a geometric $1 - \mathbf{n}\cdot\mathbf{v}_s/c$ factor in evaluation.

Layered penetration diagram (molecules → cores)

A qualitative "onion" sketch to visualize which excitations typically penetrate which structural layers. This helps readers see what's excluded and what isn't.

Legend: [+] passes, [~] depends (energy/frequency/geometry), [x] mostly blocked/strongly attenuated

Layer	Photons	Neutrinos	Charged $\pm\epsilon$	Dark-matter-like neutral
L4: Bulk molecular wake surface (solids/liquids; many-body opacity)	[~] material window; optical opaque, IR/UV/X/γ vary	[+] nearly transparent	[x] bind/deflect; do not traverse as free particles	[+] very weak coupling
L3: Atomic electron distribution (bound electrons)	[~] photoelectric/Compton; X/γ penetrate better	[+]	[x] Coulomb-coupled; captured/scattered	[+]
L2: Nuclear layer (nucleons; femtoscopic scale)	[~] γ can interact; strong attenuation in bulk	[+] weak interaction; mostly pass	[x] excluded as free traversers	[+]
L1: Nested shell braid shielding (nested shell binaries; shielded)	[x] far-field cancels; no corridor capture	[~] tiny axial coupling only	[x] self/partner couplings dominate; no transit	[+] by hypothesis: minimal coupling
L0: Axial corridors / flux-tube loci (coherent geometry)	[+] guided along corridor	[~] weak corridor coupling; alignment matters	[x] no cross-product forces; not a transit channel	[~] minimal, geometry-dependent

Notes (interpretation):

- "Dark-matter-like neutral" denotes very weakly coupled, neutral meta-assemblies consistent with this framework; included here as a hypothesis for qualitative comparison.
- Entries marked [~] depend on spectrum, thickness, coherence, and alignment (e.g., γ vs optical photons; corridor alignment for neutrinos).

- The diagram is about penetration (transit). Local interactions, capture, or re-binding are separate processes governed by geometry and delay.

Analytic Baselines

Purpose:

- State the delay differential equations (DDEs) that govern canonical interactions under the delayed line-of-action law with Jacobian-weighted magnitude.
- Record exact analytical solutions only where they exist; otherwise, state solvability status without approximations.

Models:

- Fixed center (test particle, source stationary):
  - DDE reduces exactly to the ODE $\ddot{r} = -K/r^2$ with $K = \kappa|qq'| > 0$; exact closed forms exist.
- Two-body mutual interaction (opposite or equal charges):
  - Coupled DDEs with causal roots t_0 defined by $|x_i(t) - x_j(t_0)| = t - t_0$ (v=1); accelerations superpose as $\pm\kappa\epsilon^2/(r^2|J|)$ along the line of action.
  - No exact closed-form solutions are presently known for the coupled DDEs in general.

Methodological priority:

- Treat the two-point-potential problem as the canonical first laboratory for the delayed theory.
- Any proposed energy, momentum, virial-like, or kinetic/potential closure claim should be checked here before being generalized to assemblies or Noether sea response arguments.
- In practice this means: solve the fixed-center and symmetric two-body cases first, then ask which familiar ODE identities survive, which acquire delay corrections, and which fail outright.
- For the nontrivial electrino:positrino binary, use the finite-η closure packet in Binary Dynamics and the constructive residuals in Delay-Dynamics Energy. A claimed branch must report

$$\mathrm{Res}_{2B}^{(\eta)} = \left(\mathcal{R}_{\mathrm{EOM}}^{2B}, \mathcal{R}_{\mathrm{per}}^{2B}, \mathcal{R}_{\mathrm{bal}}^{2B}, \nu_J^{2B}, \Delta_{\mathrm{gap}}^{2B}, \lambda_{\mathrm{sec}}^{2B}, \epsilon_E^{(\eta)}, \Delta_{\mathrm{E,cross}}^{(\eta)}, \mathcal{R}_\omega^{2B}\right).$$

Until these entries are computed on the same window, regulator, and branch chart, the binary remains an existence candidate rather than a validated closure result.

- The first constructive energy baseline for such a branch is the branch-local work reconstruction

$$U_{b,\mathrm{work}}^{(\eta)}(t) = U_b(t_*) - \int_{t_*}^{t} \sum_i \mu_{\mathrm{arch}}\, \mathbf{a}_{i,b}^{(\eta)}(t') \cdot \mathbf{v}_i(t')\, dt'$$

with the same replacement by $\mu_K(\|\mathbf{v}_i\|)$ when the primitive kinetic scalar is used. For a circular branch, the period-averaged integrand reduces to $\mu_{\mathrm{arch}} s_b \langle A_{\eta,b}^{\tan}\rangle_{P_b}$ in the quadratic proxy.

- The adiabatic consistency check is branch preservation under slow drift. Along a quasi-static path $\gamma : \lambda \mapsto (R(\lambda), s(\lambda), b)$ that does not cross a root-ledger threshold, the work-integral energy change should match the energy difference inferred from the neighboring solved branch family:

$$\Delta_{\mathrm{ad},E}^{2B}(\gamma) = \frac{\left|\Delta_\gamma U_{b,\mathrm{work}}^{(\eta)} - \left(E_b^{(\eta)}(\lambda_1) - E_b^{(\eta)}(\lambda_0)\right)\right|}{\left|\Delta_\gamma U_{b,\mathrm{work}}^{(\eta)}\right| + \left|E_b^{(\eta)}(\lambda_1) - E_b^{(\eta)}(\lambda_0)\right| + \varepsilon}$$

Here $E_b^{(\eta)}(\lambda)$ denotes the candidate branch energy extracted at fixed λ by the same declared construction route. The test is valid only while the same signed causal-root ledger persists with positive Jacobian and inactive-root gap floors. A jump in the ledger is a bifurcation, not a failure of adiabatic energy consistency.

- Branch-virial theorem target: separate the kinematic virial identity from the stronger classical potential virial theorem. On a fixed finite-η branch chart b over an averaging window $W = [t_a, t_b]$, define the branch virial diagnostic

$$\mathcal{G}_b^{(\eta)}(t) = \sum_i \mu_{\mathrm{arch}}\, \mathbf{x}_i(t) \cdot \mathbf{v}_i(t)$$

and the quadratic kinetic bookkeeping scalar

$$T_{\mu,b}^{(\eta)}(t) = \frac{1}{2} \sum_i \mu_{\mathrm{arch}}\, \|\mathbf{v}_i(t)\|^2$$

When the branch is differentiable after mollification and the same signed causal-root ledger is retained, direct differentiation gives the finite-window identity

$$\left\langle 2T^{(\eta)}_{\mu,b} + \sum_i \mu_{\text{arch}}\,\mathbf{x}_i(t)\cdot\mathbf{a}^{(\eta)}_{i,b}(t)\right\rangle_W = \frac{\mathcal{G}^{(\eta)}_b(t_b) - \mathcal{G}^{(\eta)}_b(t_a)}{t_b - t_a}$$

The branch-virial closure target is the special bounded or periodic case in which the right-hand side is zero or below the declared tolerance:

$$\mathcal{R}^{(\eta)}_{\text{vir},b}(W) = \left|\left\langle 2T^{(\eta)}_{\mu,b} + \sum_i \mu_{\text{arch}}\,\mathbf{x}_i(t)\cdot\mathbf{a}^{(\eta)}_{i,b}(t)\right\rangle_W\right| \le \epsilon_{\text{vir}}$$

This is not yet the classical potential statement. The reduction to $\langle 2T - pU\rangle = 0$ additionally requires a branch-local potential $U^{(\eta)}_b$ whose scale variation is controlled by a homogeneity degree p,

$$U^{(\eta)}_b(\lambda\mathbf{x}) = \lambda^p U^{(\eta)}_b(\mathbf{x})$$

together with a proof that the same branch acceleration is generated by that potential over W. A scale/virial residual that contains zero is therefore diagnostic only until it supplies the same-domain scale generator, homogeneity degree, and branch coordinate needed for this stronger reduction.

- Velocity-regime scope for the branch-virial target:
  - Strict sub-field-speed branch windows are the closest to the classical comparison because nontrivial self-hit roots are excluded on the strictly sub-field-speed interval; delayed partner hits and Jacobian weighting still remain in the acceleration term.
  - Field-speed or near-field-speed windows are threshold-sensitive. They require an explicit Jacobian floor, inactive-root gap floor, and unchanged causal-root ledger before the virial residual is meaningful.
  - Super-field-speed history requires the retained self-hit and multi-root rows to be included in $\mathbf{a}^{(\eta)}_{i,b}(t)$. A speed label alone never certifies the branch; root existence, transversality, and bounded endpoint virial drift do the work.
- Failure modes:
  - $\mathcal{G}^{(\eta)}_b$ has unbounded secular drift on W.
  - The causal-root ledger changes, an inactive-root gap closes, or the Jacobian floor fails.
  - Collision support or the $\eta \to 0$ limit is not controlled.
  - No branch-local potential, scale generator, or homogeneity degree is supplied, so the classical potential virial theorem has not been recovered.

Symmetric two-body on a line (exact DDE; challenges):

- Let $x_1(t) = +\frac{1}{2}r(t)$ and $x_2(t) = -\frac{1}{2}r(t)$ with $r(t) > 0$ and $v = 1$. The causal-time condition implies

$$\frac{r(t) + r(t_0)}{2} \;=\; t - t_0, \qquad t_0 < t$$

or, writing $\tau(t) = t - t_0 > 0$ implicitly,

$$r(t) + r\big(t - \tau(t)\big) \;=\; 2\,\tau(t)$$

- For opposite polarities, the exact relative-coordinate equation is the state-dependent DDE

$$\ddot{r}(t) \;=\; -\frac{8\,\kappa\,\epsilon^2}{\big(r(t) + r(t-\tau(t))\big)^2\,|J(t)|}$$

with $\tau(t)$ determined by the implicit constraint above. For equal charges, the sign is reversed.

Integral (delta) form selecting the causal root:

- For particle 1 one may write

$$a_1(t) \;=\; -\,\kappa\,\epsilon^2\int_0^\infty \frac{\delta\big(|x_1(t) - x_2(t-\tau)| - \tau\big)\,\operatorname{sgn}\big(x_1(t) - x_2(t-\tau)\big)}{|x_1(t) - x_2(t-\tau)|^2}\,d\tau$$

whose evaluation reduces exactly to finding the causal delay $\tau(t)$; in the symmetric 1D case this yields the DDE above.

Why closed-form solutions are unlikely (even with symmetry):

- The delay is state-dependent: the unknown $r(t)$ appears both in the right-hand side and in the implicit constraint defining $\tau(t)$, making the problem a nonlinear functional equation rather than an ODE.
- Even linear constant-delay DDEs rarely admit elementary closed forms; state-dependent delays are generically non-integrable. The fixed-center problem is a special case that collapses to an ODE (see Radial Fall to Fixed Center).

Solution techniques (toolbox for delayed, radial DDEs):

- Method of steps (constant delays): for problems with fixed delay τ and a given history $x(t) = \phi(t)$ on $t \in [-\tau, 0]$, integrate an ODE on successive intervals, using the known past segment on each step.
- State-dependent delay root-tracking: treat $\tau(t)$ as an algebraic unknown constrained by the causal-time equation (e.g., $r(t) + r(t - \tau) = 2\tau$). On each step, solve the coupled system with a Newton corrector for $\tau(t)$; ensures consistency of the delay with the evolving state.
- Collocation / implicit Runge–Kutta with history interpolation: represent the recent history by Hermite/spline polynomials; at each step solve stage equations together with the causal constraint(s), updating a continuous extension of the history.
- Shooting and continuation for periodic motions: pose a boundary-value problem over one period with delay constraints; solve by Newton shooting or collocation and continue solutions via pseudo-arclength. Useful for detecting limit cycles and their stability.
- Spectral-in-time methods: on (quasi-)periodic windows, expand in Fourier/Chebyshev bases; constant delays enter as phase factors, while state-dependent delays are handled by iterating a frozen-delay linearization.
- Stability analysis (qualitative): Lyapunov–Krasovskii and Razumikhin functionals yield sufficient conditions for stability without solving trajectories; applicable to history classes with bounded delays.
- PDE embeddings (transport representation): introduce an auxiliary history field $y(t, \theta)$ on $\theta \in [-\tau_{\max}, 0]$ with $y_t + y_\theta = 0$ and boundary $y(t, 0) = x(t)$; discretize in θ (method of lines). For state-dependent delays, use a moving boundary; aligns with the project's radial-transport perspective.
- Green's-function / hit-integral formulations: write per-hit actions as delta-weighted time integrals selecting causal roots; evaluate by robust root-finding and quadrature. This matches the event-driven law used here.
- Measure-driven/event-driven solvers with mollification: replace surface deltas by narrow Gaussians ($\eta > 0$) to obtain C^1 trajectories; take $\eta \to 0$ in the weak sense after validating work–energy over resolved windows.
- Linear constant-delay benchmarks: for linear DDEs (e.g., $x' = ax + bx(t - \tau)$) use Laplace transforms/characteristic equations and Lambert W; helpful for validation and step-size/error control, even though the canonical two-body problems here are nonlinear and state-dependent.
- A posteriori error control: use defect/residual of collocation, step halving with history re-interpolation, and event-time error estimates for adaptive step and tolerance selection.
- Fixed-point frameworks: establish local existence/uniqueness by contraction on history spaces $C([-\tau_{\max}, 0])$ (or their mollified variants); use Picard iterations as a solver preconditioner.

Deliverables:

- Precise DDE forms and causal-root conditions for use in analysis and computation.
- Cross-references to sections with exact solutions (fixed source) and status notes (mutual interaction).
- A minimal benchmark ladder for closure tests:
  - fixed-center ODE recovery,
  - symmetric two-body delayed dynamics,
  - finite-η two-body binary closure packet with branch floors and characteristic frequency extraction,
  - work-energy balance on resolved windows,
  - branch-virial residuals where periodic, quasi-periodic, or bounded-drift regimes exist.

Plain language: We give only the exact delayed equations; where an exact solution exists (fixed source), we present it, and where it does not (mutual interaction), we say so without approximations.

Attraction

Setup:

- Two architrinos with polarities $q_1 = -\epsilon$ and $q_2 = +\epsilon$.
- Initial velocities v1≈0, v2≈0; initial separation r0 ≫ 1 (in v=1 units).
- For all examples, we restrict motion to a single geometrical line.

Objectives:

- Delay-only formulation of the equations of motion (DDEs).
- Exact analytic solutions if available; otherwise, status of solvability.

Canonical delayed-law considerations:

- Delay enters through the implicit emission times t_0 satisfying $|x_1(t) - x_2(t_0)| = t - t_0$ (and its counterpart).
- All per-hit actions are radial along the line of action and carry the branch Jacobian factor $|J|^{-1}$; $H(0) = 0$ excludes $t_0 = t$.

Equations of motion (canonical delayed law; two-body, v=1):

- Definitions:
  - Polarities: $q_1 = -\epsilon$ (particle 1), $q_2 = +\epsilon$ (particle 2); $\epsilon > 0$ is the polarity-unit magnitude.
  - Coupling: $\kappa > 0$ is the universal coupling constant; we work in units with field speed $v = 1$.
  - Separation: $r(t) = |x_1(t) - x_2(t)| > 0$.
- Causal (path-history) times:
  - $t_0^{(2\to1)} \in \mathcal{C}_2(t)$ solves $|x_1(t) - x_2(t_0)| = t - t_0$.
  - $t_0^{(1\to2)} \in \mathcal{C}_1(t)$ solves $|x_2(t) - x_1(t_0)| = t - t_0$.
- Per-particle accelerations (sum over all causal roots if multiple exist):

$$a_1(t) \;=\; \sum_{t_0 \in \mathcal{C}_2(t)} -\kappa\,\epsilon^2\,\frac{\operatorname{sgn}\big(x_1(t) - x_2(t_0)\big)}{r_{12}^2\,|J_{12}(t;t_0)|}, \quad r_{12} = \big|x_1(t) - x_2(t_0)\big|$$

$$a_2(t) \;=\; \sum_{t_0 \in \mathcal{C}_1(t)} +\kappa\,\epsilon^2\,\frac{\operatorname{sgn}\big(x_2(t) - x_1(t_0)\big)}{r_{21}^2\,|J_{21}(t;t_0)|}, \quad r_{21} = \big|x_2(t) - x_1(t_0)\big|$$

Here $\sigma_{q_2q_1} = \sigma_{q_1q_2} = -1$ (unlike polarities attract), J_{12} and J_{21} are the corresponding causal-root Jacobians, $H(0) = 0$ excludes $t_0 = t$, and $\operatorname{sgn}(\cdot)$ denotes the sign function.

Relative-coordinate DDE:

- Define $r(t) = x_1(t) - x_2(t) > 0$. Then

$$\ddot{r}(t) \;=\; a_1(t) - a_2(t) \;=\; -\kappa\,\epsilon^2 \sum_{t_0 \in \mathcal{C}_2(t)} \frac{\operatorname{sgn}\big(r_{12}\big)}{r_{12}^2\,|J_{12}(t;t_0)|} - \kappa\,\epsilon^2 \sum_{t_0 \in \mathcal{C}_1(t)} \frac{\operatorname{sgn}\big(r_{21}\big)}{r_{21}^2\,|J_{21}(t;t_0)|}$$

with $r_{12} = |x_1(t) - x_2(t_0)|$ and $r_{21} = |x_2(t) - x_1(t_0)|$ defined by their respective causal-root conditions. No exact closed-form solution is presently known for the coupled DDE system.

Nonlinear history-anchored form (vector notation for clarity):

$$\mathbf{a}_1(t) \;=\; -\kappa\,\epsilon^2\,\frac{\mathbf{s}_1(t) - \mathbf{s}_2\big(t_0^{(2\to1)}\big)}{\big\|\mathbf{s}_1(t) - \mathbf{s}_2\big(t_0^{(2\to1)}\big)\big\|^3\,|J_{12}(t;t_0^{(2\to1)})|}, \qquad \mathbf{a}_2(t) \;=\; +\kappa\,\epsilon^2\,\frac{\mathbf{s}_2(t) - \mathbf{s}_1\big(t_0^{(1\to2)}\big)}{\big\|\mathbf{s}_2(t) - \mathbf{s}_1\big(t_0^{(1\to2)}\big)\big\|^3\,|J_{21}(t;t_0^{(1\to2)})|}$$

The attachment points are the partners' path-history locations at their respective causal emission times; linearizations and small-parameter expansions are intentionally omitted.

Central-origin kinematics (1D positions and velocities; symmetric two-body frame)

- Choose a fixed origin at the geometric midpoint. With equal-magnitude charges and symmetric initial data, this midpoint remains at rest by symmetry.
- Define the separation

$$r(t) \equiv x_1(t) - x_2(t) > 0$$

Positions relative to the central origin are then

$$x_1(t) = \tfrac{1}{2}\,r(t), \qquad x_2(t) = -\,\tfrac{1}{2}\,r(t)$$

- Velocities follow by differentiation:

$$v_1(t) = \dot{x}_1(t) = \tfrac{1}{2}\,\dot{r}(t), \qquad v_2(t) = \dot{x}_2(t) = -\,\tfrac{1}{2}\,\dot{r}(t)$$

- Symmetric initial conditions (example):

$$x_1(0) = \tfrac{r_0}{2}, \quad x_2(0) = -\tfrac{r_0}{2}, \quad v_1(0) = v_2(0) = 0$$

Deliverables:

- Exact DDE statements and causal-root definitions suitable for analysis and computation.
- Solvability status: no known closed-form solution; numerical integration requires robust root-finding and event-aware stepping.

Plain language: Start very far apart and nearly at rest—motion remains on the initial line. Delay enters through the partner's past position via the causal-time condition; there is no sideways component in this example.

Background and Simple Action

The dynamics of an architrino are governed by a simple action: acceleration occurs when the receiver intersects a delayed causal wake surface emitted by a source architrino.

The background is fixed absolute time times Euclidean space. Free paths are straight. Accelerations come only from delayed causal hits, with line-of-action direction and Jacobian-weighted magnitude, never from background curvature.

Dynamical Geometry

- Background kinematics (Newton-Cartan/Galilean):
  - The arena is absolute time × Euclidean space, $\mathcal{M} = \mathbb{R} \times \mathbb{R}^3$, with simultaneity slices $\Sigma_t = \{t\} \times \mathbb{R}^3$ carrying the flat spatial metric $h_{ij} = \delta_{ij}$.
  - "Geodesics are straight" means: in the absence of any interaction, a worldline $\mathbf{s}(t)$ satisfies $\mathbf{a}(t) = d^2\mathbf{s}/dt^2 = \mathbf{0}$; motion is uniform and rectilinear in each slice Σ_t. The background is fixed; there is no curvature to encode forces.
- Wake geometry as a continuous causal flux:
  - Each architrino streams potential continuously. At any observation time t, the contribution emitted at past time t_0 sits on the **causal wake surface** (spherical isochron) $r = v(t - t_0)$ centered on $\mathbf{s}(t_0)$, with surface density $\propto 1/r^2$ so the integrated flux remains q.
  - The potential wake is the superposition of all such causal isochrons from past emissions. The flux never shuts off; the surfaces are bookkeeping devices isolating portions of the path history whose intersection with a receiver delivers acceleration.
- Intersection as the driver of acceleration:
  - The receiver's worldline is $\mathbf{s}_{o'}(t)$. An intersection at time t means some earlier emission time $t_0 < t$ satisfies the causal-distance condition

$$\|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\| = v(t - t_0)$$

```
That event is a causal hit from source o's past to the receiver's present.
```

- At a hit, the acceleration impulse is directed along

$$\hat{\mathbf{r}} = \frac{\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)}{\|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\|}$$

```
No cross products or right-hand-rule terms appear; the action is collinear with r̂. Its magnitude is weighted by
the branch Jacobian |J|^{-1}, which captures causal-flux bunching or dilation due to source motion.
```

- "Simple action" in precise terms:
  - The law is event-driven: acceleration is a sum of per-hit line-of-action contributions, each scaled by $1/(r^2|J|)$. Between hits (as $\eta \to 0$) motion is inertial; with mollification ($\eta > 0$) the impulses become short, smooth pushes.
  - The background adds no force; departures from straight motion arise only from these intersections with emitted causal wakes, including self-hits when kinematics allow.
- Physical picture:
  - Picture many continuously expanding wake surfaces (causal isochrons). A push occurs whenever one of those surfaces intersects the receiver, directed straight along the radius back to its emission point, with inverse-square geometric decay and an additional Jacobian weight set by the source motion on that branch.

Causal Set and Delay Geometry

The receiver o' at time t interacts with a source o through the possibly multi-valued set of causal emission times

$$\mathcal{C}_o(t) = \left\{ t_0 < t \mid \|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\| = t - t_0 \right\}$$

For $\|\mathbf{v}_o(t_0)\| < 1$ locally, $\mathcal{C}_o(t)$ is generically a singleton; for $\|\mathbf{v}_o\| > 1$, it may contain multiple solutions, including self-hits when $o' = o$.

Clarification: "Multi-valued" means that, for a fixed observation time t, there can be more than one emission time t_0 that satisfies the causal-distance condition; i.e., $\mathcal{C}_o(t)$ may contain multiple causal roots when $\|\mathbf{v}_o\| > 1$ or when same-source roots exist for $o' = o$. This multiplicity can occur only if the transmitter/source has exceeded field speed at least once; if $\|\mathbf{v}_o\| < 1$ everywhere, $F(t_0; t)$ is strictly increasing in t_0 and the causal root is unique.

Terminology note: the `causal set` in this simulation note is the causal interaction set $\mathcal{C}_o(t)$: a set of delayed emission times that reach a receiver now. It is not Causal Set Theory, the external quantum-gravity program that treats discrete spacetime events and partial order as fundamental. That outside program remains useful as a comparison for causal ordering and continuum emergence, but the native object here is a path-history root set inside absolute timespace.

Geometry of Delay and Roots

- Root condition as an expanding causal isochron intersection:
    - Define $F(t_0; t) \equiv \|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\| - (t - t_0)$ (with $v = 1$ units). Causal roots satisfy $F(t_0; t) = 0$ with $t_0 < t$ and $H(t - t_0)$.
- Geometrically: the source point $\mathbf{s}_o(t_0)$ must lie on the causal wake surface (isochron) of radius $\tau = t - t_0$ centered at the receiver's current position $\mathbf{s}_{o'}(t)$.
- Local uniqueness (sub-field-speed, transverse crossing):
    - If the source speed is locally sub-field-speed ($\|\mathbf{v}_o(t_0)\| < 1$) and the derivative $\partial_{t_0} F(t_0; t) = -\hat{\mathbf{r}} \cdot \mathbf{v}_o(t_0) + 1$ is nonzero at the root, then the implicit function theorem guarantees a unique, smooth root branch near t.
    - Intuition: the expanding causal isochron intersects the moving source path transversely.
- Multiple roots (require super-field-speed):
    - When $\|\mathbf{v}_o\| > 1$ at some emission times, the source can outpace its recent wake surfaces, allowing several distinct historical points to satisfy the same distance–time constraint (multi-hit regime). If $\|\mathbf{v}_o\| < 1$ everywhere, $F(t_0; t)$ is strictly increasing in t_0, so at most one causal root exists.
- Conventions at singular cases:
    - We adopt $H(0) = 0$ so the instantaneous emission at $t_0 = t$ does not produce an immediate self-kick.
    - No $r = 0$ causal roots beyond $\tau = 0$: because $r = v(t - t_0)$, $r = 0$ implies $\tau = 0$; the $\tau = 0$ case is excluded by $H(0) = 0$. Under mollification, the symmetric limit as $r \to 0$ yields zero net push.

Plain language: a receiver is pushed only by earlier source moments whose causal isochrons currently pass through it. Usually there is one such moment; if the source is very fast or its path loops around, there can be several.

Non-technical visualization — outrunning your own wake (speedboat analogy):

- Picture a speedboat continuously laying down circular wake ridges that spread outward across the water at a fixed wave speed c_w (analogy variable: wake ridge expansion speed). If the boat stays slower than c_w, it remains inside its newest ridge and will never meet it again, no self-hits. Once the boat exceeds c_w, it moves ahead of its freshest ridge. Later, if it curves or slows, it can run into older ridges it created earlier. Each crossing delivers a brief shove normal to the ridge (straight outward from the ridge's center), mirroring the model's line-of-action push. The ridge "drop rate" never changes, but the received shove is stronger or weaker depending on how the boat's earlier motion bunches or dilates the ridge spacing along the crossing direction, mirroring the model's Jacobian weighting. This is an analogy: real Kelvin wakes are dispersive; we idealize to circular ridges expanding at one speed to match the model's fixed-speed causal isochrons.

Four self-hits in one maneuver (storyboard):

1. Sprint phase (exceed the field speed): The boat accelerates to a speed strictly greater than c_w and holds it for several ticks. During this super-speed run it lays down several concentric ridges that it immediately outruns.
2. Set up spacing: Maintain the super-speed for long enough to create at least four successive ridges with noticeable gaps (their radii grow at $c_w \cdot \Delta t$ while the boat advances faster than c_w).
3. Curving return: Bank into a broad, smooth turn (a teardrop/U-turn or a gentle outward spiral) that arcs back toward the track laid moments earlier.
4. Crossings: As the boat's curved path cuts across the expanding circles, it re-enters first the outermost of those recent ridges, then the next three in sequence. With a steady arc and timing, four distinct ridge crossings occur in quick succession—four self-hits. The shove at each crossing points straight away from the center of that ring (the boat's earlier position).
5. Tuning intuition: to make four hits likely, use a fast straight run $(|v| > c_w)$ to lay multiple rings, then a wide-radius turn whose chord length is comparable to the ring spacing. Tighter loops and longer super-speed runs increase the chance of multiple crossings; without exceeding c_w, this multi-hit pattern cannot occur.

Delay Dynamics Energy

This chapter isolates the energy problem created by causal-delay dynamics. It is foundations-adjacent because it states what kind of energy object the substrate law is allowed to use before later chapters invoke conservation, no-runaway arguments, event ledgers, or Noether sea exchange.

The core warning is simple: time-translation invariance of a state-dependent delay equation does not by itself supply the familiar local Noether energy of finite-dimensional mechanics. In $\mathbb{AAA}$, any term written as E_{wake} must be constructed from the same causal-history law, regularization, branch chart, and boundary convention that generate the force row. Otherwise it is a diagnostic label, not a conserved charge.

Energy Construction Problem

Fix a finite retained system over a time window $W = [t_a, t_b]$, a spatial window $\Omega \subset \Sigma_t$ when boundary flux is relevant, memory depth $h < \infty$, causal-surface width $\eta > 0$, optional core cutoff $\epsilon_c > 0$, and branch chart

$$\mathfrak{B}(\Gamma, \mathcal{S}; h, \eta, \epsilon_c)$$

for the same active causal-root rows used by the Master Equation. The retained history at time t is the segment

$$X_t = \{\mathbf{x}_a(t+\theta), \mathbf{v}_a(t+\theta), q_a : a \in A_\Omega,\ -h \le \theta \le 0\}$$

with any excluded rows, endpoint conventions, and boundary crossings recorded explicitly.
Here A_Ω is the retained architrino index set for the window, not a new kind of assembly.

A promoted delay-energy functional has the form

$$E^{(\eta)}_{\text{delay}}[X_t; \mathfrak{B}, \Omega] = K^{(\eta)}_\mu(t) + E^{(\eta)}_{\text{wake},\mathfrak{B}}(t) + E^{(\eta)}_{\text{sea},\Omega}(t)$$

where $K^{(\eta)}_\mu$ is the declared mechanical kinetic bookkeeping proxy, $E^{(\eta)}_{\text{wake},\mathfrak{B}}$ is the causal-history interaction contribution, and $E^{(\eta)}_{\text{sea},\Omega}$ is included only when retained Noether sea degrees of freedom are part of the window. None of these terms is allowed to absorb an unreported boundary flux or unresolved reaction channel.

Accepted Construction Routes

There are three admissible ways to define the wake-energy term. A calculation may use one route directly, but a theorem-level conservation claim must also state why the other routes are equivalent or irrelevant on the declared chart.

Action-Boundary Route

If a symmetry-preserving nonlocal action supplies the force row, then the energy term is the time-boundary charge induced by absolute-time translation. With causal-delay interaction kernel $\mathcal{K}^E_{ij}(t_1, t_0)$ chosen by the same action as the force residual,

$$E^{(\eta)}_{\text{wake},\mathfrak{B}}(t) = \frac{1}{2}\sum_{i,j}\int_{-\infty}^{t} dt_0 \int_t^\infty dt_1\, \partial_{t_1}\mathcal{K}^{E,\eta}_{ij,\mathfrak{B}}(t_1, t_0)$$

is the candidate in-flight causal-history charge. This is the route developed in Master Equation and Effective Lagrangian. It becomes theorem-level only when the same action also gives the accepted acceleration law and the endpoint leakage residual vanishes.

Work-Integral Route

For a realized trajectory, one may reconstruct a compatible interaction contribution by integrating the delivered power:

$$U_{\mathfrak{B}}(t) = U_* - \int_{t_*}^{t} \sum_i \mu_{\text{arch}}\, \mathbf{a}^{(\eta)}_{i,\mathfrak{B}}(t') \cdot \mathbf{v}_i(t')\, dt'$$

This route is trajectory-local. It is useful for simulations and branch replay, but it is not an off-shell conserved charge unless the same action and boundary convention have already been declared.

Binary Branch Work Ledger

For a solved two-body branch chart b, the work-integral route has a concrete first test. Let $\mathbf{a}^{(\eta)}_{i,b}(t)$ be the acceleration row obtained from exactly the active causal roots retained by the binary branch chart. With the quadratic kinetic proxy, define the delivered branch power by

$$P^{(\eta)}_{b,\text{work}}(t) = \sum_{i=1}^{2} \mu_{\text{arch}}\, \mathbf{a}^{(\eta)}_{i,b}(t) \cdot \mathbf{v}_i(t)$$

and reconstruct the compatible causal-history interaction contribution by

$$U^{(\eta)}_{b,\text{work}}(t) = U_b(t_*) - \int_{t_*}^{t} P^{(\eta)}_{b,\text{work}}(t')\, dt'$$

For a primitive kinetic scalar, replace $\mu_{\rm arch}$ by $\mu_K(\|\mathbf{v}_i\|)$ inside the sum. This is the operational binary definition: the wake-history row is whatever balances the delivered branch work along the realized trajectory, after the window, regulator, and branch ledger have been declared.

On a circular benchmark with speed s_b, the radial component is orthogonal to the receiver velocity, so the branch power is the tangential row:

$$\left\langle P^{(\eta)}_{b,\rm work}\right\rangle_{P_b} = \mu_{\rm arch}\, s_b\, \left\langle A^{\tan}_{\eta,b}\right\rangle_{P_b}$$

for the quadratic proxy. A nonzero value is not by itself an energy-conservation failure; it is the quantity that the boundary flux, recoil row, or constructed wake-history term must balance. A stable binary claim must therefore compute this row on the same branch chart as the motion residuals before invoking a Noether-style conserved energy.

Boundary-Flux Route

For finite retained windows, missing energy must be routed to boundary exchange rather than hidden in $E_{\rm wake}$. The finite-window balance target is

$$\frac{dE^{(\eta)}_\Omega}{dt} + \int_{\partial\Omega} \mathbf{J}^{(\eta)}_E \cdot \hat{\mathbf{n}}\, dA = P^{(\eta)}_{{\rm ext},\Omega} + \mathcal{R}^{(\eta)}_{E,\Omega}$$

where $\mathbf{J}^{(\eta)}_E$ records causal-wake escapement, assembly crossings, and declared medium exchange through the retained boundary. The flux term is not a new substrate field; it is the boundary part of the retained causal-history ledger.

Crosswalk Residual

The three routes must not define three different energies for the same branch. On any chart where more than one construction is available, use the crosswalk residual

$$\Delta^{(\eta)}_{\rm E,cross}(W;\mathfrak{B}) = \frac{\left|\Delta_W E^{(\eta)}_{\rm wake,act} - \Delta_W U_{\mathfrak{B}} - \Phi^{(\eta)}_{\partial\Omega,E}(W)\right|}{\left|\Delta_W E^{(\eta)}_{\rm wake,act}\right| + |\Delta_W U_{\mathfrak{B}}| + \left|\Phi^{(\eta)}_{\partial\Omega,E}(W)\right| + \varepsilon}$$

where $\Phi^{(\eta)}_{\partial\Omega,E}(W) = \int_W \int_{\partial\Omega} \mathbf{J}^{(\eta)}_E \cdot \hat{\mathbf{n}}\, dA\, dt$ is the declared boundary energy flux. The chart promotes only if $\Delta^{(\eta)}_{\rm E,cross} \to 0$ under the same refinement limit used for the force residual.

Conservation Residual

Let $\mathbf{R}^{(\eta)}_i$ be the Euler or force residual of the declared action-derived model, and let $\mathcal{B}^{(\eta)}_E$ collect endpoint leakage, period cuts, excluded self-coincidence boundaries, and omitted branch rows. The finite-window conservation residual is

$$\mathcal{R}^{(\eta)}_E(W;\mathfrak{B}) = \Delta_W\left(K^{(\eta)}_\mu + E^{(\eta)}_{\rm wake,\mathfrak{B}} + E^{(\eta)}_{\rm sea,\Omega}\right) - \int_W \sum_i \mathbf{v}_i \cdot \mathbf{R}^{(\eta)}_i\, dt - \int_W \mathcal{B}^{(\eta)}_E\, dt - W^{(\eta)}_{\partial\Omega}$$

The normalized diagnostic is

$$\epsilon^{(\eta)}_E(W;\mathfrak{B}) = \frac{\left|\mathcal{R}^{(\eta)}_E(W;\mathfrak{B})\right|}{\left|\Delta_W K^{(\eta)}_\mu\right| + \left|\Delta_W E^{(\eta)}_{\rm wake,\mathfrak{B}}\right| + \left|\Delta_W E^{(\eta)}_{\rm sea,\Omega}\right| + \left|W^{(\eta)}_{\partial\Omega}\right| + \varepsilon}$$

An exact isolated conservation claim requires $\epsilon^{(\eta)}_E \to 0$, $\Delta^{(\eta)}_{\rm E,cross} \to 0$ when applicable, and stable branch floors as η and the numerical/history-window resolution are refined.

No-Double-Counting Rule

The interaction contribution may be carried by $E_{\rm wake}$, by an equivalent work-integral reconstruction, or by an explicitly retained near-field decomposition, but not by all of them at once. If a pairwise $U_{\rm int}$ term is used inside an assembly, the wake-energy term must omit the same near-field content. If a Noether sea update is retained inside $E_{\rm sea,\Omega}$, it must not also appear as an outgoing event-ledger channel. The same rule is used by Emergence and Energy.

Promotion and Failure Conditions

A delay-energy construction is promotable only when the branch chart names:

1. the retained history window h and memory truncation residual;
2. the causal-surface regularization η and any core cutoff ϵ_c;
3. active causal roots, inactive-root gaps, and the active Jacobian floor;
4. the exact route used for $E_{\rm wake}$;
5. boundary flux, endpoint leakage, period-cut terms, and excluded self-coincidence rows;
6. the crosswalk residual whenever more than one energy construction is invoked;

7. the lower-bound condition needed for no-runaway arguments.

The construction fails if conservation is recovered only by changing the energy definition per observable, if $E^{(\eta)}_{\rm wake}$ has no lower bound on the admitted chart, if endpoint leakage is silently discarded, if the regulator is not the same regulator used by the force law, or if the branch chart loses its causal-root floors. In those cases $E_{\rm wake}$ remains a diagnostic placeholder and cannot be used to close energy bookkeeping, stability, or no-runaway claims.

Downstream Use

This chapter is the shared energy standard for Master Equation, Effective Lagrangian, Energy, Binary Dynamics, and event-ledger uses in Emergence. The two-body binary closure packet must report $\epsilon^{(\eta)}_E(W;\mathfrak{B})$, $\Delta^{(\eta)}_{\rm E,cross}(W;\mathfrak{B})$, and the lower-bound entry on the same branch chart as its motion, branch-floor, stability, and frequency residuals. Existence and stability are not enough unless the accepted branch also carries a constructive energy ledger.

Informational Ambiguity

From the perspective of the receiving architrino, the information carried by an intersecting causal wake surface is limited. The receiver has direct access only to two local facts:

1. The net strength of the potential at the point of intersection.
2. The unoriented line of action through its current position. Orientation along that line remains ambiguous.

Degeneracies and Inference Limits

- Many-to-one mapping:
    - Different combinations of source identity, polarity magnitudes, distances, and emission timing/geometry can yield the same instantaneous hit magnitude and direction at the receiver.
- Sign ambiguity across a line:
    - Attraction from a positive-polarity source on one side is indistinguishable, at an instant, from repulsion by a negative-polarity source located at the diametrically opposite point along the same line.
- Consequence for reconstruction:
    - Instantaneous local data at the receiver are insufficient to invert for sources; this remains true even for an $\mathbb{U}_{\rm now}$ universe-state perspective who knows the universal clock t and the Euclidean rest frame. The $\mathbb{U}_{\rm now}$ universe-state perspective can eliminate coordinate uncertainty (perfect synchronization and alignment) but not the physical ambiguities below.
    - Irreducible ambiguities at an instant:
        - Sign/side ambiguity: attraction from a positive-polarity source on one side is indistinguishable from repulsion by a negative-polarity source on the diametrically opposite side along the same line.
        - Superposition along a line: multiple sources aligned on the same unoriented line of action can sum to the same net magnitude and direction at one instant.
        - Self-hit confound: a self-interaction and an external source can yield identical instantaneous data if they lie on the same line with compensating magnitudes.
        - Continuum of surrogate locations: for any instantaneous hit there exists a continuum of stationary surrogate source positions along the same unoriented line of action, each with a correspondingly adjusted emission time t_0, that reproduces the same instantaneous data; hence instantaneous inversion is severely underdetermined.
    - What helps (over time or with more views):
        - Track the time series of the line of action $\hat{\mathbf{r}}(t)$ and separation proxy $r(t)$ inferred from timing and geometry; curvature and rotation of $\hat{\mathbf{r}}$ constrain source trajectories.
        - Use multiple receivers (an array) to triangulate unoriented lines at the same t; intersecting rays narrow candidate locations (two-sided).
        - Actively vary the receiver path to sample different directions and ranges, turning the inverse problem into a controlled experiment.
        - Impose priors: polarity inventories, speed bounds, and assembly templates reduce degeneracy space.
        - Use surrogate-location recasts: for instantaneous hits, place a stationary surrogate source somewhere along the same unoriented line of action and adjust only the emission time; this simplifies hypothesis testing without altering per-wavefront amplitude.
    - Absolute-observer note: Access to absolute time and a common Euclidean frame enables global correlation of events across receivers, but unique inversion at an instant would require hidden information (the full emission ledger $\{(t_0, \mathbf{s}_j(t_0), q_j, \mathbf{v}_j(t_0))\}_j$).

Practical reconstruction is therefore necessarily temporal, statistical, and multi-view.

Plain language: a hit reports magnitude and line of action, not source identity or distance. Many different source histories can fit the same momentary push. A null action at an instant conveys no information about sources; superposition can cancel perfectly even in a non-empty universe.

Numerical Recipe and Stability

Event-aware integration (practical algorithm):

1. Root finding:
    - For each source o (including $o' = o$ for potential self-hits), solve $F(t_0; t) = \|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\| - (t - t_0) = 0$ for $t_0 < t$.
    - Discard non-physical roots by convention $H(0) = 0$ (exclude $\tau = 0$); note $r = 0$ occurs only at $\tau = 0$ and is thus excluded.
2. Per-hit accumulation:
    - For each accepted root, compute r, $\hat{\mathbf{r}}$, and

$$\mathbf{a}_{o' \leftarrow o}(t; t_0) = \kappa \, \sigma_{q_o q_{o'}} \, \frac{|q_o q_{o'}|}{r^2 \, |J_{o' \leftarrow o}(t; t_0)|} \, \hat{\mathbf{r}}$$

- Sum over all sources and all roots (superposition).

3. Time stepping:
    - Impulsive mode: advance velocities with jumps at hit times (measure-driven ODE with velocity of bounded variation).
    - Mollified mode: replace $\delta(\cdot)$ by $\delta_\eta(\cdot)$ and integrate with a standard ODE solver; choose η small relative to local geometric scales.
4. Stability tips:
    - Use event bracketing or root trackers for continuity of $t'(t)$ across steps.
    - Limit step size so that at most one (or a controlled number of) mollified wake surfaces overlap significantly per step.
    - Monitor invariants over resolved windows (work–energy balance with Φ_η) to validate settings.
5. Units:
    - Use $v = 1$ nondimensionalization throughout. Remember: emission cadence and per-wavefront amplitude are constant; receiver speed influences only power via v_r.
6. Two-body closure run packet:
    - For a candidate electrino:positrino binary, emit the signed branch ledger b, regulator η, step or collocation scale h, candidate period P_b, and the residual tuple

$$\text{Run}_{2\text{B}}^{(\eta)} = \left(\mathcal{R}_{\text{EOM}}^{2\text{B}}, \mathcal{R}_{\text{per}}^{2\text{B}}, \mathcal{R}_{\text{bal}}^{2\text{B}}, \nu_J^{2\text{B}}, \Delta_{\text{gap}}^{2\text{B}}, \lambda_{\text{sec}}^{2\text{B}}, \epsilon_E^{(\eta)}, \Delta_{\text{E,cross}}^{(\eta)}, \mathcal{R}_\omega^{2\text{B}}\right).$$

- Fail closed if the signed ledger changes during the reported period, an active Jacobian floor or inactive-root gap vanishes, the projected return-map spectrum is not computed, the energy residuals use a different window or branch chart than the motion residuals, or the extracted frequency is not stable under refinement.
- Treat a visually periodic orbit without these entries as a search hit only. It is not a binary closure certificate.

Plain language: At each time, find which past emissions can reach the receiver now, sum their radial pushes with $1/r^2$ falloff, and step forward either with sharp kicks at exact hit times or with thin mollified wake surfaces for smooth integration.

Radial Attraction

Setup:

- A test architrino with polarity q' falls radially toward a fixed center with polarity q.
- The interaction is delayed; the causal emission time exists uniquely for a fixed source, but the acceleration depends only on the current separation because the source position is time-independent.

Objectives:

- Closed-form relations for r(t), v®, and time-to-fall from r0 to r.
- Energy balance and integral expressions suitable for comparison.

Delay differential equation and exact reduction:

- With field speed normalized to $v = 1$ and a fixed source location x_c, the causal root satisfies $|x(t) - x_c| = t - t_0$ with $t_0 < t$.
- The per-hit law yields a line-of-action acceleration whose magnitude depends on the current separation $r(t) = |x(t) - x_c|$:

$$\ddot{x}(t) \;=\; -\,\kappa\,\sigma_{qq'}\,\frac{|qq'|}{r(t)^2\,|J(t)|}\,\operatorname{sgn}\big(x(t) - x_c\big)$$

Writing $K = \kappa\,|qq'| > 0$ and $r = |x - x_c|$, the radial ODE is

$$\ddot{r}(t) \;=\; -\,\frac{K}{r(t)^2\,|J(t)|}$$

Exact solution (closed form):

- Energy integral: $\frac{1}{2}\dot{r}^2 - K/r = \text{const.}$
- For release from rest at $r(0) = r_0$ with $\dot{r}(0) = 0$,

$$r(t) \;=\; r_0 \cos^2\eta, \qquad t \;=\; \sqrt{\frac{r_0^3}{2K}}\;\big(\eta + \sin\eta\cos\eta\big), \quad \eta \in [0, \tfrac{\pi}{2}]$$

with fall time $T_{\text{fall}} = \frac{\pi}{2}\sqrt{r_0^3/(2K)}$.

Notes:

- For a fixed source, the source velocity vanishes, so $J(t) = 1$. The delayed formulation therefore reduces exactly to the inverse-square ODE above; the causal root determines only the emission time, not the instantaneous acceleration magnitude or direction.

Use:

- A ground-truth closed form against which delayed-law simulations can be benchmarked in the fixed-source case.

Plain language: With a stationary center, the Jacobian is trivial and the delayed law simplifies to the familiar inverse-square fall, which has an exact, closed-form solution.

Receiver Velocity and Work

Because $\mathbf{a}_{o'\leftarrow o}(t;t_0) \parallel \hat{\mathbf{r}}$, a single hit changes only the radial velocity component:

$$\frac{d}{dt}\mathbf{v}_\perp = \mathbf{0} \quad \text{from this hit,} \qquad \frac{d}{dt}v_r = \mathbf{a}_{o'\leftarrow o}(t;t_0)\cdot\hat{\mathbf{r}} = \frac{\kappa\,\sigma_{q_o q_{o'}}\,|q_o q_{o'}|}{r^2}$$

Decomposition and Energetics

- Decomposition at a hit:
  - Write $\mathbf{v} = v_r\,\hat{\mathbf{r}} + \mathbf{v}_\perp$, where $v_r = \mathbf{v}\cdot\hat{\mathbf{r}}$ and $\mathbf{v}_\perp\cdot\hat{\mathbf{r}} = 0$.
  - A single hit changes v_r but not $\mathbf{v}_\perp$ instantaneously.
- Power and work:
  - Instantaneous power is $\mathbf{a}\cdot\mathbf{v} = |\mathbf{a}|\,v_r$.
  - Orthogonal motion does no instantaneous work; only radial motion exchanges kinetic and potential energy at a hit.
- Local trend via $1/r^2$:
  - If $v_r < 0$ (moving inward), near-future hits tend to be stronger because r shrinks between events; if $v_r > 0$, they tend to weaken.

Plain language: Each hit only changes your along-the-line speed right then; sideways speed is untouched. Energy transfer happens only through the along-the-line part.

Repulsion

Setup:

- Two identical charges (e.g., q1=q2=+ϵ) placed at separation r0 with v1=v2=0 and symmetry about the midpoint.

Objectives:

- Delay-only formulation of the equations of motion (DDEs).
- Exact analytic solutions if available; otherwise, status of solvability.

Delay differential equations (two-body, v=1):

- Causal times:
  - $t_0^{(2\to 1)} \in \mathcal{C}_2(t)$ solves $|x_1(t) - x_2(t_0)| = t - t_0$.
  - $t_0^{(1\to 2)} \in \mathcal{C}_1(t)$ solves $|x_2(t) - x_1(t_0)| = t - t_0$.

- Accelerations (sum over all causal roots if multiple exist):

$$a_1(t) \;=\; \sum_{t_0 \in \mathcal{C}_2(t)} + \kappa\, \epsilon^2 \, \frac{\operatorname{sgn}\!\big(x_1(t) - x_2(t_0)\big)}{r_{12}^2\, |J_{12}(t;t_0)|}, \quad r_{12} = \big|x_1(t) - x_2(t_0)\big|$$

$$a_2(t) \;=\; \sum_{t_0 \in \mathcal{C}_1(t)} - \kappa\, \epsilon^2 \, \frac{\operatorname{sgn}\!\big(x_2(t) - x_1(t_0)\big)}{r_{21}^2\, |J_{21}(t;t_0)|}, \quad r_{21} = \big|x_2(t) - x_1(t_0)\big|$$

- J_{12} and J_{21} are the corresponding causal-root Jacobians. A root with a failed Jacobian floor is a branch-transition or caustic case, not an ordinary stable row of this two-body DDE.
- Symmetry implies $x_1(t) = -x_2(t)$ and $a_1(t) = -a_2(t)$ for all t given symmetric initial data.

Solvability status:

- No exact closed-form solution is presently known for the coupled DDE system under mutual repulsion with delay.

Deliverables:

- Exact DDE statements and causal-root definitions suitable for analysis and computation.
- Notes on symmetry and qualitative properties without invoking approximations.

Plain language: Two like polarities at rest push apart along the line under the delayed law; the governing equations are implicit in the causal times, and no closed-form solution is currently known.

Self-Energy

Purpose: explain why classical "point-charge self-energy" divergences do not arise in this framework, and summarize the role of measure-valued causal surfaces, the $H(0) = 0$ convention, and η-mollification.

Classical self-energy pathology (contrast)

In classical electrostatics, a static $1/r$ potential yields an electric field $\mathbf{E} \propto 1/r^2$ with energy density proportional to $\|\mathbf{E}\|^2 \propto 1/r^4$. Integrating $1/r^4$ over a ball produces a divergent $\int (1/r^2)\, dr$ near $r \to 0$, the textbook "infinite self-energy of a point charge." This is an artifact of modeling the source as an enduring, everywhere-filled near field.

Why the zero-radius divergence is quarantined here

This project does not posit a static near field. Instead:

- Measure-valued expanding causal surfaces (no static $1/r$ near field):
  - Each emission is a razor-thin causal isochron with surface density $q/(4\pi r^2)$, represented by $\rho(t,s) = (q/(4\pi r^2))\delta(r - c_f\tau)H(\tau)$. The support at fixed t is a causal wake surface S_r, not a three-dimensional $1/r^2$ fill down to $r = 0$. See Background and Simple Action.
- $H(0) = 0$ (no coincident self-kick):
  - The instantaneous emission $(\tau = 0)$ contributes nothing to the force on the emitter; $r = 0$ roots beyond $\tau = 0$ do not exist because $r = c_f(t - t_0)$. This removes the only event where a literal $r = 0$ could enter. See Causal Set and Delay Geometry.
- η-mollification (finite, well-defined work over resolved windows):
  - Replace $\delta(r - c_f\tau)$ by a narrow Gaussian δ_η with width $\eta > 0$ when differentiability is required. Potentials Φ_η and forces $-\nabla(q'\Phi_\eta)$ are then regular functions; on any resolved interval the work-energy identity holds:
    $\Delta E_k = -\Delta U$, with $U = q'\Phi_\eta$,
    and remains finite. As $\eta \to 0$, integrals converge in the weak sense to the impulsive model without introducing infinities. See Well-posedness and Regularization.
- Event-driven geometry (self-hits occur at $r > 0$):
  - Self-interaction requires outrunning recent wake surfaces $(\|\mathbf{v}\| > c_f)$. Self-hits are intersections with one's own earlier wakes at strictly positive radius $r > 0$, yielding finite $1/r^2$ impulses (repulsive, like-on-like). There is no accumulation of divergent near-field energy at $r \to 0$.

Net effect: within a declared admissible branch chart, the canonical ontology (moving surface measures, H(0)=0, mollification for analysis) removes the zero-radius event that creates the classical point-charge self-energy divergence. Any remaining divergence claim must enter through a failed branch floor, failed window limit, or failed $\eta \to 0$ convergence test rather than through an assumed static near field.

Practical guidance (numerics and analysis)

- Choose η small relative to local geometry (path curvature radius, inter-source spacing) for smooth ODE integration; verify $\Delta E_k = -\Delta U$ on resolved windows.
- Calibrate κ using stationary/slow benchmarks (Method 2) and use the event-driven law (Method 3) for many-body dynamics; no per-hit emitter-speed amplitude weighting is introduced.
- Treat self-hits as ordinary finite r>0 events; ensure H(0)=0 in implementation to exclude coincident-time artifacts.

Sign-resolved bookkeeping

An additional numerical caution is worth stating explicitly: a Noether sea region or assembly may carry a large internal action budget even when its coarse far-wake potential appears weak.

- Positive and negative sectors can superpose so that the net far-field potential is small.
- That cancellation does **not** imply the underlying kinetic work or stored interaction content is individually small in each sector.
- For this reason, diagnostics should track sign-resolved contributions whenever possible rather than relying only on net-potential summaries.

This matters especially for shielding claims. A strongly shielded assembly may look energetically modest from afar while still containing substantial internal positive/negative activity whose cancellation is only effective after superposition. Sign-resolved ledgers therefore help distinguish true low-energy states from high-content states hidden by cancellation.

Plain language: We don't keep a permanent 1/r field glued to the point. Instead we use thin expanding causal surfaces, ignore the instant of emission for self-push, and (when needed) slightly thicken those wake surfaces so calculus works—so nothing ever "blows up" at r=0.

Self-Interaction Switch

An architrino can intersect an expanding causal isochron that it emitted earlier in its own history. Self-hit occurs when the same-source causal-root set is nonempty, $\mathcal{C}_{aa}(t) \neq \varnothing$. Super-field-speed history is a necessary warning condition for simple nontrivial roots, but it is not sufficient by itself; curvature, branch geometry, and the transversality floor determine whether the worldline actually intersects its own causal wake. The like-polarity self-hit contribution is repulsive and plays a key role in the stability of emergent structures.

Conditions and Effects

- Root multiplicity and self-roots:
    - The simulation should open the self-hit channel only when it finds same-source roots

$$\mathcal{C}_{aa}(t) = \{\, s < t : \|\mathbf{x}_a(t) - \mathbf{x}_a(s)\| = c_f(t-s) \,\}$$

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A speed excursion above c_f flags a candidate interval; it is not an acceptance test without root existence and a nonzero Jacobian/transversality margin.
```

- Repulsive character:
    - For like-on-like (self) interaction, $\sigma_{q_a q_a} = +1$ ensures the self-contribution points outward along $+\hat{\mathbf{r}}$, opposing further collapse.
- Stabilization and scale selection:
    - In binaries and nested assemblies, delayed attraction competes with self-repulsion. On a closed branch chart, that balance is the candidate mechanism that can set a minimal sustainable radius d_0 and a fastest natural frequency $2\pi/t_0$.

Plain language: A fast interval can make self-hit possible, but the code must still solve the same-source root equation; only actual same-source hits push outward and help set the smallest sizes and fastest rhythms of stable structures.

Superposition and Locality

Potential wake contributions from all sources superpose linearly. The net potential at any point is the sum of the individual contributions:

$$\Phi_{\text{net}} = \sum_i \Phi_i$$

The total acceleration on a particle at any instant is the vector sum of the contributions from every intersecting causal wake surface. Operationally, every architrino is continuously immersed in the superposed wakes of all others and, when the same-source root condition permits, its own. Calculating the path-history integral is tractable by isolating each causal emission event, evaluating the Jacobian-weighted $1/r^2$ kernel at that emission, and then summing under a declared finite active horizon, screening rule, cancellation argument, or summation prescription.

Why Nearby Wakes Dominate

- Linear addition at the causal-surface level:
  - Because each source contributes a distribution supported on its causal wake surfaces, the total wake measure is a sum of these measures; the acceleration law is linear in the summed contributions.
- Locality from $1/r^2$ plus convergence control:
  - The surface density on each causal wake surface scales as $1/r^2$, so nearby coherent hits contribute disproportionately compared to distant ones. In an infinite three-dimensional source population this does not by itself guarantee convergence, because the number of sources in a radial layer grows like $r^2\,dr$. Random phases, angular cancellation, screening, finite active horizons, or explicit mean-field/principal-value subtraction must be part of the branch prescription.
- Practical consequence:
  - Simulations can prioritize nearby sources and recent roots only after declaring the far-field treatment: cutoff error, multipole cancellation, screened background, sampled mean field, or principal-value subtraction.

Plain language: Add the pushes from all causal wake surfaces, but do not assume one over distance squared makes an infinite universe automatically finite; the simulation must say how distant wakes cancel, screen, or get summarized.

Units and Constants

This note fixes the unit and symbol conventions used by the action-energy simulation notes. We work in units with field speed $v = 1$ unless stated otherwise, use $\kappa > 0$ for the universal coupling, and use $\eta > 0$ as the default regularization thickness for causal isochrons.

Core symbols:

- $v = 1$: field speed in normalized units.
- $\kappa > 0$: universal coupling constant.
- $\eta > 0$: causal-isochron thickness.
- $\epsilon > 0$: polarity-unit magnitude; Electrino $q = -\epsilon$, Positrino $q = +\epsilon$.
- $\sigma_{qq'} = \mathrm{sign}(q\,q') \in \{+1, -1\}$.
- $r = \|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\|$, with $\hat{\mathbf{r}} = (\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0))/r$.

Dynamical Geometry

- Field-speed units ($v = 1$):
  - Choosing L_0, T_0 with $v = L_0/T_0 = 1$ fixes a conversion between spatial and temporal scales so that all speeds are dimensionless ratios to the field speed. This is akin to “setting c=1,” but the reference is the model’s field speed. Kinematics still lives on absolute time × Euclidean space; we have not mixed time and space into a 4D line element.
  - Consequence: every velocity appears as a pure number $\|\mathbf{v}\|$; the symmetry point $\|\mathbf{v}\| = v$ becomes $\|\mathbf{v}\| = 1$. Rescaling L_0 and T_0 together leaves all dimensionless predictions invariant.
- Coupling constant ($\kappa > 0$):
  - κ sets the overall scale of per-hit acceleration. In the canonical law,
    $$\mathbf{a}_{o'\leftarrow o} = \kappa\,\sigma_{q_o q_{o'}}\,\dfrac{|q_o q_{o'}|}{r^2\,|J_{o'\leftarrow o}|}\,\hat{\mathbf{r}},$$
    larger κ uniformly strengthens every interaction.
  - Scaling insight: if you scale $\kappa \mapsto \alpha\kappa$ while keeping (ϵ, η) fixed, accelerations scale by α. Characteristic assembly scales such as the minimal binary radius d_0 and period t_0 shift accordingly through the dynamical balance that defines them.
- Regularization width ($\eta > 0$):
  - η is the width applied to each causal isochron (wake surface) to mollify the surface delta $\delta(r - \tau)$. It converts impulsive hits into brief, smooth pushes so that standard ODE integration applies and pointwise quantities (like gradients) are well-defined.
  - Geometric guidance: choose η small relative to local geometric scales (e.g., the receiver’s instantaneous curvature radius along its path and the local inter-source separation) so the regularized dynamics approximate the ideal path-history picture while remaining numerically stable.
- Polarity-unit magnitude ($\epsilon > 0$):
  - ϵ is the fundamental polarity scale of an architrino (Electrino $q = -\epsilon$, Positrino $q = +\epsilon$). In this framework ϵ is often identified with $|e|/6$, making observer-level quark electric charges integer multiples of ϵ.

    - Per-wavefront amplitude and emission cadence are constant at the source. The received force magnitude is additionally modulated by the branch Jacobian $|J|^{-1}$, which depends on source motion along the line of action.
- Sign of interaction ($\sigma_{qq'}$):
    - $\sigma_{qq'} = \mathrm{sign}(q\,q')$ selects attraction vs repulsion while keeping the acceleration strictly collinear with $\hat{\mathbf{r}}$. Like-on-like (σ=+1) points along +$\hat{\mathbf{r}}$ (repulsion); unlike (σ=-1) points along -$\hat{\mathbf{r}}$ (attraction).
- Line of action (r, $\hat{\mathbf{r}}$, J):
    - $r = \|\mathbf{s}_{o'}(t) - \mathbf{s}_o(t_0)\|$ is the separation between the receiver "now" and the source at its causal emission time. $\hat{\mathbf{r}}$ is the corresponding unit vector, and $J = 1 - \mathbf{v}_o(t_0)\cdot\hat{\mathbf{r}}/v$ is the causal Jacobian. All per-hit actions are directed along this line; no transverse or right-hand-rule terms appear.
- Combined role in assembly scales:
    - The trio (κ, ϵ, η), together with the $1/r^2$ law, determines emergent scales such as the smallest sustainable orbit d_0 and fastest natural frequency $2\pi/t_0$. Intuitively, stronger coupling (larger $\kappa\epsilon^2$) and sharper wake surfaces (smaller η) favor tighter, faster structures until self-interaction and delay balance inward trends.
- Dimensionless branch-scan controls:
    - Simulation sweeps should report dimensionless controls rather than only raw choices of $(\kappa, \epsilon, \eta, L_0, T_0)$. Choose a reference length $L_\star$ and the corresponding reference time $T_\star = L_\star/c_f$; in field-speed units, $c_f = 1$ and $T_\star = L_\star$.
    - **Speed ratio:** use

$$\beta_i(t) = \frac{\|\mathbf{v}_i(t)\|}{c_f}$$

```
and, for circular binary scans, the existing speed factor
```

$$s = \frac{R\omega}{c_f}$$

```
A branch scan must state whether the sampled histories remain below, cross, or remain above the self-hit onset
```
$\beta = 1$.

- **Delay/window ratio:** use

$$\Theta_\tau = \frac{\tau_{\max}}{T_{\text{win}}}$$

```
where
```
$\tau_{\max}$ is the longest active causal lookback time and T_{win} is the averaging, diagnostic, or return-map window. The stored history horizon h must satisfy $h \geq \tau_{\max}$ on the scanned branch chart.

- **Regularization thickness:** use

$$\hat{\eta} = \frac{\eta}{L_\star}$$

with local checks such as $\eta/r_{\min}$ against the smallest resolved separation. A scan is numerically meaningful only when branch counts and averaged observables stabilize as $\hat{\eta}$ is reduced while the causal wakes remain resolved.

- **Coupling scale:** compare the per-hit acceleration scale with the reference acceleration $L_\star/T_\star^2$:

$$g_\kappa = \frac{\kappa\epsilon^2 T_\star^2}{L_\star^3} = \frac{\kappa\epsilon^2}{c_f^2 L_\star}$$

In field-speed units this reduces to $g_\kappa = \kappa\epsilon^2/L_\star$.

- **Branch/root tolerances:** for the causal-root residual

$$g_{ij}(\tau, \phi) = \|\phi_i(0) - \phi_j(-\tau)\| - c_f\tau$$

```
accept a root only when $|g_{ij}|/L_\star \le \varepsilon_{\rm root}$, keep distinct roots separated by $|\tau_a-\tau_b|/T_\star > \varepsilon_{\rm sep}$, and treat $|J|\le\varepsilon_J$ as a branch-birth or caustic zone rather than an ordinary stable branch.
```

- A branch-scan report should therefore include at least

$$(\beta_{\max}\ \text{or}\ s,\ \Theta_\tau,\ \hat\eta,\ g_\kappa,\ \varepsilon_{\rm root},\ \varepsilon_{\rm sep},\ \varepsilon_J)$$

```
together with the active causal-root ledger. This prevents a change in units, regularization, or root finder tolerance from masquerading as a new physical branch.
```

Plain language: We measure speeds in units where the field speed is one, use κ to set how hard every hit pushes, use η to slightly thicken the razor-thin isochrons so calculus works, and use ϵ as the basic unit of polarity. The push is always straight along the line back to where the isochron was emitted, but its received strength is also shaped by the Jacobian factor $|J|^{-1}$; like polarities push out, unlike polarities pull in.

Well-Posedness and Regularization

The regularized simulation replaces each sharp causal-surface delta by a narrow mollifier while preserving total emission q:

$$\delta(r-\tau)\longrightarrow \frac{1}{\sqrt{2\pi}\,\eta}\exp\!\left(-\frac{(r-\tau)^2}{2\eta^2}\right)$$

Impulses Versus Smooth Pushes

- Measure-driven dynamics:
  - With exact surface deltas, dynamics are impulsive: velocities are functions of bounded variation with jump discontinuities at hit times.
- Mollified isochron surfaces:
  - Replacing $\delta(\cdot)$ by a narrow Gaussian of width $\eta>0$ spreads each causal surface's intersection into a short, smooth push, yielding classical C^1 trajectories for standard ODE solvers.
- Choosing η:
  - Select η small relative to local geometric scales (path curvature radius, inter-source spacing) to approximate the event-driven picture while maintaining numerical stability.
- Distributional wake-surface normalization:
  - Treat $\delta(r-v\tau)$ and $\delta_\eta(r-v\tau)$ as distributions, so the invariant statement is an integrated statement against a test function, not the sampled height of the spike. For $\tau=t-t_0$ and $r=\|\mathbf{s}-\mathbf{s}_0\|$,

$$\rho_\eta(t,\mathbf{s})=\frac{q}{4\pi r^2}\,\delta_\eta(r-v\tau)\,H(\tau)$$

```
must satisfy
```

$$\lim_{\eta\to0}\int_{\Sigma_t} f(\mathbf{s})\,\rho_\eta(t,\mathbf{s})\,dV=\frac{qH(\tau)}{4\pi}\int_{S^2} f(\mathbf{s}_0+v\tau\,\hat{\boldsymbol\omega})\,d\Omega$$

- In particular, $f\equiv1$ gives the total-emission check

$$\int_{\Sigma_t}\rho_\eta(t,\mathbf{s})\,dV\longrightarrow qH(\tau)$$

```
On a finite annulus $R_-\le r\le R_+$, the expected retained amount is
```

$$Q_\eta^{\rm ann}(R_-,R_+;t)=qH(\tau)\int_{R_-}^{R_+}\delta_\eta(r-v\tau)\,dr$$

```
The annular residual is therefore
```

$$R_N(R_-,R_+;t)\equiv\frac{\left|\int_{R_-\le r\le R_+}\rho_\eta(t,\mathbf{s})\,dV-Q_\eta^{\rm ann}(R_-,R_+;t)\right|}{|q|+\varepsilon_q}$$

```
This catches missing $4\pi r^2$ factors, lost radial Jacobians, and mollifiers that do not preserve total emission.
```

- Curvilinear-coordinate hygiene:
  - Operator checks in spherical or cylindrical charts must use the Euclidean metric scale factors, not Cartesian component formulas applied to curvilinear components. For spherical coordinates (r, θ, φ) centered on the emission point,

$$dV = r^2 \sin\theta \, dr \, d\theta \, d\varphi, \qquad dS_R = R^2 \sin\theta \, d\theta \, d\varphi$$

```
and a radial diagnostic channel $F_r(r)\hat{\mathbf{r}}$ obeys
```

$$\nabla \cdot \big(F_r(r)\hat{\mathbf{r}}\big) = \frac{1}{r^2}\frac{\partial}{\partial r}\big(r^2 F_r(r)\big)$$

```
For a radial scalar $f(r)$,
```

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right)$$

```
The invalid shortcut $\nabla \cdot (F_r\hat{\mathbf{r}}) = \partial_r F_r$ breaks the conservation normalization of causal wake surfaces.
```

- Finite-limit discipline:
  - Treat finite source count, finite memory depth, finite step size, finite domain/window, and finite $\eta > 0$ as the first proof or simulation regime.
  - Promote large-system, continuum, or $\eta \to 0$ statements only after the retained observables converge under the declared refinement path.
  - Do not replace arbitrarily large finite systems with an actual infinite medium unless the limit preserves the causal-root count, Jacobian floors, work-energy residuals, and thermodynamic summaries being claimed.
- State-dependent branch-transition discipline:
  - State-dependent delay systems can lose classical branch continuation at transition points where a delayed argument crosses a branch boundary, a causal-root count changes, or a derivative-sensitive row enters a fold-layer. A finite-η run must therefore record how the regularized trajectory crosses each such window rather than treating the crossing as ordinary time-step noise.
  - For every declared transition window $I_* = [t_* - \Delta_*, t_* + \Delta_*]$, emit

$$\mathcal{T}_{\eta,*} = \left(I_*, \mathcal{L}_{\mathrm{root}}|_{I_*}, \mathsf{status}_{\eta,*}, \mathsf{regularization}_{\eta,*}, \mathsf{window\_scale}_{\eta,*}, \mathcal{Y}_{\eta,*}, \mathcal{E}_{\mathrm{trans},*}\right)$$

```
where $\mathsf{status}_{\eta,*}$ is the candidate branch status, chosen from the existing simple-root, fold-layer, inactive-gap, or rejected statuses, $\mathsf{regularization}_{\eta,*}$ names the finite-$\eta$ route used through the window, $\mathsf{window\_scale}_{\eta,*}$ records the declared transition scaling, and $\mathcal{Y}_{\eta,*}$ is the set of observables promoted through that window.
```

- For each promoted observable $Y \in \mathcal{Y}_{\eta,*}$, define

$$E_{\mathrm{trans}}(Y; \eta, \eta/2; I_*) = \frac{\|R(Y_{\eta/2}|_{I_*}) - Y_\eta|_{I_*}\|_{L^2(I_*, \{x_k\})}}{\|R(Y_{\eta/2}|_{I_*})\|_{L^2(I_*, \{x_k\})} + \varepsilon_0}$$

- The transition passes only if

$$\mathsf{status}_{\eta,*} = \mathsf{status}_{\eta/2,*}, \qquad E_{\mathrm{trans}}(Y; \eta, \eta/2; I_*) \le \tau_{\mathrm{trans},Y} \quad \text{for every } Y \in \mathcal{Y}_{\eta,*}$$

```
and every root-ledger row in $I_*$ keeps source identity, branch class, and status metadata under the same matching rule used by $\Delta_{\eta,\mathrm{root}}$.
```

- If the branch status flips under η refinement, route the run to branch_root_instability. If the status is stable but the promoted transition observables fail the tolerance, route it to regulator_dependence. If the transition record is missing, route it to artifact_incomplete.
- For nonsmooth windows, the transition record must include jump-location rows

$$\mathcal{D}_{\mathrm{jump}} = \{(\xi_a, k_a, \ell_a, \xi_{\pi(a)}, R_{\mathrm{jump},a})\}, \qquad R_{\mathrm{jump},a} = \frac{|t_{0,\ell_a}(\xi_a) - \xi_{\pi(a)}|}{\max(\Delta t, \Delta h, \eta/c_f, \varepsilon_0)}$$

```
Unstable jump identity routes to branch_root_instability; unresolved jump or interpolation convergence routes to mesh_nonconvergence.
```

- Fold-layer status is only a transition classification. A stable fold-layer row may preserve branch identity through η refinement, but it does not prove branch-equation balance. When the run claims a corrected one-period carrier, the acceleration-balance residual for that period must also pass before the result can proceed to monodromy, $\Delta_{\mathbf{k}}$, or η-ladder persistence.
- Energetic consistency:
  - On resolved intervals, the work–energy relation holds with Φ_η; as $\eta \to 0$, interval integrals converge to the impulsive model.

Formal $\eta > 0$ Continuation Package

The regularization package for a promoted run family is

$$\mathsf{Reg}_\eta = (\delta_\eta, \mathcal{A}_\eta, \mathsf{WP}_\eta, \mathsf{NR}_\eta, \mathsf{Cont}_\eta, \partial\mathcal{A}_\eta)$$

where δ_η is the mollified causal-wake kernel, $\mathcal{A}_\eta$ is the admissible history set, WP_η is the existence-uniqueness statement, NR_η is the no-runaway bound, Cont_η is the continuation criterion, and $\partial\mathcal{A}_\eta$ is the failure boundary.

On a finite interval $[0, T]$, the admissible history set is

$$\mathcal{A}_\eta(T; V, d, \nu, B) = \left\{ S_{\eta,t} : \sup_{t\le T} \|\mathbf{v}(t)\| \le V, \quad \inf r_{ij,\ell}(t) \ge d, \quad \inf |\partial_\tau g_{ij,\ell}(t)| \ge \nu, \quad \sup B_{ij}^{\text{active}}(t) \le B \right\}$$

Existence and uniqueness mean that every declared initial history $S_{\eta,0} \in \mathcal{A}_\eta(T; V, d, \nu, B)$ generates a unique $S_\eta(t)$ on $[0, T]$ in the declared history class, and that the emitted root ledger is generated by that solution rather than by a post-hoc branch choice.

The no-runaway condition is a lower bound on the regularized wake energy:

$$E_{\text{tot}}^{(\eta)}(t) = K_\mu(t) + E_{\text{wake}}^{(\eta)}(t), \qquad E_{\text{wake}}^{(\eta)}(t) \ge U_{\min}^{(\eta)} > -\infty$$

When the regularization preserves the relevant time-translation symmetry, this gives

$$K_\mu(t) \le E_{\text{tot}}^{(\eta)}(0) - U_{\min}^{(\eta)}$$

on the isolated run window.

The continuation criterion is

$$S_\eta([0,T]) \subset \mathcal{A}_\eta(T; V, d, \nu, B) \quad \Longrightarrow \quad \text{the run may be extended past } T$$

using the same local well-posedness constants after refreshing the history segment at T. The failure boundary is

$$\partial\mathcal{A}_\eta = \{\|\mathbf{v}\| = V\} \cup \{r_{ij,\ell} = d\} \cup \{|\partial_\tau g_{ij,\ell}| = \nu\} \cup \{B_{ij}^{\text{active}} = B\} \cup \{E_{\text{wake}}^{(\eta)} \downarrow -\infty\}$$

Crossing any component of $\partial\mathcal{A}_\eta$ changes the promotion status to `eta_continuation_failure` unless a stricter replacement bound is proved in the same artifact packet.

For the finite-η pathology theorem target in Master Equation, a promoted run family must report the same boundary components as observables, not only as solver diagnostics. Divergent self-energy is routed through the d or ϵ_c row, runaway behavior through the $E_{\text{wake}}^{(\eta)}$ lower-bound row, pre-acceleration through the retained-history and endpoint-convention row, and caustic blow-up through the ν and transition-status rows. The minimum residual packet is:

- root residual and root-transport residual for every retained row,
- active Jacobian floor and inactive-root gap,
- finite-memory coverage and endpoint or period-cut leakage,
- energy, momentum, and angular-momentum residuals computed with the same η, window, and endpoint convention,
- transition-observable refinement residuals $E_{\text{trans}}(Y; \eta, \eta/2; I_*)$ for every fold-layer or caustic transit promoted through the window,
- $\Delta_{\eta,\text{root}}$ for every active branch ledger in the η ladder.

If any row is missing, the artifact status is `artifact_incomplete`. If a row is present but fails under refinement, the status is the corresponding continuation, regulator-dependence, or branch-root instability failure already defined above.

The $\eta \to 0^+$ claim boundary is

$$\limsup_{\eta\to 0^+} E_\eta(Y;\eta,\eta/2)=0, \qquad \limsup_{\eta\to 0^+} \Delta_{\eta,\mathrm{root}}=0$$

for every promoted observable and active branch ledger. Otherwise the result remains finite-η evidence only.

Plain language: The ideal model gives instantaneous kicks; a tiny thickening turns them into brief, smooth nudges that ordinary ODE solvers can integrate. Large-system or zero-width claims have to be earned by convergence, not assumed from the finite calculation.

Entropy

Entropy enters $\mathbb{AAA}$ as a record-coarse-graining concept. It is not a primitive substance, not a field in the Euclidean void, not the generator of absolute time, and not an independent gravitational mechanism. It is a functional of the histories a declared observer, apparatus, simulation packet, or effective description retains after the complete deterministic state has been projected into a finite record.

This chapter collects the entropy rule used across time, energy, measurement, computation, horizon, and cosmology discussions. The central discipline is the same-record rule: a packet may not fit entropy, temperature, flux, probability weights, apparatus cost, or horizon labels from separate hidden ensembles. If a thermal, quantum, horizon, or computational comparison is claimed, the entropy appearing in that comparison must be a projection of the same record that supplies the other quantities.

Plain-Language Reading

A simple way to read entropy is: entropy measures how many hidden detailed stories could produce the same thing a record can see. A room does not contain an entropy substance. Rather, there are many exact arrangements of dust, air, books, and clothing that a coarse observer would still record as the same messy room. Entropy counts or measures those compatible detailed arrangements after the level of detail has been fixed.

This is also why visible disorder is only a shortcut, not the definition. A jagged, broken, or visually mixed object can still have lower entropy than a smoother thermal state if fewer complete histories are compatible with its retained record. In this chapter, disorder language is acceptable only when it tracks the declared measure, macrostate partition, and unresolved history count.

In $\mathbb{AAA}$ terms, inherited entropy language usually measures unresolved path history without naming it that way. Heat spreading, phase scrambling, apparatus irreversibility, and horizon bookkeeping all express the same pressure: a finite record no longer retains enough exact architrino, assembly, causal-wake, boundary, and Noether sea history to reconstruct one unique detailed past.

Entropy remains useful because it audits coarse descriptions. It asks whether a measurement record is really stable, whether heat and work bookkeeping close, whether computation or memory reset has a physical cost, whether a horizon label count comes from real boundary records, and whether a packet is using one hidden record for entropy while using another for temperature, flux, or probability. In that sense, entropy is not fundamental ontology, but it is a powerful test of whether an effective description is physically honest.

For a fixed coarse-graining and access window, entropy is determined by universe path history. The complete path history determines the retained record, the compatible alternatives, and the boundary exchanges. The entropy value is not just a bare property of the universe by itself; it is the value obtained after declaring which histories are being distinguished and which histories are being grouped together.

The retained record may also include incoming causal-wake and potential data. A wake-inclusive entropy measures how many complete source and path-history configurations could produce the same incoming potential record, boundary-wake record, or apparatus response in the declared window. This is the form needed when measurement, radiation, horizon, or Noether sea thermodynamic bookkeeping depends on incoming causal structure rather than only on material state variables inside the window.

A common thermodynamic lesson can therefore be restated without changing the ontology: the same amount of energy can be more or less usable depending on how concentrated, phase-organized, spectrally sharp, or gradient-bearing the retained record is. Energy conservation belongs to the full same-record ledger. Entropy asks how much of that ledger has been dispersed into unresolved alternatives.

The quantum version says the same thing in a sharper language. A complete comparison state may remain pure or measure-preserving, while a subsystem looks mixed after the rest of the entangled record has been placed outside the access window. In $\mathbb{AAA}$, that is not proof that information is a primitive substance. It is another record-coarse-graining: the retained channel cannot carry the full compatible path-history and correlation record.

Core Definition

Let μ_t be a measure on complete deterministic histories compatible with a declared preparation. In a deterministic substrate this measure is not fundamental randomness. It is the pushforward of preparation-limited ignorance over the unresolved initial history and incoming-wake data. If the preparation fixes the present record at t_0 only up to a retained history depth h, let ν_{prep} be the measure on the unfixed segment $[t_0-h,t_0]$ and let $\mathcal{F}_{t_0\to t}$ be the deterministic delayed-flow map. Then

$$\mu_t=(\mathcal{F}_{t_0\to t})_*\,\nu_{\mathrm{prep}}$$

This is the official reading of μ_t in this chapter: probabilities describe unresolved retained history under a declared preparation, not stochastic substrate law. Deterministic multistability becomes important because ν_{prep} can spread over multiple basins before the flow sharpens it into a record-limited outcome distribution.

Let $W(t)$ be the access window and let $\mathcal{Q}$ be the coarse-graining used by a Physical Observer, apparatus, or simulation packet. The record projection

$$\Pi_{\mathcal{Q},W} : \Gamma_t \longrightarrow \mathcal{Z}_{\mathcal{Q},W}$$

maps complete histories into retained record variables. Here Γ_t is the preparation-conditioned complete-history space at time t, and $\mathcal{Z}_{\mathcal{Q},W}$ is the retained record-state space selected by the coarse-graining and access window. The pushed-forward record measure is

$$\nu_{\mathcal{Q},W,t} = (\Pi_{\mathcal{Q},W})_* \mu_t$$

and the corresponding observer-window entropy is

$$S_{\Pi,W}(t) = k_B\, \mathcal{H}(\nu_{\mathcal{Q},W,t})$$

where $\mathcal{H}$ is the entropy functional appropriate to the retained record measure.

For a discrete coarse partition with probabilities p_α, this reduces to the familiar Gibbs/Shannon form

$$S_{\mathcal{Q}} = -k_B \sum_\alpha p_\alpha \log p_\alpha$$

For a microcanonical retained window, the same idea is written as

$$S_{\mathcal{Q},W}(t) = k_B \log \mu\big(\Gamma_{\mathcal{Q},W(t)}\big)$$

where $\Gamma_{\mathcal{Q},W(t)}$ is the set of complete microhistories compatible with the retained macroscopic records in that window.

Plain language: entropy is not counted over reality in the abstract. It is counted over the alternatives left unresolved after the record map, measure, coarse-graining, and access window have been specified.

The exact-record limit is useful as a guardrail. If the retained partition distinguishes one complete deterministic history from every other complete deterministic history, then the active cell has probability one and the corresponding entropy is zero. That does not mean thermodynamics has disappeared from the world. It means the record has been refined until it no longer asks a thermodynamic question. A thermodynamic macrostate is a physically declared grouping of histories: a pressure, temperature, density, spectral, boundary, apparatus, or control-relevant record that a real system can retain and use.

Receiver Inference Fibers And Provenance Graphs

The wake-inclusive form has a canonical substrate construction. For a receiver i at event $(\mathbf{x}_i(t), t)$, let the retained hit record be

$$\mathcal{H}_i^{\mathrm{hit}}(t) = \{(\ell_a, \|\mathbf{a}_a\|)\}_{a \in A_i(t)}$$

where ℓ_a is the retained unoriented line of action and $\|\mathbf{a}_a\|$ is the retained hit strength for an active received root. If an apparatus retains oriented directions or source tags, those data are added to $\mathcal{H}_i^{\mathrm{hit}}$ explicitly. The receiver inference fiber is

$$\Gamma_i^{\mathrm{hit}}(t) = \big\{\gamma \in \Gamma_t : \text{the delayed branch sum of } \gamma \text{ reproduces } \mathcal{H}_i^{\mathrm{hit}}(t)\big\}$$

and the receiver-hit entropy is

$$S_i^{\mathrm{hit}}(t) = k_B\, \mathcal{H}\left(\mu_t\big|_{\Gamma_i^{\mathrm{hit}}(t)}\right)$$

This is the entropy of the receiver's inference fiber. The electrino/positrino antipode ambiguity and the surrogate-location recast described in Master Equation are then measure-preserving involutions on $\Gamma_i^{\mathrm{hit}}(t)$ whenever the retained hit record is unchanged by the recast. Measurement uncertainty at this level is therefore a computable fiber multiplicity, not a slogan added after the dynamics.

When $\mathcal{H}$ is evaluated as a probability entropy, the restricted measure is normalized on $\Gamma_i^{\mathrm{hit}}(t)$. If the fiber has zero or undefined measure under the declared preparation, the receiver-hit entropy is not licensed for that packet.

For windows with many retained roots, define the causal-wake provenance graph $G_{\mathrm{prov}}(W)$: vertices are retained causal roots in W, and two vertices are joined when their roots trace to a common emitter worldline segment in the compatible complete histories. This graph is the common native carrier for three entropy uses below: its connectedness supplies history-backed concordance, its edge cuts supply access-cut entropy, and its boundary-crossing edges supply the wake-escapement contribution to the arrow-of-time ledger.

Minimum Specification

Every entropy statement in $\mathbb{AAA}$ should declare five ingredients before the number is treated as physical. First, it should name the preparation and measure μ_t on compatible deterministic histories. Second, it should name the access window $W(t)$ and retained record carrier: apparatus state, boundary wake data, Noether sea state, Physical Observer record, or simulation packet. Third, it should name the coarse-graining $\mathcal{Q}$ and the projection $\Pi_{\mathcal{Q},W}$. Fourth, it should state the comparison job: work availability, heat flow, coding, measurement locking, horizon label counting, cosmology, or another defined use. Fifth, for open windows, it should include boundary flux and record-change residuals rather than silently treating the window as isolated.

This checklist is not extra ontology. It is the minimum context needed for an entropy claim to say something definite. Without these ingredients, a phrase such as "the entropy increased," "the system is maximally entropic," or "information was lost" has not yet specified which alternatives were unresolved, which record retained them, or which comparison class made the claim meaningful.

Work Availability And Energy Spread

Traditional thermodynamics often introduces entropy as a measure of energy spread. In this chapter that is the work-availability face of record coarse-graining. A hot reservoir, chemical store, coherent photon-channel stream, or gravitational potential gradient is low entropy only relative to a work channel and comparison record that can use the concentration. After the same energy is distributed among many thermal, boundary, or wake-history microrecords, the total energy ledger may still close, but the retained record supports less extractable work.

For a fixed reference bath or readout channel with temperature T_R declared by the same record, define the availability diagnostic

$$A^{(T_R)}_{\mathcal{Q},W}(t) = E_{\mathcal{Q},W}(t) - T_R S_{\mathcal{Q},W}(t)$$

This is a diagnostic, not a new substrate property. In a closed isothermal comparison, the useful work extractable from the retained packet is bounded by the decrease of this same-record availability,

$$W_{\text{useful}} \le A^{(T_R)}_{\mathcal{Q},W}(t_i) - A^{(T_R)}_{\mathcal{Q},W}(t_f)$$

up to declared control and boundary residuals. A packet that conserves $E_{\mathcal{Q},W}$ while increasing $S_{\mathcal{Q},W}$ has not lost energy. It has lost retained work availability in that comparison channel.

For resource-theory uses, the maximum work must also be indexed by the allowed apparatus control and readout class. Let $\mathcal{C}^{\text{ctrl}}_W$ denote the declared controls, measurements, feedback operations, and reset operations available in the window, and let R_f denote the required final record. Then the same physical packet supports the diagnostic

$$W_{\max}\big(\theta_W; \mathcal{C}^{\text{ctrl}}_W, R_f\big) = \sup_{\alpha \in \mathcal{C}^{\text{ctrl}}_W} \Delta E_{\text{weight}}[\alpha : \theta_W \to R_f]$$

The value can change when the apparatus record contains which pressure side, molecule class, isotope channel, or other controllable distinction is present. That is not a psychological addition to physics. It is a different physical record and a different control/readout channel. In $\mathbb{AAA}$, an entropy that claims to measure available work must therefore declare θ_W, $\mathcal{C}^{\text{ctrl}}_W$, and R_f together.

This also disciplines heat-death language. A claim that a universe window has no usable work left is not a bare statement about the complete microstate; it is a statement about a declared class of controls, readouts, reservoirs, and final records. For a control family $\mathcal{C}^{\text{ctrl}}$ over admissible windows, the remaining work-availability envelope can be written schematically as

$$\mathcal{A}_{\text{use}}(t; \mathcal{C}^{\text{ctrl}}) = \sup_{W, R_f} W_{\max}\big(\theta_W(t); \mathcal{C}^{\text{ctrl}}_W, R_f\big)$$

For $\mathbb{AAA}$ this envelope must distinguish exposed gradients from shielded internal assembly energy. If the declared control class includes operations that can change shielding, write schematically

$$\mathcal{A}_{\text{use}} = \mathcal{A}_{\text{exposed}} + \mathcal{A}_{\text{deshield}}, \qquad \mathcal{A}_{\text{deshield}} = \sup_{\alpha \in \mathcal{C}^{\text{ctrl}}_{\text{shield}}} \sum_A (1 - \zeta_\alpha(A))\, E_{\text{internal}}(A)$$

Here $\mathcal{C}^{\text{ctrl}}_{\text{shield}}$ is the possibly empty class of operations that can lower an assembly's shielding factor in the declared window. If topological assembly protection forbids such operations, $\mathcal{A}_{\text{deshield}}$ is not available to the control family. If shielding is reversible or partially controllable, heat-death language is stronger than exposed-gradient exhaustion and must include the shielded-reservoir term.

A heat-death statement for that control family means $\mathcal{A}_{\text{use}}$ tends to zero or below the declared operational threshold. It does not prove that every possible future record system, assembly class, or Noether sea access channel has no usable distinction. It proves only the exhaustion of usable gradients and accessible shielded reservoirs for the stated comparison class.

For open windows, the same point must be stated with boundary records. A planetary, biological, or engineered window may receive and emit nearly equal total energy while still being driven by low-entropy input. The relevant record distinguishes incoming concentrated photon-channel packets, chemical gradients, or potential-gradient data from outgoing lower-frequency radiation, heat, and boundary-wake history:

$$\mathcal{B}_{\partial W}^{\text{therm}} = \left(\dot{E}_{\text{in}}, \rho_{\text{in}}(\nu,\Omega), \dot{E}_{\text{out}}, \rho_{\text{out}}(\nu,\Omega), \mathcal{B}_{\text{wake}}\right)$$

where ρ_{in} and ρ_{out} are retained spectral/angular records, not new ontological fluids. The entropy claim is physical only when this boundary record is the same one used for energy flux, internal work, heat, and observer readout.

Complexity And Driven Intermediate Windows

Entropy and complexity answer different questions. Entropy compares how many compatible histories remain unresolved after a coarse-graining. Complexity asks whether the path between low-entropy and high-entropy records passes through organized intermediate structures. Low entropy can be simple, high entropy can be simple, and the interesting dynamics often occur in a driven window between them.

For a locally organized window W, the same-record statement is not that organization defeats the second law. It is that internal record maintenance is paid for by boundary exchange:

$$\Delta S_{\mathcal{Q},W}^{\text{inside}} + \Delta S_{\mathcal{Q},\partial W+\text{env}}^{\text{export}} \geq 0, \qquad \Delta S_{\mathcal{Q},W}^{\text{inside}} \leq 0$$

where both terms are computed from the same access window, coarse-graining, and boundary record. The first term may describe the maintained organization of a cell, reaction network, engineered refrigerator, or other open subsystem. The second term records the exported heat, lower-grade radiation, reaction byproducts, wake-boundary history, and environmental disorder that make the local organization possible.

Origin-of-life and metabolism-first arguments are useful comparison pressure at this level. They do not show that entropy creates life, and they do not add biological ontology to $\mathbb{AAA}$. They say that a plausible prebiotic reaction window must name usable gradients, compartment-like retention, reaction throughput, and entropy export. In native terms, that becomes a finite-window reaction-ledger problem: the source record must show how low-entropy chemical, photon-channel, geothermal, or potential-gradient input is converted into persistent organized records while the boundary ledger exports a larger entropy burden.

For assembly formation, the driven intermediate window can be made into an order parameter rather than only a qualitative contrast. Split the retained wake record in W into a coherent phase-locked part and an incoherent exported or background part under the same coarse-graining $\mathcal{Q}$. Define

$$\mathcal{C}_W = \frac{S_{\mathcal{Q},W}^{\text{incoh}}}{S_{\mathcal{Q},W}^{\max}} \left(1 - \frac{S_{\mathcal{Q},W}^{\text{coh}}}{S_{\mathcal{Q},W}^{\max}}\right)$$

The quantity peaks when a sharply organized coherent core coexists with substantial exported or surrounding incoherent entropy. Stable assemblies are therefore candidate local maxima or ridges of $\mathcal{C}_W$ under the second-law export constraint above. The collinear-breather and nested shell braid programs can test this directly by asking whether phase-locked trajectory bundles sit on such ridges while the surrounding Noether sea and wake-boundary ledger pay the entropy cost.

Mapping In From Standard Entropies

Legacy entropy formulas survive as effective projections with different prerequisites.

Clausius entropy, $dS = \delta Q_{\text{rev}}/T$, is licensed only in a regime where the reversible comparison class, heat channel, and temperature channel are defined by the same physical record. Without that record, the formula is a comparison mnemonic rather than a substrate claim.

The Clausius definition also has a direction of dependence that must not be reversed. A cycle statement can be made without entropy:

$$\oint \frac{\delta Q}{T} \leq 0, \qquad \oint_{\text{rev}} \frac{\delta Q_{\text{rev}}}{T} = 0$$

for the declared heat reservoirs, temperature scale, and reversible comparison class. Only after that integrability condition is available does the entropy difference

$$\Delta S_{\text{Cl}} = \int_A^B \frac{\delta Q_{\text{rev}}}{T}$$

become path-independent. In this chapter, a claim that “entropy broke the second law” must therefore specify which entropy is being used. If the Clausius integrability condition fails, the thermodynamic entropy used in that comparison was not well-defined in the first place.

The framework also predicts where this integrability fails. Let the Noether sea retuning lag on a thermodynamic cycle be

$$\Lambda_{\text{sea}}(W) = \frac{\tau_{\text{retune}}(\theta_{\text{sea}})}{\tau_{\text{cycle}}}$$

where τ_{retune} is the relaxation time for the Noether sea response variables retained by the packet and τ_{cycle} is the duration of the reversible-comparison cycle. Clausius entropy is expected to be path-independent only in the regime $\Lambda_{\text{sea}} \ll 1$. When $\Lambda_{\text{sea}} \gtrsim 1$, the sea carries cycle-scale hysteresis, the heat channel is history-dependent, and $\oint \delta Q_{\text{rev}}/T$ is not a well-defined state function for that record.

Boltzmann entropy, $S = k_B \log \Omega$, maps to the count or measure of complete architrino and assembly histories compatible with the retained macrostate. The textbook counting form is the uniform-weight special case of Gibbs/Shannon entropy:

$$p_\alpha = \frac{1}{\Omega} \quad \Longrightarrow \quad -k_B \sum_{\alpha=1}^{\Omega} p_\alpha \log p_\alpha = k_B \log \Omega$$

Thus the count is not licensed by cardinality alone. It also assumes the measure that gives the compatible microstates equal weight, usually through an isolated equilibrium comparison or another declared physical preparation. The macrostate partition is part of the claim. Changing the partition or the measure changes the entropy statement. A singleton partition over exact complete histories would assign zero Boltzmann entropy to every cell, but it would also erase the thermodynamic question. Useful Boltzmann entropy requires retained macrostates tied to measurable, controllable, or dynamically stable distinctions.

Elementary thermal examples often count energy-quanta arrangements: one macrostate may specify only how much energy lies in each body, while many bond-level or molecule-level allocations remain unresolved. The $\mathbb{AAA}$ replacement is the same mathematical role with a deeper state space: count complete deterministic histories compatible with the retained energy, wake, boundary, apparatus, and Noether sea records.

Gibbs and Shannon entropies map to pushed-forward measures over unresolved alternatives. They are useful for apparatus states, basin weights, branch records, and coding descriptions, but they become thermodynamic only when the apparatus, environment, boundary exchange, and work or heat ledger are physical parts of the same packet. Gibbs entropy is the natural comparison when the retained measure encodes uncertainty over alternatives that change available work under a declared control class; Boltzmann entropy is tied to the retained macrostate partition itself. Both are valid only with their intended job stated.

At deterministic-multistability points, the same measure gives the effective branch weights a record-limited observer must assign. If the unresolved preparation fiber is Γ_{prep} and the deterministic basins $\{B_k\}$ partition the post-event branch outcomes, define

$$w_k = \frac{\mu_t\big(\mathcal{F}^{-1}_{t\to t_+}(B_k) \cap \Gamma_{\text{prep}}\big)}{\mu_t(\Gamma_{\text{prep}})}$$

and the effective outcome entropy is $-k_B \sum_k w_k \log w_k$. Entropy does not select which branch the actual complete microstate takes. The measure μ_t predicts the branch weights that any record-limited observer must assign before the missing path-history distinctions are recovered. This is the direct entropy handoff to Born-rule closure.

Von Neumann and entanglement entropies map to a declared quantum comparison record, factorization, and access cut. For a retained sector A and unresolved complement $\bar{A}$, the standard reduced record is

$$\rho_A(\theta) = \mathrm{Tr}_{\bar{A}} \rho_{A\bar{A}}(\theta)$$

with entropy

$$S_A(\theta) = -k_B \, \mathrm{Tr}(\rho_A(\theta) \log \rho_A(\theta))$$

Even when the full comparison state is pure, reversible, or measure-preserving, S_A can be nonzero because correlations with $\bar{A}$ have been excluded from the retained record. In $\mathbb{AAA}$ this is an access-cut entropy: the same mathematical role must be recovered as coarse-graining over unresolved path-history, apparatus, boundary-wake, and Noether sea correlations that cross the declared cut.

The native carrier is the provenance graph across the access cut. For a cut Σ separating retained sector A from complement $\bar{A}$, build

$$G_{\text{prov}}(\Sigma) = (V_A \sqcup V_{\bar{A}}, E_\Sigma)$$

where vertices are retained roots on the two sides and an edge records that two roots share a compatible emitter worldline segment. The record entropy across the cut is governed by the number of emitter-history assignments compatible with the same boundary hit record,

$$S^{\text{rec}}_\Sigma \sim k_B \log \left|\mathrm{Assign}\left(G_{\text{prov}}(\Sigma), \mathcal{B}_\Sigma\right)\right|$$

The global record can remain pure or closed because G_{prov} is connected in the complete history, while the retained subregion is mixed because the edge cut has hidden the complementary provenance.

For a coding record with source distribution $P = \{p_i\}$, the Shannon entropy in bits is

$$H_2(P) = -\sum_i p_i \log_2 p_i$$

This has a precise compression interpretation: for any prefix-free code $\mathcal{C}$ that encodes symbols drawn from P, the average code length obeys

$$\bar{L}_{\mathcal{C}}(P) \geq H_2(P)$$

with block codes able to approach the bound under the usual coding assumptions. In $\mathbb{AAA}$, this is not free-floating information. It is an entropy of a declared symbol record, model class, and decoding channel.

Cross-entropy makes the model dependence explicit. If an encoding or prediction model uses $Q = \{q_i\}$ while the retained source record is distributed as P, the expected code length is

$$H_2(P, Q) = -\sum_i p_i \log_2 q_i$$

The excess over $H_2(P)$ measures model mismatch, not a new substrate ingredient. This is why next-symbol prediction and compression can be equivalent for a language model or other coding apparatus while remaining an observer-level modeling statement. The entropy becomes physical only after the symbol carrier, probability source, encoder, decoder, training or update channel, and device/boundary cost are part of the same record.

Record entropy maps to durable alternatives in an apparatus or observer channel. A record is not merely a symbolic label. It is an assembly/environment state that persists long enough to be read, copied, or reset within a declared window.

Horizon entropy maps to observer-accessible boundary or horizon-interface label capacity. It is not a literal statement that the Euclidean void is made of area bits. The label count must be derived from strong-field Noether sea and nested shell braid records.

Computation entropy maps to implemented device cost. Bit logic alone does not create a thermodynamic cost. A cost claim is physical only after the device state space, success criterion, reset operation, heat/work ledger, and boundary exchange have been declared.

Mapping Out To Effective Physics

The outward map from $\mathbb{AAA}$ to effective entropy has five steps.

First, choose the physical window W and the record carrier: apparatus, boundary wake data, Noether sea state, simulation domain, or Physical Observer record. When work extraction is in view, also choose the allowed control/readout class. Second, choose the coarse-graining $\mathcal{Q}$ that defines which complete histories count as the same retained state. Third, push the complete-history measure forward through $\Pi_{\mathcal{Q},W}$. Fourth, compute the entropy functional on the retained measure. Fifth, compare that result to the relevant effective law only with the same record still in force.

For open or cosmological windows, entropy bookkeeping must expose production, boundary flux, and record-change residuals:

$$\frac{dS_{\mathcal{Q},W}}{dt} = \sigma_W(t) - \int_{\partial W(t)} (\mathbf{J}_S - s_{\mathcal{Q}} \mathbf{v}_{\partial W}) \cdot \hat{\mathbf{n}}\, dA + \mathcal{R}_{\mathcal{Q}}(t)$$

Here σ_W is local production inside the retained window, $\mathbf{J}_S$ is entropy flux through the boundary, $s_{\mathcal{Q}}$ is the retained entropy density, $\mathbf{v}_{\partial W}$ is the velocity of a moving window boundary, and $\mathcal{R}_{\mathcal{Q}}$ records changes in the coarse-graining or retained record set. For a fixed window, $\mathbf{v}_{\partial W} = 0$ and the expression reduces to the ordinary boundary-flux form. A monotone entropy statement is therefore conditional:

$$\frac{dS_{\mathcal{Q},W}}{dt} \geq 0 \quad \Longleftrightarrow \quad \sigma_W(t) + \mathcal{R}_{\mathcal{Q}}(t) \geq \int_{\partial W(t)} (\mathbf{J}_S - s_{\mathcal{Q}} \mathbf{v}_{\partial W}) \cdot \hat{\mathbf{n}}\, dA$$

for the declared record. The phrase "entropy of the universe" is not a complete claim unless it supplies the measure, window, boundary, and residual terms.

The entropy-arrow theorem target ties this boundary term to wake escapement. Let $\mathcal{E}_{\text{esc}}(W)$ be the wake-escapement set defined in Energy, and let $\Sigma_{\text{esc}}(\mathcal{E}_{\text{esc}}(W), t)$ be the rate at which retained path-history distinctions leave W on causal wakes that no longer hit a retained receiver. The structural target is

$$\frac{d}{dt} S_{\Pi,W}(t) = k_B\, \sigma_W^{\text{int}}(t) + k_B\, \Sigma_{\text{esc}}\left(\mathcal{E}_{\text{esc}}(W), t\right) + \mathcal{R}_{\Pi,W}(t)$$

on a fixed coarse-graining and boundary convention. In words: observer-window entropy production is bounded below by the retained-history distinctions lost to escaping wakes, up to declared interior production and projection residuals. The thermodynamic arrow is therefore a theorem target about the same causal-wake boundary ledger used by finite-window energy bookkeeping, not a second primitive arrow.

Second Law And Same-Record Monotonicity

The traditional second law has several equivalent-looking forms only after the comparison class has been fixed. Clausius uses a cycle or reversible-comparison statement, Kelvin-Planck forbids a cyclic device from converting heat from one reservoir wholly into work, Boltzmann says overwhelmingly many compatible microstates lie in larger macrostates, and Maxwell-demon analyses require memory and reset costs to be included. These are not four independent substances called entropy. They are four projections of the same discipline: the complete thermodynamic packet must not shrink the retained compatible-history record for free.

The traditional slogan that entropy increases is therefore a shorthand. The safer statement is that, for an admissible isolated comparison with fixed record class and no hidden boundary or apparatus reset, the retained entropy must not decrease beyond the allowed finite-window fluctuation. It can remain constant in an ideal reversible comparison, and it can be exactly zero for a singleton exact-history partition that has stopped asking a thermodynamic question. Irreversibility enters when the retained macrostate loses access to distinctions that the complete deterministic history still contains.

In $\mathbb{AAA}$, the second law is therefore not the source of absolute time and not a primitive command that a substance called entropy must always rise. It is a finite-window typicality and bookkeeping claim over a declared record. For a fixed window, coarse-graining, boundary record, and apparatus/control class, the same-record second-law diagnostic is

$$\Delta S_{\mathcal{Q},\mathrm{tot}}(\theta_W;t_i,t_f)=\Delta S_{\mathcal{Q},W}+\Delta S_{\mathcal{Q},\partial W+\mathrm{env}}+\int_{t_i}^{t_f}\mathcal{R}_{\mathcal{Q}}(t)\,dt\geq-\epsilon_{\mathrm{fluc}}(W,\mathcal{Q},t_f-t_i)$$

Here $\Delta S_{\mathcal{Q},W}$ is the retained entropy change inside the window, $\Delta S_{\mathcal{Q},\partial W+\mathrm{env}}$ is the boundary and environmental entropy change assigned by the same record, $\mathcal{R}_{\mathcal{Q}}$ records changes in the retained coarse-graining or record set, and ϵ_{fluc} allows finite-window statistical fluctuations. In the macroscopic thermodynamic regime, ϵ_{fluc} is negligible for ordinary comparisons. In microscopic or short-time windows it is not.

This formula explains how the familiar readings fit together. For an isolated macroscopic packet with fixed coarse-graining and no boundary term, it reduces to the usual effective statement $\Delta S\gtrsim 0$. For a refrigerator, cell, planet, or reaction network, $\Delta S_{\mathcal{Q},W}$ may be negative while the boundary and environment term is larger and positive. For an ideal reversible comparison, the inequality is saturated. For an irreversible comparison, the residual is positive. For a Maxwell-demon packet, the memory, actuator, partition, target, and reset channel must all be included in the same θ_W, or the apparent violation is a split-record error.

Record-circularity pressure lands exactly here. The second law does not by itself prove that a present record descends from a low-entropy past; it uses a low-defect boundary condition and ordinary history-backed records to make the second-law inference trustworthy. In this chapter that burden is not hidden. The path-history measure, boundary-condition prior, and observer record all belong in θ_W, and the Boltzmann-brain residual $\mathcal{R}_{\mathrm{BB}}(\theta)$ below is the extreme test of whether isolated observer-fluctuation records have been suppressed relative to shared history-backed records.

What survives from the traditional interpretation is strong: thermodynamics is not being rejected. Heat engines, irreversible mixing, work availability, macrostate dominance, memory reset, decoherence, and horizon bookkeeping remain real effective constraints. What changes is the level assignment. The second law is a theorem target about projected deterministic histories and declared records, not an ontological rival to the substrate dynamics.

Same-Record Closure Rule

A thermodynamic packet has one declared record. In a local horizon, measurement, computation, or near-equilibrium simulation, write that record schematically as

$$\theta_W=\left(\mathcal{H}^W,\mathcal{B}^{(O)}(W),\mathcal{N}_{\mathrm{sea}}|_W,O_W,\Pi_{\mathrm{eff}},\mu_\theta\right)$$

where $\mathcal{H}^W$ is retained path-history data, $\mathcal{B}^{(O)}(W)$ is observer-accessible boundary or apparatus record data, $\mathcal{N}_{\mathrm{sea}}|_W$ is the resolved Noether sea state on the window, O_W is the observer clock/ruler/readout state when an observer is part of the comparison, Π_{eff} is the effective projection, and μ_θ is the conditional measure over unresolved deterministic histories.

The admissible comparison has the form

$$(S,\,T,\,dQ,\,\Delta E,\,\{p_i\},\,\mathcal{B}_{\mathrm{rec}})=\mathcal{P}_{\mathcal{Q},W}(\theta_W)$$

where all listed quantities are projections of the same θ_W. If entropy is computed from one θ_W, temperature from another, Born-style basin weights from a third, and flux from a fourth, the packet has not derived a closure. It has fitted separate descriptions.

This rule is why entropy appears as a discipline across many chapters. It protects the Born-rule program from using one ensemble for outcome weights and another for apparatus thermodynamics. It protects horizon thermodynamics from assigning independent entropy, temperature, and stress records. It protects computation-cost claims from treating logical form as a free physical process.

Entropy And Absolute Time

Absolute time is the ordering parameter of the substrate law. Entropy does not create it. The causal arrow enters the dynamics through delayed causal wakes: only emissions from $t_0<t$ can contribute to a receiver at t. Thermodynamic, biological, measurement, and cosmological arrows are finite-window consequences of dynamics, boundary conditions, and retained records.

Even if the complete deterministic dynamics preserve the underlying measure, the observer-window entropy $S_{\Pi,W}$ can increase when $\Pi_{\mathcal{Q},W}$ discards path-history, boundary-wake, or apparatus-record information. That increase is a projection effect inside the declared record. It is not evidence that time itself is generated by entropy.

The arrow-of-time closure problem is therefore sharper than a generic second-law slogan. A mature account must explain why the admissible early record is low-defect or low-entropy in the relevant coarse-graining, and why later macroscopic reversal would require reconstruction of path-history and wake-phase detail no finite observer or apparatus can retain.

The past-hypothesis comparison also depends on the partition. A nearly uniform early matter record can look high entropy under a non-gravitating gas coarse-graining and low entropy under a gravitational coarse-graining, because gravitational clumping, potential-energy release, and horizon labels open far larger compatible records later. In $\mathbb{AAA}$ the lesson is not that gravity is entropy. It is that a cosmological entropy statement must name whether its macrostate includes Noether sea state, potential gradients, causal-wake boundary data, and horizon-interface records.

Boltzmann-brain pressure exposes the same rule in extreme form. A retained observer record cannot certify the low-entropy history that is then used to certify the retained observer record. The packet must separate the internal consistency of a memory record from the boundary-condition claim that the record descends from a shared low-defect universe path history. Let Γ_{hist} denote compatible complete histories in which observer records, cosmological traces, and low-defect boundary data descend from one shared path-history record. Let Γ_{BB} denote compatible complete histories in which an observer record is an isolated high-entropy fluctuation with no shared supporting cosmological record. The corresponding fluctuation residual is

$$\mathcal{R}_{\text{BB}}(\theta) = \frac{\mu_\theta(\Gamma_{\text{BB}})}{\mu_\theta(\Gamma_{\text{hist}})}$$

for the same declared measure, coarse-graining, and access window. A mature same-record entropy cosmology requires $\mathcal{R}_{\text{BB}}(\theta) \ll 1$ or an explicit reason why the comparison class is not admissible. Otherwise the theory has only renamed the circularity: records infer a low-entropy past, while the assumed low-entropy past is what made the records trustworthy.

The delayed dynamics supply a sharper discriminator than the bare measure ratio. For a candidate observer record O_W, define the wake-concordance order parameter

$$\mathcal{K}(O_W) = \frac{\#\left\{\text{roots at } O_W \text{ sharing an emitter worldline with roots at neighboring receivers}\right\}}{\#\left\{\text{incoming roots at } O_W\right\}}$$

with the denominator restricted to the retained incoming roots in the declared window. History-backed records are expected to have $\mathcal{K} \to 1$ because the same matter and Noether sea emitters illuminate a neighborhood with correlated causal timing. Isolated fluctuation records have $\mathcal{K} \to 0$ unless they also fabricate shared-emitter concordance across neighboring receivers. Thus low-$\mathcal{K}$ configurations are dynamically suppressed by provenance mismatch, and high-$\mathcal{K}$ fluctuation records are costly because they require coherent emitter-history coincidences, not only a memory snapshot.

Measurement And Computation

Measurement records require entropy locking. For a declared apparatus/environment channel,

$$\Delta S^{\text{app+env}}_{\mathcal{Q},W} = S^{\text{app+env}}_{\mathcal{Q},W}(t_0 + T_{\text{rec}}) - S^{\text{app+env}}_{\mathcal{Q},W}(t_0)$$

is the entropy change associated with the candidate record. A strong record candidate satisfies

$$\Delta S^{\text{app+env}}_{\mathcal{Q},W} \geq S_{\text{lock}} > 0$$

with S_{lock} fixed by the apparatus class and readout channel. This is not a collapse law. It is the requirement that the branch has exported enough unresolved apparatus/environment history that coherent reversal is no longer part of the retained measurement window.

Resetting a memory-bearing apparatus with N distinguishable retained record classes also requires a physical entropy ledger:

$$\Delta S^{\text{app+env}}_{\mathcal{Q},W} \geq k_B \log N - k_B \varepsilon_\mu, \qquad N \geq 2$$

For a non-uniform retained distribution, $k_B \log N$ is replaced by $-k_B \sum_i p_i \log p_i$.

A Maxwell-demon packet therefore has two admissible readings. If the demon does not reset, it spends a low-entropy blank-memory record as a resource and converts that resource into a pressure, temperature, or sorting record. If the demon is required to act cyclically, the memory, actuator, partition, target system, and boundary environment must return to the same physical record. A cyclic packet that claims to sort a broad complete-history region into a narrower one while preserving the same boundary and memory record is not a thermodynamic miracle. It is a failed same-record closure.

The same logic applies to computation. For an implemented step s with completion probability p_s, a lower-bound claim must attach to the device and boundary records:

$$\Delta S_{\text{env},s} + \Delta S_{\text{target},s} + \Delta S_{\text{boundary},s} \geq k_B \log(1/p_s) - \epsilon_s$$

The inequality is not a new law of symbols. It states the burden that the same physical record defining success must also supply the entropy, heat, work, and boundary terms used to claim a cost.

Horizons And Emergent Gravity

Horizon entropy is the most stringent test of this mapping because it connects record counting, effective geometry, energy flux, and unitarity pressure. The useful comparison target is not that gravity is “really entropy.” The target is that one strong-field Noether sea record supplies the observer-level entropy, temperature, flux, and metric response together.

For an observer-accessible local horizon patch $\partial\Omega$, the boundary-label entropy target is

$$S^{(O)}_{\partial\Omega}(\theta; W) = k_B \log \left|\mathcal{B}^{(O)}_{\partial\Omega}(\theta; W)\right|$$

with $\mathcal{B}^{(O)}_{\partial\Omega}$ defined by retained boundary-wake labels readable by the same Physical Observer record. The local Clausius comparison becomes a residual or variation target:

$$\delta_\ell Q^{(O)}_{\partial\Omega} = T^{(O)}_U \delta_\ell S^{(O)}_{\partial\Omega} + \mathcal{O}(k_B T^{(O)}_U \epsilon_{\mathrm{local}})$$

This is a recovery target, not a postulate. The proof fails if $S^{(O)}_{\partial\Omega}$, $T^{(O)}_U$, $dQ^{(O)}_{\partial\Omega}$, and the effective metric are assigned independent records.

For black holes, the area-law coefficient must come from terminal nested shell braid alignment and horizon-interface label compatibility. For a connected block U of alignment-area patches,

$$s_{\mathrm{align}}(\theta) = \lim_{|U|\to\infty} \frac{1}{|U|} \log |\mathcal{L}_U(\theta)|$$

when the limit exists after boundary corrections. The required coefficient is area-normalized:

$$\frac{s_{\mathrm{align}}(\theta)}{a_\theta} \longrightarrow \frac{1}{4}$$

This target avoids a false one-patch interpretation. The coefficient is a block entropy density and patch-area normalization, not a literal independent count on one microscopic patch.

The label set is not arbitrary. At terminal alignment the tri-binary collapses its orbital-plane normals onto one interface axis, so the surviving discrete labels are the handedness assignment and the causal-root ledger index still carried by the aligned branch. In a block U,

$$|\mathcal{L}_U(\theta)| = \prod_{u\in U} \#\left\{(\chi_u, N_{s,u}, M_{p,u}) : \text{admissible at patch } u\right\}$$

where χ_u is the retained terminal-alignment handedness label and $(N_{s,u}, M_{p,u})$ is the local self-hit and partner-hit root-ledger index. The $1/4$ coefficient is therefore a falsifiable statement about the per-patch admissible ledger multiplicity and the patch area a_θ in the accepted alignment units, not a coefficient to fit after the fact.

Page-curve, island, replica-wormhole, Ryu-Takayanagi, and AdS/CFT calculations remain high-value comparison mathematics. They sharpen the required entropy and unitarity bookkeeping. They do not provide the $\mathbb{AAA}$ mechanism unless their constraints are recovered from horizon-interface labels, path-history bookkeeping, Noether sea storage, and release-channel selection.

Entanglement-entropy calculations sharpen the access-cut side of that same problem. A horizon partition can make an outside retained record mixed while the full comparison record remains closed. The native target is therefore not to import an abstract inaccessible complement as ontology, but to derive which horizon-interface labels, Noether sea storage modes, and release-channel selections play the role of the traced-out complement.

Failure Modes

The most common failure mode is an unqualified entropy statement. An entropy claim is incomplete when it omits the measure, coarse-graining, access window, retained record, or boundary terms.

Another failure mode is disembodied information. Shannon uncertainty over symbols is not automatically heat, work, Clausius entropy, or physical record cost. A compression or prediction score is a statement about a declared symbol model and record channel. It becomes thermodynamic only when implemented through an apparatus and boundary ledger.

A third failure mode is entropic gravity as a substitute for the mass mechanism. Thermodynamic or entropic derivations of force are comparison benchmarks, but inertial mass remains open until the assembly ledger supplies closed internal causal history, shielding extraction, Noether sea response, and acceleration response.

A fourth failure mode is fitted horizon bookkeeping. If a black-hole or local-horizon packet uses one record for entropy, another for temperature, another for stress, and another for release channels, the apparent agreement is not a native closure.

The fifth failure mode is promoting entropy into time. Entropy can diagnose an emergent arrow inside a stated physical and inferential window. It does not supply the absolute ordering parameter t.

A sixth failure mode is confusing entropy with complexity. Low entropy can be simple, maximum entropy can be simple, and complex organized structures normally belong to driven intermediate windows that export more entropy than they locally suppress. For $\mathbb{AAA}$, biological or self-organizing examples are open-window bookkeeping, not exceptions to deterministic dynamics.

A seventh failure mode is record circularity. If a packet uses retained records to infer a low-entropy past while also using that inferred low-entropy past to justify the reliability of the retained records, it has not closed the arrow-of-time problem. It must expose the same-record path-history measure and boundary-condition prior that suppress isolated observer-fluctuation records relative to history-backed observer records.

An eighth failure mode is treating "entropy never decreases" as a primitive second law. Clausius entropy depends on a reversible-cycle integrability condition, statistical entropy can fluctuate in small or finite windows, and resource entropy depends on the declared control/readout class. A packet must state which second-law form it is invoking before monotonicity has content.

A ninth failure mode is quoting entanglement entropy without declaring the factorization and access cut. A subsystem entropy is not automatically entropy of the whole universe. It is a statement about what remains after a complement has been excluded from the retained record.

A tenth failure mode is promoting present human or laboratory macrostates into the final state-space partition of the universe. Heat-death claims, order claims, and "maximum entropy" claims must declare the manipulation class and record system for which usable gradients are exhausted. Otherwise they have converted a useful thermodynamic extrapolation into an unsupported ontology of all future access.

Interfaces

The energy-side residuals are stated in Energy. The time-side arrow distinction is stated in Absolute Time. Measurement locking is stated in Measurement Ontology. Computation cost is treated in Information / Computation. Local-horizon recovery is stated in Emergent Metric, with the simulation-facing scaffold in Thermodynamic Residual. The strong-field horizon target is stated in Black Holes, and the dynamics-side label-count target is stated in Nested Shell Braid Dynamics.

The consolidated rule is simple: entropy is accepted only as a declared projection of retained deterministic histories, and every effective entropy claim must name the record that makes the projection physical.

Binary Dynamics

This chapter develops two-body architrino dynamics from the appearance of self-hit to candidate stable binaries and their conditional role as measurement standards. It then formalizes the maximum-curvature attractor analysis and closes with the state-space and conservation-law foundations needed for well-posed dynamics. **Status:** (1) self-hit makes the dynamics non-Markovian (path-history dependent), and (2) stability/attractor claims are conjectural unless explicitly established.

It is the foundational precursor to Nested Shell Braid Dynamics, Dyadic Resonance Lock, Master Equation, and the assembly-level Noether Braid.

This chapter is the canonical home for two-body wake regimes, partner-hit versus self-hit behavior, circular anti-damping, non-circular spiral hypotheses, and maximum-curvature binary analysis. The primitive-entity ontology in Architrino should point here once the discussion becomes a behavioral regime or assembly-stability mechanism.

The Spiral Orbiting Binary and the Contraction Phase

An orbiting binary is the simplest emergent assembly, consisting of two architrinos of opposite polarity: an Electrino and a Positrino. With polarities $-\epsilon$ and $+\epsilon$, the assembly is electrically neutral overall. This system is the first teaching case for delayed causal wakes, partner-hit contraction, and the self-hit onset boundary.

Consider the ideal case of a symmetric orbit in a universe with no other architrinos. In general, each architrino is subject to a superposition of external causal wake contributions from all other sources; the analysis below isolates the binary by setting those external contributions to zero.

Let the Electrino be architrino 1 and the Positrino be architrino 2.

- **Positions:** $\mathbf{s}_1(t)$ and $\mathbf{s}_2(t)$
- **Polarities:** $q_1 = -\epsilon$ and $q_2 = +\epsilon$

The motion of each architrino is determined by the wake emitted by the other at a delayed time. The acceleration of the Electrino (architrino 1) at time t is caused by the Positrino's (architrino 2) wake emitted at an emission time t_0. This is governed by the interaction condition:

$$\|\mathbf{s}_1(t) - \mathbf{s}_2(t_0)\| = c_f(t - t_0)$$

The acceleration vector for the Electrino is attractive, pointing towards the Positrino's delayed position:

$$\mathbf{a}_1(t) \propto -\hat{\mathbf{r}}_{21} = -\frac{\mathbf{s}_1(t) - \mathbf{s}_2(t_0)}{\|\mathbf{s}_1(t) - \mathbf{s}_2(t_0)\|}$$

A symmetric set of equations governs the Positrino's motion based on the Electrino's emissions.

In the strictly sub-field-speed regime (no self-interaction, $\|\mathbf{v}\| \le c_f$), a stable, circular orbit is impossible. Because the attractive force on each architrino points to the *past* position of its partner, it is not a true central force. The principal circular branch proves a sharper fact: the partner line of action has a forward tangential projection, so the partner-only near-circular ledger is anti-damped rather than a contraction proof. A logarithmic inward spiral can still be used as a separate non-circular ansatz or capture target, but its radial tightening must be certified by solving that branch chart; it is not implied by the principal circular sign.

Standard central-force mechanics conserves angular momentum because the force at time t is collinear with the equal-time separation vector. The partner-hit branch does not have that geometry. Define the equal-time separation and delayed line of action by

$$\mathbf{r}_{12}^{\mathrm{eq}}(t) \equiv \mathbf{s}_1(t) - \mathbf{s}_2(t), \qquad \hat{\mathbf{r}}_{12}(t; t_0) = \frac{\mathbf{s}_1(t) - \mathbf{s}_2(t_0)}{\|\mathbf{s}_1(t) - \mathbf{s}_2(t_0)\|}$$

The delayed partner branch carries the angular-momentum-change direction

$$\mathbf{r}_{12}^{\mathrm{eq}}(t) \times \hat{\mathbf{r}}_{12}(t; t_0)$$

which is generically nonzero because $\mathbf{s}_2(t_0)$ is not the partner's equal-time position. Therefore the usual angular-momentum barrier and the instantaneous effective potential

$$V_{\mathrm{eff}}(r) = V(r) + \frac{ml^2}{2r^2}$$

cannot be imported as the binary's governing reduction. A conserved angular-momentum-like quantity, if present, must include the causal-wake history term that balances the delayed torque.

Lemma (No stable circular orbit for $\|\mathbf{v}\| < c_f$). In units with $c_f = 1$, the circular speed is $s = R\omega$. In the partner-only regime, the per-hit tangential component satisfies

$$T_p \propto \frac{\sin(\delta_p/2)}{\cos^2(\delta_p/2)} > 0 \quad (0 < \delta_p < \pi)$$

where δ_p is the partner delay angle. The time-averaged tangential acceleration cannot vanish; a constant-speed circular orbit is impossible.

- The tangential component of the delayed force does positive work in the partner-only circular ledger.
- The radial component points inward, but inward radial pull plus positive tangential work does not by itself prove a tightening spiral.

With perfectly symmetric initial conditions, the paths of the electrino and positrino are distinct but mirror-related. If the branch begins as a radial fall or enters a non-circular capture basin, it may still contract, but that is a separate branch-history statement. Emission cadence and intrinsic per-wavefront amplitude remain constant, while the **received** force is velocity-dependent because the causal-delay Jacobian compresses or dilates the causal flux along each active branch. The evolution is therefore driven by delay geometry, branch bunching, and, once active, self-interaction.

Initially, and as long as the speeds of both architrinos are less than or equal to the wake propagation speed c_f, they are only influenced by their partner's attractive wake. The total acceleration is simply the attractive force:

$$\mathbf{a}_{1,\mathrm{total}}(t) = \mathbf{a}_{1,2}(t) \quad \text{and} \quad \mathbf{a}_{2,\mathrm{total}}(t) = \mathbf{a}_{2,1}(t)$$

During this partner-only phase, the retained force has an inward radial component and a forward tangential work row. In the circular or near-circular reduction that combination is anti-damping: it accelerates the orbiting motion and prevents a partner-only constant-speed circle. It does not, by itself, prove a tightening spiral. Any sub-field-speed contraction claim must come from a certified non-circular branch, a capture basin, or an explicit finite-window wake/recoil ledger.

Ideal Symmetric Spiral Ansatz

The ideal binary spiral used in this opening analysis is not the same geometry as the later maximum-curvature circular benchmark. It is a **symmetric logarithmic-spiral ansatz**: the electrino and positrino follow two distinct planar curves related by the binary symmetry. At equal absolute time they remain opposite about the midpoint in the ideal center frame, but each architrino's path is the mirror-conjugate of the other's path rather than the same curve traced by both architrinos.

This matters because the ideal spiral is a **transient, scale-similar contraction ansatz**, not a consequence of the principal circular calculation. Within a fixed velocity regime and fixed active-root ledger, the model assumes that the local force geometry repeats after a scale change and phase advance: radii shrink by a common factor, speeds rise according to the same delayed-geometry rule, and the partner/self branch structure is symmetric between the two architrinos. When the trajectory crosses a threshold such as $\|\mathbf{v}\| = c_f$ or a higher root-birth boundary, that scale-similar description must be re-matched on a new branch chart.

By contrast, the maximum-curvature binary section studies a **uniform circular benchmark**: fixed R, fixed s, and a single circular path geometry used to compute closed-form delay angles, branch Jacobians, and per-hit force components. That circular model is useful as a limiting or diagnostic case, and it now gives the anti-damping obstruction that any non-circular contraction story must beat. The detailed non-circular benchmark for the symmetric logarithmic spiral belongs in Master Equation; this chapter uses it only as the conceptual two-body entry point.

Spiral Momentum Budget Across the Hinge (Speculative)

This subsection records a modeling hypothesis rather than a derived law. The desired closure would link the spiral path, the per-hit force law, and the angular-momentum budget across the full velocity range. Below the wake speed, the binary feels only partner hits, and the principal circular branch has positive tangential work. A contraction ansatz must therefore explain how radial tightening survives that anti-damping row through non-circular geometry, wake-flux export, recoil, or a later multi-root ledger. We introduce a per-cycle gain parameter ΔL_c only as a provisional bookkeeping variable for that unresolved branch-history calculation.

Branch-birth jump target: a smooth doubling rule is too strong unless the active causal-root ledger stays unchanged. On a fixed signed branch chart $b(s)$, the per-cycle escaped angular-momentum entry should instead be written

$$\Delta L_{\text{cycle}}(s) = \sum_{\rho \in b(s)} \ell_{\rho}^{\text{esc}}(s),$$

where ρ ranges over the active partner and self rows that actually send wake angular momentum through the window boundary. At a branch birth the ledger changes, so the cycle budget has a jump law rather than an automatic smooth continuation. At the principal self-hit hinge,

$$\Delta L_{\text{cycle}}(1^+) - \Delta L_{\text{cycle}}(1^-) = \ell_{\text{self},0}^{\text{esc}}(1^+),$$

with the right-hand side evaluated in the same finite-η chart that regularizes the caustic. The older heuristic $\Delta L_c \mapsto 2\Delta L_c$ is recovered only in the special case where the newly born principal self row exports exactly the same cycle increment as the pre-hinge partner ledger.

This section treats an exponential-in-angle spiral (logarithmic spiral) as a **modeling assumption** rather than a derived law. It sets the bookkeeping target: a path-history force sum whose signed branch-birth increments and boundary wake fluxes yield the spiral contraction. Near $s = 1^+$ the principal self row inherits the caustic onset already displayed below; after finite-window averaging its branch-entry impulse should still report the $(s-1)^{-3/2}$ wall inherited from the self radial coefficient, while the sharper raw tangential coefficient must be regularized before it is used in a cycle ledger.

Spiral Binary Symmetry-Breaking Point ($\|\mathbf{v}\| = c_f$)

The binary system's evolution is organized around the **field-speed symmetry point** $\|\mathbf{v}\| = c_f$. This is a **hinge** where the causal structure changes: below c_f only partner-delay forces exist, while above c_f self-hit roots appear. The hinge is not a hard barrier; it is the birth of the principal self branch. In the symmetric circular geometry the self-delay equation is

$$\delta_s = 2s\sin(\delta_s/2), \qquad s = \frac{\|\mathbf{v}\|}{c_f}$$

Writing $s = 1 + \mu$ with $\mu > 0$ small, the principal root satisfies

$$\delta_s \sim \sqrt{24\mu}, \qquad \sin(\delta_s/2) \sim \sqrt{6\mu}$$

The associated branch Jacobian is

$$J_s = 1 - s\cos(\delta_s/2) = 1 - \frac{\delta_s}{2}\cot(\delta_s/2) \sim 2\mu$$

Therefore the self radial and tangential magnitudes scale as

$$\frac{1}{\sin(\delta_s/2)\,|J_s|} \sim \mu^{-3/2}, \qquad \frac{1}{\sin^2(\delta_s/2)\,|J_s|} \sim \mu^{-2}$$

This is the first major consequence of restoring the causal Jacobian: the hinge is not merely a change in root count but a genuine **caustic onset**. The principal self branch turns on with a sharply amplified outward radial response and an even more singular tangential drive. Any candidate maximum-curvature balance must therefore confront a near-threshold Jacobian wall before appealing to higher-winding smoothing.

Self-Hit: Definition and Diagnostics

Self-hit is the key non-Markovian feature of architrino dynamics. It occurs when an architrino interacts with potential it emitted earlier along its own worldline.

Geometric condition (absolute coordinates): For a given architrino with trajectory $\mathbf{x}(t)$, a self-hit event is a pair of times $(t_{\text{emit}}, t_{\text{hit}})$ with $t_{\text{hit}} > t_{\text{emit}}$ such that

$$\|\mathbf{x}(t_{\text{hit}}) - \mathbf{x}(t_{\text{emit}})\| = c_f(t_{\text{hit}} - t_{\text{emit}})$$

and the architrino is the source of the causal wake surface emitted at t_{emit}.

Terminology split: Hit type is determined by **source identity**. A **self-hit** has the same source and receiver; a **partner hit** has a different source and receiver. Root count is a separate question: either source can contribute one active causal root or multiple active roots at the same reception time. Thus "self-hit" does not mean "multi-hit," and "partner hit" does not mean "single-hit."

Dynamical role:

- On any interval with strict sub-field-speed motion, self-hit is absent by the triangle-inequality root test, unless older path-history emissions from a prior super-field-speed interval remain active.
- As velocities exceed c_f on curved histories, emission isochrons can catch up with the emitter's future positions, generating candidate nonlocal feedback and effective restoring or destabilizing forces depending on configuration.
- In generic trajectories, once an architrino has exceeded c_f and emitted wakes in that regime, it can later slow below c_f and still experience self-hits from those earlier emissions (see **Status** at top for the non-Markovian/path-history caveat).
- For binary and Noether braid assemblies, repeated self-hit events are the proposed mechanism that can prevent collapse, lock in stable radii and frequencies, and create new limit cycles and attractors.

For the circular-geometry details (principal angles, winding numbers, discrete self-hit branches), see **Setup and Notation (Symmetric Frame)** in **Maximum-Curvature Binary — Circular**.

Post-Threshold Self-Hit Phase

Once the circular branch admits same-source roots, the architrinos interact with their own earlier, repulsive wakes. The total acceleration on each architrino then becomes a superposition of attraction from its partner and self-repulsion. For the electrino:

$$\mathbf{a}_{1,\text{total}}(t) = \mathbf{a}_{1,2}(t) + \mathbf{a}_{1,1}(t)$$

In the circular benchmark, the principal self-hit branch ($m = 0$) becomes available only on the super-field-speed side; at higher speeds, additional self-hit and partner-hit roots can turn on (see **Root Multiplicity vs. Speed**). The new self-repulsive term, $\mathbf{a}_{1,1}(t)$, grows rapidly as the path curvature increases and changes the tangential ledger. On the same-sheet principal chart that tangential contribution is forward; in the full signed ledger, older sheets can contribute with the opposite tangential sign. This post-threshold phase is therefore a branch-certificate target, not a generic tightening law: any radial arrest or continued contraction must be decided by the signed multi-root ledger, wake-flux/recoil accounting, and stability certificate described below.

Maximum-Curvature Binary — Circular

Once self-hit turns on, the natural question is whether the dynamics converge to a limiting curvature. We call the candidate limit the **maximum-curvature binary (MCB)**. This section collects the full two-body, self-hit analysis for that candidate, including delay geometry, force components, and stability criteria. It is the canonical reference for MCB attractor status.

MCB stability claims rely on the well-posedness of the regularized SD-NDDE. In this chapter we treat $\eta > 0$ as fixed; any $\eta \to 0$ statement is outside the claims established here unless a weak-limit argument is explicitly supplied. The formal state-space framework appears in **State Space and Well-Posedness of the Two-Body Delay System**.

Goal: Characterize the circular, constant-speed, constant-radius configuration of two opposite-polarity architrinos and investigate where curvature $1/R$ is maximized. We work in units with field speed $c_f = 1$ and use the canonical delayed per-hit law with radial line of action and Jacobian-weighted magnitude.

Plain language: We seek the tightest (smallest-R) steady circle an opposite-polarity pair can trace when the only forces come from delayed, Jacobian-weighted line-of-action interactions with the partner (partner hits, possibly multiple at higher speed) and from each architrino's own past emissions (self-hits, accepted by same-source roots; in the circular branch these require the super-field-speed side).

Foundational Context (Ontological Clarification)

The Maximum-Curvature Binary (MCB) as Fundamental Unit

The architecture hypothesizes that the **maximum-curvature binary (MCB)** would be reachable first by the **inner binary** of a nested shell braid assembly, stabilized by certified same-source self-hit roots on the super-field-speed circular branch. Contingent on Conjectures A/B, it would supply candidate **fundamental physical units** (length and time); see **Emergent Properties and Measurement Standards** below for the explicit definitions.

Universal cap target (explicit): If a stable MCB branch is certified, it would define a single limit state with one radius/speed pair. Binaries may sit below that limit, but the claim that no binary can exceed the MCB curvature or pass beyond its defining radius/speed remains conditional on the full signed-root ledger and stability certificate.

If realized, the MCB radius $r_{\min}$ is expected to be determined by the balance of:

1. opposite-polarity causal-wake attraction, with the stripped inverse-square surrogate scaling as ϵ^2/r^2,
2. self-hit repulsion (non-Markovian feedback when same-source roots exist; super-field-speed circular history is the relevant branch),
3. Centripetal requirement for stable circular orbit.

Dynamical priority (attractor status): The architecture hypothesizes the MCB is a **robust attractor**, not a finely tuned periodic orbit. Only if the multipliers lie strictly inside the unit circle and the basin is non-trivial do we have the attractor the architecture relies on. If neutrality or instability is found, the nested shell braid ladder and Noether braid claims must be downgraded or the interaction law revised (e.g., additional damping/medium effects).

Setup and Notation (Symmetric Frame)

- **Two architrinos** with polarity bookkeeping labels $q_1 = -\epsilon$ and $q_2 = +\epsilon$ (where $\epsilon = |e|/6$).
- **Equal-time positions** (in absolute time t) are diametrically opposite on a circle of radius R about the midpoint.
- **Uniform circular motion**: Angular speed ω, constant tangential speed $s = R\omega$.
- **Non-translating binary**: Circle center (midpoint) is fixed in Euclidean 3D space; no net translation.

Translating Binary Handoff to Lorentz Closure

The circular maximum-curvature benchmark is also the rest-frame boundary condition for the first material clock/ruler test. The translating ansatz keeps absolute time and the primitive wake speed explicit:

$$\mathbf{x}_\sigma(t) = ut\,\hat{\mathbf{e}} + \sigma\,\boldsymbol{\rho}_u(\theta(t)), \qquad \sigma \in \{+1, -1\}$$

where $\boldsymbol{\rho}_0$ is the circular branch studied here and $\boldsymbol{\rho}_u$ is the deformed periodic orbit, if it exists, on the retained moving branch chart.

This is a direct delayed-root calculation, not a coordinate boost imposed on the answer. The root equations must be solved again with the source positions, source velocities, partner-hit rows, self-hit rows, and Jacobian factors evaluated on the translating history. The decisive outputs are the moving period T_u and the projected size ratio $L_\parallel(u)/L_\perp(u)$. In primitive units the Lorentz target is

$$\frac{T_u}{T_0} = \gamma_f(u), \qquad \frac{L_\parallel(u)}{L_\perp(u)} = \frac{1}{\gamma_f(u)}, \qquad \gamma_f(u) = \left(1 - \frac{u^2}{c_f^2}\right)^{-1/2}$$

The exact residual definitions and Theorem G role are recorded in Lorentz Kinematics. A Lorentzian result would make the two-body branch the first derived substrate clock. A non-Lorentzian residual would be equally informative because it would identify the first place where the primitive two-body kernel pressures the larger Lorentz-closure program.

The moving-branch test also has a root-starvation obligation. If a forward source row has minimum forward separation $d_{\min}$ in the direction of motion, then the causal delay needed to receive that row obeys the elementary bound

$$\tau_{\text{forward}}(u) \geq \frac{d_{\min}}{c_f - u}.$$

This divergence is stronger than the Lorentz factor divergence,

$$\gamma_f(u) \sim (c_f - u)^{-1/2},$$

as $u \to c_f^-$. Therefore a bare translating binary cannot be promoted to the Lorentz handoff merely by showing that one clock period stretches. It must also show that the locked branch retains enough memory depth to supply the forward roots it claims. One diagnostic target is

$$\mathcal{R}_{\text{Lor-root}}(u) = \frac{\tau_{\text{forward}}(u)/T_u}{M_b^{\text{mem}}(u) + \epsilon_h}, \qquad M_b^{\text{mem}}(u) = \frac{h_b^{\text{lock}}(u)}{T_u},$$

where h_b^{lock} is the declared retained-history depth of the moving branch and $\epsilon_h > 0$ is a fixed normalization floor. If this residual diverges on the finite-η moving chart, the two-body branch has run out of retained causal roots before it has derived Lorentz closure; the handoff must then move to a Noether-sea or larger assembly response rather than being booked as a bare-binary result.

Let $C_i(t_{\text{emit}})$ denote the causal wake surface emitted by architrino i at emission time t_{emit}. For uniform circular motion, self-hit events are discrete intersections between the worldline and its own wake surfaces. Define the **principal self-delay angle** $\tilde{\delta}_s \in (0, \pi]$ as the minimal angular separation between the current position and the emission point that yields a hit. Additional self-hits occur at longer delays indexed by winding number $m \geq 0$, giving a discrete family $\delta_s(m) = \tilde{\delta}_s + 2\pi m$.

Phase Angles and Delays

Let δ_s and δ_p denote the angular phase separations (measured along the circle) between:

- **Self** (same architrino): Current position -> its own past emission position that hits "now."
  - Delay time: τ_s; angular separation: $\delta_s = \omega\tau_s$.
  - Chord length: $r_s = 2R\sin(\delta_s/2)$.
- **Partner** (other architrino): Current position -> partner's past emission position that hits "now."
  - Delay time: τ_p; angular separation: $\delta_p = \omega\tau_p$.
  - Chord length: $r_p = 2R\cos(\delta_p/2)$.

Causal-Time Constraints (Field Speed $c_f = 1$)

For a signal to travel from emission point to reception point:

$$r = c_f \cdot \tau \quad \Rightarrow \quad r = \tau \quad \text{(in units where } c_f = 1\text{)}$$

This yields two delay equations:

1. **Self-hit**:

$$\delta_s = \omega\tau_s = \omega \cdot r_s = \omega \cdot 2R\sin(\delta_s/2) = 2s\sin(\delta_s/2)$$

2. **Partner hit**:

$$\delta_p = \omega\tau_p = \omega \cdot r_p = \omega \cdot 2R\cos(\delta_p/2) = 2s\cos(\delta_p/2)$$

These two transcendental equations determine (δ_s, δ_p) as functions of speed s.

Circular-branch threshold: On this uniform circular branch, self-hit roots exist only when $s > 1$ (i.e., $\|\mathbf{v}\| > c_f$). For $s \le 1$, no self-hit roots occur on the circular chart. This is a branch-specific root result, not a general speed-only criterion for arbitrary histories.

Principal Partner-Root Certificate

For the partner branch, write the full delay angle as

$$\phi = \omega\Delta$$

and the chapter speed ratio as

$$\beta = \frac{\omega R}{c_f}$$

The principal partner-root equation is

$$2\beta\cos\frac{\phi}{2} = \phi, \qquad 0 < \phi < \pi$$

The function $F(\phi) = 2\beta\cos(\phi/2) - \phi$ satisfies $F(0) = 2\beta > 0$, $F(\pi) = -\pi$, and

$$F'(\phi) = -\beta\sin\frac{\phi}{2} - 1 < 0$$

on $(0, \pi)$. Therefore the principal partner root exists and is unique for every $\beta > 0$.

The same conclusion gives a derived transversality floor. On the principal partner branch,

$$J_p = 1 + \beta\sin\frac{\phi}{2}$$

so the dimensional root-transversality quantity is

$$\kappa_{\text{hit}}^{\text{bin}} \equiv |c_f - \hat{\mathbf{r}} \cdot \mathbf{v}_j(t - \Delta)| = c_f\left(1 + \beta\sin\frac{\phi}{2}\right) > c_f$$

This floor is not an admissibility parameter for the principal branch; it is a computed property of the circular geometry. It certifies that the simple-root chart cannot fail by partner-root tangency on the principal partner branch.

The instantaneous radial-balance equation is also closed form. Setting the inward partner radial acceleration equal to the required centripetal acceleration gives

$$\frac{\beta^2 c_f^2}{R} = \frac{\kappa\epsilon^2}{4R^2\cos(\phi/2)\,(1+\beta\sin(\phi/2))}$$

and therefore, with $R_* = \kappa\epsilon^2/c_f^2$,

$$\frac{R}{R_*} = \frac{1}{4\beta^2\cos(\phi/2)\,(1+\beta\sin(\phi/2))}$$

As $\beta \to 0$, the root satisfies $\phi \sim 2\beta$, and the balance reduces to

$$\omega^2 R^3 = \frac{\kappa\epsilon^2}{4}$$

which is the delayed Coulomb-Kepler scaling for the isolated opposite-polarity pair.

The same principal branch still cannot be a uniform orbit. The delayed partner line of action has a forward tangential projection, so

$$a_\theta^{(\mathrm{part})} = \frac{\kappa\epsilon^2\sin(\phi/2)}{4R^2\cos^2(\phi/2)\,(1+\beta\sin(\phi/2))} > 0$$

and the instantaneous work rate satisfies $a_\theta^{(\mathrm{part})} R\omega > 0$. Thus the principal branch has positive tangential work in the partner-only circular reduction: it gives the radial family above, but it also pumps tangential energy. A partner-only constant-speed circular binary therefore requires a signed multi-root tangential residual

$$\sum_{t_0\in\mathcal{C}_{12}(t)} a_\theta^{(12)}(t;t_0) + \sum_{t_0\in\mathcal{C}_{11}(t)} a_\theta^{(11)}(t;t_0) = 0$$

on the retained ledger, or an explicitly retained wake-flux/recoil channel in the finite-window energy ledger. Since circular self-hit roots require super-field-speed history on this branch, a self-hit-stabilized MCB candidate must live on the super-field-speed side of the circular ledger rather than on the principal partner branch alone.

Additional partner roots are not speculative. In the full delay-angle representation, partner roots can occur only in positive-cosine windows

$$W_k = (4\pi k - \pi,\ 4\pi k + \pi)\,, \qquad k = 0, 1, 2, \ldots$$

with the principal root in $W_0 \cap (0,\pi)$. For $k \geq 1$, the root pair appears when the window maximum reaches zero:

$$\sqrt{\beta^2 - 1} + \arcsin\frac{1}{\beta} = 2\pi k$$

At equality the two roots are born at a tangency; above it they thicken the partner-hit ledger. The root census is therefore a computed branch diagram rather than an independent conjecture.

Terminology: Roots and Winding Numbers

Root: An emission time $t_0 < t$ (from either self or partner) that satisfies the causal constraint $r = c_f(t - t_0)$ and produces a hit at reception time t
.

Integer-indexed older roots (winding numbers):

Let $\tilde{\delta}_s \in (0,\pi]$ and $\tilde{\delta}_p \in (0,\pi]$ denote the **minimal (principal) angular separations** that determine the chord lengths and force directions.

In the same-sheet convention used for the first circular no-go, the full families of causal delays are:

- **Self**:

$$\delta_s(m) = \tilde{\delta}_s + 2\pi m = 2s\sin(\tilde{\delta}_s/2), \quad m = 0, 1, 2, \ldots$$

- **Partner**:

$$\delta_p(m) = \tilde{\delta}_p + 2\pi m = 2s\cos(\tilde{\delta}_p/2), \quad m = 0, 1, 2, \ldots$$

Geometric interpretation:

- The minimal separations $\tilde{\delta}_s, \tilde{\delta}_p$ determine the **same-sheet principal geometry** (chord lengths, force directions).
- The winding index m affects **timing/ordering** of multiple hits inside that same-sheet convention.

Signed-sheet completion: A full circular root certificate must also track whether the full delay angle is represented as $2\pi m + \alpha$ or $2\pi m - \alpha$ for a minimal chord angle $\alpha \in (0, \pi]$. Opposite signed sheets can reverse the tangential projection of a self-hit line of action. The sign-invariant statements below are therefore certified only on the same-sheet principal branch chart unless the signed sheet has been explicitly included in the root ledger.

For the full signed ledger, write

$$\Delta_s^{\sigma,m} = 2\pi m + \sigma\alpha_s, \qquad \Delta_p^{\sigma,m} = 2\pi m + \sigma\alpha_p, \qquad \sigma \in \{+1, -1\}$$

with $\sigma = -1$ requiring $m \geq 1$. The signed circular root equations become

$$2\pi m + \sigma\alpha_s = 2s\sin(\alpha_s/2), \qquad 2\pi m + \sigma\alpha_p = 2s\cos(\alpha_p/2)$$

The corresponding tangential signs are $\sigma\cos(\alpha_s/2)$ for self roots and $\sigma\sin(\alpha_p/2)$ for partner roots, up to positive branch weights. The signed sheet is therefore not a cosmetic ledger choice: it is the first place the bare circular kernel can acquire a same-source tangential contribution with the opposite sign from the same-sheet no-go row. The first negative self sheet, $m = 1, \sigma = -1$, obeys

$$2\pi - \alpha = 2s\sin(\alpha/2)$$

and appears at $s = \pi/2$ with $\alpha = \pi$. For $s > \pi/2$ it contributes negative tangential drive. This does not prove circular closure, but it makes the $\sigma = -1$ sheet a necessary closure channel rather than a caveat. A useful floor conjecture is:

> No isolated, bare, constant-speed circular MCB branch can close for $s < \pi/2$, because the first negative same-source sheet is absent and the same-sheet tangential cohomology class has no internal cancellation channel.

For $s \geq \pi/2$, closure is still not automatic. The negative sheet must survive the finite-η branch chart, satisfy the Jacobian and inactive-gap floors, and balance the remaining tangential class through signed-root cancellation, wake escapement, recoil, or multi-body exchange.

Per-Hit Directions and Force Components

Local Coordinate Frame at Receiver

- **Radial outward**: $\hat{e}_r$ (from rotation center toward receiver).
- **Tangential**: $\hat{e}_t$ (direction of motion along circle).

Unit Directions of Lines of Action (Emission -> Reception)

Self-hit:

$$\hat{u}_s = \sin(\delta_s/2)\,\hat{e}_r + \cos(\delta_s/2)\,\hat{e}_t$$

Partner hit (geometric chord across circle):

$$\hat{u}_p = \cos(\delta_p/2)\,\hat{e}_r - \sin(\delta_p/2)\,\hat{e}_t$$

Canonical Per-Hit Accelerations

Using the delayed law with line-of-action direction and Jacobian-weighted magnitude (where κ is a coupling constant and $\epsilon = |e|/6$), define branch Jacobians

$$J_s \equiv 1 - \frac{\mathbf{v}_{\text{self}}(t_0) \cdot \hat{u}_s}{c_f}, \qquad J_p \equiv 1 - \frac{\mathbf{v}_{\text{partner}}(t_0) \cdot \hat{u}_p}{c_f}$$

These encode the geometric bunching or dilation of the received causal flux along the active self and partner branches.

Self-hit (like polarities -> repulsive):

$$\mathbf{a}_s = +\kappa\epsilon^2 \frac{1}{r_s^2\,|J_s|}\hat{u}_s$$

Partner hit (opposite polarities -> attractive):

$$\mathbf{a}_p = -\kappa\epsilon^2 \frac{1}{r_p^2\,|J_p|}\hat{u}_p$$

Explicit Circular Jacobians

For the symmetric circular geometry, the emitter velocities can be resolved exactly against the line-of-action directions:

$$\mathbf{v}_{\text{self}}(t_0)\cdot\hat{u}_s = s\cos(\delta_s/2), \qquad \mathbf{v}_{\text{partner}}(t_0)\cdot\hat{u}_p = -s\sin(\delta_p/2)$$

Hence the branch Jacobians reduce to

$$J_s = 1 - s\cos(\delta_s/2), \qquad J_p = 1 + s\sin(\delta_p/2)$$

Using the delay constraints gives equivalent forms

$$J_s = 1 - \frac{\delta_s}{2}\cot(\delta_s/2), \qquad J_p = 1 + \frac{\delta_p}{2}\tan(\delta_p/2)$$

These formulas make the asymmetry between the two branch types explicit:

- The partner branch always satisfies $J_p > 1$, so delay geometry **dilutes** the received partner flux relative to the static inverse-square value.
- The self branch can satisfy $J_s \to 0^+$, producing the causal bunching that sharpens self-hit into a null-separatrix wall.

Radial and Tangential Components

Define **inward radial** as positive (toward center) and **tangential** as positive in direction of motion.

Chord lengths:

$$r_s = 2R\sin(\delta_s/2), \quad r_p = 2R\cos(\delta_p/2)$$

Inward radial components:

- **Self** (repulsive -> outward -> negative):

$$A_{s,\text{rad}} = -\kappa\epsilon^2\frac{\sin(\delta_s/2)}{r_s^2\,|J_s|} = -\frac{\kappa\epsilon^2}{4R^2\sin(\delta_s/2)\,|J_s|}$$

- **Partner** (attractive -> inward -> positive):

$$A_{p,\text{rad}} = +\kappa\epsilon^2\frac{\cos(\delta_p/2)}{r_p^2\,|J_p|} = +\frac{\kappa\epsilon^2}{4R^2\cos(\delta_p/2)\,|J_p|}$$

Net inward radial acceleration:

$$A_{\text{rad}} = \frac{\kappa\epsilon^2}{4R^2}\left(\frac{1}{\cos(\delta_p/2)\,|J_p|} - \frac{1}{\sin(\delta_s/2)\,|J_s|}\right)$$

Tangential components (both non-negative for $0 < \delta_s, \delta_p < \pi$):

- **Self**:

$$T_s = +\kappa\epsilon^2\frac{\cos(\delta_s/2)}{r_s^2\,|J_s|} = \frac{\kappa\epsilon^2\cos(\delta_s/2)}{4R^2\sin^2(\delta_s/2)\,|J_s|}$$

- **Partner**:

$$T_p = +\kappa\epsilon^2\frac{\sin(\delta_p/2)}{r_p^2\,|J_p|} = \frac{\kappa\epsilon^2\sin(\delta_p/2)}{4R^2\cos^2(\delta_p/2)\,|J_p|}$$

Net tangential acceleration:

$$T = T_s + T_p \geq 0$$

Sub-Field-Speed Simplification ($s \leq 1$; No Self-Hits)

When $s \leq 1$, self-hits do not occur (δ_s has no solution). Only the partner contributes, so the tangential drive remains strictly positive, consistent with the lemma above:

$$T(s<1) = T_p = \frac{\kappa\epsilon^2}{4R^2}\frac{\sin(\delta_p/2)}{\cos^2(\delta_p/2)\,|J_p|}$$

Using the delay relation $\delta_p = 2s\cos(\delta_p/2)$:

$$T(s<1) = \frac{\kappa\epsilon^2 s^2}{R^2}\frac{\sin(\delta_p/2)}{\delta_p^2\,|J_p|} > 0$$

Because $J_p = 1 + s\sin(\delta_p/2) > 1$, the delay geometry weakens the partner contribution relative to bare $1/r^2$, but it never changes its sign. Therefore even at sub-field speeds there is always a **net positive tangential force** (accelerating the binary), which prevents a truly stable, constant-speed circular orbit.

Requirements for True Circular Orbit (Working Hypothesis)

For uniform circular motion at fixed radius R and constant speed s:

1. **Centripetal balance**:

$$A_{\text{rad}} = \frac{s^2}{R}$$

2. **Finite-window energy balance**:

$$\left\langle \frac{dK_\mu}{dt} \right\rangle_W + \langle \Phi_{\text{wake},\partial W} + P_{\text{recoil}} \rangle_W = 0$$

Here K_μ is the chosen quadratic kinetic proxy, $\Phi_{\text{wake},\partial W}$ is the causal-wake energy flux through the boundary of the local window, and P_{recoil} is any retained local wake-emission resistance term. The older shorthand $\langle T \rangle = 0$ is valid only for a particle-only closed window with no boundary wake flux and no recoil term.

On a declared branch chart b, this balance has an operational work row:

$$P^{(\eta)}_{b,\text{work}}(t) = \sum_i \mu_{\text{arch}}\, \mathbf{a}^{(\eta)}_{i,b}(t) \cdot \mathbf{v}_i(t)$$

For a circular constant-speed benchmark, $\mathbf{v}_i$ is tangent to the orbit and the radial row does no instantaneous work, so

$$\left\langle P^{(\eta)}_{b,\text{work}} \right\rangle_{P_b} = \mu_{\text{arch}}\, s_b\, \left\langle A^{\tan}_{\eta,b} \right\rangle_{P_b}$$

for the quadratic proxy. Thus the tangential term is not merely a geometric nuisance; it is the first constructive entry in the binary wake-energy ledger. If the primitive kinetic scalar is used instead, replace μ_{arch} by $\mu_K(\|\mathbf{v}_i\|)$ inside the summed power.

Tangential Drive and Wake Escapement

Theorem (Same-sheet no-go for constant-speed circular orbit in the bare two-body kernel).
In the symmetric, non-translating circular binary with canonical delayed radial forces only, and with active roots restricted to the same-sheet principal branch chart defined above, the net tangential acceleration is strictly positive whenever at least one causal root contributes.

$$T_{\text{net}} = \sum_{m \in \mathcal{M}_p} w_{p,m} T_{p,m} \; + \sum_{m \in \mathcal{M}_s} w_{s,m} T_{s,m} > 0$$

where $w_{p,m}, w_{s,m} \geq 0$ are branch weights induced by regularization/time averaging, and $\mathcal{M}_p, \mathcal{M}_s$ are active partner/self root sets.

Proof.
For any active partner branch, the tangential contribution is

$$T_{p,m} = \frac{\kappa\epsilon^2}{4R^2} \frac{\sin(\tilde{\delta}_{p,m}/2)}{\cos^2(\tilde{\delta}_{p,m}/2)\,|J_{p,m}|} > 0, \qquad \tilde{\delta}_{p,m} \in (0,\pi)$$

and for any active self branch (when present),

$$T_{s,m} = \frac{\kappa\epsilon^2}{4R^2} \frac{\cos(\tilde{\delta}_{s,m}/2)}{\sin^2(\tilde{\delta}_{s,m}/2)\,|J_{s,m}|} > 0, \qquad \tilde{\delta}_{s,m} \in (0,\pi)$$

The sign is branch-invariant on this same-sheet chart because winding changes timing, not chord orientation. Therefore each summand in T_{net} is nonnegative, and at least one is strictly positive whenever any hit exists. Hence $T_{\text{net}} > 0$ on the certified chart. □

Corollary.
Within the same-sheet bare isolated two-body kernel, an exact constant-speed circular orbit with no boundary wake flux and no recoil term is impossible. Any MCB-like steady state must therefore close a finite-window balance: signed-ledger cancellation may reduce the local tangential drive, but the remaining forward power must be assigned either to wake escapement through ∂W, to a local recoil term, or to genuinely multi-body nested shell braid exchange.

Interpretation. The positive tangential component is not merely an obstruction to be erased. In a finite local window, partner and self wakes are continually emitted while only a subset of their causal isochrons later hit a local receiver. The unreceived portion exits the local window as wake-history flux. The same-sheet tangential drive is therefore the mechanical pump that can replace the interaction energy exported by those escaping causal wakes. A local binary can look particle-only conservative only if the outgoing wake record, recoil channel, and retained branch ledger are all included in the same balance law.

Cohomology reading. On a closed circular branch, write θ for the receiver phase and let

$$\omega_T^{(b)} = R\,T_{\mathrm{net}}^{(b)}(\theta)\,d\theta$$

be the tangential torque one-form on the retained signed ledger b. Same-sheet rows give a positive period integral, so $\omega_T^{(b)}$ is not an exact derivative of a single-valued mechanical angular-momentum potential on the particle-only chart. Closure requires a coboundary supplied by retained non-particle channels:

$$\left[\omega_T^{(b)} + \omega_{\partial W}^{(b)} + \omega_{\mathrm{recoil}}^{(b)} + \omega_{\mathrm{multi}}^{(b)}\right] = 0$$

in the cycle cohomology of the branch chart. A compact escaped-action diagnostic is

$$N_{\mathrm{esc}}^{(b)} = \frac{1}{h}\oint R\,T_{\mathrm{net}}^{(b)}(\theta)\,d\theta,$$

where h is the declared action unit used by the branch packet. A bare two-body circular closure can pass only when this class is cancelled by an explicitly retained signed sheet, wake-boundary, recoil, or multi-body exchange row.

Plain language: On the same-sheet chart, the isolated pair shows persistent tangential drive at the per-hit level; cancellation is hard because every certified root pushes the same way. The stable-branch question is not “how can the drive disappear?” but “which wake flux, recoil, or multi-body channel balances the drive without destroying the retained branch?” This is a primary test of the MCB attractor hypothesis.

What “Maximum Curvature” Demands

Mechanism summary (self-hit balance): once $s > 1$, each self-hit contributes a **repulsive acceleration away from its own past emission point**. In the symmetric circular geometry that repulsion has a **radial outward component** (opposing further contraction) and a **positive tangential component** (continuing to speed up the architrino). As the radius shrinks, both partner attraction and self-hit repulsion scale like $1/R^2$, while the decisive extra effect is the Jacobian weighting: the self-hit response can sharpen dramatically as an active branch approaches its null-separatrix geometry and because **new self-hit roots appear** at higher s. Maximum curvature would require the **outward self-hit radial component** to balance the inward partner pull without the still-positive tangential drive destroying constant-speed closure.

From the radial component formula:

$$A_{\mathrm{rad}} = \frac{\kappa\epsilon^2}{4R^2}\left(\frac{1}{\cos(\delta_p/2)\,|J_p|} - \frac{1}{\sin(\delta_s/2)\,|J_s|}\right)$$

Increasing curvature ($1/R$ larger, so R smaller) requires **stronger inward radial force**. This occurs when:

1. δ_p **increases** -> $\cos(\delta_p/2)$ decreases -> partner term $1/\cos(\delta_p/2)$ **increases** (stronger inward pull).
2. δ_s **increases** -> $\sin(\delta_s/2)$ increases -> the geometric part of the self term decreases, while the full outward response still depends on how rapidly the Jacobian factor $|J_s|^{-1}$ grows along the active branch.

Two distinct balance mechanisms are now mathematically visible:

1. **Near-threshold Jacobian wall.**
   On the principal self branch, $|J_s|^{-1}$ turns on singularly as $s \downarrow 1^+$, with radial magnitude scaling like $(s-1)^{-3/2}$. This is the earliest possible obstruction to continued contraction.

2. **Higher-speed multi-branch redistribution.**
   At larger s, additional self branches turn on and redistribute the outward response across several winding sectors. In that regime the detailed balance depends on the full weighted sum over all active branches rather than on the principal branch alone.

However: Due to the same-sheet per-hit $T > 0$ result, this “maximum curvature” state remains unverified for the isolated two-body system. Its stability must be tested by the full, signed, multi-root time-averaged dynamics.

Because the desired MCB branch is expected to graze the $J = 0$ wall, the stability target is not only a smooth Floquet calculation. On smooth arcs with a fixed ledger, Floquet multipliers are the right local test. At the null separatrix itself, the branch is a caustic-grazing limit cycle: the appropriate theorem target is an isolating block in history space that straddles the $J = 0$ wall and has a persistent Conley index under finite-η continuation. In that reading, the MCB branch is stable only if the orbit returns through the grazing chart without escaping the isolating block or changing its declared signed ledger except at the certified fold rows.

Emergent Properties and Measurement Standards

If a stable MCB exists, it provides a concrete **rod** and **clock** defined entirely by the two-body delay dynamics. Let

$$d_0 := R_{\mathrm{MCB}}, \qquad T_0 := \frac{2\pi}{\omega_{\mathrm{MCB}}}$$

The natural Layer-I two-body units are

$$R_* = \frac{\kappa\epsilon^2}{c_f^2}, \qquad T_* = \frac{R_*}{c_f}$$

so the first MCB outputs are the dimensionless ratios

$$\frac{R_{\mathrm{MCB}}}{R_*}, \qquad \frac{T_0}{T_*}, \qquad \beta_{\mathrm{MCB}}$$

rather than additional fitted constants. Once (c_f, κ, ϵ) fixes the length, time, and polarity units, the signed-root ledger and stability problem must compute those ratios as pure numbers.

Then d_0 is the candidate fundamental length scale of the architecture, and T_0 is the candidate fundamental time scale. Their comparison with the wake propagation speed is the dimensionless MCB speed factor

$$\beta_{\mathrm{MCB}} = \frac{R_{\mathrm{MCB}}\omega_{\mathrm{MCB}}}{c_f} = \frac{2\pi d_0}{c_f T_0}$$

so the wake propagation speed is not an imposed architrino-speed limit. It is the propagation reference used to compare the MCB rod and clock, while individual architrinos may enter super-field-speed regimes with

$$\|\mathbf{v}\| > c_f$$

In this view, any ruler or clock built from architrino assemblies ultimately reduces to multiples of (d_0, T_0). Measurement standards are therefore **dynamical invariants** of the two-body attractor: they persist because the underlying limit cycle (if realized) is stable and reproducible across assemblies.

A certified MCB would also define the first handedness marker. In the binary plane set

$$\hat{\mathbf{n}}_{\mathrm{MCB}} = \hat{\mathbf{r}} \times \hat{\mathbf{v}},$$

with $\hat{\mathbf{r}}$ pointing from the center to one chosen polarity row and $\hat{\mathbf{v}}$ its direction of motion. The two signs of $\hat{\mathbf{n}}_{\mathrm{MCB}}$ label two branch basins, B_+ and B_-, not two coordinate conventions. A branch-preserving deformation can rotate the plane, but it cannot flip this $\mathbb{Z}_2$ label without passing through a degeneracy where the circular plane, source order, or signed causal-root ledger changes. Thus chirality is carried by the joint path-history and signed-root framing of the branch, not by a freely chosen drawing orientation.

If the MCB does not exist as a stable attractor, these emergent standards must be replaced by whatever stable limit structure the dynamics actually support.

Root Multiplicity vs. Speed

This section separates the two terminology axes used throughout the chapter:

- **Source identity**: self-hit ($j = i$) or partner hit ($j \neq i$).
- **Root count**: single-root or multi-root on the current branch chart.

The self-hit onset is dynamically special because it introduces same-source feedback and an outward self-repulsive channel. Partner multi-hit is still part of the same super-field-speed root topology: at higher speeds, older partner wake surfaces can also satisfy the causal-root condition and contribute additional inward channels.

In the same-sheet uniform circular, non-translating geometry, admissible self-roots are indexed by winding number $m \geq 0$ and minimal angular separation $\tilde{\delta}_s \in (0, \pi]$:

$$\delta_s = \tilde{\delta}_s + 2\pi m = 2s\sin(\tilde{\delta}_s/2)$$

Counting Self-Hits by Winding Index

For fixed winding $m \geq 0$, define

$$f_m(\delta; s) = 2s\sin(\delta/2) - \delta - 2\pi m, \qquad \delta \in (0, \pi]$$

An m-branch same-sheet self-hit exists exactly when $f_m(\delta; s) = 0$ has a solution in $(0, \pi]$.

- For the principal branch $m = 0$, the threshold is sharp:

$$s_0^\star = 1$$

- For higher winding numbers $m \geq 1$, the appearance threshold is determined by the tangency condition at the interior maximizer $f_m'(\delta; s) = 0$, namely

$$\cos(\delta_m^\star/2) = \frac{1}{s}, \qquad \sqrt{(s_m^\star)^2 - 1} - \arccos\left(\frac{1}{s_m^\star}\right) = \pi m$$

Thus the higher same-sheet self branches do not turn on at equally spaced speeds. Their onset is governed by a nonlinear sequence of tangencies of the delayed self-intersection curve. A full signed-root ledger must add the $\sigma = -1$ sheets described above; the first such negative self sheet appears at $s = \pi/2$, earlier than the first higher same-sheet self branch.

For large winding number m, the threshold has the asymptotic form

$$s_m^\star = \pi m + \frac{\pi}{2} + O\left(\frac{1}{m}\right)$$

so the old equally spaced picture is recovered only as a high-speed approximation.

Note: Straight-line motion admits **no self-hits** even if $s > 1$; **curvature is required**. The above statements apply specifically to uniform circular, non-translating geometry.

The self-hit root count is therefore a genuine branch-bifurcation diagram for the circular benchmark. Here

$$s = \frac{\|\mathbf{v}\|}{c_f}$$

is the chapter's speed ratio, equivalent to β in the usual notation. Between neighboring branch-birth thresholds, the active self-root ledger $N_s(s)$ is constant and the same root labels can be transported. At the thresholds, the delay equation has a tangency and the newly born circular root lies on a Jacobian-null boundary. Thus the root census, the caustic locations, and the ledger-transition speeds are one computed object rather than three separate assumptions.

Root Ledger as a One-Parameter Morse Complex

For a fixed reception event on a one-parameter family of branch histories, write the root function as

$$F_{ij}(t_0; s) = \|\mathbf{x}_i(t; s) - \mathbf{x}_j(t_0; s)\| - c_f(t - t_0).$$

Active causal roots are the zeros of F_{ij}. A branch birth or death is a fold row:

$$F_{ij} = 0, \qquad \partial_{t_0} F_{ij} = 0, \qquad \partial_{t_0 t_0} F_{ij} \neq 0.$$

Away from those folds, the signed degree

$$D_{ij}(s) = \sum_{t_0 \in \mathcal{C}_{ij}(t;s)} \operatorname{sign}(\partial_{t_0} F_{ij}(t_0; s))$$

is locally constant, while the unsigned counts N_s and M_p track the ranks of the same-source and partner-root rows. This is the binary version of the assembly topological charge: the later tri-binary label (N_s, M_p, c_1) uses the two root-complex integers from this chapter and the phase-bundle Chern data from the resonance-lock chart. A solver that reports only raw root counts loses the signed degree needed to distinguish a true branch fold from a harmless relabeling of rows.

Parameter-Free Circular Branch Packet

The circular two-body benchmark can now be stated as a parameter-free branch packet. Use the Layer-I units

$$R_* = \frac{\kappa \epsilon^2}{c_f^2}, \qquad \rho = \frac{R}{R_*}, \qquad s = \frac{R\omega}{c_f}$$

and factor out the acceleration scale c_f^2/R_*. The remaining equations depend only on the dimensionless radius ρ, the speed ratio s, and the signed causal-root ledger.

For the principal partner branch, let $\xi_p = \delta_p/2$. The delay equation is

$$\cos \xi_p = \frac{\xi_p}{s}, \qquad 0 < \xi_p < \frac{\pi}{2}$$

with

$$J_p = 1 + s\sin\xi_p$$

and branch coefficients

$$P_{\text{rad}}(\xi_p, s) = \frac{1}{\cos\xi_p\,|J_p|}, \qquad P_{\tan}(\xi_p, s) = \frac{\sin\xi_p}{\cos^2\xi_p\,|J_p|}$$

where radial is measured inward and tangential is measured in the direction of motion.

Each partner row uses the same coefficient form with its own half-angle. For a signed self branch $\alpha_s = (\xi, \sigma)$ in the full circular ledger, use

$$\sigma\sin\xi = \frac{\xi}{s}, \qquad \sigma = \text{sign}(\sin\xi)$$

with

$$J_s(\xi, \sigma; s) = 1 - s\sigma\cos\xi$$

The outward radial and signed tangential coefficients are

$$S_{\text{rad}}(\xi, \sigma; s) = \frac{s}{\xi|J_s|}, \qquad S_{\tan}(\xi, \sigma; s) = \frac{s^2\sigma\cos\xi}{\xi^2|J_s|}$$

Higher self-root births occur at tangencies:

$$\tan\xi^\star = \xi^\star, \qquad s^\star = |\sec\xi^\star|$$

and these births are also Jacobian-null events, $J_s = 0$.

On a fixed signed ledger b, the dimensionless circular MCB candidate equations are therefore

$$\mathcal{G}^{(b)}_{\text{rad}}(\rho, s) = \frac{1}{4\rho^2}\left(\sum_{\alpha_p\in b_p} P_{\text{rad}}(\alpha_p; s) - \sum_{\alpha_s\in b_s} S_{\text{rad}}(\alpha_s; s)\right) - \frac{s^2}{\rho} = 0$$

and

$$\mathcal{G}^{(b)}_{\tan}(\rho, s) = \frac{1}{4\rho^2}\left(\sum_{\alpha_p\in b_p} P_{\tan}(\alpha_p; s) + \sum_{\alpha_s\in b_s} S_{\tan}(\alpha_s; s)\right) = 0$$

Here b_p and b_s are the partner-hit and self-hit rows in the signed causal-root ledger. The equations are parameter-free because κ, ϵ, and c_f have already been absorbed into R_* and the acceleration scale. A common zero of these two residuals is only an algebraic circular MCB candidate; promotion to a stable branch still requires the finite-window return-map certificate, positive Jacobian floors, and energy packet described below.

Where Do Causal Hits Come From on the Circle? (Discrete Azimuth Pattern)

Context: Non-translating, uniform circular binary at fixed speed s. Receiver "now" at azimuth $\theta = 0$.

The emission points on the circle that can produce hits "now" form a **finite, discrete set** of azimuths determined by the delay equations–**not arbitrary locations**. Because roots are indexed by winding number m and, in the full ledger, sheet sign σ, multiple hits at the same "now" can occur for different signed windings, but the admissible azimuths remain a finite comb and never fill the circle.

Partner Hits

- Minimal angular separation: $\tilde{\delta}_p \in (0, \pi]$.
- Causal delays:

$$\delta_p(m) = \tilde{\delta}_p + 2\pi m = 2s\cos(\tilde{\delta}_p/2), \quad m = 0, 1, 2, \ldots$$

- **Emission azimuth** at reception:

$$\varphi_p(m; s) = \pi - \tilde{\delta}_p(m; s)$$

- **Existence thresholds**: For each $m \geq 0$, a solution exists only if $s > m\pi$.
- As m increases, $\tilde{\delta}_p$ decreases -> φ_p drifts monotonically toward π (diametrically opposite point).

- Partner multi-hit means $M_p(s) > 1$: the base partner branch plus one or more older partner roots. These additional roots affect the inward partner-root ledger, but they do not create same-source feedback.

Self-Hits

- Minimal angular separation: $\tilde{\delta}_s \in (0, \pi]$.
- Causal delays:

$$\delta_s(m) = \tilde{\delta}_s + 2\pi m = 2s\sin(\tilde{\delta}_s/2), \quad m = 0, 1, 2, \dots$$

- **Emission azimuth** at reception:

$$\varphi_s(m; s) = -\tilde{\delta}_s(m; s)$$

- **Existence windows**:
- Principal branch ($m = 0$): exists for every $s > 1$, with $\tilde{\delta}_s \to 0^+$ as $s \downarrow 1$.
- For $m \geq 1$: the branch appears only when the self-delay equation develops an interior tangency. The exact threshold $s_m^\star$ is determined in **Counting Self-Hits by Winding Index** above.
- Within each branch, $\tilde{\delta}_s$ initially enters at a tangency angle and then decreases with s, so φ_s drifts toward $-\pi$ at high speed.

Super-Field-Speed Root Ledgers and Resonance Lock

The super-field-speed regime is not merely the same spiral at a larger speed. It changes the root topology of the binary. Once

$$\|\mathbf{v}\| > c_f$$

the receiver can intersect multiple older causal wake surfaces from both its own path and its partner's path. In the circular reduced model, these intersections are counted by two integer ledgers:

$$N_s(s) \equiv \#\{(m, \sigma) : \text{self branch } (m, \sigma) \text{ is active at speed } s\}$$

$$M_p(s) \equiv \#\{(m, \sigma) : \text{partner branch } (m, \sigma) \text{ is active at speed } s\}$$

The self-ledger

$$N_s$$

tracks outward self-hit channels. The partner-ledger

$$M_p$$

tracks inward partner-hit channels. Both are integer-valued because a causal root either exists or it does not. As

$$s$$

varies, these counts change only at branch birth/death thresholds where a causal delay equation develops a tangency.

A candidate stable super-field-speed bound state therefore cannot be described by a single smooth force curve alone. It must satisfy a finite root-ledger balance:

$$\sum_{m \in \mathcal{M}_p(s)} A^{\mathrm{rad}}_{p,m}(R, s) - \sum_{m \in \mathcal{M}_s(s)} A^{\mathrm{rad}}_{s,m}(R, s) = \frac{s^2}{R}$$

together with whatever tangential closure condition is supplied by the full regularized dynamics. The radial equation says that partner-root accumulation supplies inward pull while self-root accumulation supplies outward response. On a fixed signed branch ledger b, the corresponding constant-speed closure target has the form

$$\left\langle \sum_{\rho \in b} T_\rho(R, s; \eta) \right\rangle_{P_b} = 0$$

where the average is taken over one candidate period P_b of the regularized history. The tangential condition remains the hard part: in the same-sheet bare isolated two-body kernel, the no-go result above shows that every active branch contributes positive tangential drive; in the full signed ledger, negative sheets must be included before any global no-go or closure theorem is claimed.

Equivalently, on a fixed signed ledger b, the circular MCB search is the intersection problem

$$G^{(b)}_{\mathrm{rad}}(R,s)=0, \qquad G^{(b)}_{\mathrm{tan}}(R,s)=0$$

where

$$G^{(b)}_{\mathrm{rad}}(R,s)\equiv\sum_{\alpha_p\in b_p}A^{\mathrm{rad}}_{\alpha_p}(R,s)-\sum_{\alpha_s\in b_s}A^{\mathrm{rad}}_{\alpha_s}(R,s)-\frac{s^2}{R}$$

and

$$G^{(b)}_{\mathrm{tan}}(R,s)\equiv\left\langle\sum_{\alpha\in b}T_\alpha(R,s;\eta)\right\rangle_{P_b}$$

with b_p and b_s denoting the partner-hit and self-hit rows inside the signed ledger b.
The first curve enforces inward/outward radial balance, while the second enforces finite-window tangential closure. In the natural Layer-I units, the search lives in $(R/R_*,s)$, so any intersection is a parameter-free candidate point for that ledger. It is still only an algebraic MCB candidate until the fixed-ledger return map proves stability, positive Jacobian floors, and persistence under perturbation.

This gives a precise, conditional meaning to binary resonance lock. A stable slot would be a region of history space in which the integer pair

$$(N_s,M_p)$$

is fixed, the branch Jacobians stay transversal, and perturbations that approach a root threshold are pushed back into the same ledger rather than escaping to a neighboring one. If such a self-map certificate exists, the discreteness of

$$N_s \quad \text{and} \quad M_p$$

would provide a deterministic mechanism for quantized bound-state geometry: allowed radii and frequencies would be selected by integer causal-root ledgers rather than by a continuum of arbitrary circular orbits.

This statement is deliberately conditional. The present chapter derives the discrete root ledgers and the radial balance target, but the stability and quantization claims require the missing full-history certificate: finite active branches, positive Jacobian floors, returned-history closure, and a monodromy or boundary-trapping argument. In practice, that certificate may close first in a collinear breather or nested shell braid setting rather than in the bare circular two-body kernel.

Branch Stability Target (Hessian Bridge)

The standard equilibrium test in central-force mechanics uses the Hessian of an instantaneous effective potential. If $q_\star$ is an equilibrium, the matrix

$$H_{ab}(q_\star)=\partial_a\partial_b V_{\mathrm{eff}}(q_\star)$$

tests local stiffness in the non-symmetry directions. This is useful as comparison language, but it is not yet a stability proof for an architrino binary because the force law depends on path-history, the active signed causal-root ledger, and the branch Jacobian floors.

The $\mathbb{AAA}$ branch-stability target is therefore a cycle-averaged stiffness matrix on a fixed branch chart. Let b denote a fixed signed causal-root ledger and let $\mathbf{X}_b(t)$ be a candidate periodic history with period P_b. For reduced branch coordinates y^a transverse to time shift, period reparameterization, Euclidean motions, and any phase-locked flat-connection moduli retained by an enclosing assembly chart, define the diagnostic stiffness target

$$K^{(b)}_{ab}=\frac{1}{P_b}\int_0^{P_b}\left.\frac{\delta^2 U^{\mathrm{hist}}_{\eta,b}}{\delta y^a\,\delta y^b}\right|_{\mathbf{x}_t=\mathbf{X}_{b,t}}dt$$

where $U^{\mathrm{hist}}_{\eta,b}$ is the action-compatible history potential, or the corresponding diagnostic reconstruction when the regularization has not yet been derived from the delayed action. Negative stiffness in this matrix is a local instability signal; positive stiffness is only a necessary reduced-coordinate check, not a certificate.

The actual branch certificate must be delayed-history and Floquet-style. Let

$$\mathcal{P}_b:\mathcal{N}_b\subset\mathcal{H}\to\mathcal{H}$$

advance an admissible history by one candidate cycle while the signed causal-root ledger remains fixed. A stable branch requires the return map to stay inside the same branch neighborhood,

$$\mathcal{P}_b(\mathcal{N}_b)\subset\mathcal{N}_b, \qquad \inf_{\phi\in\mathcal{N}_b}|J(\phi)|\geq J_{\min}>0$$

and the non-symmetry Floquet multipliers of $D\mathcal{P}_b[\mathbf{X}_b]$ to satisfy

$$|\mu_\alpha| < 1$$

Only that return-map condition would upgrade the Hessian-style stiffness picture into branch stability. If the candidate touches a branch-fold or $J = 0$ wall, this smooth Floquet test must be supplemented by the Conley-index isolating-block certificate named above; otherwise the multiplier calculation has evaluated the smooth arcs while missing the grazing transition. Until those certificates are supplied, MCB stability remains a conditional target rather than a completed proof.

Finite-dimensional projection caveat

The circular formulas below use reduced coordinates; stability in the full history space remains a separate proof obligation.

Two-Body Closure Packet (Theorem Target)

A nontrivial electrino:positrino binary is promoted only by a replayable finite-η packet, not by the circular ansatz alone. For a fixed signed causal-root ledger b, the binary closure packet is

$$\mathfrak{C}_{2\mathrm{B}}^{(\eta)} = (b, \mathbf{X}_b, P_b, R_b, s_b, \mathfrak{B}_b, \mathcal{P}_b, \mathcal{E}_b)\,,$$

where $\mathbf{X}_b(t)$ is the two-body history, P_b is its return period, R_b and s_b are the circular benchmark radius and speed when that reduction is valid, $\mathfrak{B}_b$ is the branch chart of active and excluded roots, $\mathcal{P}_b$ is the history-space return map, and $\mathcal{E}_b$ is the constructive energy packet of Delay-Dynamics Energy. The packet must report the following residuals before the branch can be used as a closed result.

The equation-of-motion residual is

$$\mathcal{R}_{\mathrm{EOM}}^{2\mathrm{B}}(b,\eta) = \frac{1}{P_b}\int_0^{P_b} \frac{\left\|\ddot{\mathbf{X}}_b(t) - F_{\eta,b}[\mathbf{X}_{b,t}]\right\|}{1 + \|F_{\eta,b}[\mathbf{X}_{b,t}]\|}\,dt,$$

where $F_{\eta,b}$ is the regularized two-body branch force obtained from the active self and partner rows in b. The period residual is

$$\mathcal{R}_{\mathrm{per}}^{2\mathrm{B}}(b,\eta) = \frac{\|\mathbf{X}_{b,P_b} - \mathbf{X}_{b,0}\|_{\mathcal{H}}}{\|\mathbf{X}_{b,0}\|_{\mathcal{H}} + \epsilon_{\mathcal{H}}},$$

with $\mathcal{H}$ the declared history norm and $\epsilon_{\mathcal{H}} > 0$ a fixed normalization floor.

The branch-chart admissibility certificate is

$$\nu_J^{2\mathrm{B}}(b,\eta) = \inf_{\rho\in b,\, 0\le t\le P_b} |J_\rho(t)| > 0, \qquad \Delta_{\mathrm{gap}}^{2\mathrm{B}}(b,\eta) = \inf_{\rho\in b^{\mathrm{off}},\, 0\le t\le P_b} |g_\rho(t)| > 0.$$

Here J_ρ is the root Jacobian for an active row and g_ρ is the signed gap of a declared inactive row in the finite branch complement b^{off}. The certificate fails if either floor tends to zero under refinement or under the advertised η-continuation.

For a circular benchmark the radial and tangential balance residual is

$$\mathcal{R}_{\mathrm{bal}}^{2\mathrm{B}} = \frac{\left|\left\langle A_{\eta,b}^{\mathrm{rad}}(R_b,s_b)\right\rangle_{P_b} - s_b^2/R_b\right|}{1 + s_b^2/R_b + \left|\left\langle A_{\eta,b}^{\mathrm{rad}}\right\rangle_{P_b}\right|} + \frac{\left|\left\langle A_{\eta,b}^{\tan} + A_{\partial W}^{\tan} + A_{\mathrm{recoil}}^{\tan}\right\rangle_{P_b}\right|}{1 + \left\langle |A_{\eta,b}^{\tan}| + |A_{\partial W}^{\tan}| + |A_{\mathrm{recoil}}^{\tan}|\right\rangle_{P_b}}.$$

The two added tangential channels are not optional bookkeeping terms: they are the boundary and recoil entries required by the constructive wake-energy ledger. If they are absent, the packet must fail closed rather than hiding tangential work in an undefined reservoir.

The stability certificate is a secular Floquet margin in history space,

$$\lambda_{\mathrm{sec}}^{2\mathrm{B}}(b,\eta) = 1 - \rho\Big(D\mathcal{P}_b[\mathbf{X}_b]\big|_{E_\perp}\Big) > 0,$$

where $E_\perp$ removes the neutral phase and symmetry directions. A numerical orbit without this projected return-map certificate is an existence candidate, not a stable binary certificate.

For a standalone circular binary, the neutral quotient includes the global time phase, the period-reparameterization direction, and Euclidean translations and rotations of the complete history. When the same two-body packet is embedded into a phase-locked tri-binary or larger assembly chart, $E_\perp$ must also quotient the flat-connection moduli declared by that enclosing chart; otherwise a slow drift of relative phase can be hidden as an allowed symmetry even though it breaks the lock.

The energy packet is

$$\mathcal{E}_b = \left(\epsilon_E^{(\eta)}(W_b;\mathfrak{B}_b), \Delta_{\mathrm{E,cross}}^{(\eta)}(W_b;\mathfrak{B}_b), U_{b,\mathrm{work}}^{(\eta)}(t), U_{\min,b}^{(\eta)}\right),$$

and must satisfy

$$\epsilon_E^{(\eta)}(W_b;\mathfrak{B}_b) \le \epsilon_E^\star, \qquad \Delta_{\mathrm{E,cross}}^{(\eta)}(W_b;\mathfrak{B}_b) \le \epsilon_{\mathrm{cross}}^\star, \qquad E_{\mathrm{wake},b}^{(\eta)}(t) \ge U_{\mathrm{min},b}^{(\eta)}$$

on the same window, branch chart, and regulator used for the motion residuals. The work reconstruction is

$$U_{b,\mathrm{work}}^{(\eta)}(t) = U_b(t_*) - \int_{t_*}^{t} \sum_i \mu_{\mathrm{arch}}\, \mathbf{a}_{i,b}^{(\eta)}(t') \cdot \mathbf{v}_i(t')\, dt'$$

for the quadratic proxy, with $\mu_K(\|\mathbf{v}_i\|)$ replacing μ_{arch} when the primitive kinetic scalar is used. The lower-bound entry applies to the constructed action-level wake charge when that route is available, or to the compatible work reconstruction when that is the declared route. This is the handoff point to the constructive delay-energy chapter: ordinary Noether language is not sufficient until $E_{\mathrm{wake},b}^{(\eta)}$ or its compatible work-integral reconstruction has been constructed for the chosen chart.

Finally, the characteristic frequency is extracted from the return period,

$$\omega_b = \frac{2\pi}{P_b}, \qquad \mathcal{R}_\omega^{2\mathrm{B}} = \frac{|2\pi/P_b - s_b/R_b|}{|2\pi/P_b| + |s_b/R_b| + \epsilon_\omega},$$

when the circular reduction is claimed. For a noncircular branch, $\omega_b = 2\pi/P_b$ remains the fundamental return frequency, but the s_b/R_b comparison is inadmissible unless an effective radius and speed have been independently defined. A breather or spiral candidate must instead report its harmonic-extraction rule on the retained history record and compare the extracted fundamental or locked harmonic to $2\pi/P_b$.

The theorem target is therefore:

> If a finite-η branch supplies $\mathfrak{C}_{2\mathrm{B}}^{(\eta)}$ with $\mathcal{R}_{\mathrm{EOM}}^{2\mathrm{B}}$, $\mathcal{R}_{\mathrm{per}}^{2\mathrm{B}}$, $\mathcal{R}_{\mathrm{bal}}^{2\mathrm{B}}$, and $\mathcal{R}_\omega^{2\mathrm{B}}$ below declared tolerances, $\nu_J^{2\mathrm{B}}$ and $\Delta_{\mathrm{gap}}^{2\mathrm{B}}$ bounded away from zero, $\lambda_{\mathrm{sec}}^{2\mathrm{B}} > 0$, and the constructive energy residuals closed on the same branch chart, then that branch is a certified local electrino:positrino two-body binary at that finite regulator.

No such finite-η packet is supplied in this chapter yet. The status is a theorem target and simulation closure contract, not a closed proof. The $\eta \to 0$ limit, the basin measure of the branch, and the later use of the binary as a universal clock or matter standard remain separate obligations.

State Space and Well-Posedness of the Two-Body Delay System

Introduction and Scope

The master equation of motion for the architrino system constitutes a system of **State-Dependent Neutral Delay Differential Equations (SD-NDDEs)**. Unlike ordinary differential equations (ODEs) where the state is a point in $\mathbb{R}^{6N}$, the state of this system is a **function segment** representing the past history of the architrinos.

We denote the position of the i-th architrino as $\mathbf{x}_i(t) \in \mathbb{R}^3$. We work in the **Euclidean void** with fixed metric δ_{ij}.

Functional Phase Space

To define the evolution at time t, we require knowledge of the trajectory over an interval $[t-\tau_{\max}, t]$, where $\tau_{\max}$ is the maximum causal lookback time relevant to the current dynamics.

Definition 1 (The History Space)

Let $h > 0$ be a history horizon (sufficiently large to capture all active causal roots). On a smooth simple-root branch, the **smooth history space** $\mathcal{H}_{\mathrm{sm}}$ is the Banach space of continuously differentiable functions mapping the delay interval to the configuration space:

$$\mathcal{H}_{\mathrm{sm}} = C^1\left([-h,0];(\mathbb{R}^3)^N\right).$$

For a trajectory $\mathbf{x} : [-h,\infty) \to (\mathbb{R}^3)^N$, the **state at time** t, denoted $\mathbf{x}_t$, is the element of $\mathcal{H}_{\mathrm{sm}}$ on smooth charts, or of $\mathcal{H}_*$ on caustic-extension charts, given by:

$$\mathbf{x}_t(\theta) = \mathbf{x}(t+\theta), \quad \theta \in [-h,0]$$

The norm on the smooth chart is the standard C^1 sup-norm: $\|\phi\|_{\mathcal{H}_{\mathrm{sm}}} = \sup_{\theta\in[-h,0]}(\|\phi(\theta)\| + \|\dot\phi(\theta)\|)$.

Remark: We require C^1 rather than C^0 because the delay τ depends on the state (state-dependent delay). In such systems, the vector field is typically not Lipschitz continuous in the C^0 topology, endangering uniqueness.

For caustic-grazing packets this smooth space is not the whole story. The working extension is

$$\mathcal{H}_* = W^{1,\infty}\left([-h,0];(\mathbb{R}^3)^N\right),$$

with C^1 regularity retained on smooth arcs and finite impulse transitions handled by the finite-η kernel before any $\eta \to 0$ statement is made. The existence theorem below is a smooth-chart theorem. A branch that crosses a $J = 0$ wall must supply a separate impulse lemma or isolating-block continuation certificate showing that the finite-η solutions converge in $\mathcal{H}_*$ with bounded velocity and finite total impulse.

Below, $\mathcal{H}$ denotes the declared history chart for the packet being tested. Unless a caustic-extension certificate is explicitly named, $\mathcal{H} = \mathcal{H}_{\rm sm}$.

The Regularized Interaction Functional

We formalize the force term derived in the master equation.

Definition 2 (Causal Constraint Functional)

For a target architrino i at time t and source j, the delay $\tau_{ij}(t)$ is implicitly defined by the causal-isochron condition. Let $\phi \in \mathcal{H}$ be the history. A **causal root** is a value $\tau > 0$ satisfying:

$$g_{ij}(\tau, \phi) \equiv \|\phi_i(0) - \phi_j(-\tau)\| - c_f \tau = 0$$

Lemma 1 (Regularity of the Delay Map)

Assumption: The velocities are sub-field-speed relative to the separation, i.e., $\|\mathbf{v}_j\| < c_f$ (single-root regime) OR we isolate a specific branch of the multi-root solution where the relative radial velocity is not c_f.

Statement: If $\phi \in \mathcal{H}$ and τ^* is a simple root of $g_{ij}(\tau, \phi) = 0$ (i.e., $\partial_\tau g_{ij} \neq 0$), then there exists a neighborhood $U \subset \mathcal{H}$ of ϕ and a continuously differentiable functional $\tau : U \to \mathbb{R}^+$ such that $\tau(\phi) = \tau^*$.

Proof.
Define

$$g_{ij}(\tau, \phi) = \|\phi_i(0) - \phi_j(-\tau)\| - c_f \tau$$

Because $\phi \in C^1$, the evaluation maps $\phi \mapsto \phi_i(0)$ and $(\tau, \phi) \mapsto \phi_j(-\tau)$ are C^1, hence g_{ij} is C^1 on $\mathbb{R}^+ \times \mathcal{H}$. At a root (τ^*, ϕ),

$$\partial_\tau g_{ij} = -\hat{\mathbf{r}}_{ij} \cdot \dot{\phi}_j(-\tau^*) - c_f, \quad \hat{\mathbf{r}}_{ij} \equiv \frac{\phi_i(0) - \phi_j(-\tau^*)}{\|\phi_i(0) - \phi_j(-\tau^*)\|}$$

The simple-root condition is exactly $\partial_\tau g_{ij} \neq 0$, i.e. no delayed tangency/causal-shock degeneracy. Therefore, by the Banach-space Implicit Function Theorem, there exist a neighborhood U of ϕ and a unique C^1 map $\tau : U \to \mathbb{R}^+$ with $g_{ij}(\tau(\psi), \psi) = 0$ and $\tau(\phi) = \tau^*$. □

Definition 3 (Regularized Acceleration Functional)

To ensure the vector field is Lipschitz, we replace the distributional Dirac delta of the master equation with the mollifier δ_η (see Master Equation). The acceleration functional $F_i : \mathcal{H} \to \mathbb{R}^3$ is:

$$F_i(\phi) = \sum_j \kappa \sigma_{ij} |q_i q_j| \int_{-h}^{0} \frac{\phi_i(0) - \phi_j(\theta)}{\|\phi_i(0) - \phi_j(\theta)\|^3} \, \delta_\eta \left(\|\phi_i(0) - \phi_j(\theta)\| + c_f \theta \right) \, d\theta$$

Crucial Property: For $\eta > 0$ and smooth δ_η, this integral operator maps C^1 histories to continuous accelerations.

On $\mathcal{H}_*$ this same formula is interpreted through the finite-η integral first. The admissibility claim is weaker: the packet must show bounded velocity and finite total impulse across the grazing chart before it can pass to the $\eta \to 0$ limit.

Local Well-Posedness

Theorem 1 (Local Existence and Uniqueness)

Assumptions:

1. $\eta > 0$, and δ_η is C^1 with bounded value and bounded derivative.
2. Initial history $\phi^0 \in \mathcal{H}$ is admissible: there exists $d_{\min} > 0$ such that all interaction channels used by Definition 3 satisfy

$$\|\phi_i(0) - \phi_j(\theta)\| \geq d_{\min}, \qquad \theta \in [-h, 0]$$

on a neighborhood of ϕ^0.
3. Delay roots used in channel construction are simple (transversal), i.e. no causal-shock degeneracy (Lemma 1).
4. Active branches are uniformly finite on the considered history neighborhood.
5. Couplings and polarity magnitudes are finite.
6. Optional higher-smoothness gluing condition at $t = 0$ (needed for C^2 at the junction, not for C^1 well-posedness).

Statement:
Let $\mathbf{Y} = (\mathbf{x}, \mathbf{v})$ and write the system in first-order form

$$\dot{\mathbf{Y}}(t) = \mathcal{G}(\mathbf{Y}_t), \qquad \mathbf{Y}_{t_0} = \phi^0$$

Then there exists $T > 0$ and a unique C^1 solution on $[t_0 - h, t_0 + T)$.
Equivalently, there is a unique maximal solution interval

$$[t_0 - h, t_{\max}), \qquad t_{\max} > t_0$$

If the optional gluing condition holds, the solution is C^2 at t_0.

Proof.
Define

$$\mathcal{G}(\phi) = (\phi_v(0), F(\phi))$$

with F from Definition 3.

1. By Assumption 2, every denominator in the interaction kernel is bounded away from zero on the admissible neighborhood; therefore the map

$$(\mathbf{u}, \mathbf{w}) \mapsto \frac{\mathbf{u} - \mathbf{w}}{\|\mathbf{u} - \mathbf{w}\|^3}$$

is C^1 there with bounded derivative.
2. By Assumption 1, composition with δ_η preserves C^1 regularity and bounded derivatives.
3. By Lemma 1 and Assumption 3, delay branches (where used) depend C^1 on history; thus branch-evaluation maps are locally Lipschitz in ϕ.
4. Finite sums over channels and integration over finite interval $[-h, 0]$ preserve local Lipschitz continuity; hence $\mathcal{G}$ is locally Lipschitz on an open subset of $\mathcal{H}$ containing ϕ^0.
5. Apply the standard Banach-space existence/uniqueness theorem for state-dependent DDEs: a unique local C^1 solution exists and extends uniquely to a maximal interval.

Therefore Theorem 1 holds. □

Global Existence vs. Blow-Up

Unlike Newtonian gravity, global existence is **not guaranteed** simply by avoiding collisions, because the delay equation can harbor "runaway" modes where self-acceleration diverges.

Theorem 2 (Continuation Principle)

The solution $\mathbf{x}(t)$ can be extended as long as the state $\mathbf{x}_t$ remains within a compact subset of the phase space where causal roots are simple.

Definition 4 (Blow-Up Criteria)

The solution ceases to exist at finite time T^* if:

1. **Collision:** $\inf_{i,j} \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| \to 0$ inside the regularization kernel support.
2. **Infinite Speed:** $\sup_i \|\mathbf{v}_i(t)\| \to \infty$.
3. **Causal Shock:** The derivative of the delay $\dot{\tau}(t)$ diverges (Doppler factor becomes singular). This occurs if an architrino moves directly toward a receiver at speed $\|\mathbf{v}\| = c_f$.

Symmetry, Conservation, and Lyapunov Functionals

Introduction

Standard conservation laws (energy, momentum, angular momentum) rely on the application of Noether's theorem to local Lagrangian densities. In this delayed setting, the force at time t depends on the phase-space trajectory over the interval $[t - h, t]$.

For an action-derived, symmetry-preserving delayed model, symmetries of the substrate (Euclidean void + absolute time) imply conservation laws, but the conserved quantities are no longer simple functions of the instantaneous state $(\mathbf{x}, \mathbf{v})$. Instead, they are **functionals on the history**

space $\mathcal{H}$. For a working regularized kernel not yet derived from an action, the same expressions function as validation diagnostics rather than established Noether charges.

This section derives these functionals, establishes the exact symmetry group of the regularized dynamics ($\eta > 0$), and provides the *a priori* bounds required to ensure physical well-posedness (preventing unphysical runaway acceleration).

The Global Symmetry Group

We consider the regularized two-body system in the Euclidean void $\mathbb{R}^3$ with metric δ_{ij} and absolute time t.

Definition 1 (The Fundamental Symmetry Group)

The background substrate and the master equation interaction kernel

$$\mathbf{a}_{ij}(t) \propto \frac{\mathbf{x}_i(t) - \mathbf{x}_j(t_0)}{\|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|^2 \, |J_{ij}(t; t_0)|}$$

(regularized by η) respect the group:

$$G_{\text{fund}} = E(3) \times \mathbb{R}_{\text{time}}$$

where $E(3) = \mathbb{R}^3 \rtimes O(3)$ is the Euclidean group of spatial translations and rotations, and $\mathbb{R}_{\text{time}}$ denotes time translation.

Theorem 1 (Invariance of the Equations of Motion)

Let $\mathbf{x}(t)$ be a solution to the master equation.

1. **Time Translation:** For any $\tau \in \mathbb{R}$, $\mathbf{y}(t) = \mathbf{x}(t + \tau)$ is also a solution.
2. **Spatial Isometry:** For any $R \in O(3)$ and $\mathbf{b} \in \mathbb{R}^3$, $\mathbf{y}(t) = R\mathbf{x}(t) + \mathbf{b}$ is also a solution.

Proof.
For time translation, set $\mathbf{y}_i(t) = \mathbf{x}_i(t + \tau)$. If $t_0 \in \mathcal{C}_{ij}^x(t + \tau)$ for the original solution, then $t_0 - \tau \in \mathcal{C}_{ij}^y(t)$ because

$$\|\mathbf{y}_i(t) - \mathbf{y}_j(t_0 - \tau)\| = \|\mathbf{x}_i(t + \tau) - \mathbf{x}_j(t_0)\| = c_f[(t + \tau) - t_0] = c_f[t - (t_0 - \tau)]$$

Hence the same branch contributions appear with shifted times, and $\ddot{\mathbf{y}}_i(t) = \ddot{\mathbf{x}}_i(t + \tau)$ satisfies the same force law.

For spatial isometries, set $\mathbf{y}_i(t) = R\mathbf{x}_i(t) + \mathbf{b}$, $R \in O(3)$. Distances are preserved:

$$\|\mathbf{y}_i(t) - \mathbf{y}_j(t_0)\| = \|R(\mathbf{x}_i(t) - \mathbf{x}_j(t_0))\| = \|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|$$

so causal-root times are unchanged. Unit directions transform covariantly: $\hat{\mathbf{r}}_{ij}^y = R\hat{\mathbf{r}}_{ij}^x$. Therefore each force term transforms as $\mathbf{a}_{ij}^y = R\mathbf{a}_{ij}^x$, and

$$\ddot{\mathbf{y}}_i(t) = R\ddot{\mathbf{x}}_i(t) = \sum_j \sum_{t_0 \in \mathcal{C}_{ij}(t)} \kappa \sigma_{ij} \frac{|q_i q_j|}{r_{ij}^2 \, |J_{ij}(t; t_0)|} \hat{\mathbf{r}}_{ij}^y$$

Thus $\mathbf{y}$ solves the same equations. □

Implication: In an action-derived regularization, these symmetries correspond to exact history-space integrals of motion. Because the interaction is non-local in time, those integrals must account for momentum and energy carried by causal wake surfaces rather than only by the instantaneous mechanical coordinates.

Conservation of Generalized Momentum

In a delay system, Newton's Third Law ($\mathbf{F}_{12}(t) = -\mathbf{F}_{21}(t)$) fails instantaneously because $\mathbf{F}_{12}(t)$ originates from architrino 2 at $t - \tau_1$, while $\mathbf{F}_{21}(t)$ originates from architrino 1 at $t - \tau_2$.

Definition 2 (Mechanical Momentum)

The instantaneous mechanical momentum is:

$$\mathbf{P}_{\text{mech}}(t) = \sum_i \mu_{\text{arch}} \mathbf{v}_i(t)$$

Because of the delay, $\frac{d}{dt}\mathbf{P}_{\text{mech}} \neq 0$ generally.

Conservation Target 2 (Total Momentum Functional)

For an action-derived delayed model with translation symmetry, there exists a functional $\mathbf{P}_{\text{wake}}[\mathbf{x}_t]$ representing the momentum flux encoded in the active causal wake surfaces such that the total momentum:

$$\mathbf{P}_{\text{tot}} = \mathbf{P}_{\text{mech}}(t) + \mathbf{P}_{\text{wake}}[\mathbf{x}_t]$$

is conserved. For working regularized models, this same expression is a validation diagnostic unless the chosen regularization preserves the translation symmetry of the underlying action.

Explicit Form (Weak Coupling Limit):
For $\eta \to 0$, the wake momentum can be approximated by integrating the force impulse over the delay time:

$$\mathbf{P}_{\text{wake}} \approx \sum_{i \neq j} \int_{t-\tau_{ij}(t)}^{t} \mathbf{F}_{ij}^{\text{emit}}(s)\, ds$$

Physical interpretation: The "missing" momentum is accounted for by the causal wake surfaces currently traversing the space between sources and receivers in an action-derived model; otherwise this balance is the momentum diagnostic to verify.

Corollary (Center of Mass Motion):
For an isolated binary, the center of mass $\mathbf{x}_{\text{cm}}$ need not move at constant velocity in the mechanical coordinates alone. Instead, it can oscillate around a mean trajectory while wake momentum carries the compensating history term. This is the two-body version of the center-of-response theorem target: in an exactly symmetric circular binary, the exposed-energy response center $\mathbf{X}_{\text{resp}}$ is pinned to the circle center by symmetry, while the particle-only mechanical center can still show finite-window oscillatory bookkeeping if wake momentum is not included. A runaway center-of-mass self-acceleration is forbidden only in an action-derived model whose regularization preserves translation symmetry; in working regularized models this is a conservation diagnostic to be checked.

Energy and The Lyapunov Functional

Energy conservation is the critical constraint preventing runaway solutions (MCB-09).

Definition 3 (The History Hamiltonian)

For an action-derived delayed model with time-translation symmetry, the target conserved quantity $\mathcal{H}$ is a history functional. For state-dependent delays, the useful comparison object is a **Lyapunov-Krasovskii-style functional**:

$$\mathcal{H}(\mathbf{x}_t) = K(\mathbf{v}(t)) + \mathcal{U}_{\text{history}}(\mathbf{x}_t)$$

1. **Kinetic Energy:** $K(t) = \sum \frac{1}{2}\mu_{\text{arch}} \|\mathbf{v}_i(t)\|^2$.
2. **Potential Functional:** $\mathcal{U}_{\text{history}}$ accumulates the work done by the conservative forces. Unlike an instantaneous potential $V(r)$, this depends on the configuration of all active wake surfaces.

Theorem 3 (Energy Balance Equation)

$$\frac{dK}{dt} = \sum_i \mathbf{v}_i(t) \cdot \mathbf{F}_i(t)$$

We define the **Interaction Potential Functional** $\mathcal{W}(t)$ such that:

$$\mathcal{W}(t) = -\int_{t_0}^{t} \sum_i \mathbf{v}_i(s) \cdot \mathbf{F}_i(s)\, ds$$

This functional is nonlocal in time: it accumulates deferred work along the path-history of wakes and is not an instantaneous potential $U(r)$. Then, by construction along the realized trajectory, $\mathcal{E}_{\text{tot}} = K(t) + \mathcal{W}(t)$ is constant. It is an exact Noether charge only when $\mathcal{W}$ is the boundary term of the same symmetry-preserving delayed action; otherwise it is a diagnostic reconstruction.

Lemma 1 (Boundedness of the Potential)
Assumption: The interaction is regularized with width $\eta > 0$ such that the maximum force is bounded: $\|\mathbf{F}_{ij}\| \le F_{\max}(\eta)$.
Statement: For a bound system (architrinos confined to a finite volume V), the rate of work is bounded by $N F_{\max} v_{\max}$.

Conditional Target 4 (No-Runaway Criterion)

This criterion is not a completed theorem until the same symmetry-preserving regularized action supplies $\mathcal{W}$ on the retained branch chart and a lower bound is proven for that branch. Under those hypotheses, in an action-derived master-equation branch with fixed $\eta > 0$, an isolated binary cannot undergo runaway acceleration ($\|\mathbf{v}\| \to \infty$) *unless* the action-compatible potential energy functional $\mathcal{W}(t)$ diverges to $-\infty$.

Proof Logic:
Since $\mathcal{E}_{\mathrm{tot}}$ is constant:

$$K(t) = \mathcal{E}_{\mathrm{tot}} - \mathcal{W}(t)$$

For $K(t)$ to diverge, $\mathcal{W}(t)$ must decrease without bound.

1. **Partner attraction:** $q_1 q_2 < 0$. The potential is negative (attractive). As $r \to 0$, $V \to -\infty$. Collapse leads to infinite kinetic energy in the standard Kepler singularity pattern; in this architecture, self-hit is the proposed counter-channel.
2. **Self-hit repulsion:** $q_1 q_1 > 0$. The force is **repulsive**. The potential contribution is **positive**.

- Work done by self-hit: If an architrino is pushed “from behind” by its own wake, it gains K.
- However, this energy must come from the $\mathcal{W}$ term.
- Since self-hit potential is repulsive (positive energy hill), converting it to kinetic energy lowers the total potential.
- **Crucial bound:** The deferred work encoded in a self-wake is finite when the emitted causal-wake budget is finite. An architrino cannot extract infinite energy from its own past unless the history functional has already assigned an infinite budget to that causal wake.

Conclusion: A self-acceleration runaway, where an architrino accelerates itself indefinitely using self-forces, is excluded only on branches satisfying the action-derived conservation and lower-bound hypotheses. In other working models, the same statement is a validation target: the system can oscillate or settle, but an apparent explosion to $\|\mathbf{v}\| = \infty$ must be traced either to singular collapse, transversality loss, or a broken conservation diagnostic.

Tri-Binary Configuration Space

This chapter gives the general dynamics-facing search space for tri-binary Noether braid candidates. It comes before any named configuration such as the dyadic `4:2:1` lock, an equal-frequency candidate, or a middle-hinge candidate. A tri-binary branch is first a three-layer retained state whose energies, phase offsets, angular-momentum rows, plane orientations, causal-root ledgers, frequencies, radii, speeds, and whole-branch group velocity must be solved together.

This is a search architecture and theorem target, not a completed classification theorem. The goal is to find which regions of the tri-binary configuration space support stable retained branches in a Noether sea populated by like assemblies, identify which of those branches are eigen-braid candidates, and then use those branches as the entry point for assembly topological charge, energy differentials, shielding, and accessory-architrino capture.

Scope Of The Hypothesis

The tri-binary hypothesis is a decomposition strategy, not an exhaustion theorem. There may be stable Noether braid configurations that do not admit a clean split into three persistent binary rows. The reason to study tri-binaries first is that three independent angular-momentum directions are enough to span the orientation data of Euclidean three-space. In that sense, a tri-binary is the minimal exact-binary decomposition that can test whether a stable assembly carries a full three-dimensional internal frame.

This also means that the word `binary` names a retained angular-momentum row, not necessarily a perfectly circular two-body orbit at every instant. A certified row may have a conserved or slowly bounded angular-momentum ledger while the actual architrino paths on the retained support are quasiperiodic, braided, or chaotic. On such a row, f_a is a return or winding frequency, r_a is a characteristic lever arm, s_a is a speed row or speed statistic, and E_a is the retained branch-energy row. A circular carrier chart is the cleanest comparison case, not the only admissible path geometry.

Why Three Binaries

The reason to begin with tri-binaries is geometric. Euclidean space has three independent spatial directions, and a stable three-dimensional assembly needs enough internal direction data to define an orientation frame rather than only a planar cycle. A single binary supplies one orbital plane and one plane normal. Two binaries can define a relative angle, but they do not by themselves supply a full nondegenerate three-axis frame. Three binaries can, when their plane normals are independent, define a local three-dimensional frame.

Let the three retained binary planes have unit normals

$$\hat{\mathbf{n}}_1,\ \hat{\mathbf{n}}_2,\ \hat{\mathbf{n}}_3.$$

The plane-orientation nondegeneracy measure is

D_{\mathrm{plane}} = \det \left[\hat{\mathbf n}_1 & \hat{\mathbf n}_2 & \hat{\mathbf n}_3 \right].

The branch is genuinely three-dimensional only when $D_{\mathrm{plane}} \neq 0$. Near $|D_{\mathrm{plane}}| = 1$, the three planes are close to mutually orthogonal. Near $D_{\mathrm{plane}} = 0$, the tri-binary degenerates toward a coplanar or lower-dimensional support. This determinant is therefore a natural order parameter for the transition between a volumetric Noether braid branch and a planar or horizon-aligned branch.

This is a statement about a derived orientation frame, not a claim that the constituent architrino paths are axial. The actual six paths may be braided, quasiperiodic, chaotic, shell-supported, or otherwise noncircular while still emitting retained angular-momentum rows from which principal directions can be extracted. Axis language in this chapter therefore means a ledger or envelope direction derived from the branch record, not a primitive path pattern.

The claim is not that every stable assembly must be an exact tri-binary. The broader Noether braid class permits six-body branches before exact binary grouping is certified. The tri-binary search is the minimal exact-binary architecture that can test full three-dimensional frame closure.

General Branch State

Use generic layer labels $a \in \{1,2,3\}$ before assigning nested `I:M:0` roles. These labels are bookkeeping labels only. They do not imply an ordering of frequency, radius, energy, speed, phase, plane orientation, or root-ledger complexity. The minimal tri-binary branch record is

$$\mathcal{T}_{3B} = \{(f_a,\, r_a,\, E_a,\, s_a,\, \phi_a,\, \hat{\mathbf{n}}_a,\, \mathcal{L}_a)\}_{a=1}^{3}\,.$$

Here f_a is the layer frequency or return rate, r_a is the characteristic radius or retained lever arm, E_a is the retained branch-energy row, $s_a = \|\mathbf{v}_a\|$ is the scalar tangential speed or speed statistic, ϕ_a is the phase origin or offset, $\hat{\mathbf{n}}_a$ is the orbital-plane normal, and $\mathcal{L}_a$ is the active causal-root ledger data for that layer. On a circular carrier chart,

$$s_a = 2\pi f_a r_a.$$

This identity is kinematic only. It does not select the frequencies, radii, speeds, energies, phase offsets, plane orientations, or causal-root ledgers.

The practical search should treat the branch energy row E_a, angular-momentum row, phase data, and causal-root ledger $\mathcal{L}_a$ as primary retained data. The radius and speed are then constrained by the selected carrier chart, conservation laws, and the branch's energy closure. In simple circular rows, fixed f_a and E_a may determine an admissible r_a and s_a after the kinetic, binding, and wake-energy terms are specified. In noncircular rows, the same energy may correspond to a bounded family of paths with the same return frequency but different local speed profile. Thus energy is central, but it is not by itself a complete coordinate on the tri-binary configuration space.

Branch Group Velocity

The internal plane data do not encode group velocity. The plane normals $\hat{\mathbf{n}}_a$ describe the assembly's internal angular-momentum frame. The group velocity is the drift of the retained branch envelope or response center through the local Noether sea:

$$\mathbf{V}_{\text{grp}} = \frac{d\mathbf{X}_{\text{resp}}}{dt} \quad \text{relative to the declared Noether sea record.}$$

The full branch record should therefore be read as

$$B_{3B} = (\mathcal{T}_{3B},\, \mathbf{X}_{\text{resp}},\, \mathbf{V}_{\text{grp}},\, \mathbf{P}_{\mathfrak{B}},\, \mathbf{J}_{\mathfrak{B}},\, \theta_{\text{sea}})\,,$$

where $\mathbf{P}_{\mathfrak{B}}$ and $\mathbf{J}_{\mathfrak{B}}$ are the branch-total momentum and angular-momentum ledgers, and θ_{sea} is the local Noether sea response record used to compare moving branches.

This distinction matters for the equivalence-principle and Lorentz-closure programs. In a validated low-energy regime, uniform group velocity should not become an observable composition-dependent force merely because two assemblies carry different internal plane orientations. That is an effective recovery target: the moving branch must retune its clock, ruler, and signal rows so that preferred-frame leakage stays below the declared bounds. It is not a reason to omit $\mathbf{V}_{\text{grp}}$ from the dynamics. The correct statement is that $\mathbf{V}_{\text{grp}}$ is a separate branch-transport variable whose observable leakage must be suppressed by common-channel closure.

Eigen-Braid Candidates

An **eigen-braid** is a theorem-target status for a Noether braid branch, not a new primitive substance. A branch is an eigen-braid candidate when its full retained record returns to itself under the delayed dynamics, up to declared symmetries. Since the dynamics are nonlinear and path-history dependent, `eigen` here means a relative fixed point or relative periodic orbit of the branch return map, not an eigenvector of a linear quantum operator.

The retained record is not an arbitrary internal diary and it is not an arbitrary collection of architrinos. It is the finite branch chart for one Noether braid: the six-body polarity-neutral inventory of three positive-polarity and three negative-polarity architrinos, together with the path-history rows, causal-root ledger, wake-tail rows, energy/action rows, momentum and angular-momentum rows, phase data, plane-orientation data, response-center data, group-velocity row, and Noether sea record that can still affect the next delayed update of that same six-body branch. A path-history segment belongs to the retained record only while it can still enter a self-hit, partner-hit, wake-tail, boundary, or branch-return row on the declared memory window.

Let $P_T^{(\mathbf{V})}$ be the finite-memory return map over one branch period T, including translation by the branch group velocity $\mathbf{V}_{\mathrm{grp}}$. Let $\mathcal{G}_{\mathrm{sym}}$ contain only declared neutral symmetries such as global phase shift, rigid spatial rotation, translation of the response center, and permitted S_3 layer relabeling. A tri-binary branch B_{3B} is an eigen-braid candidate when there exists $g \in \mathcal{G}_{\mathrm{sym}}$ such that

$$\mathcal{R}_{\mathrm{eig}} = d_{\mathfrak{B}} \left(P_T^{(\mathbf{V})}(B_{3B}),\, g \cdot B_{3B} \right) \le \epsilon_{\mathrm{eig}},$$

on the same retained branch chart $\mathfrak{B}$, with the non-symmetry return directions carrying a positive stability margin. The metric $d_{\mathfrak{B}}$ must compare the same branch rows: causal-root ledger, energy/action ledger, angular-momentum rows, phase data, plane-orientation data, response-center motion, group velocity, Noether sea record, and assembly topological charge.

This definition keeps Lorentz behavior downstream. The branch-intrinsic conserved record may later be exported to observer components through a derived moving-assembly map,

$$C_{\mathrm{obs}} = \Lambda_{\mathrm{eff}} \left(\mathbf{V}_{\mathrm{grp}}, \theta_{\mathrm{sea}} \right) C_{\mathrm{branch}} + O(\epsilon_{\mathrm{LV}}),$$

when Lorentz closure applies. The export may dress energy-momentum, angular-momentum components, clock rates, and ruler geometry, but it does not define the eigen-braid itself. Topological rows such as assembly topological charge remain branch-intrinsic invariants unless the branch crosses a fold, reconnection, or other declared surgery event.

Momentum And Principal-Direction Decomposition

An eigen-braid candidate should also say how its internal tri-binary rows align with the conserved momentum ledgers. A branch whose retained record returns but whose axes do not align with branch-total momentum and angular momentum remains a return-map candidate, not a promoted eigen-braid. The branch-total momentum and angular momentum should be computed on the same finite window as the return map:

$$\mathbf{P}_{\mathfrak{B}} = \mathbf{P}_{\mathrm{mech}} + \mathbf{P}_{\mathrm{wake}}, \qquad \mathbf{J}_{\mathfrak{B}} = \mathbf{J}_{\mathrm{mech}} + \mathbf{J}_{\mathrm{wake}}.$$

The mechanical and wake terms must use the same endpoint convention as the retained branch chart; otherwise the axis comparison is only a visualization.

When $\|\mathbf{P}_{\mathfrak{B}}\| > 0$, the unit vector

$$\hat{\mathbf{e}}_P = \frac{\mathbf{P}_{\mathfrak{B}}}{\|\mathbf{P}_{\mathfrak{B}}\|}$$

is the transport axis. When $\|\mathbf{J}_{\mathfrak{B}}\| > 0$, the unit vector

$$\hat{\mathbf{e}}_J = \frac{\mathbf{J}_{\mathfrak{B}}}{\|\mathbf{J}_{\mathfrak{B}}\|}$$

is the branch's total angular-momentum axis. The tri-binary plane normals $\hat{\mathbf{n}}_a$ should then be read as a principal-direction decomposition of $\mathbf{J}_{\mathfrak{B}}$, not as arbitrary visual decoration and not as a claim that the paths themselves lie on axes. A simple diagnostic is the angular-momentum closure vector

$$\mathcal{R}_{J-\mathrm{axis}} = \left\| \hat{\mathbf{e}}_J - \frac{\sum_{a=1}^{3} w_a \hat{\mathbf{n}}_a}{\left\| \sum_{a=1}^{3} w_a \hat{\mathbf{n}}_a \right\|} \right\|,$$

where the weights w_a are declared branch-action, branch-angular-momentum, or energy-row weights and the weighted normal sum is required to be nonzero. This is not yet a theorem: it is the axis-alignment row a solver must populate before claiming that the tri-binary rows faithfully decompose the assembly's conserved angular momentum.

The oblate spheroidal envelope is the coarse geometry associated with this decomposition. In the rest branch, $\mathbf{P}_{\mathfrak{B}} = 0$, so the internal angular-momentum axes and plane determinant describe the retained three-dimensional support. In a moving branch, $\hat{\mathbf{e}}_P$ marks the drift direction relative to the Noether sea, and Lorentz-closure asks whether the envelope deforms with a longitudinal-to-transverse ratio

$$\xi = \frac{R_{\parallel}}{R_{\perp}}, \qquad R_{\parallel} \text{ measured along } \hat{\mathbf{e}}_P,$$

while the same internal angular-momentum ledger remains retained. Thus the tri-binary picture is also a disciplined way to visualize an oblate spheroidal Noether braid: the three retained rows decompose the internal angular momentum into principal directions, while group velocity and total momentum select the moving-envelope axis.

Unordered Layer Semantics

The search must not assume that one binary is inner, middle, outer, high-frequency, low-frequency, high-energy, low-energy, fast, slow, or geometrically privileged before the retained branch supplies that role. The raw search domain is therefore the labeled but unordered product

$$\tilde{\mathcal{C}}_{3B} = \{(\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3) : \mathcal{T}_a = (f_a, r_a, E_a, s_a, \phi_a, \hat{\mathbf{n}}_a, \mathcal{L}_a)\}\,.$$

The symmetric group S_3 acts on this space by permuting the three binary records:

$$\pi \cdot (\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3) = (\mathcal{T}_{\pi^{-1}(1)}, \mathcal{T}_{\pi^{-1}(2)}, \mathcal{T}_{\pi^{-1}(3)}), \qquad \pi \in S_3.$$

Two rows may therefore be the same physical candidate up to a relabeling even when they appear as distinct solver outputs.

The default search policy is to keep $\tilde{\mathcal{C}}_{3B}$ unquotiented. Repeated S_3-related solutions are useful confirmation that the solver is finding a symmetric sector rather than a one-off artifact. An analysis tool may later isolate one representative sector by computing a permutation-invariant key,

$$\mathrm{key}(B) = \mathrm{sort}_{a=1}^{3}\,\mathrm{fingerprint}(\mathcal{T}_a),$$

but that quotient is an analysis summary, not the search domain. No branch is rejected merely because a symmetric relabeling has already appeared.

The general configuration ratios are

$$f_1 : f_2 : f_3, \qquad r_1 : r_2 : r_3, \qquad E_1 : E_2 : E_3, \qquad s_1 : s_2 : s_3.$$

These ratios are reported in the current layer labels. They are not sorted ratios and they carry no inequality unless a retained branch later assigns a role order.

The branch-search problem is to find retained stable states

$$\mathcal{T}_{3B} \in \tilde{\mathcal{C}}_{3B}$$

over this full variable set, then compare their energy differentials

$$\Delta E_{ab} = E_a - E_b$$

and ledger decompositions on the same retained row set. The dyadic, equal-frequency, and offset-hinge families are subfamilies of $\tilde{\mathcal{C}}_{3B}$, not definitions of it.

Super-Field-Speed Carrier Rows

The general search naturally includes carrier speeds above the causal wake propagation speed. Since

$$s_a = 2\pi f_a r_a,$$

fixing one row of the search does not fix the others. Even an equal-frequency family

$$f_1 = f_2 = f_3$$

can have different radii, energies, speeds, phases, and active root ledgers:

$$r_1 : r_2 : r_3 \neq 1 : 1 : 1, \qquad s_1 : s_2 : s_3 \neq 1 : 1 : 1.$$

If one retained lever arm is large enough at the common frequency, then that layer has $s_a > c_f$.

This is not a signal-speed claim. The primitive causal wake still propagates at c_f. A row with $s_a > c_f$ is a carrier-trajectory row in the retained branch chart. Its importance is dynamical: it changes the causal-root inventory. Super-field-speed carrier motion can create additional self-hit and partner-hit roots, force Jacobian sign changes, and move the branch into the fold and caustic regimes that feed the causal-root ledger. The possibility of one or more super-field-speed layers is therefore a reason to scan the full tri-binary configuration space rather than preselecting a single speed hierarchy.

Stability In A Sea Of Like Assemblies

An isolated tri-binary return map is not enough for Noether braid promotion. A retained branch must also remain stable when embedded in a Noether sea containing like assemblies. The relevant stability question is not only whether one branch closes, but whether a population of similar branches can coexist without destroying the retained ledgers.

For a candidate branch B over a window W, write the stability target schematically as

$$\mathrm{Stable}_{3B}(B; W, \mathcal{N}_{\mathrm{sea}}) \Longleftrightarrow P_{\mathrm{root}} \wedge P_{\mathrm{phase}} \wedge P_{\mathrm{energy}} \wedge P_{\mathrm{return}} \wedge P_{\mathrm{sea}}.$$

Here P_{root} requires persistent causal-root ledgers with positive root floors except at declared caustic transits, P_{phase} requires bounded phase-offset drift, P_{energy} requires a closed branch-energy row, P_{return} requires a Floquet, Conley, or comparable return certificate, and P_{sea} requires the same branch to remain coherent under the background Noether sea response generated by like assemblies. This last predicate is the bridge from an isolated branch search to a stable medium of assemblies.

The result of this search should be an atlas of stable regions in $\tilde{\mathcal{C}}_{3B}$, not a single preferred row. Patterns may include dyadic locks, equal-frequency families, offset-hinge families, caustic-hinge families, planar degenerations, and mixed regimes where one or more layers run above c_f while the whole assembly remains a retained delayed branch. If a stable region is S_3-symmetric, the atlas may also report the corresponding quotient-sector representative, but the unquotiented evidence should remain available.

Toward A Periodic Table Of The Noether Braid

The phrase “periodic table of the Noether braid” names the classification program, not an already completed table. The proposed atlas should classify retained branches by:

1. The compact assembly topological charge $[\mathfrak{B}]_{\mathrm{top}} = (N_s, M_p, c_1)$ and its signed-degree refinement.
2. The frequency, radius, energy, and speed ratios of $\mathcal{T}_{3B}$.
3. The plane-orientation determinant D_{plane} and handedness data.
4. The energy differentials ΔE_{ab} and their wake-history decomposition.
5. The response of the branch to a sea of like assemblies.
6. The capture or exclusion behavior of additional architrinos near the branch.

The classification is topological only where the entries are invariant under branch-preserving deformation. It is dynamical where the entries depend on energy balance, phase locking, sea response, and return-map stability. A promoted table must therefore carry both topological labels and dynamical margins.

Accessory-Architrino Capture

After a stable tri-binary core has been retained, the next search level asks whether ordinary architrinos can become bound to that core without destroying the core ledger. In this search-stage sense, an **accessory architrino** is not a new ontological species. It is an architrino whose trajectory becomes coupled to an already retained core branch.

For a core branch B, define a capture site as a region of phase-position-history space where an added architrino can acquire a bounded return ledger:

$$\mathcal{C}_{\mathrm{cap}}(B) = \{(\mathbf{x}, \mathbf{v}, q, \phi) : \mathrm{Retain}_{\mathrm{acc}}(B; \mathbf{x}, \mathbf{v}, q, \phi) = 1\}\,.$$

The capture predicate must use the same causal-root, action, energy, and return-map conventions as the core branch. A site is not merely a low potential region. It must preserve the core ledger while giving the added architrino a persistent delayed-return row, finite energy exchange, and bounded phase drift.

The architectural question is therefore:

$$B \longrightarrow (\mathcal{C}_{\mathrm{cap}}(B), \#\mathrm{captured}, \mathrm{capture\ pattern}, \Delta E_{\mathrm{capture}})\,.$$

This gives the next level of search after core tri-binary stability: how many accessory architrinos can couple to the retained branch, which phase windows and polar regions they occupy, and how their capture changes the energy ledger. If the captured population becomes the six-site fermion organization, the canonical language is axial architrino, axial layer, polar site, polar dyad, and axial inventory.

Relation To The Dyadic Chapter

Dyadic Resonance Lock studies one restricted family inside this broader configuration space. It asks whether a nested `I:M:O` frequency triplet, especially the dyadic `4:2:1` candidate, can close as an integer phase-bundle lock with a stable return map and controlled caustic behavior.

The dyadic chapter should therefore be read as a specialized search row:

$$\mathcal{C}_{\mathrm{dyadic}} \subset \tilde{\mathcal{C}}_{3B}.$$

Equal-frequency, unequal-radius candidates occupy a different row:

$$\mathcal{C}_{f=f=f} = \{B \in \tilde{\mathcal{C}}_{3B} : f_1 = f_2 = f_3\}.$$

Both rows are legitimate until the retained-branch certificates decide which, if either, survives. The general tri-binary search keeps the mathematics wide enough for the solver to discover stable configurations rather than forcing every stable Noether braid into a preselected frequency pattern.

Dyadic Resonance Lock

This chapter studies resonance lock for the nested Outer, Middle, and Inner binaries as a restricted family inside the broader Tri-Binary Configuration Space. Its immediate goal is specific: identify the relationship between frequency, scalar tangential speed, and radius in a reduced branch where the middle binary caustic-grazes the field-speed hinge and the three rings form an exact integer phase-locked cycle.

It should be read together with Binary Dynamics, Nested Shell Braid Dynamics, Mapping the Planck Scale, and Noether Braid, which provide the assembly geometry and scale-setting context for the lock relations derived here.

The level distinctions matter throughout. Ontologically, the Outer, Middle, and Inner binaries are assembly layers built from architrino constituents. Dynamically, the reduced model replaces their full delayed causal-wake history by a finite-η branch chart. Effectively, low-order multipoles and potentials are comparison summaries of that branch behavior. Inferentially, an integer lock is selected only after the phase-bundle holonomy, cancellation score, and stability gap all favor the same branch.

This chapter keeps the field speed c_f explicit rather than setting it to one. We work with branch labels $k \in \{O, M, I\}$. Here r_k is the characteristic layer radius and $v_k = \|\mathbf{v}_k\|$ is the scalar tangential speed of one member of layer k around that layer's center.

General Tri-Binary Branch State

Before a dyadic, equal-frequency, or other special configuration is selected, a tri-binary branch is a three-layer retained state. The general search program is defined in Tri-Binary Configuration Space; this section records the variables needed locally for the dyadic specialization. Use generic layer labels $a \in \{1, 2, 3\}$ before assigning the canonical `I:M:O` roles. These labels are not sorted by f_a, r_a, E_a, s_a, or any other parameter; permutation-related rows remain valid search evidence until an explicit quotient-sector analysis is declared. The minimal branch variables are

$$\mathcal{T}_{3B} = \{(f_a,\, r_a,\, E_a,\, s_a,\, \phi_a,\, \hat{\mathbf{n}}_a,\, \mathcal{L}_a)\}_{a=1}^{3}\,.$$

Here f_a is the layer frequency, r_a the characteristic radius or retained lever arm, E_a the retained branch-energy row, $s_a = \|\mathbf{v}_a\|$ the scalar tangential speed, ϕ_a the phase origin or offset, $\hat{\mathbf{n}}_a$ the orbital-plane normal, and $\mathcal{L}_a$ the active causal-root ledger data for that layer. On a circular layer chart the kinematic identity is

$$s_a = 2\pi f_a r_a.$$

This identity is only a constraint among three variables. It does not by itself select the frequency ratios, energy placement, radii, speeds, or phase offsets.

The branch-search objective is therefore

$$\text{find retained, stable } \mathcal{T}_{3B} \text{ over } (f_a, r_a, E_a, s_a, \phi_a, \hat{\mathbf{n}}_a, \mathcal{L}_a),$$

then compare the energy differentials

$$\Delta E_{ab} = E_a - E_b$$

and their ledger decomposition on the same retained row set. A dyadic candidate, an offset middle-hinge candidate, and an equal-frequency candidate are special subfamilies of this same state space. In particular, the equal-frequency condition

$$f_1 = f_2 = f_3$$

still permits different r_a, s_a, and E_a, because the radii or retained lever arms can differ. Different phase offsets and different active root ledgers can then carry the branch distinction even when the frequency row is common.

For nested shell braid prose, specialize the generic labels to canonical `I:M:O` order only after the retained branch supplies the role assignment. The later dyadic lock discussion studies one restricted family inside this broader tri-binary branch state; it is not the default assumption for all stable tri-binary configurations.

Status and Assumptions

The logic of the chapter is organized around one exact identity and four explicit assumptions. This separation prevents a kinematic formula from being mistaken for a dynamical selection principle.

Exact Kinematic Identity

For each ring,

$$v_k = 2\pi f_k r_k = \beta_k c_f, \qquad 0 < \beta_k, \qquad c_f > 0$$

Equivalently,

$$f_k = \frac{v_k}{2\pi r_k}, \qquad r_k = \frac{v_k}{2\pi f_k}, \qquad v_k = 2\pi f_k r_k$$

Plain language: for any one ring, if we know any two of frequency, tangential speed, and radius, then the third is fixed.

This identity is exact. It is not an assumption, and it does not select a lock by itself.

Assumption 1 (Middle Caustic-Grazing Closure)

In the reduced exterior and horizon-transition branch studied here, the middle binary is not pinned exactly on an infinite-force surface. It is modeled as a caustic-grazing carrier whose cycle-averaged hinge value is the field speed:

$$v_M^{\mathrm{car}} = c_f, \qquad \beta_M^{\mathrm{car}} = 1$$

For compact notation, the algebra below writes $v_M = c_f$ and $\beta_M = 1$ for this carrier value.

The branch-level motion may have microscopic crossings

$$v_M(t) = c_f + \delta v_M(t), \qquad \langle \delta v_M \rangle_W = 0$$

over the declared window W. Each regularized crossing of the $J = 0$ boundary is a caustic transit with finite impulse

$$\Delta \mathbf{v}_{M,n} = \int_{t_n^-}^{t_n^+} \mathbf{a}_M^{(\eta)}(t)\, dt, \qquad \|\Delta \mathbf{v}_{M,n}\| < \infty$$

rather than an infinite-force constraint. These impulse events are candidate mechanical origins for the discrete causal-root ledger steps used in the energy bookkeeping.

This is the main regime assumption of the chapter. The speed c_f is the propagation speed of causal isochrons in the reduced dynamics, not an observer-level claim about an effective metric.
It is not a claim that every Noether braid regime has the middle binary exactly at c_f; ordinary weak-stress operation may keep the middle layer only near the hinge scale, while the caustic-grazing carrier belongs to the reduced exterior/horizon-transition branch.

Assumption 2 (Exact Integer Phase Closure)

Let the outer period be $T_O = \frac{1}{f_O}$. Assume that when the outer ring completes one full cycle, the middle and inner rings also land exactly at the beginning of their own cycles. Equivalently, there exist integers

$$m, n \in \mathbb{N}, \qquad 1 < m < n$$

such that

$$\theta_O(t + T_O) = \theta_O(t) + 2\pi$$

$$\theta_M(t + T_O) = \theta_M(t) + 2\pi m$$

$$\theta_I(t + T_O) = \theta_I(t) + 2\pi n$$

Therefore the canonical `I:M:O` frequency triplet is $f_I : f_M : f_O = n : m : 1$. Equivalently, in outer-normalized order, $f_O : f_M : f_I = 1 : m : n$, with $f_M = m f_O$ and $f_I = n f_O$.

Plain language: after one outer revolution, the middle and inner rings have completed whole numbers of revolutions as well, so the three-ring pattern closes exactly.

This is the reduced constant-frequency carrier model. It is a branch-level closure assumption, not a statement that the assembly has only three degrees of freedom. In the full Noether braid closure problem, the simple phases $\theta_k = q_k \Omega t + \phi_k$ are replaced by integrated winding, causal-root, and frame-phase ledgers over the accepted branch chart.

Assumption 3 (Fixed Relative Phase Lock)

The lock is not just commensurate in frequency. It also carries fixed relative phase offsets over time. One convenient formulation is

$$\phi_{MO}(t) \equiv \theta_M(t) - m\theta_O(t) = \phi_{MO}^*$$

$$\phi_{IO}(t) \equiv \theta_I(t) - n\theta_O(t) = \phi_{IO}^*$$

with constants ϕ_{MO}^*, ϕ_{IO}^*.

Plain language: the rings keep the same timing relationship cycle after cycle rather than drifting through one another.

Bundle Holonomy Reading

Assumptions 2 and 3 can be restated as a phase-bundle condition. Let the outer phase be the base cycle and define the relative connection one-forms

$$\vartheta_{MO} = d\theta_M - m\,d\theta_O, \qquad \vartheta_{IO} = d\theta_I - n\,d\theta_O$$

Exact integer phase closure says the covering degrees over one outer cycle are

$$\frac{1}{2\pi}\oint_{T_O} d\theta_M = m, \qquad \frac{1}{2\pi}\oint_{T_O} d\theta_I = n$$

or equivalently

$$\oint_{T_O} \vartheta_{MO} = 0, \qquad \oint_{T_O} \vartheta_{IO} = 0 \pmod{2\pi}$$

on the locked branch. Fixed relative phase then says these one-forms are flat on the retained return chart: their integrated values do not drift, and the constants ϕ^*_{MO}, ϕ^*_{IO} are the residual flat-connection data. In the language of Effective Lagrangian, the integers (m, n) are the phase-bundle winding data that make the reduced action-angle chart globally replayable rather than merely local.

The phase-bundle picture also requires genuine three-dimensional layer independence. Let $\hat{\mathbf{n}}_O, \hat{\mathbf{n}}_M, \hat{\mathbf{n}}_I$ be the orbital-plane normals of the three layer binaries and define

$$D_{\text{plane}} = \det\left[\hat{\mathbf{n}}_O, \hat{\mathbf{n}}_M, \hat{\mathbf{n}}_I\right]$$

The reduced T^3 lock is nondegenerate only while $D_{\text{plane}} \neq 0$. Mutual orthogonality gives $|D_{\text{plane}}| = 1$, while horizon-alignment or coplanar degeneration drives $D_{\text{plane}} \to 0$ and collapses the three-circle bundle to a lower-dimensional projection. The determinant is therefore the natural order parameter for the loss of dyadic precession at alignment.

Assumption 4 (Bundle-Flatness and Cancellation Selection Principle)

Among the admissible integer locks $(1 : m : n)$, the physically selected lock is assumed to be the one whose phase bundle admits the flattest replayable connection while minimizing exposed causal-wake leakage. The cycle-averaged cancellation of a low-order causal-wake multipole or effective potential signal is the effective diagnostic for that deeper bundle condition.

This is a selection principle, not yet a theorem. Its role is to explain why one exact integer lock might be preferred over nearby commensurate alternatives. The primary object is the branch bundle; the cancellation score is accepted only when it is computed from the same holonomy data, middle-caustic impulse record, and finite-η return map.
The admissible class must be declared before minimization: positive radii, $1 < m < n$, a fixed finite-η branch chart, nonzero branch-Jacobian floors, and the speed bounds assigned to the exterior/horizon regime.

In this branch, the middle binary is the curvature carrier. Between caustic events the locked triple is modeled as flat phase transport. At the regularized middle caustics, the connection acquires concentrated curvature,

$$\Omega_{\text{phase}} = \sum_n \mathcal{F}_n\, \delta_\eta(\theta_M - \theta^*_{M,n})\, d\theta_M \wedge d\theta_O + \Omega_{\text{reg}}$$

where $\theta^*_{M,n}$ are the middle caustic phases and $\mathcal{F}_n$ is proportional to the finite caustic impulse $\Delta\mathbf{v}_{M,n}$ and its wake-history increment on the retained branch. The fulcrum statement is therefore geometric: outer/inner energy routing changes only at the middle-caustic phases where the phase-bundle connection is not flat. This is the same ledger event class used by the self-hit echo bookkeeping.

A minimal test functional can be written before committing to a particular lock. Let $q_O = 1$, $q_M = m$, and $q_I = n$, with phase variables $\theta_k = q_k\Omega t + \phi_k$. For a low-order truncation depth L, define

$$S_L(t) = \sum_{k\in\{O,M,I\}} \sum_{a=1}^{L} A_{k,a}(\beta_k, r_k, \eta, J)\, e^{ia(q_k\Omega t + \phi_k)}$$

The coefficients $A_{k,a}$ are not free fit parameters. They must be extracted from the same finite-η branch-strength, branch-Jacobian, and causal-wake ledger used to test the candidate lock.
They therefore belong to the dynamics of the causal-wake branch chart, even when the resulting signal is later summarized as an effective potential.
For the caustic-grazing middle carrier this extraction is not an ordinary smooth Fourier coefficient. A middle harmonic must carry the Jacobian weight of the caustic window, schematically

$$A_{M,a} = \int_0^{2\pi} \frac{w_{M,a}(\theta_M)}{|J_M(\theta_M)| + \eta_J} e^{-ia\theta_M}\, d\theta_M$$

with η_J the declared Jacobian-floor regularization and $w_{M,a}$ the branch-derived numerator for that harmonic channel. As η_J is lowered, the coefficient is dominated by neighborhoods of the caustic phases $\theta^*_{M,n}$, while the integrated impulse remains finite under the simple-caustic rule in Master Equation. Thus the selection question is not whether three generic Fourier amplitudes cancel, but whether the finite middle-caustic impulse deposits the right spectral weight into the first common resonance block.
The cycle-averaged cancellation score is

$$C_L(m,n;\phi)=\frac{1}{T}\int_0^T |S_L(t)|^2\,dt=\sum_\nu\left|\sum_{(k,a):\,aq_k=\nu} A_{k,a}e^{ia\phi_k}\right|^2$$

The dyadic claim becomes a theorem target only if $(m,n)=(2,4)$ minimizes this score under the admissible branch equations and retains a positive stability gap.

Harmonic-overlap lemma. The score decomposes into resonance blocks labeled by ν. A phase choice can affect cancellation between two layers only when their finite harmonic supports overlap:

$$\nu\in q_k\{1,\ldots,L\}\cap q_j\{1,\ldots,L\}$$

If a block has no overlap, its contribution to C_L is phase-independent and cannot select an integer lock. For the dyadic candidate $(m,n)=(2,4)$, the first Outer/Middle overlap is $\nu=2$ via $(O,a=2)$ and $(M,a=1)$; the first all-layer overlap is

$$\nu=4$$

via $(O,a=4)$, $(M,a=2)$, and $(I,a=1)$. Thus this functional can select $1:2:4$ only if $L\geq 4$ and the $\nu=4$ block has nontrivial branch-derived amplitudes. A complete cancellation of that all-layer block additionally requires the amplitude magnitudes to satisfy the polygon condition

$$\max(|A_{O,4}|,|A_{M,2}|,|A_{I,1}|)\leq \text{sum of the other two}$$

The lemma is only a harmonic support statement. It shows where cancellation is possible; it does not show that the branch-derived amplitudes or the return-map stability actually select the dyadic lock.

Topologically, the same $\nu=4$ statement says the dyadic lock is the first common cover of the three phase circles. The covering maps can be written

$$T^O \xleftarrow{\times m} T^M \xleftarrow{\times n/m} T^I$$

when m divides n. The dyadic case $m=2,\ n=4$ is the minimal nontrivial self-similar cover because each layer double-covers the one above it. More generally, self-similar covers obey $n=m^2$; after 1:2:4, the next such comparison family is 1:3:9, not 1:2:3 or 1:3:6. This does not prove the dyadic branch wins dynamically, but it explains why 1:2:4 is the first topologically clean candidate before the amplitude calculation begins.

Non-Assumptions

This chapter does **not** assume:

- common-speed closure $v_O=v_M=v_I$,
- self-similar radii $r_M=r_O/s$, $r_I=r_O/s^2$,
- or the specific frequency lock $1:2:4$ at the outset.

Those are possible special cases or later outcomes, not starting axioms here.
This chapter studies exact integer closure. Rational or self-similar locks can be compared only after clearing denominators or constructing a separate branch map.

Immediate Consequences

This section is pure algebra from the exact identity and the first two assumptions. It does not use the cancellation principle.

From Assumptions 1-2 and the exact identity, the middle carrier radius is fixed by the outer frequency:

$$r_M=\frac{c_f}{2\pi f_M}=\frac{c_f}{2\pi m f_O}$$

For the outer ring,

$$r_O=\frac{v_O}{2\pi f_O}=\frac{\beta_O c_f}{2\pi f_O}$$

Hence

$$\frac{r_M}{r_O} = \frac{1}{m\beta_O}, \qquad r_M = \frac{r_O}{m\beta_O}$$

For the inner ring,

$$r_I = \frac{v_I}{2\pi f_I} = \frac{\beta_I c_f}{2\pi n f_O}$$

so

$$\frac{r_I}{r_O} = \frac{\beta_I}{n\beta_O}, \qquad r_I = \frac{\beta_I}{n\beta_O} r_O$$

These are the core radius relations of the chapter:

$$r_M = \frac{r_O}{m\beta_O}, \qquad r_I = \frac{\beta_I}{n\beta_O} r_O$$

They show that once the canonical integer lock $(n : m : 1)$, equivalently outer-normalized $(1 : m : n)$, is fixed, the remaining geometry depends on the outer and inner speed factors β_O and β_I. Thus a frequency hierarchy is not yet a radius hierarchy.

Proposition 1 (Exterior Integer Lock Formulas)

Under Assumptions 1-2,

$$f_I : f_M : f_O = n : m : 1$$

equivalently, $f_O : f_M : f_I = 1 : m : n$ in outer-normalized order, and

$$r_O : r_M : r_I = 1 : \frac{1}{m\beta_O} : \frac{\beta_I}{n\beta_O}$$

Proof. The frequency ratio is exactly Assumption 2. The radius ratios follow from

$$r_k = \frac{\beta_k c_f}{2\pi f_k}$$

together with the carrier value $\beta_M = 1$, $f_M = mf_O$, and $f_I = nf_O$. $\square$

The geometry is controlled by integer phase closure plus the middle caustic-grazing carrier condition. The proposition makes no claim about which integer pair is dynamically preferred.

Could $1{:}2{:}4$ Be a Solution?

If one later chooses the dyadic integers

$$m = 2, \qquad n = 4$$

then

$$f_I : f_M : f_O = 4 : 2 : 1$$

equivalently, $f_O : f_M : f_I = 1 : 2 : 4$ in outer-normalized order,
but the radius ratios become

$$r_O : r_M : r_I = 1 : \frac{1}{2\beta_O} : \frac{\beta_I}{4\beta_O}$$

So the dyadic frequency lock is a viable candidate pattern, but it does **not** by itself imply equal-speed geometry, and it does **not** by itself imply a self-similar radius law unless further assumptions are added.

What Exact Periodicity Gives, and What It Does Not

Exact periodicity naturally supports rational or integer commensurability, but it does not by itself choose the integers m, n.

What exact lock gives:

- outer, middle, and inner frequencies lie on a commensurate lattice,
- the three-ring configuration repeats after one outer period,
- fixed relative phases become meaningful dynamical observables,
- the covering data (m, n) become phase-bundle winding data for the retained branch chart.

What exact lock does not give by itself:

- that the preferred lock is dyadic,
- that the branch speeds are equal,
- that the radii are self-similar,
- or that cancellation is actually maximal for one specific integer pair (m, n).

The bundle-flatness and cancellation principle is the extra ingredient intended to select among the many admissible integer locks.

Interpreting the Cancellation Principle

The motivation for Assumption 4 is that a cycle-closing integer lock can support persistent superposition over repeated outer periods only when the relative phase connection stays flat enough to replay. If the phase organization is favorable, the low-order causal-wake multipole or effective potential contribution can cancel more effectively over one full return cycle.

At the substrate level, the relevant quantity is exposed causal-wake leakage. At the effective level, the same organization may be reported as reduced low-order potential signal. At the inference level, the reduced model is allowed to select a lock only if the cancellation gap survives the declared truncation and stability tests.

In that sense, the selection principle is closer to a flat-bundle replay test than to a bare numerology of integer ratios. The intuition is that a physically preferred lock should minimize exposed wake leakage, phase-slip variance, and residual phase curvature subject to the delayed dynamics. If the bundle-flatness diagnostic and the cancellation score disagree, the cancellation score is only an effective summary and cannot by itself overrule a holonomy or return-map failure.

This does not yet prove which pair (m, n) wins. It states the criterion that the reduced model should test.

RG-Style Truncation Test

The cancellation functional uses a finite harmonic depth

$$L$$

That truncation must be certified rather than assumed. The useful analogy from renormalization-group reasoning is not that $\mathbb{AAA}$ inherits a field-theory RG flow, but that discarded modes must be shown irrelevant for the decision being made.

The branch geometry predicts which modes are most dangerous. Smooth flat-connection layers should have rapidly decaying coefficients,

$$|A_{O,a}|,\ |A_{I,a}| \le Ce^{-ca}$$

on an analytic replayable chart. The middle caustic layer instead has an algebraic pre-cutoff tail because its impulse is phase-localized:

$$|A_{M,a}| \lesssim C_\eta\, a^{-p_{\mathrm{fold}}}$$

with p_{fold} fixed by the caustic normal form and the regulator. The finite-depth proof must therefore report the middle-caustic spectral exponent or cutoff, not only assert that high harmonics are small. In the RG analogy, the flat outer and inner harmonics are irrelevant tails, while the middle caustic block is the marginal channel that can still affect selection beyond the first all-layer block.

For a candidate lock (m, n), define the tail score

$$T_L(m,n) \equiv \sum_{\nu > L_{\mathrm{eff}}} \left| \sum_{(k,a):\, aq_k = \nu} A_{k,a} e^{ia\phi_k} \right|^2$$

where

$$L_{\mathrm{eff}}$$

is the largest resonance block retained in the selection audit. The finite-depth proof must supply a bound

$$T_L(m,n) \le \varepsilon_L$$

uniformly over the admissible branch chart and then compare the winner gap

$$\Delta C_L \equiv \min_{(m,n)\neq(m_*,n_*)} \big(C_L(m,n) - C_L(m_*,n_*)\big)$$

against the truncation error. A lock is selected by the finite calculation only if

$$\Delta C_L > 2\varepsilon_L$$

This turns “higher harmonics are small” into a checkable theorem target tied to the same branch-derived amplitudes used in

$$C_L$$

Reduced-Theorem Target

The right theorem target is not “prove $1:2:4$ from kinematics alone.” The stronger target is a proof route that keeps kinematics, branch dynamics, phase-bundle topology, effective cancellation, and inference separate:

1. classify the admissible canonical integer locks $(n:m:1)$, equivalently outer-normalized $(1:m:n)$, under exact delayed phase closure,
2. compute the corresponding radius relations under $\beta_M = 1$,
3. require nondegenerate orbital-plane data $D_{\text{plane}} \neq 0$ so the retained phase bundle is genuinely three-dimensional,
4. define the phase-bundle curvature and caustic-weighted cancellation functional for the low-order causal-wake multipole or effective potential,
5. determine which integer lock minimizes residual curvature and exposed leakage in the exterior/horizon regime,
6. and verify the selected lock by a finite-η return map with a positive Floquet gap on the complement of the flat moduli.

Equivalently, for each candidate (m, n) one should construct a return map

$$P_{\eta,m,n} : \mathcal{S}_{m,n} \to \mathcal{S}_{m,n}$$

on the retained branch chart and require

$$\Delta_{m,n} = 1 - \max_{i \notin G} |\mu_i(P_{\eta,m,n})| > 0$$

off the neutral symmetry directions G.

Here $\mathcal{S}_{m,n}$ is a finite-η reduced phase-amplitude branch chart: it retains the layer phases, relative phase offsets, orbital-plane normals, radii, speeds, active branch data, branch-Jacobian floors, caustic-impulse rows, and history variables needed to evaluate one outer-period return. The neutral directions G are not an arbitrary hand list. They are the tangent directions that preserve the same flat connection and branch identity:

$$G = T_{\text{global}} \oplus \mathfrak{so}(3)_{\text{rot}} \oplus T_{\text{flat}} \oplus G_{\text{rel}}$$

where T_{global} is the global time or phase shift, $\mathfrak{so}(3)_{\text{rot}}$ is the rigid spatial-rotation tangent space, $T_{\text{flat}} = \text{span}\{(\delta\phi_{MO}, \delta\phi_{IO})\}$ is the flat-connection moduli space, and G_{rel} contains any declared relabeling symmetry of the retained branch chart. A lock is dynamically stable only if the return map contracts on the complement of G and the flat-modulus directions remain genuinely neutral. If a flat-modulus direction becomes unstable, the frequency commensurability may remain while Assumption 3 fails through relative-phase drift.

If the minimizer turns out to be the outer-normalized lock $1{:}2{:}4$, equivalently $(m, n) = (2, 4)$, then the dyadic hierarchy would be a derived selection result rather than a starting assumption.

In the invariant language of Assembly Topological Charge, the reduced theorem target is to find an admissible topological sector

$$[\mathfrak{B}]_{\text{tri}} = (N_s,\, M_p,\, c_1) = (N_s,\, M_p,\, (m, n))$$

with flat phase connection, positive Floquet gap off G, and $|D_{\text{plane}}|$ bounded away from zero outside the horizon-alignment locus. The dyadic conjecture is the sharper claim that $(N_s, M_p, (2, 4))$ is the minimal-curvature such class in the exterior/horizon-transition regime.

Recurrence Diagnostic

The finite-η return-map test should also reject transient near-locks. For a sampled returned-branch trajectory

$$z_i = (\phi_i, a_i, \nu_i, \ell_i, \hat{\mathbf{n}}_{O,i}, \hat{\mathbf{n}}_{M,i}, \hat{\mathbf{n}}_{I,i}) \in \mathcal{S}_{m,n}$$

define a recurrence matrix

$$Q_{ij}^{(\epsilon)} = \mathbf{1}\left[d_{\mathcal{S}}(z_i, z_j) < \epsilon \;\wedge\; \|\Theta_i - \Theta_j\| < \epsilon_\Theta \;\wedge\; |D_{\text{plane},i} - D_{\text{plane},j}| < \epsilon_D\right]$$

where $d_{\mathcal{S}}$ is the declared branch-chart distance after quotienting the neutral symmetries in G, while the holonomy-defect coordinate is not quotiented:

$$\Theta(t) = (\phi_{MO}(t) - \phi^*_{MO},\, \phi_{IO}(t) - \phi^*_{IO})$$

A candidate $1{:}2$ row, or a chained $1{:}2{:}4$ row, is recurrence-positive only if returned-section hits recur at the declared outer-period multiples, the recurrence period agrees with the winding and active-branch ledger, the relative-phase defect Θ recurs to zero, the plane determinant stays in the nondegenerate domain, the recurrence structure persists under timestep, history-resolution, and η refinement, and nearby trials that fail the non-symmetry Floquet gap do not pass this recurrence check. This separates point recurrence from true phase-lock recurrence.

Ancillary Symmetry Check

The older $\mathbb{Z}_3$ dipole-cancellation identity belongs to a different assembly sector. It can still be kept as a planar symmetry test:

$$1 + e^{i2\pi/3} + e^{i4\pi/3} = 0$$

This is an in-plane cancellation for three equal phases separated by 120°. It is therefore naturally associated with coplanar, boson-like stealth arrangements rather than with the near-orthogonal tri-binary bundle studied in this chapter. In compact form:

$$\mathbb{Z}_3 \text{ stealth} \longleftrightarrow \text{coplanar cyclic sector}$$

whereas

$$1{:}2{:}4 \text{ dyadic cover} \longleftrightarrow \text{near-orthogonal } T^3 \text{ sector}$$

The two mechanisms can both reduce exposed causal-wake leakage, but they do it through different topology. Planar cyclic symmetry cancels inside one plane; the dyadic tri-binary lock distributes the phase-bundle covering across three independent orbital planes. The $\mathbb{Z}_3$ identity should therefore not be used as evidence for or against the frequency-selection assumptions above.

For neighboring closure problems, see Planar Bridge Closure and Horizon Chirality.

Assembly Topological Charge

This chapter gives a first-class home to the candidate topological label of an architrino assembly. The label combines the causal-root ledger of the delayed dynamics with the phase-bundle winding of a resonance-locked nested shell braid. Its purpose is to state what can be computed from a retained branch chart, what is invariant inside a nondegenerate branch domain, and what remains a theorem target before the label can serve as a topological periodic table of assemblies. The general search domain that emits candidate tri-binary branch charts is developed in Tri-Binary Configuration Space.

The compact notation is

$$[\mathfrak{B}]_{\text{top}} = (N_s,\, M_p,\, c_1)$$

where N_s counts active self-hit roots, M_p counts active partner-hit roots, and c_1 denotes the phase-bundle winding data of the retained resonance lock. For a tri-binary lock this last entry is usually a pair

$$c_1 = (m, n) \in \mathbb{Z}^2$$

rather than a scalar integer: m and n are the middle and inner winding numbers over one outer period.

This compact form records the count data most directly emitted by a branch solver. The conserved refinement is

$$[\mathfrak{B}]_{\text{deg}} = (D_s,\, D_p,\, c_1)\,,$$

where D_s and D_p are signed root degrees. The unsigned counts N_s and M_p can change by opposite-sign fold-pair birth or death, while D_s and D_p are the degree-like data preserved by generic fold surgery. A promoted report should therefore carry both the compact assembly topological charge and its signed-degree refinement.

This is a definition and closure target, not a completed classification theorem. It becomes a physical assembly label only after the same retained branch chart supplies positive root floors, finite memory, finite local-to-global gluing, stable return data, and a closed wake-history boundary ledger.

In the terminology of Tri-Binary Configuration Space, an eigen-braid candidate is the dynamical return-map status of the full retained branch. The assembly topological charge is the branch-intrinsic topological label carried by that candidate. It is not a Lorentz-dressed observer component: moving-assembly export may transform energy-momentum and angular-momentum readouts, but $[\mathfrak{B}]_{\text{top}}$ changes only when the retained branch crosses a fold, reconnection, or declared surgery event.

Source Of The Three Entries

The first two entries come from the causal-root complex of the Master Equation. On a retained branch chart, active roots are split by source identity and by Jacobian sign. For the self-hit sector,

$$C_{s,+}(\mathfrak{B}) = \text{span}\{s_\ell : \text{self root},\ J_\ell > 0\}, \qquad C_{s,-}(\mathfrak{B}) = \text{span}\{s_\ell : \text{self root},\ J_\ell < 0\}.$$

For the partner-hit sector,

$$C_{p,+}(\mathfrak{B}) = \mathrm{span}\{s_\ell : \text{partner root},\ J_\ell > 0\}, \qquad C_{p,-}(\mathfrak{B}) = \mathrm{span}\{s_\ell : \text{partner root},\ J_\ell < 0\}.$$

The unsigned ledgers are

$$N_s = \dim C_{s,+} + \dim C_{s,-}, \qquad M_p = \dim C_{p,+} + \dim C_{p,-}.$$

The signed degrees

$$D_s = \dim C_{s,+} - \dim C_{s,-}, \qquad D_p = \dim C_{p,+} - \dim C_{p,-}$$

are not extra entries in the compact assembly topological charge, but they are required side data and form the conserved-degree refinement $[\mathfrak{B}]_{\mathrm{deg}}$. A solver that reports only N_s and M_p has counted roots without proving which opposite-sign pairs can be born, die, or persist under deformation.

The geometric reading is intersection-theoretic. On a lifted finite-memory strip, each connected retained causal-locus component has an oriented intersection number with a generic receiver-time fiber. Summing those oriented intersections in the self and partner sectors gives D_s and D_p; summing their absolute values gives N_s and M_p. This is the bridge to Causal Action Functional: the same causal-locus components that carry action-counting weight also supply the signed root degrees used by the assembly topological charge.

The third entry comes from the phase bundle of a resonance-locked tri-binary. Let $\theta^O, \theta^M, \theta^I$ be the outer, middle, and inner phase coordinates on the retained return chart. Exact integer closure over one outer period T_O means

$$\theta_O(t + T_O) = \theta_O(t) + 2\pi,$$

$$\theta_M(t + T_O) = \theta_M(t) + 2\pi m, \qquad \theta_I(t + T_O) = \theta_I(t) + 2\pi n.$$

Equivalently, the relative-phase one-forms

$$\vartheta_M = d\theta^M - m\, d\theta^O, \qquad \vartheta_I = d\theta^I - n\, d\theta^O$$

have integer holonomy and become flat on a promoted phase-locked branch. The shorthand

$$c_1[\theta^O, \theta^M, \theta^I] = (m, n)$$

records this phase-bundle winding data. The dyadic candidate is the outer-normalized case $(m, n) = (2, 4)$, equivalently canonical `I:M:O` frequency order $4 : 2 : 1$.

The notation c_1 is justified only when the phase-bundle construction is part of the retained chart. The base is the outer phase circle $S^1_O = \mathbb{R}/T_O\mathbb{Z}$. The middle and inner phases define two S^1 fibers over that base through the return map, with clutching data

$$\theta^M \mapsto \theta^M + 2\pi m, \qquad \theta^I \mapsto \theta^I + 2\pi n.$$

Equivalently, the retained chart carries two circle bundles over S^1_O, and m, n are their Euler/Chern winding numbers. Without this bundle-and-return-map construction, (m, n) is only a frequency-ratio report, not a phase-bundle entry of the assembly topological charge.

Candidate Definition

For a finite-η branch chart $\mathfrak{B}$, the assembly topological charge is admissible only when the following data are present on the same retained row set:

1. Active root rows split by source identity: self-hit and partner-hit.
2. Jacobian-sign grading for those rows: $C_{s,+}, C_{s,-}, C_{p,+}, C_{p,-}$.
3. Positive transversality floors away from declared finite caustic transits.
4. Finite memory depth and positive inactive-root gaps.
5. A finite local-to-global gluing result for the branch chart, or an explicit finite multistability family.
6. For a tri-binary, integer phase closure and flat relative-phase connection.
7. A return-map stability certificate, such as a Floquet or Conley-style branch certificate, after quotienting only true symmetry directions.
8. If the middle layer is treated as a caustic-grazing carrier, regulator-stable middle-caustic rows showing that the reported root degrees and phase-bundle entry do not depend on the finite-η convention in the promoted limit.

Under those conditions the compact assembly topological charge is

$$[\mathfrak{B}]_{\mathrm{top}} = \big(N_s,\ M_p,\ c_1[\theta^O, \theta^M, \theta^I]\big) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0} \times \mathbb{Z}^2.$$

For a non-tri-binary branch, the partial assembly topological charge (N_s, M_p) may be recorded, but c_1 is not assigned until a phase-bundle chart exists.

A useful refinement is a branch-preserving chirality label

$$\chi_{\mathrm{fr}} \in \mathbb{Z}_2.$$

This is not part of the base triple until the branch chart supplies a deformation-stable handed marker, such as framed self-linking parity or a certified maximal-curvature-binary circulation sign. It is the natural place to record handedness, but it must not be substituted for the root and phase-bundle data.

Invariance And Allowed Transitions

The assembly topological charge is designed to be locally invariant. Between branch boundaries, the implicit-function theorem transports each simple active root continuously, so N_s, M_p, D_s, and D_p remain constant. At a generic fold, one positive and one negative root are created or annihilated. Therefore

$$\Delta N_s \in 2\mathbb{Z} \quad \text{or} \quad \Delta M_p \in 2\mathbb{Z}, \qquad \Delta D_s = \Delta D_p = 0$$

for an ordinary fold-pair event in the corresponding sector.

Cusp or higher singular strata are not automatically governed by the generic fold law. They require a separate regularized normal form before their ledger surgery can be promoted. Likewise, $c_1 = (m, n)$ remains fixed under deformation only while the phase connection stays flat and the phase torus returns without monodromy. A loss of resonance lock, a plane-degeneracy transition, or a branch-fold event that changes the return chart can change the phase-bundle entry.

The transition catalogue therefore has a native form:

Event	Codimension	Assembly topological charge effect	Required certificate
Branch-preserving deformation	0 on the retained chart	No change to (N_s, M_p, c_1) or (D_s, D_p, c_1)	Positive floors, finite memory, stable gluing
Self-root fold	1 generically	$\Delta N_s = \pm 2$, $\Delta D_s = 0$ generically	Fold normal form and post-transit chart
Partner-root fold	1 generically	$\Delta M_p = \pm 2$, $\Delta D_p = 0$ generically	Fold normal form and post-transit chart
Phase-lock jump	1 for a resonance crossing	$\Delta c_1 \neq 0$	Holonomy change and return-map transition
Plane-degeneracy transition	1 generically, higher with imposed symmetry	Phase-bundle chart may lose rank before c_1 can be compared	Orbital-plane determinant and return-chart continuation
Framing or chirality flip	1 or higher, depending on the framing chart	$\Delta \chi_{\mathrm{fr}} \neq 0$	Framed-linking or handedness transition certificate
Cusp or deeper singular stratum	2 or higher generically	Not inferred from fold law	Singular-stratum chart and regulator-stable transition data

This is why the triple belongs in one object. The root ledgers describe which delayed causal channels are live, while the phase-bundle entry describes how the multi-layer branch returns to itself. Both are characteristic data of the same retained causal-root sheaf: local root sections, overlap gluing, and phase holonomy must agree before an assembly label is promoted.

Role In The Assembly Atlas

The topological atlas of assemblies should not classify objects by visual similarity alone. It should classify retained branches by deformation-stable integers that can be extracted from the same simulation record used to test the dynamics. The candidate atlas entry for a stable assembly is therefore

$$\mathcal{Q}_{\mathrm{asm}} = (N_s,\ M_p,\ c_1,\ \chi_{\mathrm{fr}} \text{ when certified})$$

together with its stability margins, energy/wake ledger, and gluing status.

The intended use is constrained:

- (N_s, M_p) records the binding-channel census: self-hit channels, partner-hit channels, and their signed degrees.
- $c_1 = (m, n)$ records the resonance-lock winding of a promoted tri-binary branch.
- χ_{fr} records handedness only after a framed handed marker is certified.
- Physical particle identity, generation structure, spin-statistics, exclusion, and Standard Model quantum numbers are downstream mappings, not consequences of the notation alone.

Thus (N_s, M_p, c_1) is the candidate conserved label that says when two assemblies occupy the same topological sector. It is not yet a proof that a given sector is an electron analogue, photon analogue, or quark analogue.

Strictly, the compact count triple is locally conserved only inside one nondegenerate branch domain. Across generic fold-pair surgery the degree-refined data (D_s, D_p, c_1) are the conserved part, while N_s and M_p record how many live channels the retained branch currently carries.

Simulation Extraction

A branch solver should extract the assembly topological charge in this order:

1. Build the finite-η retained branch chart and declare its memory window.
2. Find active causal roots on the same retained row set.
3. Label each root by source identity: self or partner.
4. Record the Jacobian sign and compute $C_{s,+}, C_{s,-}, C_{p,+}, C_{p,-}$.
5. Compute N_s, M_p, D_s, and D_p.
6. Compute the lifted-strip fiber-intersection degrees that realize D_s and D_p whenever the causal-locus chart is available.
7. Track fold, caustic, cusp, or inactive-gap transition metadata.
8. For tri-binary branches, compute phase holonomy (m, n) from the returned phase chart and verify that it comes from the phase-bundle return map rather than from frequency ratios alone.
9. If a middle caustic-grazing carrier is used, test that the signed degrees and phase-bundle entry are stable under the declared η refinement.
10. Test gluing and finite continuation cardinality for the local charts.
11. Test the return-map stability gap off true symmetry directions.
12. Report $[\mathfrak{B}]_{\mathrm{top}}$ only after the same retained rows pass these checks.

The failure modes are equally important. A candidate is not promoted if the roots are counted without signs, if self and partner rows are mixed, if the phase lock is inferred from frequency ratios without holonomy recurrence, if local branch charts do not glue, or if the continuation family is empty, infinite, or unlabeled.

Current Status

The established pieces are local:

- The delay-map theorem pack in Master Equation proves signed degree invariance on regular families and the generic opposite-sign fold-pair law.
- The signed causal-root complex in Master Equation supplies the local chain-complex reading of active roots.
- Binary Dynamics supplies the self-hit and partner-hit ledger notation used by (N_s, M_p).
- Dyadic Resonance Lock supplies the integer phase-closure data whose winding is recorded as $c_1 = (m, n)$.
- Effective Lagrangian uses the same topological sector in the action and mass-gap theorem target.

The open proof burden is global:

- prove that a stable assembly realizes a fixed assembly topological charge over a finite branch domain;
- prove gluing of the local causal-root charts into a finite labeled continuation family;
- prove a positive stability gap for the assembly topological charge sector;
- determine whether the entries are independent or constrained by radial balance, phase flatness, and Noether sea response, starting with the reachable theorem target that $c_1 = (m, n)$ constrains the parity or degree pattern of the layerwise self-hit ledger through the lifted-strip fiber-intersection formula;
- prove that caustic-grazing middle-carrier rows have regulator-stable signed degrees and phase-bundle entries, so the assembly topological charge does not depend on the finite-η convention used to regularize the hinge;
- map any certified sectors to observer-level particle quantum numbers without fitting the labels afterward.

The chapter should therefore be read as the canonical definition and proof target for assembly topological charge, not as the completed topological periodic table.

Causal Action Functional

This chapter develops the action-counting complement to the master-equation treatment of dynamics. Its job is to define a scalar causal-hit statistic that compares delayed worldline structures, labels candidate assembly classes, and supplies one geometric input to later mass, shielding, and medium-response closure. It is not the exact variational action for the Master EOM; action-derived dynamics require the variation residual to vanish under the test stated in Effective Lagrangian and Master Equation.

The current scope is mixed. Some statements are theorem-backed in the regularized setting, while the larger closure program remains open. The chapter therefore begins with the problem statement and core functional definitions, then separates the controlled theorem spine from benchmarks, implementation notes, and longer-range closure targets.

Problem Statement and Goal

The broad objective is to explain why only certain assemblies are stable and discrete, and to treat observer-level mass as an exposed response of a closed internal causal-history ledger, shielding, and Noether sea coupling rather than an externally assigned input. The target in this chapter is narrower: a geometric causal-locus statistic derived from the causal-wake kernel that can be evaluated on periodic orbits, compared across topological classes, and tested against dynamical stability.
Canonical dynamics are defined in The Master Equation (Canonical Form); this chapter provides the complementary action-functional lens.

The level separation is essential:

1. **Ontology:** architrino histories emit causal wakes in absolute time and the Euclidean void.
2. **Dynamics:** the master equation sums delayed, Jacobian-weighted line-of-action hits.
3. **Statistic:** the functional in this chapter removes direction and counts weighted causal intersections.
4. **Effective/inferential use:** mass response, effective geometry, and branch spectra are later reconstructions that must be checked against the actual delayed dynamics.

Core Functional Definitions

Scalar causal-hit counting functional:

$$\mathcal{A}_{\text{self}}[\gamma] = \iint_{\gamma\times\gamma} \frac{\delta\big(\|\mathbf{x}(t)-\mathbf{x}(t')\| - c_f|t-t'|\big)}{\|\mathbf{x}(t)-\mathbf{x}(t')\|^2\, J_\gamma(t,t')}\, dt\, dt'$$

We introduce a scalar causal-hit counting functional to make stability searches comparable across trajectories. This is not the exact Fokker-type variational action of Effective Lagrangian; it is the branch-density statistic obtained after retaining the received inverse-square and Jacobian weights while discarding line-of-action direction. Its appropriate use is to nominate dynamically preferred worldline classes, then test those nominations with the master-equation flow before interpreting them as discrete observer-level particle states.

This integrates over all nontrivial pairs of points on a single worldline and counts only those pairs that are causally connected by a wake moving at speed c_f. The trivial diagonal $t = t'$ is excluded, either by a punctured domain or by a cutoff $|t-t'| \geq \tau_{\min} > 0$. The inverse-square factor weights nearby self-hits more strongly than distant ones, while J_γ^{-1} accounts for the geometric bunching or dilation of the delayed flux along the active branch.
Convention: this document distinguishes the compact symmetric selector $|t-t'|$ from the lifted delayed selector $\Delta_m = t - t' + mT$. The symmetric form is useful on one-period charts; the lifted delayed form is required when multi-period causal roots are active.
Here $J_\gamma(t,t')$ denotes the branch Jacobian induced by the causal constraint. In lifted delay coordinates one may use the absolute root Jacobian $J_\gamma = \big|\partial_{t'}\big(\|\mathbf{x}(t)-\mathbf{x}(t')\| - c_f\Delta\big)\big|$; when comparing to the Master Equation, the dimensionless received-flux factor is the corresponding $1 - \mathbf{v}\cdot\hat{\mathbf{r}}/c_f$, with constant factors absorbed into the declared normalization. This branch Jacobian is distinct from the coarea factor $\|\nabla F_\gamma\|$ that appears when the two-time integral is reduced to a one-dimensional causal locus.

Interpretation:

1. **Object:** The full worldline γ is treated as a single geometric object.
2. **Constraint:** The delta function enforces the causal-isochron condition, selecting causally connected pairs.
3. **Measure:** The inverse-square weight emphasizes close self-hits over distant ones, while the Jacobian factor converts constant source emission into the correct received causal flux.

Lifted normalized periodic self-action statistic:

$$\Delta_m(t,t') = t - t' + mT, \qquad F_m(t,t') = r(t,t') - c_f\Delta_m(t,t')$$

For a T-periodic orbit, finite memory depth h, and nontrivial-branch cutoff $\tau_{\min} > 0$, use

$$\bar{\mathcal{A}}_{\text{self},\eta,h,\tau_{\min}}[\gamma] = \frac{1}{T}\int_0^T \sum_{m\in\mathbb{Z}} \int_0^T \mathbf{1}_{\tau_{\min}\leq\Delta_m\leq h} \frac{\delta_\eta\big(F_m(t,t')\big)}{r(t,t')^2\, J_m(t,t')}\, dt'\, dt$$

with $r(t,t') = \|\mathbf{x}(t)-\mathbf{x}(t')\|$, δ_η a mollified delta, and $J_m(t,t') = |\partial_{t'}F_m(t,t')|$ on a simple delayed branch. This lifted form captures multi-period circular roots and avoids the trivial diagonal. A symmetric $|t-t'|$ selector is equivalent only after the diagonal is excluded and the delayed half-domain normalization is corrected; otherwise it misses high-winding branches or double-counts them.
Dimensional status depends on the chosen time/length units and normalization by T, h, and c_f; use a declared dimensionless rescaling before comparing this statistic to mass or action coefficients.

Lifted finite-memory bound. If the lifted statistic is restricted to $\tau_{\min} \leq \Delta_m \leq h$, the active support satisfies $r \geq r_{\min} > 0$, and the simple-branch floor $J_m \geq J_{\min} > 0$ holds, then

$$0 \leq \bar{\mathcal{A}}_{\text{self},\eta,h,\tau_{\min}} \leq \frac{(h-\tau_{\min})\|\delta_\eta\|_\infty}{r_{\min}^2 J_{\min}}$$

The reason is that, for each fixed t, the lifted intervals selected by m partition the delay line over the retained memory window:

$$\sum_m \int_0^T \mathbf{1}_{\tau_{\min} \le \Delta_m \le h}\, dt' = h - \tau_{\min}$$

Under transversality, the weak coarea limit becomes

$$\frac{1}{T} \sum_m \int_{\mathcal{L}_m} \frac{1}{r^2 J_m \|\nabla F_m\|}\, d\ell, \qquad \mathcal{L}_m = \{F_m = 0,\ \tau_{\min} \le \Delta_m \le h\}$$

Therefore simulations comparing lifted action-density values must report h, $\tau_{\min}$, the retained m range, $r_{\min}$, $J_{\min}$, the transversality floor, and inactive-root gaps.

Total scalar action-counting statistic (multi-assembly):

$$\bar{\mathcal{A}}_{\text{total}}[\{\gamma_i\}] = \frac{1}{T^2} \left[\sum_i \int_0^T \int_0^T \frac{\delta_\eta\big(r_{ii}(t,t') - c_f|t-t'|\big)}{r_{ii}(t,t')^2\, J_{ii}(t,t')}\, dt\, dt' \;+\; \frac{1}{2} \sum_{i \ne j} \int_0^T \int_0^T \frac{\delta_\eta\big(r_{ij}(t,t') - c_f|t-t'|\big)}{r_{ij}(t,t')^2\, J_{ij}(t,t')}\, dt\, dt' \right]$$

This single-period symmetric form aggregates self-terms and cross-terms between components, with the $\frac{1}{2}\sum_{i\ne j}$ convention ensuring unordered pairs are counted once. Self-terms inherit the same nontrivial-branch exclusion used above. When multi-period branches are active, replace each symmetric selector by the lifted finite-memory form before comparing totals across branch charts.

Definitions: $r(t,t') = \|\mathbf{x}(t) - \mathbf{x}(t')\|$, $r_{ij}(t,t') = \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\|$, $\Delta t = t - t'$, and $J_{ij}(t,t') = \big|\partial_{t'}\big(r_{ij}(t,t') - c_f|t-t'|\big)\big|$ is the branch Jacobian induced by the delayed causal constraint.

Kernel comparison:

$$\text{Force kernel: } \left[\frac{\hat{\mathbf{r}}(t,t')}{r^2\, J}, \delta\big(r - c_f \Delta t\big)\right] \qquad \text{Scalar statistic kernel: } \left[\frac{1}{r^2\, J}, \delta\big(r - c_f \Delta t\big)\right]$$

The force kernel retains direction via $\hat{\mathbf{r}}$, while the scalar statistic kernel keeps only the magnitude. This is the minimal change that turns a vector interaction into a scalar comparison functional while preserving the same causal Jacobian geometry as the master equation. It should not be read as the exact Fokker-type action whose variation derives the force law.

Causal-set action constructions provide a useful external comparison at this point. Their lesson is that interval counts can be arranged so that a discrete causal-order statistic approximates continuum curvature or action in a suitable large-scale regime. The analogous $\mathbb{AAA}$ question is whether causal-wake and causal-root statistics built from the master-equation kernel admit a coarse-grained operator or action statistic that matches the required GR and QFT recovery targets. That benchmark does not license importing causal-set dynamics; it only sharpens the test for any proposed scalar action-counting functional.

As a scalar, $\mathcal{A}_{\text{self}}$ summarizes the total strength of causal self-hits along a worldline. It is derived directly from the interaction structure, but with the directional information removed.

For reference, the self-interaction term in the master equation uses the same kernel:

$$\mathbf{a}_{\text{self}}(t) = \kappa q^2 \int dt'\, \frac{\hat{\mathbf{r}}(t,t')}{r^2(t,t')\, J_\gamma(t,t')} \delta\big(r(t,t') - c_f(t-t')\big)$$

Regularized Mathematical Setting (Explicit Regime)

To separate what is already controlled from what remains conjectural, we work in the regularized regime $\eta > 0$ and state all claims on one period.

Define

$$\phi_\eta(u) \equiv \delta_\eta(u), \qquad F_\gamma(t,t') \equiv r(t,t') - c_f|t-t'|, \qquad r(t,t') = \|\mathbf{x}(t) - \mathbf{x}(t')\|$$

For a T-periodic C^2 trajectory $\mathbf{x}(t)$ with no collisions or trivial self-support on the sampled domain ($r(t,t') \ge r_{\min} > 0$, $|t-t'| \ge \tau_{\min} > 0$ on self terms, and $J_\gamma(t,t') \ge J_{\min} > 0$ on support of ϕ_η), define

$$\bar{\mathcal{A}}_{\text{self},\eta}[\gamma] = \frac{1}{T^2} \int_0^T \int_0^T \frac{\phi_\eta(F_\gamma(t,t'))}{r(t,t')^2\, J_\gamma(t,t')}\, dt\, dt'$$

This is the single-period symmetric object for proofs and numerics when one period contains the full relevant causal memory. It is therefore a controlled chart, not the most general causal-memory functional. When high-winding or multi-period branches are active, replace it by the lifted statistic above with the same lower-bound and Jacobian assumptions. The unregularized $\eta \to 0^+$ limit is treated only after bounds are established.

Axioms and Admissibility Assumptions

We use the following minimal assumption set for theorem-level statements:

- **(A1) Regularity:** $\mathbf{x} \in C^2(\mathbb{R}; \mathbb{R}^3)$ and is T-periodic.
- **(A2) Finite-speed causality:** The causal selector is $F_\gamma(t, t') = 0$ with field speed $c_f > 0$.
- **(A3) Collision and trivial-diagonal exclusion on support:** $r(t, t') \geq r_{\min} > 0$ whenever $\phi_\eta(F_\gamma(t, t')) \neq 0$, with $|t - t'| \geq \tau_{\min} > 0$ on self terms unless a separate core regularization is declared.
- **(A3b) Jacobian nondegeneracy on support:** $J_\gamma(t, t') \geq J_{\min} > 0$ whenever $\phi_\eta(F_\gamma(t, t')) \neq 0$.
- **(A4) Uniform transversality (generic branch):** on the retained compact domain, the selected causal set has a floor

$$\|\nabla F_\gamma\| \geq \nu > 0$$

in a neighborhood of the zero level. Pointwise nonvanishing is the geometric condition; the uniform floor is the operative form used in coarea limits and simulations.

- **(A5) Fixed topological class:** Deformations are taken inside one homotopy class on T^2 unless a bifurcation condition is crossed.
- **(A6) Isolated system bookkeeping:** When connecting to dynamics, energy/momentum use the same η and history window conventions as the master-equation diagnostics.

These assumptions are deliberately local and testable. If any assumption fails, the corresponding theorem is not claimed.

For the finite-η pathology theorem target, (A3) and (A3b) are the action-statistic side of the self-energy and caustic quarantine. They bound the scalar causal-hit statistic on the retained chart, but they do not by themselves prove the Master EOM, no-runaway behavior, or exact conservation. Those stronger claims require the same branch chart to pass the force residual, action residual, and energy-momentum residuals stated in Master Equation.

Rationale for the Functional

- **Action-like comparison candidate:** If a motion class is stationary or extremal for this statistic, the result gives a candidate branch label. It does not by itself prove attraction, rest mass, or a variational derivation of the master equation.
- **Bridge to geometric analysis and knot theory:** Showing that simple periodic motions, such as maximum-curvature self-hit orbits, locally minimize $\mathcal{A}_{\text{self}}$ within a topological class would give a geometric reason to test those orbits as preferred branches.
- **Simulation-friendly statistic:** Given any numerically computed orbit, one can sample (t, t'), test the causal-isochron condition, and estimate $\mathcal{A}_{\text{self}}[\gamma]$ to compare geometries. This makes the "stable = local minimum" heuristic a testable claim rather than a definition.
- **Statistical-invariant candidate:** Because the functional is built from the master-equation kernel and can be estimated from simulated histories, it is a candidate input to invariant-measure or basin-measure studies of attractor selection.

Geometric/Topological Framework

Causal locus and lifted-strip degree: For a periodic orbit the compact coordinates $(t, t') \in [0, T]^2$ form a torus before the trivial diagonal is removed. The causal locus

$$\mathcal{L}_{\text{causal}} = \{(t, t') \in T^2 \mid \|\mathbf{x}(t) - \mathbf{x}(t')\| = c_f |t - t'|\}$$

is the set of self-hits. Once the diagonal collar $|t - t'| < \tau_{\min}$ is excluded, the working domain is not the closed torus but a torus with boundary. A regular component that closes on the unpunctured torus carries a winding class $(p, q) \in H_1(T^2, \mathbb{Z})$, but a near-threshold component may instead be an arc with endpoints on the excluded collar. The primary invariant on the retained finite-memory chart is therefore the lifted-strip fiber-intersection number.

On the lifted strip

$$S_{\tau,h} = \{(t, t') : \tau_{\min} \leq \Delta_m(t, t') \leq h\}, \qquad \Delta_m = t - t' + mT,$$

let $\mathcal{L}_a$ be a connected component of $F_m^{-1}(0)$ and define

$$\iota_a = \#_{\text{alg}}\big(\mathcal{L}_a \cap \{t = t_0\}\big)$$

for a generic vertical fiber $\{t = t_0\}$. The algebraic sign is the branch-orientation sign, equivalently the sign convention used for the delayed-root Jacobian on that component. The unsigned sum $\sum_a |\iota_a|$ recovers the per-period self-hit count on the retained strip, while the signed sum recovers the signed degree used by the causal-root ledger. If a component also closes on the unpunctured torus with winding class (p_a, q_a), then $\iota_a = q_a$ after the same orientation convention is fixed. The winding language is therefore valid for closed components, while the fiber-intersection degree is the bridge to the root-ledger entries used in the assembly topological charge.

As geometric control parameters change, the locus can undergo reconnection or fold events; these are the bifurcations where families appear or disappear, giving a branch-topology mechanism for discrete self-hit patterns. In the circular benchmark below, sub-c_f motion leaves the retained causal locus empty after the trivial diagonal is removed, while super-c_f motion creates branches whose lifted-strip intersections determine the integer self-hit count per period.
The self-action integral is the **weighted arc length** of the retained causal locus with weight $1/(r^2 J_\gamma)$, so topology and metric weight enter together.

Causal writhe (chirality):

$$Wr_c[\gamma] = \iint_{\mathcal{L}_{\text{causal}}} \operatorname{sign}\big(\mathbf{v}(t) \times \mathbf{v}(t') \cdot \mathbf{r}(t,t')\big)\, d\ell$$

where $\mathbf{r}(t,t') = \mathbf{x}(t) - \mathbf{x}(t')$ and $d\ell$ is the induced line measure on the causal locus. This is a candidate signed measure of handedness for the self-interaction pattern. Nonzero Wr_c is a possible topological handle for chirality/spin closure; it is not yet a proof that spin is fixed by the causal locus alone.
The symmetric torus chart counts the same unordered self-hit pair twice. A chirality comparison must therefore either restrict the integral to the delayed half-domain $\Delta_m > 0$ or divide the symmetric quotient value by two, with the orientation convention stated before comparing handedness between branches.

Topological vs Noether data: Continuous symmetries (time shifts, rotations) identify Noether-charge targets: energy from time-translation symmetry and total angular momentum from rotational symmetry. In a closed symmetry-preserving delayed action these would become conserved history functionals. The winding class of closed components and the lifted-strip intersection degree supply candidate inputs to the assembly topological charge program. A generation-level claim would require a branch that is both Noether-stationary and topologically locked; dissociation would then require changing the causal-locus sector, i.e., a reconnection or fold transit of the retained causal locus.

Causal-locus sector as a comparison invariant: The soliton comparison teaches a useful restraint: a sector label does not by itself select a representative inside that sector. For this chapter the native data are the homology classes of closed causal-locus components together with lifted-strip intersection degrees. Write

$$Q_{\text{causal}}(\gamma) = \big(\{[(\mathcal{L}_{\text{causal}})_a]\}_{a \in C_{\text{closed}}}, \{\iota_b\}_{b \in C_{\text{strip}}}\big)$$

the multiset of closed-component winding classes and retained lifted-strip fiber-intersection degrees, optionally refined by source identity and chirality sign. The comparison rule is:

$$Q_{\text{causal}}(\gamma_0) = Q_{\text{causal}}(\gamma_1)$$

means the two trajectories lie in the same branch-topology sector; it does not imply equal action, equal mass response, or stability. A stability claim additionally needs either a constrained critical-point test for

$$\bar{\mathcal{A}}_{\text{total}}$$

or a return-map spectrum for the actual delayed dynamics.

Instanton-style path competition: When two branch-topology sectors are connected only by passing through a transversality failure, the useful comparison object is not a new force law but a minimal regularized barrier in path space. For a one-parameter path of histories

$$\Gamma : [0,1] \to \mathcal{H}_h, \qquad \Gamma(0) = \gamma_0, \qquad \Gamma(1) = \gamma_1$$

define the barrier proxy

$$B_{\eta,h}(\gamma_0 \to \gamma_1) \equiv \inf_{\Gamma} \max_{s \in [0,1]} \bar{\mathcal{A}}_{\text{total},\eta,h}[\Gamma(s)]$$

The infimum is taken over paths whose endpoints lie in the declared sectors and whose intermediate histories obey the same regularization convention. This is an instanton-like comparison only in the variational sense: it measures the least regularized action-counting barrier between sectors. It does not assert tunneling, supersymmetry, or Euclidean field-theory ontology.
For this expression to be more than formal, take $\mathcal{H}_h$ to be a declared history space such as $C^2([-h,0];\mathbb{R}^{3N})$ with the C^2 norm, or a Sobolev closure strong enough to preserve the delayed root map. Under the finite-η bounds above, $\bar{\mathcal{A}}_{\text{total},\eta,h}$ is continuous on an admissible chart. The physical mountain-pass hypothesis is then that different causal-locus sectors are separated by a transversality wall where J_γ or $\|\nabla F_\gamma\|$ loses its floor. Across such a caustic wall the unregularized barrier is expected to diverge, while $B_{\eta,h}$ records the regulator-controlled cost of crossing the wall. This is the action-counting version of topological protection, not an additional force law.

Multi-component topology: For assemblies, project the spatial trajectories over one period, classify the resulting link, and when hyperbolic, use the volume of the link complement as a comparison measure. Brunnian or highly knotted complements are evidence for strong causal interlocking; higher action density remains a dynamical/statistical claim to be measured with the same kernel.

Theorem Spine (Provable Core under A1-A5)

In this section we also assume standard approximate-identity properties:
$\phi_\eta \geq 0$, $\int_{\mathbb{R}} \phi_\eta(s)\, ds = 1$, $\|\phi_\eta\|_\infty < \infty$ for fixed $\eta > 0$, and $\phi_\eta \to \delta$ weakly as $\eta \to 0^+$. Compact support or sufficient decay may be used; the estimates below require boundedness on the sampled domain.

Assumptions Checklist (Use Before Citing a Theorem)

Claim	A1	A2	A3	A3b	A4	A5
Theorem 1 (finiteness/nonnegativity)	required	required	required	required	not required	not required
Theorem 2 (coarea limit)	required	required	required	required	required	not required
Corollary 2.1 (integer labels)	required	required	required	not required	required	required
Theorem 3 (bifurcation criterion)	required	required	required	not required	required (except at critical value)	required
Theorem 4 (two-sided bounds)	required	required	required	required	not required	not required

Theorem 1 (Well-defined finite regularized action)

Under (A1)-(A3b), $\bar{\mathcal{A}}_{\mathrm{self},\eta}[\gamma]$ is finite and nonnegative.

Proof. Write

$$\bar{\mathcal{A}}_{\mathrm{self},\eta} = \frac{1}{T^2} \int_{[0,T]^2} \frac{\phi_\eta(F_\gamma(t,t'))}{r(t,t')^2 J_\gamma(t,t')}\, dt\, dt'$$

The integrand is nonnegative because $\phi_\eta \geq 0$, $r^{-2} > 0$, and $J_\gamma^{-1} > 0$, so $\bar{\mathcal{A}}_{\mathrm{self},\eta} \geq 0$.
By (A3) and (A3b), on the support of $\phi_\eta(F_\gamma)$ we have $r \geq r_{\min} > 0$ and $J_\gamma \geq J_{\min} > 0$, hence $r^{-2} J_\gamma^{-1} \leq r_{\min}^{-2} J_{\min}^{-1}$. Therefore

$$0 \leq \bar{\mathcal{A}}_{\mathrm{self},\eta} \leq \frac{1}{T^2}\, r_{\min}^{-2} J_{\min}^{-1}\, \|\phi_\eta\|_\infty\, |[0,T]^2| = \frac{\|\phi_\eta\|_\infty}{r_{\min}^2 J_{\min}} < \infty$$

So the functional is finite and nonnegative.

Theorem 2 (Coarea reduction to causal locus)

Under (A1)-(A4), the $\eta \to 0^+$ limit of
$\bar{\mathcal{A}}_{\mathrm{self},\eta}$ is the weighted 1D measure of the causal locus:

$$\lim_{\eta \to 0^+} \bar{\mathcal{A}}_{\mathrm{self},\eta} = \frac{1}{T^2} \int_{\mathcal{L}_{\mathrm{causal}}} \frac{1}{r(t,t')^2\, J_\gamma(t,t')\, \|\nabla F_\gamma(t,t')\|}\, d\ell$$

where $\mathcal{L}_{\mathrm{causal}} = \{(t,t') \in T^2 : F_\gamma(t,t') = 0\}$.

Proof. Apply the coarea formula on $[0,T]^2$ with level function F_γ:

$$\int_{[0,T]^2} \frac{\phi_\eta(F_\gamma)}{r^2 J_\gamma}\, dt\, dt' = \int_{\mathbb{R}} \phi_\eta(s)\, H(s)\, ds$$

with

$$H(s) \equiv \int_{F_\gamma^{-1}(s)} \frac{1}{r^2\, J_\gamma\, \|\nabla F_\gamma\|}\, d\ell$$

By (A4), $\|\nabla F_\gamma\| \geq \nu > 0$ on a tubular neighborhood of the retained zero level, so nearby level sets are regular 1-manifolds and $H(s)$ is continuous near $s = 0$. By (A3) and (A3b), both r^{-2} and J_γ^{-1} are bounded on the active support, and the diagonal collar prevents accumulation on the excluded trivial branch. Hence $H(s)$ is locally bounded. Since ϕ_η is an approximate identity, $\int \phi_\eta(s) H(s)\, ds \to H(0)$ as $\eta \to 0^+$. Dividing by T^2 yields the claimed limit.

Corollary 2.1 (Discrete branch labels)

Closed connected components of $\mathcal{L}_{\mathrm{causal}}$ carry winding numbers $(p,q) \in \mathbb{Z}^2$ on the unpunctured torus. Components retained on the lifted strip carry algebraic fiber-intersection degrees ι_a with generic vertical fibers. These labels are unchanged under smooth deformations that preserve (A4), keep the same diagonal-collar and memory-window convention, and remain inside one homotopy class (A5).

Proof. Under (A4), each retained component of the level set $F_\gamma = 0$ is a smooth embedded 1-manifold in the chosen chart. If it closes on the unpunctured torus, it defines a homology class in $H_1(T^2, \mathbb{Z}) \cong \mathbb{Z}^2$, whose coordinates are the winding numbers (p,q). If the diagonal collar or finite-memory boundary cuts the component, the component need not define a closed torus class, but its algebraic intersection number with a generic vertical fiber is well-defined as long as the component remains transverse to that fiber and endpoints do not cross the declared boundary. Under a smooth deformation preserving regularity, boundary convention, and homotopy class, these data evolve by isotopy, so the winding classes and fiber-intersection degrees are unchanged.

Theorem 3 (Bifurcation criterion for quantized branch changes)

For a smooth one-parameter family γ_λ (equivalently F_λ), component count and winding labels can change only at parameter values λ_* where transversality fails:

$$F_{\lambda_*}(t,t') = 0, \qquad \nabla F_{\lambda_*}(t,t') = 0$$

for some $(t,t') \in T^2$.

Proof. Fix λ_0 such that $F_{\lambda_0}^{-1}(0)$ is regular (A4). By the implicit function theorem, near every point of $F_{\lambda_0}^{-1}(0)$ the zero set is a smooth curve varying smoothly with λ. Compactness of T^2 gives a finite cover, so the full causal locus varies by isotopy for λ in a neighborhood of λ_0. Isotopy preserves component count and homology labels. Therefore these quantities are locally constant on regular parameter intervals. Any change between two regular intervals must pass through a non-regular parameter where $\nabla F = 0$ at a zero-level point.

Theorem 4 (Two-sided bounds useful for validation)

Under (A1)-(A3b), for any fixed $\eta > 0$:

$$0 \le \bar{\mathcal{A}}_{\mathrm{self},\eta} \le \frac{\|\phi_\eta\|_\infty}{r_{\min}^2 J_{\min}}$$

If additionally $r \le r_{\max}$ and $J_\gamma \le J_{\max}$ on support, then

$$\bar{\mathcal{A}}_{\mathrm{self},\eta} \ge \frac{1}{r_{\max}^2 J_{\max} T^2} \int_{[0,T]^2} \phi_\eta(F_\gamma)\, dt\, dt'$$

Proof. The upper bound is exactly the estimate used in Theorem 1. For the lower bound, if $r \le r_{\max}$ and $J_\gamma \le J_{\max}$ on support, then $r^{-2} \ge r_{\max}^{-2}$ and $J_\gamma^{-1} \ge J_{\max}^{-1}$ on support, hence

$$\bar{\mathcal{A}}_{\mathrm{self},\eta} = \frac{1}{T^2} \int_{[0,T]^2} \frac{\phi_\eta(F_\gamma)}{r^2 J_\gamma}\, dt\, dt' \ge \frac{1}{r_{\max}^2 J_{\max} T^2} \int_{[0,T]^2} \phi_\eta(F_\gamma)\, dt\, dt'$$

Meaning: numerical pipelines can assert hard pass/fail envelopes before any physical interpretation is attempted.

Analytic Benchmarks (Circular Orbit)

For a circular orbit of radius R and speed $v = \beta c_f$:

$$2R\left|\sin\left(\frac{\omega\Delta}{2}\right)\right| = c_f\Delta, \quad \text{with } \omega = \frac{v}{R}$$

Define $\xi = \frac{\omega\Delta}{2}$, giving the root condition:

$$\sin\xi = \frac{\xi}{\beta}$$

The nontrivial self-hit threshold is

$$\beta^\star = 1$$

For $\beta \le 1$, the only solution is the trivial coincidence $\xi = 0$, so the circular self-action vanishes. For $\beta > 1$, each admissible root ξ_n determines a concrete branch datum:

$$\Delta_n = \frac{2\xi_n}{\omega}, \qquad r_n = c_f\Delta_n = \frac{2R\xi_n}{\beta}, \qquad J_n = 1 - \beta\cos\xi_n = 1 - \xi_n\cot\xi_n$$

The derivative of the root function is

$$g_\beta'(\xi_n) = \cos\xi_n - \frac{1}{\beta} = \cos\xi_n - \frac{\sin\xi_n}{\xi_n}$$

which is the additional coarea factor controlling branch weight when the two-time integral is collapsed onto the circular causal locus.

Near threshold, write $\beta = 1 + \mu$ with $\mu > 0$ small. The principal root then satisfies

$$\xi_0 \sim \sqrt{6\mu}, \qquad r_0 \sim 2R\sqrt{6\mu}, \qquad J_0 \sim 2\mu, \qquad g_\beta'(\xi_0) \sim -2\mu$$

Hence the principal branch contribution to the circular action density scales like

$$\frac{1}{r_0^2\,|J_0|\,|g'_\beta(\xi_0)|} \sim \frac{1}{96R^2\,\mu^3}$$

Here J_n is the dimensionless Master Equation received-flux Jacobian, while $|g'_\beta(\xi_n)|$ is the root-density/coarea factor from reducing the circular causal locus to discrete roots. They are distinct geometric factors, not a double count, and both scale as μ on the principal near-threshold branch. The denominator arithmetic is $r_0^2|J_0||g'_\beta| \sim (24R^2\mu)(4\mu^2) = 96R^2\mu^3$. This is the action-functional expression of the same circular caustic seen in the force law: the onset of self-hit is already singular once the Jacobian and coarea reduction are both kept.

At high speed, all admissible roots lie in $(0, \beta)$, so the branch count grows only linearly with β. The circular toy therefore gives a controlled benchmark: discrete branch creation, explicit near-threshold asymptotics, and a root-by-root action density that can be compared directly to numerical orbit scans.

Circular Benchmark as a Branch-Count Theorem

Define

$$g_\beta(\xi) = \sin\xi - \frac{\xi}{\beta}$$

Admissible circular self-hit branches are zeros of g_β in $(0, \beta)$.

Proposition 5.1 (Discrete Root Count and Branch-Change Criterion)

Fix a compact admissible interval $I_{\text{branch}} = [a, b] \subset (0, \beta)$ with boundary regularity $g_\beta(a) \neq 0$, $g_\beta(b) \neq 0$.

1. For fixed $\beta > 0$, the admissible root set $\{\xi \in I_{\text{branch}} : g_\beta(\xi) = 0\}$ is finite.
2. In a smooth one-parameter scan $\beta = \beta(\lambda)$, the root count in I_{branch} is locally constant except when

$$g_\beta(\xi) = 0, \qquad \partial_\xi g_\beta(\xi) = 0$$

at some interior point $\xi \in (a, b)$.

Proof. For fixed β, g_β is real-analytic on $(0, \beta)$, hence zeros are isolated unless the function is identically zero on an interval. That cannot occur here because g_β is not identically zero. A discrete subset of a compact interval is finite, proving (1).

For (2), if ξ_* is a simple root ($\partial_\xi g_\beta(\xi_*) \neq 0$), the implicit function theorem gives a unique smooth continuation of that root under small parameter changes, so simple roots cannot be created or destroyed locally. Root-count change can therefore occur only when simplicity fails, i.e. when $g_\beta = 0$ and $\partial_\xi g_\beta = 0$ simultaneously (multiple/tangent root). Boundary-root events are excluded by the boundary-regularity condition.

For the principal circular branch, the bifurcation point occurs at $(\beta, \xi) = (1, 0)$ in the limiting sense. Higher branches appear at interior tangencies where both equations hold with $\xi > 0$. This is the 1D analog of Theorem 3 and provides an explicit, checkable bifurcation condition for the circular toy model.

Dynamical Interpretation

- Candidate stable periodic orbits should first appear as **critical points** of $\bar{\mathcal{A}}_{\text{total}}$ constrained within a winding class. The delay flow need not be a gradient flow of this functional, so extremality is a branch-selection test, not a proof of asymptotic stability.
- **Existence vs. stability:** Topology of $\mathcal{L}_{\text{causal}}$ constrains which families can exist by identifying bifurcations where branches reconnect. Linear spectra of the delay equation decide which of those families persist or attract. The causal locus gives the branch skeleton; Lyapunov exponents and return-map spectra test dynamical survival.
- **Discreteness:** Each winding class gives an integer self-hit count; moving between classes requires a reconnection event. This supplies a candidate mechanism for mass gaps and generation-like families, but the actual mass map still requires shielding, partner terms, and Noether sea response.
- **Conservation with memory:** In the symmetry-preserving delayed action, time-translation and rotational symmetry imply conserved total energy and total angular momentum as history functionals. In regularized working models, these same quantities become validation diagnostics. Energy includes the history contribution stored in active causal wakes.
- **Gradient vs. symplectic:** The master equation is conservative; critical points of $\bar{\mathcal{A}}$ should be compared with KAM-style persistence islands, not with dissipative sinks. If a separate Noether sea coupling introduces dissipation, minima could become attractors, but absent that extra channel, stability means orbital persistence rather than asymptotic convergence.

Emergent Geometry Constraints

Define the coarse-grained hit density

$$\mathcal{I}(t,\mathbf{x}) = \sum_j \int_{-\infty}^{t} \frac{\delta_\eta\big(\|\mathbf{x}-\mathbf{x}_j(t')\| - c_f(t-t')\big)}{\|\mathbf{x}-\mathbf{x}_j(t')\|^2\, J_j(t,\mathbf{x};t')}\, dt'$$

where

$$J_j(t,\mathbf{x};t') = \left|1 - \frac{\mathbf{v}_j(t')\cdot\hat{\mathbf{n}}(t,\mathbf{x};t')}{c_f}\right|, \qquad \hat{\mathbf{n}}(t,\mathbf{x};t') = \frac{\mathbf{x}-\mathbf{x}_j(t')}{\|\mathbf{x}-\mathbf{x}_j(t')\|}$$

This scalar hit density is an effective coarse-grained summary of causal-wake intersections, not a substrate metric and not by itself the observer-level effective metric. At most, it supplies one provisional scalar channel feeding the ADM/Cartan effective-metric handoff. A restricted isotropic subcase may be written as

$$g^{\rm eff}_{\mu\nu}dx^\mu dx^\nu = -N^2(\mathcal{I})\, c_\star^2 dt^2 + \Omega_s^2(\mathcal{I})\, h_{ij}dx^i dx^j$$

with small couplings $N = 1+\lambda_t\mathcal{I}$ and $\Omega_s = 1+\lambda_s\mathcal{I}$ in the weak-field regime. Here $c_\star$ must be declared: primitive branch charts may set $c_\star = c_f$, while observer-level metric comparisons normally use the dressed asymptotic channel speed. The full geometry program must also include the Noether sea lapse, shift/medium velocity, spatial metric response, stress, and PPN decision variables used by the spacetime chapters. Bianchi identities and weak-equivalence demands constrain the admissible scalar subcase; otherwise the emergent geometry reduces to a scalar-tensor approximation with potentially observable fifth forces. Matching the long-range limit of test-assembly motion to geodesics in $g^{\rm eff}_{\mu\nu}$ is the consistency check linking microscopic causal hits to macroscopic effective curvature.

Here, “fifth force” means an additional long-range interaction mediated by the scalar sector encoded in $\mathcal{I}$, on top of the shared effective-metric response. If that scalar coupling is not sufficiently constrained, test assemblies can acquire composition-dependent accelerations, producing weak-equivalence-principle violations and post-Newtonian deviations that are tightly bounded experimentally.

Numerical check: evolve two assemblies with different internal $\bar{\mathcal{A}}_{\rm total}$ through the same prescribed $\mathcal{I}(t,\mathbf{x})$ background and verify their centers follow the same geodesic to numerical tolerance.

Mean-field view: in a dilute limit with many architrinos, the closure target is to derive a Vlasov equation for $f(t,\mathbf{x},\mathbf{v})$ whose force term is induced by the coarse-grained hit density. That derivation would provide the statistical bridge from causal-wake microdynamics to continuum geometry.

Implementation Notes (Appendix)

- Use the same δ_η and η for force and action estimators.
- For periodic orbits, normalize by T^2 and enforce periodic boundary conditions.
- For circular-orbit calibration, compute ξ_n roots numerically and sum with the Jacobian factor.
- Handle the $\beta = 1$ onset caustic with care; the unregularized circular action is singular there once both Jacobian and coarea factors are retained.
- Keep $\eta > 0$ during variation: $\nabla\delta$ terms appear in $\delta\mathcal{A}$; regularization makes the Euler–Lagrange equations well-posed. Take $\eta \to 0$ only after solving or bounding solutions.

Simulation Protocol (Minimal Theorem-Backed Checks)

For each simulated orbit family:

1. Verify (A1)-(A4) numerically (regularity, no-support collisions, transversality).
2. Compute $\bar{\mathcal{A}}_{\text{self},\eta}$ at multiple η and confirm boundedness by Theorem 4.
3. Extract $\mathcal{L}_{\rm causal}$, its closed-component winding labels (p,q) where available, and its lifted-strip fiber-intersection degrees ι_a.
4. Scan one control parameter (e.g., β or radius ratio) and confirm labels change only at detected transversality failures.
5. In the circular benchmark, verify Proposition 5.1 double-root condition at branch transitions.

Limitations and Caveats

- **Rest mass is not just self-action:** $\mathcal{A}_{\rm self}$ needs careful units; observer-level rest energy also depends on partner interactions, shielding, Noether sea coupling, and the medium-response tensor.
- **Minima ≠ stability without dynamics:** Stability depends on the full DDE flow; the functional must be windowed/normalized (e.g., one period) to avoid divergences and to compare orbits meaningfully.
- **Topology needs precision:** Time is monotone; periodic motion yields a spatially closed path but a helical curve in absolute timespace. Be explicit about which projection or linking notion defines the “topological class.”
- **Cohomology language is aspirational:** A cochain complex over the moduli of periodic orbits is not yet constructed; treat “cohomology of causal interaction” as a research direction, not a result.

Closure Extension: Spin Bundle and Confinement Energy Law

Two downstream theorem targets can be stated on top of the existing causal-locus spine. They are not used by the theorems above; they mark what would have to be proven before spin and confinement language becomes native rather than comparative.

(T5.1) Spinor lift target

Construct a framed configuration bundle for nested shell braid ordered axes and prove that the relevant internal-orientation transport lifts through

$$\widetilde{R} : SU(2) \simeq \mathrm{Spin}(3) \to SO(3)$$

so the internal phase distinguishes 2π and 4π loops.

(T5.2) Open-vs-closed braid energy target

Define an effective color-braid energy law:

$$E_{\mathrm{open}}(L) = \sigma_{\mathrm{eff}} L + E_0 + \mathcal{O}(1/L), \qquad \sigma_{\mathrm{eff}} > 0$$

$$E_{\mathrm{closed}}(L) \to E_\infty < \infty \quad (L \to \infty)$$

Combined with causal-locus class constraints, this would give a quantitative separation between confined open sectors and screened singlet sectors after the color-braid and singlet-sector proof is supplied. Until that proof is supplied, the equations are an effective closure target rather than a result of this chapter.

Integration map

- causal-locus topology and bifurcation class invariants: **this chapter**
- color-algebra and singlet braid structure: assemblies/fermions/color-charge-su3.md
- gauge-covariant effective layer and failure criteria: assemblies/gauge-symmetries.md

Reduced Branch-Certificate Targets

The theorem spine proves that the scalar statistic is finite, has a coarea limit, and carries branch labels that remain invariant under the stated deformations. The next question is stronger: whether a retained branch chart also behaves like a conservative reduced action system. The following residuals are therefore validation targets, not additional theorems of this chapter.

Branch return-map symplectic residual. The scalar statistic can identify candidate stationary branch classes, but a stationary value of $\bar{A}$ is not yet a Hamiltonian closure claim. On a retained branch chart $\mathfrak{B}$ with reduced section coordinates $z = (Q^a, \Pi_a)$, let

$$\mathcal{P}_{\mathfrak{B}} : z_n \mapsto z_{n+1}$$

be the one-cycle return map and let

$$M_{\mathfrak{B}}(z) = D\mathcal{P}_{\mathfrak{B}}(z)$$

be its linearized monodromy. If the reduced chart is genuinely inherited from a symmetry-preserving delayed action, then after the retained constraints and section condition are solved there must be a pulled-back symplectic form $\Omega_{\mathfrak{B}}$ for which

$$\mathcal{R}_\Omega(\mathfrak{B}) \equiv \sup_{z \in U} \left\| M_{\mathfrak{B}}(z)^T \Omega_{\mathfrak{B}}(z) M_{\mathfrak{B}}(z) - \Omega_{\mathfrak{B}}(\mathcal{P}_{\mathfrak{B}}(z)) \right\|$$

is small on the tested neighborhood U of the branch. The companion phase-volume residual

$$\mathcal{R}_{\mathrm{vol}}(\mathfrak{B}) \equiv \sup_{z \in U} |\det M_{\mathfrak{B}}(z) - 1|$$

is weaker but easier to compute. The closure direction is therefore:

$$\nabla_\gamma \bar{A} = 0 \quad \text{within a winding class} \qquad \mathcal{R}_\Omega(\mathfrak{B}) \le \epsilon_\Omega \qquad \lambda_{\mathrm{sec}} > 0$$

The first condition marks a candidate branch class, the second tests whether the retained return map has the canonical structure expected of an action-derived conservative reduction, and the third checks local section persistence. A failure of $\mathcal{R}_\Omega$ does not falsify the Master EOM; it says that the scalar action-counting extremum has not yet been promoted to a reduced Hamiltonian branch certificate.

Hamilton-Jacobi branch phase target. If a retained branch chart passes the action-derived return-map tests, one can ask for a Hamilton-Jacobi description of the same reduced motion. This is only a comparison target until the delayed action residual closes. Let $H_{\mathfrak{B}}(Q, \Pi, t)$ be the reduced Hamiltonian on the certified chart. A branch principal function $W_{\mathfrak{B}}(Q, t)$ should satisfy

$$\mathcal{R}_{\mathrm{HJ}}(Q, t) \equiv \partial_t W_{\mathfrak{B}}(Q, t) + H_{\mathfrak{B}}(Q, \partial_Q W_{\mathfrak{B}}(Q, t), t)$$

with $\mathcal{R}_{\mathrm{HJ}} \to 0$ on the retained window. The associated momentum reconstruction is

$$\Pi_a = \partial_{Q^a} W_{\mathfrak{B}}$$

and the first-order branch motion is

$$\dot{Q}^a = \left.\frac{\partial H_{\mathfrak{B}}}{\partial \Pi_a}\right|_{\Pi=\partial_Q W_{\mathfrak{B}}}$$

For a time-independent reduced chart, the separated form

$$W_{\mathfrak{B}}(Q,t) = W^0_{\mathfrak{B}}(Q) - E_{\mathfrak{B}} t$$

turns the energy label $E_{\mathfrak{B}}$ into a branch-family parameter. In $\mathbb{AAA}$ this would not replace the causal-root ledger; it would be a compact phase-function certificate that the retained ledger, wake-history charge, and reduced canonical coordinates are mutually consistent.

Summary and Status

- The chapter defines causal self-hit and total action-counting statistics from the Jacobian-weighted inverse-square delayed kernel, plus normalized forms for periodic orbits.
- The causal locus $\mathcal{L}_{\mathrm{causal}} \subset T^2$ supplies discrete branch labels such as winding class, writhe candidate, and link type; those labels segment orbit families but do not by themselves prove stability or mass.
- The circular-orbit benchmark gives an analytic threshold at $\beta = 1$, explicit branchwise Jacobians, and controlled near-threshold asymptotics, anchoring numerical calibrations.
- Under explicit assumptions (A1-A5), the theorem spine establishes finiteness, coarea reduction, topological invariance away from critical points, and a bifurcation condition for branch changes.
- The emergent-metric ansatz from coarse-grained hit density $\mathcal{I}$ remains conjectural until weak-field, equivalence, and PPN constraints are met.
- Overall, the causal-locus action-counting route is theorem-level in the regularized regime, while mass mapping, asymptotic stability, branch Hamiltonian certification, and emergent metric closure remain open.

Effective Lagrangian

This chapter formalizes the conditional variational scaffold used by $\mathbb{AAA}$. Its purpose is to connect the exact, path-history-dependent microdynamics of discrete architrinos to coarse-grained effective descriptions of macroscopic assembly behavior in the Noether sea.

The bridge is deliberately conditional. The Master EOM remains the primary dynamics at the substrate level; an action or Lagrangian chart becomes theorem-grade only after its variation, boundary, and conservation residuals close on the retained branch chart. Until then, the effective Lagrangian is a disciplined inference device rather than an independent ontology.

Ordinary Lagrangian Orientation

In ordinary local mechanics, one chooses generalized coordinates $q^a(t)$ and writes a Lagrangian $L(q, \dot{q}, t)$, often in the simple form

$$L = T - V$$

where T is kinetic energy and V is potential energy. The corresponding action is

$$S[q] = \int_{t_a}^{t_b} L(q, \dot{q}, t)\, dt$$

and fixed-endpoint stationarity,

$$\delta S = 0$$

gives the Euler-Lagrange equation

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^a} - \frac{\partial L}{\partial q^a} = 0$$

for each coordinate q^a. This equation is not a separate force postulate. It is the recovery condition that the chosen scalar L must satisfy if the action is to generate the equations of motion.

Operationally, stationarity is tested by nearby trial paths
$q^a_\epsilon(t) = q^a(t) + \epsilon\xi^a(t)$ with
$\xi^a(t_a) = \xi^a(t_b) = 0$. Because ξ^a is otherwise arbitrary,

setting the first variation of S to zero forces the Euler-Lagrange expression itself to vanish. The action is therefore a history functional with units of energy times time, not an instruction to minimize instantaneous energy.

A minimal recovery check is the one-dimensional harmonic oscillator. For a mass m attached to an ideal spring of stiffness k with displacement $x(t)$,

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

so the Euler-Lagrange equation gives

$$\frac{d}{dt}(m\dot{x}) - (-kx) = 0$$

or equivalently

$$m\ddot{x} = -kx$$

which is the same equation obtained from Newton's law and Hooke's law. The value of the example is not that Lagrangian mechanics replaces the tested motion, but that it recovers the same equation from an energy scalar and generalizes cleanly to many coordinates.

The $\mathbb{AAA}$ correction to this toy example is more informative than the recovery itself. A real assembly-level spring is a delayed restoring channel, so the first effective model is not exactly $m\ddot{x} = -kx$ but

$$m\ddot{x}(t) = -k\, x(t - \tau_{\text{eff}}) + \cdots$$

for an effective causal-wake delay τ_{eff} across the assembly. Expanding the delayed displacement gives

$$\left(m + \frac{1}{2}k\tau_{\text{eff}}^2\right)\ddot{x} - k\tau_{\text{eff}}\dot{x} + kx = O(k\tau_{\text{eff}}^3 x^{(3)})$$

on a slowly varying branch. The leading correction is therefore sign-definite once the branch delay orientation is fixed. For the causal restoring convention displayed above it is anti-damping, the same local pattern that appears as positive tangential work in the circular binary. The mass-like coefficient is also shifted by the delayed response. This does not prove the full assembly mass map, but it shows in the simplest chart why inertia and dissipation-like terms are delayed-response quantities rather than primitive architrino constants.

Historically, the route into this form matters. Newtonian force balance can be projected along fixed-endpoint variations as virtual work. For conservative interactions, $\mathbf{F} = -\nabla V$ turns the work term into a variation of potential energy, while the inertial term supplies a variation of kinetic energy plus an endpoint term. When the endpoint variation vanishes, Hamilton's construction turns that differential relation into the stationary action of $T - V$. The useful condition is therefore stationarity of the action, not a literal minimum in every case.

The same idea survives in $\mathbb{AAA}$ only after changing the object being varied. The Master EOM is not local in the instantaneous variables $(\mathbf{x}_i(t), \dot{\mathbf{x}}_i(t))$: receiver acceleration depends on delayed source coordinates, causal-root branches, Jacobian weights, and the retained causal-wake history. A local expression $L(\mathbf{x}, \dot{\mathbf{x}}, t)$ therefore cannot be the substrate-level action for the exact law. The appropriate candidate is a multi-time path-history functional whose variation must reproduce the delayed, Jacobian-weighted branch law.

The operational bridge is:

1. ordinary mechanics uses $L(q, \dot{q}, t)$ and tests $\delta S = 0$;
2. $\mathbb{AAA}$ uses a regularized delayed action $S_\eta[\{\mathbf{x}_i\}]$ over path history;
3. the action is promoted only if its variation yields the Master EOM on the retained branch chart;
4. failure is measured by the variation residual $\mathbf{R}_i^{(\eta)}(t)$ and the window diagnostic $\epsilon_{\text{var}}^{(\eta)}(W)$ defined below.

Thus the Lagrangian question in $\mathbb{AAA}$ is not whether one can write a familiar-looking $T - V$ expression. The question is whether a delayed action with the same causal-root, Jacobian, boundary, and wake-history conventions as the Master EOM has a stationary variation whose residual closes. Only then do Noether-style energy, momentum, and angular-momentum statements become theorem-grade rather than diagnostic.

Ordinary Hamiltonian Orientation

Hamiltonian mechanics repackages the same local dynamics into coordinates and canonical momenta. Starting from a local Lagrangian $L(q, \dot{q}, t)$, define the canonical momentum

$$p_a \equiv \frac{\partial L}{\partial \dot{q}^a}$$

and, when the velocity-momentum map can be inverted, define the Hamiltonian by the Legendre transform

$$H(q,p,t) = p_a \dot{q}^a - L(q,\dot{q},t)$$

with the velocities rewritten in terms of (q,p,t). Hamilton's equations are

$$\dot{q}^a = \frac{\partial H}{\partial p_a}, \qquad \dot{p}_a = -\frac{\partial H}{\partial q^a}$$

so one second-order equation in q^a becomes a first-order flow on phase space (q^a, p_a). In simple time-independent mechanical systems H is often the total energy $T+V$, but the defining statement is the Legendre transform and the canonical flow, not the energy slogan by itself.

The same equations can also be read from the phase-space action

$$S_H[q,p] = \int_{t_a}^{t_b} \left(p_a \dot{q}^a - H(q,p,t)\right) dt$$

when variations in both q^a and p_a are admitted and endpoint variations of q^a vanish. Variation with respect to p_a gives $\dot{q}^a = \partial H/\partial p_a$, while variation with respect to q^a gives $\dot{p}_a = -\partial H/\partial q^a$. This is the action-level form of the canonical flow, and it is the part that matters when asking whether a reduced $\mathbb{AAA}$ chart is genuinely Hamiltonian rather than only an energy-like fit.

The conjugate momenta are more than bookkeeping in ordinary mechanics. When a coordinate is cyclic, the corresponding conjugate momentum is conserved; the same coordinate-momentum pairing later becomes the classical object used in Bohr-Sommerfeld action integrals and in canonical commutation rules. In $\mathbb{AAA}$ these are recovery targets for a reduced effective chart, not permission to quantize the substrate variables directly.

This matters for $\mathbb{AAA}$ because the exact Master EOM is a delayed path-history law, not an ordinary finite-dimensional phase-space law. The instantaneous pair $(\mathbf{x}_i(t), \mathbf{p}_i(t))$ does not contain all active causal-root, boundary, and wake-history data. A Hamiltonian chart is therefore an effective reduction: it is admissible only when a coarse-graining compresses the retained path history into coordinates and momenta while preserving the comparison invariants. The test is not merely that an expression called H_{eff} can be written, but that the induced return map preserves the relevant measure, symplectic form, or Poisson-bracket structure to the declared tolerance.

For $\mathbb{AAA}$, the strongest phase-space use case is therefore not an arbitrary instantaneous snapshot. It is a replayable or phase-locked branch chart: a reduced description in which the retained internal motion returns to a comparable section despite bounded surrounding influences. If a retained assembly is modeled by coarse coordinates Q^A and by phase-bearing sub-assemblies indexed by $\alpha = 1, \ldots, N_{\text{ph}}$, the candidate chart has the form

$$z_{\mathfrak{B}} = (Q^A, \Pi_A, \theta^\alpha, I_\alpha)$$

where θ^α records the sub-assembly phase and I_α is the conjugate action variable for that phase. Each retained phase-locked sub-assembly adds its own phase-action pair. Surrounding influences are admissible only when they are represented as fixed branch data, slow parameters, or additional coordinates over the comparison window; otherwise the chart is a driven open system rather than a closed Hamiltonian phase space.

The action variables are local objects unless the phase torus is globally unobstructed. For a three-layer nested shell braid chart, the phase circles of the outer, middle, and inner binaries need not form a trivial T^3 bundle over the retained branch family. A cycle that exchanges two orbital planes can carry an integer phase-bundle winding

$$c_1[\theta^O, \theta^M, \theta^I] = \frac{1}{2\pi} \oint_\gamma d(\text{relative phase}) \in \mathbb{Z}$$

when the relative phase closes on the branch. This is the topological content of integer resonance lock: the lock ratios (m,n) in Dyadic Resonance Lock make the phase-bundle data integral rather than irrationally drifting. The effective Hamiltonian chart is therefore globally promotable only on resonance-locked branches where the returned phase torus and causal-root ledger close together. Off-lock, the same I_α may exist on a local patch but acquires monodromy under return, so quantization and measure preservation become local fitting statements rather than global chart facts.

Regularized Nonlocal Action and Variation

The Master Equation of Motion for architrinos is non-Markovian, driven by intersections between receiver trajectories and past causal wake surfaces. Consequently, any action-level scaffold for this law cannot be a local integral over instantaneous states. It must be a multi-time functional over path history, and its variation residual must be identified before the scaffold is treated as an exact action derivation. A scale-only derivation requires that residual to vanish or become a boundary term; a recoil-inclusive derivation may instead retain it as a mechanical wake-emission resistance term.

For a finite, isolated set of architrinos parameterized by absolute time t in the Euclidean void, use the $\eta > 0$ regularized delayed action below. The exact causal wake kernel is recovered in the weak branch limit as $\eta \to 0^+$. The admissible interaction sum excludes trivial self-coincidence: $i \neq j$ terms are retained, and $i = j$ terms are retained only on nontrivial self-hit branches with $t - t_0 \geq \tau_{\min} > 0$ or with an explicitly declared core regularization.

The $\eta \to 0^+$ statement is a weak or distributional scaling claim over declared observables unless a stronger topology is explicitly supplied. A finite-regulator trend supports this action scaffold only after the observable map, normalization, admissible test functions, and uniform control needed for the limit are stated. It is not by itself a proof of the exact causal-wake action.

$$S_\eta[\{\mathbf{x}_i\}] = \int dt \sum_i \frac{1}{2}\mu_{\text{arch}} \|\dot{\mathbf{x}}_i(t)\|^2 - \frac{1}{2}\sum_{i,j}^{\text{adm}} \frac{\kappa\, \sigma_{ij} |q_i q_j|}{c_f} \int dt \int_{-\infty}^{t} dt_0\, \frac{\phi_\eta\big(g_{ij}(t,t_0)\big)}{r_{ij}(t;t_0)}$$

$$g_{ij}(t,t_0) \equiv t - t_0 - \frac{r_{ij}(t;t_0)}{c_f}, \qquad r_{ij}(t;t_0) = \|\mathbf{x}_i(t) - \mathbf{x}_j(t_0)\|, \qquad \phi_\eta \equiv \delta_\eta$$

Here:

- $\mathbf{x}_i(t)$ is the trajectory of architrino i.
- μ_{arch} is the universal force/energy bookkeeping constant, not a particle-specific inertial mass.
- $r_{ij}(t;t_0)$ is the Euclidean separation between reception and emission events.
- δ_η is a mollified delta function of width $\eta > 0$. It supports Lipschitz control only together with the collision floor, finite-branch, transversality, and integrability assumptions below.
- $\sigma_{ij} = \text{sign}(q_i q_j)$ enforces attraction for opposite polarities and repulsion for like polarities.

Regularization and Admissibility Assumptions

The derivation below is valid under:

- **(EL1)** $\mathbf{x}_i \in C^2([t_a,t_b];\mathbb{R}^3)$ and variations $\boldsymbol{\xi}_i$ are C^1 with $\boldsymbol{\xi}_i(t_a) = \boldsymbol{\xi}_i(t_b) = 0$.
- **(EL2)** $\phi_\eta \in C_c^1(\mathbb{R})$, $\phi_\eta \geq 0$, $\int \phi_\eta(s)\, ds = 1$.
- **(EL3)** Collision and trivial-self exclusion on active support: $r_{ij}(t;t_0) \geq r_{\min} > 0$ whenever $\phi_\eta(g_{ij}(t,t_0)) \neq 0$, and for $i = j$ the active support also satisfies $t - t_0 \geq \tau_{\min} > 0$ unless a separate core regularization supplies the same lower-bound control.
- **(EL4)** Delay-root transversality on active branches: $\partial_{t_0} g_{ij}(t,t_0) \neq 0$ when $g_{ij}(t,t_0) = 0$.
- **(EL5)** Integrability on the chosen history window, either by finite support or sufficient tail falloff, so differentiation under the time integrals is justified.
- **(EL6)** Delayed branch convention: only $t_0 \leq t$ contributes (equivalently, the $\Theta(t - t_0)$ branch of the causal selector).

Kernel Variation and Branch Reduction

This subsection isolates the exact step at which a variational scaffold can fail. Set $\mathbf{x}_i^\varepsilon = \mathbf{x}_i + \varepsilon \boldsymbol{\xi}_i$ and differentiate at $\varepsilon = 0$.

Kinetic term:

$$\delta S_{\eta,\text{kin}} = \sum_i \int_{t_a}^{t_b} \mu_{\text{arch}} \dot{\mathbf{x}}_i \cdot \dot{\boldsymbol{\xi}}_i\, dt = -\sum_i \int_{t_a}^{t_b} \mu_{\text{arch}} \ddot{\mathbf{x}}_i \cdot \boldsymbol{\xi}_i\, dt$$

For the interaction kernel

$$\mathcal{K}_{ij}(t,t_0) \equiv \frac{\phi_\eta(g_{ij}(t,t_0))}{r_{ij}(t;t_0)}, \qquad \hat{\mathbf{r}}_{ij} \equiv \frac{\mathbf{x}_i(t) - \mathbf{x}_j(t_0)}{r_{ij}(t;t_0)}$$

the receiver-coordinate gradient is

$$\nabla_{\mathbf{x}_i(t)} \mathcal{K}_{ij} = -\hat{\mathbf{r}}_{ij} \left[\frac{\phi_\eta(g_{ij})}{r_{ij}^2} + \frac{\phi_\eta'(g_{ij})}{c_f\, r_{ij}} \right]$$

This receiver-side gradient is one ingredient in the full first variation, but it is not the complete Euler-Lagrange expression. In the double-time action, each varied worldline appears both as a receiver coordinate $\mathbf{x}_i(t)$ and as a source coordinate inside transposed kernels. The full branch-resolved variation is carried out in master-equation. The term proportional to $\phi'_\eta(g_{ij})$ is not an algebraic nuisance to discard: on a purely delayed branch it is the local signature of wake-emission recoil. If a chart proves that this term is boundary-only, the scale term below gives the scale-only Master EOM; if not, the same variation points to a recoil-inclusive force law.

On charts where the constraint-variation residual is boundary-only, or is cancelled by an explicitly declared regularized action-level term, the scale-only result is the delayed force law

$$\mu_{\rm arch}\ddot{\mathbf{x}}_i(t) = \sum_j \kappa\,\sigma_{ij}|q_iq_j| \sum_{t_0\in\mathcal{C}_{ij}(t)} \frac{\hat{\mathbf{r}}_{ij}(t;t_0)}{r_{ij}(t;t_0)^2\ |1-\hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)/c_f|}$$

including self-hit branches $j=i$ when the trivial coincidence root is excluded.

The branch collapse used here is an $\eta\to 0^+$ simple-root statement, not an identity at arbitrary finite η. Since

$$\partial_{t_0} g_{ij}(t,t_0) = -\left(1-\frac{\hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)}{c_f}\right)$$

any branch-local smooth f satisfies

\lim_{\eta\to0^+}\int_{-\infty}^{t} f(t_0)\phi_\eta\!\big(g_{ij}(t,t_0)\big)\,dt_0 = \sum_{t_0\in\mathcal{C}_{ij}(t)} \frac{f(t_0)} \left|1-\hat{\mathbf{r}}_{ij}(t;t_0)\cdot\mathbf{v}_j(t_0)/c_f\right|}

provided the active roots are simple and separated from collision support.

Equivalently, in the finite-η branch-selector form one may write

$$\mu_{\rm arch}\ddot{\mathbf{x}}_i(t) = \sum_j \kappa\,\sigma_{ij}|q_iq_j| \int_{-\infty}^{t} dt_0\,\frac{\hat{\mathbf{r}}_{ij}(t;t_0)}{r_{ij}(t;t_0)^2}\,\phi_\eta\big(g_{ij}(t,t_0)\big)$$

with the understanding that the displayed finite-η integral is a branch-selector surrogate whose weak limit is the Jacobian-weighted branch law above. The derivative term in $\nabla_{\mathbf{x}_i}\mathcal{K}_{ij}$ is absorbed only after the full delayed variation is assembled and the branch reduction is performed. In a recoil-inclusive reading, this sentence is replaced by a stronger requirement: the derivative term is retained as $\mathbf{C}_{ij}^{(\eta)}$ and tested as part of the force and conservation ledger rather than being forced to zero.

A derivation, reduction, or simulation that claims action-derived dynamics must therefore report the variation residual

$$\mathbf{R}_i^{(\eta)}(t) = \mu_{\rm arch}\ddot{\mathbf{x}}_i(t) - \sum_j \kappa\,\sigma_{ij}|q_iq_j| \left(\mathbf{F}_{ij,\rm scale}^{(\eta)}(t) + \mathbf{C}_{ij}^{(\eta)}(t)\right)$$

using the scale term and constraint residual defined in Master Equation. The dimensionless window diagnostic is

$$\epsilon_{\rm var}^{(\eta)}(W) = \frac{\sum_i \int_W \|\mathbf{R}_i^{(\eta)}(t)\|\,dt}{\sum_i \int_W \left(\mu_{\rm arch}\|\ddot{\mathbf{x}}_i(t)\| + \|\mathbf{F}_{i,\rm act}^{(\eta)}(t)\|\right) dt + \varepsilon}$$

The scale-only branch law is theorem-grade on W only when this residual tends to zero with the declared branch floors and boundary convention. The broader action-derived dynamics may instead be theorem-grade with nonzero $\mathbf{C}_{ij}^{(\eta)}$ if that term is retained as mechanical recoil and the same action closes the energy, momentum, and angular-momentum ledgers. If neither condition is reported, the local effective Lagrangian remains a fitted chart.

The current status is therefore a conditional theorem schema, not a universal action theorem. The pure scalar $1/r$ action is not a universal exact action for the scale-only Master EOM; it is valid as that derivation only on residual-closed charts. On charts where the interior residual survives, $\mathbf{C}_{ij}^{(\eta)}$ is the strict mechanical recoil (wake-emission resistance) required by a purely delayed action. It is the same bookkeeping channel that balances the positive tangential drive and wake escapement described in Binary Dynamics and Energy.

The recoil-inclusive reading also supplies the native seed of effective gauge structure. The scale term is a spatial gradient of the causal scale kernel and coarse-grains into an effective scalar wake potential. The derivative-of-constraint term is different: it differentiates the causal phase function g_{ij} itself. On an effective product chart with coordinates $(t,\mathbf{x})$, write the recoil current schematically as

$$\mathcal{A}_\mu^{\rm wake}(\mathbf{x},t) \propto \left\langle \phi'_\eta(g_{ij})\,\partial_\mu g_{ij}\right\rangle_{\rm cg}$$

where μ indexes the absolute-time component and the three spatial components of the effective chart; this is not a substrate Lorentz four-vector. The point is structural: the scalar/vector split $(\Phi_{\rm wake},\mathbf{A}_{\rm wake})$ introduced in the continuum reduction is forced by the scale/recoil split of the first variation. The scale term is the scalar-potential channel, while the retained recoil current is the vector-transport channel.

Thus a chart that keeps $\mathbf{C}_{ij}^{(\eta)}$ should not treat it as noise to be hidden in a residual. It should compute the effective field-strength candidate

$$F_{\mu\nu}^{\text{wake}} = \partial_\mu \mathcal{A}_\nu^{\text{wake}} - \partial_\nu \mathcal{A}_\mu^{\text{wake}}$$

as the curl of the coarse recoil current and test whether its spatial and temporal components reproduce the effective electric-like and magnetic-like response. The no-go against same-support scalar cancellation then has a positive corollary: on a recoil-inclusive action branch, magnetic-like response is not an optional extra law. It is the effective expression of the non-cancellable derivative-of-causal-phase channel. If a scale-only repair cancels this channel by a characteristic-tail kernel or richer invariant counterterm, that repair must also explain where the corresponding vector-potential response has gone.

The same-support local scalar route and its finite delta-jet extension are ruled out under the restricted assumptions in master-equation: cancelling the derivative residual forces the counterterm to change the accepted inverse-square scale term. The remaining minimal scale-only repair is the delayed-interior characteristic-tail kernel stated there. With

$$u = g + \frac{r}{c_f}$$

the endpoint-clear candidate is

$$K_{\text{eff}}^{(\eta)}(r, g) = \int_{-\infty}^{g} \frac{\delta_\eta(s)}{c_f (u - s)^2} ds$$

or the finite-endpoint variant with lower limit $-h_+$ after the characteristic gauge has cancelled the endpoint-clearance term. It satisfies

$$\left(\partial_r - \frac{1}{c_f}\partial_g\right) K_{\text{eff}}^{(\eta)} = -\frac{\delta_\eta(g)}{r^2}$$

so it cancels the derivative-of-constraint residual without changing the accepted inverse-square scale term. Effective Lagrangian reductions should still inherit the Master EOM directly unless they explicitly choose the normalized characteristic-tail kernel and carry its boundary-increment convention on the retained chart.

The normalized characteristic-tail kernel now carries explicit energy, momentum, and angular-momentum wake-history increments in master-equation. An effective Lagrangian reduction may therefore choose that kernel only when it also carries the same boundary-increment convention and reports the corresponding variation and conservation residuals on its branch chart. Without those residuals, the reduced Lagrangian remains a scaffold for the Master EOM rather than an independent proof of the branch force.

Symmetries and History-Aware Conservation Laws

The regularized action S_η is invariant under the fundamental symmetry group of the substrate when the mollifier, history window, and self-branch cutoff preserve those symmetries: the Euclidean group $E(3)$ and absolute time translations $\mathbb{R}_{\text{time}}$; the exact statement is recovered in the $\eta \to 0^+$ limit. If the regularization is inserted only at the equation-of-motion level or uses a non-invariant window, the associated energy, momentum, and angular-momentum expressions become diagnostics rather than proved Noether charges.

Because the Lagrangian is nonlocal in time, the corresponding Noether charges are path-history functionals tracking interactions that are still carried by causal wakes between emission and reception.

Energy Functional:
Invariance under absolute time translation yields a conserved total energy only for the symmetry-preserving action-derived model:

$$E_{\text{tot}}(t) = K(t) + E_{\text{wake}}(t)$$

where the action-level nonlocal Noether charge can be written with the weighted causal kernel from master-equation. To avoid confusing the receiver-gradient kernel above with the Noether-energy kernel, write

$$\mathcal{K}_{ij}^{E}(t_1, t_0) = \frac{\kappa\, \sigma_{ij}\, |q_i q_j|}{c_f} \Theta(t_1 - t_0) \frac{\delta\big(g_{ij}(t_1, t_0)\big)}{r_{ij}(t_1, t_0)}$$

For the delayed-interior characteristic-tail candidate, the Noether-energy kernel must instead be built from the same normalized action kernel,

$$\mathcal{K}_{ij,\text{eff}}^{E}(t_1, t_0) = \frac{\kappa\, \sigma_{ij}\, |q_i q_j|}{c_f} \Theta(t_1 - t_0) K_{\text{eff}}^{(\eta)}(r_{ij}(t_1, t_0), g_{ij}(t_1, t_0))$$

The scalar $1/r$ expression remains the diagnostic scaffold only when this replacement has not been declared for the chart.
Then:

$$E_{\text{wake}}(t) = \frac{1}{2} \sum_{i,j} \int_{-\infty}^{t} dt_0 \int_{t}^{\infty} dt_1\, \partial_{t_1} \mathcal{K}_{ij}^{E}(t_1, t_0)$$

For compatible trajectory reconstruction one may use the work-integral form

$$U(t) = U_* - \int_{t_*}^{t} \sum_i \mu_{\mathrm{arch}}\, \mathbf{a}_i(t') \cdot \mathbf{v}_i(t')\, dt'$$

when it is derived from the same action-level force and boundary convention. Otherwise $U(t)$ is a diagnostic history functional, not an independently proved Noether charge.

The corresponding finite-window energy residual is

$$\epsilon_E^{(\eta)}(W) = \frac{\left|\Delta_W \left(K + E_{\mathrm{wake}}^{(\eta)}\right) - \int_W \sum_i \mathbf{v}_i \cdot \mathbf{R}_i^{(\eta)}\, dt - \int_W \mathcal{B}_E^{(\eta)}\, dt\right|}{|\Delta_W K| + \left|\Delta_W E_{\mathrm{wake}}^{(\eta)}\right| + \varepsilon}$$

Here $\mathcal{B}_E^{(\eta)}$ is the declared endpoint or period-cut leakage. For isolated period-matched tests, $\epsilon_{\mathrm{var}}^{(\eta)} \to 0$, $\mathcal{B}_E^{(\eta)} \to 0$, and $\epsilon_E^{(\eta)} \to 0$ are the minimal conservation checks before the effective Hamiltonian is promoted beyond a diagnostic fit.

For a branch chart that explicitly chooses the normalized delayed-interior characteristic-tail kernel, the conservation object is not the generic scalar $1/r$ scaffold above but the pullback

$$K_{\mu,\mathfrak{B}} + E_{\mathrm{wake,eff},\mathfrak{B}}^{(\eta)}, \qquad \mathbf{P}_{\mathrm{mech},\mathfrak{B}} + \mathbf{P}_{\mathrm{wake,eff},\mathfrak{B}}^{(\eta)}, \qquad \mathbf{J}_{\mathrm{mech},\mathfrak{B}} + \mathbf{J}_{\mathrm{wake,eff},\mathfrak{B}}^{(\eta)}$$

defined on the same retained branch rows that enter the force residual. The energy residual above is theorem-level only after this chart declares the action-level g, endpoint convention, branch floors, and endpoint or period-cut leakage terms. The work-integral reconstruction $U(t)$ remains a trajectory diagnostic unless it is derived from that same normalized kernel and boundary convention.

Generalized Momentum:
Spatial translation invariance guarantees the conservation of total momentum, $\mathbf{P}_{\mathrm{tot}} = \mathbf{P}_{\mathrm{mech}}(t) + \mathbf{P}_{\mathrm{wake}}(t)$, where the mechanical momentum of the architrinos is balanced by the momentum flux propagating within the causal wake surfaces. Boundedness of the history-aware energy is therefore the natural diagnostic against runaway behavior, not a separate postulate.

For an effective reduction to promote a retained chart rather than fit it, it must also report vector residuals for the same branch pullback:

$$\epsilon_P^{(\eta)}(W) = \frac{\left\|\Delta_W \left(\mathbf{P}_{\mathrm{mech}} + \mathbf{P}_{\mathrm{wake,eff}}^{(\eta)}\right) - \int_W \sum_i \mathbf{R}_i^{(\eta)}\, dt - \int_W \mathcal{B}_P^{(\eta)}\, dt\right\|}{\|\Delta_W \mathbf{P}_{\mathrm{mech}}\| + \left\|\Delta_W \mathbf{P}_{\mathrm{wake,eff}}^{(\eta)}\right\| + \varepsilon}$$

and

$$\epsilon_J^{(\eta)}(W) = \frac{\left\|\Delta_W \left(\mathbf{J}_{\mathrm{mech}} + \mathbf{J}_{\mathrm{wake,eff}}^{(\eta)}\right) - \int_W \sum_i \mathbf{x}_i(t) \times \mathbf{R}_i^{(\eta)}\, dt - \int_W \mathcal{B}_J^{(\eta)}\, dt\right\|}{\|\Delta_W \mathbf{J}_{\mathrm{mech}}\| + \left\|\Delta_W \mathbf{J}_{\mathrm{wake,eff}}^{(\eta)}\right\| + \varepsilon}$$

Small $\epsilon_E^{(\eta)}$, $\epsilon_P^{(\eta)}$, and $\epsilon_J^{(\eta)}$ are conservation diagnostics when the regularization is inserted at the equation-of-motion level. They become Noether-charge tests only when the action regularization itself preserves time translation, spatial translation, and rotation symmetry on the retained chart.

Coarse-Graining: The Effective Continuum Lagrangian

The continuum Lagrangian belongs to the effective level. To describe emergent behavior of the Noether sea and complex assemblies, the description passes from discrete trajectories to continuum densities. Define a coarse-grained architrino polarity density $\rho_q(\mathbf{x}, t)$ and current density $\mathbf{j}_q(\mathbf{x}, t)$, smoothed over a scale much larger than the nested shell braid scale but smaller than macroscopic gradients. This notation is deliberately distinct from Noether braid density variables such as ρ_{NS} and n.

At the level of a branch-collapsed delayed causal action, the exact multi-time interaction double sum suggests the continuum delayed functional

$$S_{\mathrm{int}}^{\mathrm{cg}} = -\frac{\kappa}{2c_f} \int dt \int d^3x \int d^3x'\, \frac{\rho_q(\mathbf{x}, t)\rho_q(\mathbf{x}', t - \|\mathbf{x} - \mathbf{x}'\|/c_f)}{\|\mathbf{x} - \mathbf{x}'\|\, J_{\mathrm{eff}}(\mathbf{x}, t; \mathbf{x}', t')}$$

with delayed source time

$$t' = t - \frac{\|\mathbf{x} - \mathbf{x}'\|}{c_f}$$

propagation direction

$$\hat{\mathbf{n}}(\mathbf{x}, \mathbf{x}') = \frac{\mathbf{x} - \mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|}$$

coarse transport velocity

$$\mathbf{u}(\mathbf{x}',t') = \frac{\mathbf{j}_q(\mathbf{x}',t')}{\rho_q(\mathbf{x}',t')} \qquad (\rho_q \neq 0)$$

and effective Jacobian

$$J_{\mathrm{eff}}(\mathbf{x},t;\mathbf{x}',t') = \left|1 - \frac{\mathbf{u}(\mathbf{x}',t')\cdot\hat{\mathbf{n}}(\mathbf{x},\mathbf{x}')}{c_f}\right|$$

This functional is the continuum inheritance of the discrete delayed causal $1/r$ action kernel together with the same Jacobian branch weight that appears in the Master EOM. Source emission remains isotropic at the microscopic level, but the received coarse flux is compressed or dilated by delayed transport geometry. Differentiating this delayed action with respect to receiver coordinates produces the corresponding Jacobian-weighted inverse-square force density plus velocity-dependent correction terms. In the quasi-static limit $\|\mathbf{u}\|/c_f \to 0$, one recovers $J_{\mathrm{eff}} \to 1$ and the leading force law reduces to the familiar inverse-square form.

The J_{eff} denominator is also the continuum location of the per-hit third-law defect. It is evaluated using the delayed source velocity $\mathbf{u}(\mathbf{x}',t')$, so the receiver/source exchange is not represented by a symmetric mechanical stress alone. Translation invariance still protects total momentum when the wake momentum is included, but the mechanical current must split as

$$\Pi_q^{ij} = \Pi_{q,\mathrm{sym}}^{ij} + \Pi_{q,J}^{ij}, \qquad \Pi_{q,J}^{[ij]} \equiv \frac{1}{2}\left(\Pi_{q,J}^{ij} - \Pi_{q,J}^{ji}\right)$$

where $\Pi_{q,J}^{[ij]}$ is generated by the source-only velocity dependence of the Jacobian factor. The antisymmetric part is not a new force; it is the continuum expression of wake momentum that the particle-only mechanical ledger has omitted. This identifies three equivalent diagnostics of the same effective channel: the discrete recoil term $\mathbf{C}_{ij}^{(\eta)}$, the vector transport potential $\mathbf{A}_{\mathrm{wake}}$, and the antisymmetric stress contribution $\Pi_{q,J}^{[ij]}$. A continuum simulation can therefore test magnetic-like emergence by measuring whether these three reconstructions agree on the same branch window.

The continuum variables are admitted only through balance laws inherited from resolved histories. A coarse polarity density and current must satisfy

$$\partial_t \rho_q + \nabla\cdot\mathbf{j}_q = R_\rho^{\mathrm{cg}}$$

and the first two kinetic moments must close through a declared momentum-current tensor and energy-flux vector,

$$\partial_t(\rho_q u^i) + \partial_j \Pi_q^{ij} = f_q^i + R_{P,q}^i$$

$$\partial_t e_q + \nabla\cdot\mathbf{J}_{e,q} = \mathbf{f}_q\cdot\mathbf{u} + R_{E,q}$$

Here Π_q^{ij} and $\mathbf{J}_{e,q}$ are coarse-history summaries of the retained causal-wake record, not new substrate fields. The effective action is a promoted continuum chart only when R_ρ^{cg}, $R_{P,q}^i$, and $R_{E,q}$ are small under history, spatial, and regulator refinement. Otherwise the chart has reproduced only low-order moments while leaving unresolved memory in the omitted kinetic hierarchy.

For near-equilibrium reductions, a constitutive response may be written schematically as

$$\Pi_q^{ij} = \Pi_{\mathrm{rev}}^{ij} - 2\eta_{\mathrm{cg}}\left(E^{ij} - \frac{1}{3}(\nabla\cdot\mathbf{u})h^{ij}\right) - \zeta_{\mathrm{cg}}(\nabla\cdot\mathbf{u})h^{ij} + \Pi_{\mathrm{mem}}^{ij}$$

where $E^{ij} = \frac{1}{2}(\partial^i u^j + \partial^j u^i)$. This is a comparison form borrowed from continuum mechanics and kinetic theory. In $\mathbb{AAA}$ it becomes native only after η_{cg}, ζ_{cg}, and Π_{mem}^{ij} are derived from the same delayed branch record that supplies the force law.

The constructive route is to read the transport coefficients as low-frequency moments of the delayed response kernel, not as independent material constants. If $\widetilde{K}_{\mathrm{shear}}(\omega)$ and $\widetilde{K}_{\mathrm{bulk}}(\omega)$ are the shear and bulk projections of the same branch-derived causal kernel, then the leading near-equilibrium coefficients have the schematic form

$$\eta_{\mathrm{cg}} \sim \lim_{\omega\to 0}\frac{\mathrm{Im}\,\widetilde{K}_{\mathrm{shear}}(\omega)}{\omega}, \qquad \zeta_{\mathrm{cg}} \sim \lim_{\omega\to 0}\frac{\mathrm{Im}\,\widetilde{K}_{\mathrm{bulk}}(\omega)}{\omega}$$

and Π_{mem}^{ij} carries the finite-frequency remainder. This makes the viscosity-like channel an odd-frequency readout of the delayed force kernel. The ratio between η_{cg}, ζ_{cg}, and the force-law coupling κ is therefore a kernel-shape consequence on a certified branch, not an additional parameter family.

The corresponding dissipation residual is

$$\mathcal{R}_{\mathrm{diss}}(W) = \frac{\left|\Delta_W K_{\mathrm{cg}} + \int_W 2\eta_{\mathrm{cg}} E_{ij}E^{ij} + \zeta_{\mathrm{cg}}(\nabla\cdot\mathbf{u})^2\,dt\,dV + \Delta_W E_{\mathrm{wake}}\right|}{|\Delta_W K_{\mathrm{cg}}| + \int_W \left(2\eta_{\mathrm{cg}} E_{ij}E^{ij} + \zeta_{\mathrm{cg}}(\nabla\cdot\mathbf{u})^2\right)dt\,dV + |\Delta_W E_{\mathrm{wake}}| + \varepsilon}$$

This residual prevents ordinary viscous loss language from replacing the exact wake-history energy ledger. A nonzero positive quadratic term is allowed as a coarse channel for coherent-to-incoherent transfer, but the transferred content must appear in the retained wake, heat, or medium-response record.

By defining an effective scalar potential $\Phi_{\text{wake}}(\mathbf{x}, t)$ and a vector transport potential $\mathbf{A}_{\text{wake}}(\mathbf{x}, t)$ that track the integrated causal wakes of the continuous medium, the system maps locally onto an effective field theory. These potentials are bookkeeping variables for delayed transport, not additional ontological primitives. The resulting local Lagrangian density $\mathcal{L}_{\text{eff}}$ therefore belongs to a further closure step beyond the exact delayed causal action.

Effective Hamiltonian Domain Gate

A local Hamiltonian or local Lagrangian description is admissible only after the path-history law has been reduced to a finite set of coarse variables that preserve the relevant state-counting measure over the comparison window. This is an inference condition: it tests whether exact histories can be represented by local canonical coordinates without losing the invariants under comparison. Let $\mathcal{Q}$ be the coarse-graining from exact histories $\Gamma(t)$ to effective coordinates $z = (\rho_q, \mathbf{j}_q, \ldots)$, and let $\mathcal{P}^{\text{eff}}_{\Delta t}$ be the induced effective flow. The local canonical approximation must supply a measure $\mu_{\mathcal{Q}}$ such that

$$(\mathcal{P}^{\text{eff}}_{\Delta t})_* \mu_{\mathcal{Q}} = \mu_{\mathcal{Q}} + O(\epsilon_{\mathcal{Q}})$$

on the retained regime. This measure condition is necessary but not sufficient for canonical mechanics. The same handoff must also control a bracket or symplectic residual, for example

$$\left\|(\mathcal{P}^{\text{eff}}_{\Delta t})^* \omega_{\mathcal{Q}} - \omega_{\mathcal{Q}}\right\| \le \epsilon_\omega$$

for the retained two-form $\omega_{\mathcal{Q}}$, or an equivalent Poisson-bracket residual on the admitted observables. If $\epsilon_{\mathcal{Q}}$ or ϵ_ω is not controlled, the local Hamiltonian is only a fitting chart, not a derived mechanics.

For a replayable branch chart, this measure condition is the $\mathbb{AAA}$ analogue of Liouville's theorem. In ordinary finite-dimensional Hamiltonian mechanics the phase-space flow is divergence-free, $\nabla_z \cdot \dot{z} = 0$, so a phase-space volume element may stretch and fold but is not compressed by the exact flow. In the delayed setting, the analogous statement is valid only after $\mathcal{Q}$ retains the phase variables, causal-root ledger, wake-history record, and surrounding context that actually control the return map. Dropping an active sub-assembly phase can make a closed chart look dissipative or probabilistic merely because the chart has thrown away one of the variables that carries the recurrence.

The preserved two-form is likewise not the naive instantaneous form alone. On a delayed branch the candidate symplectic structure has a memory correction,

$$\omega_{\mathcal{Q}} = \omega_0 + \omega_{\text{mem}}, \qquad \omega_0 = \sum_A dQ^A \wedge d\Pi_A + \sum_\alpha d\theta^\alpha \wedge dI_\alpha$$

with

$$\omega_{\text{mem}} = \int_{-h}^{0} \mathcal{K}_{\text{symp}}(\vartheta)\, \delta\mathbf{x}(\vartheta) \wedge \delta\dot{\mathbf{x}}(\vartheta)\, d\vartheta$$

where h is the retained memory depth and $\mathcal{K}_{\text{symp}}$ is built from the same branch causal kernel that supplies the force. The residual ϵ_ω is small only when this memory term is replayable: after one return, the retained history segment $[-h, 0]$ must map to a congruent segment with the same branch rows and boundary convention. This is why phase-locked branches are the natural Hamiltonian domain. They replay the history window that carries ω_{mem}, while off-lock branches leak symplectic content through the memory boundary and can look dissipative after projection.

This gate keeps the exact and effective levels separate. The Master Equation owns the delayed causal dynamics; the effective Hamiltonian owns only those regimes where internal wake memory, branch changes, and unresolved Noether sea exchange have been compressed without losing the observer-level invariants being compared.

The same domain restriction applies before translating an effective Hamiltonian chart into quantum operators. The admissible observable set in Quantum Operator Mapping must be derived from this retained coarse-graining and record window, not chosen afterward as a free quantization convention.

The positive selection rule is that the quantizable algebra is generated by the globally defined branch variables,

$$\{Q^A, \Pi_A\} \cup \{e^{i\theta^\alpha}, I_\alpha\}$$

not by arbitrary functions on a projected chart. The phases enter through the single-valued observables $e^{i\theta^\alpha}$, while I_α records the corresponding action. A Bohr-Sommerfeld-like integer is admissible only when it is forced by single-valuedness around the resonance-locked phase bundle,

$$\oint_{\gamma_\alpha} \Pi \, dQ \in 2\pi\hbar_{\text{eff}}\mathbb{Z}$$

with the integer tied to the phase-bundle winding above. Thus quantization in this reduction is a topological single-valuedness condition on a retained phase-locked bundle, not a global quantization convention imposed on every smooth effective observable.

Topological Constraints and Assembly Stability

The delayed action, after branch reduction to causal-locus and root-ledger data, constrains the allowed topological configurations of architrino assemblies in the Noether sea. Stable assemblies, such as nested maximal-curvature candidates inside nested shell braids, should therefore be treated as theorem targets for localized, phase-locked causal-locus classes rather than as already-proved vortices or continuum topological defects.

The stability of these assemblies must be checked by the nonlinear self-hit feedback embedded in the interaction functional. When internal circulation velocities exceed c_f, the non-Markovian repulsion supplies a candidate branch-trapping mechanism; it becomes a robust geometric attractor only after a branch chart, Lyapunov or Floquet diagnostic, and history-aware energy bound are supplied. Likewise, mass-gap language is a closure target tied to discrete admissible branch classes, not an automatic consequence of writing the effective action.

The native topological sector is the stabilized causal-root ledger, not a borrowed field-theory vortex number. The canonical definition is given in Assembly Topological Charge; in the effective-action chart, the same assembly topological charge is the retained sector

$$[\mathfrak{B}] = \left(N_s,\, M_p,\, c_1[\theta^O, \theta^M, \theta^I]\right) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0} \times \mathbb{Z}^2$$

where N_s counts active self-hit roots, M_p counts active partner-hit roots, and $c_1[\theta^O, \theta^M, \theta^I]$ is the phase-bundle winding of the resonance-locked tri-binary chart. This class is deformation-stable only inside the nondegenerate branch domain: a causal-root fold, reconnection, or loss of phase-bundle closure changes the sector.

The corresponding mass-gap target is therefore native and computable. The gap is the minimum action cost to change $[\mathfrak{B}]$ by an admissible branch transition, such as a $\Delta N_s = \pm 2$ root birth or death under the same Jacobian floor and boundary convention. In a caustic-grazing transition this cost should be estimated from the finite impulse and wake-history increment across the fold. If that minimum vanishes under refinement, the action chart has no protected assembly gap; if it remains positive, the gap is a property of the branch ledger and delayed action, not an imported continuum-defect assumption.

Closure Interface: Action-to-Envelope Reduction

This chapter supplies the variational bridge used by the quantum closure chain. The bridge remains effective and comparative: it tests when a signed polarity/current history can be compressed into a nonnegative envelope without erasing memory terms.

From the regularized nonlocal action, the first step is to derive a continuum effective action in terms of coarse variables $(\rho_q, \mathbf{j}_q)$. The second step tests a phase-amplitude closure ansatz for the retained nonnegative envelope channel:

$$\rho_{\text{env}} = |\psi|^2, \qquad \mathbf{j}_{\text{env}} = \frac{\hbar_{\text{eff}}}{m_{\text{eff}}} \Im(\psi^* \nabla \psi)$$

Here m_{eff} is the retained envelope mass parameter of the benchmark chart, not a primitive architrino mass. The projection from the signed polarity/current data $(\rho_q, \mathbf{j}_q)$ to the nonnegative envelope channel must be declared before ρ_{env} is interpreted as $|\psi|^2$.

That projection has a topological cost. The signed polarity density carries a polarity-sign sheet

$$\sigma(\mathbf{x}, t) = \operatorname{sign} \rho_q(\mathbf{x}, t) \qquad (\rho_q \neq 0)$$

and the interfaces $\rho_q = 0$ are polarity domain walls. The envelope map is faithful only on a region where σ is constant. When a loop γ encloses domain-wall crossings, the phase chart must carry the lost sign sheet as a $\mathbb{Z}_2$ bundle datum:

$$\oint_\gamma \nabla S_{\text{env}} \cdot d\boldsymbol{\ell} = \pi \, N_{\text{wall}}(\gamma) \pmod{2\pi}$$

where $N_{\text{wall}}(\gamma)$ is the parity count of enclosed polarity-domain-wall intersections in the retained projection. The memory current is therefore not generic residue in this regime. Its circulation classifies the polarity-domain-wall topology that the nonnegative envelope has forgotten. A spin-$\frac{1}{2}$-like double-valued envelope can be promoted only if this $\mathbb{Z}_2$ sign-sheet circulation is recovered from the same $(\rho_q, \mathbf{j}_q)$ history and persists under branch-preserving deformation.

The handoff must report the continuity residual

$$R_{\mathrm{cg}} = \partial_t \rho_{\mathrm{env}} + \nabla \cdot \mathbf{j}_{\mathrm{env}}, \qquad \epsilon_{\mathrm{cg}} = \frac{\|R_{\mathrm{cg}}\|}{\|\partial_t \rho_{\mathrm{env}}\| + \|\nabla \cdot \mathbf{j}_{\mathrm{env}}\| + \varepsilon}$$

and keep the memory current

$$\mathbf{j}_{\mathrm{mem}} = \mathbf{j}_q - \mathbf{j}_{\mathrm{env}}$$

as an explicit residual rather than absorbing it into fitted constants. Equivalently, with $\Delta\rho = \rho_q - \rho_{\mathrm{env}}$,

$$\partial_t \rho_q + \nabla \cdot \mathbf{j}_q = R_{\mathrm{cg}} + \partial_t \Delta\rho + \nabla \cdot \mathbf{j}_{\mathrm{mem}}$$

Thus a small R_{cg} by itself does not prove envelope closure; the projection mismatch and memory-current divergence must be controlled as well.

For the non-relativistic, fixed-particle-number benchmark, the same envelope must also admit a phase chart

$$\psi = \sqrt{\rho_{\mathrm{env}}}\, e^{iS_{\mathrm{env}}/\hbar_{\mathrm{eff}}}, \qquad \mathbf{j}_{\mathrm{env}} = \frac{\rho_{\mathrm{env}}}{m_{\mathrm{eff}}} \nabla S_{\mathrm{env}}$$

Define

$$K_{\mathrm{env}} = \frac{\|\nabla S_{\mathrm{env}}\|^2}{2m_{\mathrm{eff}}}, \qquad Q_{\mathrm{env}} = -\frac{\hbar_{\mathrm{eff}}^2}{2m_{\mathrm{eff}}} \frac{\nabla^2 \sqrt{\rho_{\mathrm{env}}}}{\sqrt{\rho_{\mathrm{env}}}}$$

and test the corresponding Hamilton-Jacobi residual

$$R_{\mathrm{HJ}} = \partial_t S_{\mathrm{env}} + K_{\mathrm{env}} + V_{\mathrm{eff}} + Q_{\mathrm{env}}$$

The effective Schrödinger/Madelung chart is licensed on a retained window only when

$$\mathcal{R}_{\mathrm{env}} = \max\left(\epsilon_{\mathrm{cg}}, \frac{\|R_{\mathrm{HJ}}\|}{\|\partial_t S_{\mathrm{env}}\| + \|K_{\mathrm{env}}\| + \|V_{\mathrm{eff}}\| + \|Q_{\mathrm{env}}\| + \varepsilon}, \frac{\|\mathbf{j}_{\mathrm{mem}}\|}{\|\mathbf{j}_q\| + \varepsilon}\right) \le \epsilon_{\mathrm{env}}$$

This is a comparison residual, not a new ontology. If it fails, the wave function and Hamiltonian remain useful fitting charts for that window rather than promoted quantum closure.

The interface is closed only when:

- the Euler-Lagrange equations of the coarse action reproduce the effective envelope equation used in pilot-wave-character;
- the phase-amplitude chart reports $\mathcal{R}_{\mathrm{env}}$ rather than assuming the Schrödinger limit;
- memory contributions $\mathbf{j}_{\mathrm{mem}}$ remain explicit as controlled correction terms rather than hidden parameter absorbs.

Proof Programs

Master Equation Breather

This chapter sits between the canonical delayed law in master-equation.md and the one-dimensional reference scaffold in collinear-breather.md. It is a theorem-program atlas, not the proof itself. Its purpose is to extract the transportable theorem architecture and to state clearly which replacement lemmas would be required before a genuine breather theorem can be pursued at the level of the master equation.

The strategic point is simple. The proof should first close in the collinear dual-mollified model by producing a candidate cycle, a finite branch chart, a closed convex certificate, a return self-map, and the Schauder fixed point. Only after that closure is certified should the higher-dimensional sections below be reused as dependency maps. The next task in this chapter is therefore abstraction: identify what part of the collinear scaffold belongs to the general delayed dynamics, what part uses the ordered geometry of the line, and what new geometry must replace those 1D-only moves in higher dimension.

Purpose

The breather problem for the full delayed master equation is not a single obstruction. It is the conjunction of five distinct analytic burdens:

- choosing a codimension-one return section in history space,
- controlling active delayed-root topology along one cycle,
- proving that delayed self-drive does not defeat global recapture,
- packaging the resulting trajectories into a single closed convex tame self-map domain,
- and closing the fixed-point step on that domain.

The collinear scaffold shows that these burdens can be separated cleanly. In particular, it shows that the final existence theorem should be organized around a history-space return map rather than around a scalar speed closure alone. This chapter records that abstraction in a form suitable for later use in the master-equation stack.

Position in the Dynamics Stack

The intended division of labor is:

- master-equation.md gives the exact delayed law, the branch-sum form, the path-history integral form, and the causal Jacobians.
- collinear-breather.md supplies the live reduced proof target in a geometry where ordering on the line eliminates tangential drift.
- this chapter translates the 1D scaffold into a master-equation theorem atlas by separating portable structure from collinear-specific arguments.

Accordingly, this document should be read neither as a replacement for the master equation nor as a second reduced-model note. It is a bridge chapter: a theorem blueprint for transporting the 1D existence architecture into higher-dimensional delayed dynamics after the collinear certificate has closed.

Portable Return-Map Architecture

Let

$$\mathbf{X}(t) \in \mathcal{Q}$$

denote the full configuration variable of the master equation on its configuration manifold

$$\mathcal{Q}$$

For a fixed memory horizon

$$h > 0$$

write the history space as

$$\mathcal{H}_h \equiv C^1([-h, 0]; \mathcal{Q})$$

Choose a codimension-one section function

$$G : \mathcal{Q} \to \mathbb{R}$$

and let

$$\mathscr{S} \equiv \{\mathbf{Y} \in \mathcal{Q} \mid G(\mathbf{Y}) = 0\}$$

The raw outbound and inbound section classes are then

$$\Sigma_{\mathscr{S}}^{+} \equiv \left\{\Phi \in \mathcal{H}_h \,\middle|\, G(\Phi(0)) = 0, \qquad \nabla G(\Phi(0)) \cdot \dot{\Phi}(0) > 0\right\}$$

$$\Sigma_{\mathscr{S}}^{-} \equiv \left\{\Phi \in \mathcal{H}_h \,\middle|\, G(\Phi(0)) = 0, \qquad \nabla G(\Phi(0)) \cdot \dot{\Phi}(0) < 0\right\}$$

This section anchoring removes the absolute time-translation symmetry of the continuous delayed flow. As in the collinear reference model, the periodic-orbit question is therefore recast as a fixed-point question for a returned history.

For the dual-mollified dynamics with shell width

$$\eta > 0$$

and short-distance core scale

$$\epsilon_c > 0$$

the exact history-space return map should be written on its natural domain:

$$P_\eta : \mathrm{Dom}(P_\eta) \subseteq \Sigma_{\mathscr{S}}^{-} \to \Sigma_{\mathscr{S}}^{-}, \qquad P_\eta(\Phi) = \mathbf{X}_{T(\Phi)}$$

where

$$T(\Phi) > 0$$

is the first later return time to the same inbound section after one full excursion.

This is the first master-equation lesson from the 1D scaffold: the correct object is the full history-space return map. Scalar diagnostics can still be useful, but they are only projections of

$$P_\eta$$

not substitutes for it.

Absolute-time evolution law for certification

The return-map program does not require an elementary closed-form orbit. It requires a precise evolution law, one candidate cycle, and a finite certificate showing that the same closed convex tame domain is preserved by the return map.

For the dual-mollified master equation, the certification-level evolution law may be written directly in absolute time as

$$\ddot{\mathbf{x}}_i(t) = \kappa\epsilon^2 \sum_j \sigma_{ij} \int_{t-h}^{t} \frac{\hat{\mathbf{r}}_{ij}(t,s)}{\|\mathbf{r}_{ij}(t,s)\|^2 + \epsilon_c^2}\, \delta_\eta\big(\|\mathbf{r}_{ij}(t,s)\| - c_f(t-s)\big)\, ds$$

where

$$\mathbf{r}_{ij}(t,s) \equiv \mathbf{x}_i(t) - \mathbf{x}_j(s), \qquad \hat{\mathbf{r}}_{ij}(t,s) \equiv \frac{\mathbf{r}_{ij}(t,s)}{\|\mathbf{r}_{ij}(t,s)\|}$$

Here

$$h$$

is the certified memory horizon,

$$\eta$$

is the causal-isochron width, and

$$\epsilon_c$$

is the short-distance core scale.

Branch-sum formulas are derived from this absolute-time integral only on simple-root charts, where the causal isochron has isolated transversal roots. They are therefore local analytic reductions, not the global definition of the dynamics through fold transit or certified topology.

The proof burden is consequently finite-certificate closure:

$$\text{candidate cycle} \to \text{null-coordinate pre-ledger closure} \to \text{finite branch chart} \to \text{closed convex certificate} \to \text{return self-map} \to \text{Schauder}$$

The collinear pre-ledger is a falsification gate, not bookkeeping: finite parent-complement coverage must close before any branch chart is authorized.

In concrete terms, instantiate a candidate history

$$\Phi_{\text{cyc}}$$

choose the certified domain around it, and prove that

$$P_\eta$$

is continuous, precompact, and self-mapping on that one domain.

The planar and many-body sections below should therefore be read as roadmap layers. They record dependencies that must eventually be discharged, but they are not themselves completed proofs.

What the 1D Reference Model Already Settled

The frozen collinear chapter contributes five structural lessons that should now be treated as stable.

1. Dual mollification is structural, not cosmetic

The shell width

$$\eta$$

and the core cutoff

$$\epsilon_c$$

play different roles and should not be conflated. The first regularizes caustic transit across the causal isochron. The second regularizes the short-distance amplitude divergence. The 1D scaffold works precisely because those two pathologies are separated rather than blurred into a single smoothing parameter.

2. Convex Banach bounds and tame delayed geometry must be split

The 1D chapter now makes a clean distinction between:

- a convex section envelope

$$\mathcal{C}_{x_*,\eta}$$

- and a later closed convex tame envelope

$$\mathcal{K}_{x_*,\eta} \subseteq \mathcal{C}_{x_*,\eta}$$

That split is portable. In the master-equation setting, one should not expect Jacobian lower bounds, branch-count bounds, or root-persistence conditions to be convex by inspection. The convex Banach-space box and the tame delayed-root package are different theorem objects.

3. Seed nonvacuity must be explicit

The 1D scaffold no longer leaves the tame class abstractly nonempty. It builds an explicit affine seed history, then thickens it to a section-side tame neighborhood, and only afterward seeks full-cycle propagation. That logic is also portable. One should not expect higher-dimensional tame classes to be nonempty merely because their defining inequalities look plausible.

4. The fixed-point capstone needs one matching domain

The final Schauder route only becomes legitimate after continuity, precompactness, and the self-map property all live on one and the same closed convex tame domain. This is now explicit in the 1D manuscript and should remain explicit in every higher-dimensional formulation.

5. Parameter solvability is coupled

The collinear audit showed that local recapture margins and global envelope constants cannot be treated as algebraically independent when the crossing-speed bounds depend on envelope-scale collapse estimates. The master-equation analogue should therefore be phrased from the beginning as a coupled admissible-regime problem rather than as a sequential parameter-picking exercise.

Abstract Envelope Hierarchy

The general master-equation breather program should inherit the same three-layer hierarchy:

1. the raw inbound section

$$\Sigma^-_{\mathscr{S}}$$

2. a convex Banach envelope

$$\mathcal{C}_{\mathscr{S},\eta} \subseteq \Sigma^-_{\mathscr{S}}$$

3. and a closed convex tame envelope

$$\mathcal{K}_{\mathscr{S},\eta} \subseteq \mathcal{C}_{\mathscr{S},\eta}$$

The master-equation analogue of

$$\mathcal{C}_{\mathscr{S},\eta}$$

should carry only visibly convex constraints such as:

- section anchoring,
- uniform position bounds,
- uniform speed bounds,
- uniform acceleration bounds,
- and a causal-memory-depth bound.

The analogue of

$$\mathcal{K}_{\mathscr{S},\eta}$$

must not be defined by a raw intersection of delayed-root predicates. Persistence of active branches, Jacobian floors, memory-depth bounds, and caustic exclusions are generally not convex conditions when imposed pointwise on histories. Instead, the standard method throughout this chapter is:

1. choose a finite affine or sampled certificate around a candidate cycle;
2. make the defining inequalities of the certificate visibly convex in the stored history data;
3. prove that those convex certificate inequalities imply the delayed-geometry package:

- persistence of active partner and self roots,
- lower bounds on the causal Jacobians,
- branch-count control,
- and exclusion of root birth, root collision, and other topological degeneracies along the controlled cycle.

Convexity lives in the certificate. The delayed-root topology is a theorem-level consequence of the certificate, not the definition of the convex set itself. This is the natural packaging in which Arzela-Ascoli and Schauder can later be used. Anything looser risks repeating the old domain/codomain mismatch.

1D-Only Mechanisms and Their Replacement Obligations

The transport from the collinear reference model to the master equation is not literal. Several key moves in the 1D note use the ordered geometry of the line and therefore cannot simply be quoted in higher dimension.

Ordered sorting maps

The 1D scaffold organizes delayed topology around the scalar sorting maps

$$w(t) = x(t) + c_f t \qquad \text{and} \qquad z(t) = x(t) - c_f t$$

Those maps force deep-past roots into rigid order intervals and make descent arguments explicit.

No direct higher-dimensional analogue exists merely by replacing

$$x$$

with

$$\mathbf{x}$$

What is needed instead is a replacement coercive functional or ordered comparison geometry that can play the same role:

- it must isolate which delayed branches are geometrically admissible;
- it must prevent uncontrolled root proliferation;
- and it must turn far-past delayed roots into a controlled subset that can be either excluded or transported into a regime of uniform transversality.

Exact scalar Jacobians

In 1D the causal Jacobians reduce to explicit signed scalars. In the master equation they retain the exact form

$$J_{ij}(t;t_0) = 1 - \frac{\mathbf{v}_j(t_0) \cdot \hat{\mathbf{r}}_{ij}(t;t_0)}{c_f}$$

but the sign bookkeeping is no longer exhausted by line ordering.

The replacement burden is therefore a vector transversality theorem: one must find geometric hypotheses that keep

$$|J_{ij}|$$

uniformly away from zero along the relevant branch family, even in the presence of tangential motion and changing line-of-sight direction.

Deep-past self-root relocation

In the 1D reference note, deep-past outward self-roots on the apocenter window are forced back onto the pre-crossing inbound leg, where they become unique and automatically transversal. This is a genuinely strong collinear mechanism, but it is also genuinely one-dimensional.

The higher-dimensional replacement cannot rely on line order. It must instead produce one of two outcomes:

- either a geometric exclusion theorem showing that such remote self-roots cannot occur in the chosen regime,
- or a transport theorem showing that any such roots must lie on a controlled branch family with uniform Jacobian bounds and finite multiplicity.

Affine seed simplicity

The affine seed in 1D works because a linearly inbound history on the line automatically suppresses same-side exact self roots when its speed is strictly sub-field-speed. In higher dimension an affine seed is no longer automatically tame, because direction and curvature matter.

The replacement burden is therefore an explicit section-anchored seed packet in the full configuration geometry. It must come with:

- a strict sub-field-speed margin where needed,
- a finite controlled set of delayed partner roots,
- exclusion or control of same-source self intersections on the stored interval,
- and a small section neighborhood that preserves those properties in the

$$C^1$$

topology.

Outer-turn closure without tangential loss

The 1D problem has no tangential channel. In the master equation, any breather theorem must account for tangential drift, angular deflection, or other transverse escape directions that the line simply does not possess.

The replacement burden is therefore a recapture theorem with a genuinely vector coercive quantity. One needs a higher-dimensional analogue of the 1D inner and outer force margins, but measured against all escape channels rather than against a single signed radial variable.

Master-Equation Theorem Spine

The portable theorem ladder should therefore look as follows.

> **Target Proposition (Sectioned tame well-posedness).**
> For a chosen return section

$$> \mathscr{S} >$$

> and dual-mollified parameters

$$> (\eta, \epsilon_c), >$$

> there exists a nonempty section-side tame class

$$> \mathcal{H}^{\mathrm{adm}}_{\mathscr{S},\eta} \subseteq \Sigma^-_{\mathscr{S}} >$$

> on which the active delayed branches are well defined, simple, and stable under small

$$> C^1 >$$

> perturbations.

> **Target Proposition (Seeded nonvacuity in the full geometry).**
> There exists an explicit history

$$> \Phi_{\mathrm{seed}} \in \Sigma^-_{\mathscr{S}} >$$

> and a nonempty section neighborhood

$$> \mathcal{C}^{\mathrm{seed}}_{\mathscr{S},\eta} >$$

> on which the delayed-root topology and the chosen section transversality remain controlled.

> **Target Theorem (One-cycle tame propagation).**
> A sufficiently small nonempty subclass of

$$> \mathcal{C}^{\mathrm{seed}}_{\mathscr{S},\eta} >$$

> propagates through one full delayed cycle while preserving the branch-control, Jacobian, memory-depth, and section-return bounds needed for the return map.

> **Target Proposition (Closed convex tame envelope).**
> There exists a nonempty closed convex set

$$> \mathcal{K}_{\mathscr{S},\eta} \subseteq \mathcal{C}_{\mathscr{S},\eta} \subseteq \Sigma^-_{\mathscr{S}} >$$

> defined by finite affine or sampled certificate inequalities. It contains the propagated tame class, and its certificate implies the relevant branch labels, Jacobian floors, memory-depth bounds, and caustic exclusions needed to carry a well-defined return map

$$> P_\eta : \mathcal{K}_{\mathscr{S},\eta} \to \mathcal{K}_{\mathscr{S},\eta}. >$$

> **Target Proposition (Continuity and precompactness).**
> On

$$> \mathcal{K}_{\mathscr{S},\eta}, >$$

> the return map is continuous in the

$$> C^1 >$$

> topology and its image is precompact.

> **Target Theorem (Master-equation breather via Schauder).**
> Under the preceding inputs, the return map has a fixed point

$$> \Phi_\eta^* \in \mathcal{K}_{\mathscr{S},\eta}, > \qquad > P_\eta(\Phi_\eta^*) = \Phi_\eta^*. >$$

> If the section is posed in absolute configuration space with no quotient gauge reset, the associated delayed trajectory is a bounded periodic solution of the dual-mollified master equation.

This is the abstract endpoint once no quotient gauge reset is being used. In any reduced or gauge-fixed planar setting, the same fixed-point statement first gives a relative breather in the quotient variables. An absolute periodic trajectory in the fixed Euclidean void requires an additional zero-holonomy reconstruction condition. The real work is not the formal Schauder step itself, but the geometric production and certification of the tame self-map domain on which Schauder is allowed to act.

Collective-coordinate and zero-mode bookkeeping

Higher-dimensional transport must also separate genuine deformation directions from neutral collective coordinates. For a candidate cycle

$$\mathbf{X}_{\mathrm{cyc}}(t;\alpha)$$

with finite parameters

$$\alpha^a$$

define the tangent histories

$$Z_a(\theta) \equiv \partial_{\alpha^a} \mathbf{X}_{\mathrm{cyc}}(\theta;\alpha)$$

Each

$$Z_a$$

must be assigned to one of three roles before the monodromy or Floquet data are used:

- removed by section or gauge fixing, such as time translation or rigid rotation;
- retained as a physical collective coordinate whose return is tested by a zero-holonomy or phase-closure equation;
- transverse to the branch and therefore part of the stability or returned-sample certificate.

This is the master-equation analogue of moduli and zero-mode bookkeeping in soliton theory, but it remains an $\mathbb{AAA}$ certificate rule. It does not add supersymmetry or gauge-theory ontology. Its concrete use is to prevent neutral drift from contaminating the finite certificate: the return-map derivative should be interpreted on the quotient chart, while any retained collective coordinate must satisfy its own closure residual

$$\mathcal{H}_a(\Phi_{\mathrm{cyc}}) = 0$$

Immediate Geometric Research Burdens

The first master-equation work should now concentrate on four concrete questions.

1. Choice of section

One needs a return section that is both dynamically natural and analytically stable under perturbation. In the 1D chapter this role is played by

$$x = x_*$$

with fixed crossing sign. In higher dimension the corresponding section should separate one cycle cleanly and eliminate time-shift symmetry without introducing artificial coordinate singularities.

2. Branch-topology control

The master equation already gives the exact causal root equations and the delay-map Jacobians. What is missing is a theorem that packages them into a finite tame branch family on a full excursion, rather than only at isolated times. This is the higher-dimensional replacement for the collinear root-sorting technology.

3. Vector recapture margins

The 1D scaffold reduces the inner and outer turns to explicit inequalities

$$\mathfrak{M}_{\mathrm{in}} > 0 \qquad \text{and} \qquad \mathfrak{M}_{\mathrm{out}} > 0$$

The master-equation replacement must be a vector coercive margin that beats all escape channels, not merely a scalar outward radial speed.

4. Coupled regime closure

The full theorem program must close a coupled algebraic and geometric regime in which:

- the local delayed geometry is tame,
- the recapture inequalities are strict,
- the one-cycle bounds fit inside a convex envelope,
- and the resulting return image stays inside the same tame domain.

This coupled closure problem should be stated honestly from the outset. It is where the global theorem will either succeed or fail.

Recommended Next Regime

The most sensible continuation after the frozen collinear model is not the completely unconstrained many-body master equation. It is the first higher-dimensional regime in which line-order arguments fail but a strong symmetry reduction still survives.

The natural candidate is a reflection-symmetric planar binary with a codimension-one return section chosen to control both radial and tangential escape. That regime is still close enough to the collinear reference model to inherit much of the return-map architecture, but it is already far enough from the line to force genuinely new geometry.

The active working chapter for that bridge is now Planar Bridge Closure. The present document remains the larger roadmap and dependency spine.

If that planar bridge regime also resists tame-envelope closure, then the obstruction will be informative: it will show exactly where the collinear proof architecture ceases to transport.

First Planar Bridge Regime

Work on the reflection-symmetric planar two-body subclass

$$\mathbf{x}_1(t) = -\mathbf{r}(t), \qquad \mathbf{x}_2(t) = \mathbf{r}(t), \qquad \mathbf{r}(t) \in \Pi \cong \mathbb{R}^2, \qquad q_1 = -\epsilon, \qquad q_2 = +\epsilon$$

Write

$$\rho(t) \equiv \|\mathbf{r}(t)\|, \qquad \hat{\mathbf{e}}_r(t) \equiv \frac{\mathbf{r}(t)}{\rho(t)}, \qquad \hat{\mathbf{e}}_\theta(t) \equiv R_{\pi/2}\hat{\mathbf{e}}_r(t)$$

away from the collision set, and decompose the planar velocity as

$$\dot{\mathbf{r}}(t) = u_r(t)\hat{\mathbf{e}}_r(t) + u_\theta(t)\hat{\mathbf{e}}_\theta(t)$$

When a polar-angle coordinate is convenient, write

$$\mathbf{r}(t) = \rho(t)(\cos\vartheta(t), \sin\vartheta(t))$$

The first technical lesson is that the planar bridge should be written on a rotationally reduced chart. If one keeps the full planar rotation symmetry visible, then the most natural fixed-radius section is not affine in the ambient Banach space and the convex-envelope step is obscured from the start. The clean approach is therefore to quotient rigid planar rotations locally at the section by choosing the representative with

$$\mathbf{r}(0) = \rho_* \mathbf{e}_1, \qquad \mathbf{e}_2 \cdot \dot{\mathbf{r}}(0) > 0$$

After this gauge choice, the remaining return question is codimension one in the reduced history space: the only equality that defines the section is the fixed section radius

$$\rho(0) = \rho_*$$

This regime is the first honest transport problem beyond the line. It preserves reflection symmetry, center-of-mass reduction, and a single binary degree of freedom, but it no longer permits scalar ordering arguments to suppress tangential drift or delayed-root wrapping by inspection.

Raw Section and Envelope Hierarchy in the Planar Regime

Let

$$\mathcal{H}_h^{\Pi} \equiv C^1([-h,0];\Pi)$$

denote the reduced planar history space in the above gauge.

The raw outbound and inbound sections should be taken as

$$\Sigma^{+}_{\rho_*,\Pi} \equiv \left\{ \Phi \in \mathcal{H}_h^{\Pi} \,\middle|\, \Phi(0) = \rho_* \mathbf{e}_1, \quad \mathbf{e}_1 \cdot \dot{\Phi}(0) > 0, \quad \mathbf{e}_2 \cdot \dot{\Phi}(0) > 0 \right\}$$

$$\Sigma^{-}_{\rho_*,\Pi} \equiv \left\{ \Phi \in \mathcal{H}_h^{\Pi} \,\middle|\, \Phi(0) = \rho_* \mathbf{e}_1, \quad \mathbf{e}_1 \cdot \dot{\Phi}(0) < 0, \quad \mathbf{e}_2 \cdot \dot{\Phi}(0) > 0 \right\}$$

The fixed position removes the reduced rotational freedom, the sign of

$$\mathbf{e}_1 \cdot \dot{\Phi}(0)$$

selects outbound versus inbound passage, and the sign of

$$\mathbf{e}_2 \cdot \dot{\Phi}(0)$$

fixes the orientation branch so that the returned history does not flip across the reflection symmetry.

The planar analogue of the convex Banach envelope should then be a visibly convex subset

$$\mathcal{C}^{\Pi}_{\rho_*,\eta} \subseteq \Sigma^{-}_{\rho_*,\Pi}$$

defined only by affine or norm-convex bounds. A workable target is

$$\mathcal{C}^{\Pi}_{\rho_*,\eta} \equiv \left\{ \Phi \in \Sigma^{-}_{\rho_*,\Pi} \,\middle|\, \begin{array}{l} \|\Phi(\theta) - \rho_* \mathbf{e}_1\| \le R_{\max}, \\ \|\dot{\Phi}(\theta)\| \le U_{\max}, \\ \|\dot{\Phi}(\theta) - \dot{\Phi}(\theta')\| \le A_{\max}|\theta - \theta'|, \\ U_{\theta,\min} \le \mathbf{e}_2 \cdot \dot{\Phi}(0) \le U_{\theta,\max} \end{array} \quad \text{for all } \theta, \theta' \in [-h,0] \right\}$$

Its role is exactly the role played by the section envelope in the frozen collinear chapter: it carries only the convex bookkeeping needed for compactness and for section-side control. It should not yet contain any delayed-root topology.

The tame envelope must then be posed as a stricter target on the same section:

$$\mathcal{K}^{\Pi}_{\rho_*,\eta} \subseteq \mathcal{C}^{\Pi}_{\rho_*,\eta}$$

The point is not to redefine

$$\mathcal{K}^{\Pi}_{\rho_*,\eta}$$

as an arbitrary nonconvex tame subclass, but to seek a closed convex set on which the delayed geometry can be packaged by quantitative common bounds. Concretely, the intended theorem object is a closed convex set

$$\mathcal{K}^{\Pi}_{\rho_*,\eta}$$

and constants

$$N_{\mathrm{br}}, \quad \nu_J, \quad \delta_{\mathrm{sep}}, \quad I_{\mathrm{cau}}, \quad m_{\mathrm{in}}, \quad m_{\mathrm{out}}$$

such that every

$$\Phi \in \mathcal{K}^{\Pi}_{\rho_*,\eta}$$

admits one-cycle continuation with:

- at most

$$N_{\mathrm{br}}$$

active delayed partner and self branches on the controlled cycle;

- branch separation at least

$$\delta_{\mathrm{sep}}$$

- causal Jacobian bound

$$|J_{ij}(t;t_0)| \geq \nu_J$$

- dual-mollified caustic-transit impulse bounded by

$$I_{\mathrm{cau}}$$

- and vector recapture margins at least

$$m_{\mathrm{in}} \qquad \text{and} \qquad m_{\mathrm{out}}$$

on the inner and outer return windows.

This is the first concrete higher-dimensional target for a legitimate Schauder route.

Planar Replacement Obligations

The reduced planar bridge now breaks into six concrete packages:

- sectorized directional sorting, replacing the 1D

$$w/z$$

order by directional support control;

- deep-past sector relocation, replacing line-order transport of remote self roots;
- sectorwise cone transversality, replacing scalar Jacobian sign bookkeeping;
- an explicit planar seed packet, replacing the 1D affine-seed nonvacuity step;
- bounded planar caustic transit, integrating the compulsory inbound self-branch birth as a controlled fold impulse;
- and unified vector recapture criteria, replacing the scalar inner and outer turn inequalities.

The package details below carry the actual theorem statements.

First theorem package: sectorized directional sorting

The first concrete planar task should be to replace the scalar 1D order by a finite directional atlas.

Fix a sector half-width

$$0 < \alpha_{\mathrm{sort}} < \frac{\pi}{4}$$

and choose a finite family of unit directions

$$\mathcal{U} = \{\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_M\} \subset S^1$$

such that the closed sectors

$$\mathfrak{S}_k \equiv \left\{\hat{\mathbf{u}} \in S^1 \;\middle|\; \angle(\hat{\mathbf{u}}, \hat{\mathbf{u}}_k) \leq \alpha_{\mathrm{sort}}\right\}$$

cover

$$S^1$$

The sector family is fixed once and for all for the chosen tame class. What varies from trajectory to trajectory is only which subfamily is active on a given controlled window.

For each direction, distinguish the instantaneous directional phases

$$\zeta^-_{\hat{\mathbf{u}}}(t) \equiv \mathbf{r}(t) \cdot \hat{\mathbf{u}} - c_f t, \qquad \zeta^+_{\hat{\mathbf{u}}}(t) \equiv \mathbf{r}(t) \cdot \hat{\mathbf{u}} + c_f t$$

from the automatic running envelopes

$$\underline{\zeta}^-_{\hat{\mathbf{u}}}(t) \equiv \inf_{\theta \le t} \zeta^-_{\hat{\mathbf{u}}}(\theta), \qquad \overline{\zeta}^+_{\hat{\mathbf{u}}}(t) \equiv \sup_{\theta \le t} \zeta^+_{\hat{\mathbf{u}}}(\theta)$$

The running infimum

$$\underline{\zeta}^-_{\hat{\mathbf{u}}}$$

is non-increasing by definition, and the running supremum

$$\overline{\zeta}^+_{\hat{\mathbf{u}}}$$

is non-decreasing by definition. The nontrivial theorem burden is therefore not monotonicity of those running envelopes. It is the stronger active-sector assertion that, on the windows where a delayed chord is actually active, the running envelope is achieved by the current time and the instantaneous phase has a strict derivative margin.

Two cycle windows should then be named explicitly:

$$I_{\mathrm{in}} = [t^-_{\mathrm{in}}, t_{\mathrm{x}}], \qquad I_{\mathrm{ap}} = [t^-_{\mathrm{ap}}, t^+_{\mathrm{ap}}]$$

Here

$$I_{\mathrm{in}}$$

is the final inbound window ending at the first center crossing time

$$t_{\mathrm{x}}$$

and

$$I_{\mathrm{ap}}$$

is the late-apocenter window on the later outbound branch where the outer-turn problem is analyzed.

For any two times

$$s < t$$

define the exact self and partner chord directions

$$\hat{\mathbf{u}}^{\mathrm{self}}_{s,t} \equiv \frac{\mathbf{r}(t) - \mathbf{r}(s)}{\|\mathbf{r}(t) - \mathbf{r}(s)\|}, \qquad \hat{\mathbf{u}}^{\mathrm{part}}_{s,t} \equiv \frac{\mathbf{r}(t) + \mathbf{r}(s)}{\|\mathbf{r}(t) + \mathbf{r}(s)\|}$$

whenever the denominators are nonzero. The sector label of a root is the unique index

$$k$$

for which the corresponding chord direction lies in

$$\mathfrak{S}_k$$

after shrinking the tame class so that active directions stay a positive angular distance away from sector overlaps.

> **Target Proposition (Two-tier windowed directional monotonicity).**
> There exist positive constants

$$> \sigma_{\mathrm{in}}, > \qquad > \sigma_{\mathrm{ap}}, >$$

> and active sector subfamilies

$$> \mathcal{U}_{\mathrm{in}}, > \qquad > \mathcal{U}_{\mathrm{ap}} > \subseteq > \mathcal{U} >$$

> such that:
>
> 1. every active self or partner chord direction on

$$> I_{\mathrm{in}} >$$

> lies in some sector from

$$> \mathcal{U}_{\mathrm{in}}; >$$

> 2. every active self or partner chord direction on

$$> I_{\mathrm{ap}} >$$

> lies in some sector from

$$> \mathcal{U}_{\mathrm{ap}}; >$$

> 3. these active subfamilies are stable on their windows: active chord directions stay a positive angular distance away from sector boundaries, except inside the later certified sector-boundary or fold tubes;
> 4. for every

$$> \hat{\mathbf{u}}_k \in \mathcal{U}_{\mathrm{in}}, > \qquad > t \in I_{\mathrm{in}}, >$$

> the support function obeys

$$> \underline{\zeta}^-_{\hat{\mathbf{u}}_k}(t) >=> \zeta^-_{\hat{\mathbf{u}}_k}(t), > \qquad > \frac{d}{dt}\zeta^-_{\hat{\mathbf{u}}_k}(t) \le -\sigma_{\mathrm{in}}; >$$

> 5. for every

$$> \hat{\mathbf{u}}_k \in \mathcal{U}_{\mathrm{ap}}, > \qquad > t \in I_{\mathrm{ap}}, >$$

> one has

$$> \underline{\zeta}^-_{\hat{\mathbf{u}}_k}(t) >=> \zeta^-_{\hat{\mathbf{u}}_k}(t), > \qquad > \frac{d}{dt}\zeta^-_{\hat{\mathbf{u}}_k}(t) \le -\sigma_{\mathrm{ap}}, > \qquad > \overline{\zeta}^+_{\hat{\mathbf{u}}_k}(t) >=> \zeta^+_{\hat{\mathbf{u}}_k}(t), > \qquad > \frac{d}{dt}\zeta^+_{\hat{\mathbf{u}}_k}(t) \ge \sigma_{\mathrm{ap}}. >$$

> 6. for inactive sector centers

$$> \hat{\mathbf{u}}_k \notin \mathcal{U}_{\mathrm{in}} > \quad \text{or} \quad > \hat{\mathbf{u}}_k \notin \mathcal{U}_{\mathrm{ap}}, >$$

> only the automatic running-envelope monotonicity is required. No strict derivative margin is imposed on transverse directions with no active branch on the corresponding window.

The first monotonicity statement is the planar replacement for the inbound ordered fall that, in the collinear chapter, feeds collapse and root control. The second is the higher-dimensional descendant of the late

$$z$$

-descent package: it separates the outgoing and incoming directional support levels on the apocenter window instead of relying on a single global scalar order.

This two-tier form is intentionally weaker than universal sectorwise descent. Tangential motion can make transverse sector centers move with the wrong instantaneous sign even while the active chord geometry remains controlled. The theorem target only asks for strict descent in the finite active sector subfamily

$$\mathcal{U}_{\mathrm{in}} \quad \text{or} \quad \mathcal{U}_{\mathrm{ap}}$$

and asks the certificate to prove that this subfamily does not migrate continuously around

$$S^1$$

during the window.

The exact root equations now reveal why these support functions are the right replacement objects. If

$$s < t$$

is a self root, then with

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}^{\mathrm{self}}_{s,t}$$

one has

$$\mathbf{r}(t) - \mathbf{r}(s) = c_f(t-s)\hat{\mathbf{u}}$$

hence

$$\zeta^-_{\hat{\mathbf{u}}}(t) = \zeta^-_{\hat{\mathbf{u}}}(s)$$

If

$$s < t$$

is a partner root, then with

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}^{\text{part}}_{s,t}$$

one has

$$\mathbf{r}(t) + \mathbf{r}(s) = c_f(t - s)\hat{\mathbf{u}}$$

hence

$$\zeta^-_{\hat{\mathbf{u}}}(t) = -\zeta^+_{\hat{\mathbf{u}}}(s)$$

> **Target Corollary (Sector-labeled branch family).**
> Assume the windowed directional monotonicity proposition. Then on each of the windows

$$> I_{\text{in}} > \qquad > \text{and} > \qquad > I_{\text{ap}}, >$$

> every active delayed root belongs to a unique labeled family

$$> \beta_k^{\text{self}} > \qquad > \text{or} > \qquad > \beta_k^{\text{part}}, > \qquad > \hat{\mathbf{u}}_k \in \mathcal{U}, >$$

> with the following consequences:
>
> 1. for fixed

$$> t >$$

> and fixed sector

$$> \mathfrak{S}_k, >$$

> there is at most one earlier self-root time

$$> s < t >$$

> in that sector on the window under consideration;
> 2. for fixed

$$> t >$$

> and fixed sector

$$> \mathfrak{S}_k, >$$

> there is at most one earlier partner-root time

$$> s < t >$$

> in that sector on the window under consideration;
> 3. the total number of active self and partner roots on either window is therefore bounded by

$$> 2M_{\text{act},W} > \leq > 2M, >$$

> where

$$> M_{\text{act},W} >$$

> is the number of active sector centers on the window

$$> W \in \{I_{\text{in}}, I_{\text{ap}}\}; >$$

> 4. branch birth, branch death, or branch relabeling can occur only when an active chord direction meets a sector boundary or when the relevant monotonicity margin

$$> \sigma_{\text{in}} > \quad > \text{or} > \quad > \sigma_{\text{ap}} >$$

> degenerates, which defines the planar caustic tube that later propositions must control.

This corollary is the exact branch-labeling consequence needed for the rest of the bridge program. It converts the delayed-root picture from an a priori moving continuum of planar chord directions into a finite labeled branch family that can be propagated, bounded, and inserted into the tame-envelope construction.

The next burden is deep-past sector relocation: remote late-apocenter self roots must either be excluded or forced into a pre-crossing inbound cone where uniqueness and transversality return.

Second theorem package: deep-past sector relocation

The next concrete step is to convert the sector labels from the first package into a genuine exclusion-versus-relocation theorem on the late-apocenter window.

Let

$$I_{\text{out}} = [t_{\text{x}}, t_{\text{ap}}^{-}]$$

denote the earlier outbound interval between the first center crossing and the start of the late-apocenter window. For each active apocenter sector

$$\mathfrak{S}_k, \qquad \hat{\mathbf{u}}_k \in \mathcal{U}_{\text{ap}}$$

define the sector support envelopes

$$\zeta_{k,\max}^{-}(t) \equiv \sup_{\hat{\mathbf{u}} \in \mathfrak{S}_k} \zeta_{\hat{\mathbf{u}}}^{-}(t), \qquad \zeta_{k,\min}^{-}(t) \equiv \inf_{\hat{\mathbf{u}} \in \mathfrak{S}_k} \zeta_{\hat{\mathbf{u}}}^{-}(t)$$

The higher-dimensional replacement for the collinear outbound-level exclusion is the sectorwise gap

$$\Delta_k^{\text{out}} \equiv \inf_{s \in I_{\text{out}}} \zeta_{k,\min}^{-}(s) - \sup_{t \in I_{\text{ap}}} \zeta_{k,\max}^{-}(t)$$

If

$$\Delta_k^{\text{out}} > 0$$

then no self root on

$$I_{\text{ap}}$$

whose chord direction lies in

$$\mathfrak{S}_k$$

can have its source time on

$$I_{\text{out}}$$

Indeed, a self root with exact direction

$$\hat{\mathbf{u}} \in \mathfrak{S}_k$$

would satisfy

$$\zeta_{\hat{\mathbf{u}}}^{-}(t) = \zeta_{\hat{\mathbf{u}}}^{-}(s)$$

hence

$$\zeta_{k,\max}^{-}(t) \geq \zeta_{k,\min}^{-}(s)$$

contradicting the strict gap.

The remaining source candidate is therefore the pre-crossing inbound interval. To record that geometry, define the pre-crossing source cones

$$\mathfrak{C}_{\mathrm{in},k} \equiv \{\lambda \hat{\mathbf{u}} \mid \lambda \geq 0, \quad \hat{\mathbf{u}} \in \mathfrak{S}_k\}, \qquad \mathfrak{C}_{\mathrm{in}} \equiv \bigcup_{\hat{\mathbf{u}}_k \in \mathcal{U}_{\mathrm{ap}}} \mathfrak{C}_{\mathrm{in},k}$$

The intended meaning is not that the full inbound leg lies in one cone. The intended meaning is that once a deep-past root is assigned a sector label

$$k$$

its admissible pre-crossing source point should be confined to the matching inbound cone

$$\mathfrak{C}_{\mathrm{in},k}$$

The first package should also be strengthened sectorwise: after shrinking

$$\alpha_{\mathrm{sort}}$$

and the tame class if necessary, the directional monotonicity bounds should hold uniformly for every

$$\hat{\mathbf{u}} \in \mathfrak{S}_k, \qquad \hat{\mathbf{u}}_k \in \mathcal{U}_{\mathrm{in}} \cup \mathcal{U}_{\mathrm{ap}}$$

not only for the sector centers. This is the form actually needed for uniqueness and relocation.

> **Target Proposition (Sectorwise outbound exclusion).**
> Assume the windowed directional monotonicity package and suppose that for every active apocenter sector

$$> \hat{\mathbf{u}}_k \in \mathcal{U}_{\mathrm{ap}} >$$

> one has

$$> \Delta_k^{\mathrm{out}} > 0. >$$

> Let

$$> t \in I_{\mathrm{ap}} >$$

> and let

$$> s < t >$$

> be a self-root time whose chord direction belongs to

$$> \mathfrak{S}_k. >$$

> Then

$$> s \notin I_{\mathrm{out}}. >$$

This is the direct planar analogue of the 1D statement that late apocenter levels fall below the entire earlier outbound range. The point is the same as in the frozen scaffold, but the comparison is now sectorwise and uses support envelopes rather than a single scalar

$$z$$

> **Target Proposition (Pre-crossing sector relocation).**
> Assume the sectorwise outbound exclusion proposition, and assume in addition that:
>
> 1. every self root on

$$> I_{\mathrm{ap}} >$$

> with delay at least

$$> \tau_{\mathrm{dp}} >$$

> has source time

$$> s \leq t_{\mathrm{ap}}^{-}; >$$

> 2. for every active apocenter sector

$$> \mathfrak{S}_k, >$$

> the pre-crossing inbound history intersects the matching cone

$$> \mathfrak{C}_{\text{in},k} >$$

> in a connected interval on which

$$> \frac{d}{ds}\zeta_{\hat{\mathbf{u}}}^{-}(s) \le -\sigma_{\text{in}}^{\sharp} < 0 > \qquad > \text{for every } > \hat{\mathbf{u}} \in \mathfrak{S}_k; >$$

> 3. outside that connected inbound interval, the pre-crossing history has no source point whose self chord to the late-apocenter window lies in

$$> \mathfrak{S}_k. >$$

> Then every deep-past self root on

$$> I_{\text{ap}} >$$

> is either absent or has a unique source time

$$> s \in I_{\text{in}} >$$

> with

$$> \mathbf{r}(s) \in \mathfrak{C}_{\text{in},k} >$$

> for the matching sector label

$$> k. >$$

This is the correct replacement for the collinear relocation lemma. The source is not forced onto one signed interval because there is no global line order. It is forced into a sector-matched pre-crossing cone where directional monotonicity restores uniqueness.

> **Target Corollary (Deep-past sector suppression).**
> Assume the pre-crossing sector relocation proposition and, in addition, a uniform Jacobian lower bound

$$> |J_s(t;s)| \ge \nu_{J,\text{dp}} > 0 >$$

> on the relocated deep-past branches. Then the total self contribution from all deep-past late-apocenter roots satisfies

$$> \|\mathbf{a}_s^{\text{deep}}(t)\| > \le > \frac{M_{\text{ap}}\kappa\epsilon^2}{> (c_f^2\tau_{\text{dp}}^2 + \epsilon_c^2)\nu_{J,\text{dp}}} > \qquad > \text{for every } > t \in I_{\text{ap}}, >$$

> where

$$> M_{\text{ap}} = |\mathcal{U}_{\text{ap}}|. >$$

This corollary is the exact output needed later for the outer-turn comparison argument. Once each deep-past sector contributes at most one transversal branch, the full delayed self-drive is reduced to a finite sector count times a single branch amplitude bound.

The next burden is cone transversality: active branch families need velocity-cone hypotheses that keep the relevant Jacobians uniformly positive throughout the controlled cycle.

Third theorem package: sectorwise cone transversality

The transversality problem should be stated sectorwise, because the line of sight is already sector-labeled by the first package and the deep-past relocation theorem returns the source to a sector-matched inbound cone.

For each sector

$$\mathfrak{S}_k$$

and each controlled window

$$W \in \{I_{\text{in}}, I_{\text{ap}}\}$$

introduce closed velocity cones

$$\mathfrak{V}^{\text{self}}_{k,W} \subset \Pi, \qquad \mathfrak{V}^{\text{part}}_{k,W} \subset \Pi$$

These are not spatial source cones. They live in velocity space and encode the admissible emitter velocities for source points whose active chord directions lie in

$$\mathfrak{S}_k$$

For each such cone define the projection ceilings

$$\Gamma^{\text{self}}_{k,W} \equiv \sup_{\mathbf{v} \in \mathfrak{V}^{\text{self}}_{k,W}} \ \sup_{\hat{\mathbf{u}} \in \mathfrak{S}_k} \mathbf{v} \cdot \hat{\mathbf{u}}$$

$$\Gamma^{\text{part}}_{k,W} \equiv \sup_{\mathbf{v} \in \mathfrak{V}^{\text{part}}_{k,W}} \ \sup_{\hat{\mathbf{u}} \in \mathfrak{S}_k} \mathbf{v} \cdot \hat{\mathbf{u}}$$

The associated dimensionless Jacobian floors are

$$\nu^{\text{self}}_{J,k,W} \equiv 1 - \frac{\Gamma^{\text{self}}_{k,W}}{c_f}, \qquad \nu^{\text{part}}_{J,k,W} \equiv 1 - \frac{\Gamma^{\text{part}}_{k,W}}{c_f}$$

Thus any theorem that produces

$$\Gamma^{\text{self}}_{k,W} < c_f, \qquad \Gamma^{\text{part}}_{k,W} < c_f$$

automatically yields positive transversality margins.

> **Target Proposition (Windowwise velocity-cone realization).**
> There exist closed cones

$$> \mathfrak{V}^{\text{self}}_{k,W}, > \qquad > \mathfrak{V}^{\text{part}}_{k,W}, > \qquad > \hat{\mathbf{u}}_k \in \mathcal{U}_{\text{in}} \cup \mathcal{U}_{\text{ap}}, > \qquad > W \in \{I_{\text{in}}, I_{\text{ap}}\}, >$$

> such that every active labeled branch family

$$> \beta^{\text{self}}_k > \qquad > \text{or} > \qquad > \beta^{\text{part}}_k >$$

> on

$$> W >$$

> has its emitter velocity in the corresponding cone and satisfies

$$> \Gamma^{\text{self}}_{k,W} \le c_f (1 - \nu^{\text{self}}_{J,k,W}), > \qquad > \Gamma^{\text{part}}_{k,W} \le c_f (1 - \nu^{\text{part}}_{J,k,W}) >$$

> for some positive constants

$$> \nu^{\text{self}}_{J,k,W}, > \qquad > \nu^{\text{part}}_{J,k,W}. >$$

> Consequently every active branch on those windows obeys

$$> J_{ij}(t; t_0) \ge \min\{\nu^{\text{self}}_{J,k,W}, \nu^{\text{part}}_{J,k,W}\} > 0. >$$

The content of this proposition is geometric rather than algebraic. One has to prove that admissible emitter velocities stay inside cones whose forward projection onto every active line-of-sight sector remains strictly sub-field-speed. That is the planar replacement for the scalar statement that the 1D Jacobian sign never approaches zero on the controlled branch family.

For later use it is convenient to compress the windowwise floors into

$$\nu_{J,\text{cyc}} \equiv \min_{k,W} \{\nu^{\text{self}}_{J,k,W}, \nu^{\text{part}}_{J,k,W}\}$$

Then the entire controlled cycle satisfies

$$|J_{ij}(t; t_0)| \ge \nu_{J,\text{cyc}} > 0$$

on every labeled active branch.

The relocated deep-past self branches should carry a stronger inbound version of the same statement. On the pre-crossing source interval the desirable geometry is not merely sub-field-speed projection, but strictly negative projection onto the sector direction.

> **Target Proposition (Inbound cone strengthening for relocated deep-past branches).**
> Assume the pre-crossing sector relocation theorem. Then for every active apocenter sector

$$> \mathfrak{S}_k >$$

> there exists a closed inbound velocity cone

$$> \mathfrak{V}_{k,\mathrm{in}}^{\mathrm{dp}} \subset \Pi >$$

> and a constant

$$> \mu_{J,\mathrm{dp},k} > 0 >$$

> such that every relocated deep-past source point on the matching inbound interval satisfies

$$> \dot{\mathbf{r}}(s) \in \mathfrak{V}_{k,\mathrm{in}}^{\mathrm{dp}}, > \qquad > \dot{\mathbf{r}}(s) \cdot \hat{\mathbf{u}} \le -\mu_{J,\mathrm{dp},k} > \qquad > \text{for every } > \hat{\mathbf{u}} \in \mathfrak{S}_k. >$$

> Therefore every relocated deep-past self branch satisfies

$$> J_s(t;s) >=> 1 - \frac{\dot{\mathbf{r}}(s) \cdot \hat{\mathbf{u}}_{s,t}^{\mathrm{self}}}{c_f} >\ge> 1 + \frac{\mu_{J,\mathrm{dp},k}}{c_f}. >$$

This is the precise higher-dimensional analogue of the collinear fact that a pre-crossing inbound source automatically gives

$$J_s > 1$$

The proof burden is now cone separation on the pre-crossing source interval rather than a one-line sign argument.

> **Target Corollary (Deep-past Jacobian floor from inbound cone separation).**
> Assume the inbound cone strengthening proposition and define

$$> \nu_{J,\mathrm{dp}} >\equiv> 1 + \frac{1}{c_f} \min_k \mu_{J,\mathrm{dp},k}. >$$

> Then every relocated deep-past self branch on

$$> I_{\mathrm{ap}} >$$

> obeys

$$> |J_s(t;s)| \ge \nu_{J,\mathrm{dp}} > 1. >$$

This is the exact Jacobian input promised in the deep-past sector suppression corollary above. Once it is available, the late-apocenter deep-past amplitude bound is fully reduced to sector count, delay separation, and the inbound cone geometry.

The next burden is section-side nonvacuity: the planar seed must realize strict chord defect and finite transversal partner geometry on the reduced section.

Fourth theorem package: explicit planar seed packet

The section-side nonvacuity problem should now be made explicit in the reduced planar chart rather than left as an abstract existence claim.

Fix positive constants

$$\rho_* > 0, \qquad u_{r,\mathrm{seed}} > 0, \qquad u_{\theta,\mathrm{seed}} > 0$$

and define

$$U_{\mathrm{seed}} \equiv \sqrt{u_{r,\mathrm{seed}}^2 + u_{\theta,\mathrm{seed}}^2}$$

Assume

$$U_{\mathrm{seed}} < c_f$$

Set

$$\mathbf{v}_{\mathrm{seed}} \equiv -u_{r,\mathrm{seed}} \mathbf{e}_1 + u_{\theta,\mathrm{seed}} \mathbf{e}_2$$

and define the explicit affine planar seed by

$$\Phi_{\text{seed}}(\theta) \equiv \rho_* \mathbf{e}_1 + \theta \mathbf{v}_{\text{seed}}, \qquad \theta \in [-h, 0]$$

This is the minimal planar analogue of the frozen 1D affine inbound seed. It is still affine in history time, but it already carries the genuinely planar datum that the tangential component is positive at the section.

> **Target Proposition (Explicit planar affine seed history).**
> Fix

$$> \rho_* > 0, > \qquad > u_{r,\text{seed}} > 0, > \qquad > u_{\theta,\text{seed}} > 0, > \qquad > U_{\text{seed}} < c_f, >$$

> and let

$$> \Phi_{\text{seed}}(\theta) >=> \rho_* \mathbf{e}_1 + \theta \mathbf{v}_{\text{seed}} >$$

> as above. Define

$$> \sigma_{p,\text{seed}} >\equiv> \frac{2\rho_*}{> \sqrt{c_f^2 - u_{\theta,\text{seed}}^2} > - > u_{r,\text{seed}}}. >$$

> If

$$> h > \sigma_{p,\text{seed}}, >$$

> then:
>
> 1. the section conditions hold:

$$> \Phi_{\text{seed}}(0) = \rho_* \mathbf{e}_1, > \qquad > \mathbf{e}_1 \cdot \dot{\Phi}_{\text{seed}}(0) = -u_{r,\text{seed}} < 0, > \qquad > \mathbf{e}_2 \cdot \dot{\Phi}_{\text{seed}}(0) = u_{\theta,\text{seed}} > 0; >$$

> 2. the stored path has constant speed and zero acceleration:

$$> \sup_{\theta \in [-h,0]} > \|\dot{\Phi}_{\text{seed}}(\theta)\| >=> U_{\text{seed}}, > \qquad > \ddot{\Phi}_{\text{seed}}(\theta) = 0; >$$

> 3. there are no exact same-source self roots on

$$> [-h, 0); >$$

> 4. there is exactly one partner root on the stored interval, located at

$$> \theta_{p,\text{seed}} = -\sigma_{p,\text{seed}}, >$$

> and its Jacobian obeys

$$> J_{p,\text{seed}} >\geq> 1 - \frac{U_{\text{seed}}}{c_f} >> 0. >$$

The proof is the planar version of the 1D seed argument, but the partner root is now genuinely vectorial. Writing

$$\sigma = -\theta > 0$$

the partner root equation at the section time is

$$\|\rho_* \mathbf{e}_1 + \Phi_{\text{seed}}(-\sigma)\| = \sqrt{(2\rho_* + u_{r,\text{seed}}\sigma)^2 + u_{\theta,\text{seed}}^2 \sigma^2} = c_f \sigma$$

Because

$$U_{\text{seed}} < c_f$$

this equation has the unique positive solution

$$\sigma = \sigma_{p,\text{seed}}$$

Equivalently, the scalar function

$$H_{\text{seed}}(\sigma) \equiv \|\rho_* \mathbf{e}_1 + \Phi_{\text{seed}}(-\sigma)\| - c_f \sigma$$

starts from

$$H_{\text{seed}}(0) = 2\rho_* > 0$$

and satisfies

$$H'_{\text{seed}}(\sigma) \le U_{\text{seed}} - c_f < 0$$

so the root is unique once

$$h > \sigma_{p,\text{seed}}$$

For exact self roots on the stored interval one has

$$\|\Phi_{\text{seed}}(0) - \Phi_{\text{seed}}(\theta)\| = U_{\text{seed}}|\theta| < c_f|\theta| \qquad \text{for every } \theta \in [-h, 0)$$

which is the desired chord-defect inequality. Thus the seed carries no exact same-source self roots at all. For the unique partner root, the source particle velocity has norm

$$U_{\text{seed}}$$

so the causal Jacobian satisfies the uniform lower bound

$$J_{p,\text{seed}} = 1 - \frac{\mathbf{v}_1(\theta_{p,\text{seed}}) \cdot \hat{\mathbf{r}}_{p,\text{seed}}}{c_f} \ge 1 - \frac{U_{\text{seed}}}{c_f} > 0$$

Consequently, if the convex-envelope constants satisfy

$$R_{\max} \ge U_{\text{seed}} h, \qquad U_{\max} \ge U_{\text{seed}}, \qquad A_{\max} > 0, \qquad 0 < U_{\theta,\min} \le u_{\theta,\text{seed}} \le U_{\theta,\max}$$

then

$$\Phi_{\text{seed}} \in \mathcal{C}^{\Pi}_{\rho_*,\eta}$$

> **Target Corollary (Nonempty planar section-side tame neighborhood).**
> Under the hypotheses of the explicit planar affine seed proposition, there exist

$$> \varepsilon_{\text{seed}} > 0, > \qquad > \nu_{\text{seed}} > 0, >$$

> and a sector label

$$> \hat{\mathbf{u}}_{k_{\text{seed}}} \in \mathcal{U}_{\text{in}} >$$

> such that the set

$$> \mathcal{C}^{\Pi,\text{seed}}_{\rho_*,\eta} > \equiv > \left\{ > \Phi \in \mathcal{C}^{\Pi}_{\rho_*,\eta} > \;\middle|\; > \Phi(0) = \rho_* \mathbf{e}_1, > \quad > \mathbf{e}_1 \cdot \dot{\Phi}(0) \le -\frac{u_{r,\text{seed}}}{2}, > \quad > \mathbf{e}_2 \cdot \dot{\Phi}(0) \ge \frac{u_{\theta,\text{seed}}}{2}, > \quad > \|\Phi - \Phi_{\text{seed}}\|_{C^1([-h,0])} \le \varepsilon_{\text{seed}} > \right\} >$$

> is nonempty and has the following properties:
>
> 1. every

$$> \Phi \in \mathcal{C}^{\Pi,\text{seed}}_{\rho_*,\eta} >$$

> satisfies

$$> \sup_{\theta \in [-h,0]} \|\dot{\Phi}(\theta)\| > \le > \frac{U_{\text{seed}} + c_f}{2} > < c_f; >$$

> 2. hence no member of the class has an exact same-source self root on

$$> [-h, 0); >$$

> 3. the stored partner root persists uniquely, remains simple, and satisfies

$$> |J_p| \ge \nu_{\text{seed}}; >$$

> 4. the corresponding partner chord direction remains inside one fixed sector

$$> \mathfrak{S}_{k_{\text{seed}}}. >$$

This is the first honest nonvacuity statement for the planar bridge. It says that the reduced section, the chord-defect self-root exclusion, and the sector-labeled partner geometry are simultaneously realizable on a nonempty open patch of the planar history space.

The next burden is bounded planar caustic transit: the first inbound self-branch birth must be integrated through as a controlled shell impulse rather than excluded as a pathology.

Fifth theorem package: bounded planar caustic transit

The first seed-side dynamical obstruction is the birth of the principal inbound self branch. In the planar regime this should be formulated as a controlled fold in the exact self-delay equation, not as a global transversality statement that pretends the fold never occurs.

For

$$s < t$$

define the self-delay defect by

$$G_s(t, s) \equiv \|\mathbf{r}(t) - \mathbf{r}(s)\| - c_f(t - s)$$

Self roots satisfy

$$G_s(t, s) = 0$$

Whenever

$$\mathbf{r}(t) \neq \mathbf{r}(s)$$

one has

$$\partial_s G_s(t, s) = c_f - \dot{\mathbf{r}}(s) \cdot \hat{\mathbf{u}}_{s,t}^{\text{self}} = c_f J_s(t; s)$$

Thus a self-root caustic is exactly the loss of source transversality

$$J_s(t; s) = 0$$

along the same defect equation.

Fix a nonempty seed-generated tame inbound class

$$\mathcal{C}_{\rho_*,\eta}^{\Pi,\text{in}} \subseteq \mathcal{C}_{\rho_*,\eta}^{\Pi,\text{seed}}$$

on which the directional sorting, deep-past sector relocation, and cone-transversality packages all hold on the pre-crossing leg.

The planar caustic event should then be organized around one birth pair

$$(t_{\text{cau}}, s_{\text{cau}}), \qquad s_{\text{cau}} < t_{\text{cau}}$$

with associated sector label

$$\hat{\mathbf{u}}_{k_{\text{cau}}} \in \mathcal{U}_{\text{in}}$$

For a tube radius

$$\delta_{\text{cau}} > 0$$

write the caustic tube in the

$$(t, s)$$

plane as

$$\mathcal{T}_{\text{cau}}(\delta_{\text{cau}}) \equiv \{(t, s) \mid |t - t_{\text{cau}}| + |s - s_{\text{cau}}| \leq \delta_{\text{cau}}, \quad s < t\}$$

and let the associated reception-time window be

$$W_{\text{cau}} \equiv [t_{\text{cau}} - \delta_{\text{cau}}, t_{\text{cau}} + \delta_{\text{cau}}]$$

> **Target Theorem (Inbound planar caustic transit and recovery).**
> Assume there exist class constants

$$> \nu_{p,\mathrm{cau}} > 0, > \qquad > \lambda_{\mathrm{cau}} > 0, > \qquad > \chi_{\mathrm{cau}} > 0, > \qquad > \nu_{s,\mathrm{rec}} > 0, > \qquad > N_{s,\mathrm{cau}} \in \mathbb{N}, > \qquad > I_{\mathrm{cau}} < \infty, >$$

> such that for every inbound trajectory issued from

$$> \mathcal{C}^{\mathrm{II,in}}_{\rho_s,\eta} >$$

> the following hold:
>
> 1. **partner-branch safety:** every active partner branch on the pre-crossing leg remains simple and satisfies

$$> |J_p| \geq \nu_{p,\mathrm{cau}}; >$$

> 2. **single fold birth of the principal self branch:** there is exactly one pair

$$> (t_{\mathrm{cau}}, s_{\mathrm{cau}}) >$$

> with sector label

$$> k_{\mathrm{cau}} >$$

> such that

$$> G_s(t_{\mathrm{cau}}, s_{\mathrm{cau}}) = 0, > \qquad > \partial_s G_s(t_{\mathrm{cau}}, s_{\mathrm{cau}}) = 0, >$$

> while the fold is nondegenerate:

$$> \partial_{ss} G_s(t_{\mathrm{cau}}, s_{\mathrm{cau}}) \geq \lambda_{\mathrm{cau}}, > \qquad > \partial_t G_s(t_{\mathrm{cau}}, s_{\mathrm{cau}}) \leq -\chi_{\mathrm{cau}}; >$$

> 3. **controlled branch count through the tube:** outside

$$> \mathcal{T}_{\mathrm{cau}}(\delta_{\mathrm{cau}}) >$$

> all active self branches are simple and sector-labeled, and the total number of active self branches on the pre-crossing leg is bounded by

$$> N_{s,\mathrm{cau}}; >$$

> 4. **bounded dual-mollified caustic impulse:** if

$$> \mathbf{a}^{\mathrm{self}}_{\eta,\mathrm{cau}}(t) >$$

> denotes the self contribution coming from the branch family intersecting

$$> \mathcal{T}_{\mathrm{cau}}(\delta_{\mathrm{cau}}), >$$

> then

$$> \left\| > \int_{W_{\mathrm{cau}}} > \mathbf{a}^{\mathrm{self}}_{\eta,\mathrm{cau}}(t)\, dt > \right\| > \leq > I_{\mathrm{cau}}; >$$

> 5. **post-tube Jacobian recovery:** once

$$> t \geq t_{\mathrm{cau}} + \delta_{\mathrm{cau}}, >$$

> the born self branch persists in a fixed sector family and satisfies

$$> |J_s(t; s(t))| \geq \nu_{s,\mathrm{rec}}; >$$

> 6. **sorting-gap handoff:** the exiting post-tube history still lies in the same directional-sorting regime needed for the subsequent post-crossing window.

> Then the compulsory inbound self-branch birth contributes only a finite controlled velocity impulse, does not destroy the tame branch package, and hands the trajectory to the later recapture stage with restored Jacobian control.

This is the exact higher-dimensional analogue of the resolved 1D pivot. The theorem does not deny the caustic. It isolates it to one fold tube, integrates the dual-mollified effect across that tube, and demands recovery of the simple-branch regime immediately afterward.

The proof architecture should be split into three bounded tasks.

1. **Fold-birth localization.**
   Show that the first loss of self transversality on the pre-crossing leg occurs at one sector-labeled fold pair and not by uncontrolled simultaneous births in several sectors.
2. **Shell-impulse estimate.**
   Use dual mollification and the fold normal form to prove integrability of the self contribution across

$$W_{\mathrm{cau}}$$

with a class-uniform bound by

$$I_{\mathrm{cau}}$$

3. **Recovery outside the tube.**
   Prove that the born self branch exits the caustic tube into the same sectorwise cone-transversality regime already used elsewhere, yielding

$$|J_s| \geq \nu_{s,\mathrm{rec}}$$

> **Target Corollary (Planar caustic handoff).**
> Under the inbound planar caustic-transit theorem, the pre-crossing leg issued from

$$> \mathcal{C}^{\Pi,\mathrm{in}}_{\rho_*,\eta} >$$

> reaches the post-tube inbound window with:
>
> 1. the same partner-branch control,
> 2. one sector-labeled principal self branch with recovered Jacobian floor

$$> \nu_{s,\mathrm{rec}}, >$$

> 3. total self impulse error bounded by

$$> I_{\mathrm{cau}}, >$$

> 4. and no uncontrolled change in branch labels.

This corollary is the precise output needed before any vector recapture margin can be trusted. Without it, the post-crossing comparison problem would start from a branch topology that may already have disintegrated at the first self-birth event.

The next burden is vector recapture: the planar turn criteria must beat self drive and centrifugal leakage while keeping tangential forcing bounded.

Sixth theorem package: unified vector recapture criteria

The planar comparison problem should be written on two explicit windows:

$$W^{\mathrm{turn}}_{\mathrm{in}} \equiv [t^{\mathrm{turn}}_{\mathrm{in}},\, t^{\mathrm{turn}}_{\mathrm{in}} + \tau_{\mathrm{in}}], \qquad W^{\mathrm{turn}}_{\mathrm{out}} \equiv I_{\mathrm{ap}} = [t^{-}_{\mathrm{ap}},\, t^{+}_{\mathrm{ap}}]$$

The first is the short post-crossing window issued by the caustic handoff. The second is the late-apocenter window on the later outbound branch.

Write the net acceleration in the moving polar frame as

$$\mathbf{a}_{\mathrm{net}}(t) = a_r(t)\hat{\mathbf{e}}_r(t) + a_\theta(t)\hat{\mathbf{e}}_\theta(t), \qquad a_r(t) = \hat{\mathbf{e}}_r(t)\cdot\mathbf{a}_{\mathrm{net}}(t), \qquad a_\theta(t) = \hat{\mathbf{e}}_\theta(t)\cdot\mathbf{a}_{\mathrm{net}}(t)$$

Then

$$\ddot{\rho}(t) = a_r(t) + \rho(t)\dot{\vartheta}(t)^2$$

The scalar 1D turn inequalities are therefore replaced by a competition among three windowwise quantities:

- an inward partner floor,
- an outward self ceiling,
- and a centrifugal leakage ceiling.

> **Target Proposition (Windowwise vector-force split).**
> There exist nonnegative constants

$$> \underline{A}_p^{\mathrm{in}}, > \quad > \overline{A}_s^{\mathrm{in}}, > \quad > \Theta_{\mathrm{in}}, > \quad > \Gamma_{\theta,\mathrm{in}}, >$$

> and

$$> \underline{A}_p^{\mathrm{out}}, > \quad > \overline{A}_s^{\mathrm{out}}, > \quad > \Theta_{\mathrm{out}}, > \quad > \Gamma_{\theta,\mathrm{out}}, >$$

> such that on

$$> W_{\mathrm{in}}^{\mathrm{turn}} >$$

> one has

$$> -a_r^{\mathrm{part}}(t) \geq \underline{A}_p^{\mathrm{in}}, > \qquad > a_r^{\mathrm{self}}(t) \leq \overline{A}_s^{\mathrm{in}}, > \qquad > \rho(t)\dot{\vartheta}(t)^2 \leq \Theta_{\mathrm{in}}, > \qquad > |a_\theta(t)| \leq \Gamma_{\theta,\mathrm{in}}, >$$

> and on

$$> W_{\mathrm{out}}^{\mathrm{turn}} >$$

> one has

$$> -a_r^{\mathrm{part}}(t) \geq \underline{A}_p^{\mathrm{out}}, > \qquad > a_r^{\mathrm{self}}(t) \leq \overline{A}_s^{\mathrm{out}}, > \qquad > \rho(t)\dot{\vartheta}(t)^2 \leq \Theta_{\mathrm{out}}, > \qquad > |a_\theta(t)| \leq \Gamma_{\theta,\mathrm{out}}. >$$

Define the reference inward accelerations

$$a_{\mathrm{ref}}^{\mathrm{in}} \equiv \underline{A}_p^{\mathrm{in}} - \overline{A}_s^{\mathrm{in}} - \Theta_{\mathrm{in}}, \qquad a_{\mathrm{ref}}^{\mathrm{out}} \equiv \underline{A}_p^{\mathrm{out}} - \overline{A}_s^{\mathrm{out}} - \Theta_{\mathrm{out}}$$

Whenever these are positive, the radial comparison inequality becomes

$$\ddot{\rho}(t) \leq -a_{\mathrm{ref}}^{\mathrm{in}} < 0 \qquad \text{on } W_{\mathrm{in}}^{\mathrm{turn}}$$

and

$$\ddot{\rho}(t) \leq -a_{\mathrm{ref}}^{\mathrm{out}} < 0 \qquad \text{on } W_{\mathrm{out}}^{\mathrm{turn}}$$

This is the exact place where the planar bridge differs from the 1D proof scaffold. In the collinear note the comparison race is partner attraction versus self drive. Here the same race acquires the additional positive leakage term

$$\rho\dot{\vartheta}^2$$

and the tangential forcing bound is needed precisely so that the leakage ceiling

$$\Theta_{\mathrm{in}} \quad \text{or} \quad \Theta_{\mathrm{out}}$$

can be maintained throughout the comparison window.

> **Target Theorem (Unified planar vector recapture criterion).**
> Assume the windowwise vector-force split proposition and define the initial outward radial speeds

$$> V_{\mathrm{in},0} > \equiv > \dot{\rho}(t_{\mathrm{in}}^{\mathrm{turn}}), > \qquad > V_{\mathrm{out},0} > \equiv > \dot{\rho}(t_{\mathrm{ap}}^{-}). >$$

> If

$$> a_{\mathrm{ref}}^{\mathrm{in}} > 0, > \qquad > |W_{\mathrm{in}}^{\mathrm{turn}}| > \geq > \frac{V_{\mathrm{in},0}}{a_{\mathrm{ref}}^{\mathrm{in}}}, >$$

> then the first post-crossing radial velocity reaches zero on or before the end of

$$> W_{\mathrm{in}}^{\mathrm{turn}}. >$$

> If, in addition,

$$> a_{\mathrm{ref}}^{\mathrm{out}} > 0, > \qquad > |W_{\mathrm{out}}^{\mathrm{turn}}| > \geq > \frac{V_{\mathrm{out},0}}{a_{\mathrm{ref}}^{\mathrm{out}}}, >$$

> then the late outbound radial velocity also reaches zero on or before the end of

$$> W_{\mathrm{out}}^{\mathrm{turn}}. >$$

> Consequently the trajectory makes both the inner post-crossing turn and the final outer turn inside the controlled planar windows.

The proof is a direct comparison argument once the windowwise floors and ceilings are in hand. The real work is therefore not the last line of the proof. It is the production of the constants

$$\underline{A}_p^{\mathrm{in}}, \quad \overline{A}_s^{\mathrm{in}}, \quad \Theta_{\mathrm{in}}, \quad \underline{A}_p^{\mathrm{out}}, \quad \overline{A}_s^{\mathrm{out}}, \quad \Theta_{\mathrm{out}}$$

from the earlier bridge packages.

The dependency chain should be recorded explicitly:

1. the caustic handoff theorem supplies the controlled entry data for

$$W_{\mathrm{in}}^{\mathrm{turn}}$$

2. the directional sorting and transversality packages keep the partner floor and local self ceiling stable on the inner window;
3. the deep-past relocation and suppression packages produce the outer self ceiling on

$$W_{\mathrm{out}}^{\mathrm{turn}}$$

4. the tangential forcing bounds are what keep

$$\Theta_{\mathrm{in}} \quad \text{and} \quad \Theta_{\mathrm{out}}$$

from becoming uncontrolled.

> **Target Corollary (Inner and outer planar turn margins).**
> Under the unified planar vector recapture criterion, define

$$> \mathfrak{M}_{\mathrm{in}}^{\Pi} >\equiv> a_{\mathrm{ref}}^{\mathrm{in}}, > \qquad > \mathfrak{M}_{\mathrm{out}}^{\Pi} >\equiv> a_{\mathrm{ref}}^{\mathrm{out}}. >$$

> Then the bridge note has precise higher-dimensional replacements for the 1D quantities

$$> \mathfrak{M}_{\mathrm{in}} > 0 > \qquad > \text{and} > \qquad > \mathfrak{M}_{\mathrm{out}} > 0. >$$

> The only remaining burden is to prove those margins from the earlier sector, cone, caustic, and deep-past packages.

Synthesis of the Reduced Planar Bridge

At this point the reduced-planar bridge layer is not treating the completely general planar master equation. It records a symmetry-reduced planar binary dependency map with:

- reflection symmetry,
- center-of-mass reduction,
- a rotationally fixed section representative,
- and one relative planar degree of freedom.

That distinction should remain explicit. The present bridge is the first higher-dimensional transport problem beyond the line, but it is still a reduced regime. The note has not yet advanced to arbitrary 2D delayed trajectories, arbitrary planar many-body branch topology, or the full master equation without symmetry reduction.

Within this reduced planar regime, the theorem ladder now has a definite shape:

1. a nonempty section-side seed packet

$$\mathcal{C}_{\rho_*,\eta}^{\Pi,\mathrm{seed}}$$

provides explicit nonvacuity;
2. directional sorting organizes active delayed roots into finitely many sector-labeled branch families;
3. deep-past sector relocation or exclusion removes the late-apocenter analogue of uncontrolled remote self roots;
4. sectorwise cone transversality supplies cycle-wide Jacobian floors together with the stronger deep-past bound

$$\nu_{J,\mathrm{dp}} > 1$$

5. bounded planar caustic transit integrates the compulsory inbound self-branch birth through one controlled fold tube;
6. unified vector recapture criteria replace the 1D inner and outer turn inequalities by the planar margins

$$\mathfrak{M}_{\mathrm{in}}^{\Pi} > 0, \qquad \mathfrak{M}_{\mathrm{out}}^{\Pi} > 0$$

This synthesis also isolates the exact difference from the frozen 1D scaffold. The collinear note relies on total order and scalar comparison. The reduced planar bridge replaces those by sector atlases, velocity cones, fold tubes, and a radial comparison law that must pay the centrifugal leakage term

$$\rho\dot{\vartheta}^2$$

So the present note is not “the general 2D case.” It is the first reduced 2D bridge in which tangential escape is already real, but still symmetry-controlled.

What remains missing before the Schauder route can be executed as a serious closure theorem is now sharply delimited:

- one closed convex tame self-map domain

$$\mathcal{K}^{\Pi}_{\rho_*,\eta}$$

- continuity and precompactness of the reduced planar return map on that same domain;
- and one coupled parameter regime in which all sorting, relocation, transversality, caustic, and recapture constants are simultaneously realizable.

Seventh theorem package: reduced planar tame-envelope closure

The final reduced planar closure step should now be stated on the gauge-fixed section itself, not merely on the ungauged physical return slice.

If

$$\Phi \in \Sigma^{-}_{\rho_*,\Pi}$$

admits a first full-cycle return time

$$T^{\Pi}(\Phi) > 0$$

to the physical radius

$$\rho(T^{\Pi}(\Phi)) = \rho_*$$

with the same orientation branch

$$\mathbf{e}_2 \cdot \dot{\mathbf{r}}(T^{\Pi}(\Phi)) > 0$$

let

$$\mathcal{R}_{\Phi} \in SO(2)$$

be the unique rigid rotation such that

$$\mathcal{R}_{\Phi}\mathbf{r}(T^{\Pi}(\Phi)) = \rho_*\mathbf{e}_1, \qquad \mathbf{e}_2 \cdot \mathcal{R}_{\Phi}\dot{\mathbf{r}}(T^{\Pi}(\Phi)) > 0$$

The reduced planar return map should therefore be defined by

$$P^{\Pi}_{\eta}(\Phi)(\theta) \equiv \mathcal{R}_{\Phi}\mathbf{r}(T^{\Pi}(\Phi) + \theta), \qquad \theta \in [-h, 0]$$

This is the correct reduced return map. Without the gauge reset

$$\mathcal{R}_{\Phi}$$

the returned history would not land back in the fixed representative section

$$\Sigma^{-}_{\rho_*,\Pi}$$

The price of this gauge reset is a reconstruction condition. A fixed point of

$$P^{\Pi}_{\eta}$$

is periodic in the reduced representative section, but in the fixed Euclidean void it reconstructs as

$$\mathbf{r}(T^{\Pi} + \theta) = \mathcal{R}^{-1}_{\Phi^*}\mathbf{r}(\theta), \qquad \theta \in [-h, 0]$$

Thus the physical trajectory is an absolute periodic orbit only when the rotational holonomy is trivial:

$$\mathcal{R}_{\Phi^*} = \mathrm{Id}$$

Without that additional condition, the result is a relative breather modulo the chosen

$$SO(2)$$

gauge.

The planar envelope target should now be phrased in terms of the constants already produced by the local packages:

$$R_{\max}, \quad U_{\max}, \quad A_{\max}, \quad T_{\mathrm{cyc,max}}, \quad N_{\mathrm{br}}, \quad \delta_{\mathrm{sep}}, \quad \nu_{J,\mathrm{cyc}}, \quad \nu_{J,\mathrm{dp}}, \quad I_{\mathrm{cau}}, \quad \mathfrak{M}^{\Pi}_{\mathrm{in}}, \quad \mathfrak{M}^{\Pi}_{\mathrm{out}}$$

The point is not to define

$$\mathcal{K}^{\Pi}_{\rho_*,\eta}$$

by naively intersecting

$$\mathcal{C}^{\Pi}_{\rho_*,\eta}$$

with all root-label predicates pointwise. That would generally destroy convexity. The correct target is the existence of one closed convex subset on which these constants imply the whole delayed-geometry package uniformly.
Equivalently, the reduced planar bridge should use the same finite certificate convention as the collinear proof: a finite affine or sampled certificate defines the convex set, and the branch labels, Jacobian floors, memory-depth bounds, and caustic exclusions are consequences proved from that certificate.

> **Target Proposition (Closed convex tame envelope in the reduced planar section).**
> There exists a nonempty closed convex set

$$> \mathcal{K}^{\Pi}_{\rho_*,\eta} > \subseteq > \mathcal{C}^{\Pi}_{\rho_*,\eta} > \subseteq > \Sigma^{-}_{\rho_*,\Pi} >$$

> such that:
>
> 1. the seed-generated class

$$> \mathcal{C}^{\Pi,\mathrm{seed}}_{\rho_*,\eta} >$$

> lies in

$$> \mathcal{K}^{\Pi}_{\rho_*,\eta}; >$$

> 2. every

$$> \Phi \in \mathcal{K}^{\Pi}_{\rho_*,\eta} >$$

> admits a unique forward continuation on

$$> [0, T_{\mathrm{cyc,max}}] >$$

> with uniform position, speed, acceleration, and memory-depth bounds inside the convex envelope;
> 3. along that full cycle, the directional sorting, deep-past relocation, cone transversality, caustic-transit, and vector-recapture packages all hold with the same class constants

$$> N_{\mathrm{br}}, > \quad > \delta_{\mathrm{sep}}, > \quad > \nu_{J,\mathrm{cyc}}, > \quad > \nu_{J,\mathrm{dp}}, > \quad > I_{\mathrm{cau}}, > \quad > \mathfrak{M}^{\Pi}_{\mathrm{in}}, > \quad > \mathfrak{M}^{\Pi}_{\mathrm{out}}; >$$

> 4. the first full-cycle return time

$$> T^{\Pi}(\Phi) >$$

> is uniformly transverse on the physical return circle, and the rotated return

$$> P^{\Pi}_{\eta}(\Phi) >$$

> satisfies the same section anchoring and orientation constraints that define

$$> \Sigma^{-}_{\rho_*,\Pi}. >$$

This proposition is the reduced planar analogue of the 1D envelope-construction target. Its role is to place the entire cycle geometry on one domain before asking for a self-map theorem.

> **Target Theorem (Reduced planar invariant-envelope closure).**
> Assume the closed convex tame envelope proposition and suppose, in addition, that:
>
> 1. the rotated returned histories satisfy the same convex envelope bounds

$$> \|\Phi(\theta)-\rho_*\mathbf{e}_1\| \le R_{\max}, > \qquad > \|\dot\Phi(\theta)\| \le U_{\max}, > \qquad > \mathrm{Lip}(\dot\Phi) \le A_{\max}; >$$

> 2. the returned finite certificate implies the same sector labels, branch-count bound, and separation margin;
> 3. the returned finite certificate implies the same Jacobian floors

$$> \nu_{J,\mathrm{cyc}}, > \qquad > \nu_{J,\mathrm{dp}}; >$$

> 4. the returned inner and outer recapture windows satisfy the same strict planar margins

$$> \mathfrak{M}^{\Pi}_{\mathrm{in}} > 0, > \qquad > \mathfrak{M}^{\Pi}_{\mathrm{out}} > 0. >$$

> Then

$$> P^{\Pi}_{\eta}\big(\mathcal{K}^{\Pi}_{\rho_*,\eta}\big) > \subseteq > \mathcal{K}^{\Pi}_{\rho_*,\eta}. >$$

This is the self-map statement the entire bridge has been building toward. It says that after one full physical excursion and one gauge reset back to the reduced section, the same finite certificate remains valid and no envelope constant is lost.

> **Target Proposition (Continuity and precompactness of the reduced planar return map).**
> On

$$> \mathcal{K}^{\Pi}_{\rho_*,\eta}, >$$

> the map

$$> P^{\Pi}_{\eta} : \mathcal{K}^{\Pi}_{\rho_*,\eta} \to \mathcal{K}^{\Pi}_{\rho_*,\eta} >$$

> is continuous in the

$$> C^1([-h,0];\Pi) >$$

> topology, and its image is precompact.

The argument should parallel the 1D continuity step, but with one additional ingredient: continuity of the gauge-reset rotation

$$\mathcal{R}_{\Phi}$$

with respect to the returned section data. Once the return time, returned position, and returned velocity vary continuously and transversely, the rotation back to

$$\rho_*\mathbf{e}_1$$

also varies continuously.

> **Target Theorem (Reduced planar Schauder capstone).**
> Assume:
>
> 1. the closed convex tame envelope proposition in the reduced planar section;
> 2. the reduced planar invariant-envelope closure theorem;
> 3. the continuity and precompactness proposition for

$$> P^{\Pi}_{\eta} >$$

> on

$$> \mathcal{K}^{\Pi}_{\rho_*,\eta}; >$$

> 4. and the nonempty seed-generated class

$$> \mathcal{C}^{\Pi,\mathrm{seed}}_{\rho_*,\eta} \ne \varnothing. >$$

> Then there exists

$$> \Phi_\eta^* >\in> \mathcal{K}^{\Pi}_{\rho_*,\eta} >$$

> such that

$$> P_\eta^{\Pi}(\Phi_\eta^*) = \Phi_\eta^*. >$$

> The corresponding reduced planar trajectory is a bounded relative breather of the dual-mollified master equation within the reflection-symmetric planar binary regime. It is an absolute periodic solution in the fixed Euclidean void only if the reconstructed rotational holonomy satisfies

$$> \mathcal{R}_{\Phi_\eta^*} = \mathrm{Id}. >$$

This is the honest endpoint of the current bridge note. It is still conditional, but it is now conditional on one sharply identified reduced planar closure problem rather than on a diffuse collection of unresolved local lemmas.

Precise Failure Alternative for the Planar Bridge

If the planar bridge fails, the failure should be recorded as a theorem-level obstruction rather than as a vague expression of difficulty. The meaningful obstruction alternatives are:

1. no rotationally reduced affine section produces a nonempty convex section envelope

$$\mathcal{C}^{\Pi}_{\rho_*,\eta}$$

together with a well-defined gauge-reset return map

$$P_\eta^{\Pi}$$

2. every candidate directional sorting or cone-transversality package loses a positive Jacobian floor, either on the full controlled cycle

$$\nu_{J,\mathrm{cyc}} \le 0$$

or on the relocated deep-past branches

$$\nu_{J,\mathrm{dp}} \le 1$$

so branch persistence cannot be maintained;
3. every candidate tame class loses one of the delayed-geometry controls needed for one common return domain: bounded branch count

$$N_{\mathrm{br}}$$

branch separation

$$\delta_{\mathrm{sep}}$$

or bounded caustic impulse

$$I_{\mathrm{cau}} < \infty$$

4. every candidate inner or outer comparison window satisfies

$$\mathfrak{M}^{\Pi}_{\mathrm{in}} \le 0 \qquad \text{or} \qquad \mathfrak{M}^{\Pi}_{\mathrm{out}} \le 0$$

so partner attraction cannot beat self-drive plus centrifugal leakage on the reduced planar windows;
5. the rotated full-cycle return fails to preserve the same section anchoring, envelope bounds, or tame constants, so that

$$P_\eta^{\Pi}(\mathcal{K}^{\Pi}_{\rho_*,\eta}) \not\subseteq \mathcal{K}^{\Pi}_{\rho_*,\eta}$$

6. or the reduced return map fails to be continuous or precompact on the same closed convex domain, so the Schauder capstone has no legitimate domain of application.

Any one of these constitutes a precise statement that the frozen 1D scaffold does not transport to the reflection-symmetric planar binary without an additional invariant, symmetry, or coercive mechanism. That is the obstruction that should be written down if the planar program breaks.

Beyond the Reduced Planar Bridge

The next genuinely new 2D regime would not yet be the full many-body master equation. It would already arise if one dropped the reflection-symmetric binary reduction while staying in a single plane. Even that smaller step introduces new burdens that are absent from the present bridge.

First, the section and gauge problem becomes genuinely multicomponent. The present note fixes one relative planar degree of freedom and resets the return by a single rotation back to

$$\rho_* \mathbf{e}_1$$

In a less constrained planar regime, the return section would have to be posed on a higher-dimensional shape space modulo rigid Euclidean symmetries, and the gauge reset would no longer be a one-angle correction.

Second, the delayed-root topology ceases to be a single sector-labeled branch family over one relative chord geometry. One should expect a finite branch graph rather than one labeled list: several inequivalent source-receiver chord types, several sector atlases, and potentially competing branch births on the same window.

Third, the deep-past relocation mechanism would need a new replacement. In the reduced planar bridge, remote late-apocenter self roots are pushed back into one pre-crossing inbound cone family. Without that reduction, there may be no single inbound cone or even one distinguished pre-crossing leg. A more general 2D regime would therefore need either a global provenance graph for delayed branches or a new topological exclusion theorem.

Fourth, the escape geometry is no longer exhausted by the scalar leakage term

$$\rho\dot{\vartheta}^2$$

That term is the correct planar correction for one relative polar degree of freedom. A less constrained 2D regime would require a genuinely multi-channel escape estimate, with several tangential or shear-like directions that can steal coercivity from the radial comparison argument.

Fifth, the tame-envelope problem itself becomes harder. The reduced planar note asks for one closed convex tame domain in a fixed

$$C^1([-h,0];\Pi)$$

chart. Beyond that regime, even the correct Banach chart and the right convex section envelope may become part of the theorem burden rather than fixed background data.

The burden growth is cumulative:

Burden	Reduced planar binary	Unreduced planar binary	Many-body bridge
Section and gauge	one radius section plus one rotation reset	quotient chart with explicit $SE(2)$ selector and holonomy	atlas-level gauge selector with chart-stability theorem
Delayed-root topology	finite sector-labeled branch family	canonical finite branch graph with fold edges	finite active delay hypergraph
Deep-past control	sector relocation or exclusion	branch-graph provenance or exclusion	cluster-valued ancestry or exclusion
Recapture	one radial channel plus one angular leakage term	finite leakage-channel comparison with resonance control	finite escape-observable family with channel margins
Closure domain	closed convex tame envelope in one reduced chart	quotient-space convex envelope preserving graph and holonomy data	atlas-level convex core preserving hypergraph, ancestry, and recapture windows

For that reason, the present chapter should be read as the first 2D bridge, not as the general planar theorem program. Only after these additional burdens are isolated and given their own theorem targets would it be honest to say that the work has moved beyond the reduced planar bridge.

First unreduced planar theorem targets

The next honest regime should still remain modest: a planar binary without the reflection-symmetric return reduction, but also without yet opening the full many-body master equation. Even there, the bridge should be restarted with a new theorem ladder rather than by informal analogy with the reduced planar case.

Write

$$\mathcal{Q}_{\mathrm{pl}}^{\sharp} \equiv \mathcal{Q}_{\mathrm{pl}}/SE(2)$$

for the planar shape quotient after rigid Euclidean symmetries are removed. The first unreduced planar bridge should then be organized around the following targets.

1. A section-and-gauge package on

$$C^1([-h,0];(\mathbb{R}^2)^2)$$

represented in the quotient chart

$$\mathcal{Q}^{\sharp}_{\mathrm{pl}}$$

with one codimension-one return condition and one explicit gauge selector that chooses a canonical representative of each returned history. In the reduced planar bridge the gauge reset is one rotation angle. Here the theorem target must identify the correct quotient chart and prove that the returned history depends continuously on that gauge choice.

2. A finite branch-graph package for the active delayed roots. Instead of one sector-labeled family of self and partner branches, one should expect a finite graph

$$\mathscr{G}^{\sharp}_{\mathrm{br}}$$

whose vertices encode source-receiver chord types and whose edges encode admissible fold births, fold deaths, or branch handoffs between windows. The replacement theorem must bound that graph uniformly and exclude uncontrolled simultaneous fold accumulation.

3. A provenance or exclusion package for deep-past roots. The reduced planar argument pushes late self roots into one pre-crossing inbound cone family. The unreduced planar target should instead prove that every remote active root either relocates into a finite provenance class on the earlier branch graph or is excluded by a topological obstruction principle. Without such a theorem there is no honest replacement for deep-past relocation.

4. A multi-channel recapture package. The reduced planar comparison law pays one scalar leakage term

$$\rho\dot{\vartheta}^2$$

In the unreduced planar regime the replacement must control several escape channels at once: rotational, tangential, and shear-like components in the quotient dynamics. The correct theorem target is therefore not one scalar inequality, but a coercive inward comparison that dominates every nonradial leakage channel on the chosen inner and outer windows.

5. A new closure package on one quotient-space convex envelope

$$\mathcal{C}^{\sharp}_{\rho_*,\eta}$$

and one closed convex tame envelope

$$\mathcal{K}^{\sharp}_{\rho_*,\eta} \subseteq \mathcal{C}^{\sharp}_{\rho_*,\eta}$$

The returned histories must land back in the same gauge-fixed quotient section, preserve the branch-graph, provenance, leakage, and recapture constants, and define a continuous precompact self-map on that same domain.

Section-and-gauge target for the first unreduced planar binary

The first step in that ladder should be made fully explicit. For a labeled planar binary write

$$\mathbf{Y}(t) \equiv (\mathbf{x}_1(t),\mathbf{x}_2(t)) \in (\mathbb{R}^2)^2$$

and decompose the instantaneous configuration into midpoint and chord variables

$$\mathbf{m}(t) \equiv \frac{\mathbf{x}_1(t)+\mathbf{x}_2(t)}{2}, \qquad \mathbf{q}(t) \equiv \mathbf{x}_2(t)-\mathbf{x}_1(t)$$

This is the reason the unreduced planar binary is the correct next regime. The quotient by translations is still explicit through

$$\mathbf{m}$$

and the quotient by rotations is still explicit through the present chord

$$\mathbf{q}(0)$$

even though the dynamics no longer collapse to one reflection-symmetric relative trajectory.

Fix a return radius

$$\rho_* > 0$$

The raw inbound section should be posed on full binary histories by

$$\Sigma^{-,\sharp}_{\rho_*} \equiv \left\{\Phi = (\Phi_1,\Phi_2) \in C^1([-h,0];(\mathbb{R}^2)^2) \;\middle|\; \|\mathbf{q}_\Phi(0)\| = \rho_*, \quad \mathbf{q}_\Phi(0)\cdot\dot{\mathbf{q}}_\Phi(0) < 0\right\}$$

where

$$\mathbf{q}_\Phi(\theta) \equiv \Phi_2(\theta) - \Phi_1(\theta)$$

This is the physical inbound crossing condition before any quotient representative is chosen.

For

$$\Phi \in \Sigma^{-,\sharp}_{\rho_*}$$

let

$$\mathbf{a}_\Phi \equiv \mathbf{m}_\Phi(0), \qquad \mathbf{m}_\Phi(\theta) \equiv \frac{\Phi_1(\theta) + \Phi_2(\theta)}{2}$$

and let

$$\mathcal{R}_\Phi \in SO(2)$$

be the unique rotation such that

$$\mathcal{R}_\Phi \mathbf{q}_\Phi(0) = \rho_* \mathbf{e}_1$$

Define the gauge selector by

$$\mathfrak{G}_{\rho_*}(\Phi)(\theta) \equiv \left(\mathcal{R}_\Phi\big(\Phi_1(\theta) - \mathbf{a}_\Phi\big), \mathcal{R}_\Phi\big(\Phi_2(\theta) - \mathbf{a}_\Phi\big)\right), \qquad \theta \in [-h, 0]$$

The corresponding gauge-fixed section is

$$\widehat{\Sigma}^{-,\sharp}_{\rho_*} \equiv \left\{\Psi \in C^1([-h,0];(\mathbb{R}^2)^2) \;\middle|\; \mathbf{m}_\Psi(0) = 0, \quad \mathbf{q}_\Psi(0) = \rho_* \mathbf{e}_1, \quad \mathbf{e}_1 \cdot \dot{\mathbf{q}}_\Psi(0) < 0\right\}$$

> **Target Proposition (Gauge-fixed inbound section for the first unreduced planar bridge).**
> There exists a nonempty class

$$> \mathcal{H}^{\sharp,\mathrm{sec}}_{\rho_*,\eta} > \subseteq > \Sigma^{-,\sharp}_{\rho_*} >$$

> such that:
>
> 1. every

$$> SE(2) >$$

> orbit in

$$> \mathcal{H}^{\sharp,\mathrm{sec}}_{\rho_*,\eta} >$$

> meets the gauge-fixed section

$$> \widehat{\Sigma}^{-,\sharp}_{\rho_*} >$$

> in exactly one history;
> 2. the selector

$$> \mathfrak{G}_{\rho_*} :> \mathcal{H}^{\sharp,\mathrm{sec}}_{\rho_*,\eta} > \to > \widehat{\Sigma}^{-,\sharp}_{\rho_*} >$$

> is continuous in the

$$> C^1 >$$

> topology;
> 3. every

$$> \Psi \in \widehat{\Sigma}^{-,\sharp}_{\rho_*} > \cap > \mathfrak{G}_{\rho_*}\big(> \mathcal{H}^{\sharp,\mathrm{sec}}_{\rho_*,\eta} > \big) >$$

> admits a first later return time

$$> T^\sharp(\Psi) > 0 >$$

> to the raw section

$$> \Sigma^{-,\sharp}_{\rho_*} >$$

> with uniform transversality

$$ > \big| > \mathbf{q}(T^\sharp(\Psi)) > \cdot > \dot{\mathbf{q}}(T^\sharp(\Psi)) > \big| >\geq> \lambda^\sharp_{\mathrm{sec}} >> 0; > $$

> 4. the gauge-reset return map

$$ > P^\sharp_\eta(\Psi) >\equiv> \mathfrak{G}_{\rho_*}\big(> \mathbf{Y}_{T^\sharp(\Psi)} > \big) > $$

> is therefore well defined and lands again in

$$ > \widehat{\Sigma}^{-,\sharp}_{\rho_*}. > $$

This is the first unreduced-planar replacement for the reduced planar section anchoring. Its purpose is to make the quotient representative, the return section, and the gauge reset part of the theorem burden before any branch-graph or recapture package is attempted.

> **Target Proposition (Holonomy reconstruction for quotient fixed points).**
> As in the reduced planar bridge, a fixed point of

$$ > P^\sharp_\eta > $$

> is first a quotient-space fixed point. Let the full-cycle gauge reset be represented by the Euclidean holonomy

$$ > H_{\Psi^*} >=> (\mathbf{a}_{\Psi^*}, \mathcal{R}_{\Psi^*}) >\in SE(2), > $$

> where

$$ > \mathbf{a}_{\Psi^*} > $$

> is the translation removed by the selector and

$$ > \mathcal{R}_{\Psi^*} > $$

> is the rotation back to the canonical chord. The reconstructed physical motion is absolute periodic only if

$$ > H_{\Psi^*} = (0, \mathrm{Id}). > $$

> Otherwise the fixed point is a relative breather modulo

$$ > SE(2). > $$

> In symmetry-reduced regimes this condition may be forced by the symmetry itself. For example, the reflection-symmetric reduced planar bridge can force the rotational part to be

$$ > \mathcal{R}_{\Psi^*} = \mathrm{Id} > $$

> once the gauge is chosen compatibly with the reflection. In the unreduced planar bridge there is no such automatic cancellation. The absolute-periodic problem is therefore a separate finite-dimensional zero-holonomy reconstruction problem, with two effective holonomy components after the section-and-gauge constraints are imposed.

This proposition should be treated as part of every quotient-space Schauder capstone. Schauder on the quotient gives a relative breather; zero holonomy is an additional reconstruction condition, analogous in role to a characteristic multiplier condition in Floquet theory.

Finite branch-graph target for the unreduced planar bridge

Once the gauge-fixed section is available, the next missing object is the delayed-root replacement for the reduced planar sector-labeled branch family. In the unreduced planar binary one no longer follows a single relative chord geometry. Instead one must track several source-receiver chord types at once.

For

$$ \tau = (i \leftarrow j) \in \{1,2\} \times \{1,2\} $$

define the corresponding delayed-root defect by

$$ G_\tau(t,s) \equiv \big\|\mathbf{x}_i(t) - \mathbf{x}_j(s)\big\| - c_f(t-s), \qquad s < t $$

and let

\hat{\mathbf{u}}_{\tau}(t,s) \equiv \frac{\mathbf{x}_i(t)-\mathbf{x}_j(s)} \|\mathbf{x}_i(t)-\mathbf{x}_j(s)\|}

be the associated chord direction whenever the denominator is nonzero. The active delayed roots over one candidate cycle should be studied only after the cycle is partitioned into a finite family of windows

$$\mathcal{W}^\sharp = \{W_1, \dots, W_{L^\sharp}\}$$

chosen so that no window straddles more than one geometric event of the candidate excursion: section entry, near-crossing passage, outbound expansion, late turn, or returned entry into the section.

Fix also a finite sector atlas

$$\mathcal{U}^\sharp = \{\mathfrak{S}_1^\sharp, \dots, \mathfrak{S}_{M^\sharp}^\sharp\}$$

on

$$S^1$$

with positive overlap margins removed so that every active chord direction remains a definite angular distance away from sector boundaries except inside the later caustic neighborhoods. For each type

$$\tau$$

window

$$W_\ell$$

and sector

$$\mathfrak{S}_k^\sharp$$

an active local branch should mean a simple root curve

$$s = \beta_{k,\ell,m}^\tau(t), \qquad t \in I_{k,\ell,m}^\tau \subseteq W_\ell$$

satisfying

$$G_\tau\big(t, \beta_{k,\ell,m}^\tau(t)\big) = 0, \qquad \hat{\mathbf{u}}_\tau\big(t, \beta_{k,\ell,m}^\tau(t)\big) \in \mathfrak{S}_k^\sharp$$

and

$$\big|J_\tau\big(t, \beta_{k,\ell,m}^\tau(t)\big)\big| > 0$$

The branch graph

$$\mathscr{G}_{\mathrm{br}}^\sharp$$

should then be defined by one canonical convention.

- A vertex is one maximal simple local branch segment

$$v = (\tau, k, \ell, m)$$

- The multiplicity label

$$m$$

is assigned by the source-time order of the simple roots inside the fixed type-sector-window cell after the fold tubes and sector-boundary tubes have been removed.

- Folds are represented as edges, not as additional vertices. A fold edge carries the fold-tube label, the incoming and outgoing local branch labels, and the parity data for the root-count jump.
- Continuation across adjacent windows is also represented by an edge. Thus a window-boundary fold is never represented by a vertex split; it is one labeled fold edge between the adjacent simple-branch vertices that enter and exit the certified tube.

With this convention, the graph is uniquely determined by a gauge-fixed history, the fixed window partition, the fixed sector atlas, and the listed caustic tubes: vertices are maximal connected components of the simple-root set after the controlled tubes are removed, and edges record the unique admissible continuation or fold transition through the adjacent tube.

This is the correct replacement object. In the reduced planar bridge the active roots could be recorded as a finite list because one relative geometry and one sector family were enough. In the unreduced planar bridge the natural finite object is instead a graph whose vertices remember both the chord type and the local window.

> **Target Proposition (Finite active branch graph on the unreduced planar cycle).**
> There exist:

$$> L^\sharp, > \qquad > M^\sharp, > \qquad > N^\sharp_{\mathrm{br}}, > \qquad > N^{\max}_{\mathrm{mult}}, > \qquad > \delta^\sharp_{\mathrm{sep}}, > \qquad > \nu^\sharp_J, > \qquad > M^\sharp_{\mathrm{cau}}, >$$

> a cycle partition

$$> \mathcal{W}^\sharp, >$$

> a sector atlas

$$> \mathcal{U}^\sharp, >$$

> and a finite family of pairwise separated caustic tubes

$$> \mathfrak{T}^\sharp_{\mathrm{cau},1}, > \dots, > \mathfrak{T}^\sharp_{\mathrm{cau},M^\sharp_{\mathrm{cau}}}, >$$

> such that for every gauge-fixed history

$$> \Psi \in > \widehat{\Sigma}^{-,\sharp}_{\rho_*} > \cap > \mathfrak{G}_{\rho_*}\big(> \mathcal{H}^{\sharp,\mathrm{sec}}_{\rho_*,\eta} > \big) >$$

> the active delayed roots on one full returned cycle satisfy:
>
> 1. every active root of every chord type

$$> \tau >$$

> belongs to exactly one vertex

$$> (\tau, k, \ell, m) >$$

> of a finite graph

$$> \mathscr{G}^\sharp_{\mathrm{br}}(\Psi), >$$

> and the total number of vertices is bounded by

$$> N^\sharp_{\mathrm{br}} > \leq > 4M^\sharp L^\sharp N^{\max}_{\mathrm{mult}}; >$$

> here the factor

$$> 4 >$$

> is the number of source-receiver chord types

$$> |\{1,2\} \times \{1,2\}|, >$$

> and

$$> N^{\max}_{\mathrm{mult}} >$$

> is the certified maximum number of simple root branches in one fixed type-sector-window cell;
> 2. on every vertex domain outside the caustic tubes, the root branch is

$$> C^1 >$$

> in

$$> t >$$

> and satisfies the uniform Jacobian lower bound

$$> \big|J_\tau\big(t, \beta^\tau_{k,\ell,m}(t)\big)\big| > \geq > \nu^\sharp_J >> 0; >$$

> 3. two distinct active vertices with the same chord type

$$> \tau >$$

> and the same sector label

$$> k >$$

> on the same window are separated by a delay gap at least

$$> \delta^{\sharp}_{\mathrm{sep}} >> 0; >$$

> 4. every edge of

$$> \mathscr{G}^{\sharp}_{\mathrm{br}}(\Psi) >$$

> is of exactly one of the following kinds under the canonical convention:
> a continuation edge across adjacent windows,
> a fold birth/death edge labeled by one caustic tube,
> or one admissible branch-handoff edge labeled by the same tube;
> 5. outside the union of the caustic tubes the graph is locally constant in

$$> t, >$$

> so no uncontrolled simultaneous fold accumulation or instantaneous infinite relabeling can occur along the cycle.
> 6. the graph construction is canonical: no admissible fold tube may be encoded alternatively as a vertex split, and the source-time ordering convention fixes the multiplicity labels.
>
> In particular, the active delayed-root topology of the unreduced planar binary is encoded by one finite graph rather than by an a priori continuum of chord directions.

This proposition is the unreduced-planar replacement for the reduced planar branch-count and branch-labeling package. The main difference is not merely higher notation. It is that the theorem now has to control branch continuation across several chord types and windows, not just uniqueness inside one scalar or sector-labeled family.

Deep-past provenance-or-exclusion target for the unreduced planar bridge

The next burden is the unreduced-planar replacement for deep-past relocation. In the reduced planar bridge, a remote late-turn self root is pushed back into one pre-crossing inbound cone. That statement is too rigid once several chord types and several window families are active. The correct replacement is weaker in form but still finite: every remote late-turn root must either trace backward to one of finitely many earlier provenance classes in the branch graph, or be excluded altogether.

Let

$$\mathcal{W}^{\sharp}_{\mathrm{lt}} \subseteq \mathcal{W}^{\sharp}$$

denote the late-turn receiver windows on which outer recapture versus escape is analyzed, and let

$$\mathcal{W}^{\sharp}_{\mathrm{out}}, \qquad \mathcal{W}^{\sharp}_{\mathrm{prov}} \subseteq \mathcal{W}^{\sharp}$$

denote, respectively, the intermediate outbound windows and the earlier provenance windows that precede

$$\mathcal{W}^{\sharp}_{\mathrm{lt}}$$

in the receiver-time order of the cycle partition. A deep-past active root on a late-turn window should mean one with delay

$$t - s \geq \tau^{\sharp}_{\mathrm{dp}}$$

For a vertex

$$v = (\tau, k, \ell, m)$$

of

$$\mathscr{G}^{\sharp}_{\mathrm{br}}(\Psi)$$

write

$$S(v) \equiv \beta^{\tau}_{k,\ell,m}\big(I^{\tau}_{k,\ell,m}\big)$$

for its source-time trace. If

$$W_\ell \in \mathcal{W}^\sharp_{\rm lt}$$

call

$$v$$

a late-turn vertex. Define the backward ancestry

$$\mathrm{Anc}^-(v)$$

to be the set of vertices reachable from

$$v$$

by a chain of continuation or handoff edges that never moves to a later receiver window in the cycle order. This is the graph-theoretic replacement for “going back to the inbound leg.”

The finite provenance objects should now be defined at the graph level rather than at the level of individual root times. A provenance class should mean a connected subgraph

$$\mathfrak{P}^\sharp_a \subseteq \mathscr{G}^\sharp_{\rm br}(\Psi)$$

whose vertices all lie in provenance windows

$$W_\ell \in \mathcal{W}^\sharp_{\rm prov}$$

share one chord type

$$\tau$$

and one sector label

$$k$$

and have connected source traces

$$\bigcup_{v\in\mathfrak{P}^\sharp_a} S(v)$$

on which the source parameterization is simple and carries one uniform Jacobian floor

$$\nu^\sharp_{J,\rm dp} > 0$$

The unreduced-planar exclusion step must also be stated graphwise. For each chord type

$$\tau$$

and each sector

$$\mathfrak{S}^\sharp_k$$

that appears on a late-turn vertex, define the corresponding outbound exclusion slab

$$\mathfrak{E}^\sharp_{\tau,k} \equiv \big\{(t,s)\,\big|\,t\in W_\ell,\ s\in W_{\ell'},\ W_\ell\in\mathcal{W}^\sharp_{\rm lt},\ W_{\ell'}\in\mathcal{W}^\sharp_{\rm out},\ \hat{\mathbf{u}}_\tau(t,s)\in\mathfrak{S}^\sharp_k\big\}$$

The intended theorem input is that

$$G_\tau$$

has fixed nonzero sign on every such slab, so no active late-turn root can draw its source from the intermediate outbound block.

> **Target Proposition (Deep-past provenance or exclusion on the unreduced planar branch graph).**
> Assume the finite active branch-graph proposition and suppose, in addition, that:
>
> 1. for every late-turn type-sector pair

$$> (\tau, k), >$$

> the defect

$$> G_\tau >$$

> has fixed nonzero sign on the outbound exclusion slab

$$> \mathfrak{E}^\sharp_{\tau,k}, >$$

> so no active root with receiver in

$$> \mathcal{W}^\sharp_{\mathrm{lt}} >$$

> can have source in

$$> \mathcal{W}^\sharp_{\mathrm{out}}; >$$

> 2. there exists a finite family of provenance classes

$$> \mathfrak{P}^\sharp_1, > \dots, > \mathfrak{P}^\sharp_{P^\sharp_{\mathrm{prov}}} >$$

> with the connected-source and Jacobian-floor properties described above;
> 3. every connected component of

$$> \mathscr{G}^\sharp_{\mathrm{br}}(\Psi) >$$

> that meets a late-turn vertex contains at most one provenance class;
> 4. any backward ancestry chain from a late-turn vertex that avoids all provenance classes must remain trapped in the union of late-turn vertices and caustic tubes, and such a trapped component is forbidden by the branch graph parity rule for fold births and deaths.
>
> Then every deep-past active root on a late-turn window is either absent or belongs to a late-turn vertex

$$> v >$$

> whose backward ancestry

$$> \mathrm{Anc}^-(v) >$$

> meets exactly one provenance class

$$> \mathfrak{P}^\sharp_a. >$$

> In particular, the total family of deep-past late-turn roots is controlled by the finite provenance count

$$> P^\sharp_{\mathrm{prov}}. >$$

This is the correct unreduced-planar replacement for deep-past relocation. A remote root is no longer forced onto one literal inbound cone. Instead it is forced into one finite earlier provenance class of the branch graph, and the only alternative is a precise obstruction: a trapped late-turn component unsupported by any earlier provenance source.

> **Target Corollary (Deep-past suppression from finite provenance classes).**
> Assume the deep-past provenance-or-exclusion proposition. Then on every late-turn receiver window one has the uniform bound

$$> \|> \mathbf{a}^{\sharp,\mathrm{deep}}(t) >\| > \le > \frac{> P^\sharp_{\mathrm{prov}} > \kappa\epsilon^2 >}{> \left(c_f^2(\tau^\sharp_{\mathrm{dp}})^2 + \epsilon_c^2\right) > \nu^\sharp_{J,\mathrm{dp}} >}, >$$

> for every

$$> t \in \bigcup_{W \in \mathcal{W}^\sharp_{\mathrm{lt}}} W, >$$

> because each admissible provenance class contributes at most one uniformly transversal deep-past branch at the chosen delay scale.

This is the quantity the later unreduced-planar recapture package should consume. Once the remote self contribution is reduced to a finite provenance count times one branch amplitude bound, the deep-past part of self-drive is again a controlled term rather than an open-ended graph-combinatorial hazard.

Multi-channel recapture target for the unreduced planar bridge

The next replacement burden is the turn mechanism itself. In the reduced planar bridge the recapture inequality was written in polar form and paid the single leakage term

$$\rho\dot{\vartheta}^2$$

That is no longer the honest object once the dynamics are organized on the unreduced quotient chart. The correct target is a coercive primary escape coordinate together with a finite family of leakage channels that measure how the remaining quotient modes steal inward control.

Choose a gauge-fixed quotient neighborhood

$$\mathscr{U}^{\sharp}_{\mathrm{turn}} \subseteq C^1([-h,0];\mathcal{Q}^{\sharp}_{\mathrm{pl}})$$

that contains the relevant post-crossing and late-turn windows. The comparison argument should now be phrased in terms of one smooth escape observable

$$\rho^{\sharp}(t) \geq 0$$

and finitely many nonnegative leakage channels

$$\Lambda^{\sharp}_1(t), \ldots, \Lambda^{\sharp}_{Q^{\sharp}_{\mathrm{esc}}}(t)$$

The intended meaning is:

- $\rho^{\sharp}$ measures excursion size in the primary escape direction on the quotient chart;
- each

$$\Lambda^{\sharp}_{\alpha}$$

measures one distinct noncoercive way in which the remaining quotient modes can feed outward motion;

- the number

$$Q^{\sharp}_{\mathrm{esc}}$$

should be finite on the chosen chart.

Let

$$W^{\sharp}_{\mathrm{in,turn}}$$

be the first post-crossing recapture window and let

$$W^{\sharp}_{\mathrm{out,turn}} \subseteq \bigcup_{W \in \mathcal{W}^{\sharp}_{\mathrm{lt}}} W$$

be the late-turn window. The quotient comparison law should then be organized as

$$\ddot{\rho}^{\sharp}(t) = -A^{\sharp}_{p}(t) + A^{\sharp}_{s,\mathrm{loc}}(t) + A^{\sharp}_{s,\mathrm{deep}}(t) + \sum_{\alpha=1}^{Q^{\sharp}_{\mathrm{esc}}} \Lambda^{\sharp}_{\alpha}(t)$$

where:

- $A^{\sharp}_{p}(t) \geq 0$ is the inward partner contribution in the primary escape coordinate;
- $A^{\sharp}_{s,\mathrm{loc}}(t) \geq 0$ is the ceiling coming from non-deep active self branches on the controlled windows;
- $A^{\sharp}_{s,\mathrm{deep}}(t) \geq 0$ is the deep-past ceiling supplied by the provenance-suppression package.

> **Target Proposition (Windowwise multi-channel quotient-force split).**
> There exist nonnegative constants

$$> \underline{A}^{\sharp}_{p,\mathrm{in}}, > \qquad > \overline{A}^{\sharp}_{s,\mathrm{loc,in}}, > \qquad > \overline{A}^{\sharp}_{s,\mathrm{deep,in}}, > \qquad > \Theta^{\sharp}_{\alpha,\mathrm{in}} > \quad > (1 \leq \alpha \leq Q^{\sharp}_{\mathrm{esc}}), >$$

> and

$$> \underline{A}^{\sharp}_{p,\mathrm{out}}, > \qquad > \overline{A}^{\sharp}_{s,\mathrm{loc,out}}, > \qquad > \overline{A}^{\sharp}_{s,\mathrm{deep,out}}, > \qquad > \Theta^{\sharp}_{\alpha,\mathrm{out}} > \quad > (1 \leq \alpha \leq Q^{\sharp}_{\mathrm{esc}}) >$$

> such that on

$$> W^{\sharp}_{\mathrm{in,turn}} >$$

> one has

$$> A^{\sharp}_{p}(t) \ge \underline{A}^{\sharp}_{p,\mathrm{in}}, > \qquad > A^{\sharp}_{s,\mathrm{loc}}(t) \le \overline{A}^{\sharp}_{s,\mathrm{loc,in}}, > \qquad > A^{\sharp}_{s,\mathrm{deep}}(t) \le \overline{A}^{\sharp}_{s,\mathrm{deep,in}}, > \qquad > \Lambda^{\sharp}_{\alpha}(t) \le \Theta^{\sharp}_{\alpha,\mathrm{in}}, >$$

> for every

$$> \alpha, >$$

> and on

$$> W^{\sharp}_{\mathrm{out,turn}} >$$

> one has

$$> A^{\sharp}_{p}(t) \ge \underline{A}^{\sharp}_{p,\mathrm{out}}, > \qquad > A^{\sharp}_{s,\mathrm{loc}}(t) \le \overline{A}^{\sharp}_{s,\mathrm{loc,out}}, > \qquad > A^{\sharp}_{s,\mathrm{deep}}(t) \le \overline{A}^{\sharp}_{s,\mathrm{deep,out}}, > \qquad > \Lambda^{\sharp}_{\alpha}(t) \le \Theta^{\sharp}_{\alpha,\mathrm{out}}, >$$

> for every

$$> \alpha. >$$

> **Target Hypothesis (Leakage-channel independence and resonance control).**
> The leakage-channel ceilings in the preceding force split are usable in the recapture margin only under one of two certified conditions:
>
> 1. **simultaneous pointwise ceiling:** the inequalities

$$> 0 \le \Lambda^{\sharp}_{\alpha}(t) \le \Theta^{\sharp}_{\alpha,\bullet} >$$

> are proved on the same controlled tube, at the same times, for all

$$> \alpha = 1, \ldots, Q^{\sharp}_{\mathrm{esc}}; >$$

> 2. **non-resonant channel decomposition:** the leakage subsystem obtained by linearizing the quotient dynamics along the candidate cycle has Floquet exponents or channel frequencies

$$> \omega_1, \ldots, \omega_{Q^{\sharp}_{\mathrm{esc}}} >$$

> satisfying an explicit finite-order non-resonance gap

$$> |n \cdot \omega| >\ge> \gamma^{\sharp}_{\mathrm{res}} > 0 > \qquad > \text{for every } 0 \ne n \in \mathbb{Z}^{Q^{\sharp}_{\mathrm{esc}}}, > \quad > |n|_1 \le N^{\sharp}_{\mathrm{res}}, >$$

> or equivalently a monodromy certificate with non-resonant Floquet multipliers on the leakage block.
>
> If neither condition is certified, the scalar leakage budget

$$> \sum_{\alpha} \Theta^{\sharp}_{\alpha,\bullet} >$$

> is not an admissible proof input. It must be replaced by a coupled leakage budget

$$> \Theta^{\sharp}_{\mathrm{coupled},\bullet} >$$

> computed from the full leakage block, including possible parametric resonance between channels.

Define the unreduced-planar recapture margins

$$\mathfrak{M}^{\sharp}_{\mathrm{in}} \equiv \underline{A}^{\sharp}_{p,\mathrm{in}} - \overline{A}^{\sharp}_{s,\mathrm{loc,in}} - \overline{A}^{\sharp}_{s,\mathrm{deep,in}} - \sum_{\alpha=1}^{Q^{\sharp}_{\mathrm{esc}}} \Theta^{\sharp}_{\alpha,\mathrm{in}}$$

$$\mathfrak{M}^{\sharp}_{\mathrm{out}} \equiv \underline{A}^{\sharp}_{p,\mathrm{out}} - \overline{A}^{\sharp}_{s,\mathrm{loc,out}} - \overline{A}^{\sharp}_{s,\mathrm{deep,out}} - \sum_{\alpha=1}^{Q^{\sharp}_{\mathrm{esc}}} \Theta^{\sharp}_{\alpha,\mathrm{out}}$$

These displayed margins are the simultaneous-ceiling form. In the non-resonant averaged form, or in any case where the channels are only bounded after coupling, replace the sums by the certified coupled budgets

$$\Theta^{\sharp}_{\mathrm{coupled,in}}, \qquad \Theta^{\sharp}_{\mathrm{coupled,out}}$$

Whenever these are positive, the primary escape observable obeys the comparison inequalities

$$\ddot{\rho}^\sharp(t) \le -\mathfrak{M}^\sharp_{\text{in}} < 0 \qquad \text{on } W^\sharp_{\text{in,turn}}$$

and

$$\ddot{\rho}^\sharp(t) \le -\mathfrak{M}^\sharp_{\text{out}} < 0 \qquad \text{on } W^\sharp_{\text{out,turn}}$$

> **Target Theorem (Unreduced-planar multi-channel recapture criterion).**
> Assume the windowwise multi-channel quotient-force split proposition and define the initial outward escape speeds

$$> V^\sharp_{\text{in},0} >\equiv> \dot{\rho}^\sharp(t^\sharp_{\text{in}}), > \qquad > V^\sharp_{\text{out},0} >\equiv> \dot{\rho}^\sharp(t^\sharp_{\text{out}}), >$$

> at the entrance times of

$$> W^\sharp_{\text{in,turn}} > \qquad > \text{and} > \qquad > W^\sharp_{\text{out,turn}}. >$$

> If

$$> \mathfrak{M}^\sharp_{\text{in}} > 0, > \qquad > |W^\sharp_{\text{in,turn}}| >\ge> \frac{V^\sharp_{\text{in},0}}{\mathfrak{M}^\sharp_{\text{in}}}, >$$

> then the primary escape speed reaches zero on or before the end of

$$> W^\sharp_{\text{in,turn}}. >$$

> If, in addition,

$$> \mathfrak{M}^\sharp_{\text{out}} > 0, > \qquad > |W^\sharp_{\text{out,turn}}| >\ge> \frac{V^\sharp_{\text{out},0}}{\mathfrak{M}^\sharp_{\text{out}}}, >$$

> then the late-turn outward escape speed also reaches zero on or before the end of

$$> W^\sharp_{\text{out,turn}}. >$$

> Consequently the candidate excursion makes both the post-crossing recapture turn and the final late turn inside the controlled unreduced-planar windows.

This is the exact place where the unreduced planar bridge separates from the reduced planar one. The reduced planar note had one leakage ceiling,

$$\rho\dot{\vartheta}^2$$

The unreduced planar bridge requires a finite leakage budget

$$\sum_{\alpha=1}^{Q^\sharp_{\text{esc}}} \Theta^\sharp_{\alpha,\bullet}$$

because coercivity can now be lost through several quotient modes rather than through one scalar angular channel.
That sum is proof-valid only when the preceding resonance-control hypothesis has been discharged. Otherwise the recapture criterion must consume the coupled leakage budget produced by the full leakage monodromy block.

The dependency chain should be stated explicitly here:

1. the gauge-fixed section package supplies the chart in which

$$\rho^\sharp$$

and the leakage channels are defined;
2. the finite branch graph supplies the local active-branch ceilings that define

$$\overline{A}^\sharp_{s,\text{loc,in}} \quad \text{and} \quad \overline{A}^\sharp_{s,\text{loc,out}}$$

3. the provenance-or-exclusion package supplies

$$\overline{A}^\sharp_{s,\text{deep,in}} \quad \text{and} \quad \overline{A}^\sharp_{s,\text{deep,out}}$$

4. the remaining new burden is to prove that all quotient leakage channels admit finite ceilings

$$\Theta^\sharp_{\alpha,\mathrm{in}}, \qquad \Theta^\sharp_{\alpha,\mathrm{out}}$$

on the same windows, together with either simultaneous pointwise validity or the non-resonant leakage-block certificate.

Why the unreduced planar object is still called a breather

The word **breather** does not mean that the motion remains one-dimensional. It means that one controlled size or escape observable repeatedly contracts and expands while the full delayed trajectory stays bounded and returns to a section. In the reduced planar bridge that observable was the relative radius

$$\rho(t)$$

In the unreduced planar bridge it is the quotient escape coordinate

$$\rho^\sharp(t)$$

So the intended dynamical picture is still a breathing one:

- an inbound contraction toward the near-crossing regime;
- entry into the self-hit-capable or even super-field-speed regime, where active delayed self branches and caustic tubes can appear;
- a post-crossing outward excursion;
- a controlled recapture turn;
- a later outward excursion;
- and a final late turn that sends the trajectory back toward the return section.

What is being claimed is therefore a **turnaround**, not a hard elastic bounce. The theorem targets do not say that the architrinos strike a wall and reverse instantaneously. They say that after the super-field-speed episode has activated the relevant delayed geometry, the outward escape observable still reaches zero on controlled windows and changes sign again under the net inward delayed force balance. In that sense the configuration “breathes”: it expands, fails to escape, contracts again, and repeats.

Super-field-speed motion is therefore not the endpoint of the story. It is the regime in which self-hit branches are born and the causal geometry becomes singular enough to require caustic control. The breather claim is precisely that these episodes can be integrated through without converting the trajectory into scattering. Instead they feed a later recapture-and-return mechanism.

Unreduced-planar tame-envelope and closure targets

The remaining bridge step is now the same structural one that appeared in the 1D and reduced planar programs: place the entire cycle on one closed convex tame self-map domain.

The unreduced-planar envelope should be phrased in terms of the constants already produced by the local packages:

$$R^\sharp_{\max}, \quad U^\sharp_{\max}, \quad A^\sharp_{\max}, \quad T^\sharp_{\mathrm{cyc},\max}, \quad N^\sharp_{\mathrm{br}}, \quad \delta^\sharp_{\mathrm{sep}}, \quad \nu^\sharp_J, \quad \nu^\sharp_{J,\mathrm{dp}}, \quad P^\sharp_{\mathrm{prov}}, \quad Q^\sharp_{\mathrm{esc}}, \quad \mathfrak{M}^\sharp_{\mathrm{in}}, \quad \mathfrak{M}^\sharp_{\mathrm{out}}$$

The point is again not to define the tame class by pointwise intersection of every branch-graph or provenance predicate. That would generally fail to preserve convexity. The correct theorem target is one finite certificate whose affine or sampled inequalities define a closed convex subset; the branch graph, provenance count, leakage channels, Jacobian floors, and recapture margins must then be proved from that certificate with uniform slack.

Write

$$\mathcal{C}^\sharp_{\rho_*,\eta} \subseteq \widehat{\Sigma}^{-,\sharp}_{\rho_*}$$

for the unreduced-planar convex section envelope in the gauge-fixed chart and

$$\mathcal{K}^\sharp_{\rho_*,\eta} \subseteq \mathcal{C}^\sharp_{\rho_*,\eta}$$

for the desired closed convex tame envelope.

> **Target Proposition (Closed convex tame envelope in the unreduced planar section).**
> There exists a nonempty closed convex set

$$> \mathcal{K}^\sharp_{\rho_*,\eta} >\subseteq> \mathcal{C}^\sharp_{\rho_*,\eta} >\subseteq> \widehat{\Sigma}^{-,\sharp}_{\rho_*} >$$

> such that:
>
> 1. a nonempty propagated gauge-fixed class

$$> \mathcal{C}^{\sharp,\mathrm{prop}}_{\rho_*,\eta} >$$

> lies in

$$> \mathcal{K}^{\sharp}_{\rho_*,\eta}; >$$

> 2. every

$$> \Psi \in \mathcal{K}^{\sharp}_{\rho_*,\eta} >$$

> admits a unique forward continuation on

$$> [0, T^{\sharp}_{\mathrm{cyc,max}}] >$$

> with uniform quotient-chart position, speed, acceleration, and memory-depth bounds inside the convex envelope;
> 3. along that full cycle, the gauge-fixed section, finite branch graph, deep-past provenance-or-exclusion, and multi-channel recapture packages all hold with the same class constants

$$> N^{\sharp}_{\mathrm{br}}, > \quad > \delta^{\sharp}_{\mathrm{sep}}, > \quad > \nu^{\sharp}_{J}, > \quad > \nu^{\sharp}_{J,\mathrm{dp}}, > \quad > P^{\sharp}_{\mathrm{prov}}, > \quad > Q^{\sharp}_{\mathrm{esc}}, > \quad > \mathfrak{M}^{\sharp}_{\mathrm{in}}, > \quad > \mathfrak{M}^{\sharp}_{\mathrm{out}}; >$$

> 4. the first full-cycle return time

$$> T^{\sharp}(\Psi) >$$

> is uniformly transverse on the raw section, and the gauge-reset return

$$> P^{\sharp}_{\eta}(\Psi) >$$

> satisfies the same gauge-fixed section anchoring that defines

$$> \widehat{\Sigma}^{-,\sharp}_{\rho_*}. >$$

This is the unreduced-planar analogue of the earlier envelope-construction targets. Its role is to put the entire quotient-space cycle geometry on one domain before asking for a self-map theorem.

> **Target Theorem (Unreduced-planar invariant-envelope closure).**
> Assume the closed convex tame envelope proposition and suppose, in addition, that:
>
> 1. the returned gauge-fixed histories satisfy the same convex envelope bounds

$$> \|\Psi(\theta)\|_{(\mathbb{R}^2)^2} \le R^{\sharp}_{\max}, > \qquad > \|\dot{\Psi}(\theta)\|_{(\mathbb{R}^2)^2} \le U^{\sharp}_{\max}, > \qquad > \mathrm{Lip}(\dot{\Psi}) \le A^{\sharp}_{\max}; >$$

> 2. the returned finite certificate implies the same branch-graph bound, separation margin, provenance count, and leakage-channel count

$$> N^{\sharp}_{\mathrm{br}}, > \qquad > \delta^{\sharp}_{\mathrm{sep}}, > \qquad > P^{\sharp}_{\mathrm{prov}}, > \qquad > Q^{\sharp}_{\mathrm{esc}}; >$$

> 3. the returned finite certificate implies the same Jacobian floors

$$> \nu^{\sharp}_{J}, > \qquad > \nu^{\sharp}_{J,\mathrm{dp}}; >$$

> 4. the returned post-crossing and late-turn windows satisfy the same strict recapture margins

$$> \mathfrak{M}^{\sharp}_{\mathrm{in}} > 0, > \qquad > \mathfrak{M}^{\sharp}_{\mathrm{out}} > 0. >$$

> Then

$$> P^{\sharp}_{\eta}\big(> \mathcal{K}^{\sharp}_{\rho_*,\eta} > \big) > \subseteq > \mathcal{K}^{\sharp}_{\rho_*,\eta}. >$$

This is the self-map statement the unreduced-planar bridge is aiming at. It says that after one full physical excursion and one quotient gauge reset, no branch-graph, provenance, leakage, or recapture constant is lost.

> **Target Proposition (Continuity and precompactness of the unreduced-planar return map).**
> On

$$> \mathcal{K}^{\sharp}_{\rho_*,\eta}, >$$

> the map

> $P^\sharp_\eta :\mathcal{K}^\sharp_{\rho_*,\eta} \to \mathcal{K}^\sharp_{\rho_*,\eta}$

> is continuous in the

> $C^1([-h,0];\mathcal{Q}^\sharp_{\mathrm{pl}})$

> topology, and its image is precompact.

The extra burden beyond the reduced planar case is that continuity must now absorb the quotient gauge selector together with the returned branch graph. So the proof target is continuity not only of return time and returned history, but also of the canonical representative and the finite graph data used to define the tame class.

> **Target Theorem (Unreduced-planar Schauder capstone).**
> Assume:
>
> 1. the closed convex tame envelope proposition in the unreduced planar section;
> 2. the unreduced-planar invariant-envelope closure theorem;
> 3. the continuity and precompactness proposition for

> $P^\sharp_\eta$

> on

> $\mathcal{K}^\sharp_{\rho_*,\eta};$

> 4. and the nonempty propagated gauge-fixed class

> $\mathcal{C}^{\sharp,\mathrm{prop}}_{\rho_*,\eta} \neq \varnothing.$

> Then there exists

> $\Psi^*_\eta \in \mathcal{K}^\sharp_{\rho_*,\eta}$

> such that

> $P^\sharp_\eta(\Psi^*_\eta) = \Psi^*_\eta.$

> The corresponding gauge-fixed trajectory is a bounded relative breather modulo the chosen

> $SE(2)$

> gauge. By Proposition `Holonomy reconstruction for quotient fixed points`, it reconstructs to an absolute periodic solution in the fixed Euclidean void only if the full-cycle holonomy satisfies

> $H_{\Psi^*_\eta} = (0,\mathrm{Id}).$

The unreduced-planar bridge can therefore now be read as one explicit ladder:
gauge-fixed section and return map

$$\longrightarrow$$

finite active branch graph

$$\longrightarrow$$

deep-past provenance or exclusion

$$\longrightarrow$$

multi-channel recapture

$$\longrightarrow$$

closed convex tame envelope and self-map

$$\longrightarrow$$

Schauder closure.
This is the precise resumable order of proof burden for the first non-reflection-symmetric planar binary.

If the unreduced-planar bridge fails, the failure should now be recorded in closure-stage terms rather than as a generic expression of difficulty. The meaningful obstruction alternatives are:

1. no gauge-fixed quotient section produces a nonempty convex envelope

$$\mathcal{C}^{\sharp}_{\rho_*,\eta}$$

together with a well-defined return map

$$P^{\sharp}_{\eta}$$

2. every candidate active branch graph loses one of the finite-control bounds

$$N^{\sharp}_{\text{br}} < \infty, \qquad \delta^{\sharp}_{\text{sep}} > 0, \qquad \nu^{\sharp}_{J} > 0$$

3. every candidate deep-past provenance package either forces

$$P^{\sharp}_{\text{prov}} = \infty$$

or loses the deep-past Jacobian floor

$$\nu^{\sharp}_{J,\text{dp}} > 0$$

so the remote self contribution is no longer uniformly controlled;
4. every candidate quotient comparison law either forces

$$Q^{\sharp}_{\text{esc}} = \infty$$

or yields nonpositive recapture margins

$$\mathfrak{M}^{\sharp}_{\text{in}} \le 0 \qquad \text{or} \qquad \mathfrak{M}^{\sharp}_{\text{out}} \le 0$$

so partner attraction cannot dominate self-drive plus the full leakage budget on the controlled windows;
5. or every closed convex tame candidate domain fails one of the closure requirements

$$P^{\sharp}_{\eta}(\mathcal{K}^{\sharp}_{\rho_*,\eta}) \subseteq \mathcal{K}^{\sharp}_{\rho_*,\eta}$$

continuity, or precompactness.

Any one of these is the honest statement that the frozen 1D scaffold and the reduced planar bridge do not yet transport to the first non-reflection-symmetric planar binary without an additional invariant, symmetry, or coercive mechanism.

Beyond the Unreduced Planar Binary

The next genuinely many-body regime does not require an immediate jump to full spatial generality. It already appears when one allows a third active body while remaining in a single plane. So the first honest step beyond the present bridge is not yet the unrestricted master equation on arbitrary configurations. It is the first planar many-body regime in which binary-relative coordinates are no longer sufficient.

That step introduces new burdens that are absent even from the unreduced planar binary.

First, the quotient section and gauge problem becomes higher-rank. In the binary bridge, one present chord is enough to anchor orientation after translation is removed. In a planar many-body regime there is no single distinguished chord that canonically fixes the whole shape. The return section would have to be posed on a higher-dimensional quotient shape space

$$\mathcal{Q}^{\text{mb}}_{\text{pl}} \equiv \mathcal{Q}_{N,\text{pl}}/SE(2)$$

and the gauge selector would have to control stabilizers, near-collinear degeneracies, and possible relabeling ambiguities among bodies that play symmetric roles.

Second, the delayed-root combinatorics ceases to be a finite graph over one receiver-source family. Each receiver now sees several active source families at once, and several branch events can interact through shared bodies. The natural replacement object is therefore a finite active delay hypergraph

$$\mathscr{H}^{\text{mb}}_{\text{br}}$$

not merely a graph

$$\mathscr{G}^{\sharp}_{\mathrm{br}}$$

Its vertices would encode receiver index, source index, sector data, and window data, while its higher-order incidences would record coupled fold events or exchange events that cannot be represented by pairwise edges alone.

Third, provenance control becomes cluster-valued rather than branch-valued. In the unreduced planar binary, a remote late-turn root is forced into one finite provenance class of the earlier branch graph. In a many-body regime, a late remote contribution may pass through several interacting source families before it is geometrically understood. The next theorem target would therefore need a finite ancestry complex for delayed branches, together with an exclusion principle preventing indefinite migration through ever-new source clusters.

Fourth, the recapture problem ceases to be a one-observable comparison. The binary bridge can still organize the turn mechanism around one primary escape coordinate

$$\rho^{\sharp}$$

and a finite leakage budget. In a many-body regime there are several genuine escape channels: pair separation, cluster separation, shear between subclusters, and exchange of the body that is farthest from the current core. The next honest theorem target would therefore require a finite family of escape observables

$$\rho^{\mathrm{mb}}_1, \ldots, \rho^{\mathrm{mb}}_{K_{\mathrm{esc}}}$$

and a coercive comparison law that dominates all open scattering channels at once, not only one preferred outward mode.

Fifth, the tame-envelope problem becomes atlas-level. It is no longer enough to preserve one branch graph, one provenance count, and one leakage count on one fixed quotient chart. The many-body closure problem must preserve the active delay hypergraph, the cluster ancestry data, the recapture margins for all escape observables, and the choice of gauge representative on one closed convex tame self-map domain, or else state precisely why no such single chart exists.

For that reason, the present chapter should still be read as a binary bridge note, even after the unreduced planar extension. Only after the many-body section, hypergraph, ancestry, multi-observable recapture, and closure targets are isolated in the same theorem-level way would it be honest to say that the breather program has moved from the binary bridge toward the full master-equation setting.

First planar three-body bridge regime

The next concrete target should now be fixed more narrowly than the generic

$$N\text{-body}$$

language above. The first honest many-body bridge is best posed as a planar three-body model with one distinguished opposite-sign body and a same-sign outer pair. Up to global sign reversal, write

$$q_1 = +\epsilon, \qquad q_2 = -\epsilon, \qquad q_3 = +\epsilon$$

with all three trajectories constrained to one plane.

This seed is not polarity-neutral:

$$q_1 + q_2 + q_3 = +\epsilon$$

It should therefore be read as a local nonneutral three-body subsystem, or as a compensated subsystem inside a larger neutral assembly whose compensating polarity remains outside the reduced bridge model. A globally neutral many-body theorem would need a different seed, for example a neutral four-body packet, and should not be inferred from the present

$$(+,-,+)$$

three-body bridge.

This compensation is a dynamical hypothesis, not only an ontological interpretation. A nonneutral local subsystem may carry a long-wavelength radiation or reaction channel that slowly drains or injects energy relative to the local bridge variables. The three-body bridge must therefore use one of the following two conventions:

1. replace the seed by a globally neutral four-body packet, for example

$$(+,-,+,-)$$

in a quadrupole-like arrangement, before claiming a neutral many-body theorem;
2. keep the local

$$(+,-,+)$$

subsystem but add an external compensation budget

$$E_{\mathrm{ext}}$$

from the surrounding neutral assembly, and subtract that budget from every local recapture margin that could be weakened by uncompensated radiation or far-field reaction.

Under the second convention, every many-body margin below should be read in compensated form, for example

$$\mathfrak{M}^{\mathrm{mb}}_{m,\bullet} \quad \rightsquigarrow \quad \mathfrak{M}^{\mathrm{mb}}_{m,\bullet} - E_{\mathrm{ext}}$$

unless a sharper channel-specific external budget has been certified. Without one of these conventions, the planar three-body bridge is only a local nonneutral transport test and cannot be promoted to a globally neutral breather theorem.

This is the smallest regime in which the binary-relative chart fails for structural rather than cosmetic reasons. It preserves enough symmetry to permit a clean gauge discussion, but it already introduces the genuinely new burdens that the binary bridge cannot see:

- no single present chord canonically fixes orientation;
- each receiver sees more than one source family at once;
- delayed provenance can pass through changing pair or cluster organization;
- and recapture must dominate more than one outward channel.

The theorem objective is not yet a classification of all planar three-body bounded motions. It is the first transport test for the breather architecture itself:

> **Planar-three-body bridge objective.**
> Construct a history-space return map for the compensated local

$$> (+,-,+) >$$

> planar three-body delayed subsystem and isolate a nonempty closed convex tame domain on which that return map is well defined. If this succeeds, the corresponding Schauder capstone becomes the first local many-body breather theorem in the master-equation stack. If it fails, the obstruction should be written down in section/gauge, hypergraph, ancestry, recapture, or atlas-closure terms. A globally neutral theorem is a later four-body-or-larger closure problem.

The package ladder below records the dependency map for a later many-body proof attempt. It should not be treated as active proof work until the collinear certificate chain has closed.

Seed-side leading-order geometry for the planar three-body bridge

Before tightening the full delay geometry, one should record one explicit planar seed family for which the leading partner attraction is already stronger than the obvious geometric leakage. That is the many-body analogue of the frozen 1D affine seed packet: it does not prove the breather by itself, but it shows that the first recapture margins are not vacuous.

Work in the center-of-mass chart and write

$$\mathbf{x}_1 = \frac{1}{2}\mathbf{a} - \frac{1}{3}\mathbf{b}, \qquad \mathbf{x}_2 = \frac{2}{3}\mathbf{b}, \qquad \mathbf{x}_3 = -\frac{1}{2}\mathbf{a} - \frac{1}{3}\mathbf{b}$$

These factors are the standard equal-mass Jacobi coordinates for the chosen labels:

$$\mathbf{a} = \mathbf{x}_1 - \mathbf{x}_3, \qquad \mathbf{b} = \mathbf{x}_2 - \frac{\mathbf{x}_1 + \mathbf{x}_3}{2}, \qquad \mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3 = 0$$

Thus the factors

$$-\frac{1}{3}\mathbf{b}$$

in

$$\mathbf{x}_1 \quad \text{and} \quad \mathbf{x}_3$$

are intentional: the same-sign outer pair lies on the base line with midpoint

$$-\frac{1}{3}\mathbf{b}$$

the opposite-sign body lies at

$$\frac{2}{3}\mathbf{b}$$

and the center of mass remains at the origin on the axis of symmetry.

Choose seed parameters

$$A_* > 0, \qquad B_* > 0, \qquad u_{a,\text{seed}} > 0, \qquad u_{b,\text{seed}} > 0, \qquad V_{x,\text{seed}} > 0$$

and define the Jacobi seed history by

$$\mathbf{a}_{\text{seed}}(\theta) \equiv \big(A_* - u_{a,\text{seed}}\theta\big)\mathbf{e}_1$$

$$\mathbf{b}_{\text{seed}}(\theta) \equiv -V_{x,\text{seed}}\theta\,\mathbf{e}_1 + \big(B_* - u_{b,\text{seed}}\theta\big)\mathbf{e}_2, \qquad \theta \in [-h, 0]$$

Since

$$\theta \le 0$$

the same-sign pair is farther apart in the recent past and the opposite-sign body sits higher in the recent past. At the section time

$$\theta = 0$$

one has

$$\mathbf{a}_{\text{seed}}(0) = A_*\mathbf{e}_1, \qquad \mathbf{b}_{\text{seed}}(0) = B_*\mathbf{e}_2, \qquad \dot{\mathbf{b}}_{\text{seed}}(0) = -V_{x,\text{seed}}\mathbf{e}_1 - u_{b,\text{seed}}\mathbf{e}_2$$

so the gauge conditions

$$\mathbf{e}_2 \cdot \mathbf{b}(0) > 0, \qquad \mathbf{e}_1 \cdot \dot{\mathbf{b}}(0) < 0$$

are automatic.

The associated seed body velocities are

$$\dot{\mathbf{x}}_{1,\text{seed}} = -\left(\frac{u_{a,\text{seed}}}{2} - \frac{V_{x,\text{seed}}}{3}\right)\mathbf{e}_1 + \frac{u_{b,\text{seed}}}{3}\mathbf{e}_2$$

$$\dot{\mathbf{x}}_{2,\text{seed}} = -\frac{2V_{x,\text{seed}}}{3}\mathbf{e}_1 - \frac{2u_{b,\text{seed}}}{3}\mathbf{e}_2$$

$$\dot{\mathbf{x}}_{3,\text{seed}} = \left(\frac{u_{a,\text{seed}}}{2} + \frac{V_{x,\text{seed}}}{3}\right)\mathbf{e}_1 + \frac{u_{b,\text{seed}}}{3}\mathbf{e}_2$$

Hence one sufficient sub-field-speed condition is

$$U^{\text{mb}}_{\text{seed}} \equiv \max\left\{\left\|\dot{\mathbf{x}}_{1,\text{seed}}\right\|, \left\|\dot{\mathbf{x}}_{2,\text{seed}}\right\|, \left\|\dot{\mathbf{x}}_{3,\text{seed}}\right\|\right\} < c_f$$

At the section time define the common partner distance

$$R_{\text{pair}} \equiv \left\|\frac{1}{2}A_*\mathbf{e}_1 - B_*\mathbf{e}_2\right\| = \sqrt{\frac{A_*^2}{4} + B_*^2}$$

Then the instantaneous Coulomb-like partner projections satisfy

$$\ddot{\mathbf{a}}^{\text{part}}_{\text{seed}} = -\frac{\kappa\epsilon^2}{R^3_{\text{pair}}}\,\mathbf{a}_{\text{seed}}(0)$$

while the direct same-sign pair repulsion contributes

$$\ddot{\mathbf{a}}^{\text{same}}_{\text{seed}} = \frac{2\kappa\epsilon^2}{A_*^3}\,\mathbf{a}_{\text{seed}}(0)$$

Therefore the leading pair-axis seed forcing is

$$\Lambda^{\text{mb}}_{1,\text{seed}} \equiv -\hat{\mathbf{a}}_{\text{seed}}(0) \cdot \left(\ddot{\mathbf{a}}^{\text{part}}_{\text{seed}} + \ddot{\mathbf{a}}^{\text{same}}_{\text{seed}}\right) = \kappa\epsilon^2\left(\frac{A_*}{R^3_{\text{pair}}} - \frac{2}{A_*^2}\right)$$

Likewise, the leading midpoint-axis attraction from the opposite-sign body against the outer-pair midpoint is

$$\ddot{\mathbf{b}}_{\text{seed}}^{\text{part}} = -\frac{3\kappa\epsilon^2}{R_{\text{pair}}^3}\,\mathbf{b}_{\text{seed}}(0)$$

so

$$\Lambda_{2,\text{seed}}^{\text{mb}} \equiv -\hat{\mathbf{b}}_{\text{seed}}(0)\cdot\ddot{\mathbf{b}}_{\text{seed}}^{\text{part}} = \frac{3\kappa\epsilon^2 B_*}{R_{\text{pair}}^3}$$

At this symmetric seed, the direct same-sign pair force does not contribute to

$$\mathbf{b}$$

at leading order.

The leading geometric leakage ceilings at the section are equally explicit:

$$L_{1,\text{geom,seed}}^{\text{mb}} = 0$$

because

$$\dot{\mathbf{a}}_{\text{seed}}(0) = -u_{a,\text{seed}}\mathbf{e}_1$$

is parallel to

$$\mathbf{a}_{\text{seed}}(0)$$

and

$$L_{2,\text{geom,seed}}^{\text{mb}} = \frac{V_{x,\text{seed}}^2}{B_*}$$

because

$$\hat{\mathbf{b}}_{\text{seed}}(0) = \mathbf{e}_2$$

while the transverse component of

$$\dot{\mathbf{b}}_{\text{seed}}(0)$$

is exactly

$$V_{x,\text{seed}}\mathbf{e}_1$$

These formulas already show the right geometric sweet spot: choose

$$B_*$$

not too large compared with

$$A_*$$

and choose

$$V_{x,\text{seed}}$$

small relative to

$$\kappa\epsilon^2 B_*/R_{\text{pair}}^3$$

Then the first two inward channels are positive before any refined delay bookkeeping is invoked.

> **Target Proposition (Explicit symmetric planar-three-body seed packet with positive leading principal margins).**
> Fix

$$> A_* > 0, > \qquad > B_* > 0 >$$

> such that

$$> B_*^2 ><> \left(> 2^{-2/3} - \frac{1}{4} >\right) > A_*^2, > \qquad > B_* ><> \frac{\sqrt{3}}{2} A_*. >$$

Then:

1. the pair-axis seed attraction is strictly inward,

$$> \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > 0; >$$

2. the midpoint-axis seed attraction is strictly inward,

$$> \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} > 0; >$$

3. the role gap at the seed is strictly positive,

$$> \|\mathbf{x}_{1,\mathrm{seed}}(0)\| >=> \|\mathbf{x}_{3,\mathrm{seed}}(0)\| >>> \|\mathbf{x}_{2,\mathrm{seed}}(0)\|; >$$

4. and if

$$> 0 < V_{x,\mathrm{seed}} ><> \sqrt{> \frac{3\kappa\epsilon^2 B_*^2}{2R_{\mathrm{pair}}^3} >}, >$$

then the leading midpoint leakage satisfies

$$> L_{2,\mathrm{geom,seed}}^{\mathrm{mb}} ><> \frac{1}{2} \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}}. >$$

Consequently the first two principal margins and the role-gap floor are nonvacuous on a nonempty open neighborhood of this seed, provided the delayed fold and deep-past ceilings are chosen smaller than the remaining slack.

Proof.
The inequality

$$> \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > 0 >$$

is equivalent to

$$> \frac{A_*}{R_{\mathrm{pair}}^3} >>> \frac{2}{A_*^2}, >$$

which is

$$> R_{\mathrm{pair}}^3 < \frac{A_*^3}{2} >$$

and therefore

$$> R_{\mathrm{pair}}^2 < 2^{-2/3} A_*^2. >$$

Since

$$> R_{\mathrm{pair}}^2 = \frac{A_*^2}{4} + B_*^2, >$$

the stated bound on

$$> B_* >$$

is exactly the desired condition.

The positivity of

$$> \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} >$$

is immediate from

$$> B_* > 0. >$$

The role-gap inequality becomes

$$> \frac{A_*^2}{4} + \frac{B_*^2}{9} >>> \frac{4B_*^2}{9}, >$$

> which is equivalent to

$$> B_* < \frac{\sqrt{3}}{2} A_*. >$$

> Finally,

$$> L^{\mathrm{mb}}_{2,\mathrm{geom,seed}} >=> \frac{V^2_{x,\mathrm{seed}}}{B_*} ><> \frac{1}{2} \cdot > \frac{3\kappa\epsilon^2 B_*}{R^3_{\mathrm{pair}}} >=> \frac{1}{2} \Lambda^{\mathrm{mb}}_{2,\mathrm{seed}} >$$

> is exactly the displayed upper bound on

$$> V_{x,\mathrm{seed}}. >$$

> These inequalities are strict, so they persist on one nonempty

$$> C^1 >$$

> neighborhood of the seed history.

> **Target Corollary (Seed-neighborhood realization of the leading planar-three-body principal margins).**
> Under the hypotheses of the explicit symmetric seed proposition, there exists

$$> \varepsilon^{\mathrm{mb}}_{\mathrm{seed}} > 0 >$$

> and a nonempty

$$> C^1 >$$

> neighborhood

$$> \mathcal{C}^{\mathrm{mb,seed}}_{A_*,B_*,\eta} >$$

> of

$$> \Phi^{\mathrm{mb}}_{\mathrm{seed}} >$$

> inside the gauge-fixed section such that every

$$> \Phi \in \mathcal{C}^{\mathrm{mb,seed}}_{A_*,B_*,\eta} >$$

> retains:
>
> 1. the same-sign pair axis in one fixed narrow cone around

$$> \mathbf{e}_1; >$$

> 2. the opposite-sign midpoint axis in one fixed upper-half-plane cone;
> 3. a strictly positive role gap;
> 4. and strictly positive leading-order seed-side margins for

$$> \rho_1^{\mathrm{mb}} > \quad > \text{and} > \quad > \rho_2^{\mathrm{mb}}. >$$

This is the many-body analogue of the old seed-neighborhood realization step in the frozen 1D chapter. Its role is only to certify that the principal inward hierarchy is not attached to one isolated affine history, but persists on one genuine local seed packet from which the later delayed and hypergraph packages may start.

The remaining seed-side burden is to pass from these Coulomb-like proxy margins to the true delayed branch-sum law. On the affine seed this should be a perturbative step, because every emitter speed is strictly sub-field-speed and the recent history is exactly linear.

For the affine seed and its small

$$C^1$$

thickenings, denote by

$$r_{ij}^{\text{inst}}(t) \equiv \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|$$

the instantaneous pair distances and by

$$s_{ij}(t)$$

the exact causal-delay partner/source times on the preserved seed-side branch families, whenever those branches exist on a short controlled window.

> **Lemma (Delayed seed-margin persistence on the symmetric planar seed packet).**
> Assume the explicit symmetric planar-three-body seed proposition and fix one seed-side neighborhood

$$> \mathcal{C}^{\text{mb,seed}}_{A_*,B_*,\eta} >$$

> with

$$> U^{\text{mb}}_{\text{seed}} < c_f. >$$

> After replacing this neighborhood by a smaller nonempty seed packet if necessary, there exist:
>
> 1. a short controlled seed window

$$> [0, \tau^{\text{mb}}_{\text{seed}}], >$$

> 2. a finite branch-delay ceiling

$$> \Delta^{\text{mb}}_{\text{seed}}, >$$

> 3. local velocity and acceleration tolerances

$$> \varepsilon_V, > \qquad > \varepsilon_A, >$$

> satisfying

$$> \varepsilon_V + \varepsilon_A \Delta^{\text{mb}}_{\text{seed}} > \leq > \min\left\{ > U^{\text{mb}}_{\text{seed}}, > \frac{c_f - U^{\text{mb}}_{\text{seed}}}{2} > \right\}, >$$

> 4. a finite list of seed-side branch families

$$> \mathscr{B}^{\text{mb}}_{\text{seed}}, >$$

> 5. and constants

$$> C^{\text{mb}}_{\text{delay}}, > \qquad > C^{\text{mb}}_{J}, > \qquad > C^{\text{mb}}_{\Lambda,1}, > \qquad > C^{\text{mb}}_{\Lambda,2} >$$

> depending only on the seed parameters and the local seed neighborhood,
>
> such that every history

$$> \Phi \in \mathcal{C}^{\text{mb,seed}}_{A_*,B_*,\eta} >$$

> satisfies, on that window and on the listed source intervals:
>
> 1. each active seed-side branch in

$$> \mathscr{B}^{\text{mb}}_{\text{seed}} >$$

> is unique and simple;
> 2. the exact causal-delay distances obey

$$> \left| > r_{ij}(t; s_{ij}(t)) > - > r_{ij}^{\text{inst}}(t) > \right| > \leq > C^{\text{mb}}_{\text{delay}} > \frac{U^{\text{mb}}_{\text{seed}}}{c_f} > r_{ij}^{\text{inst}}(t); >$$

> 3. the exact causal Jacobians obey

$$> \left| > J_{ij}(t; s_{ij}(t)) - 1 > \right| > \leq > C^{\text{mb}}_{J} > \frac{U^{\text{mb}}_{\text{seed}}}{c_f}; >$$

> 4. the exact delayed principal inward terms satisfy

$$> |> \Lambda_1^{\mathrm{mb}}(t) - \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} >| >\le> C_{\Lambda,1}^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}, >$$

$$> |> \Lambda_2^{\mathrm{mb}}(t) - \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} >| >\le> C_{\Lambda,2}^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}; >$$

5. and therefore, if

$$> \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} >>> 2C_{\Lambda,1}^{\mathrm{mb}} \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}, > \qquad > \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} >>> 2C_{\Lambda,2}^{\mathrm{mb}} \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}, >$$

then the true delayed inward terms satisfy the uniform seed-side bounds

$$> \Lambda_1^{\mathrm{mb}}(t) \ge \frac{1}{2}\Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > 0, > \qquad > \Lambda_2^{\mathrm{mb}}(t) \ge \frac{1}{2}\Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} > 0 >$$

on

$$> [0, \tau_{\mathrm{seed}}^{\mathrm{mb}}]. >$$

Proof.
Work first on the affine seed history. The seed-side branch list

$$> \mathscr{B}_{\mathrm{seed}}^{\mathrm{mb}} >$$

is chosen to be precisely the finite collection of source-receiver branches whose Coulomb-like contributions form

$$> \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > \qquad > \text{and} > \qquad > \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}}. >$$

For one such branch write

$$> \beta = (i,j), > \qquad > \Delta t_\beta(t) = t - s_\beta(t), >$$

and set

$$> g_\beta(t;s;\Phi) >=> \|\mathbf{x}_i(t;\Phi) - \mathbf{x}_j(s;\Phi)\| - c_f(t-s). >$$

On the affine seed, the source derivative is

$$> \partial_s g_\beta(t;s;\Phi_{\mathrm{seed}}) >=> c_f - \dot{\mathbf{x}}_{j,\mathrm{seed}} \cdot > \hat{\mathbf{r}}_\beta(t;s;\Phi_{\mathrm{seed}}) >\ge> c_f - U_{\mathrm{seed}}^{\mathrm{mb}} >> 0. >$$

Hence every seed branch is simple. Since the seed branch set is finite and the seed roots stay in compact source subintervals inside

$$> [-h, 0], >$$

the implicit-function theorem gives, after reducing

$$> \tau_{\mathrm{seed}}^{\mathrm{mb}} >$$

if necessary, one source-time graph

$$> s_{\beta,\mathrm{seed}}(t) >$$

for each

$$> \beta \in \mathscr{B}_{\mathrm{seed}}^{\mathrm{mb}} >$$

on

$$> [0, \tau_{\mathrm{seed}}^{\mathrm{mb}}]. >$$

Define

$$> \Delta_{\mathrm{seed}}^{\mathrm{mb}} >$$

to dominate both the listed seed branch delays and the receiver-window length on this compact set.
Shrink the seed packet so that these roots remain inside the same source intervals and so that, on those intervals,

$$> \|\dot{\mathbf{x}}_k(0;\Phi) - \dot{\mathbf{x}}_{k,\mathrm{seed}}(0)\| >\le> \varepsilon_V, > \qquad > \|\ddot{\mathbf{x}}_k(\theta;\Phi)\| >\le> \varepsilon_A >$$

> for every body

$$> k >$$

> and for almost every source time

$$> \theta. >$$

> The local Lipschitz-velocity bound gives the uniform speed ceiling

$$> \|\dot{\mathbf{x}}_k(\theta;\Phi)\| >\le> U^{\mathrm{mb}}_{\mathrm{seed}} >+> \varepsilon_V >+> \varepsilon_A \Delta^{\mathrm{mb}}_{\mathrm{seed}} >\le> \frac{U^{\mathrm{mb}}_{\mathrm{seed}} + c_f}{2} >< c_f, >$$

> and it is also bounded by

$$> 2U^{\mathrm{mb}}_{\mathrm{seed}} >$$

> for perturbative estimates in powers of

$$> U^{\mathrm{mb}}_{\mathrm{seed}}/c_f. >$$

> Thus

$$> \partial_s g_\beta(t;s;\Phi) >\ge> c_f - > \left(> U^{\mathrm{mb}}_{\mathrm{seed}} >+> \varepsilon_V >+> \varepsilon_A \Delta^{\mathrm{mb}}_{\mathrm{seed}} >\right) >\equiv> \nu^{\mathrm{mb}}_{\mathrm{seed}} >> 0. >$$

> The defect

$$> g_\beta(t;s;\Phi) >$$

> is therefore strictly increasing in source time on each listed source interval. The seed root persists by the implicit-function theorem, and strict monotonicity excludes any second root in the same seed-side interval. This proves uniqueness and simplicity for every branch in

$$> \mathscr{B}^{\mathrm{mb}}_{\mathrm{seed}}. >$$

> The causal-delay length estimate follows from the root identity

$$> c_f \Delta t_\beta(t) >=> r_\beta(t; s_\beta(t)) >$$

> and the source displacement formula

$$> \mathbf{x}_j(t) - \mathbf{x}_j(s_\beta(t)) >=> \dot{\mathbf{x}}_{j,\mathrm{seed}}(0)\Delta t_\beta(t) >+> O\left(> \varepsilon_V \Delta t_\beta(t) >+> \varepsilon_A \Delta t_\beta(t)^2 >\right). >$$

> Equivalently, with

$$> U_* >\equiv> U^{\mathrm{mb}}_{\mathrm{seed}} >+> \varepsilon_V >+> \varepsilon_A \Delta^{\mathrm{mb}}_{\mathrm{seed}}, >$$

> one has

$$> \left\|> \mathbf{x}_j(t) - \mathbf{x}_j(s_\beta(t)) >\right\| >\le> U_* \Delta t_\beta(t). >$$

> By the reverse triangle inequality,

$$> \left|> r_\beta(t; s_\beta(t)) > - > r^{\mathrm{inst}}_\beta(t) >\right| >\le> U_* \Delta t_\beta(t) >=> \frac{U_*}{c_f} > r_\beta(t; s_\beta(t)). >$$

> Since

$$> r_\beta(t; s_\beta(t)) >\le> r^{\mathrm{inst}}_\beta(t) >+> \frac{U_*}{c_f} > r_\beta(t; s_\beta(t)), >$$

> and

$$> U_* < c_f, >$$

> it follows that

$$> r_\beta(t; s_\beta(t)) >\le> \frac{1}{1 - U_*/c_f} > r^{\mathrm{inst}}_\beta(t). >$$

Combining the two inequalities gives

$$> \left|> r_\beta(t;s_\beta(t)) > - > r_\beta^{\mathrm{inst}}(t) >\right| >\le> \frac{U_*/c_f}{1-U_*/c_f} > r_\beta^{\mathrm{inst}}(t). >$$

The shrinkage condition

$$> \varepsilon_V + \varepsilon_A \Delta_{\mathrm{seed}}^{\mathrm{mb}} >\le> \min\left\{> U_{\mathrm{seed}}^{\mathrm{mb}}, > \frac{c_f - U_{\mathrm{seed}}^{\mathrm{mb}}}{2} >\right\} >$$

and the fixed strict gap

$$> U_{\mathrm{seed}}^{\mathrm{mb}} < c_f >$$

absorb the factor on the right into the advertised constant

$$> C_{\mathrm{delay}}^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}. >$$

The same speed ceiling gives the Jacobian estimate directly. For every listed branch,

$$> J_\beta(t;s_\beta(t)) >=> 1 - \frac{\dot{\mathbf{x}}_j(s_\beta(t)) \cdot > \hat{\mathbf{r}}_\beta(t;s_\beta(t))}{c_f}, >$$

so

$$> \bigl|J_\beta(t;s_\beta(t)) - 1\bigr| >\le> \frac{U_*}{c_f} >\le> C_J^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}. >$$

In particular

$$> J_\beta(t;s_\beta(t)) >\ge> 1 - \frac{U_*}{c_f} >=> \frac{\nu_{\mathrm{seed}}^{\mathrm{mb}}}{c_f} >> 0, >$$

so the reciprocal Jacobian remains uniformly bounded on the seed packet.

It remains to compare the projected branch sums with the seed projections. On the compact seed-side set just constructed, all instantaneous distances are bounded below by one positive number

$$> d_{\mathrm{seed}}^{\mathrm{mb}} > 0, >$$

all causal Jacobians are bounded below by

$$> \nu_{\mathrm{seed}}^{\mathrm{mb}}/c_f, >$$

and the branch list is finite. The dual-mollified branch contribution is therefore a smooth function of

$$> \mathbf{r}_\beta, > \qquad > J_\beta^{-1}, > \qquad > \hat{\mathbf{a}}, > \qquad > \hat{\mathbf{b}}, >$$

on this compact set. The causal-delay estimate controls the difference between delayed and instantaneous chords. The present-time drift over the short seed window satisfies

$$> \|\mathbf{x}_k(t) - \mathbf{x}_{k,\mathrm{seed}}(0)\| >\le> U_* \tau_{\mathrm{seed}}^{\mathrm{mb}}, >$$

and the window was chosen so that

$$> \tau_{\mathrm{seed}}^{\mathrm{mb}} >\le> d_{\mathrm{seed}}^{\mathrm{mb}}/c_f. >$$

Thus the present-time axis drift and the causal-delay chord drift are both

$$> O\bigl(U_{\mathrm{seed}}^{\mathrm{mb}}/c_f\bigr) >$$

relative to the seed distances. The mean-value theorem on the compact branch-geometry set gives constants

$$> C_{\Lambda,1}^{\mathrm{mb}}, > \qquad > C_{\Lambda,2}^{\mathrm{mb}} >$$

such that

$$> |> \Lambda_1^{\mathrm{mb}}(t) - \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} >| >\le> C_{\Lambda,1}^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}, >$$

$$> |> \Lambda_2^{\mathrm{mb}}(t) - \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} >| >\le> C_{\Lambda,2}^{\mathrm{mb}} > \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f}. >$$

> Finally, if the two seed margins dominate twice these perturbation ceilings, subtracting the corresponding estimate gives

$$> \Lambda_1^{\mathrm{mb}}(t) \ge \frac{1}{2}\Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > 0, > \qquad > \Lambda_2^{\mathrm{mb}}(t) \ge \frac{1}{2}\Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} > 0 >$$

> throughout

$$> [0, \tau_{\mathrm{seed}}^{\mathrm{mb}}], >$$

> which is the claimed seed-side persistence.

> **Corollary (Delayed realization of the first seed-side principal margins).**
> Under the hypotheses of the delayed seed-margin persistence lemma, if

$$> \Lambda_{1,\mathrm{seed}}^{\mathrm{mb}} > - > 2C_{\Lambda,1}^{\mathrm{mb}} \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f} >>> 0, >$$

$$> \Lambda_{2,\mathrm{seed}}^{\mathrm{mb}} > - > 2C_{\Lambda,2}^{\mathrm{mb}} \frac{U_{\mathrm{seed}}^{\mathrm{mb}}}{c_f} >>> 2L_{2,\mathrm{geom,seed}}^{\mathrm{mb}}, >$$

> and the seed-side fold and deep-past ceilings are smaller than the remaining slack, then the true delayed principal margins for

$$> \rho_1^{\mathrm{mb}} > \qquad > \text{and} > \qquad > \rho_2^{\mathrm{mb}} >$$

> are strictly positive on the first seed-side controlled window.
>
> **Proof.**
> The delayed seed-margin persistence lemma gives positive lower bounds for the two leading seed-side inward terms after subtracting the causal-delay perturbation ceilings. The remaining fold and deep-past contributions enter the principal-margin inequalities only through their stated ceilings. If those ceilings are smaller than the remaining slack in the two displayed inequalities, subtracting them leaves both principal margins strictly positive on the same controlled seed window.

This is the missing bridge from the explicit geometric seed packet to the real delayed master equation. Once this perturbative upgrade is available, the many-body seed no longer lives only in the Coulomb-like proxy model; it enters the exact branch-sum dynamics with quantitative slack.

This is the first genuine many-body seed-side margin calculation in the chapter. It does not yet prove the full delayed recapture theorem, but it identifies one concrete planar geometry in which the desired inward hierarchy is already visible in the bare Jacobi dynamics.

First many-body theorem package: section and gauge fixing on planar shape space

Let

$$\mathbf{x}(t) = \big(\mathbf{x}_1(t), \mathbf{x}_2(t), \mathbf{x}_3(t)\big) \in \mathcal{Q}_{3,\mathrm{pl}} \equiv \left(\mathbb{R}^2\right)^3 \setminus \Delta_{\mathrm{coll}}$$

denote the ordered planar three-body configuration away from collisions, where

$$\Delta_{\mathrm{coll}}$$

is the collision locus. Remove translations by imposing the center-of-mass condition

$$\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3 = 0$$

The remaining quotient by rigid planar rotations defines the reduced shape space

$$\mathcal{Q}_{\mathrm{pl}}^{\mathrm{mb}} \equiv \mathcal{Q}_{3,\mathrm{pl}}/SE(2)$$

Before fixing any gauge, the first analytic object should be the unreduced history space

$$\widetilde{\mathcal{H}}_h^{\mathrm{mb}} \equiv C^1([-h,0]; \mathcal{Q}_{3,\mathrm{pl}})$$

together with the center-of-mass-compatible admissible class

$$\widetilde{\mathcal{H}}_h^{\mathrm{adm,mb}} \equiv \left\{ \widetilde{\Phi} \in \widetilde{\mathcal{H}}_h^{\mathrm{mb}} \;\middle|\; \begin{array}{l} \widetilde{\Phi}_1(\theta) + \widetilde{\Phi}_2(\theta) + \widetilde{\Phi}_3(\theta) = 0 \text{ for all } \theta \in [-h, 0], \\ \min_{i \neq j} \|\widetilde{\Phi}_i(\theta) - \widetilde{\Phi}_j(\theta)\| \geq d_{\min} > 0, \\ \left|\det(\mathbf{a}_{\widetilde{\Phi}}(0), \mathbf{b}_{\widetilde{\Phi}}(0))\right| \geq \delta_{\mathrm{col}} > 0, \\ \|\dot{\widetilde{\Phi}}(\theta)\| \leq U_{\max} \end{array} \right\}$$

This is the space on which local existence, uniqueness, and continuous dependence should first be posed. Only after that should one quotient by rigid rotations and choose a canonical representative.

For the first three-body bridge it is useful to distinguish the opposite-sign body from the same-sign pair, but not to hard-code a full ordering beyond that role split. Accordingly, the gauge selector should be permutation-rigid only with respect to the two same-sign outer bodies.

Choose the present-time Jacobi vectors

$$\mathbf{a}(t) \equiv \mathbf{x}_1(t) - \mathbf{x}_3(t), \qquad \mathbf{b}(t) \equiv \mathbf{x}_2(t) - \frac{\mathbf{x}_1(t) + \mathbf{x}_3(t)}{2}$$

The first vector tracks the same-sign pair separation, while the second tracks the opposite-sign body's displacement from the outer-pair midpoint. Away from the near-collinear set

$$\mathfrak{D}_{\mathrm{col}} \equiv \left\{(\mathbf{a}, \mathbf{b}) \;\middle|\; \left|\det(\mathbf{a}, \mathbf{b})\right| \leq \delta_{\mathrm{col}}\right\}$$

one may impose the canonical gauge

$$\mathbf{a}(0) = A_* \mathbf{e}_1, \qquad \mathbf{e}_2 \cdot \mathbf{b}(0) > 0, \qquad \mathbf{e}_1 \cdot \dot{\mathbf{b}}(0) < 0$$

with fixed

$$A_* > 0$$

The first condition removes rotation by anchoring the same-sign pair on the horizontal axis, the second chooses the upper-half-plane representative for the opposite-sign body, and the third picks the inbound branch across the section.

The resulting gauge-fixed history space is

$$\mathcal{H}_h^{\mathrm{mb}} \equiv C^1([-h, 0]; \mathcal{Q}_{\mathrm{pl}}^{\mathrm{mb}})$$

and the raw inbound section should be written as

$$\Sigma_{A_*}^{-,\mathrm{mb}} \equiv \left\{ \Phi \in \mathcal{H}_h^{\mathrm{mb}} \;\middle|\; \mathbf{a}_\Phi(0) = A_* \mathbf{e}_1, \quad \mathbf{e}_2 \cdot \mathbf{b}_\Phi(0) \geq B_{\min}, \quad \mathbf{e}_1 \cdot \dot{\mathbf{b}}_\Phi(0) \leq -V_{\mathrm{in}} \right\}$$

for fixed positive constants

$$B_{\min}, \qquad V_{\mathrm{in}}$$

This is the many-body analogue of the binary section packages: it removes time translation through a codimension-one section, removes planar rotation through a canonical representative, and keeps the same-sign pair quotient visible without pretending that the full three-body shape can be controlled by one scalar radius.

> **Target Proposition (Unreduced local well-posedness and maximal-time alternative for the planar three-body bridge).**
> For sufficiently small dual-mollification scales

$$> \eta > 0, > \qquad > \epsilon_c > 0, >$$

> and admissible initial histories

$$> \widetilde{\Phi} \in \widetilde{\mathcal{H}}_h^{\mathrm{adm,mb}}, >$$

> the dual-mollified master equation defines a unique local solution

$$> \widetilde{\mathbf{X}}(t) >$$

> depending continuously on the initial history in the

$$> C^1 >$$

> topology.

> Moreover, there exists a maximal continuation time

$$> T_{\max} = T_{\max}(\widetilde{\Phi}) \in (0,\infty] >$$

> such that if

$$> T_{\max} < \infty, >$$

> then at least one explicit control quantity must fail as

$$> t \uparrow T_{\max} :>$$

> 1. a pair distance reaches the collision threshold

$$> d_{\min}; >$$

> 2. the non-collinearity margin

$$> |\det(\mathbf{a}(t), \mathbf{b}(t))| >$$

> falls to

$$> \delta_{\mathrm{col}}; >$$

> 3. a Jacobian floor or branch-separation floor from the later hypergraph package collapses;
> 4. or the required

$$> C^1 >$$

> norm bound for the controlled cycle blows up.

This is the analytic entry point the later theorem ladder should consume. The bridge program should no longer speak of one-cycle continuation without speaking about distance from this explicit bad set.

> **Target Proposition (Gauge selector and sectioned chart stability for the planar three-body bridge).**
> For sufficiently small dual-mollification scales

$$> \eta > 0, > \qquad > \epsilon_c > 0, >$$

> and for section constants

$$> A_*, > \quad > B_{\min}, > \quad > V_{\mathrm{in}}, > \quad > \delta_{\mathrm{col}}, > \quad > d_{\min} >$$

> in an admissible regime, there exists a nonempty open subset

$$> \mathcal{H}^{\mathrm{adm,mb}}_{A_*,\eta} >\subseteq> \Sigma^{-,\mathrm{mb}}_{A_*} >$$

> such that:
>
> 1. the gauge selector is unique and continuous on

$$> \mathcal{H}^{\mathrm{adm,mb}}_{A_*,\eta}; >$$

> 2. every history in that class comes from one unreduced admissible history

$$> \widetilde{\Phi} \in \widetilde{\mathcal{H}}^{\mathrm{adm,mb}}_h >$$

> through the canonical gauge map;
> 3. as long as the unreduced solution stays a positive distance away from the bad set named in the maximal-time alternative, the gauge-fixed representative remains in the same chart and depends continuously on the underlying unreduced solution;
> 4. the first-return time to the same gauge-fixed inbound section is well defined whenever the later recapture packages succeed;
> 5. the only local gauge-level failure alternatives are collision approach, near-collinear gauge degeneracy, section-tangency loss, or role-exchange ambiguity of the same-sign pair.

This is the correct chart-level theorem target because all later many-body bookkeeping depends on having one honest chart in which the delayed branches can even be named, but only after unreduced local well-posedness has already been secured.

Second many-body theorem package: quantitative branch regularity and no-accumulation

Before the active delay hypergraph can be treated as a finite object, one needs an analytic theorem ruling out Zeno-type accumulation of folds, sector crossings, or source-cluster exchanges inside one controlled cycle.

For each ordered receiver-source pair

$$(i,j) \in \{1,2,3\}^2$$

including

$$i = j$$

define the exact delay defect

$$g_{ij}(t;s) \equiv \|\mathbf{x}_i(t) - \mathbf{x}_j(s)\| - c_f(t-s)$$

A fold event is a pair

$$(t, s_*)$$

with

$$g_{ij}(t;s_*) = 0, \qquad \partial_s g_{ij}(t;s_*) = 0$$

Write

$$\mathbf{r}_{ij}(t;s) \equiv \mathbf{x}_i(t) - \mathbf{x}_j(s), \qquad r_{ij}(t;s) \equiv \|\mathbf{r}_{ij}(t;s)\|, \qquad \hat{\mathbf{r}}_{ij}(t;s) \equiv \frac{\mathbf{r}_{ij}(t;s)}{r_{ij}(t;s)}$$

Then

$$\partial_s g_{ij}(t;s) = c_f - \dot{\mathbf{x}}_j(s) \cdot \hat{\mathbf{r}}_{ij}(t;s)$$

$$\partial_t g_{ij}(t;s) = \dot{\mathbf{x}}_i(t) \cdot \hat{\mathbf{r}}_{ij}(t;s) - c_f$$

$$\partial_s^2 g_{ij}(t;s) = -\ddot{\mathbf{x}}_j(s) \cdot \hat{\mathbf{r}}_{ij}(t;s) + \frac{\|\dot{\mathbf{x}}_j(s)\|^2 - \left(\dot{\mathbf{x}}_j(s) \cdot \hat{\mathbf{r}}_{ij}(t;s)\right)^2}{r_{ij}(t;s)}$$

and

$$\partial_s^3 g_{ij}(t;s)$$

is a finite linear combination of terms built from

$$\dddot{\mathbf{x}}_j(s), \qquad \dot{\mathbf{x}}_j(s), \qquad \ddot{\mathbf{x}}_j(s), \qquad r_{ij}(t;s)^{-1}, \qquad r_{ij}(t;s)^{-2}$$

so it is controlled by the same

$$U_{\max}, \quad A_{\max}, \quad L_A, \quad d_{\min}$$

envelope.
The first analytic bridge theorem should impose one quantitative nondegeneracy regime:

- a uniform acceleration bound

$$\|\ddot{\mathbf{x}}_i\| \le A_{\max}$$

- a uniform acceleration-Lipschitz bound

$$\|\ddot{\mathbf{x}}_i(t) - \ddot{\mathbf{x}}_i(t')\| \le L_A |t - t'|$$

- a strict fold curvature floor

$$|\partial_s^2 g_{ij}(t;s_*)| \ge \gamma_{\text{fold}} > 0$$

at every admissible fold;

- and a uniform Jacobian floor

$$|\partial_s g_{ij}(t;s)| \geq \nu_J^{\mathrm{mb}} > 0$$

away from the explicitly listed fold tubes.

Under these hypotheses the delay defects should satisfy one explicit derivative hierarchy on the controlled cycle:

$$|\partial_s g_{ij}(t;s)| \leq C_{1,g}^{\mathrm{mb}}, \qquad |\partial_s^2 g_{ij}(t;s)| \leq C_{2,g}^{\mathrm{mb}}, \qquad |\partial_s^3 g_{ij}(t;s)| \leq C_{3,g}^{\mathrm{mb}}$$

for constants determined only by

$$U_{\max}, \quad A_{\max}, \quad L_A, \quad d_{\min}$$

The point of this hierarchy is that it makes the fold geometry quantitative rather than qualitative: once

$$\partial_s g_{ij} = 0$$

is known to be a genuinely curved zero with bounded third derivative, the root geometry cannot oscillate arbitrarily fast nearby.

> **Lemma (Uniform isolation of admissible folds).**
> Fix one receiver-source family

$$> (i,j) >$$

> and one admissible fold

$$> (t,s_*) >$$

> with

$$> \partial_s g_{ij}(t;s_*) = 0, > \qquad > |\partial_s^2 g_{ij}(t;s_*)| \geq \gamma_{\mathrm{fold}} > 0. >$$

> If

$$> |\partial_s^3 g_{ij}(t;s)| \leq C_{3,g}^{\mathrm{mb}} >$$

> on the corresponding fold tube, then

$$> \partial_s g_{ij}(t;s) \neq 0 >$$

> for every

$$> 0 < |s - s_*| < \delta_{\mathrm{fold}}, > \qquad > \delta_{\mathrm{fold}} > \equiv > \frac{\gamma_{\mathrm{fold}}}{C_{3,g}^{\mathrm{mb}}}. >$$

> In particular, two distinct fold roots for the same receiver-source family in the same receiver-time slice cannot have source-time separation smaller than

$$> \delta_{\mathrm{fold}}. >$$

> **Proof.**
> Taylor-expand

$$> \partial_s g_{ij}(t;s) >$$

> about

$$> s = s_*. >$$

> Since

$$> \partial_s g_{ij}(t;s_*) = 0, >$$

> one has

$$> \partial_s g_{ij}(t;s) > = > \partial_s^2 g_{ij}(t;s_*)(s - s_*) > + > \frac{1}{2}\partial_s^3 g_{ij}(t;\xi_s)(s - s_*)^2 >$$

> for some

$$\xi_s$$

between

$$s$$

and

$$s_*.$$

Therefore

$$|\partial_s g_{ij}(t;s)| \geq |s-s_*| \left(\gamma_{\mathrm{fold}} - \frac{1}{2} C^{\mathrm{mb}}_{3,g} |s-s_*| \right),$$

and the bracket stays strictly positive throughout the stated interval.

Lemma (Uniform receiver-time passage through admissible folds).
Fix one admissible fold

$$(t_*, s_*)$$

and assume, in addition, that the receiver-time derivative is transversal there:

$$|\partial_t g_{ij}(t_*; s_*)| \geq \chi_{\mathrm{fold}} > 0.$$

Let

$$s_{\mathrm{fold}}(t)$$

be the local fold sheet obtained from

$$\partial_s g_{ij}(t; s_{\mathrm{fold}}(t)) = 0$$

by the curvature floor, and define the scalar fold-passage function

$$G_{\mathrm{fold}}(t) \equiv g_{ij}(t; s_{\mathrm{fold}}(t)).$$

If

$$|\ddot{G}_{\mathrm{fold}}(t)| \leq C^{\mathrm{mb}}_{2,tg}$$

on the corresponding controlled fold tube, then the same receiver-source family cannot produce a second fold event with receiver-time separation smaller than

$$\delta_{\mathrm{fold},t} \equiv \frac{\chi_{\mathrm{fold}}}{C^{\mathrm{mb}}_{2,tg}}.$$

Proof.
The curvature floor gives

$$\partial_s^2 g_{ij}(t_*; s_*) \neq 0,$$

so the implicit-function theorem gives the local fold sheet

$$s_{\mathrm{fold}}(t).$$

A receiver-time fold passage occurs exactly at a zero of

$$G_{\mathrm{fold}}(t).$$

At the given fold,

$$\dot{G}_{\mathrm{fold}}(t_*) = \partial_t g_{ij}(t_*; s_*) + \partial_s g_{ij}(t_*; s_*) \dot{s}_{\mathrm{fold}}(t_*) = \partial_t g_{ij}(t_*; s_*),$$

hence

$$|\dot{G}_{\mathrm{fold}}(t_*)| \geq \chi_{\mathrm{fold}}.$$

Taylor expansion gives, for some

$$\xi_t$$

between

$$t$$

and

$$t_*,$$

$$\dot G_{\mathrm{fold}}(t) = \dot G_{\mathrm{fold}}(t_*) + \ddot G_{\mathrm{fold}}(\xi_t)(t-t_*).$$

Therefore

$$|\dot G_{\mathrm{fold}}(t)| \ge \chi_{\mathrm{fold}} - C^{\mathrm{mb}}_{2,tg}|t-t_*|,$$

which stays strictly positive whenever

$$0<|t-t_*|<\chi_{\mathrm{fold}}/C^{\mathrm{mb}}_{2,tg}.$$

Thus

$$G_{\mathrm{fold}}$$

is strictly monotone throughout that receiver-time interval and cannot have a second zero there. Therefore the same fold family cannot complete a second receiver-time passage through a fold inside that interval.

Proposition (Quantitative no-accumulation of many-body delay events).
Assume the unreduced local well-posedness package and suppose, in addition, that on the controlled cycle:

1. each defect

$$g_{ij}(t;s)$$

obeys the derivative hierarchy

$$C^{\mathrm{mb}}_{1,g}, \qquad C^{\mathrm{mb}}_{2,g}, \qquad C^{\mathrm{mb}}_{3,g};$$

2. every admissible fold is nondegenerate with curvature floor

$$\gamma_{\mathrm{fold}}>0;$$

3. away from the listed fold tubes one has the simple-branch floor

$$|\partial_s g_{ij}(t;s)| \ge \nu^{\mathrm{mb}}_J > 0.$$

4. every admissible fold is also transversal in receiver time:

$$|\partial_t g_{ij}(t_*;s_*)| \ge \chi_{\mathrm{fold}} > 0,$$

and the controlled cycle carries one receiver-time fold-passage ceiling

$$|\ddot G_{\mathrm{fold}}(t)| \le C^{\mathrm{mb}}_{2,tg}$$

on each local fold sheet

$$G_{\mathrm{fold}}(t)=g_{ij}(t;s_{\mathrm{fold}}(t)).$$

Then there exists a strict minimum event gap

$$\Delta\tau_{\mathrm{evt}}>0$$

depending only on the derivative bounds and the floors

$$C^{\mathrm{mb}}_{1,g}, \qquad C^{\mathrm{mb}}_{2,g}, \qquad C^{\mathrm{mb}}_{3,g}, \qquad \gamma_{\mathrm{fold}}, \qquad \nu^{\mathrm{mb}}_J, \qquad \chi_{\mathrm{fold}}, \qquad C^{\mathrm{mb}}_{2,tg},$$

such that:

> 1. two distinct fold roots for the same receiver-source family in one receiver-time slice cannot occur with source-time separation smaller than

$$> \Delta\tau_{\mathrm{evt}}; >$$

> 2. two fold passages for the same receiver-source family are separated in receiver time by at least

$$> \Delta\tau_{\mathrm{evt}}; >$$

> 3. sector-boundary crossings and admissible source-cluster exchanges are likewise separated by at least

$$> \Delta\tau_{\mathrm{evt}} >$$

> along every controlled branch family;
> 4. hence no fold, relabeling, or exchange event can accumulate in finite time on the controlled cycle;
> 5. consequently the total number of admissible event hyperedges on one cycle is bounded by

$$> N_{\mathrm{edge}}^{\mathrm{mb}} > \leq > \left\lceil \frac{T_{\mathrm{cyc}}}{\Delta\tau_{\mathrm{evt}}} \right\rceil > N_{\mathrm{fam}}^{\mathrm{mb}}, >$$

> where

$$> T_{\mathrm{cyc}} >$$

> is the controlled cycle length and

$$> N_{\mathrm{fam}}^{\mathrm{mb}} >$$

> is the finite number of receiver-source-sector families allowed by the gauge chart and directional atlas.

One admissible quantitative choice is

$$\Delta\tau_{\mathrm{evt}} \equiv \min\left\{ \frac{\gamma_{\mathrm{fold}}}{C_{3,g}^{\mathrm{mb}}}, \frac{\chi_{\mathrm{fold}}}{C_{2,tg}^{\mathrm{mb}}}, \frac{\nu_J^{\mathrm{mb}}}{2C_{2,g}^{\mathrm{mb}}}, \Delta\tau_{\mathrm{sec}}, \Delta\tau_{\mathrm{exc}} \right\}$$

where

$$\Delta\tau_{\mathrm{sec}}, \qquad \Delta\tau_{\mathrm{exc}}$$

are the corresponding sector-boundary and admissible exchange isolation scales produced by the same derivative hierarchy. The first two terms are the source-time and receiver-time fold-isolation scales. The third term is the uniform simple-branch persistence scale away from fold tubes: if

$$|\partial_s g_{ij}(t; s_0)| \geq \nu_J^{\mathrm{mb}}$$

then

$$|\partial_s g_{ij}(t; s)| \geq \frac{1}{2}\nu_J^{\mathrm{mb}}$$

whenever

$$|s - s_0| < \frac{\nu_J^{\mathrm{mb}}}{2C_{2,g}^{\mathrm{mb}}}$$

> **Corollary (Finite event count on one controlled cycle).**
> Under the hypotheses of the quantitative no-accumulation proposition,

$$> N_{\mathrm{edge}}^{\mathrm{mb}} > \leq > \left\lceil \frac{T_{\mathrm{cyc}}}{\Delta\tau_{\mathrm{evt}}} \right\rceil > N_{\mathrm{fam}}^{\mathrm{mb}}, >$$

> so every admissible fold, relabeling, or exchange family is finite on one controlled cycle.

The remaining local input is simple-branch persistence away from fold tubes.

> **Lemma (Uniform persistence of simple branches away from fold tubes).**
> Fix one receiver-source family

$$> (i, j) >$$

> and one point

$$> (t, s_0) >$$

> outside the controlled fold tubes with

$$> |\partial_s g_{ij}(t; s_0)| \geq \nu_J^{\mathrm{mb}}. >$$

> If

$$> |\partial_s^2 g_{ij}(t; s)| \leq C_{2,g}^{\mathrm{mb}} >$$

> on the corresponding branch neighborhood, then

$$> |\partial_s g_{ij}(t; s)| \geq \frac{1}{2} \nu_J^{\mathrm{mb}} >$$

> whenever

$$> |s - s_0| < \delta_{\mathrm{simp}}, > \qquad > \delta_{\mathrm{simp}} >\equiv> \frac{\nu_J^{\mathrm{mb}}}{2C_{2,g}^{\mathrm{mb}}}. >$$

> **Proof.**
> By the mean-value theorem,

$$> |\partial_s g_{ij}(t; s) - \partial_s g_{ij}(t; s_0)| >\leq> C_{2,g}^{\mathrm{mb}} |s - s_0|. >$$

> So on the stated interval one has

$$> |\partial_s g_{ij}(t; s)| >\geq> \nu_J^{\mathrm{mb}} - C_{2,g}^{\mathrm{mb}} |s - s_0| >\geq> \frac{1}{2} \nu_J^{\mathrm{mb}}. >$$

The remaining event types are not detected by

$$\partial_s g_{ij} = 0$$

alone, so their proof burden should also be stated explicitly. For each fixed branch family in one atlas chart, let

$$\Theta_{\mathrm{sec}}(t)$$

denote the signed distance of the active direction to the nearest sector boundary, and let

$$\Theta_{\mathrm{exc}}(t)$$

denote the signed gap function that distinguishes the currently active admissible source-cluster channel from the next admissible exchange channel in the same local event block. On the controlled cycle these should satisfy one derivative hierarchy

$$|\dot{\Theta}_{\mathrm{sec}}(t)| \leq C_{1,\mathrm{sec}}^{\mathrm{mb}}, \qquad |\ddot{\Theta}_{\mathrm{sec}}(t)| \leq C_{2,\mathrm{sec}}^{\mathrm{mb}}$$

$$|\dot{\Theta}_{\mathrm{exc}}(t)| \leq C_{1,\mathrm{exc}}^{\mathrm{mb}}, \qquad |\ddot{\Theta}_{\mathrm{exc}}(t)| \leq C_{2,\mathrm{exc}}^{\mathrm{mb}}$$

together with transversality floors at admissible relabeling and exchange events:

$$|\dot{\Theta}_{\mathrm{sec}}(t_*)| \geq \gamma_{\mathrm{sec}} > 0, \qquad |\dot{\Theta}_{\mathrm{exc}}(t_*)| \geq \gamma_{\mathrm{exc}} > 0$$

These are the sector and exchange analogues of the fold-curvature floor. Under those hypotheses one may define

$$\Delta\tau_{\mathrm{sec}} \equiv \frac{\gamma_{\mathrm{sec}}}{C_{2,\mathrm{sec}}^{\mathrm{mb}}}, \qquad \Delta\tau_{\mathrm{exc}} \equiv \frac{\gamma_{\mathrm{exc}}}{C_{2,\mathrm{exc}}^{\mathrm{mb}}}$$

and the same Taylor argument shows that sector relabelings and admissible exchanges are isolated in cycle time by at least those scales. Along any active branch away from the fold tubes one also has

$$\dot{s}(t) = -\frac{\partial_t g_{ij}(t; s(t))}{\partial_s g_{ij}(t; s(t))}$$

so the simple-branch persistence lemma supplies the denominator floor needed to keep

$$\dot{s}(t)$$

uniformly bounded there. That is the analytic input behind the bounded first and second derivatives of

$$\Theta_{\mathrm{sec}}(t), \qquad \Theta_{\mathrm{exc}}(t)$$

on the corresponding branch segments.

Proof.
Fix one controlled cycle and one admissible receiver-source-sector family.

1. **Fold isolation.**
   For each admissible fold

$$> (t, s_*), >$$

the source-time fold-isolation lemma yields one local branch parameterization of the fold tube, while the receiver-time passage lemma yields a forbidden cycle-time interval

$$> (t_* - \delta_{\mathrm{fold},t},\ t_* + \delta_{\mathrm{fold},t}) >$$

containing no second fold event of the same family. Hence fold events for that family are discrete in receiver time with separation at least

$$> \delta_{\mathrm{fold},t}. >$$

2. **Simple-branch persistence between folds.**
   Outside the union of those fold neighborhoods, the Jacobian floor

$$> |\partial_s g_{ij}| \geq \nu_J^{\mathrm{mb}} >$$

and the simple-branch persistence lemma imply that every active root branch remains uniformly transversal on intervals of source-time size

$$> \delta_{\mathrm{simp}} >==> \frac{\nu_J^{\mathrm{mb}}}{2C_{2,g}^{\mathrm{mb}}}. >$$

In particular, no hidden tangency can arise between two successive fold neighborhoods.

3. **Sector-event isolation.**
   On each active branch family, a sector relabeling occurs precisely when

$$> \Theta_{\mathrm{sec}}(t) = 0. >$$

The sector transversality floor

$$> |\dot{\Theta}_{\mathrm{sec}}(t_*)| \geq \gamma_{\mathrm{sec}} >$$

and the second-derivative bound

$$> |\ddot{\Theta}_{\mathrm{sec}}(t)| \leq C_{2,\mathrm{sec}}^{\mathrm{mb}} >$$

imply, by the same Taylor argument, that two successive sector relabelings on the same branch family are separated by at least

$$> \Delta\tau_{\mathrm{sec}}. >$$

4. **Exchange-event isolation.**
   Likewise, an admissible source-cluster exchange occurs only when

$$> \Theta_{\mathrm{exc}}(t) = 0. >$$

The exchange transversality floor

$$> |\dot{\Theta}_{\mathrm{exc}}(t_*)| \geq \gamma_{\mathrm{exc}} >$$

and the derivative bound

$$> |\ddot{\Theta}_{\mathrm{exc}}(t)| \leq C_{2,\mathrm{exc}}^{\mathrm{mb}} >$$

imply separation by at least

$$> \Delta\tau_{\mathrm{exc}}. >$$

> 5. **Common event gap.**
>   Set

$$> \Delta\tau_{\mathrm{evt}} >\equiv> \min\{> \delta_{\mathrm{fold}}, > \delta_{\mathrm{fold},t}, > \delta_{\mathrm{simp}}, > \Delta\tau_{\mathrm{sec}}, > \Delta\tau_{\mathrm{exc}} >\}. >$$

> Then no admissible fold, hidden Jacobian loss, sector relabeling, or admissible exchange can recur within cycle-time separation smaller than

$$> \Delta\tau_{\mathrm{evt}}. >$$

> 6. **Finite event count.**
>   The gauge chart and directional atlas allow only finitely many receiver-source-sector families

$$> N^{\mathrm{mb}}_{\mathrm{fam}}. >$$

> Since each family contributes at most

$$> \lceil T_{\mathrm{cyc}}/\Delta\tau_{\mathrm{evt}} \rceil >$$

> admissible events on one cycle, the total number of admissible hyperedges is bounded by

$$> N^{\mathrm{mb}}_{\mathrm{edge}} >\leq> \left\lceil \frac{T_{\mathrm{cyc}}}{\Delta\tau_{\mathrm{evt}}} \right\rceil > N^{\mathrm{mb}}_{\mathrm{fam}}. >$$

> This proves the proposition.

This is the missing analytic bridge from local well-posedness to finite combinatorics. Without it, the later hypergraph package is only suggestive bookkeeping rather than a theorem-level object.

Third many-body theorem package: bounded many-body caustic transit and fold ceilings

The no-accumulation package makes the fold geometry discrete. The next analytic burden is to show that traversing the corresponding fold tubes does not inject an uncontrolled velocity impulse into the many-body comparison channels.

This point is harmless in the binary scaffolds only because the dual-mollified caustic transit bounds were already available there. In the planar three-body regime one must now account not only for simple Type II fold tubes, but also for Type III shared-body coupled folds, where two or three active branch families involving one common body traverse controlled Jacobian collapse in one local event block.

For each principal escape channel

$$m \in \{1, 2, 3, 4\}$$

let

$$W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e})$$

denote one controlled fold tube associated with an admissible Type II or Type III hyperedge

$$\mathsf{e} \in \mathscr{E}^{\mathrm{mb}}_{\mathrm{br}}$$

The theorem target is that the dual-mollified branch sum contributes only a finite channelwise impulse across that tube:

$$\left| \int_{W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e})} \Pi_m(t) \cdot \ddot{\mathbf{X}}(t)\, dt \right| \leq F^{\mathrm{mb}}_{m,\mathsf{e}}$$

where

$$\Pi_m$$

is the channel projection associated with

$$\rho^{\mathrm{mb}}_m$$

and the right-hand side depends only on the dual-mollification parameters, the fold-curvature floor, the branch-separation data, and the local multiplicity of the hyperedge.

The first proof-oriented step is to reduce every admissible fold tube to one quantitative normal form.

> **Lemma (Fold-tube normal form with quantitative Jacobian control).**
> Fix one admissible Type II or Type III fold tube

$$> W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e}) >$$

> and one participating branch family

$$> (i, j, s(t)). >$$

> Assume the no-accumulation derivative hierarchy, the fold-curvature floor

$$> \gamma_{\mathrm{fold}} > 0, >$$

> and the branch-separation floor

$$> \delta^{\mathrm{mb}}_{\mathrm{sep}} > 0. >$$

> Then after translating the fold center to

$$> (t_*, s_*) >$$

> and restricting to a sufficiently small controlled tube, there exists a local source parameter

$$> u = s - s_* >$$

> such that

$$> J_{ij}(t; s) >=> \partial_s g_{ij}(t; s) >=> \alpha_{ij}(t)\, u > + > \mathcal{R}_{ij}(t, u), >$$

> with

$$> |\alpha_{ij}(t)| \geq \frac{1}{2}\gamma_{\mathrm{fold}}, > \qquad > |\mathcal{R}_{ij}(t, u)| \leq C^{\mathrm{mb}}_{3,g} |u|^2 >$$

> throughout the tube.
> Consequently,

$$> |J_{ij}(t; s)| >\gtrsim> \gamma_{\mathrm{fold}}\, |u| >$$

> away from the fold center, with constants depending only on

$$> \gamma_{\mathrm{fold}} > \qquad > \text{and} > \qquad > C^{\mathrm{mb}}_{3,g}. >$$

> **Proof.**
> Translate the fold center to

$$> (t_*, s_*) >$$

> and write

$$> u = s - s_*. >$$

> Since

$$> \partial_s g_{ij}(t_*; s_*) = 0 >$$

> and

$$> |\partial_s^2 g_{ij}(t_*; s_*)| \geq \gamma_{\mathrm{fold}}, >$$

> Taylor expansion in

$$> s >$$

> gives

$$> \partial_s g_{ij}(t; s) >=> \partial_s^2 g_{ij}(t; s_*)\, u > + > \mathcal{R}_{ij}(t, u), >$$

> with remainder satisfying

> $$|\mathcal{R}_{ij}(t,u)| \le C^{\mathrm{mb}}_{3,g}|u|^2.$$

> Define

> $$\alpha_{ij}(t) \equiv \partial_s^2 g_{ij}(t;s_*).$$

> By continuity in

> $$t$$

> and by shrinking the receiver-time tube if necessary, the fold-curvature floor persists:

> $$|\alpha_{ij}(t)| \ge \frac{1}{2}\gamma_{\mathrm{fold}}.$$

> Choose the source-width of the tube so that

> $$C^{\mathrm{mb}}_{3,g}|u| \le \frac{1}{4}\gamma_{\mathrm{fold}}$$

> there. Then

> $$|J_{ij}(t;s)| = |\alpha_{ij}(t)u + \mathcal{R}_{ij}(t,u)| \ge \frac{1}{4}\gamma_{\mathrm{fold}}|u|,$$

> which is the required normal form and lower bound.

This is the exact local reduction needed for the dual-mollified transit bound: every admissible many-body fold tube behaves like one controlled one-dimensional Jacobian zero, up to uniformly bounded error.

Once that reduction is available, the actual impulse bound becomes a finite-dimensional bookkeeping problem.

> **Lemma (Channelwise bounded impulse across one admissible fold block).**
> Under the same hypotheses, fix one principal channel

> $$m \in \{1,2,3,4\}.$$

> Then there exists a constant

> $$\mathfrak{F}^{\mathrm{mb}}_m \equiv \mathfrak{F}^{\mathrm{mb}}_m \big(\eta, \epsilon_c, \gamma_{\mathrm{fold}}, \delta^{\mathrm{mb}}_{\mathrm{sep}}, U_{\max}, A_{\max} \big) < \infty$$

> such that for every admissible fold block

> $$\mathsf{e}$$

> one has

> $$\left| \int_{W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e})} \Pi_m(t)\cdot \ddot{\mathbf{X}}(t)\,dt \right| \le \mathfrak{F}^{\mathrm{mb}}_m \, M_{\mathrm{loc}}(\mathsf{e}),$$

> where

> $$M_{\mathrm{loc}}(\mathsf{e}) \in \{1,2,3\}$$

> is the local admissible multiplicity of the Type II or Type III event block.
>
> In particular, if

> $$M^{\mathrm{mb}}_{\max} \equiv \max_{\mathsf{e}\in\mathscr{E}^{\mathrm{mb}}_{\mathrm{br}}} M_{\mathrm{loc}}(\mathsf{e}),$$

> then the universal fold ceiling

> $$F^{\mathrm{mb}}_m \equiv \mathfrak{F}^{\mathrm{mb}}_m M^{\mathrm{mb}}_{\max}$$

> is finite.

> **Proof.**
> For a Type II fold, insert the fold-tube normal form into the dual-mollified branch kernel. After the local change of variable from

> $$s$$

to

$$> u, >$$

the singular factor is reduced to the same one-dimensional model already controlled in the earlier bridge packages, with all remaining distance and projection factors bounded by the

$$> d_{\min}, > \qquad > U_{\max}, > \qquad > A_{\max} >$$

envelope.

For a Type III fold block, sum over the finitely many participating branch families. The branch-separation floor excludes uncontrolled secondary collisions away from the common fold center, and the admissible event alphabet bounds the local multiplicity by

$$> M_{\mathrm{loc}}(\mathsf{e}) \leq 3. >$$

So the total channelwise impulse is bounded by a finite sum of the same controlled one-dimensional transit integrals.

Proposition (Bounded dual-mollified caustic transit for simple and shared-body folds).
Assume the unreduced local well-posedness package and the quantitative no-accumulation package. Suppose, in addition, that on every admissible fold tube:

1. the fold-curvature floor

$$> \gamma_{\mathrm{fold}} > 0 >$$

holds;
2. distinct active branches are separated by at least

$$> \delta_{\mathrm{sep}}^{\mathrm{mb}} > 0; >$$

3. the local fold multiplicity is restricted to the admissible Type II and Type III event blocks;
4. and the dual-mollification parameters

$$> \eta, > \qquad > \epsilon_c >$$

lie in the same small regime as the earlier binary transit lemmas.

Then for each principal channel

$$> m = 1, 2, 3, 4 >$$

there exists a finite universal fold ceiling

$$> F_m^{\mathrm{mb}} < \infty >$$

such that:

1. every admissible Type II fold tube contributes at most

$$> F_m^{\mathrm{mb}} >$$

to the channelwise velocity impulse of

$$> \rho_m^{\mathrm{mb}}; >$$

2. every admissible Type III shared-body coupled fold block contributes at most

$$> F_m^{\mathrm{mb}} >$$

after summing all participating branch families in that block;
3. the corresponding post-transit velocity and acceleration remain inside the same

$$> U_{\max}, > \qquad > A_{\max}, > \qquad > L_A >$$

envelope up to a controlled renormalization of the constants;
4. therefore the fold contribution appearing in the recapture package may be absorbed into one finite channelwise ceiling

$$> L^{\mathrm{mb}}_{m,\mathrm{fold}}(t) > \le > F^{\mathrm{mb}}_m >$$

> on every controlled recapture window.

More concretely, one may take

$$F^{\mathrm{mb}}_m \equiv \mathfrak{F}^{\mathrm{mb}}_m M^{\mathrm{mb}}_{\max}$$

where

$$\mathfrak{F}^{\mathrm{mb}}_m$$

is the single-branch transit constant from the previous lemma and

$$M^{\mathrm{mb}}_{\max} \le 3$$

is the maximal admissible local fold multiplicity in the Type II / Type III event alphabet.

The proof is written one local fold block at a time.

> **Lemma (Type II fold-tube transit estimate).**
> Fix one admissible Type II fold tube

$$> W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e}) >$$

> and one principal channel

$$> m \in \{1,2,3,4\}. >$$

> Under the fold-tube normal form, the corresponding branch contribution to the channelwise impulse satisfies

$$> \left| > \int_{W^{\mathrm{mb}}_{\mathrm{fold}}(\mathsf{e})} > \Pi_m(t)\cdot > \mathbf{a}^{(ij)}_\eta(t)\,dt > \right| > \le > \mathfrak{F}^{\mathrm{mb}}_{m,\mathrm{II}}, >$$

> for some finite constant depending only on

$$> \epsilon_c, > \qquad > \gamma_{\mathrm{fold}}, > \qquad > d_{\min}, > \qquad > U_{\max}, > \qquad > A_{\max}, > \qquad > \chi_{\mathrm{fold}}, > \qquad > U^{\mathrm{mb}}_{\mathrm{tube}}. >$$

> Here

$$> U^{\mathrm{mb}}_{\mathrm{tube}} >$$

> denotes the controlled source half-width of the fold tube in the normal-form coordinate

$$> u = s - s_*. >$$

> **Proof.**
> Write

$$> J_{ij}(t;s) = \alpha_{ij}(t)u + \mathcal{R}_{ij}(t,u), > \qquad > u = s - s_*, >$$

> as in the fold-tube normal form. The proof is then an explicit reduction to a one-dimensional singular integral.
>
> 1. **Jacobian lower bound on the tube.**
> By the normal form lemma,

$$> |J_{ij}(t;s)| \ge c_{\mathrm{fold}}|u| >$$

> with

$$> c_{\mathrm{fold}} \asymp \gamma_{\mathrm{fold}}. >$$

> 2. **Distance and projection control.**
> On the controlled tube, the pair distance obeys

$$> r_{ij}(t;s) \ge d_{\min}, >$$

> while the channel projection satisfies

$$> |\Pi_m(t) \cdot \hat{\mathbf{r}}_{ij}(t;s)| >\leq> C_m^{\mathrm{proj}} >$$

> for some

$$> C_m^{\mathrm{proj}} >=> C_m^{\mathrm{proj}}(U_{\max}, A_{\max}). >$$

> 3. **Kernel reduction.**
> The dual-mollified branch kernel is therefore bounded by

$$> \frac{C_m^{\mathrm{proj}} \kappa \epsilon^2}{> \left(d_{\min}^2 + \epsilon_c^2\right)\left(c_{\mathrm{fold}}|u| + \eta\right) >} >$$

> up to a harmless change of constants. Thus the entire fold singularity is reduced to the one-dimensional factor

$$> \frac{1}{|u| + \eta}. >$$

> 4. **Exact fold-time cancellation.**
> Since the source branch

$$> s = s(t) >$$

> is simple, differentiating the root relation

$$> g_{ij}(t; s(t)) = 0 >$$

> yields

$$> \frac{ds}{dt} >=> -\frac{\partial_t g_{ij}(t; s(t))}{\partial_s g_{ij}(t; s(t))}. >$$

> Hence

$$> dt >=> \frac{|\partial_s g_{ij}(t; s(t))|}{|\partial_t g_{ij}(t; s(t))|}\, ds >=> \frac{|J_{ij}(t;s)|}{|\partial_t g_{ij}(t;s)|}\, du. >$$

> By shrinking the controlled fold tube if necessary, the receiver-time passage floor from the no-accumulation package persists along the active root branch; write the persisted floor again as

$$> \chi_{\mathrm{fold}}. >$$

> Thus

$$> |\partial_t g_{ij}(t;s)| \geq \chi_{\mathrm{fold}} > 0. >$$

> Therefore the singular factor

$$> |J_{ij}|^{-1} >$$

> in the force kernel is exactly cancelled by the

$$> |J_{ij}| >$$

> factor in the time-volume element

$$> dt. >$$

> 5. **Uniform finite impulse bound.**
> Let

$$> U_{\mathrm{tube}}^{\mathrm{mb}} >\equiv> \sup_{(t,s) \in W_{\mathrm{fold}}^{\mathrm{mb}}(\mathfrak{e})} |u|. >$$

> The fold-tube width is already controlled by the no-accumulation and normal-form constants, so

$$> U_{\mathrm{tube}}^{\mathrm{mb}} < \infty >$$

> depends only on the same geometric data. After cancellation, the branch contribution is bounded by

$$> \frac{C_m^{\mathrm{proj}} \kappa \epsilon^2}{> \chi_{\mathrm{fold}} \left(d_{\min}^2 + \epsilon_c^2\right) >} > \int_{|u| \le U_{\mathrm{tube}}^{\mathrm{mb}}} du, >$$

hence by

$$> \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}} > \le > \frac{> 2C_m^{\mathrm{proj}} \kappa \epsilon^2 U_{\mathrm{tube}}^{\mathrm{mb}} >}{> \chi_{\mathrm{fold}} \left(d_{\min}^2 + \epsilon_c^2\right) >}. >$$

This bound is finite and remains uniformly finite as

$$> \eta \downarrow 0. >$$

Thus the fold impulse ceiling is controlled by one geometric transit width and one receiver-time passage floor rather than by a logarithmic mollifier loss. Therefore

$$> \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}} >$$

depends only on

$$> \epsilon_c, > \gamma_{\mathrm{fold}}, > d_{\min}, > U_{\max}, > A_{\max}, > \chi_{\mathrm{fold}}, > U_{\mathrm{tube}}^{\mathrm{mb}}. >$$

6. **Conclusion.**
   The resulting finite bound is

$$> \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}}, >$$

as required.

Lemma (Type III shared-body superposition estimate).
Fix one admissible Type III shared-body coupled fold block

$$> \mathsf{e}. >$$

Then for each principal channel

$$> m >$$

one has

$$> \left| > \int_{W_{\mathrm{fold}}^{\mathrm{mb}}(\mathsf{e})} > \Pi_m(t) \cdot > \ddot{\mathbf{X}}(t)\, dt > \right| > \le > M_{\mathrm{loc}}(\mathsf{e})\, \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}}, >$$

where

$$> M_{\mathrm{loc}}(\mathsf{e}) \le 3. >$$

Proof.
Decompose the block into its participating branch families

$$> \beta_1, \ldots, \beta_{M_{\mathrm{loc}}(\mathsf{e})}. >$$

The branch-separation floor

$$> \delta_{\mathrm{sep}}^{\mathrm{mb}} > 0 >$$

ensures that away from the common fold center these branches do not produce any additional unresolved near-collision or near-fold singularity. Each branch therefore satisfies the hypotheses of the Type II estimate on the same controlled tube. Summing the finitely many bounds gives

$$> \sum_{\ell=1}^{M_{\mathrm{loc}}(\mathsf{e})} > \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}} > = > M_{\mathrm{loc}}(\mathsf{e})\, \mathfrak{F}_{m,\mathrm{II}}^{\mathrm{mb}}, >$$

and the admissible event alphabet gives

$$> M_{\mathrm{loc}}(\mathsf{e}) \le 3. >$$

> **Corollary (Renormalized post-transit envelope constants).**
> There exist finite constants

$$> U^{+}_{\max}, > \qquad > A^{+}_{\max}, > \qquad > L^{+}_{A} >$$

> depending only on the pre-transit envelope and the fold ceilings

$$> F^{\mathrm{mb}}_{m} >$$

> such that every admissible fold block sends the controlled tube data into a post-transit state satisfying

$$> \|\dot{\mathbf{X}}\| \le U^{+}_{\max}, > \qquad > \|\ddot{\mathbf{X}}\| \le A^{+}_{\max}, > \qquad > \mathrm{Lip}(\ddot{\mathbf{X}}) \le L^{+}_{A}. >$$

> In the eventual invariant-envelope argument one must choose the kinematic box so that these renormalized constants are absorbed back into the same admissible class.
>
> **Proof.**
> The channelwise impulse bounds give a finite change in every projected velocity component across one admissible fold block, bounded by

$$> F^{\mathrm{mb}}_{m}. >$$

> Away from the fold tube, the pre-transit trajectory already obeys the ambient kinematic envelope. Therefore the total post-transit velocity bound is obtained by adding the finite fold-impulse ceilings to the pre-transit velocity box. The same localization and tube-width bounds show that acceleration and acceleration-Lipschitz norms can increase by only a finite amount depending on the same transit constants. Hence one obtains finite renormalized constants

$$> U^{+}_{\max}, > \qquad > A^{+}_{\max}, > \qquad > L^{+}_{A} >$$

> depending only on the pre-transit envelope and the fold ceilings.

> **Proof of the bounded caustic-transit proposition.**
> Fix one admissible fold block.
>
> 1. **Tube localization.**
>   By the no-accumulation package, the fold block lies in one isolated receiver-time tube of width controlled by

$$> \Delta\tau_{\mathrm{evt}}. >$$

> So no second singular event interacts with the tube at the same scale.
>
> 2. **Type II local model.**
>   For a simple fold, apply the fold-tube normal form and the Type II transit estimate. This gives one finite channelwise impulse bound

$$> \mathfrak{F}^{\mathrm{mb}}_{m,\mathrm{II}}. >$$

> By the exact fold-time cancellation, this bound is uniform in

$$> \eta \downarrow 0 >$$

> and is governed by the receiver-time passage floor and the geometric source-width of the tube.
>
> 3. **Type III superposition.**
>   For a shared-body coupled fold, decompose into the finitely many participating branch families and use the superposition lemma. This yields

$$> \mathfrak{F}^{\mathrm{mb}}_{m} >=> \mathfrak{F}^{\mathrm{mb}}_{m,\mathrm{II}} M^{\mathrm{mb}}_{\max}. >$$

> 4. **Projection to the principal channels.**
>   The projections

$$> \Pi_{m} >$$

> defining

$$> \rho^{\mathrm{mb}}_{1}, \ldots, \rho^{\mathrm{mb}}_{4} >$$

> are uniformly bounded on the smooth windows. Therefore the same finite impulse controls the fold-loss terms

$$> L^{\mathrm{mb}}_{m,\mathrm{fold}}(t) \le F^{\mathrm{mb}}_m. >$$

> 5. **Post-transit regularity.**
>   The renormalized envelope corollary shows that the post-transit trajectory still satisfies a controlled

$$> C^1 >$$

> and acceleration-Lipschitz bound, so the next no-accumulation cycle remains available.
>
> 6. **Universal ceiling.**
>   Since every admissible fold block is of Type II or Type III and the local multiplicity is bounded by

$$> M^{\mathrm{mb}}_{\max} \le 3, >$$

> the universal choice

$$> F^{\mathrm{mb}}_m >=> \mathfrak{F}^{\mathrm{mb}}_{m,\mathrm{II}} M^{\mathrm{mb}}_{\max} >$$

> controls every block on the cycle. This is now a genuine geometric transit invariant of the controlled fold alphabet, and it is exactly the ceiling consumed later by the recapture margins.

This is the last missing analytic bridge between finite branch combinatorics and the concrete recapture inequalities. Without it, the fold ceilings in the principal margins remain formal placeholders.

Fourth many-body theorem package: finite active delay hypergraph

Once the gauge-fixed section is chosen, the next burden is to replace the binary branch graph by a finite active delay hypergraph over one controlled cycle.

Fix a finite family of cycle windows

$$\mathcal{W}^{\mathrm{mb}} = \{W_1, \dots, W_{N_W}\}$$

chosen to separate the inbound compression stage, the crossing or near-core stage, the post-crossing recapture stage, and the late-turn stage. For each ordered receiver-source pair

$$(i,j) \in \{1,2,3\}^2, \qquad i \ne j$$

and for the self family

$$(i,i)$$

let

$$g_{ij}(t;s) \equiv \|\mathbf{x}_i(t) - \mathbf{x}_j(s)\| - c_f(t-s)$$

denote the exact delayed-root equation on a window

$$W_\alpha \in \mathcal{W}^{\mathrm{mb}}$$

As in the reduced planar bridge, choose a fixed finite directional atlas on

$$S^1$$

but now each active root must carry receiver index, source index, sector label, and window label.

Define the active delay hypergraph

$$\mathscr{H}^{\mathrm{mb}}_{\mathrm{br}} = \left(\mathscr{V}^{\mathrm{mb}}_{\mathrm{br}}, \mathscr{E}^{\mathrm{mb}}_{\mathrm{br}}\right)$$

as follows.

- A vertex

$$\mathsf{v} = (i, j, \alpha, k, \ell)$$

records one active branch family with receiver

$$i$$

source

$$j$$

window

$$W_\alpha$$

directional sector

$$\mathfrak{S}_k$$

and branch multiplicity label

$$\ell$$

- A hyperedge

$$\mathsf{e} \in \mathscr{E}_{\mathrm{br}}^{\mathrm{mb}}$$

joins a finite set of such vertices whenever the corresponding root families can meet in one coupled fold event, one sector-exchange event, or one receiver-source role-exchange event forced by the same body geometry.

The point of the hypergraph language is that in a three-body regime, several branch births can share one body and one geometric degeneracy. Pairwise edges are therefore not enough to encode the admissible local branch moves.

For the first planar-three-body bridge, the admissible hyperedges should be restricted to a finite list of local event types:

- **Type I: single-branch continuation hyperedge.**
  A one-vertex or two-vertex hyperedge recording continuation of the same simple branch family across adjacent windows with unchanged receiver, source, and sector data.
- **Type II: simple fold birth or fold death hyperedge.**
  A hyperedge supported inside one controlled fold tube for one receiver-source pair

$$(i,j)$$

where exactly one simple branch is created or annihilated.

- **Type III: shared-body coupled fold hyperedge.**
  A hyperedge joining two or three vertices when several branch families involving one common body meet one common degeneracy and their births or deaths must be recorded together.
- **Type IV: sector relabeling hyperedge.**
  A hyperedge recording passage of one active branch across one sector boundary of the fixed directional atlas without changing the underlying receiver-source family.
- **Type V: source-cluster exchange hyperedge.**
  A hyperedge recording one admissible cluster move from the ancestry package:

$$\{i\} \leftrightarrow \{i,j\}, \qquad \{i,j\} \leftrightarrow \{i\}, \qquad \{i,j\} \leftrightarrow \{i,k\}$$

always through one local shared-body event already visible in the delayed geometry.

No other hyperedge type should be allowed in the first bridge regime. In particular, there should be no hyperedge representing simultaneous creation of arbitrarily many fresh branches, no instantaneous jump to a source cluster outside

$$\mathfrak{C}_{\mathrm{src}}^{\mathrm{mb}}$$

and no event that changes receiver, source cluster, sector family, and window label all at once without passing through one of the listed local types.

> **Target Proposition (Finite active hypergraph control on the planar three-body cycle).**
> Assume the unreduced local well-posedness package, the gauge-selector package, and the quantitative no-accumulation package. On a sufficiently small section-side tame subclass of

$$> \mathcal{H}_{A_*,\eta}^{\mathrm{adm,mb}}, >$$

> there exist finite constants

$$> N^{\mathrm{mb}}_{\mathrm{vert}}, > \qquad > N^{\mathrm{mb}}_{\mathrm{edge}}, > \qquad > \nu^{\mathrm{mb}}_{J}, > \qquad > \delta^{\mathrm{mb}}_{\mathrm{sep}}, > \qquad > \Delta\tau_{\mathrm{evt}} >$$

> such that every one-cycle continuation admits an active delay hypergraph

$$> \mathscr{H}^{\mathrm{mb}}_{\mathrm{br}} >$$

> with:
>
> 1. at most

$$> N^{\mathrm{mb}}_{\mathrm{vert}} >$$

> active branch vertices and at most

$$> N^{\mathrm{mb}}_{\mathrm{edge}} >$$

> admissible coupled fold hyperedges;
> 2. branch separation at least

$$> \delta^{\mathrm{mb}}_{\mathrm{sep}}; >$$

> 3. causal Jacobian bound

$$> |J_{ij}(t;s)| \geq \nu^{\mathrm{mb}}_{J} >$$

> away from the explicitly controlled fold tubes;
> 4. root birth, root death, sector relabeling, and source-exchange events occurring only through the listed hyperedges and separated in cycle order by at least

$$> \Delta\tau_{\mathrm{evt}}; >$$

> 5. no uncontrolled branch proliferation outside those hypergraph-coded events.

> More concretely:
>
> 1. every hyperedge belongs to one of the five admissible types above;
> 2. every fold tube supports only finitely many Type II or Type III hyperedges, with multiplicity bounded by the no-accumulation event-gap constant

$$> \Delta\tau_{\mathrm{evt}}; >$$

> 3. every sector boundary crossing supports only one Type IV relabeling event per active branch family at the chosen scale;
> 4. every source-cluster exchange belongs to one Type V hyperedge compatible with the ancestry-package move list;
> 5. hence every backward ancestry chain and every forward recapture chain passes through a finite event alphabet rather than an open-ended combinatorial explosion.

This proposition should also be read together with the smooth-window floors from the recapture package:

$$\|\mathbf{a}\| \geq a_{\min}, \qquad \|\mathbf{b}\| \geq b_{\min}, \qquad |\det(\mathbf{a},\mathbf{b})| \geq \Delta_{\min}, \qquad \delta_{\mathrm{role}} \geq \delta_{\mathrm{role,min}}$$

whenever one asks the same active hypergraph to feed the principal recapture margins. The hypergraph theorem itself does not differentiate the observables

$$\rho^{\mathrm{mb}}_{m}$$

but the event structure it produces must be compatible with the smooth windows on which those derivatives are later taken.

> **Proof draft of the finite active hypergraph proposition.**
> Fix one controlled one-cycle continuation in the admissible gauge chart.
>
> 1. **Finite event alphabet.**
> By construction, every allowed topological change of an active branch family belongs to one of the five admissible local types: continuation, simple fold, shared-body coupled fold, sector relabeling, or source-cluster exchange. No other event type is permitted.
>
> 2. **Finite event times.**
> The no-accumulation package gives one common cycle-time gap

$$> \Delta\tau_{\mathrm{evt}} > 0 >$$

> between any two admissible events on the same controlled family. Hence the total number of event times on one cycle is finite.
>
> 3. **Finite vertex set.**
> Away from event times, every active branch is simple, sector-labeled, and carried by one receiver-source-window-sector family. Since the receiver index, source index, window label, and sector label all range over finite sets, and since the event count is finite, only finitely many branch segments can occur over the full cycle. This yields

$$> N_{\mathrm{vert}}^{\mathrm{mb}} < \infty. >$$

> 4. **Finite hyperedge set.**
> Each admissible event time contributes exactly one local hyperedge of Type I-V. Since the admissible event times are finite and every event block has multiplicity bounded by the admissible alphabet, the total hyperedge count is finite and obeys

$$> N_{\mathrm{edge}}^{\mathrm{mb}} > \leq > \left\lceil \frac{T_{\mathrm{cyc}}}{\Delta\tau_{\mathrm{evt}}} \right\rceil > N_{\mathrm{fam}}^{\mathrm{mb}}. >$$

> 5. **Branch separation and Jacobian control.**
> The simple-branch persistence lemma and the caustic-transit package guarantee that outside the explicitly controlled fold tubes one retains the uniform Jacobian floor

$$> |J_{ij}(t;s)| \geq \nu_J^{\mathrm{mb}} >$$

> and branch-separation floor

$$> \delta_{\mathrm{sep}}^{\mathrm{mb}} > 0. >$$

> Therefore no uncontrolled branch proliferation can occur between hypergraph-coded event blocks.
>
> 6. **Compatibility with later packages.**
> Because the hypergraph changes only through the listed event alphabet and does so at finitely many isolated times, every later backward ancestry chain and every later forward recapture chain may be read on one finite combinatorial object. That is exactly the content needed by the ancestry and recapture packages.

This is the many-body replacement for the binary branch graph packages. Once it is proved, later ancestry and recapture arguments can consume a finite combinatorial object rather than an open-ended moving family of delayed roots.

Fifth many-body theorem package: cluster-valued ancestry and deep-past exclusion

The next burden is the many-body replacement for deep-past provenance. In the unreduced planar binary, every remote late-turn root was traced back to one finite provenance class on an earlier branch graph. In the planar three-body regime that is no longer the right object, because a delayed influence may pass through changing pair or cluster organization before its geometry becomes simple enough to compare.

Let

$$\mathcal{W}_{\mathrm{lt}}^{\mathrm{mb}} \subseteq \mathcal{W}^{\mathrm{mb}}$$

denote the late-turn receiver windows, and let

$$\mathcal{W}_{\mathrm{mid}}^{\mathrm{mb}}, \qquad \mathcal{W}_{\mathrm{prov}}^{\mathrm{mb}} \subseteq \mathcal{W}^{\mathrm{mb}}$$

denote, respectively, the intermediate post-crossing windows and the earlier provenance windows that precede the late-turn block in the cycle order. A deep-past active branch on a late-turn window should mean one with delay

$$t - s \geq \tau_{\mathrm{dp}}^{\mathrm{mb}}$$

For the first planar-three-body bridge, the source-cluster alphabet should be fixed as

$$\mathfrak{C}_{\mathrm{src}}^{\mathrm{mb}} \equiv \Big\{\{1\}, \{2\}, \{3\}, \{1,3\}, \{1,2\}, \{2,3\}\Big\}$$

and only the following backward exchange moves should be admissible:

- singleton-to-pair attachment

$$\{i\} \leftrightarrow \{i,j\}$$

through one hypergraph-coded fold or sector exchange;

- pair-to-singleton detachment

$$\{i,j\} \leftrightarrow \{i\}$$

through one listed fold exit;

- pair swap

$$\{i,j\} \leftrightarrow \{i,k\}$$

through one shared-body exchange hyperedge.

No other source-cluster jump should be allowed in the ancestry relation.

For each late-turn hypergraph vertex

$$\mathsf{v}_{\mathrm{late}} \in \mathscr{V}_{\mathrm{br}}^{\mathrm{mb}}$$

define its backward ancestry set

$$\mathrm{Anc}(\mathsf{v}_{\mathrm{late}})$$

to be the sub-hypergraph reached by following admissible backward continuation through the hyperedges of

$$\mathscr{H}_{\mathrm{br}}^{\mathrm{mb}}$$

into earlier windows, never moving to a later receiver window in the cycle order. A cluster ancestry complex should then mean a finite family

$$\mathfrak{A}^{\mathrm{mb}} = \{\mathfrak{a}_1, \ldots, \mathfrak{a}_{N_{\mathrm{anc}}}\}$$

of connected ancestry components, each tagged by:

- the receiver body,
- the active source cluster, meaning either a single body or an ordered pair acting as the current effective delayed source family,
- the provenance windows in which that ancestry lives,
- and the admissible exchange moves by which one source cluster can pass to another.

More concretely, each ancestry component

$$\mathfrak{a}_m$$

should lie entirely in provenance windows

$$W_\alpha \in \mathcal{W}_{\mathrm{prov}}^{\mathrm{mb}}$$

carry one fixed receiver body

$$i_m$$

one fixed sector label

$$k_m$$

and one connected source-cluster trace on which the relevant branch parameterizations remain simple with one common Jacobian floor

$$\nu_{J,\mathrm{anc}}^{\mathrm{mb}} > 0$$

The point is that a remote contribution should now be forced into one finite ancestry complex rather than into one literal earlier branch family.

> **Target Proposition (Deep-past cluster ancestry or exclusion for the planar three-body bridge).**
> Assume the preceding section, no-accumulation, and finite-hypergraph packages. Then on a sufficiently small tame subclass, with the same event-gap and branch-regularity data

$$> \Delta\tau_{\mathrm{evt}}, > \qquad > \nu_J^{\mathrm{mb}}, > \qquad > \delta_{\mathrm{sep}}^{\mathrm{mb}}, >$$

> there exists a finite ancestry complex

$$> \mathfrak{A}^{\mathrm{mb}} >$$

> such that every late-turn active delayed branch satisfies exactly one of the following:

> 1. its backward ancestry meets one unique cluster ancestry component

$$> \mathfrak{a}_m \in \mathfrak{A}^{\mathrm{mb}}; >$$

> 2. its backward ancestry enters a controlled fold tube or exchange tube already listed in the active hypergraph;
> 3. or it is excluded by a no-migration alternative asserting that an infinite chain of fresh source-cluster exchanges cannot occur inside the controlled cycle.

> Moreover:
>
> 1. each late-turn branch meets at most one ancestry component;
> 2. the total number of admissible ancestry components is bounded by

$$> N_{\mathrm{anc}}; >$$

> 3. each ancestry component contributes at most one uniformly transversal deep-past branch per admissible source-cluster channel on the chosen delay scale;
> 4. any backward ancestry chain from a late-turn branch that avoids all ancestry components must remain trapped in the union of late-turn windows, mid windows, and controlled fold or exchange tubes, and such a trapped component is forbidden by the finite-cycle parity rule for admissible cluster exchanges;
> 5. every admissible backward cluster exchange in such a chain is separated from the next by at least

$$> \Delta\tau_{\mathrm{evt}}, >$$

> so no ancestry chain can hide an infinite migration process inside one controlled cycle;
> 6. therefore the full deep-past contribution on the late-turn windows is bounded by a finite ancestry count times one branch-amplitude ceiling.

This is the many-body replacement for the binary deep-past relocation theorem. A remote root is no longer pushed onto one pre-crossing leg. Instead it is forced into one finite ancestry object, and the only alternative is a precise obstruction: uncontrolled migration through ever-new source clusters.

The no-migration clause should be read sharply. Because the source-cluster alphabet

$$\mathfrak{C}^{\mathrm{mb}}_{\mathrm{src}}$$

is finite and the admissible exchange moves are local and hypergraph-coded, an endless backward chain would have to revisit one earlier cluster pattern without ever entering a provenance component. The intended obstruction theorem is that no such trapped exchange cycle can persist entirely inside

$$\mathcal{W}^{\mathrm{mb}}_{\mathrm{lt}} \cup \mathcal{W}^{\mathrm{mb}}_{\mathrm{mid}}$$

and the controlled fold or exchange tubes.

This finite-cycle parity rule should now be treated as an explicit upstream geometric target, not as invisible background logic:

> **Target Lemma (No trapped admissible exchange cycles in the planar three-body ancestry graph).**
> No closed loop of admissible source-cluster exchanges supported entirely inside

$$> \mathcal{W}^{\mathrm{mb}}_{\mathrm{lt}} > \cup > \mathcal{W}^{\mathrm{mb}}_{\mathrm{mid}} >$$

> and the controlled fold or exchange tubes can avoid the provenance region indefinitely while preserving the admissible receiver, sector, and cluster bookkeeping.
>
> **Proof draft.**
> The intended mechanism is that every admissible backward exchange consumes a definite amount of source-time depth, while the late-turn and mid-window block has only finite cycle extent.
>
> 1. **Backward source-time drift on one simple branch.**
> Along one simple delayed branch

$$> g_{ij}(t; s(t)) = 0, >$$

> one has

$$> \frac{ds}{dt} >=> -\frac{\partial_t g_{ij}}{\partial_s g_{ij}}. >$$

> On the controlled late-turn and mid windows, the no-accumulation package gives the receiver-time passage floor

$$> |\partial_t g_{ij}| \geq \chi_{\text{evt}} > 0, >$$

> while away from the explicitly listed fold tubes the simple-branch Jacobian floor gives

$$> |\partial_s g_{ij}| \geq \nu_J^{\text{mb}} > 0. >$$

> Hence

$$> \left|\frac{ds}{dt}\right| >\geq> \frac{\chi_{\text{evt}}}{\sup |\partial_s g_{ij}|} >\equiv> c_{s,\text{drift}}^{\text{mb}} > 0 >$$

> on every simple branch segment in those windows. In backward continuation, the source time therefore moves strictly toward earlier history at a controlled rate.
>
> 2. **Macroscopic drop across one admissible exchange block.**
> Distinct admissible exchanges are separated in receiver time by at least

$$> \Delta\tau_{\text{evt}}. >$$

> Therefore, between two successive exchange blocks on one backward chain, the source time drops by at least

$$> \Delta s_{\min}^{\text{mb}} >\equiv> c_{s,\text{drift}}^{\text{mb}} \Delta\tau_{\text{evt}} >> 0, >$$

> up to the uniformly controlled fold-tube and exchange-tube widths already absorbed into the event constants.
>
> 3. **No indefinite cycling inside the late/mid block.**
> Suppose a closed loop of admissible cluster exchanges were supported entirely inside

$$> \mathcal{W}_{\text{lt}}^{\text{mb}} > \cup > \mathcal{W}_{\text{mid}}^{\text{mb}}. >$$

> After traversing one full loop with

$$> q \geq 1 >$$

> exchange blocks, the cumulative backward source-time drop would be at least

$$> q\,\Delta s_{\min}^{\text{mb}}. >$$

> But the receiver bookkeeping is assumed to return to the same late/mid combinatorial state, so the corresponding source time would also have to return to the same bounded portion of the history strip attached to that state. This is impossible once

$$> q\,\Delta s_{\min}^{\text{mb}} >$$

> exceeds the total width of the late/mid source-time slab.
>
> 4. **Finite-state closure contradiction.**
> Because the state alphabet

$$> (i, \alpha, k, \mathfrak{c}_{\text{src}}, \ell) >$$

> is finite, any putative infinite trapped exchange process would eventually repeat one previous state. Repeating the same state after a nonzero source-time drop contradicts the deterministic branch continuation on simple segments and the controlled local event alphabet at folds or exchanges. Therefore no trapped admissible exchange cycle can persist entirely inside the late-turn and mid-window region without entering the provenance zone.

This is now the exact topological/kinematic input consumed by the ancestry package: finite-state recurrence alone is not enough, but finite-state recurrence plus monotone source-time drift rules out the only remaining migration loophole.

> **Proof draft of the deep-past cluster-ancestry-or-exclusion proposition.**
> Fix one late-turn vertex

$$> \mathsf{v}_{\text{late}} >\in> \mathscr{V}_{\text{br}}^{\text{mb}}. >$$

> 1. **Finite backward state space.**
> By the finite-hypergraph package, every backward continuation path from

$$> \mathsf{v}_{\text{late}} >$$

runs inside one finite directed hypergraph whose event times are separated by

$$> \Delta\tau_{\mathrm{evt}}. >$$

Introduce the combinatorial state

$$> \Sigma(\mathsf{v}) >\equiv> \big(> i(\mathsf{v}), > \alpha(\mathsf{v}), > k(\mathsf{v}), > \mathfrak{c}_{\mathrm{src}}(\mathsf{v}), > \ell(\mathsf{v}) >\big), >$$

consisting of receiver label, window label, sector label, active source cluster, and branch multiplicity label. Because each factor ranges over a finite alphabet, the total number of such states is finite. Hence every backward ancestry chain either:

$$> \text{(a) terminates in an earlier window,} > \qquad > \text{(b) reaches the provenance region,} > \qquad > \text{or (c) revisits one previous state.} >$$

2. **Monotone window descent.**
   By definition of backward ancestry, continuation is only allowed into earlier receiver windows in the cycle order, except for passage through one listed fold tube or one listed exchange tube already recorded in the hypergraph. Therefore no ancestry chain can oscillate indefinitely between unrelated window families. Once the chain has entered

$$> \mathcal{W}^{\mathrm{mb}}_{\mathrm{prov}}, >$$

it stays inside the earlier part of the cycle unless one of the admissible hypergraph moves forces it out again; but such an exit would have to be encoded by one of the same finitely many state transitions above.

3. **Provenance-component extraction.**
   Collect all backward-connected sub-hypergraphs lying entirely in the provenance windows

$$> \mathcal{W}^{\mathrm{mb}}_{\mathrm{prov}} >$$

for which the receiver label and sector label remain fixed and the admissible source-cluster trace stays inside the allowed move list

$$> \{i\} \leftrightarrow \{i,j\}, > \qquad > \{i,j\} \leftrightarrow \{i\}, > \qquad > \{i,j\} \leftrightarrow \{i,k\}. >$$

On each such component the branchwise Jacobian floor

$$> \nu^{\mathrm{mb}}_{J,\mathrm{anc}} > 0 >$$

and the branch-separation floor

$$> \delta^{\mathrm{mb}}_{\mathrm{sep}} > 0 >$$

exclude secondary unresolved branching. The resulting connected pieces are the ancestry components

$$> \mathfrak{a}_1, \dots, \mathfrak{a}_{N_{\mathrm{anc}}}. >$$

Their total number is finite because both the state alphabet and the finite hypergraph are finite.

4. **Uniqueness of ancestry attachment.**
   Suppose one late-turn branch met two distinct ancestry components

$$> \mathfrak{a}_p \neq \mathfrak{a}_q. >$$

Then a backward path from

$$> \mathsf{v}_{\mathrm{late}} >$$

to

$$> \mathfrak{a}_p >$$

and one to

$$> \mathfrak{a}_q >$$

would have to split through either:

$$> \text{(a) a receiver-window reversal,} > \qquad > \text{(b) an unlisted cluster exchange,} > \qquad > \text{or (c) a repeated branch birth unsupported by the hypergraph} >$$

Each alternative contradicts the finite-hypergraph proposition. Hence every late-turn branch meets at most one ancestry component.

> 5. **Exclusion of trapped migration.**
> Assume now that one backward ancestry chain avoids all ancestry components. By Step 1, it must eventually revisit one previous combinatorial state

$$> \Sigma(\mathsf{v}). >$$

> By Step 2, this repeated state cannot be realized by wandering through infinitely many earlier windows; it must close into one trapped loop inside

$$> \mathcal{W}^{\mathrm{mb}}_{\mathrm{lt}} > \cup > \mathcal{W}^{\mathrm{mb}}_{\mathrm{mid}} >$$

> together with the explicitly controlled fold or exchange tubes. Every jump in that loop is one admissible cluster exchange separated from the next by at least

$$> \Delta\tau_{\mathrm{evt}}. >$$

> The target lemma on no trapped admissible exchange cycles rules out exactly such a closed loop by showing that each exchange block forces a definite backward drop in source time. Therefore every backward ancestry chain either reaches one unique ancestry component or terminates through one listed local event tube without generating uncontrolled migration.
>
> 6. **Componentwise deep-past amplitude bound.**
> Fix one ancestry component

$$> \mathfrak{a}_m. >$$

> On that component,

$$> t - s \geq \tau^{\mathrm{mb}}_{\mathrm{dp}}, > \qquad > |J_{ij}(t;s)| \geq \nu^{\mathrm{mb}}_{J,\mathrm{anc}}, > \qquad > r_{ij}(t;s) \geq c_f \tau^{\mathrm{mb}}_{\mathrm{dp}}, >$$

> so one branch contribution is bounded by

$$> \frac{\kappa\epsilon^2}{> \left(c_f^2 (\tau^{\mathrm{mb}}_{\mathrm{dp}})^2 + \epsilon_c^2\right) > \nu^{\mathrm{mb}}_{J,\mathrm{anc}} >}. >$$

> Because branch simplicity and the admissible source-cluster alphabet allow at most one uniformly transversal deep-past branch per source-cluster channel on the chosen delay scale, each ancestry component contributes at most a fixed finite multiple of that ceiling. Summing over at most

$$> N_{\mathrm{anc}} >$$

> ancestry components gives the claimed finite deep-past suppression bound.

> **Target Corollary (Deep-past suppression from finite cluster ancestry).**
> Assume the deep-past cluster-ancestry-or-exclusion proposition. Then on every controlled late-turn window one has a uniform bound

$$> A^{\mathrm{mb}}_{s,\mathrm{deep}}(t) > \leq > \frac{> N_{\mathrm{anc}}\, \kappa\epsilon^2 >}{> \left(c_f^2 (\tau^{\mathrm{mb}}_{\mathrm{dp}})^2 + \epsilon_c^2\right) > \nu^{\mathrm{mb}}_{J,\mathrm{anc}} >}, >$$

> for every

$$> t \in \bigcup_{W \in \mathcal{W}^{\mathrm{mb}}_{\mathrm{lt}}} W, >$$

> because each admissible ancestry component contributes at most one uniformly transversal deep-past branch at the chosen delay scale and Jacobian floor.

This is the quantity the many-body recapture inequalities should consume. Once the remote self drive is reduced to a finite ancestry count times one ceiling, and once the fold contribution is reduced to the channelwise ceilings

$$F^{\mathrm{mb}}_m$$

from the caustic-transit package, the late-turn comparison law becomes quantitative again.

For the later recapture and closure packages, abbreviate this ancestry ceiling by

$$\overline{A}^{\mathrm{mb}}_{\mathrm{deep}} \equiv \frac{N_{\mathrm{anc}}\, \kappa\epsilon^2}{\left(c_f^2 (\tau^{\mathrm{mb}}_{\mathrm{dp}})^2 + \epsilon_c^2\right) \nu^{\mathrm{mb}}_{J,\mathrm{anc}}}$$

Then

$$A^{\mathrm{mb}}_{s,\mathrm{deep}}(t) \le \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}$$

on every controlled late-turn window, and the recapture margins below may treat

$$\overline{A}^{\mathrm{mb}}_{\mathrm{deep}}$$

as one fixed arithmetic input.

Sixth many-body theorem package: finite escape-observable recapture law

The next replacement burden is the turn mechanism itself. In the planar three-body regime there is no single honest escape coordinate. The recapture theorem must instead dominate a finite family of outward channels at once.

Choose smooth quotient observables

$$\rho^{\mathrm{mb}}_1, \dots, \rho^{\mathrm{mb}}_{K_{\mathrm{esc}}} : \mathcal{Q}^{\mathrm{mb}}_{\mathrm{pl}} \to [0, \infty)$$

adapted to the Jacobi coordinates

$$\mathbf{a} = \mathbf{x}_1 - \mathbf{x}_3, \qquad \mathbf{b} = \mathbf{x}_2 - \frac{\mathbf{x}_1 + \mathbf{x}_3}{2}$$

For the first three-body bridge the natural concrete choice is

$$\rho^{\mathrm{mb}}_1(\mathbf{a}, \mathbf{b}) \equiv \|\mathbf{a}\|$$

$$\rho^{\mathrm{mb}}_2(\mathbf{a}, \mathbf{b}) \equiv \|\mathbf{b}\|$$

$$\rho^{\mathrm{mb}}_3(\mathbf{a}, \mathbf{b}) \equiv \frac{\left|\det(\mathbf{a}, \mathbf{b})\right|}{\|\mathbf{a}\| + \|\mathbf{b}\|}$$

$$\rho^{\mathrm{mb}}_4(\mathbf{a}, \mathbf{b}) \equiv \max\left\{\|\mathbf{x}_1\|, \|\mathbf{x}_2\|, \|\mathbf{x}_3\|\right\} - \|\mathbf{x}_2\|$$

Their intended meanings are:

-

$$\rho^{\mathrm{mb}}_1$$

---

measures same-sign outer-pair separation;

$$\rho^{\mathrm{mb}}_2$$

---

measures separation between the opposite-sign body and the outer-pair midpoint;

$$\rho^{\mathrm{mb}}_3$$

---

measures shear or triangle-area escape, normalized to avoid pure scale inflation dominating the shape signal;

$$\rho^{\mathrm{mb}}_4$$

measures farthest-body exchange, vanishing when the opposite-sign body remains the distinguished core body and becoming positive when one outer body threatens to take over that role.

These are not claimed to be the only possible observables. They are the first concrete family that matches the chosen gauge and the intended failure channels.

Let

$$I^{\mathrm{mb}}_{\mathrm{post}}$$

and

$$I^{\mathrm{mb}}_{\mathrm{late}}$$

denote the post-crossing and late-turn windows, respectively. For each observable

$$\rho^{\mathrm{mb}}_{m}$$

write its one-cycle comparison law abstractly as

$$\ddot{\rho}^{\mathrm{mb}}_{m}(t) \le -\Lambda^{\mathrm{mb}}_{m}(t) + L^{\mathrm{mb}}_{m}(t) + A^{\mathrm{mb}}_{s,\mathrm{deep}}(t)$$

where, for each

$$m$$

the comparison terms are understood channelwise:

-
$$\Lambda^{\mathrm{mb}}_{m}(t) \ge 0$$

is the net inward or recapturing contribution relevant to channel

$$m;$$

-
$$L^{\mathrm{mb}}_{m}(t) \ge 0$$

---

is the leakage budget from all other channels and gauge-motion terms;

$$A^{\mathrm{mb}}_{s,\mathrm{deep}}(t)$$

is the deep-past ceiling supplied by the cluster-ancestry package.

To make these comparison laws honest theorem objects, the principal channels should only be differentiated on windows where the relevant denominators and branches stay uniformly away from their singular sets. Introduce the smooth-window floors

$$\|\mathbf{a}(t)\| \ge a_{\min} > 0, \qquad \|\mathbf{b}(t)\| \ge b_{\min} > 0,$$

$$|\det(\mathbf{a}(t), \mathbf{b}(t))| \ge \Delta_{\min} > 0 \qquad \text{on sign-fixed shear windows,}$$

$$\delta_{\mathrm{role}}(t) \ge \delta_{\mathrm{role},\min} > 0 \qquad \text{on role windows.}$$

Accordingly, the smooth recapture windows should be refined as

$$I^{\mathrm{mb}}_{1,\mathrm{post}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{post}} \mid \|\mathbf{a}(t)\| \ge a_{\min} \right\},$$

$$I^{\mathrm{mb}}_{1,\mathrm{late}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{late}} \mid \|\mathbf{a}(t)\| \ge a_{\min} \right\},$$

$$I^{\mathrm{mb}}_{2,\mathrm{post}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{post}} \mid \|\mathbf{b}(t)\| \ge b_{\min} \right\},$$

$$I^{\mathrm{mb}}_{2,\mathrm{late}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{late}} \mid \|\mathbf{b}(t)\| \ge b_{\min} \right\},$$

$$I^{\mathrm{mb}}_{3,\mathrm{post}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{post}} \mid |\det(\mathbf{a}(t), \mathbf{b}(t))| \ge \Delta_{\min} \right\},$$

$$I^{\mathrm{mb}}_{3,\mathrm{late}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{late}} \mid |\det(\mathbf{a}(t), \mathbf{b}(t))| \ge \Delta_{\min} \right\},$$

$$I^{\mathrm{mb}}_{4,\mathrm{post}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{post}} \mid \delta_{\mathrm{role}}(t) \ge \delta_{\mathrm{role},\min} \right\},$$

$$I^{\mathrm{mb}}_{4,\mathrm{late}} \equiv \left\{ t \in I^{\mathrm{mb}}_{\mathrm{late}} \mid \delta_{\mathrm{role}}(t) \ge \delta_{\mathrm{role},\min} \right\}.$$

If one of these floors collapses, that should not be hidden inside the recapture inequalities. It is a separate named event: pair-collapse, midpoint-collapse, shear-sign switching, or role-tie loss. The principal channel theorems should therefore be read only on the corresponding smooth windows above.

For the first three-body bridge these terms should be read concretely as follows:

- for

$$\rho_1^{\mathrm{mb}},$$

the leading inward term should come from the delayed attraction of the opposite-sign body on the same-sign pair, while leakage includes tangential rotation of the pair axis and same-sign self-repulsive widening;

- for

$$\rho_2^{\mathrm{mb}},$$

the leading inward term should come from the combined opposite-sign attractions between body

$$2$$

and the outer pair, while leakage includes pair breathing and quotient-frame acceleration of the midpoint;

- for

$$\rho_3^{\mathrm{mb}},$$

the leading inward term should come from delayed flattening or anti-shear alignment, while leakage includes uniform breathing that preserves aspect ratio only to lower order;

- for

$$\rho_4^{\mathrm{mb}},$$

the leading inward term should come from the persistence of the distinguished opposite-sign core role, while leakage includes any near-tie in body radii that threatens a role exchange.

The point of writing the comparison law channelwise is that one does not need one scalar miracle quantity. One needs a finite family of inequalities whose common positivity excludes all geometrically natural scattering routes.

The first two channels admit a direct Jacobi-level expansion that should anchor the later recapture estimates.

For

$$\rho_1^{\mathrm{mb}} = \|\mathbf{a}\|, \qquad \hat{\mathbf{a}} \equiv \frac{\mathbf{a}}{\|\mathbf{a}\|},$$

one has on every noncollision, non-pair-collapse window

$$\dot{\rho}_1^{\mathrm{mb}} = \hat{\mathbf{a}} \cdot \dot{\mathbf{a}},$$

and

$$\ddot{\rho}_1^{\mathrm{mb}} = \hat{\mathbf{a}} \cdot \ddot{\mathbf{a}} \; + \; \frac{\|\dot{\mathbf{a}}\|^2 - \left(\hat{\mathbf{a}} \cdot \dot{\mathbf{a}}\right)^2}{\|\mathbf{a}\|}.$$

Writing

$$\ddot{\mathbf{a}} = \ddot{\mathbf{x}}_1 - \ddot{\mathbf{x}}_3,$$

the first term is the true pair-axis forcing and the second is the tangential leakage caused by rotation of the same-sign pair axis in the plane. Since the leakage term is nonnegative, the useful inward comparison law is obtained by isolating the pair-axis component

$$\Lambda_1^{\mathrm{mb}}(t) \equiv -\hat{\mathbf{a}}(t) \cdot \ddot{\mathbf{a}}(t),$$

and the geometric leakage ceiling

$$L_1^{\mathrm{mb}}(t) \equiv \frac{\|\dot{\mathbf{a}}(t)\|^2 - \left(\hat{\mathbf{a}}(t) \cdot \dot{\mathbf{a}}(t)\right)^2}{\|\mathbf{a}(t)\|} + L_{1,\mathrm{rep}}^{\mathrm{mb}}(t),$$

where

$$L_{1,\mathrm{rep}}^{\mathrm{mb}}(t)$$

is the explicit ceiling for same-sign self-driven widening and any partner-mediated outward component not absorbed into

$$\Lambda_1^{\mathrm{mb}}.$$

The recapture target for

$$\rho_1^{\mathrm{mb}}$$

is therefore

$$\ddot{\rho}_1^{\mathrm{mb}}(t) \le -\Lambda_1^{\mathrm{mb}}(t) + L_1^{\mathrm{mb}}(t) + A_{s,\mathrm{deep}}^{\mathrm{mb}}(t).$$

To connect this directly to the master equation, write

$$\ddot{\mathbf{x}}_i(t) = \sum_{j=1}^{3} \sum_{s \in \mathcal{C}_{ij}(t)} \kappa\, \sigma_{ij} \frac{|q_i q_j|}{r_{ij}(t;s)^2\, |J_{ij}(t;s)|} \hat{\mathbf{r}}_{ij}(t;s),$$

with the usual convention that the

$$j = i$$

terms are self branches. Then

$$\Lambda_1^{\mathrm{mb}}(t) = -\hat{\mathbf{a}}(t) \cdot \big(\ddot{\mathbf{x}}_1(t) - \ddot{\mathbf{x}}_3(t)\big)$$

splits into branch families

$$\Lambda_1^{\mathrm{mb}} = \Lambda_{1,\mathrm{core}}^{\mathrm{mb}} - \Lambda_{1,\mathrm{same}}^{\mathrm{mb}} - \Lambda_{1,\mathrm{self}}^{\mathrm{mb}},$$

where:

- $$\Lambda_{1,\mathrm{core}}^{\mathrm{mb}}$$

is the inward projection of the opposite-sign body

$$2$$

acting on the same-sign pair

$$(1,3)$$

- $$\Lambda_{1,\mathrm{same}}^{\mathrm{mb}}$$

collects the outward projection of the direct same-sign interaction between bodies

$$1$$

and

$$3;$$

- $$\Lambda_{1,\mathrm{self}}^{\mathrm{mb}}$$

collects the outward pair-axis projections of the self branches on bodies

$$1$$

and

$$3$$

The unresolved but now explicit theorem burden is to prove, on the recapture windows, that the opposite-sign core term dominates the same-sign and self-driven widening after all admissible fold-tube and deep-past ceilings are paid.

Accordingly, a first branch-sum ceiling for the residual widening term should be written as

$$L^{\mathrm{mb}}_{1,\mathrm{rep}}(t) \equiv \left(\Lambda^{\mathrm{mb}}_{1,\mathrm{same}}(t)\right)_+ + \left(\Lambda^{\mathrm{mb}}_{1,\mathrm{self}}(t)\right)_+ + L^{\mathrm{mb}}_{1,\mathrm{fold}}(t)$$

where

$$(\cdot)_+$$

denotes positive part and

$$L^{\mathrm{mb}}_{1,\mathrm{fold}}(t)$$

is the additional local ceiling needed near controlled coupled folds. In other words, the pair-separation channel is reduced to a contest between one inward projected source family and a finite list of outward projected branch families.

For

$$\rho^{\mathrm{mb}}_2 = \|\mathbf{b}\|, \qquad \hat{\mathbf{b}} \equiv \frac{\mathbf{b}}{\|\mathbf{b}\|}$$

one has likewise

$$\dot{\rho}^{\mathrm{mb}}_2 = \hat{\mathbf{b}} \cdot \dot{\mathbf{b}}$$

and

$$\ddot{\rho}^{\mathrm{mb}}_2 = \hat{\mathbf{b}} \cdot \ddot{\mathbf{b}} \; + \; \frac{\|\dot{\mathbf{b}}\|^2 - \left(\hat{\mathbf{b}} \cdot \dot{\mathbf{b}}\right)^2}{\|\mathbf{b}\|}$$

Since

$$\ddot{\mathbf{b}} = \ddot{\mathbf{x}}_2 - \frac{\ddot{\mathbf{x}}_1 + \ddot{\mathbf{x}}_3}{2}$$

the first term measures the net delayed pull of the opposite-sign body toward or away from the outer-pair midpoint, while the second term is the transverse leakage created by midpoint-frame rotation and shape drift. The natural channelwise definitions are therefore

$$\Lambda^{\mathrm{mb}}_2(t) \equiv -\hat{\mathbf{b}}(t) \cdot \ddot{\mathbf{b}}(t)$$

$$L^{\mathrm{mb}}_2(t) \equiv \frac{\|\dot{\mathbf{b}}(t)\|^2 - \left(\hat{\mathbf{b}}(t) \cdot \dot{\mathbf{b}}(t)\right)^2}{\|\mathbf{b}(t)\|} + L^{\mathrm{mb}}_{2,\mathrm{breath}}(t)$$

where

$$L^{\mathrm{mb}}_{2,\mathrm{breath}}(t)$$

collects the pair-breathing and quotient-frame error terms generated by the simultaneous motion of

$$\mathbf{a}$$

and

$$\mathbf{b}$$

The recapture target for

$$\rho^{\mathrm{mb}}_2$$

is thus

$$\ddot{\rho}^{\mathrm{mb}}_2(t) \leq -\Lambda^{\mathrm{mb}}_2(t) + L^{\mathrm{mb}}_2(t) + A^{\mathrm{mb}}_{s,\mathrm{deep}}(t)$$

Here the master-equation projection is

$$\Lambda^{\mathrm{mb}}_2(t) = -\hat{\mathbf{b}}(t) \cdot \left(\ddot{\mathbf{x}}_2(t) - \frac{\ddot{\mathbf{x}}_1(t) + \ddot{\mathbf{x}}_3(t)}{2}\right)$$

and this naturally decomposes into:

$$\Lambda^{\mathrm{mb}}_2 = \Lambda^{\mathrm{mb}}_{2,\mathrm{attr}} - \Lambda^{\mathrm{mb}}_{2,\mathrm{pair}} - \Lambda^{\mathrm{mb}}_{2,\mathrm{self}}$$

where:

-

$$\Lambda^{\mathrm{mb}}_{2,\mathrm{attr}}$$

is the inward projection of the combined opposite-sign attractions between body

$$2$$

---

and the outer pair;

$$\Lambda^{\mathrm{mb}}_{2,\mathrm{pair}}$$

---

is the outward projection induced by breathing of the same-sign pair midpoint itself;

$$\Lambda^{\mathrm{mb}}_{2,\mathrm{self}}$$

is the outward projection of self branches on all three bodies onto the

$$\hat{\mathbf{b}}$$

direction.

The natural branch-sum leakage ceiling is therefore

$$L^{\mathrm{mb}}_{2,\mathrm{breath}}(t) \equiv \left(\Lambda^{\mathrm{mb}}_{2,\mathrm{pair}}(t)\right)_+ + \left(\Lambda^{\mathrm{mb}}_{2,\mathrm{self}}(t)\right)_+ + L^{\mathrm{mb}}_{2,\mathrm{fold}}(t),$$

with

$$L^{\mathrm{mb}}_{2,\mathrm{fold}}(t)$$

again denoting the explicit ceiling for fold-tube amplification and local gauge-frame error terms.

So the midpoint-separation channel is likewise reduced to one concrete branch-sum competition: the combined inward opposite-sign attraction against pair breathing, self-drive, transverse rotation, and controlled fold amplification.

For the shear channel

$$\rho^{\mathrm{mb}}_3(\mathbf{a},\mathbf{b}) = \frac{|\det(\mathbf{a},\mathbf{b})|}{\|\mathbf{a}\| + \|\mathbf{b}\|},$$

write

$$\Delta_{ab} \equiv \det(\mathbf{a},\mathbf{b}), \qquad S_{ab} \equiv \|\mathbf{a}\| + \|\mathbf{b}\|.$$

On one fixed sign branch

$$\operatorname{sign}(\Delta_{ab}) = \varsigma \in \{\pm 1\},$$

the observable becomes

$$\rho^{\mathrm{mb}}_3 = \varsigma \frac{\Delta_{ab}}{S_{ab}},$$

so its first derivative may be organized as

$$\dot{\rho}^{\mathrm{mb}}_3 = \varsigma \frac{\det(\dot{\mathbf{a}},\mathbf{b}) + \det(\mathbf{a},\dot{\mathbf{b}})}{S_{ab}} - \varsigma \frac{\Delta_{ab}\,\dot{S}_{ab}}{S^2_{ab}}.$$

The theorem burden is not to memorize the full second derivative term-by-term, but to split it into

$$\ddot{\rho}^{\mathrm{mb}}_3(t) \le -\Lambda^{\mathrm{mb}}_3(t) + L^{\mathrm{mb}}_3(t) + A^{\mathrm{mb}}_{s,\mathrm{deep}}(t),$$

where

$$\Lambda^{\mathrm{mb}}_3(t)$$

is the net anti-shear projection coming from delayed flattening of the triangle relative to the chosen sign branch, and

$$L_3^{\mathrm{mb}}(t)$$

collects denominator breathing, branch-sign switching near

$$\Delta_{ab} = 0,$$

and fold-tube amplification. A first useful decomposition is

$$\Lambda_3^{\mathrm{mb}} = \Lambda_{3,\mathrm{flat}}^{\mathrm{mb}} - \Lambda_{3,\mathrm{swap}}^{\mathrm{mb}} - \Lambda_{3,\mathrm{self}}^{\mathrm{mb}},$$

where:

-

$$\Lambda_{3,\mathrm{flat}}^{\mathrm{mb}}$$

is the inward projection of delayed flattening or anti-shear alignment;

$$\Lambda_{3,\mathrm{swap}}^{\mathrm{mb}}$$

is the outward projection produced by pair breathing or source-cluster swaps that increase signed area;

$$\Lambda_{3,\mathrm{self}}^{\mathrm{mb}}$$

is the outward shear contribution of self branches.

The corresponding leakage ceiling should be written as

$$L_3^{\mathrm{mb}}(t) \equiv \left(\Lambda_{3,\mathrm{swap}}^{\mathrm{mb}}(t)\right)_+ + \left(\Lambda_{3,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ + L_{3,\mathrm{den}}^{\mathrm{mb}}(t) + L_{3,\mathrm{fold}}^{\mathrm{mb}}(t)$$

where

$$L_{3,\mathrm{den}}^{\mathrm{mb}}(t)$$

controls the derivative loss from the breathing denominator

$$S_{ab}^{-1}$$

and

$$L_{3,\mathrm{fold}}^{\mathrm{mb}}(t)$$

controls fold-tube and sign-branch switching errors near the small-area set.

For the role-exchange channel

$$\rho_4^{\mathrm{mb}} = \max\left\{\|\mathbf{x}_1\|, \|\mathbf{x}_2\|, \|\mathbf{x}_3\|\right\} - \|\mathbf{x}_2\|$$

the correct local formulation is piecewise. On any subwindow where one outer body, say

$$\mathbf{x}_{i_*}, \qquad i_* \in \{1,3\}$$

uniquely realizes the maximum radius with a gap

$$\delta_{\mathrm{role}} > 0$$

one has the smooth branch

$$\rho_4^{\mathrm{mb}} = \|\mathbf{x}_{i_*}\| - \|\mathbf{x}_2\|$$

Then, with

$$\hat{\mathbf{x}}_{i_*} \equiv \frac{\mathbf{x}_{i_*}}{\|\mathbf{x}_{i_*}\|}, \qquad \hat{\mathbf{x}}_2 \equiv \frac{\mathbf{x}_2}{\|\mathbf{x}_2\|}$$

the channelwise comparison law should be organized as

$$\ddot{\rho}_4^{\mathrm{mb}}(t) \leq -\Lambda_4^{\mathrm{mb}}(t) + L_4^{\mathrm{mb}}(t) + A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)$$

with

$$\Lambda_4^{\mathrm{mb}}(t) \equiv -\hat{\mathbf{x}}_{i_*}(t) \cdot \ddot{\mathbf{x}}_{i_*}(t) + \hat{\mathbf{x}}_2(t) \cdot \ddot{\mathbf{x}}_2(t)$$

interpreted as persistence of the distinguished opposite-sign core role, and with leakage ceiling

$$L_4^{\mathrm{mb}}(t) \equiv L_{4,\mathrm{curv}}^{\mathrm{mb}}(t) + L_{4,\mathrm{tie}}^{\mathrm{mb}}(t) + L_{4,\mathrm{fold}}^{\mathrm{mb}}(t)$$

Here:

- $$L_{4,\mathrm{curv}}^{\mathrm{mb}}$$

---

collects the usual tangential curvature terms from differentiating the two norms;

$$L_{4,\mathrm{tie}}^{\mathrm{mb}}$$

controls the loss of smoothness near a near-tie

$$\|\mathbf{x}_{i_*}\| \approx \|\mathbf{x}_2\|$$

or

$$\|\mathbf{x}_1\| \approx \|\mathbf{x}_3\|$$

- $$L_{4,\mathrm{fold}}^{\mathrm{mb}}$$

controls fold-tube amplification on the chosen role branch.

The intended theorem burden is to show that, on controlled subwindows with a strict role gap

$$\delta_{\mathrm{role}} > 0,$$

the inward role-persistence term beats curvature, tie, self-drive, and fold leakage strongly enough to keep the opposite-sign body from losing the distinguished core role.

These four formulas are the concrete comparison identities the planar-three-body bridge should use. They say where the useful inward coercivity must come from and where the leakage terms enter for all principal escape channels.

The next step is to specify the first recapture windows for the principal channels, beginning with the first two Jacobi channels for which the cleanest local turning lemma is available. Let

$$t_{\mathrm{x}}^{\mathrm{mb}}$$

denote the first near-core crossing or minimum-core event after the initial inbound section, and let

$$t_{\mathrm{turn}}^{\mathrm{mb}}$$

denote the first later time at which the distinguished opposite-sign body and the same-sign outer pair are both candidates for outward escape rather than continued compression. The first controlled windows should then be taken as

$$I_{\rho,\mathrm{post}}^{\mathrm{mb}} \equiv \left[t_{\mathrm{x}}^{\mathrm{mb}},\, t_{\mathrm{x}}^{\mathrm{mb}} + \Delta_{\rho,\mathrm{post}}\right],$$

$$I_{\rho,\mathrm{late}}^{\mathrm{mb}} \equiv \left[t_{\mathrm{turn}}^{\mathrm{mb}} - \Delta_{\rho,\mathrm{late}},\, t_{\mathrm{turn}}^{\mathrm{mb}}\right],$$

with positive window lengths

$$\Delta_{\rho,\mathrm{post}}, \qquad \Delta_{\rho,\mathrm{late}}.$$

On

$$I^{\mathrm{mb}}_{\rho,\mathrm{post}}$$

the intended role is immediate post-core recapture: the same-sign pair must not continue widening unchecked, and the opposite-sign body must not continue peeling away from the outer-pair midpoint. On

$$I^{\mathrm{mb}}_{\rho,\mathrm{late}}$$

the intended role is late-turn closure: the same observables must already be losing outward momentum before a genuine scattering configuration forms.

> **Target Proposition (Two-channel Jacobi recapture on the first planar-three-body windows).**
> Assume the section package, the bounded caustic-transit package, the finite active delay hypergraph package, and the deep-past suppression bound

$$> A^{\mathrm{mb}}_{s,\mathrm{deep}}(t) >\leq> \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}; >$$

> together with the fold ceilings produced there

$$> L^{\mathrm{mb}}_{1,\mathrm{fold}}(t) \leq F^{\mathrm{mb}}_1, > \qquad > L^{\mathrm{mb}}_{2,\mathrm{fold}}(t) \leq F^{\mathrm{mb}}_2 >$$

> on the two windows above, with

$$> F^{\mathrm{mb}}_1 >=> \mathfrak{F}^{\mathrm{mb}}_1 M^{\mathrm{mb}}_{\max}, > \qquad > F^{\mathrm{mb}}_2 >=> \mathfrak{F}^{\mathrm{mb}}_2 M^{\mathrm{mb}}_{\max}. >$$

> Suppose, in addition, that on each of the smooth windows

$$> I^{\mathrm{mb}}_{1,\mathrm{post}} \cap I^{\mathrm{mb}}_{2,\mathrm{post}}, > \qquad > I^{\mathrm{mb}}_{1,\mathrm{late}} \cap I^{\mathrm{mb}}_{2,\mathrm{late}}, >$$

> one has strict projected inequalities

$$> \Lambda^{\mathrm{mb}}_{1,\mathrm{core}} >>> \left(\Lambda^{\mathrm{mb}}_{1,\mathrm{same}}\right)_+ > + > \left(\Lambda^{\mathrm{mb}}_{1,\mathrm{self}}\right)_+ > + > \frac{\|\dot{\mathbf{a}}\|^2 - \left(\hat{\mathbf{a}} \cdot \dot{\mathbf{a}}\right)^2}{\|\mathbf{a}\|} > + > F^{\mathrm{mb}}_1 > + > \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}, >$$

$$> \Lambda^{\mathrm{mb}}_{2,\mathrm{attr}} >>> \left(\Lambda^{\mathrm{mb}}_{2,\mathrm{pair}}\right)_+ > + > \left(\Lambda^{\mathrm{mb}}_{2,\mathrm{self}}\right)_+ > + > \frac{\|\dot{\mathbf{b}}\|^2 - \left(\hat{\mathbf{b}} \cdot \dot{\mathbf{b}}\right)^2}{\|\mathbf{b}\|} > + > F^{\mathrm{mb}}_2 > + > \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}. >$$

> Then the two principal escape observables obey

$$> \ddot{\rho}^{\mathrm{mb}}_1(t) < 0, > \qquad > \ddot{\rho}^{\mathrm{mb}}_2(t) < 0 >$$

> throughout those windows.
>
> In particular:
>
> 1. if

$$> \dot{\rho}^{\mathrm{mb}}_1 >$$

> or

$$> \dot{\rho}^{\mathrm{mb}}_2 >$$

> is initially positive at the start of either window, it must strictly decrease along that window;
> 2. if the window is long enough and the initial outward rates are uniformly bounded, each observable reaches a turning time inside the window;
> 3. therefore the pair-separation and midpoint-separation channels cannot both sustain monotone outward escape across the first post-crossing or late-turn stages.

This is the first honest recapture lemma for the planar three-body bridge. It does not yet prove the full many-body return, but it reduces the first two escape channels to explicit projected inequalities on controlled windows.

> **Proof draft of the two-channel Jacobi recapture proposition.**
> On the smooth windows

$$> I^{\mathrm{mb}}_{1,\mathrm{post}} \cap I^{\mathrm{mb}}_{2,\mathrm{post}}, > \qquad > I^{\mathrm{mb}}_{1,\mathrm{late}} \cap I^{\mathrm{mb}}_{2,\mathrm{late}}, >$$

> the observables

$$> \rho_1^{\mathrm{mb}} = \|\mathbf{a}\|, > \qquad > \rho_2^{\mathrm{mb}} = \|\mathbf{b}\| >$$

> are twice differentiable and obey the comparison identities already derived above.
>
> 1. **Channel ρ_1^{mb}.**
>   Insert the decomposition

$$> \Lambda_1^{\mathrm{mb}} >=> \Lambda_{1,\mathrm{core}}^{\mathrm{mb}} > - > \Lambda_{1,\mathrm{same}}^{\mathrm{mb}} > - > \Lambda_{1,\mathrm{self}}^{\mathrm{mb}} >$$

> into the identity for

$$> \ddot{\rho}_1^{\mathrm{mb}}. >$$

> Bound the outward pieces by positive part, control the fold contribution by

$$> L_{1,\mathrm{fold}}^{\mathrm{mb}}(t) \le F_1^{\mathrm{mb}}, >$$

> and bound the deep-past term by

$$> A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \le \overline{A}_{\mathrm{deep}}^{\mathrm{mb}}. >$$

> The stated projected inequality then gives

$$> \ddot{\rho}_1^{\mathrm{mb}}(t) < 0. >$$

> 2. **Channel ρ_2^{mb}.**
>   The same argument applied to

$$> \Lambda_2^{\mathrm{mb}} >=> \Lambda_{2,\mathrm{attr}}^{\mathrm{mb}} > - > \Lambda_{2,\mathrm{pair}}^{\mathrm{mb}} > - > \Lambda_{2,\mathrm{self}}^{\mathrm{mb}} >$$

> yields

$$> \ddot{\rho}_2^{\mathrm{mb}}(t) < 0. >$$

> 3. **Turning inside the window.**
>   If either outward rate is initially positive, strict negativity of the second derivative forces that outward rate to decrease monotonically. Therefore, provided the window length dominates the initial slope divided by the corresponding margin, the observable reaches an inward-turning time before the end of the window.
>
> 4. **Joint exclusion of monotone escape.**
>   Since both

$$> \ddot{\rho}_1^{\mathrm{mb}} > \qquad > \text{and} > \qquad > \ddot{\rho}_2^{\mathrm{mb}} >$$

> are strictly negative on the same controlled windows, the pair-separation and midpoint-separation channels cannot both sustain monotone outward escape through the post-crossing or late-turn stage.

To align the abstract recapture theorem with the first concrete lemma, the principal margins for

$$m = 1, 2$$

should be defined directly from the projected branch-sum gaps above. Namely

$$\mathfrak{M}_{1,\mathrm{post}}^{\mathrm{mb}} \equiv \inf_{t \in I_{1,\mathrm{post}}^{\mathrm{mb}}} \left[\Lambda_{1,\mathrm{core}}^{\mathrm{mb}}(t) - \left(\Lambda_{1,\mathrm{same}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{1,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - \frac{\|\dot{\mathbf{a}}(t)\|^2 - \left(\hat{\mathbf{a}}(t) \cdot \dot{\mathbf{a}}(t)\right)^2}{\|\mathbf{a}(t)\|} - L_{1,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \right],$$

$$\mathfrak{M}_{1,\mathrm{late}}^{\mathrm{mb}} \equiv \inf_{t \in I_{1,\mathrm{late}}^{\mathrm{mb}}} \left[\Lambda_{1,\mathrm{core}}^{\mathrm{mb}}(t) - \left(\Lambda_{1,\mathrm{same}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{1,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - \frac{\|\dot{\mathbf{a}}(t)\|^2 - \left(\hat{\mathbf{a}}(t) \cdot \dot{\mathbf{a}}(t)\right)^2}{\|\mathbf{a}(t)\|} - L_{1,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \right],$$

$$\mathfrak{M}_{2,\mathrm{post}}^{\mathrm{mb}} \equiv \inf_{t \in I_{2,\mathrm{post}}^{\mathrm{mb}}} \left[\Lambda_{2,\mathrm{attr}}^{\mathrm{mb}}(t) - \left(\Lambda_{2,\mathrm{pair}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{2,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - \frac{\|\dot{\mathbf{b}}(t)\|^2 - \left(\hat{\mathbf{b}}(t) \cdot \dot{\mathbf{b}}(t)\right)^2}{\|\mathbf{b}(t)\|} - L_{2,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \right],$$

$$\mathfrak{M}_{2,\mathrm{late}}^{\mathrm{mb}} \equiv \inf_{t \in I_{2,\mathrm{late}}^{\mathrm{mb}}} \left[\Lambda_{2,\mathrm{attr}}^{\mathrm{mb}}(t) - \left(\Lambda_{2,\mathrm{pair}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{2,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - \frac{\|\dot{\mathbf{b}}(t)\|^2 - \left(\hat{\mathbf{b}}(t) \cdot \dot{\mathbf{b}}(t)\right)^2}{\|\mathbf{b}(t)\|} - L_{2,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \right].$$

Here

$$L_{1,\mathrm{fold}}^{\mathrm{mb}}(t) \le F_1^{\mathrm{mb}}, \qquad L_{2,\mathrm{fold}}^{\mathrm{mb}}(t) \le F_2^{\mathrm{mb}}$$

are understood as direct outputs of the bounded many-body caustic-transit package, not as independent recapture assumptions.
On the late-turn windows, the ancestry package likewise provides the fixed ceiling

$$A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \le \overline{A}_{\mathrm{deep}}^{\mathrm{mb}}.$$

So the first two principal margins may be read as explicit arithmetic inequalities with already-settled fold and deep-past inputs.

For the shear and role-exchange channels, the principal margins should likewise be defined from the concrete decompositions above

$$\mathfrak{M}_{3,\mathrm{post}}^{\mathrm{mb}} \equiv \inf_{t\in I_{3,\mathrm{post}}^{\mathrm{mb}}} \left[\Lambda_{3,\mathrm{flat}}^{\mathrm{mb}}(t) - \left(\Lambda_{3,\mathrm{swap}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{3,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - L_{3,\mathrm{den}}^{\mathrm{mb}}(t) - L_{3,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right],$$

$$\mathfrak{M}_{3,\mathrm{late}}^{\mathrm{mb}} \equiv \inf_{t\in I_{3,\mathrm{late}}^{\mathrm{mb}}} \left[\Lambda_{3,\mathrm{flat}}^{\mathrm{mb}}(t) - \left(\Lambda_{3,\mathrm{swap}}^{\mathrm{mb}}(t)\right)_+ - \left(\Lambda_{3,\mathrm{self}}^{\mathrm{mb}}(t)\right)_+ - L_{3,\mathrm{den}}^{\mathrm{mb}}(t) - L_{3,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right],$$

$$\mathfrak{M}_{4,\mathrm{post}}^{\mathrm{mb}} \equiv \inf_{t\in I_{4,\mathrm{post}}^{\mathrm{mb}}} \left[\Lambda_4^{\mathrm{mb}}(t) - L_{4,\mathrm{curv}}^{\mathrm{mb}}(t) - L_{4,\mathrm{tie}}^{\mathrm{mb}}(t) - L_{4,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right],$$

$$\mathfrak{M}_{4,\mathrm{late}}^{\mathrm{mb}} \equiv \inf_{t\in I_{4,\mathrm{late}}^{\mathrm{mb}}} \left[\Lambda_4^{\mathrm{mb}}(t) - L_{4,\mathrm{curv}}^{\mathrm{mb}}(t) - L_{4,\mathrm{tie}}^{\mathrm{mb}}(t) - L_{4,\mathrm{fold}}^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right],$$

where the smooth-window floors

$$a_{\min}, \qquad b_{\min}, \qquad \Delta_{\min}, \qquad \delta_{\mathrm{role},\min}$$

are part of the recapture hypotheses rather than hidden background assumptions.

Likewise, the fold-loss terms

$$L_{3,\mathrm{fold}}^{\mathrm{mb}}(t) \le F_3^{\mathrm{mb}}, \qquad L_{4,\mathrm{fold}}^{\mathrm{mb}}(t) \le F_4^{\mathrm{mb}}$$

come from the same bounded caustic-transit package, with

$$F_3^{\mathrm{mb}} = \mathfrak{F}_3^{\mathrm{mb}} M_{\max}^{\mathrm{mb}}, \qquad F_4^{\mathrm{mb}} = \mathfrak{F}_4^{\mathrm{mb}} M_{\max}^{\mathrm{mb}}.$$

Together with

$$A_{s,\mathrm{deep}}^{\mathrm{mb}}(t) \le \overline{A}_{\mathrm{deep}}^{\mathrm{mb}},$$

this means the principal four-channel closure theorem below consumes only fixed ceilings coming from the caustic-transit and ancestry packages, not any additional unresolved path-history term.

For the remaining channels

$$m = 5, \ldots, K_{\mathrm{esc}},$$

retain the abstract definitions

$$\mathfrak{M}_{m,\mathrm{post}}^{\mathrm{mb}} \equiv \inf_{t\in I_{\mathrm{post}}^{\mathrm{mb}}} \left(\Lambda_m^{\mathrm{mb}}(t) - L_m^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right),$$

$$\mathfrak{M}_{m,\mathrm{late}}^{\mathrm{mb}} \equiv \inf_{t\in I_{\mathrm{late}}^{\mathrm{mb}}} \left(\Lambda_m^{\mathrm{mb}}(t) - L_m^{\mathrm{mb}}(t) - A_{s,\mathrm{deep}}^{\mathrm{mb}}(t)\right).$$

With this convention

$$\mathfrak{M}_{1,\mathrm{post}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{1,\mathrm{late}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{2,\mathrm{post}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{2,\mathrm{late}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{3,\mathrm{post}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{3,\mathrm{late}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{4,\mathrm{post}}^{\mathrm{mb}} > 0, \quad \mathfrak{M}_{4,\mathrm{late}}^{\mathrm{mb}} > 0$$

are the concrete margin versions of the four principal escape-channel inequalities. Whenever all of the many-body margins are positive, each escape observable feels a strictly inward comparison law on the corresponding controlled window.

> **Target Proposition (Principal four-channel recapture closure).**
> Assume the section package, the bounded caustic-transit package, the finite active delay hypergraph package, and the deep-past cluster-ancestry suppression bound. Suppose, in addition, that

$$> \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{post}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{late}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{post}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{late}} > 0, >$$

$$> \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{post}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{late}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{post}} > 0, > \quad > \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{late}} > 0. >$$

> Then the four principal escape channels

$$> \rho_1^{\mathrm{mb}}, \rho_2^{\mathrm{mb}}, \rho_3^{\mathrm{mb}}, \rho_4^{\mathrm{mb}} >$$

> cannot sustain a simultaneous scattering drift through the post-crossing and late-turn stages.
>
> More precisely:
>
> 1. the pair-separation channel cannot keep widening monotonically;
> 2. the midpoint-separation channel cannot keep ejecting the opposite-sign body monotonically;
> 3. the shear channel cannot keep increasing triangle-area escape monotonically on one fixed sign branch;
> 4. the role-exchange channel cannot cross into a persistent outer-body takeover on any subwindow with

$$> \delta_{\mathrm{role}} > 0; >$$

> 5. therefore any remaining candidate scattering route must pass through a higher channel

$$> \rho_m^{\mathrm{mb}}, > \qquad > m \geq 5, >$$

> or else violate one of the already-listed fold, tie, or chart hypotheses.

This proposition is the bridge between the local channel calculations and the full many-body recapture theorem. It says that once the four principal channels are controlled, any residual failure is no longer hidden in the obvious geometry; it must come from either a higher auxiliary channel or an explicit closure-stage obstruction.

> **Proof draft of the principal four-channel recapture closure proposition.**
> On the corresponding smooth windows, positivity of

$$> \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{post}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{late}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{post}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{late}} >$$

> gives the two-channel Jacobi turning result already proved above. Likewise, positivity of

$$> \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{post}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{late}} >$$

> forces

$$> \ddot{\rho}_3^{\mathrm{mb}} < 0 >$$

> on the sign-fixed shear windows, and positivity of

$$> \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{post}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{late}} >$$

> forces

$$> \ddot{\rho}_4^{\mathrm{mb}} < 0 >$$

> on the role-gap windows.
> Because the principal margins were defined using

$$> F_1^{\mathrm{mb}}, \ldots, F_4^{\mathrm{mb}} >$$

> and

$$> \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}, >$$

> this already accounts for every controlled fold loss and every controlled deep-past contribution in the four principal channels.
>
> Therefore none of the four principal channels can maintain a positive outward-driving second derivative throughout the controlled post-crossing and late-turn stages. Any candidate excursion that still avoids return must therefore do so through either:
>
> 1. one auxiliary channel

$$> \rho_m^{\mathrm{mb}}, > \qquad > m \geq 5, >$$

> not already covered by the principal four-channel block; or
> 2. breakdown of one named hypothesis, namely collapse of a smooth-window floor, failure of the chart, or one unaccounted fold/tie event.
>
> This is exactly the claimed alternative.

> **Target Theorem (Planar-three-body multi-observable recapture criterion).**
> Assume the section package, the bounded caustic-transit package, the finite active delay hypergraph package, and the deep-past cluster-ancestry suppression bound, all in one coupled parameter regime in which the comparison terms are defined with common constants.
>
> Suppose, in addition, that for every

$$> m = 1, \dots, K_{\mathrm{esc}} >$$

> one has

$$> \mathfrak{M}^{\mathrm{mb}}_{m,\mathrm{post}} > 0, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{m,\mathrm{late}} > 0. >$$

> Then:
>
> 1. no escape observable can continue increasing throughout the full post-crossing window;
> 2. no escape observable can remain positive and outward-driving throughout the late-turn window;
> 3. at least one controlled return event forces the shape back toward the gauge-fixed inbound section without opening a new uncontrolled scattering channel;
> 4. consequently the candidate excursion makes both the post-crossing recapture turn and the late-turn return inside the controlled planar-three-body windows.

> Here the only nonlocal error budgets entering the comparison laws are the fixed ceilings

$$> F^{\mathrm{mb}}_m, > \qquad > m = 1, \dots, 4, >$$

> from the bounded caustic-transit package and

$$> \overline{A}^{\mathrm{mb}}_{\mathrm{deep}} >$$

> from the cluster-ancestry package. Once those constants are fixed, the recapture criterion is reduced to strict positivity of finitely many margin inequalities on the controlled windows.

This is the first honest many-body recapture theorem target. It says that the configuration does not merely avoid one preferred binary escape. It must fail to escape in every channel that the three-body quotient geometry naturally opens.

Seventh many-body theorem package: atlas-level tame-envelope closure

The remaining bridge step is now the same structural one that appeared in the 1D, reduced-planar, and unreduced-planar programs: put the whole cycle on one closed convex tame self-map domain. The only difference is that the data to be preserved are now genuinely atlas-level.

Let

$$\mathcal{C}^{\mathrm{mb}}_{A_*,\eta} \subseteq \Sigma^{-,\mathrm{mb}}_{A_*}$$

denote the convex section envelope carrying only visible Banach-space bounds, such as:

- uniform bounds on the Jacobi vectors

$$\mathbf{a}, \mathbf{b};$$

- uniform velocity and Lipschitz acceleration bounds;
- memory-depth bounds;
- and the section-side non-near-collinearity margin.

This convex envelope should remain purely kinematic. It should not be defined by fixing one hypergraph, one ancestry complex, or one exact branch-count pattern, because those are not visibly convex invariants.

The next theorem burden is therefore not yet a convex tame set, but a stability proposition on the convex Banach box:

> **Target Proposition (Hypergraph and atlas stability on the convex Banach envelope).**
> On a sufficiently small convex section envelope

$$> \mathcal{C}^{\mathrm{mb}}_{A_*,\eta}, >$$

the later branch-regularity, hypergraph, ancestry, and recapture packages remain stable in the following sense:

1. every history in

$$ > \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} > $$

that satisfies the quantitative no-accumulation floors

$$ > \gamma_{\mathrm{fold}}, > \qquad > \nu^{\mathrm{mb}}_{J}, > \qquad > \Delta\tau_{\mathrm{evt}} > $$

belongs to one finite admissible event class;
2. the corresponding active hypergraph and ancestry data vary only through the listed local event alphabet;
3. the smooth-window floors

$$ > a_{\min}, > \qquad > b_{\min}, > \qquad > \Delta_{\min}, > \qquad > \delta_{\mathrm{role,min}} > $$

and the corresponding principal windows

$$ > I^{\mathrm{mb}}_{m,\mathrm{post}}, > \qquad > I^{\mathrm{mb}}_{m,\mathrm{late}}, > \qquad > m = 1, 2, 3, 4, > $$

remain nondegenerate on one nonempty tame subregion;
4. the bounded caustic-transit package remains uniform there, with one common local multiplicity cap

$$ > M^{\mathrm{mb}}_{\max} \le 3 > $$

and one common family of transit constants

$$ > \mathfrak{F}^{\mathrm{mb}}_{1}, > \qquad > \mathfrak{F}^{\mathrm{mb}}_{2}, > \qquad > \mathfrak{F}^{\mathrm{mb}}_{3}, > \qquad > \mathfrak{F}^{\mathrm{mb}}_{4}, > $$

hence one common family of fold ceilings

$$ > F^{\mathrm{mb}}_{m} >=> \mathfrak{F}^{\mathrm{mb}}_{m} M^{\mathrm{mb}}_{\max}, > \qquad > m = 1, 2, 3, 4; > $$

5. the principal recapture margins on that tame subregion remain bounded away from zero by one common floor

$$ > \mathfrak{m}^{\mathrm{mb}}_{\mathrm{prin}} > 0; > $$

6. and there exists a nonempty convex subset

$$ > \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}} >\subseteq> \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} > $$

lying entirely inside that tame subregion and carrying the same preserved event class, smooth-window floors, fold ceilings, and principal-margin floors.

Proof draft of the hypergraph-and-atlas stability proposition.
Start from one seed history in the convex Banach envelope for which the no-accumulation, caustic-transit, ancestry, and principal recapture packages all hold. Because the envelope controls

$$ > \|\mathbf{X}\|, > \qquad > \|\dot{\mathbf{X}}\|, > \qquad > \mathrm{Lip}(\dot{\mathbf{X}}), > $$

the delay defects and the sector/exchange gap functions depend continuously on the history in the

$$ > C^1 > $$

topology.

1. **Persistence of the event class.**
   Transversal zeros of

$$ > \partial_s g_{ij}, > \qquad > \Theta_{\mathrm{sec}}, > \qquad > \Theta_{\mathrm{exc}} > $$

are structurally stable under sufficiently small

$$ > C^1 > $$

perturbations. Therefore, on a sufficiently small convex neighborhood of the seed, no new event type is created and no listed event disappears except by leaving the neighborhood.

> 2. **Persistence of the smooth windows.**
>   The floors

$$> a_{\min}, > \qquad > b_{\min}, > \qquad > \Delta_{\min}, > \qquad > \delta_{\mathrm{role},\min} >$$

> depend continuously on the trajectory. Hence on a sufficiently small tame subregion they remain positive, and the corresponding windows

$$> I^{\mathrm{mb}}_{m,\mathrm{post}}, > \qquad > I^{\mathrm{mb}}_{m,\mathrm{late}} >$$

> remain nondegenerate.
>
> 3. **Persistence of the principal margins.**
>   Every term entering

$$> \mathfrak{M}^{\mathrm{mb}}_{m,\mathrm{post}}, > \qquad > \mathfrak{M}^{\mathrm{mb}}_{m,\mathrm{late}} >$$

> varies continuously on the smooth windows, and the fold-loss terms are uniformly bounded by the same caustic-transit constants. Therefore strict positivity at the seed persists on a sufficiently small tame subregion.
>
> 4. **Convex tame core.**
>   The next proposition below shows how to realize such a subset by explicit tube-and-cone inequalities around the seed. Therefore one may choose a sufficiently small convex subset

$$> \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}} > \subseteq > \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} >$$

> lying entirely inside the tame region where the same event class, smooth-window floors, transit constants, and principal margins are preserved.

The remaining closure burden is then purely analytic: prove that the return map sends this convex tame core into itself.

To make this convex core theorem-level rather than rhetorical, one should define it by visibly convex seed-centered tube and cone conditions, not by delayed-root predicates.

Fix one seed history

$$\Phi^{\mathrm{mb}}_{\mathrm{seed}} \in \mathcal{C}^{\mathrm{mb}}_{A_*,\eta}$$

for which the no-accumulation, caustic-transit, ancestry, and principal recapture packages already hold. Let

$$W \in \left\{ I^{\mathrm{mb}}_{\mathrm{post}}, I^{\mathrm{mb}}_{\mathrm{late}} \right\}.$$

Choose:

- closed planar cones

$$\mathfrak{C}^{a}_{W}, \qquad \mathfrak{C}^{b}_{W}, \qquad \mathfrak{C}^{\mathrm{role}}_{W} \subset \Pi$$

for the pair axis

$$\mathbf{a},$$

the midpoint axis

$$\mathbf{b},$$

and the distinguished role-gap vector

$$\mathbf{x}_{i_*} - \mathbf{x}_2;$$

- closed velocity cones

$$\mathfrak{V}^{i}_{W} \subset \Pi, \qquad i = 1, 2, 3,$$

and, for each active branch family

$$\beta$$

in the preserved event class, one closed chord-direction cone

$$\mathfrak{U}_W^\beta \subset \Pi;$$

- support vectors

$$\mathbf{n}_W^a, \qquad \mathbf{n}_W^b, \qquad \mathbf{n}_W^{\mathrm{role}} \in \Pi;$$

- and positive support floors

$$\alpha_W^a, \qquad \alpha_W^b, \qquad \alpha_W^{\mathrm{role}}.$$

Choose the seed-centered tube radii small enough that every active branch family

$$\beta$$

arising from a history in that tube has its unnormalized chord vector in one fixed narrow spatial cone whose normalized directions lie in

$$\mathfrak{U}_W^\beta.$$

This induced chord-direction control is part of the geometric setup, not an extra defining predicate of the convex set.

The intended geometric compatibility conditions are:

1. for every

$$\mathbf{u}_a \in \mathfrak{C}_W^a, \qquad \mathbf{u}_b \in \mathfrak{C}_W^b,$$

one has

$$\mathbf{n}_W^a \cdot \mathbf{u}_a \ge \alpha_W^a \|\mathbf{u}_a\|, \qquad \mathbf{n}_W^b \cdot \mathbf{u}_b \ge \alpha_W^b \|\mathbf{u}_b\|,$$

and the cone pair is angle-separated so that

$$|\det(\mathbf{u}_a, \mathbf{u}_b)| \ge \varsigma_W^{ab} \|\mathbf{u}_a\| \|\mathbf{u}_b\|$$

for some

$$\varsigma_W^{ab} > 0;$$

2. for every

$$\mathbf{u}_{\mathrm{role}} \in \mathfrak{C}_W^{\mathrm{role}}$$

one has

$$\mathbf{n}_W^{\mathrm{role}} \cdot \mathbf{u}_{\mathrm{role}} \ge \alpha_W^{\mathrm{role}} \|\mathbf{u}_{\mathrm{role}}\|;$$

3. for every active branch family

$$\beta = (i, j)$$

and every

$$\mathbf{v} \in \mathfrak{V}_W^j, \qquad \hat{\mathbf{u}} \in \mathfrak{U}_W^\beta,$$

one has

$$\mathbf{v} \cdot \hat{\mathbf{u}} \le c_f(1 - \nu_{\beta,W}^{\mathrm{cvx}})$$

for some

$$\nu_{\beta,W}^{\mathrm{cvx}} > 0.$$

These are the many-body closure analogues of the earlier sectorwise cone-transversality package: they are visibly convex restrictions on unnormalized vectors, but they imply the nonconvex Jacobian, shear, and role-gap floors used later.

For the explicit symmetric seed packet above, these abstract cones and supports should be specialized rather than left anonymous. At the section time the seed directions are

$$\hat{\mathbf{a}}_{\mathrm{seed}}(0) = \mathbf{e}_1, \qquad \hat{\mathbf{b}}_{\mathrm{seed}}(0) = \mathbf{e}_2,$$

and the relevant role-gap vector may be taken as

$$\mathbf{r}^{\mathrm{role}}_{\mathrm{seed}} \equiv \mathbf{x}_{1,\mathrm{seed}}(0) - \mathbf{x}_{2,\mathrm{seed}}(0) = \frac{1}{2}A_*\mathbf{e}_1 - B_*\mathbf{e}_2.$$

Accordingly one should choose

$$\mathbf{n}^a_W = \mathbf{e}_1, \qquad \mathbf{n}^b_W = \mathbf{e}_2, \qquad \mathbf{n}^{\mathrm{role}}_W = \frac{\mathbf{r}^{\mathrm{role}}_{\mathrm{seed}}}{\|\mathbf{r}^{\mathrm{role}}_{\mathrm{seed}}\|},$$

on the seed-side windows, and one should center the corresponding cones on

$$\mathfrak{C}^a_W \text{ around } \mathbf{e}_1, \qquad \mathfrak{C}^b_W \text{ around } \mathbf{e}_2,$$

$$\mathfrak{C}^{\mathrm{role}}_W \text{ around } \mathbf{r}^{\mathrm{role}}_{\mathrm{seed}}, \qquad \mathfrak{V}^i_W \text{ around } \dot{\mathbf{x}}_{i,\mathrm{seed}}.$$

The support floors should be chosen from the seed values with explicit slack

$$\alpha^a_W = A_* - \sigma^a_{\mathrm{seed}}, \qquad \alpha^b_W = B_* - \sigma^b_{\mathrm{seed}},$$

$$\alpha^{\mathrm{role}}_W = \|\mathbf{r}^{\mathrm{role}}_{\mathrm{seed}}\| - \sigma^{\mathrm{role}}_{\mathrm{seed}},$$

for positive seed slacks

$$\sigma^a_{\mathrm{seed}}, \qquad \sigma^b_{\mathrm{seed}}, \qquad \sigma^{\mathrm{role}}_{\mathrm{seed}}$$

small enough that the corresponding inequalities still hold strictly on the whole seed packet. This is the concrete way in which the symmetric seed should sit inside the later convex core.

> **Target Proposition (Explicit convex tame core inside the planar-three-body Banach envelope).**
> There exist positive tube radii

$$> \varepsilon_X, > \qquad > \varepsilon_V, > \qquad > \varepsilon_A >$$

> and closed cones

$$> \mathfrak{C}^a_W, > \qquad > \mathfrak{C}^b_W, > \qquad > \mathfrak{C}^{\mathrm{role}}_W, > \qquad > \mathfrak{V}^i_W, > \qquad > \mathfrak{U}^\beta_W >$$

> satisfying the compatibility conditions above such that the set

$$> \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}} >$$

> of all gauge-fixed histories

$$> \mathbf{\Phi} = (\mathbf{X}, \dot{\mathbf{X}}) > \in > \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} >$$

> obeying
>
> 1. the seed-centered tube bounds

$$> \sup_t > \|\mathbf{X}(t) - \mathbf{X}_{\mathrm{seed}}(t)\| > \leq > \varepsilon_X, >$$

$$> \sup_t > \|\dot{\mathbf{X}}(t) - \dot{\mathbf{X}}_{\mathrm{seed}}(t)\| > \leq > \varepsilon_V, >$$

$$> \mathrm{Lip}(\dot{\mathbf{X}}) > \leq > A_* + \varepsilon_A; >$$

> 2. the pointwise cone conditions

$$> \mathbf{a}(t) \in \mathfrak{C}^a_W, > \qquad > \mathbf{b}(t) \in \mathfrak{C}^b_W, > \qquad > \mathbf{x}_{i_*}(t) - \mathbf{x}_2(t) \in \mathfrak{C}^{\mathrm{role}}_W, > \qquad > \dot{\mathbf{x}}_i(t) \in \mathfrak{V}^i_W >$$

> on every controlled window

$$> W, >$$

> 3. the affine support inequalities

$$> \mathbf{n}^a_W \cdot \mathbf{a}(t) \geq \alpha^a_W, > \qquad > \mathbf{n}^b_W \cdot \mathbf{b}(t) \geq \alpha^b_W, > \qquad > \mathbf{n}^{\mathrm{role}}_W \cdot > \big(\mathbf{x}_{i_*}(t) - \mathbf{x}_2(t)\big) > \geq > \alpha^{\mathrm{role}}_W >$$

on every controlled window,

is a nonempty closed convex subset of

$$> \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} >$$

and lies entirely inside the tame region where the following nonconvex data are forced with uniform floors:

1. the pair and midpoint floors

$$> \|\mathbf{a}(t)\| \geq a^{\mathrm{cvx}}_{\min}, > \qquad > \|\mathbf{b}(t)\| \geq b^{\mathrm{cvx}}_{\min}; >$$

2. the shear floor

$$> |\det(\mathbf{a}(t), \mathbf{b}(t))| >\geq> \Delta^{\mathrm{cvx}}_{\min}; >$$

3. the role-gap floor

$$> \delta_{\mathrm{role}}(t) >\geq> \delta^{\mathrm{cvx}}_{\mathrm{role,min}}; >$$

4. and the branchwise Jacobian floors

$$> |J_\beta(t)| >\geq> \nu^{\mathrm{cvx}}_{J,\beta} > \qquad > \text{for every active branch family } > \beta. >$$

Proof draft.
The defining inequalities are visibly convex: the seed-centered

$$> C^0/C^1 >$$

tube bounds are norm-convex, the Lipschitz bound is convex, pointwise membership in each closed cone is convex, and the support inequalities are affine.

1. **Closedness and convexity.**
   Every defining condition is closed under uniform convergence on the history interval, hence

$$> \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}} >$$

is closed. Intersections of closed convex tube, cone, and affine constraints remain convex.

2. **Nonemptiness.**
   Choose the cones and support vectors around the seed with aperture small enough that the seed itself satisfies them with strict slack. Then the seed belongs to

$$> \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}}. >$$

3. **Norm floors from affine support.**
   Since

$$> \mathbf{n}^a_W \cdot \mathbf{a} \geq \alpha^a_W > \qquad > \text{and} > \qquad > \|\mathbf{n}^a_W\| = 1, >$$

one gets

$$> \|\mathbf{a}\| \geq \alpha^a_W. >$$

Likewise

$$> \|\mathbf{b}\| \geq \alpha^b_W. >$$

Hence one may take

$$> a^{\mathrm{cvx}}_{\min} >=> \min_W \alpha^a_W, > \qquad > b^{\mathrm{cvx}}_{\min} >=> \min_W \alpha^b_W. >$$

4. **Shear and role-gap floors from cone separation.**
   The angular separation of

$$> \mathfrak{C}^a_W > \qquad > \text{and} > \qquad > \mathfrak{C}^b_W >$$

implies

$$> |\det(\mathbf{a},\mathbf{b})| >\geq> \varsigma_W^{ab} > \|\mathbf{a}\|\,\|\mathbf{b}\| >\geq> \varsigma_W^{ab}\alpha_W^a\alpha_W^b. >$$

Therefore

$$> \Delta_{\min}^{\mathrm{cvx}} >=> \min_W > \varsigma_W^{ab}\alpha_W^a\alpha_W^b >> 0. >$$

Likewise the role cone and support inequality give a positive lower bound

$$> \delta_{\mathrm{role,min}}^{\mathrm{cvx}} >$$

after shrinking the role-cone aperture if necessary.

5. **Jacobian floors from cone transversality.**
   On each active branch family

$$> \beta = (i,j), >$$

the induced chord-direction cone

$$> \mathfrak{U}_W^{\beta} >$$

and the emitter velocity cone

$$> \mathfrak{V}_W^{j} >$$

satisfy

$$> \dot{\mathbf{x}}_j \cdot \hat{\mathbf{r}}_\beta >\leq> c_f(1-\nu_{\beta,W}^{\mathrm{cvx}}), >$$

so

$$> J_\beta(t) >=> 1 - \frac{\dot{\mathbf{x}}_j \cdot \hat{\mathbf{r}}_\beta}{c_f} >\geq> \nu_{\beta,W}^{\mathrm{cvx}}. >$$

Taking the minimum over the finite branch family list yields one common Jacobian floor.

6. **Containment in the tame region.**
   These floors imply the smooth-window, branch-regularity, and role-gap hypotheses used in the recapture and closure packages. Therefore

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} >$$

lies entirely inside the tame region where the preserved event class, fold ceilings, and principal margins are already stable.

Target Corollary (Seed-centered realization of the explicit convex tame core).
Assume the explicit symmetric planar-three-body seed proposition, the seed-neighborhood realization of the leading principal margins, and the delayed seed-margin persistence lemma. Then one may choose:

1. seed-centered support vectors

$$> \mathbf{n}_W^a = \mathbf{e}_1, > \qquad > \mathbf{n}_W^b = \mathbf{e}_2, > \qquad > \mathbf{n}_W^{\mathrm{role}} >=> \frac{\mathbf{r}_{\mathrm{seed}}^{\mathrm{role}}}{\|\mathbf{r}_{\mathrm{seed}}^{\mathrm{role}}\|}; >$$

2. narrow closed cones centered on

$$> \mathbf{e}_1, > \qquad > \mathbf{e}_2, > \qquad > \mathbf{r}_{\mathrm{seed}}^{\mathrm{role}}, > \qquad > \dot{\mathbf{x}}_{i,\mathrm{seed}}; >$$

3. and positive support slacks

$$> \sigma_{\mathrm{seed}}^{a}, > \qquad > \sigma_{\mathrm{seed}}^{b}, > \qquad > \sigma_{\mathrm{seed}}^{\mathrm{role}}, >$$

such that the resulting convex core

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} >$$

is nonempty and contains the whole delayed seed packet

$$> \mathcal{C}_{A_*,B_*,\eta}^{\mathrm{mb,seed}} >$$

> as a strict interior subset relative to the preserved tame class.

> **Proof draft.**
> The explicit symmetric seed satisfies:
>
> 1. the support inequalities with margins

$$> A_*, > \qquad > B_*, > \qquad > \|\mathbf{r}^{\mathrm{role}}_{\mathrm{seed}}\|; >$$

> 2. the cone conditions with zero angular defect, since

$$> \mathbf{a}_{\mathrm{seed}}(0) \parallel \mathbf{e}_1, > \qquad > \mathbf{b}_{\mathrm{seed}}(0) \parallel \mathbf{e}_2; >$$

> 3. and the delayed seed-margin persistence lemma gives one seed-side window on which the same branch families remain unique and the same principal inward margins stay positive.
>
> Therefore one may choose the cone apertures and support slacks small enough that:
>
> - the whole local seed packet remains inside the same tube-and-cone inequalities;
> - the induced Jacobian, shear, and role-gap floors remain strictly positive;
> - and the corresponding convex core is nonempty because it already contains that seed packet.
>
> This is the explicit nonvacuity step for the many-body convex core. After this corollary, the convex-core package is no longer merely compatible with the seed geometry; it is concretely centered on it.

Only after this separation should one define the many-body tame target. The many-body tame envelope target should then be a closed subset

$$\mathcal{K}^{\mathrm{mb}}_{A_*,\eta} \subseteq \mathcal{C}^{\mathrm{mb}}_{A_*,\eta}$$

on which the following constants are all preserved simultaneously:

- gauge-selector continuity and uniqueness;
- finite active hypergraph bounds;
- finite cluster-ancestry bounds;
- the branch-regularity and event-gap data

$$\Delta\tau_{\mathrm{evt}}, \qquad \nu^{\mathrm{mb}}_J, \qquad \delta^{\mathrm{mb}}_{\mathrm{sep}};$$

- the bounded caustic-transit data

$$M^{\mathrm{mb}}_{\max}, \qquad \mathfrak{F}^{\mathrm{mb}}_1, \qquad \mathfrak{F}^{\mathrm{mb}}_2, \qquad \mathfrak{F}^{\mathrm{mb}}_3, \qquad \mathfrak{F}^{\mathrm{mb}}_4;$$

- the smooth-window floors

$$a_{\min}, \qquad b_{\min}, \qquad \Delta_{\min}, \qquad \delta_{\mathrm{role,min}};$$

- recapture margins for every

$$\rho^{\mathrm{mb}}_m;$$

- the concrete recapture-window lengths

$$\Delta_{\rho,\mathrm{post}}, \qquad \Delta_{\rho,\mathrm{late}};$$

- the fold ceilings

$$F^{\mathrm{mb}}_1, \qquad F^{\mathrm{mb}}_2, \qquad F^{\mathrm{mb}}_3, \qquad F^{\mathrm{mb}}_4;$$

- the first four principal margins

$$\mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{post}}, \quad \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{late}}, \quad \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{post}}, \quad \mathfrak{M}^{\mathrm{mb}}_{2,\mathrm{late}}, \quad \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{post}}, \quad \mathfrak{M}^{\mathrm{mb}}_{3,\mathrm{late}}, \quad \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{post}}, \quad \mathfrak{M}^{\mathrm{mb}}_{4,\mathrm{late}};$$

- and one fixed atlas representative on the relevant quotient chart.

The coupled parameter regime underlying this closed tame target should likewise be made explicit:

> **Target Proposition (Coupled parameter solvability for the first planar-three-body tame regime).**
> There exists a nonempty parameter region in

$$> (\eta, \epsilon_c, A_*, B_{\min}, V_{\text{in}}, a_{\min}, b_{\min}, \Delta_{\min}, \delta_{\text{role},\min}, \tau_{\text{dp}}^{\text{mb}}, \dots) >$$

such that all of the following hold simultaneously:

1. the no-accumulation floors and event-gap bounds;
2. the bounded caustic-transit ceilings

$$> F_m^{\text{mb}}; >$$

3. the ancestry and deep-past suppression bounds;
4. the strict positivity of the principal margins

$$> \mathfrak{M}_{m,\text{post}}^{\text{mb}}, > \qquad > \mathfrak{M}_{m,\text{late}}^{\text{mb}}; >$$

5. and the nondegeneracy of the smooth recapture windows.

Proof draft of the coupled parameter solvability proposition.
The constants are coupled, but their dependencies are triangular once the previous packages are ordered correctly.

1. **Choose the kinematic box first.**
   Fix

$$> A_* >$$

and the seed-centered tube radii

$$> \varepsilon_X, > \qquad > \varepsilon_V, > \qquad > \varepsilon_A >$$

so that the explicit convex core proposition applies.

2. **Choose cone apertures and support floors.**
   Pick the spatial and velocity cones narrow enough that the induced constants

$$> a_{\min}^{\text{cvx}}, > \qquad > b_{\min}^{\text{cvx}}, > \qquad > \Delta_{\min}^{\text{cvx}}, > \qquad > \delta_{\text{role},\min}^{\text{cvx}}, > \qquad > \nu_{J,\beta}^{\text{cvx}} >$$

are positive with slack relative to the seed history.

3. **Choose the mollification regime.**
   Then take

$$> \eta, > \qquad > \epsilon_c >$$

small enough for the Type II / Type III caustic-transit estimates, but not so small that the resulting fold ceilings

$$> F_m^{\text{mb}} >$$

dominate the inward recapture floors.

4. **Balance recapture against leakage.**
   Because the inward branch-sum terms and the leakage terms vary continuously with these parameters on the smooth windows, the strict seed inequalities persist on one nonempty open parameter neighborhood. Hence the principal margins

$$> \mathfrak{M}_{m,\text{post}}^{\text{mb}}, > \qquad > \mathfrak{M}_{m,\text{late}}^{\text{mb}} >$$

remain positive there.

5. **Preserve deep-past suppression.**
   Finally choose

$$> \tau_{\text{dp}}^{\text{mb}} >$$

large enough that the deep-past ceiling remains below the recapture slack already reserved in Step 4.

6. **Reserve trapping slack on the convex-core boundary.**
   Tighten the support floors and tube radii so that the seed and its one-cycle image satisfy every cone and support inequality with a strictly positive margin. The same open-parameter argument then leaves room for the inward-pointing boundary estimates used in the convex-core trapping lemma below.

> Therefore the admissible parameter region is the intersection of finitely many open inequalities, all already satisfied by the seed with slack; hence it is nonempty.

The proposition above gives the right abstract dependency structure, but the next proof burden is to write one explicit sufficient corridor analogous to the short-window recapture regime in the frozen 1D scaffold. For the first planar-three-body bridge, the natural corridor is controlled by the four principal channels together with the branch-regularity and trapping slack budgets.

For

$$m = 1, 2, 3, 4$$

and

$$W \in \{\mathrm{post}, \mathrm{late}\},$$

introduce

$$\underline{\Lambda}^{\mathrm{mb}}_{m,W} \equiv \inf_{t \in I^{\mathrm{mb}}_{m,W}} \Lambda^{\mathrm{mb}}_{m}(t),$$

$$\overline{L}^{\mathrm{mb}}_{m,W} \equiv \sup_{t \in I^{\mathrm{mb}}_{m,W}} \Big(L^{\mathrm{mb}}_{m,\mathrm{geom}}(t) + L^{\mathrm{mb}}_{m,\mathrm{fold}}(t) \Big),$$

$$\overline{A}^{\mathrm{mb}}_{\mathrm{deep}} \equiv \sup_{t \in I^{\mathrm{mb}}_{\mathrm{post}} \cup I^{\mathrm{mb}}_{\mathrm{late}}} A^{\mathrm{mb}}_{s,\mathrm{deep}}(t),$$

and the strict principal margin floors

$$\mathfrak{m}^{\mathrm{mb}}_{m,W} \equiv \underline{\Lambda}^{\mathrm{mb}}_{m,W} - \overline{L}^{\mathrm{mb}}_{m,W} - \overline{A}^{\mathrm{mb}}_{\mathrm{deep}}.$$

Here

$$L^{\mathrm{mb}}_{m,\mathrm{geom}}$$

collects all non-fold leakage already isolated in the channel decompositions, so that

$$\overline{L}^{\mathrm{mb}}_{m,W}$$

is a genuine channelwise ceiling.

Likewise define the maximal outward initial speeds

$$V^{\mathrm{mb}}_{m,W,\max} \equiv \sup_{\Phi \in \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}}} \left(\dot{\rho}^{\mathrm{mb}}_{m}\right)_{+} (t^{\Phi}_{m,W,\mathrm{in}}),$$

where

$$t^{\Phi}_{m,W,\mathrm{in}}$$

is the entry time of the history

$$\Phi$$

into the corresponding controlled window, and let

$$\ell^{\mathrm{mb}}_{m,W} \equiv |I^{\mathrm{mb}}_{m,W}|$$

denote the window length.

The trapping-region bookkeeping should then use the excursion ceilings

$$E^{\mathrm{mb}}_{m,W} \equiv \frac{\left(V^{\mathrm{mb}}_{m,W,\max}\right)^2}{2\mathfrak{m}^{\mathrm{mb}}_{m,W}},$$

whenever

$$\mathfrak{m}^{\mathrm{mb}}_{m,W} > 0,$$

because

$$\ddot{\rho}^{\mathrm{mb}}_m \le -\mathfrak{m}^{\mathrm{mb}}_{m,W}$$

on the corresponding window then bounds every outward excursion before turning by the usual one-dimensional comparison estimate.

> **Target Proposition (Explicit principal-channel parameter corridor for the first planar-three-body tame regime).**
> Assume the explicit convex tame core proposition and define the principal channel floors above. Suppose there exist positive constants

$$> \mathfrak{m}^{\mathrm{mb}}_{\mathrm{prin}}, > \qquad > s^{a}_{\mathrm{trap}}, > \qquad > s^{b}_{\mathrm{trap}}, > \qquad > s^{\Delta}_{\mathrm{trap}}, > \qquad > s^{\mathrm{role}}_{\mathrm{trap}} >$$

> such that:
>
> 1. for every

$$> m = 1,2,3,4 > \qquad > \text{and} > \qquad > W \in \{\mathrm{post}, \mathrm{late}\}, >$$

> one has

$$> \mathfrak{m}^{\mathrm{mb}}_{m,W} >\ge> \mathfrak{m}^{\mathrm{mb}}_{\mathrm{prin}} > 0; >$$

> 2. the controlled windows are long enough for turning:

$$> \ell^{\mathrm{mb}}_{m,W} >\ge> \frac{V^{\mathrm{mb}}_{m,W,\max}}{\mathfrak{m}^{\mathrm{mb}}_{m,W}} > \qquad > \text{for all } > m = 1,2,3,4; >$$

> 3. the resulting excursion ceilings satisfy

$$> E^{\mathrm{mb}}_{1,W} < s^{a}_{\mathrm{trap}}, > \qquad > E^{\mathrm{mb}}_{2,W} < s^{b}_{\mathrm{trap}}, > \qquad > E^{\mathrm{mb}}_{3,W} < s^{\Delta}_{\mathrm{trap}}, > \qquad > E^{\mathrm{mb}}_{4,W} < s^{\mathrm{role}}_{\mathrm{trap}} >$$

> on both windows;
> 4. the support and cone slack in the explicit convex core dominate those trapping budgets:

$$> a^{\mathrm{cvx}}_{\min} - \alpha^{a}_{\partial} > s^{a}_{\mathrm{trap}}, > \qquad > b^{\mathrm{cvx}}_{\min} - \alpha^{b}_{\partial} > s^{b}_{\mathrm{trap}}, >$$

$$> \Delta^{\mathrm{cvx}}_{\min} - \Delta_{\partial} > s^{\Delta}_{\mathrm{trap}}, > \qquad > \delta^{\mathrm{cvx}}_{\mathrm{role},\min} - \delta^{\mathrm{role}}_{\partial} > s^{\mathrm{role}}_{\mathrm{trap}}, >$$

> where the boundary thresholds

$$> \alpha^{a}_{\partial}, > \qquad > \alpha^{b}_{\partial}, > \qquad > \Delta_{\partial}, > \qquad > \delta^{\mathrm{role}}_{\partial} >$$

> are the defining support values of the convex-core boundary faces;
> 5. the branchwise Jacobian floor and fold ceilings satisfy

$$> \nu^{\mathrm{cvx}}_{J,\beta} \ge \nu^{\mathrm{mb}}_{J,\min} > 0, > \qquad > F^{\mathrm{mb}}_m \le \overline{F}^{\mathrm{mb}}_m >$$

> with the chosen

$$> \overline{F}^{\mathrm{mb}}_m >$$

> already absorbed into

$$> \overline{L}^{\mathrm{mb}}_{m,W}. >$$

> Then all four principal recapture channels turn inward before exhausting the convex-core slack, and the principal part of the boundary-trapping lemma follows. In particular, these inequalities are a concrete sufficient realization of the coupled parameter solvability proposition for the principal many-body corridor.

> **Proof draft.**
> On each controlled window,

$$> \ddot{\rho}^{\mathrm{mb}}_m >\le> -\mathfrak{m}^{\mathrm{mb}}_{m,W} >$$

> by definition of the principal floor. Therefore the outward speed can persist for at most

$$> V^{\mathrm{mb}}_{m,W,\max}/\mathfrak{m}^{\mathrm{mb}}_{m,W} >$$

> time, which is available by hypothesis. Integrating once more gives the excursion ceiling

$$> E^{\mathrm{mb}}_{m,W} >\le> \frac{\left(V^{\mathrm{mb}}_{m,W,\max}\right)^2}{>} 2\mathfrak{m}^{\mathrm{mb}}_{m,W}. >$$

> The slack inequalities in Step 4 then prevent the corresponding support, shear, or role boundary from being reached before turning. The Jacobian and fold hypotheses ensure that no fold or transversality loss invalidates the comparison law on the same windows. Hence the four principal channels remain trapped inside the explicit convex core.

One should also record a practical scaling corridor for later proofs. The first planar-three-body bridge should be pursued in a regime where

$$\eta, \epsilon_c \downarrow 0, \qquad \tau^{\mathrm{mb}}_{\mathrm{dp}} \uparrow \infty,$$

with:

- the branchwise Jacobian floors and cone apertures fixed away from zero;
- the fold ceilings

$$F^{\mathrm{mb}}_m$$

remaining

$$o(1)$$

relative to the inward branch-sum floors

$$\underline{\Lambda}^{\mathrm{mb}}_{m,W};$$

- the deep-past ceiling

$$\overline{A}^{\mathrm{mb}}_{\mathrm{deep}}$$

tending to zero;

- and the entry speeds

$$V^{\mathrm{mb}}_{m,W,\max}$$

small enough that the excursion ceilings stay below the fixed convex-core slack.

In that corridor the coupled parameter problem is no longer a vague compatibility hope. It is reduced to checking a finite family of explicit inequalities against one seed-centered slack budget.

Accordingly, the abstract coupled parameter solvability proposition should be read as proved once one verifies:

1. the explicit convex tame core proposition;
2. the explicit principal-channel parameter corridor above;
3. the branchwise Jacobian and no-accumulation floors on the same parameter slab;
4. and the deep-past / ancestry suppression inequalities on that slab.

This is the planar-three-body analogue of the explicit short-window recapture regime in the frozen 1D scaffold. The closure packages may now consume it as a named parameter-intersection target rather than a silent background hope.

This strengthening matters. In the planar-three-body bridge, the tame layer cannot merely remember that “some recapture theorem holds.” It must preserve the actual windows and margin constants on which the first concrete recapture lemma was proved, or else the local turning argument could be lost after one return. But those data should now be understood as structure carried on a stable tame subregion of the convex Banach box, not as the definition of convexity itself.

> **Target Proposition (Closed tame graph-stable subregion in the planar three-body section).**
> There exists a nonempty closed set

$$> \mathcal{K}^{\mathrm{mb}}_{A_*,\eta} >\subseteq> \mathcal{C}^{\mathrm{mb}}_{A_*,\eta} >\subseteq> \Sigma^{-,\mathrm{mb}}_{A_*} >$$

> such that every history in

$$> \mathcal{K}^{\mathrm{mb}}_{A_*,\eta} >$$

> admits one-cycle continuation with the same gauge, hypergraph, ancestry, recapture-window, fold-ceiling, and recapture-margin constants, and such that the corresponding returned history lands back in the same set after the canonical gauge reset.

> Concretely, the preserved data should include:
>
> 1. one common gauge-fixed representative and one common non-near-collinear chart;
> 2. the same active delay-hypergraph size bounds and cluster-ancestry bounds;
> 3. the same branch-regularity and event-gap data

$$> \Delta\tau_{\mathrm{evt}}, > \qquad > \nu_J^{\mathrm{mb}}, > \qquad > \delta_{\mathrm{sep}}^{\mathrm{mb}}; >$$

> 4. the same bounded caustic-transit data

$$> M_{\max}^{\mathrm{mb}}, > \qquad > \mathfrak{F}_1^{\mathrm{mb}}, > \qquad > \mathfrak{F}_2^{\mathrm{mb}}, > \qquad > \mathfrak{F}_3^{\mathrm{mb}}, > \qquad > \mathfrak{F}_4^{\mathrm{mb}}; >$$

> 5. the same smooth-window floors

$$> a_{\min}, > \qquad > b_{\min}, > \qquad > \Delta_{\min}, > \qquad > \delta_{\mathrm{role},\min}; >$$

> 6. the same controlled windows

$$> I_{m,\mathrm{post}}^{\mathrm{mb}}, > \qquad > I_{m,\mathrm{late}}^{\mathrm{mb}}, > \qquad > m = 1, 2, 3, 4; >$$

> 7. the same fold ceilings

$$> F_1^{\mathrm{mb}}, > \qquad > F_2^{\mathrm{mb}}, > \qquad > F_3^{\mathrm{mb}}, > \qquad > F_4^{\mathrm{mb}}; >$$

> 8. and the same strict positivity margins

$$> \mathfrak{M}_{1,\mathrm{post}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{1,\mathrm{late}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{2,\mathrm{post}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{2,\mathrm{late}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{3,\mathrm{post}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{3,\mathrm{late}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{4,\mathrm{post}}^{\mathrm{mb}}, > \quad > \mathfrak{M}_{4,\mathrm{late}}^{\mathrm{mb}}, >$$

> together with the remaining

$$> \mathfrak{M}_{m,\mathrm{post}}^{\mathrm{mb}}, > \qquad > \mathfrak{M}_{m,\mathrm{late}}^{\mathrm{mb}} >$$

> for

$$> m = 5, \dots, K_{\mathrm{esc}}. >$$

This is the correct tame-structure target because the first two Jacobi channels are no longer abstract placeholders. If their window geometry or margin positivity is not propagated through the return map, then the concrete recapture lemma has no stable domain on which to operate.

> **Proof draft of the closed tame graph-stable subregion proposition.**
> Assume the coupled parameter solvability proposition and the explicit convex tame core proposition. Let

$$> \mathcal{T}_{A_*,\eta}^{\mathrm{mb}} >$$

> be the tame subregion produced by the hypergraph-and-atlas stability proposition. Define

$$> \mathcal{K}_{A_*,\eta}^{\mathrm{mb}} >$$

> to be the closed subset of that tame region on which the same gauge chart, event class, ancestry bounds, smooth-window floors, fold ceilings, and principal margins are all preserved with the same constants.
> Choose in addition a nonempty convex subset

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \subseteq > \mathcal{K}_{A_*,\eta}^{\mathrm{mb}} >$$

> given by the explicit tube-and-cone construction and lying entirely inside the same tame region.
>
> 1. **Closedness.**
>   Each defining inequality for

$$> \mathcal{K}_{A_*,\eta}^{\mathrm{mb}} >$$

> is closed in the ambient

$$> C^1 >$$

> topology: non-strict norm bounds are closed, lower bounds on the preserved floors are closed after the margins are frozen with positive slack, and the event class is constant on the tame region by construction. Hence

$$> \mathcal{K}^{\mathrm{mb}}_{A_*,\eta} >$$

> is closed.
>
> 2. **Nonemptiness.**
> The seed history used to define the tame region belongs to

$$> \mathcal{K}^{\mathrm{mb}}_{A_*,\eta}, >$$

> and the coupled parameter solvability proposition guarantees that the defining margins and floors are simultaneously satisfiable, so the set is nonempty.
>
> 3. **Return-map invariance at the level of tame data.**
> By the earlier well-posedness, no-accumulation, caustic-transit, ancestry, and recapture packages, every history in

$$> \mathcal{K}^{\mathrm{mb}}_{A_*,\eta} >$$

> completes one controlled cycle with the same preserved constants. After the canonical gauge reset, the returned history therefore lies in the same closed tame data class.

> **Lemma (Boundary trapping for the explicit convex core).**
> Assume the explicit convex tame core proposition, the coupled parameter solvability proposition, the principal four-channel recapture closure proposition, and the closed tame graph-stable subregion proposition.
> Then every codimension-one boundary face of

$$> \mathcal{K}^{\mathrm{mb}}_{\mathrm{cvx}} >$$

> coming from one defining tube, cone, or affine support inequality is strictly inward-pointing under one controlled return.
>
> More precisely:
>
> 1. if the returned history touched one pair-axis support face

$$> \mathbf{n}^{a}_{W} \cdot \mathbf{a} = \alpha^{a}_{W} >$$

> or one boundary ray of

$$> \mathfrak{C}^{a}_{W}, >$$

> then the corresponding boundary defect would force a nonnegative outward drift in

$$> \rho^{\mathrm{mb}}_{1}, >$$

> contradicting

$$> \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{post}} > 0 > \qquad > \mathrm{or} > \qquad > \mathfrak{M}^{\mathrm{mb}}_{1,\mathrm{late}} > 0; >$$

> 2. if it touched one midpoint support face or one boundary ray of

$$> \mathfrak{C}^{b}_{W}, >$$

> the same argument contradicts the

$$> \rho^{\mathrm{mb}}_{2} >$$

> margins;
> 3. if it touched the shear-separation boundary determined by the cone pair

$$> (\mathfrak{C}^{a}_{W}, \mathfrak{C}^{b}_{W}), >$$

> then the resulting loss of signed-area slack contradicts the

$$> \rho^{\mathrm{mb}}_{3} >$$

> margin positivity;
> 4. if it touched one role-gap support face or one boundary ray of

$$> \mathfrak{C}^{\mathrm{role}}_{W}, >$$

the resulting near-tie contradicts the

$$> \rho_4^{\mathrm{mb}} >$$

margin positivity;
5. if it touched one velocity-cone face

$$> \partial\mathfrak{V}_W^i, >$$

then the corresponding chord projection reaches the Jacobian transversality threshold and contradicts the preserved branch-regularity floor

$$> \nu_{J,\beta}^{\mathrm{cvx}} > 0; >$$

6. and if it touched one outer Banach-tube face in the ambient

$$> C^0/C^1 >$$

box, then the resulting extremal history would violate the reserved post-transit and recapture slack from the coupled parameter regime.

Proof draft.
Let one returned history touch a boundary face of

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}}. >$$

Because the closed tame graph-stable subregion proposition already preserves the event class, fold ceilings, deep-past bounds, smooth windows, and principal margins, only the explicit convex-core inequalities need to be checked.

1. **Support faces for**

$$> \mathbf{a}, > \mathbf{b}. >$$

Touching

$$> \mathbf{n}_W^a \cdot \mathbf{a} = \alpha_W^a >$$

means the pair axis has exhausted its inward slack on one controlled window. But the corresponding margin positivity forces

$$> \ddot{\rho}_1^{\mathrm{mb}} < 0 >$$

there, so the returned history cannot remain on or beyond that support face. The same argument with

$$> \rho_2^{\mathrm{mb}} >$$

excludes contact with the midpoint support face.

2. **Cone rays and shear boundary.**
   Contact with a boundary ray of

$$> \mathfrak{C}_W^a > \qquad > \mathrm{or} > \qquad > \mathfrak{C}_W^b >$$

is exactly the loss of angular slack that drives

$$> |\det(\mathbf{a}, \mathbf{b})| >$$

toward its minimum. The

$$> \rho_3^{\mathrm{mb}} >$$

comparison law forbids persistent outward motion toward that shear boundary while

$$> \mathfrak{M}_{3,\mathrm{post}}^{\mathrm{mb}}, > \mathfrak{M}_{3,\mathrm{late}}^{\mathrm{mb}} > 0. >$$

3. **Role boundary.**
   Contact with one support face or boundary ray of

$$> \mathfrak{C}_W^{\mathrm{role}} >$$

is precisely the collapse of the distinguished-core role gap. But the role-exchange margin positivity forces

$$> \ddot{\rho}_4^{\mathrm{mb}} < 0 >$$

on the role windows, so this boundary cannot be reached under one controlled return.

4. **Velocity-cone faces.**
   If one returned history touched

$$> \partial\mathfrak{V}_W^i, >$$

the corresponding emitter velocity would saturate the branchwise transversality inequality against one admissible chord-direction cone. That contradicts the preserved positive Jacobian floor and hence the no-accumulation / branch-regularity package.

5. **Outer Banach-tube faces.**
   On the remaining ambient tube faces, use the reserved trapping slack from the coupled-parameter regime together with the bounded caustic-transit ceilings and the recapture margins. Any returned history touching the outer

$$> C^0/C^1 >$$

boundary would require one principal observable or one kinematic ceiling to use up all reserved slack, contradicting the previously established inward comparison laws and post-transit bounds.

Hence every defining boundary face is inward-pointing.

Lemma (One-cycle preservation of the explicit convex core inequalities).
Assume the explicit convex tame core proposition, the coupled parameter solvability proposition, the boundary trapping lemma above, and the closed tame graph-stable subregion proposition.
Then the gauge-reset return map sends every history in

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} >$$

to a returned history satisfying the same:

1. seed-centered tube bounds

$$> \varepsilon_X, > \qquad > \varepsilon_V, > \qquad > \varepsilon_A; >$$

2. pointwise cone conditions

$$> \mathfrak{C}_W^a, > \qquad > \mathfrak{C}_W^b, > \qquad > \mathfrak{C}_W^{\mathrm{role}}, > \qquad > \mathfrak{V}_W^i; >$$

3. and affine support inequalities

$$> \alpha_W^a, > \qquad > \alpha_W^b, > \qquad > \alpha_W^{\mathrm{role}}. >$$

Proof draft.
The returned history already lies in the same closed tame data class by the previous proposition, so the active event class, smooth-window floors, fold ceilings, and principal margins are preserved.

Assume for contradiction that the returned history does not lie in

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}}. >$$

Since the convex core is closed and the seed lies strictly in its interior relative to the preserved tame class, there is a first defining boundary face touched by the returned history. But the boundary trapping lemma rules out contact with every such face: pair and midpoint support faces are excluded by

$$> \rho_1^{\mathrm{mb}}, > \rho_2^{\mathrm{mb}}, >$$

the shear boundary by

$$> \rho_3^{\mathrm{mb}}, >$$

the role boundary by

$$> \rho_4^{\mathrm{mb}}, >$$

the velocity-cone faces by the preserved Jacobian floors, and the remaining ambient tube faces by the reserved trapping slack in the coupled-parameter regime. This contradiction proves

$$> \mathcal{P}_{\eta}^{\mathrm{mb}} > \left(> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \right) > \subseteq > \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}}. >$$

Proof draft of the invariant-envelope closure theorem.
Assume the coupled parameter solvability proposition and the explicit convex tame core proposition. The remaining closure burden is to show that the gauge-reset return map

$$> \mathcal{P}_{\eta}^{\mathrm{mb}} >$$

sends

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} >$$

into itself.

1. The well-posedness and continuation packages ensure the map is defined on the whole tame region.
2. The no-accumulation and caustic-transit packages preserve the event class and fold ceilings.
3. The ancestry package preserves the deep-past ceiling.
4. The recapture package preserves the principal margin positivity and the controlled windows.
5. The one-cycle convex-core preservation lemma preserves the same tube-and-cone inequalities.
6. The gauge reset preserves the chosen chart representative.

Therefore

$$> \mathcal{P}_{\eta}^{\mathrm{mb}} > \left(> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \right) > \subseteq > \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}}. >$$

Target Theorem (Planar-three-body invariant-envelope closure and Schauder capstone).
Assume the convex Banach-envelope proposition, the hypergraph-and-atlas stability proposition on that envelope, the explicit convex tame core proposition, the coupled parameter solvability proposition, the closed tame graph-stable subregion proposition, the corresponding invariant-envelope closure theorem, and continuity and precompactness of the gauge-fixed many-body return map on the relevant convex Banach box.

Then the return map has a fixed point

$$> \Phi_{\eta}^{*,\mathrm{mb}} > \in > \mathcal{K}_{A_*,\eta}^{\mathrm{mb}}, >$$

and the associated delayed trajectory is a bounded relative-periodic planar-three-body solution of the dual-mollified master equation modulo the canonical

$$> SE(2) >$$

gauge. It is an absolute periodic solution in the fixed Euclidean void only if the accumulated full-cycle holonomy is

$$> H_{\Phi_{\eta}^{*,\mathrm{mb}}}^{\mathrm{mb}} = (0, \mathrm{Id}). >$$

In particular, the fixed-point trajectory preserves one common gauge chart, one common active delay hypergraph, one common ancestry complex, and one common family of post-crossing and late-turn recapture margins through every return. That is the many-body analogue of the earlier bridge closures on one tame self-map domain.

Proof draft of the Schauder capstone.
Apply Schauder directly to the return map

$$> \mathcal{P}_{\eta}^{\mathrm{mb}} >$$

on the convex subset

$$> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \subseteq > \mathcal{C}_{A_*,\eta}^{\mathrm{mb}}. >$$

By the closure theorem,

$$> \mathcal{P}_{\eta}^{\mathrm{mb}} > \left(> \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \right) > \subseteq > \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}}, >$$

and by assumption the map is continuous and precompact there. Therefore the standard Schauder fixed-point theorem yields one fixed history

$$> \Phi_{\eta}^{*,\mathrm{mb}} > \in > \mathcal{K}_{\mathrm{cvx}}^{\mathrm{mb}} > \subseteq > \mathcal{K}_{A_*,\eta}^{\mathrm{mb}}. >$$

> The corresponding trajectory is relative-periodic by construction of the gauge-reset return map and remains bounded because it never leaves the same controlled Banach box and tame data class. Absolute periodicity is the separate zero-holonomy reconstruction condition stated above.

This is the first honest local many-body breather target in the chapter. Everything above it is there only to make this statement legitimate.

The planar-three-body bridge now has the same explicit theorem-ladder shape as the earlier binary bridges:

- gauge-fixed section and shape-space well-posedness;
- quantitative branch regularity and no-accumulation of delay events;
- bounded many-body caustic transit and fold ceilings;
- finite active delay-hypergraph control;
- cluster-valued ancestry or exclusion for deep-past branches;
- finite escape-observable recapture on explicit smooth windows;
- and atlas-level tame-envelope closure leading to the Schauder capstone.

Capstone Statement

The 1D collinear chapter should now be used as a frozen reference theorem scaffold. The present chapter records the higher-level lesson:

> **Theorem Program (Breather architecture for the master equation).**
> A master-equation breather theorem should be pursued from the dual-mollified absolute-time integral law, with branch sums used only on certified simple-root charts. The proof task is to construct a sectioned history-space return map, produce one candidate cycle with finite certificate data, separate convex Banach bounds from tame delayed-root geometry, and then close the resulting return map on one closed convex tame self-map domain. The unresolved burden is no longer the abstract fixed-point theorem or an elementary closed-form orbit. It is the geometric production and certification of that domain outside the ordered 1D setting.

Operationally, the live proof burden remains the collinear certificate chain

$$\phi_{\mathrm{cyc}} \to \text{null-coordinate pre-ledger closure} \to \text{finite branch chart} \to \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} \to P_\eta(\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}) \subseteq \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} \to \text{Schauder}.$$

Here the pre-ledger is a candidate-falsification gate: unresolved parent-complement strips stop the chain before branch-chart, certificate, self-map, or Schauder work begins.

The reduced planar, unreduced planar, and planar three-body sections should stay frozen as dependency maps until that chain is closed.

This is the correct point from which to resume work on the broader dynamics stack.

Related Chapters

- master-equation.md
- collinear-breather.md
- binary-dynamics.md
- Nested Shell Braid Dynamics
- energy.md

Collinear Breather

This chapter isolates the simplest reduced dynamical problem that can test a self-hit-assisted bounded-recapture mechanism without tangential geometry. Its purpose is to provide a mathematically tractable bridge between the full delayed master equation and the first rigorous existence question for bounded two-body motion.

The guiding idea is narrow: if delayed self-interaction can contribute to any bounded recapture mechanism at all, it should first be visible in a reflection-symmetric one-dimensional opposite-polarity binary. If it cannot be made to work there, then later claims about maximum-curvature binaries, nested shell braid locking, and assembly-level closure lose their cleanest analytic foothold.

Overview

In Lineland there is only a single endless road. Upon it travel two polarity-bearing points. From a great distance they rush toward one another, drawn together by their mutual pull. Each accelerates as it approaches the other. Yet whatever influence a point emits into the line does not act everywhere at once; it spreads along the road at a finite pace, leaving behind a wake of its past motion. For a time each point runs ahead of the disturbance it has already sent out. They meet, pass, and continue apart. But presently each encounters the older wake it cast while approaching. This delayed encounter pushes outward, while the partner, now behind, continues to pull inward. Thus the whole affair reduces to a contest on a line: a delayed push from one's own past against the present pull of the other. The purpose of this chapter is to determine whether that contest can be forced into repeated recapture rather than escape, and to state the theorem program that would make such a bounded cycle rigorous.

Formally, this chapter develops a proof scaffold for the global existence question of a periodic limit cycle in a symmetric two-body collinear system governed by a strongly nonlinear state-dependent delay differential equation. The dynamics use a dual-mollified delayed kernel, separating the short-distance $1/r^2$ singularity from the causal-isochron boundary. The main analytic difficulty is the velocity-dependent causal-fold geometry, where Jacobians can approach

$$J \to 0$$

The scaffold attacks that geometry by constructing the sorting maps

$$w(t) = x(t) + c_f t \qquad \text{and} \qquad z(t) = x(t) - c_f t$$

which isolate root birth, root exclusion, deep-past localization, and bounded caustic transit. From there the delayed dynamics are reduced to explicit conservative force margins for the inner recapture and the outer apocenter turn, and these are assembled into a closed, convex, precompact invariant-envelope program in

$$C^1([-h, 0])$$

The final fixed-point step is then delegated to Arzela-Ascoli compactness and a Schauder-type argument once the nonempty tame class is fully propagated through one cycle. At present, that capstone remains a theorem target rather than a completed proof.

Status Map

This chapter now has three different status layers, and they should be read separately:

- completed local and regional lemma packages, especially for delayed-root geometry, caustic transit, inner recapture, and trimmed-apocenter outer-turn control,
- target propositions that package those estimates into one closed convex tame self-map domain,
- and the final Schauder capstone, which remains conditional on that domain-level closure.

In particular, the manuscript already contains substantial outer-turn and apocenter material. The main remaining burden is not to invent an outer-turn mechanism from scratch, but to assemble the local theorem packages into one coupled invariant-envelope regime on which the return map is continuous, precompact, and self-mapping.

This chapter is the proof core for the breather program. Completion now means replacing the conditional finite-certificate rows with one verified certificate: an instantiated

$$\phi_{\text{cyc}}$$

a finite branch chart, a closed convex certificate, a strict coupled corridor, a monodromy diagnostic, returned-sample preservation, certified topology, and then Schauder.

Reading Map

Readers looking for the main structural bottlenecks can use the following map.

- The sign and physical interpretation caveat appears in Signed-branch caution.
- The return-map setup begins in Regularized Return Map.
- The compactness and fixed-point architecture begins in Global Existence via Arzela-Ascoli.
- The coupled-envelope bottleneck appears in Invariant-envelope closure and the later Capstone Statement.
- The outer-turn program is developed in Outer-turn recapture target, Deep-past outer self suppression target, z-map descent target, and the Capstone Statement.
- The compressed endpoint appears in Capstone Statement.

Proof-Program Dependency Map

At the highest level, the proof program now runs in the following order:

1. collapse-to-crossing control,
2. pre-crossing caustic transit and recovery,
3. local post-crossing recapture,
4. outer-turn recapture on the trimmed apocenter window,
5. turn-to-section return,
6. invariant-envelope closure on one coupled tame domain,
7. continuity and precompactness of the return map on that same domain,
8. Schauder fixed-point closure.

This is the dependency chain that should govern future edits. New local estimates are useful only insofar as they feed one of these eight loads.

Purpose

The full dynamics stack currently mixes several hard problems at once:

- state-dependent delays,
- Jacobian amplification,
- self-hit branch birth,
- tangential drift in 2D and 3D,
- and multi-scale coupling in nested shell braids.

This chapter strips away everything except the minimum ingredients needed to test a bounded delayed orbit:

- two architrinos,
- one spatial dimension,
- opposite polarities,
- exact partner hits,
- exact self-hit roots,
- and an $\eta > 0$ regularization suitable for return-map analysis.

The point is not to claim that this reduced problem is already the physical atom of the theory. The point is to identify the first model in which a breather-like bounded state could be proved or ruled out.

This chapter should therefore be read as an internal reduced model inside $\mathbb{AAA}$, not as a claim about standard electrodynamics. Its delayed kernel, self-hit bookkeeping, and dual-mollified return-map architecture are the working axioms of the present theorem program. The relation of that program to more classical delayed-interaction formalisms, such as action-based Fokker or Wheeler-Feynman-type viewpoints, belongs to the surrounding master-equation discussion rather than being assumed here as an equivalence theorem.

Exact 1D State Variables

Work on the reflection-symmetric center-of-mass subspace

$$x_1(t) = -x(t), \qquad x_2(t) = x(t), \qquad q_1 = -\epsilon, \qquad q_2 = +\epsilon$$

Here:

- $x(t) \in \mathbb{R}$ is the signed position of the right-hand architrino,
- $\dot{x}(t)$ is its signed velocity,
- the center of mass is fixed at the origin,
- and the full two-body state is recovered by reflection.

The exact delayed state lives on a history space

$$\mathcal{H}_h = C^1([-h, 0]; \mathbb{R})$$

with history segment

$$x_t(\theta) \equiv x(t + \theta), \qquad \theta \in [-h, 0]$$

for a memory horizon $h > 0$ large enough to contain all active causal roots under study.

Useful derived quantities:

$$d(t) \equiv 2|x(t)|, \qquad u(t) \equiv \dot{x}(t)$$

When $x(t) > 0$ and $u(t) < 0$, the labeled right-hand architrino is inbound on the right exterior branch. After label-preserving passage through the center, the same particle continues on the left exterior branch with $x(t) < 0$. For theorem work on full oscillations, the safest interpretation is therefore in signed coordinates $x \in \mathbb{R}$ together with the radial distance $d(t) = 2|x(t)|$.

Partner-Only Hinge Radius

Before self-hit is active, the natural zeroth-order picture is partner-dominated infall from large separation. In that reduced picture it is useful to define the first dynamically meaningful radius as the location where the inbound speed reaches the field speed.

Definition

Let $x_{c_f} > 0$ denote the **hinge radius** for the partner-only inbound benchmark:

$$|u| = c_f \qquad \text{at} \qquad x = x_{c_f}$$

This is not yet a theorem of the full delayed system. It is a reduced-model normalization tied to the inbound partner-attraction phase before the self-hit-capable regime is entered.

Dimensionless normalization

Define

$$\chi \equiv \frac{x}{x_{c_f}}, \qquad v \equiv \frac{u}{c_f}$$

Then the hinge is located at

$$\chi = 1, \qquad |v| = 1$$

This makes the reduced narrative explicit:

- $\chi \gg 1$: far-field partner-dominated infall,
- $\chi \searrow 1$: approach to the field-speed hinge,
- $\chi < 1$: self-hit-capable regime becomes dynamically relevant.

Coulomb-like zeroth-order estimate

If one uses the quadratic kinetic bookkeeping proxy with universal constant μ_{arch} and ignores delay at leading order, the partner-only effective potential is

$$U_{\text{pair}}(x) \approx -\frac{\kappa\epsilon^2}{2x}$$

since the pair separation is $d = 2x$.

Starting from rest at infinity, a zeroth-order energy balance gives

$$\frac{1}{2}\mu_{\text{arch}}u^2 \approx \frac{\kappa\epsilon^2}{2x}$$

Imposing the hinge condition $|u| = c_f$ yields

$$\mu_{\text{arch}}c_f^2 \approx \frac{\kappa\epsilon^2}{x_{c_f}}, \qquad x_{c_f} \approx \frac{\kappa\epsilon^2}{\mu_{\text{arch}}c_f^2}$$

Thus one may, if desired, choose reduced units so that

$$x_{c_f} = 1$$

This is the cleanest way to formalize the intuition that the partner-only inbound fall from infinity reaches field speed at a distinguished radius. In the bookkeeping convention just displayed, setting $x_{c_f} = 1$ fixes the dimensionless combination

$$\frac{\kappa\epsilon^2}{\mu_{\text{arch}}c_f^2} = 1$$

If one also chooses reduced units with $c_f = 1$, $\epsilon = 1$, and $\mu_{\text{arch}} = 1$, then the local benchmark has $\kappa = 1$. This is a unit convention for the partner-only hinge estimate, not a physical derivation of the coupling. An acceleration-first normalization that omits the quadratic bookkeeping constant must declare its convention separately, because numerical factors in the hinge relation then shift.

The partner-only benchmark should also not be read as a crossing formula at the origin. The same zeroth-order energy estimate gives

$$u^2 \approx \frac{\kappa\epsilon^2}{\mu_{\text{arch}}x}$$

so it predicts $|u| \to \infty$ as $x \to 0^+$. The hinge marks where the reduced partner-only model has reached the self-hit-capable regime and should hand off to the delayed root ledger rather than being extrapolated through the origin.

Delayed partner correction

The preceding estimate is intentionally only a dimensional scale. It drops the causal-root Jacobian and therefore should not be used as evidence that a released history reaches field speed at a finite exterior radius. On the simplest action-generated exterior chart, the partner source is locally affine:

$$x(s) \approx x(t) - v(t-s), \qquad v = \dot{x}(t), \qquad x(t) > 0$$

The partner causal equation gives

$$\tau_p = t - s = \frac{2x}{c_f + v}, \qquad r_p = c_f \tau_p, \qquad J_p = 1 + \frac{v}{c_f}$$

Thus the delayed partner force is not the naive conservative inverse-square force. The simple-root branch law is

$$\ddot{x} = -\frac{g}{4x^2}\left(1 + \frac{\dot{x}}{c_f}\right), \qquad g \equiv \kappa \epsilon^2$$

as long as the exterior partner chart remains valid and $-c_f < \dot{x}$.

Writing

$$\beta \equiv \frac{\dot{x}}{c_f}, \qquad \alpha \equiv \frac{g}{4c_f^2}$$

the phase equation has the exact invariant

$$\beta - \beta_0 - \ln\left(\frac{1+\beta}{1+\beta_0}\right) = \alpha\left(\frac{1}{x} - \frac{1}{x_0}\right)$$

For a released exterior branch with $\beta_0 = 0$ at $x = x_0$, this becomes

$$\beta - \ln(1+\beta) = \alpha\left(\frac{1}{x} - \frac{1}{x_0}\right)$$

The left side diverges as $\beta \to -1^+$, so this delayed partner branch approaches field speed only in the limiting approach to $x = 0$, not at a finite exterior radius. The finite-radius hinge scale x_{c_f} is therefore a normalization of the naive partner-only estimate, while the action-generated delayed partner test says that any finite-radius field-speed crossing must come from structure omitted by the affine exterior partner chart: finite-width shell effects, core-layer transit, nonaffine path history, self-image terms, or a different certified branch chart.

The corresponding exact sub-field branch can be written with the Lambert W function. Let

$$S(x) \equiv \beta_0 - \ln(1+\beta_0) + \alpha\left(\frac{1}{x} - \frac{1}{x_0}\right)$$

Then

$$\beta_k(x) = -1 - \mathrm{W}_k\left(-e^{-(S(x)+1)}\right)$$

For the inbound released branch, $k = 0$. Define

$$\beta_{\mathrm{in}}(x) \equiv -1 - \mathrm{W}_0\left(-e^{-(S(x)+1)}\right)$$

Then $-1 < \beta_{\mathrm{in}}(x) < 0$, and the time parametrization is the quadrature

$$t - t_0 = \int_x^{x_0} \frac{d\xi}{-c_f\, \beta_{\mathrm{in}}(\xi)}$$

This is the controlled sub-field-speed analytic comparison problem: it is generated by the delayed branch force rather than by prescribing a future path. A full breather theorem still has to prove how this exterior branch connects through the core layer and returns through the history-space map.

Field-Speed Head-On Caustic Test

A tempting boundary test is to place the left Electrino and right Positrino at

$$x_L(0) = -x_0, \qquad x_R(0) = +x_0$$

with inward velocities

$$\dot{x}_L(0) = +c_f, \qquad \dot{x}_R(0) = -c_f$$

This is not ordinary initial data for the simple-root branch law unless the path history is also specified. If the intended prehistory is affine field-speed infall,

$$x_L(s) = -x_0 + c_f s, \qquad x_R(s) = x_0 - c_f s, \qquad s \le 0$$

then the opposite-source wake is still in flight at $t = 0$. For the right-hand receiver before the origin meeting,

$$x_R(t) - x_L(s) = c_f(t - s)$$

reduces to

$$2x_0 - c_f(t + s) = c_f(t - s)$$

and hence

$$t = \frac{x_0}{c_f}$$

Thus there is no partner root for $0 \le t < x_0/c_f$ on this affine chart, while at $t = x_0/c_f$ all affine source times co-arrive at the origin. That event is a caustic, not a regular branch.

The same-source ledger is already singular. Along either affine field-speed history,

$$|x_i(t) - x_i(s)| = c_f(t - s)$$

for every $s < t$, and the simple-root Jacobian is

$$J = 0$$

Therefore the exact field-speed head-on seed is a fail-closed separator test. It can be studied only through the dual-mollified finite-history integral with declared shell width, core scale, emission cadence if used, and memory horizon. If that regularized limit fails to converge, the result is not a failed simulation detail; it means that exact field-speed inbound history is not a lawful seed for the collinear breather certificate without dephasing, curvature, held-release preparation, or another branch-certified regularization mechanism.

The finite-history calculation can still be stated exactly before the origin caustic. For the right Positrino, define

$$\Delta(t) \equiv 2(x_0 - c_f t), \qquad g \equiv \kappa\epsilon^2$$

The same-source continuum contributes

$$a_R^{\text{self}}(t; H, \eta, \epsilon_c) = -g\,\delta_\eta(0)\int_0^H \frac{du}{c_f^2 u^2 + \epsilon_c^2} = -\frac{g\,\delta_\eta(0)}{c_f\epsilon_c}\arctan\left(\frac{c_f H}{\epsilon_c}\right)$$

while the off-shell partner tail contributes

$$a_R^{\text{partner}}(t; H, \eta, \epsilon_c) = -g\,\delta_\eta(\Delta(t))\int_0^H \frac{du}{(\Delta(t) + c_f u)^2 + \epsilon_c^2}$$

Equivalently,

$$a_R^{\text{partner}} = -\frac{g\,\delta_\eta(\Delta(t))}{c_f\epsilon_c}\left[\arctan\left(\frac{\Delta(t) + c_f H}{\epsilon_c}\right) - \arctan\left(\frac{\Delta(t)}{\epsilon_c}\right)\right]$$

Thus the infinite-history limit is finite for fixed η and ϵ_c,

$$a_R^{\text{self}}(\infty, \eta, \epsilon_c) = -\frac{\pi g\,\delta_\eta(0)}{2c_f\epsilon_c}$$

but it is not regulator independent. For any centered mollifier with $\delta(0) > 0$ and

$$\delta_\eta(y) = \eta^{-1}\delta(y/\eta)$$

the same-source continuum scales as

$$-\frac{\pi g\,\delta(0)}{2c_f\eta\epsilon_c}$$

as $\eta, \epsilon_c \to 0$. For the compact polynomial shell candidate

$$\delta(z) = \frac{15}{16}(1 - z^2)^2$$

on $|z| \le 1$ and zero outside, this becomes

$$-\frac{15\pi g}{32 c_f \eta \epsilon_c}$$

At $t = 0$ with $x_0 = 1$, this compact shell also sets the partner term exactly to zero whenever $\eta < 2x_0$, because the partner support has not reached the receiver. The theory consequence is sharper than "the partner wake is in flight": an exact affine $v = c_f$ inbound history produces a regulator-dependent self-continuum acceleration before the partner wake can contribute. A lawful candidate must therefore break the exact continuum by preparation or branch geometry before the simple-root certificate can begin.

Prepared Held-Release Benchmark

A useful initial-data test fixes the pre-release history explicitly. Let a right-hand Positrino and left-hand Electrino be held at

$$x_2(t) = +x_0, \qquad x_1(t) = -x_0, \qquad x_0 > 0$$

with zero velocity for a holding interval long enough that the stationary partner wakes have reached the opposite side before release:

$$T_{\text{hold}} \ge \frac{2x_0}{c_f}$$

Set the release time to $t = 0$. During the held interval the holding constraint cancels the stationary partner attraction, and a stationary architrino has no nontrivial self-hit root.

For the right-hand coordinate after release, while the partner emission time still lies in the held history, the partner source position is fixed at $-x_0$ and the partner Jacobian is 1. The reduced equation is the ordinary initial-value problem

$$\ddot{x}(t) = -\frac{\kappa\epsilon^2}{(x(t) + x_0)^2}, \qquad x(0) = x_0, \qquad \dot{x}(0) = 0$$

In particular,

$$\ddot{x}(0) = -\frac{\kappa\epsilon^2}{4x_0^2}$$

This initial action-generated ODE segment has an exact energy identity. With

$$g \equiv \kappa\epsilon^2$$

one has

$$\frac{1}{2}\dot{x}^2 - \frac{g}{x + x_0} = -\frac{g}{2x_0}$$

and therefore

$$|\dot{x}|^2 = 2g\left(\frac{1}{x + x_0} - \frac{1}{2x_0}\right)$$

Extrapolating the stationary-source ODE to the origin gives the speed bound

$$|\dot{x}|^2_{\max} = \frac{g}{x_0}$$

Therefore the extrapolated held-source branch reaches field speed strictly before the origin only if

$$c_f^2 < \frac{g}{x_0}$$

with equality corresponding to field speed at the origin. In the normalized comparison $g = 1$ and $c_f = 1$, a release from $x_0 > 1$ remains sub-field-speed throughout this ODE extrapolation, and therefore throughout the actual held-source segment. Thus a starting position such as $x_0 = 1.25$ does not by itself imply a field-speed crossing under the action-generated held-release force.

The handoff from held partner history to moving partner history is also explicit. Put

$$y(t) \equiv x(t) + x_0$$

The inbound solution can be parametrized by

$$y(\theta) = 2x_0 \cos^2\theta, \qquad x(\theta) = x_0 \cos(2\theta), \qquad \dot{x}(\theta) = -\sqrt{\frac{g}{x_0}}\tan\theta$$

with

$$t(\theta) = 2x_0\sqrt{\frac{x_0}{g}}\,(\theta + \sin\theta\cos\theta)$$

The first moving-partner emission, released at $t = 0$, reaches the right-hand receiver when

$$y(t_*) = c_f t_*$$

Equivalently, if

$$\rho \equiv c_f\sqrt{\frac{x_0}{g}}$$

then the handoff angle is the unique solution of

$$\cos^2\theta_* = \rho\,(\theta_* + \sin\theta_*\cos\theta_*)\,, \qquad 0 \le \theta_* \le \frac{\pi}{4}$$

whenever the solution occurs before the origin. Since the left-minus-right side has derivative

$$-\sin(2\theta) - 2\rho\cos^2\theta < 0$$

the root is unique. It occurs before or at the origin exactly when

$$\rho \ge \frac{1}{1 + \pi/2}$$

The stronger normalized condition $x_0 > g/c_f^2$ gives $\rho > 1$, so the held-source segment both hands off before the origin and remains strictly sub-field-speed up to the handoff.

If one asks whether field speed occurs before the handoff rather than before the origin, the exact comparison is sharper. Field speed would occur at $\theta_c = \arctan\rho$, so it occurs before handoff exactly when

$$\frac{1-\rho^2}{1+\rho^2} > \rho\arctan\rho$$

The $x_0 = 1.25$, $g = 1$, $c_f = 1$ benchmark has $\rho > 1$, so it is safely outside that early-field-speed regime.

For $x_0 = 1.25$, $g = 1$, and $c_f = 1$, the handoff values are

$$\theta_* \approx 0.400048009813582, \qquad x_* \approx 0.8707972823389274, \qquad \dot{x}_* \approx -0.37820836925058077$$

Thus the lawful held-release preparation enters the moving-partner delayed chart with a finite sub-field-speed state rather than with the singular exact field-speed self-continuum.

The handoff is also a regular simple-root opening. For the moving partner emission time t_e, define

$$F(t, t_e) \equiv x(t) + x(t_e) - c_f(t - t_e)$$

At the handoff,

$$F(t_*, 0) = 0, \qquad \partial_{t_e}F(t_*, 0) = \dot{x}(0) + c_f = c_f > 0$$

Therefore the delayed emission time continues uniquely for t near t_*, with

$$\frac{dt_e}{dt} = \frac{c_f - \dot{x}(t)}{c_f + \dot{x}(t_e)}$$

The partner Jacobian at handoff is

$$J_p(t_*; 0) = 1 + \frac{\dot{x}(0)}{c_f} = 1$$

so the partner acceleration is continuous across the transition:

$$-\frac{g}{(x(t_*) + x_0)^2} = -\frac{g}{(x(t_*) + x(0))^2\,J_p(t_*; 0)}$$

Thus the held-release handoff is not a hidden caustic. It is a regular transfer from fixed partner history into the delayed moving-partner chart.

This ODE segment remains valid until the first post-release partner emission reaches the right-hand receiver. If t_* denotes that handoff time, then

$$x(t_*) + x_0 = c_f t_*$$

After that time the partner root samples the moving history and must be solved as a delayed emission time $t_e < t$:

$$x(t) + x(t_e) = c_f(t - t_e)$$

with exterior-branch partner Jacobian

$$J_p(t; t_e) = 1 + \frac{\dot{x}(t_e)}{c_f}$$

Thus the held-release benchmark supplies a simple ODE start, but it does not remove the delayed partner-root problem once the receiver begins sampling post-release history.

Partner-Hit and Self-Hit Root Equations

For the right-hand architrino $x_2(t) = x(t)$, the exact causal root conditions split naturally into partner and self branches.

Partner-hit roots

A partner-hit emission time $t_0 < t$ satisfies

$$|x(t) + x(t_0)| = c_f(t - t_0)$$

Define the partner-root set

$$\mathcal{C}_p(t) \equiv \{t_0 < t \mid |x(t) + x(t_0)| = c_f(t - t_0)\}$$

On this symmetry subspace the 1D line-of-action sign is

$$\hat{r}_p(t; t_0) = \operatorname{sgn}\big(x(t) + x(t_0)\big)$$

and the partner Jacobian becomes

$$J_p(t; t_0) = 1 - \frac{\dot{x}_1(t_0)\hat{r}_p(t; t_0)}{c_f} = 1 + \frac{\dot{x}(t_0)\hat{r}_p(t; t_0)}{c_f}$$

Because $q_1 q_2 < 0$, this branch is attractive.

Self-hit roots

A nontrivial self-hit emission time $t_0 < t$ satisfies

$$|x(t) - x(t_0)| = c_f(t - t_0), \qquad t_0 \neq t$$

Define the self-root set

$$\mathcal{C}_s(t) \equiv \{t_0 < t \mid |x(t) - x(t_0)| = c_f(t - t_0)\}$$

The self line-of-action sign is

$$\hat{r}_s(t; t_0) = \operatorname{sgn}\big(x(t) - x(t_0)\big)$$

and the self Jacobian is

$$J_s(t; t_0) = 1 - \frac{\dot{x}(t_0)\hat{r}_s(t; t_0)}{c_f}$$

Because $q_2 q_2 > 0$, each self branch is repulsive.

Reduced branch-resolved equation

On the exact root-selected model, the right-particle acceleration is

$$\ddot{x}(t) = -\kappa\epsilon^2 \sum_{t_0 \in \mathcal{C}_p(t)} \frac{\hat{r}_p(t; t_0)}{|x(t) + x(t_0)|^2 \, |J_p(t; t_0)|} + \kappa\epsilon^2 \sum_{t_0 \in \mathcal{C}_s(t)} \frac{\hat{r}_s(t; t_0)}{|x(t) - x(t_0)|^2 \, |J_s(t; t_0)|}$$

The first sum is partner attraction. The second is self-hit repulsion. Reflection symmetry gives the left-particle equation automatically.

Plain language: in 1D there is no tangential direction to hide in. The entire competition is between delayed inward attraction and delayed outward self-repulsion, with the Jacobian deciding how sharply each branch is weighted.

Regularized 1D Equation

For analysis and numerics, replace the shell delta by a smooth mollifier δ_η with width $\eta > 0$. Then the reduced equation can be written in integral form:

$$\ddot{x}(t) = -\kappa\epsilon^2 \int_{-\infty}^{t} dt_0\, \frac{\hat{r}_p(t;t_0)}{|x(t)+x(t_0)|^2}\delta_\eta\big(|x(t)+x(t_0)| - c_f(t-t_0)\big)$$

$$+\kappa\epsilon^2 \int_{-\infty}^{t} dt_0\, \frac{\hat{r}_s(t;t_0)}{|x(t)-x(t_0)|^2}\delta_\eta\big(|x(t)-x(t_0)| - c_f(t-t_0)\big)$$

with the understanding that the exact Jacobian factors reappear in the branch-sum representation when the mollified shell collapses onto isolated roots.

The normalization convention for the shell mollifier is fixed once and used throughout the estimates below. Choose a nonnegative even

$$C^1$$

function

$$\delta \in C_c^1(\mathbb{R}), \qquad \operatorname{supp}\delta \subset [-1,1], \qquad \int_{\mathbb{R}} \delta(y)\,dy = 1$$

and set

$$\delta_\eta(y) \equiv \eta^{-1}\delta(y/\eta)$$

Thus

$$\operatorname{supp}\delta_\eta \subset [-\eta,\eta], \qquad \int_{\mathbb{R}} \delta_\eta(y)\,dy = 1, \qquad \|\delta_\eta\|_\infty = \eta^{-1}\|\delta\|_\infty$$

Every shell-leakage, fold-impulse, and outer-self estimate using

$$\|\delta_\eta\|_\infty$$

uses this convention. On a simple-root chart,

$$\int f(s)\delta_\eta(g(t,s))\,ds \longrightarrow \sum_{g(t,s_k)=0} \frac{f(s_k)}{|\partial_s g(t,s_k)|}$$

and the fixed factor

$$c_f^{-1}$$

from

$$\partial_s g = c_f J$$

is absorbed into the branch-law normalization of

$$\kappa$$

For theorem work across the origin crossing, shell regularization alone is not enough to control the inverse-square amplitude. A more robust local model therefore introduces a **dual mollification**: the shell mollifier δ_η for delayed root selection together with a short-distance core mollifier $\epsilon_c > 0$ in the amplitude denominator,

$$\frac{1}{r^2} \quad \rightsquigarrow \quad \frac{1}{r^2+\epsilon_c^2}$$

This leaves the delayed shell selection controlled by η while the core mollifier caps the near-origin amplitude spike strongly enough for a clean C^1 theorem program.

For the certified finite-memory problem, the exact dual-mollified reduced evolution law is the absolute-time integral

$$\ddot{x}(t) = -\kappa\epsilon^2 \int_{t-h}^{t} \frac{\hat{r}_p(t;s)}{|x(t)+x(s)|^2+\epsilon_c^2}\, \delta_\eta\big(|x(t)+x(s)| - c_f(t-s)\big)\, ds$$

$$+\kappa\epsilon^2 \int_{t-h}^{t} \frac{\hat{r}_s(t;s)}{|x(t)-x(s)|^2+\epsilon_c^2}\, \delta_\eta\big(|x(t)-x(s)| - c_f(t-s)\big)\, ds$$

The branch-sum equations used throughout the proof scaffold are simple-root reductions of this law. Across causal folds, caustic transit, and certified topology arguments, the integral law is the primary object.

The regularized formulation is the one best suited to:

- local well-posedness,
- continuation criteria,
- numerical return-map construction,
- and eventually the controlled limit $\eta \to 0^+$.

Origin-layer continuity of the dual-mollified 1D field

On an interval that contains an origin crossing, the working equation is the absolute-time integral law above, not the branch-sum reduction. The branch-sum signs

$$\hat{r}_p, \qquad \hat{r}_s$$

are exterior-chart data. They must be reattached to the correct outgoing sheet after the crossing, and they should not be treated as a smooth scalar formula through

$$x = 0$$

> **Lemma (Origin-layer continuity of the dual-mollified 1D field).**
> Fix

$$> \eta > 0, > \qquad > \epsilon_c > 0, > \qquad > h > 0. >$$

> Let a signed collinear history have a single label-preserving origin crossing on a layer

$$> |t| \le \tau_{\mathrm{cross}}, >$$

> with fixed incoming and outgoing exterior-sheet labels. Define the sheet-projected radial acceleration on the layer by applying the absolute-time integral law in the signed coordinate and then projecting to the active radial sheet:

$$> F^{\rho}_{\eta,\epsilon_c}(t) > \equiv > \sigma_{\mathrm{out}}(t) F^{x}_{\eta,\epsilon_c}(t), > \qquad > \rho(t) = |x(t)|. >$$

> Assume the layer has a uniform velocity bound, a uniform acceleration bound, and no uncontrolled nontransverse root accumulation except the certified fold events carried by the layer chart. Then

$$> F^{\rho}_{\eta,\epsilon_c} >$$

> extends as a

$$> C^1 >$$

> function of the radial coordinate and stored history across

$$> \rho = 0. >$$

> In particular, the sign flip of the exterior scalar branch terms is absorbed by the sheet projection, and the radial equation may be continued through the origin layer without introducing a scalar force discontinuity.

Proof.
The absolute-time integral law has denominator bounded below by

$$\epsilon_c^2$$

and the shell factor

$$\delta_\eta$$

is a fixed

$$C^1$$

mollifier with compact support. Hence the layer integrand and its first variations in the stored

$$C^1$$

history are dominated by constants depending only on

$$(\eta, \epsilon_c, h)$$

and the layer tube bounds. The signed exterior direction changes when the trajectory passes through

$$x = 0$$

but the radial projection multiplies by the outgoing sheet label at the same crossing. On the two sides of the layer this converts the exterior signed direction into the same radial direction field. Any remaining sign changes occur only across the certified causal-root surfaces inside the integral; away from certified folds they are simple-root changes of variables, and at certified folds the fold tube is handled by the integral law rather than by a branch sum. Dominated convergence, with the standard one-dimensional coarea calculation on the simple-root pieces, gives continuity and the first radial derivative on the whole layer.

Thus the core scale controls the amplitude and the shell scale controls the root selection, while the sheet projection controls the origin sign flip. Exterior branch-sum formulas may be resumed after the trajectory leaves the crossing layer and the signed sheet labels are again fixed.

Inbound/Outbound Sign Structure

The first genuine dynamical question is not whether self-hit exists, but whether its sign structure permits recapture. In 1D this can be stated exactly.

Exterior-branch convention

Fix an interval on which

$$x(t) > 0$$

Then:

- **inbound** means $\dot{x}(t) < 0$,
- **outbound** means $\dot{x}(t) > 0$.

This is the natural branch on which to analyze collapse, rebound, and return to a section at $x = x_* > 0$.

Partner term

On the exterior branch, the partner source points inward only for active partner roots whose delayed source remains on the opposite side of the current right-hand particle. In the signed variables this is the branch condition

$$x(t) + x(t_0) > 0$$

The recapture estimates below use a tame exterior-root class in which all active partner roots entering the lower-bound arguments satisfy this condition. Any partner roots with

$$x(t) + x(t_0) < 0$$

are not inward partner roots; they must either be excluded by the delayed-root hypotheses or carried as a separate error channel.

Write

$$A_p(t) \equiv \kappa \epsilon^2 \sum_{t_0 \in \mathcal{C}_p(t)} \frac{\mathbf{1}_{\{x(t)+x(t_0)>0\}}}{|x(t) + x(t_0)|^2 \, |J_p(t; t_0)|} \geq 0$$

On that inward exterior partner channel the signed contribution is

$$a_p(t) = -A_p(t)$$

Therefore, when the inward exterior partner channel is active:

- on the inbound leg, a_p has the **same sign as the velocity** and speeds the collapse up,
- on the outbound leg, a_p has the **opposite sign to the velocity** and brakes the escape.

Self-hit split into outer-memory and inner-memory roots

The self term does not have a fixed sign. Split the active self roots into

$$\mathcal{C}_s^{\text{out}}(t) \equiv \{t_0 \in \mathcal{C}_s(t) \mid x(t_0) > x(t)\}$$

$$\mathcal{C}_s^{\text{in}}(t) \equiv \{t_0 \in \mathcal{C}_s(t) \mid x(t_0) < x(t)\}$$

For $t_0 \in \mathcal{C}_s^{\text{out}}(t)$ one has

$$\hat{r}_s(t; t_0) = \operatorname{sgn}(x(t) - x(t_0)) = -1$$

so that branch contributes **negative** acceleration.

For $t_0 \in \mathcal{C}_s^{\text{in}}(t)$ one has

$$\hat{r}_s(t; t_0) = \operatorname{sgn}(x(t) - x(t_0)) = +1$$

so that branch contributes **positive** acceleration.

Define the corresponding positive amplitudes

$$A_s^{\text{out}}(t) \equiv \kappa\epsilon^2 \sum_{t_0 \in \mathcal{C}_s^{\text{out}}(t)} \frac{1}{|x(t) - x(t_0)|^2 \, |J_s(t; t_0)|}$$

$$A_s^{\text{in}}(t) \equiv \kappa\epsilon^2 \sum_{t_0 \in \mathcal{C}_s^{\text{in}}(t)} \frac{1}{|x(t) - x(t_0)|^2 \, |J_s(t; t_0)|}$$

On the tame exterior-root class where all active partner roots are inward exterior roots, the total acceleration on the exterior branch is

$$\ddot{x}(t) = -A_p(t) - A_s^{\text{out}}(t) + A_s^{\text{in}}(t)$$

This is the key reduced formula.

Physical interpretation

- **Inbound** ($\dot{x} < 0$):
  - the retained inward partner channel strengthens infall,
  - outer-memory self roots also strengthen infall,
  - inner-memory self roots oppose infall.
- **Outbound** ($\dot{x} > 0$):
  - the retained inward partner channel brakes the outward motion,
  - outer-memory self roots also brake the outward motion,
  - inner-memory self roots drive further escape.

So self-hit is not a permanent outward engine. Its effect depends on where the active remembered emission points sit relative to the current position.

Signed-branch caution

The formulas above are exact on a fixed exterior slice $x(t) > 0$, but they should not be overread as proving that a physical 1D trajectory can rebound at some $x_{\min} > 0$ and then move back out on the same right-hand branch. In the current 1D delayed kernel, the pre-origin inbound leg is driven inward by partner attraction and by the self branches available on that slice. So the physically relevant oscillatory program should be formulated as an **origin-crossing** one in signed coordinates, or equivalently in the radial variable

$$\rho(t) \equiv |x(t)|$$

In that formulation, a full oscillation alternates between the right and left exterior branches with label-preserving passage through $x = 0$. The theorem targets later in this note should therefore be read as targets for post-crossing recapture of the radial distance rather than as literal pre-origin bounce statements on a single $x > 0$ branch.

Every theorem that crosses the origin must use the origin-layer integral chart from Lemma `Origin-layer continuity of the dual-mollified 1D field`. The signed branch-sum formulas are valid again only after the crossing layer has been exited and the exterior sheet labels have been fixed. In particular, local recapture estimates stated in

$$\rho(t) = |x(t)|$$

are radial post-crossing estimates; they are not proofs that the signed scalar branch-sum field is smooth at

$$x = 0$$

In particular, the present 1D geometry should not be treated as a radial simplification of the 2D circular case. Along a true collinear history, the self-hit term is naturally read as an anti-damping or positive-work contribution on the physically relevant post-crossing outbound branch: the self interaction tends to reinforce the current radial motion rather than furnish a centrifugal-style barrier. The corrected theorem program therefore asks whether partner attraction can recapture the motion **despite** that self-drive, not because self-hit itself creates the turnaround.

Necessary Recapture Condition

The breather question can now be reduced to one concrete inequality.

Fix an outbound time $t_\sharp$ on the exterior branch with

$$x(t_\sharp) > 0, \qquad \dot{x}(t_\sharp) = u_\sharp > 0$$

If the trajectory is ever to turn around and re-enter as an inbound branch, there must exist a later time $t_{\text{turn}} > t_\sharp$ such that

$$\dot{x}(t_{\text{turn}}) = 0$$

Integrating the reduced acceleration identity gives

$$0 = u_\sharp + \int_{t_\sharp}^{t_{\text{turn}}} \Big(- A_p(s) - A_s^{\text{out}}(s) + A_s^{\text{in}}(s) \Big)\, ds$$

Equivalently,

$$u_\sharp = \int_{t_\sharp}^{t_{\text{turn}}} \Big(A_p(s) + A_s^{\text{out}}(s) - A_s^{\text{in}}(s) \Big)\, ds$$

Therefore a **necessary condition for recapture** is

$$\sup_{t > t_\sharp} \int_{t_\sharp}^{t} \Big(A_p(s) + A_s^{\text{out}}(s) - A_s^{\text{in}}(s) \Big)\, ds \geq u_\sharp$$

If this inequality fails, then the total accumulated braking from partner attraction plus outer-memory self-hit is never strong enough to overcome the outbound speed and the trajectory cannot turn around.

Stronger sufficient criterion

If there exists an interval $[t_1, t_2]$ with $t_1 \geq t_\sharp$ on which

$$A_p(t) + A_s^{\text{out}}(t) - A_s^{\text{in}}(t) \geq \delta > 0$$

for all $t \in [t_1, t_2]$, and

$$\int_{t_1}^{t_2} \delta\, dt \geq \dot{x}(t_1)$$

then a turning point must occur no later than t_2.

This criterion is not expected to be the final theorem, but it gives the correct sign target for both numerics and analysis.

Regularized Return Map

To state a breather problem precisely, define a return section on the symmetric history space rather than on instantaneous phase space alone.

Fix:

- a section location $x_* > 0$,
- a memory horizon h large enough to contain all active branches on one cycle,
- and a regularization width $\eta > 0$.

Admissible history class

Work first with the raw outbound and inbound sections in the full history space

$$\mathcal{H}_h$$

$$\Sigma^{+}_{x_*,\eta} \equiv \left\{ \phi \in \mathcal{H}_h \;\middle|\; \phi(0) = x_*, \qquad \dot{\phi}(0) > 0 \right\}$$

$$\Sigma^{-}_{x_*,\eta} \equiv \left\{ \phi \in \mathcal{H}_h \;\middle|\; \phi(0) = x_*, \qquad \dot{\phi}(0) < 0 \right\}$$

Because the section histories are anchored by

$$\phi(0) = x_*$$

with prescribed crossing sign, this return section quotients out the absolute time-translation symmetry of the continuous delayed flow. A periodic trajectory therefore appears as a fixed returned history rather than as an unpinned one-parameter family of time shifts.

The first workable theorem domain should not be the full sections

$$\Sigma^{\pm}_{x_*,\eta}$$

but a controlled tame subclass on which the regularized delayed dynamics and return times are well behaved.

Let

$$\mathcal{H}^{\mathrm{adm}}_{x_*,\eta} \subset \mathcal{H}_h$$

denote an admissible reflection-symmetric history class with the following properties on $\theta \in [-h, 0]$:

- section anchoring at $\theta = 0$,
- uniform position bounds

$$x_{\min} \le \phi(\theta) \le x_{\max}$$

- uniform speed bounds

$$|\dot{\phi}(\theta)| \le u_{\max}$$

- uniform Lipschitz-velocity bounds

$$|\dot{\phi}(\theta_1) - \dot{\phi}(\theta_2)| \le a_{\max}|\theta_1 - \theta_2|, \qquad \theta_1, \theta_2 \in [-h, 0]$$

- and a transversality bound on every active partner and self root,

$$|J_p| \ge \nu, \qquad |J_s| \ge \nu, \qquad \nu > 0$$

Also require the active causal memory depth to fit inside the chosen history window:

$$\tau_{\max}(\phi) \le h \qquad \text{for every } \phi \in \mathcal{H}^{\mathrm{adm}}_{x_*,\eta}$$

The role of

$$\mathcal{H}^{\mathrm{adm}}_{x_*,\eta}$$

is simple: it isolates a tame region of history space on which the regularized vector field, root selection, and section crossings can plausibly be controlled. The Lipschitz-velocity bound is the first compactness-oriented ingredient for a later Arzela-Ascoli step in C^1; equivalently, $\ddot{\phi}$ exists almost everywhere with $|\ddot{\phi}| \le a_{\max}$ in the weak sense. The memory-depth bound ensures the delayed law really closes on the chosen history interval. Whether the eventual theorem program allows histories that approach $x = 0$ arbitrarily closely is a separate question and should not be conflated with the first well-posedness regime.

For

$$\phi \in \Sigma^{+}_{x_*,\eta}$$

for which the η-regularized dynamics is well defined up to the first later time

$$T^{-}_{\eta}(\phi) > 0$$

such that:

- the trajectory has completed one outbound excursion and recapture,
- $x(T^{-}_{\eta}(\phi)) = x_*$,
- and $\dot{x}(T^{-}_{\eta}(\phi)) < 0$.

Then define the exact outbound-to-inbound history map on its natural domain

$$Q_\eta : \mathrm{Dom}(Q_\eta) \subseteq \Sigma^+_{x_*,\eta} \to \Sigma^-_{x_*,\eta}, \qquad Q_\eta(\phi) = x_{T^-_\eta(\phi)}$$

For

$$\phi \in \Sigma^-_{x_*,\eta}$$

for which the η-regularized dynamics is well defined up to the first return time

$$T(\phi) > 0$$

such that:

- the trajectory has completed one collapse-and-rebound cycle,
- $x(T(\phi)) = x_*$,
- and $\dot{x}(T(\phi)) < 0$ again.

Then define the exact history-space return map on its natural domain

$$P_\eta : \mathrm{Dom}(P_\eta) \subseteq \Sigma^-_{x_*,\eta} \to \Sigma^-_{x_*,\eta}, \qquad P_\eta(\phi) = x_{T(\phi)}$$

This is the natural reduced object for theorem work. The core fixed-point question belongs to P_η on a controlled subset of history space, not to any scalar speed map by itself.

Projected scalar speed map

The scalar map is useful only after choosing a specific way to inject scalar speed data into the outbound history section.

Assume there is a continuous injection

$$\iota_\eta(\,\cdot\,; x_*) : I \to \Sigma^+_{x_*,\eta}, \qquad u \mapsto \phi^+_\eta(u; x_*)$$

from an interval $I \subset (0, \infty)$ of outbound speeds into admissible outbound histories, such that

$$\dot{\phi}^+_\eta(u; x_*)(0) = u$$

This injection is extra structure. It is not part of the master equation itself; it is a chosen slice through history space.

Write the corresponding trajectory as $x(t; u, x_*, \eta)$ with initial section data

$$x(0; u, x_*, \eta) = x_*, \qquad \dot{x}(0; u, x_*, \eta) = u$$

If the trajectory is recaptured and returns to the inbound section, define the projected scalar map

$$R_\eta(u; x_*) \equiv -\dot{x}\big(T^-_\eta(u; x_*); u, x_*, \eta\big) > 0$$

Thus $R_\eta(u; x_*)$ is the magnitude of the next inbound speed when the trajectory re-crosses the same section $x = x_*$.

Equivalently, if

$$\Pi : \Sigma^-_{x_*,\eta} \to (0, \infty), \qquad \Pi(\phi) \equiv -\dot{\phi}(0)$$

denotes the inbound speed projection on the section, then

$$R_\eta(\,\cdot\,; x_*) = \Pi \circ Q_\eta \circ \iota_\eta(\,\cdot\,; x_*)$$

This is the correct status of the scalar map: it is a projection of the history-space excursion map through a chosen one-parameter injection, not an autonomous closure law of the delayed system.

Now introduce the net inward braking density

$$B_\eta(t; u, x_*) \equiv A_p(t) + A^{\mathrm{out}}_s(t) - A^{\mathrm{in}}_s(t)$$

so that along the trajectory

$$\ddot{x}(t; u, x_*, \eta) = -B_\eta(t; u, x_*)$$

Integrating from the outbound crossing at $t = 0$ to the next inbound crossing at $t = T^-_\eta(u; x_*)$ gives

$$-R_\eta(u;x_*) = u + \int_0^{T_\eta^-(u;x_*)} \ddot{x}(s;u,x_*,\eta)\,ds = u - \int_0^{T_\eta^-(u;x_*)} B_\eta(s;u,x_*)\,ds$$

Equivalently,

$$R_\eta(u;x_*) = -u + \int_0^{T_\eta^-(u;x_*)} B_\eta(s;u,x_*)\,ds$$

This is the clean projected scalar map: the next inbound speed equals the total accumulated inward braking budget over the outbound-and-return excursion minus the outbound launch speed at the section.

If $T_\eta^{\mathrm{turn}}(u;x_*)$ denotes the first turning time with

$$\dot{x}\big(T_\eta^{\mathrm{turn}}(u;x_*)\big) = 0$$

then the same map splits into two exact pieces:

$$u = \int_0^{T_\eta^{\mathrm{turn}}(u;x_*)} B_\eta(s;u,x_*)\,ds$$

$$R_\eta(u;x_*) = \int_{T_\eta^{\mathrm{turn}}(u;x_*)}^{T_\eta^-(u;x_*)} B_\eta(s;u,x_*)\,ds$$

The first identity is the outbound recapture condition on the section. The second states that the next inbound speed is exactly the inward gain accumulated after the turning point.

Scalar closure condition

The map R_η is only a projection of the exact history-space map Q_η, but it is the sharpest scalar diagnostic for recapture on the fixed section $x = x_*$.

If the admissible family is symmetric enough that outbound and inbound section data are parameterized by the same scalar speed, then a scalar breather candidate satisfies

$$u_* = R_\eta(u_*;x_*)$$

However, this scalar fixed-point condition does not by itself imply periodic closure. The delayed dynamics only closes when the full history is returned:

$$\phi^* = P_\eta(\phi^*)$$

The scalar map is therefore best read as a reduced diagnostic for recapture and speed balance. The actual theorem program should proceed by finding a closed, bounded, invariant subset of the raw inbound section

$$\Sigma^-_{x_*,\eta}$$

and then packaging it inside the later convex-envelope hierarchy

$$\mathcal{C}_{x_*,\eta} \supseteq \mathcal{K}_{x_*,\eta}$$

Local recapture architecture

The scalar map is only a diagnostic for section-speed balance. The real local input to the global fixed-point route is a post-crossing recapture theorem on a uniform admissible crossing subclass. The global envelope hierarchy

$$\mathcal{C}_{x_*,\eta} \supseteq \mathcal{K}_{x_*,\eta}$$

is introduced later; at the local level the only issue is whether partner attraction can erase the first post-crossing outward radial speed before the self drive pushes the trajectory to large radius.

The sorting map

$$w(t) \equiv x(t) + c_f t$$

organizes that geometry. On the initial post-crossing branch, as long as

$$\dot{x}(t) < -c_f$$

one has

$$\dot{w}(t) = \dot{x}(t) + c_f < 0, \qquad w(0) = 0$$

and therefore

$$w(t) < 0 \qquad \text{for } 0 < t \le \tau_{\text{loc}}$$

If

$$t_{\text{zero}} < 0$$

is the earlier inbound time satisfying

$$w(t_{\text{zero}}) = 0$$

then every active self root selected by

$$w(t_s) = w(t)$$

must satisfy

$$t_s < t_{\text{zero}}$$

The active self roots are therefore forced back into the earlier sub-field-speed inbound source region, where the self Jacobian is automatically noncaustic. This is the mechanism behind the bounded self-drive estimate used in the local theorem below.

> **Theorem (Local Origin-Crossing Recapture).**
> Let the 1D kernel be dual-mollified by a shell width $\eta > 0$ and a core mollifier $\epsilon_c > 0$. Let ϕ be an admissible signed history with an origin crossing at $t = 0$ and outward radial speed

$$> V_0 \equiv V_\phi(0) > c_f. >$$

> Take ϕ from a fixed admissible crossing subclass

$$> \mathcal{K}^{\text{cross}}_{\eta,\epsilon_c}, >$$

> defined below so that the local constants used in Lemmas 1-4 are uniform on that class.
> Assume hypotheses (H1)-(H4) below, and assume either:
>
> 1. the abstract Goldilocks hypothesis (H5), or
> 2. the explicit short-window assumptions of Proposition `Explicit short-window recapture regime`.
>
> Then there exists a time window $[0, \tau_{\text{env}}]$ on which:

$$> w(t) < 0, > \qquad > A^\rho_s(t) \le \overline{A}^\rho_s, >$$

> and the radial acceleration satisfies

$$> \ddot{\rho}(t) \le -A^\rho_p(t) + A^\rho_s(t). >$$

> If, in addition,

$$> \tau_{\text{env}} \ge \tau_{\text{sep}}, > \qquad > \sigma \equiv \frac{V_0 - c_f}{2}, > \qquad > \tau_{\text{sep}} \equiv \frac{2\eta}{\sigma}, >$$

> then on the delayed subwindow

$$> [\tau_{\text{sep}}, \tau_{\text{env}}], >$$

> every active self root satisfies

$$> t_s \le t_{\text{zero}} - \frac{\eta}{\nu}, >$$

> so the caustic is separated from the active self branches there.
> If the resulting impulse margin obeys

$$> V_0 <> \int_0^{\tau_{\text{env}}} > \left(> \underline{A}^\rho_p(s) - \overline{A}^\rho_s > \right) ds, >$$

> then there exists

$$> \tau_{\mathrm{turn}} \in (0, \tau_{\mathrm{env}}] >$$

> such that

$$> \dot{\rho}(\tau_{\mathrm{turn}}) = 0. >$$

> In particular, the post-crossing branch is radially recaptured on the initial window.

This is the operative local theorem of the manuscript. The abstract form passes through (H5), while the concrete route used later is the explicit short-window proposition proved from Lemmas 1-4.

Hypotheses unpacked

The current theorem target is intended as a **local-in-time recapture theorem** on an initial post-crossing window. Its hypotheses can be organized as follows.

(H1) Origin-crossing data.

$$\phi(0) = 0, \qquad \dot{\phi}(0) = -V_0, \qquad V_0 > c_f$$

(H2) Sorting-map phase picture on the stored past.
For the history sorting map

$$w(\theta) = \phi(\theta) + c_f\theta, \qquad \theta \in [-h, 0]$$

there exist times

$$t_{\mathrm{zero}} < t_{\mathrm{hinge}} < 0$$

such that

$$w(t_{\mathrm{zero}}) = 0, \qquad \dot{\phi}(t_{\mathrm{hinge}}) = -c_f$$

and

$$w(\theta) > 0 \qquad \text{for } \theta \in (t_{\mathrm{zero}}, 0)$$

For any fixed interior margin

$$0 < \gamma_w < \min\{t_{\mathrm{hinge}} - t_{\mathrm{zero}}, -t_{\mathrm{hinge}}\}$$

continuity then gives the compact-subinterval gap

$$\delta_w \equiv \min_{\theta \in [t_{\mathrm{zero}} + \gamma_w, -\gamma_w]} w(\theta) > 0$$

(H3) Past transversality on the sub-field-speed source region.
There exists $\nu > 0$ such that

$$\dot{\phi}(\theta) \geq -c_f + \nu \qquad \text{for } \theta \in [-h, t_{\mathrm{zero}}]$$

(H4) Shell-mollifier separation from the interior sorting gap.
For the compact-subinterval gap chosen in (H2), the shell mollifier width is small enough that its support cannot bridge from the negative post-crossing values of $w(t)$ into the positive interior sorting hump:

$$\eta < \frac{\delta_w}{2}$$

(H5) Goldilocks crossing-speed / core-mollifier regime.
There exists a time window

$$[0, \tau_{\mathrm{env}}] \subseteq [0, \tau_{\mathrm{tube}}]$$

such that

$$V_0 < \int_0^{\tau_{\mathrm{env}}} \left(\underline{A}_p^{\rho}(s; \phi, V_0, \epsilon_c) - \overline{A}_s^{\rho}(\phi, \nu) \right) ds$$

At this stage this remains the abstract bottleneck hypothesis. A concrete sufficient realization is provided later by the proposition `Explicit short-window recapture regime`, which chooses

$$\tau_{\text{env}} = \tau_\epsilon \equiv \frac{\epsilon_c}{2\beta_{p,\max}}$$

on a fixed admissible crossing subclass and replaces the integral inequality by explicit algebraic bounds on

$$(\eta, \epsilon_c, V_{\max}, \kappa\epsilon^2)$$

Uniform admissible crossing subclass

To make those local constants concrete, fix positive class parameters

$$c_f < V_{\min} \le V_{\max}, \qquad \gamma_w, \qquad \delta_{w,\min}, \qquad \nu, \qquad \rho_{0,\min}, \qquad a_{\max}, \qquad a_{\text{tube}}, \qquad \tau_{\text{tube}}$$

and an integer root-count bound

$$N_s^{\max} \ge 1$$

Let

$$\mathcal{K}^{\text{cross}}_{\eta,\epsilon_c} \subset C^1([-h,0];\mathbb{R})$$

denote the class of signed crossing histories ϕ satisfying:

- the theorem hypotheses (H1)-(H4) with class-wide constants bounded by

$$V_{\min} \le V_0(\phi) \le V_{\max}, \qquad \delta_w(\phi;\gamma_w) \ge \delta_{w,\min}, \qquad \nu(\phi) \ge \nu$$

- a uniform pre-crossing Lipschitz-velocity bound,

$$|\dot\phi(\theta_1) - \dot\phi(\theta_2)| \le a_{\max}|\theta_1 - \theta_2| \qquad \text{for } \theta_1, \theta_2 \in [-h, 0]$$

- a uniform pre-caustic radius bound,

$$\rho_{\text{zero}}(\phi) \equiv -c_f t_{\text{zero}}(\phi) = \phi(t_{\text{zero}}(\phi)) \ge \rho_{0,\min}$$

- and a forward local tube condition: the dual-mollified forward continuation exists on

$$[0, \tau_{\text{tube}}]$$

satisfies

$$|\ddot x_\phi(t)| \le a_{\text{tube}} \qquad \text{for } 0 \le t \le \tau_{\text{tube}}$$

every active self root on that window obeys

$$|J_s(t; t_s)| \ge \frac{\nu}{c_f}$$

and has at most

$$N_s^{\max}$$

active self roots on the initial post-crossing window.

The shell width is chosen inside the class-uniform interior sorting gap:

$$\eta < \frac{\delta_{w,\min}}{2}$$

From these class parameters one may fix the derived constants

$$\sigma_{\min} \equiv \frac{V_{\min} - c_f}{2}, \qquad a_{\text{loc}} \equiv a_{\text{tube}}, \qquad a_* \equiv \max\{a_{\max}, a_{\text{tube}}\}, \qquad \beta_{p,\min} \equiv \frac{2c_f V_{\min}}{V_{\min} + c_f}, \qquad \beta_{p,\max} \equiv \frac{2c_f V_{\max}}{V_{\max} + c_f}, \qquad \tau_1 \equiv \min\left\{\tau_{\text{tube}}, \frac{\sigma_{\min}}{a_{\text{tube}}}\right\}$$

with τ_ρ chosen so that

$$V_{\max}\tau_\rho + \frac{a_{\text{tube}}}{2}\tau_\rho^2 \le \frac{\rho_{0,\min}}{2}$$

On this subclass:

- Lemma 1 uses the common continuation constants $(\sigma_{\min}, a_{\text{tube}}, \tau_1)$,
- Lemma 2 uses the common geometric separation data $(\rho_{0,\min}, \nu, N_s^{\max})$,
- Lemma 3 admits a common partner-root remainder constant

$$C_p = C_p(V_{\max}, c_f, a_*)$$

- and the explicit short-window proposition can be written uniformly by replacing (V_0, β_p) with the class-wide worst-case pair $(V_{\max}, \beta_{p,\max})$, while the delayed-entry time uses the lower-speed bound $\sigma_{\min}$.

This is the sense in which the later local constants are inherited by construction rather than introduced ad hoc.

The crossing subclass uses the following constants.

Constant	Bound or role	Used by
$V_{\min}$	lower crossing speed, strictly above c_f	Lemma 1, Lemma 3
$V_{\max}$	upper crossing speed and worst-case radial speed	Lemma 3, Lemma 4, explicit recapture regime
γ_w	compact pre-crossing sorting-gap trimming scale	(H2), Lemma 2
$\delta_{w,\min}$	class-wide lower sorting gap on the trimmed interval	shell-width exclusion in Lemma 2
ν	self-root Jacobian floor, scaled by c_f in the statement	Lemma 2 and tame topology
$\rho_{0,\min}$	minimum pre-caustic radius at t_{zero}	Lemma 2 delayed-source separation
$a_{\max}$	stored-history Lipschitz-velocity bound	Lemma 3 partner-root remainder
a_{tube}	forward acceleration ceiling on the local tube	Lemma 1 and Lemma 3
τ_{tube}	guaranteed forward continuation window	Lemma 1
$N_s^{\max}$	active self-root count ceiling	Lemma 2 self-drive bound
$\sigma_{\min}$	uniform super-field crossing surplus $(V_{\min} - c_f)/2$	Lemma 1 and delayed-entry time
a_*	common acceleration remainder bound $\max\{a_{\max}, a_{\text{tube}}\}$	Lemma 3
$\beta_{p,\min}$	lower partner linear coefficient from $V_{\min}$	short-window dominance checks
$\beta_{p,\max}$	upper partner linear coefficient from $V_{\max}$	explicit recapture regime
τ_1	class-uniform post-crossing monotonicity window	Lemma 1 through Lemma 4
τ_ρ	window on which delayed sources stay away from the origin layer	Lemma 2 delayed-window refinement

Lemma ladder

The theorem target naturally breaks into four lemmas.

Lemma 1: Short-time continuation and sorting-map monotonicity.

Prove that there exists $\tau_1 > 0$ such that

$$\dot{x}(t) \le -c_f \qquad \text{for } 0 \le t \le \tau_1$$

so that

$$\dot{w}(t) \le 0 \qquad \text{and hence} \qquad w(t) < 0$$

on the initial post-crossing window.

Working form:

let

$$\sigma \equiv \frac{V_0 - c_f}{2} > 0$$

Because the dual-mollified vector field is finite on the post-crossing window, there exists a local acceleration bound

$$|\ddot{x}(t)| \le a_{\text{loc}} \qquad \text{for } 0 \le t \le \tau_{\text{loc}}$$

Choose

$$\tau_1 \le \min\left\{\tau_{\text{loc}}, \frac{\sigma}{a_{\text{loc}}}\right\}$$

Then

$$\dot{x}(t) \le \dot{x}(0) + a_{\mathrm{loc}}t = -V_0 + a_{\mathrm{loc}}t \le -V_0 + \sigma = -c_f - \sigma < -c_f$$

for all $t \in [0, \tau_1]$. Consequently

$$\dot{w}(t) = \dot{x}(t) + c_f \le -\sigma$$

and integrating from $w(0) = 0$ gives

$$w(t) \le -\sigma t < 0 \qquad \text{for } 0 < t \le \tau_1$$

Proof.
The forward local tube condition in the admissible crossing subclass gives existence of the dual-mollified continuation on $[0, \tau_{\mathrm{loc}}]$ together with the bound

$$|\ddot{x}(t)| \le a_{\mathrm{loc}}$$

Because

$$\dot{x}(0) = -V_0 < -c_f$$

the fundamental theorem of calculus yields

$$\dot{x}(t) = \dot{x}(0) + \int_0^t \ddot{x}(s)\, ds \le -V_0 + a_{\mathrm{loc}}t$$

Choosing

$$\tau_1 \le \min\left\{\tau_{\mathrm{loc}}, \frac{\sigma}{a_{\mathrm{loc}}}\right\}$$

forces

$$\dot{x}(t) \le -V_0 + \sigma = -c_f - \sigma < -c_f$$

for every $t \in [0, \tau_1]$. Therefore

$$\dot{w}(t) = \dot{x}(t) + c_f \le -\sigma$$

and integration from the crossing value

$$w(0) = x(0) + c_f \cdot 0 = 0$$

gives

$$w(t) \le -\sigma t < 0$$

on $(0, \tau_1]$. This proves the lemma. On a fixed admissible crossing subclass the same argument is uniform after replacing σ by $\sigma_{\min}$ and a_{loc} by a_{tube}.

Lemma 2: Caustic isolation and uniform self-drive bound.
Use the local tube bounds to obtain a crude self-drive estimate on the full post-crossing window, and then use (H2)-(H4) together with Lemma 1 to show that on a delayed subwindow every active self root lies strictly before t_{zero} and hence stays away from the caustic hinge.

Working form:
fix $t \in (0, \tau_1]$ and suppose a self-emission time $t_s < t$ lies in the support of the shell mollifier on the left-moving post-crossing branch. If the shell mollifier has support band η, then

$$|x(t) - x(t_s) + c_f(t - t_s)| \le \eta$$

which is equivalent to

$$|w(t_s) - w(t)| \le \eta$$

On the full initial tube one only assumes the class-wide transversality and branch-count bounds. Therefore

$$A_s^\rho(t) \le N_s^{\max} \kappa \epsilon^2 \frac{c_f}{\nu} \frac{1}{\epsilon_c^2} \equiv \overline{A}_s^\rho \qquad \text{for } 0 \le t \le \tau_1$$

This is the basic bounded self-drive estimate used in the local theorem.

One obtains a stronger separated statement on a slightly delayed subwindow. Define

$$\tau_{\text{sep}} \equiv \frac{2\eta}{\sigma}$$

Then for $t \in [\tau_{\text{sep}}, \tau_1]$,

$$w(t) \le -2\eta$$

so any active self root satisfies

$$w(t_s) \le -\eta$$

Hence every active self root on that delayed subwindow satisfies

$$t_s < t_{\text{zero}}$$

Since on the sub-field-speed source region one has

$$\dot{w}(\theta) = \dot{x}(\theta) + c_f \ge \nu, \qquad \theta \in [-h, t_{\text{zero}}]$$

monotonicity gives

$$t_s \le t_{\text{zero}} - \gamma(\eta), \qquad \gamma(\eta) \equiv \frac{\eta}{\nu}$$

Thus the caustic is uniformly separated from the active self roots on the delayed subwindow.

A sharper geometric version is available only on the delayed window where the roots have already entered the sub-field-speed source region. Since

$$w(t_{\text{zero}}) = 0 \qquad \Longrightarrow \qquad x(t_{\text{zero}}) = -c_f t_{\text{zero}} \equiv \rho_{\text{zero}} > 0$$

and the pre-crossing branch is inbound, one has

$$x(t_s) \ge \rho_{\text{zero}} \qquad \text{for every } t_s \le t_{\text{zero}}$$

Choose a short window $[0, \tau_\rho]$ on which

$$\rho(t) = |x(t)| \le \frac{\rho_{\text{zero}}}{2}$$

Then every active self root on the delayed geometric window

$$t \in [\tau_{\text{sep}}, \min\{\tau_1, \tau_\rho\}]$$

satisfies

$$|x(t) - x(t_s)| = \rho(t) + x(t_s) \ge \frac{\rho_{\text{zero}}}{2}$$

Hence the same branch-count bound yields the sharper estimate

$$A_s^\rho(t) \le N_s^{\max} \kappa \epsilon^2 \frac{c_f}{\nu} \frac{4}{\rho_{\text{zero}}^2} \equiv \overline{A}_{s,\text{geom}}^\rho$$

which is independent of the core mollifier ϵ_c. This is the delayed-window version that can sharpen the Goldilocks condition once the short-time window extends beyond τ_{sep}.

Proof.
On the full initial tube, each active self branch contributes a radial acceleration of the form

$$\kappa \epsilon^2 \frac{1}{|J_s|} \frac{1}{r_s^2 + \epsilon_c^2}$$

with

$$|J_s| \geq \frac{\nu}{c_f}$$

by the class definition and

$$r_s^2 + \epsilon_c^2 \geq \epsilon_c^2$$

by core mollification. Summing over at most

$$N_s^{\max}$$

active self roots gives the crude bound

$$A_s^\rho(t) \leq \overline{A}_s^\rho$$

for

$$0 \leq t \leq \tau_1$$

For the delayed separation, Lemma 1 gives

$$w(t) \leq -\sigma t$$

Hence for

$$t \in [\tau_{\text{sep}}, \tau_1], \qquad \tau_{\text{sep}} = \frac{2\eta}{\sigma}$$

one has

$$w(t) \leq -2\eta$$

If a self root $t_s < t$ lies in the shell support, then

$$|w(t_s) - w(t)| \leq \eta$$

so

$$w(t_s) \leq -\eta < 0$$

But hypothesis (H2) states that

$$w(\theta) > 0 \qquad \text{for } \theta \in (t_{\text{zero}}, 0)$$

therefore no such t_s can lie in $(t_{\text{zero}}, 0)$ and hence

$$t_s < t_{\text{zero}}$$

On the source region

$$[-h, t_{\text{zero}}]$$

hypothesis (H3) gives

$$\dot{w}(\theta) = \dot{x}(\theta) + c_f \geq \nu$$

Applying the mean-value theorem between t_s and t_{zero} yields

$$w(t_{\text{zero}}) - w(t_s) \geq \nu\,(t_{\text{zero}} - t_s)$$

Since

$$w(t_{\text{zero}}) = 0 \qquad \text{and} \qquad w(t_s) \leq -\eta$$

it follows that

$$t_s \leq t_{\text{zero}} - \frac{\eta}{\nu} = t_{\text{zero}} - \gamma(\eta)$$

This proves the delayed caustic-separation claim.

For the geometric refinement, use that the selected source branch is inbound before the crossing, so $x(\theta)$ decreases toward the origin on the stored pre-crossing leg. Thus

$$t_s \le t_{\text{zero}} \qquad \Longrightarrow \qquad x(t_s) \ge x(t_{\text{zero}}) = \rho_{\text{zero}}$$

If in addition

$$0 \le t \le \tau_\rho \qquad \text{and} \qquad \rho(t) \le \frac{\rho_{\text{zero}}}{2}$$

then on the delayed geometric window

$$t \in [\tau_{\text{sep}}, \min\{\tau_1, \tau_\rho\}]$$

one has

$$|x(t) - x(t_s)| = \rho(t) + x(t_s) \ge \frac{\rho_{\text{zero}}}{2}$$

Replacing the crude denominator bound

$$r_s^2 + \epsilon_c^2 \ge \epsilon_c^2$$

by

$$r_s^2 + \epsilon_c^2 \ge \frac{\rho_{\text{zero}}^2}{4}$$

and summing again over at most

$$N_s^{\max}$$

branches gives

$$A_s^\rho(t) \le \overline{A}_{s,\text{geom}}^\rho$$

This proves Lemma 2.

Lemma 3: Partner-root linearization and lower bound.
Use the linearized partner root

$$t_p = -\left(\frac{V_0 - c_f}{V_0 + c_f}\right) t$$

to derive the partner-distance bound

$$r_p(t) \le \left(\frac{2c_f V_0}{V_0 + c_f}\right) t + \mathcal{O}(t^2)$$

and hence a lower bound on the core-mollified partner attraction

$$A_p^\rho(t) \ge \underline{A}_p^\rho(t)$$

Working form:
write

$$s \equiv -t_p > 0$$

for the past partner-emission time measured backward from the crossing. On the active partner branch, shell support gives

$$||x(t) + x(t_p)| - c_f(t + s)| \le \eta$$

Assume the signed trajectory is C^1 through the crossing and obeys uniform acceleration bounds on both sides of $t = 0$. Then there exists

$$a_* \equiv \max\{a_{\text{loc}}, a_{\max}\}$$

such that the Taylor remainders satisfy

$$x(t) = -V_0 t + R_+(t), \qquad |R_+(t)| \le \frac{a_*}{2} t^2$$

for $t \in [0, \tau_1]$, and

$$x(t_p) = V_0 s + R_-(s), \qquad |R_-(s)| \le \frac{a_*}{2} s^2$$

for $s \in [0, \tau_1]$.

Substituting these expansions into the partner-shell condition yields

$$|V_0(t-s) - c_f(t+s) + E_p(t,s)| \le \eta$$

where

$$|E_p(t,s)| \le \frac{a_*}{2}(t^2 + s^2)$$

Let

$$\alpha \equiv \frac{V_0 - c_f}{V_0 + c_f} \in (0,1), \qquad \beta_p \equiv \frac{2c_f V_0}{V_0 + c_f}$$

Then the linearized root is $s = \alpha t$. For sufficiently small t and η, the exact active partner root obeys the one-sided estimate

$$s \le \alpha t + C_p(t^2 + \eta)$$

for some constant C_p depending only on (V_0, c_f, a_*).

Consequently the delayed partner distance satisfies the upper bound

$$r_p(t) = c_f(t+s) \le \beta_p t + c_f C_p(t^2 + \eta)$$

This is the form needed for the theorem program: as the trajectory brakes after the crossing, the true partner distance can only become smaller than this leading linear estimate, which strengthens the partner attraction. Therefore the core-mollified partner term admits the explicit lower bound

$$A_p^\rho(t) \ge \frac{\kappa \epsilon^2}{\left(\beta_p t + c_f C_p(t^2 + \eta)\right)^2 + \epsilon_c^2} \equiv \underline{A}_p^\rho(t)$$

Proof.
Let

$$F(t,s) \equiv V_0(t-s) - c_f(t+s) + E_p(t,s)$$

The shell condition on the active partner branch is precisely

$$|F(t,s)| \le \eta$$

At the linear level,

$$F_0(t,s) = V_0(t-s) - c_f(t+s)$$

has root

$$s = \alpha t, \qquad \alpha = \frac{V_0 - c_f}{V_0 + c_f}$$

and

$$\partial_s F_0(0,0) = -(V_0 + c_f) \ne 0$$

Write

$$E_p(t,s) = -(R_+(t) + R_-(s))$$

so the absolute remainder bounds imply

$$|E_p(t, \alpha t)| \le \frac{a_*}{2}(1 + \alpha^2) t^2 \equiv C_0 t^2$$

Moreover the integral remainder formula gives

$$|\partial_s E_p(t,s)| \le a_* s$$

After shrinking the local window if necessary, assume

$$0 \le s \le \tau_1 \qquad \text{and} \qquad a_*\tau_1 \le \frac{V_0 + c_f}{2}$$

Then on that window

$$\partial_s F(t,s) = -(V_0 + c_f) + \partial_s E_p(t,s) \le -\frac{V_0 + c_f}{2} < 0$$

Hence the active partner branch is quantitatively nondegenerate. Applying the mean-value theorem in the s variable between s and αt yields a point ξ between them such that

$$F(t,s) - F(t,\alpha t) = \partial_s F(t,\xi)\,(s - \alpha t)$$

Therefore

$$\frac{V_0 + c_f}{2}\,|s - \alpha t| \le |F(t,s)| + |F(t,\alpha t)| \le \eta + C_0 t^2$$

so

$$|s - \alpha t| \le \frac{2}{V_0 + c_f}\left(\eta + C_0 t^2\right)$$

Thus there is an explicit constant

$$C_p = C_p(V_0, c_f, a_*)$$

such that

$$s \le \alpha t + C_p(t^2 + \eta)$$

Substituting into

$$r_p(t) = c_f(t + s)$$

gives

$$r_p(t) \le c_f(1+\alpha)t + c_f C_p(t^2 + \eta) = \beta_p t + c_f C_p(t^2 + \eta)$$

Because the core-mollified partner contribution is monotone decreasing in the delayed distance,

$$r_p(t) \le r_{\mathrm{ub}}(t) \qquad \Longrightarrow \qquad \frac{1}{r_p(t)^2 + \epsilon_c^2} \ge \frac{1}{r_{\mathrm{ub}}(t)^2 + \epsilon_c^2}$$

where

$$r_{\mathrm{ub}}(t) \equiv \beta_p t + c_f C_p(t^2 + \eta)$$

Multiplying by the positive prefactor

$$\kappa\epsilon^2$$

gives

$$A_p^\rho(t) \ge \underline{A}_p^\rho(t)$$

which proves the lemma. On the admissible crossing subclass the same argument is uniform after replacing

$$V_0 \mapsto V_{\max} \qquad \text{in } C_p$$

and, when desired for a conservative bound, replacing

$$\beta_p \mapsto \beta_{p,\max}$$

Lemma 4: Recapture integration.
Show that the function

$$f(t) \equiv V_0 - \int_0^t \left(\underline{A}_p^\rho(s) - \overline{A}_s^\rho \right) ds$$

has a zero on the initial window under (H5), and conclude that the true radial speed must vanish there.

Working form:
fix a window $[0, \tau]$ on which Lemma 2 and Lemma 3 both hold, and define

$$B_\tau \equiv c_f C_p(\tau^2 + \eta)$$

Then for $0 \le t \le \tau$,

$$\underline{A}_p^\rho(t) \ge \frac{\kappa \epsilon^2}{(\beta_p t + B_\tau)^2 + \epsilon_c^2}$$

Integrating this explicit lower bound gives the partner impulse estimate

$$\Delta V_p(\tau) \equiv \int_0^\tau \underline{A}_p^\rho(s)\, ds \ge \frac{\kappa \epsilon^2}{\beta_p \epsilon_c} \left[\arctan\left(\frac{\beta_p \tau + B_\tau}{\epsilon_c} \right) - \arctan\left(\frac{B_\tau}{\epsilon_c} \right) \right]$$

If the self-drive is bounded above by a constant $\overline{A}_s^\rho$ on the same window, then the total outward impulse from self-hit is at most

$$\Delta V_s(\tau) \le \overline{A}_s^\rho \tau$$

Therefore a sufficient recapture condition is

$$V_0 < \Delta V_p(\tau) - \overline{A}_s^\rho \tau$$

If, in addition, the chosen window reaches the delayed geometric regime,

$$\tau \ge \tau_{\text{sep}} \qquad \text{and} \qquad \tau \le \tau_\rho$$

then one may split the self-drive loss as

$$\Delta V_s(\tau) \le \overline{A}_s^\rho \tau_{\text{sep}} + \overline{A}_{s,\text{geom}}^\rho (\tau - \tau_{\text{sep}})$$

and therefore the sharper sufficient recapture condition becomes

$$V_0 < \Delta V_p(\tau) - \overline{A}_s^\rho \tau_{\text{sep}} - \overline{A}_{s,\text{geom}}^\rho (\tau - \tau_{\text{sep}})$$

This is the working form of the Goldilocks condition. It makes the bottleneck explicit: one must show that there exist parameters

$$(\eta, \epsilon_c, V_0, \tau)$$

for which the dual-mollified partner impulse beats the bounded self-drive loss on a nonempty initial window.

On a fixed admissible crossing subclass, the corresponding class-uniform version replaces

$$V_0 \mapsto V_{\max}, \qquad \beta_p \mapsto \beta_{p,\max}, \qquad \sigma \mapsto \sigma_{\min}$$

and uses the common remainder constant

$$C_p = C_p(V_{\max}, c_f, a_*)$$

That conservative substitution is the bridge from the single-history Lemma 4 estimate to the class-uniform proposition below.

Proof.
Let

$$V(t) \equiv \dot{\rho}(t)$$

denote the outward radial speed on the post-crossing branch. Then

$$V(0) = V_0 > 0$$

By Lemma 2 and Lemma 3, on every window $[0, \tau]$ where both lemmas hold one has

$$\ddot{\rho}(t) \le -\underline{A}_p^\rho(t) + \overline{A}_s^\rho$$

Integrating from 0 to $t \le \tau$ yields

$$V(t) = V_0 + \int_0^t \ddot{\rho}(s)\, ds \le V_0 - \int_0^t \left(\underline{A}_p^\rho(s) - \overline{A}_s^\rho\right) ds = f(t)$$

If the delayed geometric regime is available, the same integration gives the sharper estimate

$$V(t) \le V_0 - \Delta V_p(t) + \overline{A}_s^\rho\, \tau_{\text{sep}} + \overline{A}_{s,\text{geom}}^\rho\, (t - \tau_{\text{sep}})$$

for

$$t \in [\tau_{\text{sep}}, \tau]$$

Now assume the Goldilocks condition holds on $[0, \tau]$, so that

$$f(\tau) < 0$$

If V remained strictly positive on the whole interval $[0, \tau]$, then evaluating the previous bound at $t = \tau$ would give

$$0 < V(\tau) \le f(\tau) < 0$$

which is impossible. Therefore the set

$$\{t \in [0, \tau] : V(t) = 0\}$$

is nonempty. Define

$$\tau_{\text{turn}} \equiv \inf\{t \in [0, \tau] : V(t) = 0\}$$

Continuity of V implies

$$V(\tau_{\text{turn}}) = 0$$

so the outward radial speed vanishes by time τ. This is the desired recapture statement.

The class-uniform version is the same argument with the conservative substitutions

$$V_0 \mapsto V_{\max}, \qquad \beta_p \mapsto \beta_{p,\max}, \qquad \sigma \mapsto \sigma_{\min}$$

and the common remainder constant

$$C_p = C_p(V_{\max}, c_f, a_*)$$

which is precisely the form used in the explicit short-window proposition below.

Using monotonicity of the arctangent integrand gives a simpler algebraic lower bound:

$$\Delta V_p(\tau) \ge \frac{\kappa \epsilon^2\, \tau}{\epsilon_c^2 + (\beta_p \tau + B_\tau)^2}$$

Hence a cleaner sufficient recapture condition is

$$V_0 < \tau \left[\frac{\kappa \epsilon^2}{\epsilon_c^2 + (\beta_p \tau + B_\tau)^2} - \overline{A}_s^\rho\right]$$

Equivalently,

$$\kappa \epsilon^2 > \left(\frac{V_0}{\tau} + \overline{A}_s^\rho\right) \left[\epsilon_c^2 + (\beta_p \tau + B_\tau)^2\right]$$

This is the most useful practical form of (H5) in the present manuscript: once the constants in Lemma 2 and Lemma 3 are fixed, recapture reduces to a checkable algebraic inequality.

For a fixed admissible crossing subclass, the same inequality is made class-uniform by replacing

$$V_0 \mapsto V_{\max}, \qquad \beta_p \mapsto \beta_{p,\max}$$

and taking the common remainder constant

$$C_p = C_p(V_{\max}, c_f, a_*)$$

That replacement is exactly what the proposition below implements.

One can simplify further on a short window where the shell-error term is dominated by the linear partner term. If

$$B_\tau \le \beta_p \tau$$

then

$$\Delta V_p(\tau) \ge \frac{\kappa \epsilon^2 \, \tau}{\epsilon_c^2 + 4\beta_p^2 \tau^2}$$

and therefore a sufficient short-window recapture condition is

$$V_0 < \tau \left[\frac{\kappa \epsilon^2}{\epsilon_c^2 + 4\beta_p^2 \tau^2} - \overline{A}_s^\rho \right]$$

Since

$$B_\tau = c_f C_p (\tau^2 + \eta)$$

the dominance condition $B_\tau \le \beta_p \tau$ is itself a quadratic inequality in τ. A nonempty admissible interval exists whenever

$$\eta \le \frac{\beta_p^2}{4 c_f^2 C_p^2}$$

provided the corresponding roots lie inside the local validity window of Lemmas 1-3.

For class-uniform use on $\mathcal{K}^{\mathrm{cross}}_{\eta,\epsilon_c}$, the corresponding sufficient condition is obtained conservatively by replacing

$$\beta_p \mapsto \beta_{p,\min}$$

since the linear partner term must dominate uniformly for every admissible history. The explicit proposition below avoids mixing $\beta_{p,\min}$ and $\beta_{p,\max}$ in a single window estimate by choosing τ_ϵ directly from $\beta_{p,\max}$ and then bounding

$$\beta_p \tau_\epsilon + B_{\tau_\epsilon}$$

in one step.

> **Proposition (Explicit short-window recapture regime).**
> On a fixed admissible crossing subclass $\mathcal{K}^{\mathrm{cross}}_{\eta,\epsilon_c}$, choose the class-uniform window

$$> \tau_\epsilon \equiv \frac{\epsilon_c}{2\beta_{p,\max}}. >$$

> Assume

$$> \tau_\epsilon \le \tau_1, > \qquad > \eta \le \frac{\epsilon_c}{4 c_f C_p}, > \qquad > \epsilon_c \le \frac{\beta_{p,\max}^2}{c_f C_p}. >$$

> Then

$$> B_{\tau_\epsilon} > = > c_f C_p \left(\tau_\epsilon^2 + \eta \right) > \le > \frac{\epsilon_c}{2} >$$

> and since

$$> \beta_p \tau_\epsilon + B_{\tau_\epsilon} > \le > \beta_{p,\max} \tau_\epsilon + \frac{\epsilon_c}{2} > = > \epsilon_c, >$$

> one obtains

$$> \Delta V_p(\tau_\epsilon) > \ge > \frac{\kappa \epsilon^2}{4\beta_{p,\max} \epsilon_c}. >$$

> Therefore a class-uniform sufficient recapture condition is

$$> V_{\max} < > \frac{\kappa \epsilon^2}{4\beta_{p,\max} \epsilon_c} > - > \frac{\overline{A}_s^\rho \, \epsilon_c}{2\beta_{p,\max}}, >$$

> or equivalently

$$> \kappa\epsilon^2 >>> 4\beta_{p,\max} V_{\max}\,\epsilon_c > + > 2\overline{A}_s^{\rho}\,\epsilon_c^2. >$$

> If, in addition,

$$> \tau_{\text{sep,max}} \le \tau_\epsilon \le \tau_\rho, > \qquad > \tau_{\text{sep,max}} \equiv \frac{2\eta}{\sigma_{\min}}, >$$

> then Lemma 2 yields the delayed-window refinement

$$> V_{\max} <> \frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c} > - > \overline{A}_s^{\rho}\,\tau_{\text{sep,max}} > - > \overline{A}_{s,\text{geom}}^{\rho}\,(\tau_\epsilon - \tau_{\text{sep,max}}). >$$

This proposition is the first genuinely explicit realization of (H5) in the note. It converts the abstract impulse inequality into a concrete dual-mollified parameter regime.

Proof sketch:

1. Lemma 1 gives the class-uniform post-crossing monotonicity

$$w(t) < 0 \qquad \text{for } 0 < t \le \tau_1$$

2. Lemma 2 supplies the full-window self-drive bound

$$A_s^{\rho}(t) \le \overline{A}_s^{\rho} \qquad \text{for } 0 \le t \le \tau_1$$

with the delayed-window refinement available once

$$\tau_{\text{sep,max}} \le t \le \tau_\rho$$

3. Lemma 3 gives the class-uniform partner lower bound with

$$\beta_p \le \beta_{p,\max}, \qquad B_\tau \le c_f C_p(\tau^2 + \eta)$$

At

$$\tau_\epsilon = \frac{\epsilon_c}{2\beta_{p,\max}}$$

the stated assumptions force

$$\beta_p \tau_\epsilon + B_{\tau_\epsilon} \le \epsilon_c$$

so the denominator of the partner integrand is bounded above by

$$\epsilon_c^2 + \epsilon_c^2 = 2\epsilon_c^2$$

The rectangle-area lower bound therefore gives

$$\Delta V_p(\tau_\epsilon) \ge \frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c}$$

4. The stated algebraic inequality is exactly the condition that this class-uniform inward partner impulse beats the class-uniform outward self-drive loss by time τ_ϵ.
5. Lemma 4 then gives a zero of the radial speed on $[0, \tau_\epsilon]$, proving local post-crossing recapture for every history in the subclass.

The rectangle-area estimate in Step 3 is deliberately conservative. A sharper certificate should keep the exact arctangent impulse from Lemma 4, or certify a Cauchy-Schwarz lower bound on the partner integrand over the same short window. Record this improvement by a factor

$$Q_{\text{CS}} \ge 1$$

in

$$\Delta V_p(\tau_\epsilon) \ge Q_{\text{CS}}\,\frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c}$$

In the standard half-core window the target refinement is

$$Q_{\mathrm{CS}} = \sqrt{2}$$

provided the interval certificate proves the required monotone coverage of the partner-distance strip. The later corridor arithmetic should use the certified value of

$$Q_{\mathrm{CS}}$$

not assume the improvement without an interval report.

In the joint short-window regime

$$\eta = \mathcal{O}(\epsilon_c), \qquad \epsilon_c \downarrow 0$$

the right-hand side is

$$\mathcal{O}(\epsilon_c)$$

so any fixed positive coupling scale $\kappa\epsilon^2$ eventually dominates it. Subject to the local validity constraints from Lemmas 1-3, this exhibits a nonempty dual-mollified parameter regime in which local post-crossing recapture follows directly from the explicit inequality.

Local takeaway

The local bottleneck is exactly Lemma 4 together with (H5). Lemmas 1 and 2 lock down the sorting-map geometry and the bounded self drive; Lemma 3 extracts the partner lower bound; Lemma 4 converts those two ingredients into a recapture condition by integrating the net radial impulse.

Operationally, the local theorem reduces the first post-crossing turn to one explicit race: the regularized partner impulse must beat the bounded self-drive loss on a nonempty initial window. Proposition `Explicit short-window recapture regime` is the concrete form used later in the manuscript, and its strict margin is precisely the inner-cycle quantity

$$\mathfrak{M}_{\mathrm{in}} > 0$$

that enters the global invariant-envelope theorem.

Global Existence via Arzela-Ascoli

The local origin-crossing theorem supplies only the inner turnaround. The global capstone is an isolated fixed point of the full return map

$$P_\eta : \Sigma^-_{x_*,\eta} \to \Sigma^-_{x_*,\eta}$$

In the dual-mollified setting the final topological target is therefore to construct a nonempty closed convex tame envelope

$$\mathcal{K}_{x_*,\eta} \subset C^1([-h,0])$$

not a continuous family of equal-amplitude cycles. Exact energy for the dual-mollified problem remains conditional on action-level regularization, so the fixed-point route should be built from uniform bounds, continuity, and compactness rather than from a presumed conserved history functional.

The global input list is now fixed:

1. a nonempty tame inbound class propagated from the affine seed history;
2. collapse-to-crossing control and the local origin-crossing recapture theorem;
3. outer-turn and return-to-section control;
4. a convex section envelope

$$\mathcal{C}_{x_*,\eta}$$

and a closed convex tame sub-envelope

$$\mathcal{K}_{x_*,\eta} \subseteq \mathcal{C}_{x_*,\eta}$$

5. continuity and precompactness of

$$P_\eta$$

on

$$\mathcal{K}_{x_*,\eta}$$

6. and the self-map property

$$P_\eta(\mathcal{K}_{x_*,\eta}) \subseteq \mathcal{K}_{x_*,\eta}$$

Only after those inputs live on the same domain does Schauder apply.

Status of the global capstone ingredients

The theorem status of the global program should be read in three layers.

- The local and regional geometry is already organized into serious theorem packages: branch control, caustic transit, inner recapture, outer-turn recapture, and return-to-section.
- The compactness mechanism is conceptually standard once one has class-uniform bounds on one closed domain: this is the Arzela-Ascoli side of the argument.
- The active unresolved burden is domain production: the manuscript still has to place nonempty tame propagation, closed convexity, continuity, precompactness, and the self-map property on one and the same set

$$\mathcal{K}_{x_*,\eta}$$

So the true blocker is not the abstract fixed-point theorem. It is the production of one legitimate tame self-map domain carrying all of the hypotheses at once.

Convex section envelope

The visible Banach-space constraints should be separated from the delayed-root constraints. Fix constants

$$x_* \in (0, X_{\max}), \qquad U_{\max} > 0, \qquad A_{\max} > 0, \qquad h \geq \frac{2X_{\max}}{c_f}$$

and define

$$\mathcal{C}_{x_*,\eta} \subset \Sigma^-_{x_*,\eta}$$

to be the set of histories $\phi \in C^1([-h, 0])$ such that:

- section anchoring:

$$\phi(0) = x_*$$

- inbound sign at the section:

$$\dot{\phi}(0) \leq 0$$

- position envelope:

$$-X_{\max} \leq \phi(\theta) \leq X_{\max} \qquad \text{for } \theta \in [-h, 0]$$

- speed envelope:

$$|\dot{\phi}(\theta)| \leq U_{\max} \qquad \text{for } \theta \in [-h, 0]$$

- Lipschitz-velocity envelope:

$$|\dot{\phi}(\theta_1) - \dot{\phi}(\theta_2)| \leq A_{\max}|\theta_1 - \theta_2| \qquad \text{for } \theta_1, \theta_2 \in [-h, 0]$$

This set is closed and convex in the C^1 topology. The horizon condition is handled externally: if

$$|\phi(\theta)| \leq X_{\max}$$

on the stored interval, then every partner or self chord is at most

$$2X_{\max}$$

so

$$\tau_{\max}(\phi) \leq \frac{2X_{\max}}{c_f} \leq h$$

The point is to keep only affine and supremum-type constraints inside

$$\mathcal{C}_{x_*,\eta}$$

while postponing nonlocal tame delayed-root conditions to a sub-envelope.
The envelope is intentionally written in the signed coordinate

$$x \in [-X_{\max}, X_{\max}]$$

rather than in a one-sided radial coordinate. A genuine origin-crossing cycle may store data from both sign sheets inside

$$[-h, 0]$$

so a one-sided condition

$$0 \le \phi \le X_{\max}$$

would exclude valid histories whenever the memory window crosses the origin. Branch labels, exterior sheets, and origin-crossing status belong to the finite tame certificate, not to the convex Banach envelope.
Those delayed-root conditions are not visibly convex inside

$$\mathcal{C}_{x_*,\eta}$$

so the next proposition is a genuine packaging target rather than an automatic consequence of intersecting

$$\mathcal{C}_{x_*,\eta}$$

with the naive tame subclass.

> **Target Proposition (Closed Convex Tame Envelope).**
> The remaining topological task is to exhibit a nonempty closed convex set

$$> \mathcal{K}_{x_*,\eta} > \subseteq > \mathcal{C}_{x_*,\eta} >$$

> such that:
>
> 1. the propagated nonempty tame class lies inside

$$> \mathcal{K}_{x_*,\eta}; >$$

> 2. the return map

$$> P_\eta >$$

> is well defined on

$$> \mathcal{K}_{x_*,\eta}; >$$

> 3. the delayed-root persistence and Jacobian bounds defining tameness remain valid on that domain;
> 4. and the tame constraints are closed under limits in the

$$> C^1 >$$

> topology on that domain.

This is the exact topological object needed by the final fixed-point theorem. The role of

$$\mathcal{C}_{x_*,\eta}$$

is to carry the convex bounds; the role of

$$\mathcal{K}_{x_*,\eta}$$

is to put the same convex bounds and the tame delayed geometry on one matching domain. In particular, this target does not assert that Jacobian lower bounds or branch-count restrictions are convex by inspection. It isolates the additional burden of producing a closed convex subset on which those tame conditions persist. The self-map property is a separate dynamical burden supplied later by invariant-envelope closure.

The clean way to discharge that burden is not to put the nonconvex delayed-root labels directly into the definition of

$$\mathcal{K}_{x_*,\eta}$$

Instead, one should produce a finite tame certificate: a finite family of continuous affine functionals

$$\ell_\alpha : C^1([-h,0]) \to \mathbb{R}, \qquad \alpha \in \mathcal{I}_{\mathrm{cert}}$$

and constants

$$b_\alpha$$

such that the closed affine tube

$$\mathcal{K}^{\mathrm{cert}}_{x_*,\eta} \equiv \{\phi \in \mathcal{C}_{x_*,\eta} \mid \ell_\alpha(\phi) \le b_\alpha \text{ for every } \alpha \in \mathcal{I}_{\mathrm{cert}}\}$$

implies the desired finite branch chart, Jacobian floors, root-count ceilings, and memory-depth bounds.

> **Proposition (Finite certificate construction of a closed convex tame envelope).**
> Suppose there exists a finite tame certificate

$$> \{\ell_\alpha \le b_\alpha\}_{\alpha \in \mathcal{I}_{\mathrm{cert}}} >$$

> with the following properties:
>
> 1. one seed-propagated history

$$> \phi_{\mathrm{seed,cyc}} \in \mathcal{C}_{x_*,\eta} >$$

> satisfies all certificate inequalities with strict slack;
> 2. every

$$> \phi \in \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} >$$

> has the same finite active branch chart on the stored interval and on the controlled one-cycle continuation;
> this chart includes the signed exterior sheet labels and origin-crossing layer labels needed to interpret the signed

$$> x >$$

> history;
> 3. on that chart the delayed roots remain simple with uniform Jacobian floor

$$> |J| \ge \nu_{\mathrm{cert}} > 0; >$$

> 4. the active branch count, memory depth, position, speed, and Lipschitz-velocity bounds are bounded by the constants used in

$$> \mathcal{C}_{x_*,\eta}; >$$

> 5. and these certificate implications are closed under

$$> C^1 >$$

> limits inside

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}. >$$

> Then

$$> \mathcal{K}_{x_*,\eta} > \equiv > \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} >$$

> is a nonempty closed convex tame envelope.

Proof.
The set

$$\mathcal{C}_{x_*,\eta}$$

is closed and convex by its affine section condition, interval bounds, speed bounds, and Lipschitz-velocity bound. Each certificate condition

$$\ell_\alpha(\phi) \le b_\alpha$$

is a closed half-space in

$$C^1([-h,0])$$

so the finite intersection

$$\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

is closed and convex. It is nonempty because

$$\phi_{\mathrm{seed,cyc}}$$

lies in it with strict slack. Items 2-4 give the finite branch chart, Jacobian floors, branch-count ceilings, memory-depth bounds, and Banach-envelope bounds required for tameness. Item 5 says exactly that these tame properties persist under

$$C^1$$

limits inside the certified set. Hence

$$\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

is the required nonempty closed convex tame envelope.

The finite-certificate language can be made concrete by using a sampled

$$C^1$$

tube around one strictly controlled seed-cycle history. This avoids treating the nonlinear branch-chart conditions themselves as convex constraints.

For completion, the center history cannot remain schematic. The proof needs an instantiated

$$\phi_{\mathrm{cyc}}$$

with a period

$$T_{\mathrm{cyc}}$$

a finite active branch list

$$\mathcal{B}_{\mathrm{act}}$$

inactive branch complements, a mesh

$$\{\theta_j\}_{j=0}^N$$

and returned-sample residuals. Until that data packet exists, every finite-certificate statement below is conditional.

> **Proposition (Sampled seed-cycle tube gives a finite tame certificate).**
> Let

$$> \phi_{\mathrm{cyc}} \in \mathcal{C}_{x_*,\eta} >$$

> be a seed-propagated history whose one-cycle continuation is defined on

$$> [0, T_{\max}] >$$

> and has strict margins:
>
> 1. every active delayed root on the stored interval and on the controlled continuation is simple with

$$> |J| \geq 2\nu_{\mathrm{chart}} > 0; >$$

> 2. every inactive candidate root equation has gap at least

$$> 2\gamma_{\mathrm{gap}} > 0 >$$

> on the compact chart complement;
> 3. all memory depths stay at distance at least

$$> 2\gamma_h > 0 >$$

> from the boundary of the stored horizon;
> 4. the position, speed, and Lipschitz-velocity envelope bounds hold with strict slack.
>
> Assume also that the dual-mollified solution map is continuous from initial

$$> C^1([-h,0]) >$$

> data into

$$> C^1([-h,T_{\max}]) >$$

> on the corresponding branch chart. Then there are a radius

$$> r_{\text{cert}} > 0, >$$

> a finite mesh

$$> -h = \theta_0 < \theta_1 < \cdots < \theta_N = 0, >$$

> and finitely many affine sample inequalities

$$> |\phi(\theta_j) - \phi_{\text{cyc}}(\theta_j)| \leq \frac{r_{\text{cert}}}{4}, > \qquad > |\dot{\phi}(\theta_j) - \dot{\phi}_{\text{cyc}}(\theta_j)| > \leq \frac{r_{\text{cert}}}{4}, > \qquad > 0 \leq j \leq N, >$$

> such that every

$$> \phi \in \mathcal{C}_{x_*,\eta} >$$

> satisfying those finite inequalities lies in a

$$> C^1 >$$

> neighborhood on which the same active branch chart, Jacobian floor, root-count ceiling, and memory-depth bound persist through the controlled continuation. Consequently these sample inequalities form a finite tame certificate of the kind used in the previous proposition.

Proof.
By the strict branch-chart margins and compactness of the stored and controlled continuation intervals, there is a radius

$$r_{\text{chart}} > 0$$

such that any history whose controlled continuation stays within

$$r_{\text{chart}}$$

of the seed-cycle continuation in

$$C^1$$

has the same active roots, no inactive root births, Jacobian floor at least

$$\nu_{\text{chart}}$$

and the same memory-depth bound. The strict envelope slack gives a second radius

$$r_{\text{env}} > 0$$

for the position, speed, and Lipschitz-velocity constraints. By continuous dependence of the branch-chart solution map, shrink to

$$r_{\text{cert}} \leq \min\{r_{\text{chart}}, r_{\text{env}}\}$$

so that initial

$$C^1$$

distance at most

$$r_{\text{cert}}$$

from

$$\phi_{\mathrm{cyc}}$$

keeps the full controlled continuation inside the required chart tube.

Choose the mesh with maximum step

$$\Delta$$

small enough that

$$2U_{\max}\Delta \le \frac{r_{\mathrm{cert}}}{2}, \qquad 2A_{\max}\Delta \le \frac{r_{\mathrm{cert}}}{2}$$

If the displayed sample inequalities hold, then for any

$$\theta \in [-h, 0]$$

and a nearest mesh point

$$\theta_j$$

one has

$$|\phi(\theta) - \phi_{\mathrm{cyc}}(\theta)| \le |\phi(\theta) - \phi(\theta_j)| + |\phi(\theta_j) - \phi_{\mathrm{cyc}}(\theta_j)| + |\phi_{\mathrm{cyc}}(\theta_j) - \phi_{\mathrm{cyc}}(\theta)| \le r_{\mathrm{cert}}$$

and the same estimate with the Lipschitz-velocity bound gives

$$|\dot{\phi}(\theta) - \dot{\phi}_{\mathrm{cyc}}(\theta)| \le r_{\mathrm{cert}}$$

Thus the finite sample tube implies the required

$$C^1$$

tube. Each absolute-value sample condition is just two continuous affine inequalities in

$$C^1([-h, 0])$$

The previous proposition then turns their finite intersection with

$$\mathcal{C}_{x_*,\eta}$$

into a closed convex tame envelope.

For the first remaining blocker, the seed-cycle certificate should therefore be audited by the following finite margin ledger:

$$\nu_{\mathrm{seed}} \equiv \min_{\beta \in \mathcal{B}_{\mathrm{act}}} \inf_{t \in I_\beta} |J_\beta(t)|$$

$$\gamma_{\mathrm{gap}} \equiv \min_{\beta \in \mathcal{B}_{\mathrm{inact}}} \inf_{(t,\theta) \in Q_\beta} |F_\beta(t, \theta)|$$

$$\gamma_h \equiv \min_{\beta \in \mathcal{B}_{\mathrm{act}}} \inf_{t \in I_\beta} \operatorname{dist}\left(\theta_\beta(t), \{-h, 0\}\right)$$

together with the envelope slack

$$\gamma_{\mathrm{env}} \equiv \min\left\{X_{\max} - \sup|\phi_{\mathrm{cyc}}|,\ U_{\max} - \sup|\dot{\phi}_{\mathrm{cyc}}|,\ A_{\max} - \mathrm{Lip}(\dot{\phi}_{\mathrm{cyc}})\right\}$$

Here

$$F_\beta(t, \theta) = 0$$

denotes the delayed-root equation for branch candidate

$$\beta$$

the sets

$$I_\beta$$

are the compact active branch intervals, and

$$Q_\beta$$

is the compact inactive chart complement after deleting small neighborhoods of the active roots. Positivity of

$$\nu_{\mathrm{seed}}, \qquad \gamma_{\mathrm{gap}}, \qquad \gamma_h, \qquad \gamma_{\mathrm{env}}$$

is exactly the strict seed-cycle tube condition needed to choose

$$r_{\mathrm{cert}}$$

> **Proposition (Quantitative seed-cycle radius choice).**
> Assume the seed-cycle margin ledger satisfies

$$> \nu_{\mathrm{seed}} > 0, > \qquad > \gamma_{\mathrm{gap}} > 0, > \qquad > \gamma_h > 0, > \qquad > \gamma_{\mathrm{env}} > 0. >$$

> Suppose also that on the certified chart there are finite local sensitivity constants

$$> L_J, > \qquad > L_F, > \qquad > L_h, > \qquad > L_{\mathrm{env}} >$$

> such that a

$$> C^1 >$$

> perturbation of size

$$> r >$$

> changes active Jacobians by at most

$$> L_J r, >$$

> inactive root-equation gaps by at most

$$> L_F r, >$$

> active memory-depth distances by at most

$$> L_h r, >$$

> and the envelope slacks by at most

$$> L_{\mathrm{env}} r. >$$

> Then any radius satisfying

$$> 0 < r_{\mathrm{cert}} ><> \min\left\{> \frac{\nu_{\mathrm{seed}}}{2L_J}, > \frac{\gamma_{\mathrm{gap}}}{2L_F}, > \frac{\gamma_h}{2L_h}, > \frac{\gamma_{\mathrm{env}}}{2L_{\mathrm{env}}} >\right\} >$$

> produces the strict chart margins required by Proposition `Sampled seed-cycle tube gives a finite tame certificate`, after omitting any quotient with zero sensitivity because that margin is then unchanged to first order on the chart.

Proof.
For any history within

$$r_{\mathrm{cert}}$$

of

$$\phi_{\mathrm{cyc}}$$

in

$$C^1$$

the active Jacobian floor is at least

$$\nu_{\mathrm{seed}} - L_J r_{\mathrm{cert}} > \frac{\nu_{\mathrm{seed}}}{2} > 0$$

The inactive root-equation gap remains at least

$$\gamma_{\text{gap}} - L_F r_{\text{cert}} > \frac{\gamma_{\text{gap}}}{2} > 0$$

so no inactive branch is born. The active memory-depth distance remains at least

$$\gamma_h - L_h r_{\text{cert}} > \frac{\gamma_h}{2} > 0$$

so no active root reaches the stored-horizon boundary. Finally, the envelope slack remains at least

$$\gamma_{\text{env}} - L_{\text{env}} r_{\text{cert}} > \frac{\gamma_{\text{env}}}{2} > 0$$

These four strict inequalities are precisely the branch-chart, gap, memory-depth, and envelope margins required for the sampled finite certificate.

Precompactness of returned histories

> **Proposition (Precompactness of the Return Image).**
> Fix a dual-mollified inbound class

$$> \mathcal{A}_{x_*,\eta} >\subset> \Sigma^-_{x_*,\eta} >$$

> such that:
>
> 1. for every

$$> \psi \in \mathcal{A}_{x_*,\eta}, >$$

> the one-cycle return time

$$> T(\psi) >$$

> is well defined and lies in

$$> [T_{\min}, T_{\max}]; >$$

> 2. every returned history

$$> \phi = P_\eta(\psi) >$$

> satisfies the bounds

$$> -X_{\max} \le \phi(\theta) \le X_{\max}, > \qquad > |\dot\phi(\theta)| \le U_{\max}, > \qquad > |\dot\phi(\theta_1) - \dot\phi(\theta_2)| >\le> A_{\max}|\theta_1 - \theta_2|, > \qquad > \theta, \theta_1, \theta_2 \in [-h, 0], >$$

> together with

$$> \tau_{\max}(\phi) \le h. >$$

> Then

$$> P_\eta\big(\mathcal{A}_{x_*,\eta}\big) >$$

> is precompact in

$$> C^1([-h, 0]). >$$

Proof.
Take any sequence

$$\phi_n = P_\eta(\psi_n), \qquad \psi_n \in \mathcal{A}_{x_*,\eta}$$

The returned-history bounds give uniform boundedness in

$$C^0([-h, 0])$$

and the speed bound gives

$$|\phi_n(\theta_1) - \phi_n(\theta_2)| \le U_{\max}|\theta_1 - \theta_2|$$

so

$$\{\phi_n\}$$

is equicontinuous. The Lipschitz-velocity bound gives

$$|\dot{\phi}_n(\theta_1) - \dot{\phi}_n(\theta_2)| \le A_{\max}|\theta_1 - \theta_2|$$

so

$$\{\dot{\phi}_n\}$$

is uniformly bounded and equicontinuous.

Arzela-Ascoli therefore yields a subsequence, still denoted

$$\phi_n$$

such that

$$\phi_n \to \phi_* \qquad \text{and} \qquad \dot{\phi}_n \to v_*$$

uniformly on

$$[-h, 0]$$

Since

$$\phi_n(\theta) - \phi_n(0) = \int_0^\theta \dot{\phi}_n(s)\, ds$$

passing to the limit gives

$$\phi_*(\theta) - \phi_*(0) = \int_0^\theta v_*(s)\, ds$$

Hence

$$\phi_* \in C^1([-h, 0]) \qquad \text{and} \qquad \dot{\phi}_* = v_*$$

so the subsequence converges in the

$$C^1$$

norm. Therefore

$$P_\eta(\mathcal{A}_{x_*,\eta})$$

is precompact in

$$C^1([-h, 0])$$

This proposition deliberately stops short of invariance. Its role is only to show that once the return map is defined on a uniformly controlled class, its image cannot spread out arbitrarily in history space.

Certified branch-chart well-posedness

The continuity row used later should be a theorem on certified branch charts, not an informal regularity assumption. The following proposition is the local analytic input needed by the return-map proof.

> **Proposition (Local well-posedness on a certified branch chart).**
> Fix dual-mollified parameters

$$> \eta > 0, > \qquad > \epsilon_c > 0, >$$

> and a memory horizon

$$> h > 0. >$$

> Let

$$> \mathcal{U} \subset C^1([-h,0]) >$$

> be a certified branch-chart neighborhood with:
>
> 1. a finite active branch list

$$> \mathcal{B}_{\mathrm{act}}; >$$

> 2. signed sheet and crossing-layer labels fixed on the chart;
> 3. if a chart interval meets an origin-crossing layer, the vector field there is evaluated by Lemma `Origin-layer continuity of the dual-mollified 1D field`, not by an exterior branch-sum formula;
> 4. active causal roots satisfying

$$> |J_\beta| \ge \nu > 0 > \qquad > \text{for every } \beta \in \mathcal{B}_{\mathrm{act}}; >$$

> 5. uniform bounds

$$> \|\phi\|_{C^1} \le M, > \qquad > \mathrm{Lip}(\dot{\phi}) \le A_{\max}, > \qquad > \tau_\beta(t) \in [0, h - \gamma_h] >$$

> for some

$$> \gamma_h > 0; >$$

> 6. and inactive branch gaps bounded away from zero on the chart complement.
>
> Then there exists

$$> \tau_{\mathrm{wp}} > 0 >$$

> such that every

$$> \phi \in \mathcal{U} >$$

> has a unique dual-mollified forward continuation on

$$> [0, \tau_{\mathrm{wp}}], >$$

> and the solution map

$$> \phi > \longmapsto > x_\phi|_{[-h, \tau_{\mathrm{wp}}]} >$$

> is locally Lipschitz from

$$> C^1([-h,0]) >$$

> into

$$> C^1([-h, \tau_{\mathrm{wp}}]). >$$

Proof.
On exterior certified charts the active root functions persist with

$$|J| \ge \nu$$

so the implicit-function theorem makes each root time locally Lipschitz in the receiver time and in the stored history. The inactive gap prevents any additional root from entering the finite chart on the controlled interval. The dual-mollified kernel is smooth on the shell scale

$$\eta$$

and is uniformly bounded on the core scale

$$\epsilon_c > 0$$

with denominator at least

$$\epsilon_c^2$$

On an origin-crossing chart, the previous origin-layer lemma supplies the same local

$$C^1$$

radial vector-field control after the sheet projection, so the signed scalar branch-sum discontinuity is not part of the local well-posedness argument.
Together with the finite branch count and the fixed horizon

$$h$$

these bounds make the branch-chart vector field locally Lipschitz as a map from the stored

$$C^1$$

history segment to acceleration. The integral equation

$$x(t) = \phi(0) + t\dot{\phi}(0) + \int_0^t (t-s)\, F_\eta(x_s)\, ds$$

then gives local existence and uniqueness by the standard contraction argument on a short

$$C^1$$

tube. Applying the same Lipschitz estimate to two solutions and using Gronwall on the controlled interval gives local Lipschitz dependence of

$$x$$

and

$$\dot{x}$$

on the initial history.

Certified fold-event atlas

The continuity theorem must distinguish uncontrolled branch changes from certified separator events. A full origin-crossing cycle may pass through field-speed folds, so the certificate should not require a single unchanged branch list on the whole cycle.

> **Definition (Certified fold-event atlas).**
> A certified fold-event atlas for one return consists of finitely many fold layers

$$> \mathfrak{F}_1, \dots, \mathfrak{F}_{N_{\mathrm{fold}}} >$$

> together with:
>
> 1. incoming and outgoing active branch lists

$$> \mathcal{B}_k^-, > \qquad > \mathcal{B}_k^+; >$$

> 2. a local fold normal form

$$> g_k(t,s;\lambda) >=> a_k(s-s_k)^2 + b_k\lambda + \text{higher order}, > \qquad > a_k b_k \neq 0; >$$

> 3. parity data

$$> \Delta N_k \in 2\mathbb{Z}, > \qquad > \Delta D_k = 0; >$$

> 4. a finite fold-impulse ceiling and an outgoing chart on which the post-fold roots again have a positive Jacobian floor.
>
> Outside the union of the fold layers, the active roots must remain simple with the certified Jacobian floors and inactive-root gaps.

This reconciles the continuity row with the causal-fold geometry. A field-speed separator may be a genuine root-pair birth or death, but it is not an uncontrolled discontinuity if the atlas records the parity-preserving transition and hands the trajectory to a certified outgoing chart.

Continuity on the tame envelope

> **Proposition (Continuity of the Return Map on $\mathcal{K}_{x_*,\eta}$).**
> Let

$$> \mathcal{K}_{x_*,\eta} >\subseteq> \mathcal{C}_{x_*,\eta} >$$

> be a closed convex tame envelope such that:

> 1. each
>
> $$> \psi \in \mathcal{K}_{x_*,\eta} >$$
>
> admits a unique forward continuation on
>
> $$> [0, T_{\max}] >$$
>
> with class-uniform position, speed, acceleration, Jacobian, and memory-depth bounds;
> 2. on that forward tube, Proposition `Local well–posedness on a certified branch chart` applies on finitely many exterior and origin-layer chart intervals covering the continuation;
> 3. outside a certified fold-event atlas the active delayed roots persist continuously with the history, with no root birth, root collision, or Jacobian loss of transversality; across each certified fold layer, the active branch list undergoes the parity-preserving transition recorded in the atlas;
> 4. the first return to the inbound section is uniformly transverse:
>
> $$> x(T(\psi);\psi) = x_*, > \qquad > \dot{x}(T(\psi);\psi) \le -u_{\text{sec}} < 0. >$$
>
> Then
>
> $$> P_\eta : \mathcal{K}_{x_*,\eta} \to C^1([-h,0]) >$$
>
> is continuous.

Proof.
Take

$$\psi_n \to \psi \qquad \text{in } C^1([-h,0])$$

The certified branch-chart well-posedness proposition and the class-uniform tube bounds imply

$$x_n \to x, \qquad \dot{x}_n \to \dot{x}$$

uniformly on compact intervals in

$$[0, T_{\max}]$$

The tame root-persistence hypothesis prevents uncontrolled branch changes and Jacobian loss. The certified fold-event atlas covers the finitely many permitted separator transitions by integral-law fold layers with fixed incoming and outgoing charts, so the forward solution map is continuous on the entire tame envelope. For the section function

$$G(t,\psi) \equiv x(t;\psi) - x_*$$

the convergence of

$$x_n$$

to

$$x$$

is uniform in a fixed neighborhood of

$$T(\psi)$$

Uniform transversality gives

$$G(T(\psi),\psi) = 0, \qquad \partial_t G(T(\psi),\psi) = \dot{x}(T(\psi);\psi) \le -u_{\text{sec}} < 0$$

and therefore

$$T(\psi_n) \to T(\psi)$$

Indeed, for small

$$\delta > 0$$

the values

$$G(T(\psi)-\delta,\psi) \qquad \text{and} \qquad G(T(\psi)+\delta,\psi)$$

have opposite signs, and the same sign separation holds for

$$G(\cdot,\psi_n)$$

for all sufficiently large

$$n$$

The uniform transversality bound excludes a second nearby return and identifies this zero with

$$T(\psi_n)$$

Finally,

$$P_\eta(\psi_n)(\theta)=x_n(T(\psi_n)+\theta)$$

so the convergence of trajectories and return times yields

$$P_\eta(\psi_n)\to P_\eta(\psi)$$

in

$$C^1([-h,0])$$

Invariant-envelope closure

The cycle estimates now reduce to three explicit margins:

$$\mathfrak{M}_{\mathrm{in}} \equiv \frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c}-\frac{\overline{A}_s^{\rho}\,\epsilon_c}{2\beta_{p,\max}}-V_{\max}$$

coming from Proposition `Explicit short-window recapture regime`, and

$$\mathfrak{M}_{\mathrm{ent}} \equiv \underline{A}_p^{\mathrm{out}}-\overline{A}_{s,\mathrm{ent}}^{\mathrm{out}}$$

coming from Lemma 29, and

$$\mathfrak{M}_{\mathrm{out}} \equiv \underline{A}_p^{\mathrm{out}}-\frac{\kappa\epsilon^2}{c_f^2T_{\mathrm{deep}}^2+\epsilon_c^2}-\frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\epsilon_c^2}$$

coming from the unified trimmed-apocenter outer-turn criterion. The first margin forces the initial post-crossing turnaround, the second supplies the non-circular sub-field-speed apocenter-entry window, and the third forces the final apocenter turn once that window exists.

> **Theorem (Invariant-Envelope Closure from Compatible Explicit Regimes).**
> Fix

$$> x_* > 0 >$$

> and a tame inbound class

$$> \mathcal{C}_{x_*,\eta}^{\mathrm{tame}} > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

> Assume:
>
> 1. the collapse-to-crossing control theorem holds on

$$> \mathcal{C}_{x_*,\eta}^{\mathrm{tame}} >$$

> with crossing-speed upper bound

$$> V_{\max}; >$$

> 2. Proposition `Explicit short-window recapture regime` applies at every first crossing issued from this class, so that

$$> \mathfrak{M}_{\mathrm{in}} > 0; >$$

> 3. Lemma 29 applies on the outer-entry interval with

$$> \mathfrak{M}_{\mathrm{ent}} > \geq > a_{\mathrm{ent}}^{\mathrm{out}} > 0, >$$

and with enough interval length to produce either a finite outer turn or a retained strict sub-field-speed window;
4. Proposition `Unified trimmed–apocenter outer–turn criterion` applies on the final apocenter window supplied by the entry step, so that

$$> \mathfrak{M}_{\mathrm{out}} > 0; >$$

5. the turn-to-section return lemmas give class-uniform section-return bounds

$$> X_{\mathrm{out,max}}, > \qquad > U_{\mathrm{sec,max}}, > \qquad > A_{\mathrm{cyc,max}}, > \qquad > T_{\mathrm{cyc,max}}; >$$

6. the envelope parameters satisfy

$$> X_{\max} \geq \max\{x_*, X_{\mathrm{out,max}}\}, >$$

$$> U_{\max} \geq \max\{V_{\max}, V_{\mathrm{ent}}^{\mathrm{out}}, U_{\mathrm{sec,max}}\}, >$$

$$> A_{\max} \geq A_{\mathrm{cyc,max}}, > \qquad > T_{\max} \geq T_{\mathrm{cyc,max}}, > \qquad > h \geq \frac{2X_{\max}}{c_f}; >$$

7. the returned history preserves the same Jacobian and branch-count bounds used to define tameness.

Then

$$> P_\eta\left(\mathcal{C}_{x_*,\eta}^{\mathrm{tame}}\right) > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

Proof.
Collapse-to-crossing control delivers an admissible crossing with speed at most

$$V_{\max}$$

The strict inner margin

$$\mathfrak{M}_{\mathrm{in}} > 0$$

then gives the first post-crossing turn by Proposition `Explicit short–window recapture regime`. On the return half, the entry margin

$$\mathfrak{M}_{\mathrm{ent}} \geq a_{\mathrm{ent}}^{\mathrm{out}} > 0$$

activates Lemma 29. Thus either the outer turn has already occurred, or the trajectory enters a retained strict sub-field-speed apocenter window. In the second case, Proposition `Unified trimmed–apocenter outer–turn criterion` supplies the final apocenter turn because

$$\mathfrak{M}_{\mathrm{out}} > 0$$

The return lemmas then give re-entry to

$$x = x_*$$

with class-uniform position, speed, acceleration, time, and tame delayed-root bounds. The envelope inequalities in item 6 place the entire returned history back inside

$$\mathcal{C}_{x_*,\eta}$$

This is the dynamical input needed to turn

$$\mathcal{K}_{x_*,\eta}$$

into a genuine self-map domain for

$$P_\eta$$

This theorem should be read narrowly. It records the exact self-map statement obtained once the tame envelope exists and the compatibility inequalities are jointly solvable. It does not by itself close either of those two burdens.

Target Proposition (Coupled admissible parameter regime).
Fix the geometric and dynamical constants extracted from the cycle estimates:

$$> V_{\max}, > \qquad > V_{\text{ent}}^{\text{out}}, > \qquad > X_{\text{out,max}}, > \qquad > U_{\text{sec,max}}, > \qquad > A_{\text{cyc,max}}, > \qquad > T_{\text{cyc,max}}, >$$

> together with the local and outer-turn parameters

$$> \beta_{p,\max}, > \qquad > C_p, > \qquad > \tau_1, > \qquad > \tau_{\text{deep}}, > \qquad > \tau_{\text{sub}}^{\text{out}}, > \qquad > a_{\text{ent}}^{\text{out}}, > \qquad > T_{\text{ent}}^{\text{out}}, > \qquad > \sigma_{\text{out}}, > \qquad > \overline{A}_s^{\rho}, > \qquad > \overline{A}_{s,\text{ent}}^{\text{out}}, > \qquad > \underline{A}_p^{\text{out}}$$

> Assume the dual-mollified parameters

$$> (\eta, \epsilon_c) >$$

> satisfy the explicit inner-window inequalities

$$> \tau_\epsilon = \frac{\epsilon_c}{2\beta_{p,\max}} \le \tau_1, > \qquad > \eta \le \frac{\epsilon_c}{4c_f C_p}, > \qquad > \epsilon_c \le \frac{\beta_{p,\max}^2}{c_f C_p}, >$$

> and the strict margin conditions

$$> \mathfrak{M}_{\text{in}} > 0, > \qquad > \mathfrak{M}_{\text{ent}} >=> \underline{A}_p^{\text{out}} > - > \overline{A}_{s,\text{ent}}^{\text{out}} >\ge> a_{\text{ent}}^{\text{out}} > 0, > \qquad > \mathfrak{M}_{\text{out}} > 0. >$$

> Also assume the outer-entry interval budget satisfies

$$> T_{\text{ent}}^{\text{out}} >\ge> \frac{\left(V_{\text{ent}}^{\text{out}} - (c_f - \sigma_{\text{out}})\right)_+}{>} a_{\text{ent}}^{\text{out}} > + > \tau_{\text{sub}}^{\text{out}}. >$$

> Then there exist envelope constants

$$> X_{\max}, > \qquad > U_{\max}, > \qquad > A_{\max}, > \qquad > T_{\max}, > \qquad > h >$$

> satisfying

$$> X_{\max} \ge \max\{x_*, X_{\text{out,max}}\}, >$$

$$> U_{\max} \ge \max\{V_{\max}, V_{\text{ent}}^{\text{out}}, U_{\text{sec,max}}\}, >$$

$$> A_{\max} \ge A_{\text{cyc,max}}, > \qquad > T_{\max} \ge T_{\text{cyc,max}}, > \qquad > h \ge \frac{2X_{\max}}{c_f}. >$$

> The remaining compatibility task is to solve these inequalities simultaneously. In particular, the manuscript must not treat the strict local margins

$$> \mathfrak{M}_{\text{in}} > 0, > \qquad > \mathfrak{M}_{\text{ent}} \ge a_{\text{ent}}^{\text{out}} > 0, > \qquad > \mathfrak{M}_{\text{out}} > 0 >$$

> as algebraically independent of the envelope constants. The crossing-speed bound

$$> V_{\max} >$$

> enters

$$> U_{\max} \ge \max\{V_{\max}, V_{\text{ent}}^{\text{out}}, U_{\text{sec,max}}\}, >$$

> while the collapse estimates producing

$$> V_{\max} >$$

> may themselves depend on the partner acceleration floor and hence on the global position scale

$$> X_{\max}. >$$

> Likewise, the coarse entry ceiling

$$> \overline{A}_{s,\text{ent}}^{\text{out}} >$$

> and the entry speed ceiling

$$> V_{\text{ent}}^{\text{out}} >$$

> are envelope-level quantities: they depend on the same branch-count, fold-ceiling, deep-past, speed, and position bounds that define the controlled cycle.

> A valid nonemptiness proof must therefore close a coupled algebraic system in

$$> (\eta, \epsilon_c, X_{\max}, U_{\max}, A_{\max}, T_{\max}, h, > V_{\mathrm{ent}}^{\mathrm{out}}, > a_{\mathrm{ent}}^{\mathrm{out}}, > T_{\mathrm{ent}}^{\mathrm{out}}, > \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}}), >$$

> rather than verify the local margins first and choose the envelope constants afterward with arbitrary slack.

This target isolates the remaining algebraic compatibility issue. Once collapse-to-crossing bounds, the inner recapture margin, the outer-turn margin, and the envelope bookkeeping constants are packaged on one coupled regime, invariant-envelope closure becomes an actual self-map statement. Until then, simultaneous solvability of the displayed inequalities remains part of the scaffold rather than a completed proposition.

For later proof checking, the finite strict-regime list can be taken to include:

$$\tau_1 - \frac{\epsilon_c}{2\beta_{p,\max}} > 0, \qquad \frac{\epsilon_c}{4c_f C_p} - \eta > 0, \qquad \frac{\beta_{p,\max}^2}{c_f C_p} - \epsilon_c > 0$$

$$\frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c} - \frac{\overline{A}_s^{\rho}\epsilon_c}{2\beta_{p,\max}} - V_{\max} > 0$$

$$\underline{A}_p^{\mathrm{out}} - \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}} - a_{\mathrm{ent}}^{\mathrm{out}} \geq 0, \qquad a_{\mathrm{ent}}^{\mathrm{out}} > 0$$

$$\underline{A}_p^{\mathrm{out}} - \frac{\kappa\epsilon^2}{c_f^2\tau_{\mathrm{deep}}^2 + \epsilon_c^2} - \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\epsilon_c^2} > 0$$

$$T_{\mathrm{ent}}^{\mathrm{out}} - \frac{\left(V_{\mathrm{ent}}^{\mathrm{out}} - (c_f - \sigma_{\mathrm{out}})\right)_+}{a_{\mathrm{ent}}^{\mathrm{out}}} - \tau_{\mathrm{sub}}^{\mathrm{out}} \geq 0$$

together with the five envelope domination inequalities for

$$X_{\max}, \qquad U_{\max}, \qquad A_{\max}, \qquad T_{\max}, \qquad h$$

Any dependence of

$$V_{\max}, \qquad V_{\mathrm{ent}}^{\mathrm{out}}, \qquad T_{\mathrm{ent}}^{\mathrm{out}}, \qquad \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}}$$

on the envelope constants must be inserted into this list before claiming a strict slack point.

> **Proposition (Strict slack point gives a nonempty coupled regime).**
> Let

$$> p >\equiv> (\eta, \epsilon_c, X_{\max}, U_{\max}, A_{\max}, T_{\max}, h, > V_{\mathrm{ent}}^{\mathrm{out}}, > a_{\mathrm{ent}}^{\mathrm{out}}, > T_{\mathrm{ent}}^{\mathrm{out}}, > \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}}) >$$

> denote the coupled parameter tuple, and suppose the coupled-regime requirements can be written as a finite family of continuous inequalities

$$> F_q(p) > 0, > \qquad > q \in \mathcal{Q}_{\mathrm{reg}}, >$$

> together with the finite envelope domination inequalities

$$> G_r(p) \geq 0, > \qquad > r \in \mathcal{Q}_{\mathrm{env}}. >$$

> Here the list includes the inner-window inequalities, the margins

$$> \mathfrak{M}_{\mathrm{in}} > 0, > \qquad > \mathfrak{M}_{\mathrm{ent}} \geq a_{\mathrm{ent}}^{\mathrm{out}} > 0, > \qquad > \mathfrak{M}_{\mathrm{out}} > 0, >$$

> the outer-entry interval budget, and the envelope bounds for

$$> X_{\max}, > \qquad > U_{\max}, > \qquad > A_{\max}, > \qquad > T_{\max}, > \qquad > h. >$$

> If there exists one parameter tuple

$$> p_0 >$$

> such that all strict inequalities have positive slack and all envelope inequalities have nonnegative slack, with the zero-slack envelope inequalities allowed only where increasing the corresponding envelope constant preserves every other inequality, then the admissible coupled-regime set is nonempty. If the envelope inequalities also have strict slack at

$$> p_0, >$$

> then the admissible regime contains an open neighborhood of

$$> p_0. >$$

Proof.
Because the family

$$\mathcal{Q}_{\mathrm{reg}}$$

is finite and each

$$F_q$$

is continuous, positive slack at

$$p_0$$

persists on a small neighborhood of

$$p_0$$

The same argument applies to every envelope inequality with strict slack. If one envelope inequality is saturated but the corresponding envelope constant can be increased without weakening the other inequalities, enlarge that constant slightly first; this turns the saturated domination inequality into a strict one while preserving the already strict margin inequalities. After this finite adjustment, all inequalities hold with strict slack on one neighborhood. Hence the coupled admissible set is nonempty, and in the strict-slack case open.

This proposition reduces the coupled-regime problem to a finite arithmetic certificate: exhibit one tuple

$$p_0$$

at which the inner margin, apocenter-entry margin, outer margin, entry-time budget, and envelope domination inequalities all hold simultaneously.
For completion this tuple must be actual data, either concrete numbers or interval enclosures whose lower endpoints give strict positive slack. A qualitative statement that the inner margin improves as

$$\epsilon_c \downarrow 0$$

is not enough, because the outer caustic and self terms may worsen under the same parameter move.

The following nonemptiness test should be run before the full coupled-corridor certificate. It isolates the core-scale conflict between the inner recapture estimate and the outer shell-leakage estimate.

> **Proposition (Corridor nonemptiness criterion).**
> Use the fixed mollifier normalization

$$> \Lambda_\delta >\equiv> \eta\|\delta_\eta\|_\infty >=> \|\delta\|_\infty. >$$

> Suppose

$$> S_{\mathrm{in}}^{\rho} > 0, > \qquad > \sigma_{\mathrm{out}} > 0, > \qquad > P_{\mathrm{out}} > 0. >$$

> The inner coefficient condition

$$> C_{\mathrm{in}}(\epsilon_c) > 0 >$$

> is equivalent to

$$> \epsilon_c^2 ><> \frac{1}{2S_{\mathrm{in}}^{\rho}}. >$$

> The outer shell-deep coefficient condition is equivalent to

$$> P_{\mathrm{eff}}(\epsilon_c) > 0, > \qquad > \epsilon_c^2 >>> \frac{2\Lambda_\delta}{>}\sigma_{\mathrm{out}} P_{\mathrm{eff}}(\epsilon_c), >$$

> where

$$> P_{\mathrm{eff}}(\epsilon_c) >\equiv> P_{\mathrm{out}} - D_{\mathrm{deep}}(\epsilon_c). >$$

> Therefore the factorized corridor has a possible core-scale window only if there exists

$$> \epsilon_c > 0 >$$

> such that

$$> \frac{2\Lambda_\delta}{>}\sigma_{\rm out}P_{\rm eff}(\epsilon_c) ><> \epsilon_c^2 ><> \frac{1}{2S_{\rm in}^\rho}. >$$

> In the coarse audit where

$$> D_{\rm deep}(\epsilon_c) >$$

> is negligible, this reduces to the explicit window

$$> \sqrt{\frac{2\Lambda_\delta}{\sigma_{\rm out}P_{\rm out}}} >\lesssim> \epsilon_c >\lesssim> \frac{1}{\sqrt{2S_{\rm in}^\rho}}, >$$

> and the approximate nonemptiness condition

$$> \sigma_{\rm out}P_{\rm out}S_{\rm in}^\rho >>> 4\Lambda_\delta. >$$

Proof.
The first equivalence follows directly from

$$\frac{1}{4\beta_{p,\max}\epsilon_c} - \frac{S_{\rm in}^\rho\epsilon_c}{2\beta_{p,\max}} > 0$$

The second follows from

$$P_{\rm out} - D_{\rm deep}(\epsilon_c) - \frac{2\eta\|\delta_\eta\|_\infty}{\sigma_{\rm out}\epsilon_c^2} > 0$$

and the definition of

$$\Lambda_\delta$$

Combining the lower and upper core-scale requirements gives the displayed window. If this window is empty, the factorized corridor fails by parameter incompatibility before any seed-cycle residual or return-map argument is relevant.

The following sufficient corridor is the scalar form of that arithmetic certificate. It does not prove the geometric coefficients by itself; it separates the coefficient audit from the final coupling choice.

> **Proposition (Factorized corridor for a strict coupled-regime point).**
> Write

$$> g \equiv \kappa\epsilon^2. >$$

> Suppose the force bounds on a chosen envelope factor as

$$> \overline{A}_s^\rho = gS_{\rm in}^\rho, > \qquad > \underline{A}_p^{\rm out} = gP_{\rm out}, > \qquad > \overline{A}_{s,\rm ent}^{\rm out} = gS_{\rm ent}^{\rm out}, >$$

> where the coefficients are independent of

$$> g. >$$

> Define

$$> D_{\rm deep}(\epsilon_c) >\equiv> \frac{1}{c_f^2\tau_{\rm deep}^2 + \epsilon_c^2}, > \qquad > L_{\rm shell}(\eta,\epsilon_c) >\equiv> \frac{2\eta\|\delta_\eta\|_\infty}{\sigma_{\rm out}\epsilon_c^2}. >$$

> Assume the selected

$$> (\eta, \epsilon_c) >$$

> satisfy the strict short-window inequalities

$$> \frac{\epsilon_c}{2\beta_{p,\max}} < \tau_1, > \qquad > \eta < \frac{\epsilon_c}{4c_fC_p}, > \qquad > \epsilon_c < \frac{\beta_{p,\max}^2}{c_fC_p}, >$$

> and that there exists

$$> m_{\text{ent}} > 0 >$$

> with

$$> P_{\text{out}} - S^{\text{out}}_{\text{ent}} - m_{\text{ent}} > 0. >$$

> Also assume

$$> C_{\text{in}}(\epsilon_c) >\equiv> \frac{1}{4\beta_{p,\max}\epsilon_c} > - > \frac{S^{\rho}_{\text{in}}\,\epsilon_c}{2\beta_{p,\max}} >> 0, >$$

> or, if the Cauchy-Schwarz partner-impulse refinement has been interval-certified,

$$> C^{\text{CS}}_{\text{in}}(\epsilon_c) >\equiv> \frac{Q_{\text{CS}}}{4\beta_{p,\max}\epsilon_c} > - > \frac{S^{\rho}_{\text{in}}\,\epsilon_c}{2\beta_{p,\max}} >> 0. >$$

$$> P_{\text{out}} > - > D_{\text{deep}}(\epsilon_c) > - > L_{\text{shell}}(\eta, \epsilon_c) >> 0, >$$

> and

$$> T^{\text{out}}_{\text{ent}} > \tau^{\text{out}}_{\text{sub}}. >$$

> Then every coupling scale

$$> g >>> \max\left\{ > \frac{V_{\max}}{C_{\text{in}}(\epsilon_c)}, > \frac{\bigl(V^{\text{out}}_{\text{ent}} - (c_f - \sigma_{\text{out}})\bigr)_+}{>} m_{\text{ent}}\bigl(T^{\text{out}}_{\text{ent}} - \tau^{\text{out}}_{\text{sub}}\bigr) > \right\} >$$

> with

$$> C_{\text{in}}(\epsilon_c) >$$

> replaced by the certified

$$> C^{\text{CS}}_{\text{in}}(\epsilon_c) >$$

> if that refinement is used,
> gives a strict coupled-regime point by setting

$$> a^{\text{out}}_{\text{ent}} = g m_{\text{ent}}. >$$

Proof.
The short-window inequalities are strict by assumption. The inner margin becomes

$$\mathfrak{M}_{\text{in}} = g C_{\text{in}}(\epsilon_c) - V_{\max}$$

which is positive by the lower bound on

$$g$$

If the certified Cauchy-Schwarz refinement is used, the same argument replaces

$$C_{\text{in}}$$

by

$$C^{\text{CS}}_{\text{in}}$$

The entry margin satisfies

$$\mathfrak{M}_{\text{ent}} - a^{\text{out}}_{\text{ent}} = g\bigl(P_{\text{out}} - S^{\text{out}}_{\text{ent}} - m_{\text{ent}}\bigr) > 0$$

The outer margin factors as

$$\mathfrak{M}_{\text{out}} = g\bigl(P_{\text{out}} - D_{\text{deep}}(\epsilon_c) - L_{\text{shell}}(\eta, \epsilon_c)\bigr)$$

which is positive by the coefficient hypothesis. Finally, the lower bound on

$$g$$

also gives

$$\mathcal{T}_{\mathrm{ent}}^{\mathrm{out}} \geq \frac{\left(V_{\mathrm{ent}}^{\mathrm{out}} - (c_f - \sigma_{\mathrm{out}})\right)_+}{g m_{\mathrm{ent}}} + \tau_{\mathrm{sub}}^{\mathrm{out}}$$

Thus the finite strict-regime list holds. Choosing the envelope constants with strict domination slack then supplies the strict slack point required by the previous proposition.

> **Proposition (Certified self-map criterion).**
> Let

$$> \mathcal{K}_{x_*,\eta} >=> \mathcal{K}_{x_*,\eta}^{\mathrm{cert}} >$$

> be a closed convex tame envelope produced by a finite certificate

$$> \{\ell_\alpha \leq b_\alpha\}_{\alpha \in \mathcal{I}_{\mathrm{cert}}}. >$$

> Assume:
>
> 1. the invariant-envelope closure theorem applies on

$$> \mathcal{K}_{x_*,\eta}, >$$

> so that

$$> P_\eta(\phi) \in \mathcal{C}_{x_*,\eta} > \qquad > \text{for every } \phi \in \mathcal{K}_{x_*,\eta}; >$$

> 2. each certificate inequality is preserved by one return:

$$> \ell_\alpha(P_\eta(\phi)) \leq b_\alpha > \qquad > \text{for every } > \alpha \in \mathcal{I}_{\mathrm{cert}} > \text{ and every } > \phi \in \mathcal{K}_{x_*,\eta}. >$$

> Then

$$> P_\eta(\mathcal{K}_{x_*,\eta}) >\subseteq> \mathcal{K}_{x_*,\eta}. >$$

Proof.
Fix

$$\phi \in \mathcal{K}_{x_*,\eta}$$

By invariant-envelope closure,

$$P_\eta(\phi) \in \mathcal{C}_{x_*,\eta}$$

By certificate preservation,

$$\ell_\alpha(P_\eta(\phi)) \leq b_\alpha \qquad \text{for every } \alpha \in \mathcal{I}_{\mathrm{cert}}$$

Therefore

$$P_\eta(\phi) \in \mathcal{K}_{x_*,\eta}^{\mathrm{cert}} = \mathcal{K}_{x_*,\eta}$$

Since

$$\phi$$

was arbitrary, the claimed self-map inclusion follows.

For the sampled certificate above, certificate preservation has an entirely finite form.

> **Proposition (Finite sampled preservation criterion).**
> Suppose

$$> \mathcal{K}_{x_*,\eta}^{\mathrm{cert}} >$$

> is defined by the sampled seed-cycle tube in Proposition `Sampled seed-cycle tube gives a finite tame certificate`, with mesh

$$> \{\theta_j\}_{j=0}^{N} >$$

> and radius

$$> r_{\text{cert}}. >$$

> Assume invariant-envelope closure gives

$$> P_\eta(\phi) \in \mathcal{C}_{x_*,\eta} > \qquad > \text{for every } \phi \in \mathcal{K}^{\text{cert}}_{x_*,\eta}. >$$

> If, for every

$$> \phi \in \mathcal{K}^{\text{cert}}_{x_*,\eta} >$$

> and every mesh index

$$> 0 \le j \le N, >$$

> the returned history obeys

$$> |P_\eta(\phi)(\theta_j) - \phi_{\text{cyc}}(\theta_j)| > \le \frac{r_{\text{cert}}}{4}, > \qquad > |\partial_\theta P_\eta(\phi)(\theta_j) - \dot{\phi}_{\text{cyc}}(\theta_j)| > \le \frac{r_{\text{cert}}}{4}, >$$

> then

$$> P_\eta(\mathcal{K}^{\text{cert}}_{x_*,\eta}) > \subseteq > \mathcal{K}^{\text{cert}}_{x_*,\eta}. >$$

Proof.
The invariant-envelope theorem gives the returned-history membership in

$$\mathcal{C}_{x_*,\eta}$$

The displayed finite sample inequalities are exactly the certificate inequalities defining

$$\mathcal{K}^{\text{cert}}_{x_*,\eta}$$

in the sampled construction. Hence every returned history satisfies all certificate inequalities and therefore lies in

$$\mathcal{K}^{\text{cert}}_{x_*,\eta}$$

The hard part of applying this criterion is proving the finite mesh inequalities uniformly. The following budget form is the one that should be used in later proof checking.

> **Proposition (Returned-sample budget certificate).**
> In the setting of Proposition `Finite sampled preservation criterion`, suppose there are finite returned-sample budgets

$$> E^x_{j,+}, > \qquad > E^x_{j,-}, > \qquad > E^v_{j,+}, > \qquad > E^v_{j,-}, > \qquad > 0 \le j \le N, >$$

> such that for every

$$> \phi \in \mathcal{K}^{\text{cert}}_{x_*,\eta} >$$

> the returned history satisfies

$$> P_\eta(\phi)(\theta_j) - \phi_{\text{cyc}}(\theta_j) > \le E^x_{j,+}, > \qquad > \phi_{\text{cyc}}(\theta_j) - P_\eta(\phi)(\theta_j) > \le E^x_{j,-}, >$$

> and

$$> \partial_\theta P_\eta(\phi)(\theta_j) - \dot{\phi}_{\text{cyc}}(\theta_j) > \le E^v_{j,+}, > \qquad > \dot{\phi}_{\text{cyc}}(\theta_j) - \partial_\theta P_\eta(\phi)(\theta_j) > \le E^v_{j,-}. >$$

> If the strict sample-slack inequalities

$$> \max\{E^x_{j,+}, E^x_{j,-}, E^v_{j,+}, E^v_{j,-}\} > < > \frac{r_{\text{cert}}}{4} > \qquad > \text{for every } 0 \le j \le N >$$

> hold, then

$$> P_\eta(\mathcal{K}^{\text{cert}}_{x_*,\eta}) > \subseteq > \mathcal{K}^{\text{cert}}_{x_*,\eta}. >$$

Proof.
The one-sided budget inequalities imply

$$|P_\eta(\phi)(\theta_j) - \phi_{\text{cyc}}(\theta_j)| < \frac{r_{\text{cert}}}{4}$$

and

$$|\partial_\theta P_\eta(\phi)(\theta_j) - \dot{\phi}_{\text{cyc}}(\theta_j)| < \frac{r_{\text{cert}}}{4}$$

for every mesh index. Proposition `Finite sampled preservation criterion` then gives the self-map inclusion.

> **Proposition (Residual-plus-sensitivity sampled preservation).**
> In the setting above, assume the center history

$$> \phi_{\text{cyc}} >$$

> has a defined return

$$> P_\eta(\phi_{\text{cyc}}), >$$

> and define the one-sided returned residuals

$$> R^x_{j,+} >\equiv> \big(P_\eta(\phi_{\text{cyc}})(\theta_j) - \phi_{\text{cyc}}(\theta_j)\big)_+, > \qquad > R^x_{j,-} >\equiv> \big(\phi_{\text{cyc}}(\theta_j) - P_\eta(\phi_{\text{cyc}})(\theta_j)\big)_+, >$$

$$> R^v_{j,+} >\equiv> \big(\partial_\theta P_\eta(\phi_{\text{cyc}})(\theta_j) - \dot{\phi}_{\text{cyc}}(\theta_j)\big)_+, > \qquad > R^v_{j,-} >\equiv> \big(\dot{\phi}_{\text{cyc}}(\theta_j) - \partial_\theta P_\eta(\phi_{\text{cyc}})(\theta_j)\big)_+. >$$

> Suppose also that the return map sample functionals have finite local sensitivity constants on

$$> \mathcal{K}^{\text{cert}}_{x_*,\eta}, >$$

> namely

$$> |P_\eta(\phi)(\theta_j) - P_\eta(\phi_{\text{cyc}})(\theta_j)| >\le> L^x_j \|\phi - \phi_{\text{cyc}}\|_{C^1}, >$$

$$> |\partial_\theta P_\eta(\phi)(\theta_j) > - > \partial_\theta P_\eta(\phi_{\text{cyc}})(\theta_j)| >\le> L^v_j \|\phi - \phi_{\text{cyc}}\|_{C^1} >$$

> for every

$$> \phi \in \mathcal{K}^{\text{cert}}_{x_*,\eta}. >$$

> If

$$> \max\{R^x_{j,+}, R^x_{j,-}\} + L^x_j r_{\text{cert}} ><> \frac{r_{\text{cert}}}{4}, >$$

> and

$$> \max\{R^v_{j,+}, R^v_{j,-}\} + L^v_j r_{\text{cert}} ><> \frac{r_{\text{cert}}}{4} >$$

> for every

$$> 0 \le j \le N, >$$

> then the returned-sample budget certificate holds, and hence

$$> P_\eta(\mathcal{K}^{\text{cert}}_{x_*,\eta}) >\subseteq> \mathcal{K}^{\text{cert}}_{x_*,\eta}. >$$

Proof.
The sampled certificate construction gives

$$\|\phi - \phi_{\text{cyc}}\|_{C^1} \le r_{\text{cert}}$$

for every

$$\phi \in \mathcal{K}^{\text{cert}}_{x*,\eta}$$

Therefore

$$P_\eta(\phi)(\theta_j) - \phi_{\text{cyc}}(\theta_j) \le R^x_{j,+} + L^x_j r_{\text{cert}}$$

and the same triangle-inequality argument gives the other three one-sided bounds. The displayed strict inequalities therefore define returned-sample budgets satisfying the previous proposition. The self-map inclusion follows.

This criterion is only a sufficient route. If the raw local sensitivity is too large, the boundary-trapping lemma below can still prove preservation by direct inward-margin estimates at the certificate faces.

> **Lemma (Boundary trapping for the sampled certificate).**
> Assume the returned-sample budget certificate and write

$$> s_{\mathrm{sam}} >\equiv> \frac{r_{\mathrm{cert}}}{4} > - > \max_{0\le j\le N} > \max\{E^x_{j,+}, E^x_{j,-}, E^v_{j,+}, E^v_{j,-}\}. >$$

> If

$$> s_{\mathrm{sam}} > 0, >$$

> then every codimension-one sample face of

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} >$$

> is strictly inward under one return.

Proof.
The sample faces are exactly the four one-sided equalities, for each mesh index

$$j$$

obtained by replacing one of

$$|\phi(\theta_j) - \phi_{\mathrm{cyc}}(\theta_j)| \le \frac{r_{\mathrm{cert}}}{4}, \qquad |\dot\phi(\theta_j) - \dot\phi_{\mathrm{cyc}}(\theta_j)| \le \frac{r_{\mathrm{cert}}}{4}$$

with equality and choosing a sign. If a returned history touched one such face, the corresponding returned-sample defect would equal

$$\frac{r_{\mathrm{cert}}}{4}$$

But the returned-sample budget bounds that same defect by at most

$$\frac{r_{\mathrm{cert}}}{4} - s_{\mathrm{sam}}$$

a contradiction. Hence no returned history reaches any sample face; all sample faces are strictly inward.

One useful route for proving the budget hypotheses is a boundary-trapping check: for each certificate face

$$\ell_\alpha = b_\alpha$$

show that any trajectory whose returned history would otherwise touch that face is pushed strictly back toward

$$\ell_\alpha < b_\alpha$$

by one of the established cycle margins. Because the certificate family is finite, these facewise checks reduce the global self-map property to finitely many inward-pointing inequalities.

> **Theorem (Finite-certificate invariant closure package).**
> Assume:
>
> 1. the seed-cycle margin ledger is positive and the quantitative radius criterion has been used to choose

$$> r_{\mathrm{cert}}; >$$

> 2. the sampled finite certificate defines

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}; >$$

> 3. the factorized coupled-regime corridor holds, so invariant-envelope closure gives

$$> P_\eta(\phi) \in \mathcal{C}_{x_*,\eta} > \qquad > \text{for every } \phi \in \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}; >$$

> 4. and either the direct returned-sample budget certificate holds on

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}, >$$

> for example by boundary trapping, or the residual-plus-sensitivity sampled preservation criterion holds on

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}. >$$

> Then

$$> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} >$$

> is a nonempty closed convex tame self-map domain for

$$> P_\eta, >$$

> namely

$$> P_\eta\big(\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}\big) >\subseteq> \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}. >$$

Proof.
The positive seed-cycle ledger and the radius criterion give the strict branch-chart, gap, memory-depth, and envelope margins required by the sampled finite certificate. The finite certificate construction then makes

$$\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

a nonempty closed convex tame envelope. The factorized corridor supplies the coupled strict-slack point needed by invariant-envelope closure, so returned histories lie in

$$\mathcal{C}_{x_*,\eta}$$

Finally, a direct returned-sample budget certificate gives the finite sampled preservation criterion immediately. If that direct route is not used, the residual-plus-sensitivity criterion implies the same returned-sample budget certificate. In either case the finite sampled preservation criterion gives the certificate inequalities after one return. Therefore the returned history lies in the certified set itself, proving the self-map inclusion.

The finite self-map ledger has five rows. The first, second, third, and fifth rows produce the self-map certificate and well-posed variational interpretation; the fourth row is a stability diagnostic that decides whether returned-sample preservation should be attempted by sensitivities or by boundary trapping. Adding the certified topology row below gives the full six-row Schauder-ready audit.

1. **Seed-chart row.**
   Verify

$$\nu_{\mathrm{seed}} > 0, \qquad \gamma_{\mathrm{gap}} > 0, \qquad \gamma_h > 0, \qquad \gamma_{\mathrm{env}} > 0$$

and finite sensitivities

$$L_J, \qquad L_F, \qquad L_h, \qquad L_{\mathrm{env}}$$

This row chooses

$$r_{\mathrm{cert}}$$

and produces the closed convex tame certificate.
The row begins with a candidate

$$\phi_{\mathrm{cyc}}$$

a common mesh, and a null-coordinate causal pre-ledger. For each ordered receiver-source block

$$(I_\alpha, I_\beta)$$

the pre-ledger must classify the row as empty, simple-root, or fold-layer using

$$u = c_f t - x, \qquad w = c_f t + x$$

Empty rows require strict range separation; simple-root rows require a positive source-side derivative floor; fold-layer rows remain outside branch-sum reduction until the dual-mollified fold certificate supplies the parity-preserving transition

$$\Delta N \in 2\mathbb{Z}, \qquad \Delta D = 0$$

Any unresolved row blocks the seed chart before corridor, monodromy, or returned-sample work begins.
2. **Coupled-corridor row.**
Verify

$$C_{\mathrm{in}}(\epsilon_c) > 0, \qquad P_{\mathrm{out}} - S_{\mathrm{ent}}^{\mathrm{out}} - m_{\mathrm{ent}} > 0, \qquad P_{\mathrm{out}} - D_{\mathrm{deep}}(\epsilon_c) - L_{\mathrm{shell}}(\eta, \epsilon_c) > 0$$

choose

$$g = \kappa \epsilon^2$$

above the factorized threshold, and set

$$a_{\mathrm{ent}}^{\mathrm{out}} = g m_{\mathrm{ent}}$$

This row supplies the strict coupled-regime point. For a completed proof, this row must be a concrete numerical or interval certificate for one tuple

$$p_0$$

in the coupled system, not separate local parameter choices.
3. **Solution-manifold compatibility row.**
The section history must live on the compatible first-order history manifold before any variational or monodromy row is interpreted. Write the local first-order lift as

$$Y = (X, U), \qquad \mathcal{H}_h^{(1)} = C^1([-h, 0]; \mathbb{R}^2)$$

and define the admissible compatibility class

$$\mathcal{X}_\eta = \left\{ \Phi = (X, U) \in \mathcal{H}_h^{(1)} : \dot{X}(0) = U(0), \quad \dot{U}(0) = F_\eta(\Phi) \right\}$$

The candidate packet must report this endpoint row on the same packet identity as the pre-ledger, branch chart, fold atlas, and returned samples. The tangent row consumed by monodromy must satisfy

$$\dot{\Xi}(0) = V(0), \qquad \dot{V}(0) = DF_\eta(\Phi)\Psi$$

Thus monodromy differentiates certified branch maps on compatible histories; it is not a frozen-delay calculation on an arbitrary C^1 box.
4. **Monodromy diagnostic row.**
Compute an interval enclosure for the section-anchored linearized return map

$$DP_\eta(\phi_{\mathrm{cyc}})$$

on the certificate mesh. The section anchoring removes the neutral time-translation direction before the spectrum is interpreted. Record the discrete monodromy matrix

$$M_N$$

an interval spectral enclosure, and an explicit diagnostic margin

$$\delta_{\mathrm{mon}} > 0$$

If any certified eigenvalue satisfies

$$|\lambda| > 1 + \delta_{\mathrm{mon}}$$

the residual-plus-sensitivity route should be considered closed for that unstable direction, and the returned-sample row must use direct one-sided boundary trapping. If the spectrum and operator-norm enclosure are small enough to give usable constants

$$L_j^x, \qquad L_j^v$$

this row authorizes the residual-plus-sensitivity route. Schauder itself does not require linear stability; this row is a proof-strategy selector for the finite preservation audit.

The same row must also report the zero-mode quotient used for interpretation. Let

$$Z_{\mathrm{time}}(\theta) = \dot{\phi}_{\mathrm{cyc}}(\theta)$$

denote the infinitesimal time-shift direction before section anchoring. If additional ansatz or certificate parameters

$$\alpha^a$$

are carried, their tangent rows

$$Z_a(\theta) = \partial_{\alpha^a}\phi_{\mathrm{cyc}}(\theta;\alpha)$$

must be classified as neutral, constrained by the section, or transverse. This prevents a harmless collective-coordinate drift from being mistaken for an unstable return direction, and it prevents a genuine transverse instability from being hidden inside a free parameter.
5. **Returned-sample row.**
Prefer the direct one-sided budget route when local sensitivities are large: prove

$$E^x_{j,\pm}, \qquad E^v_{j,\pm}$$

by boundary trapping with strict sample slack. Equivalently, when the sensitivity constants are tame enough, verify

$$\max\{R^x_{j,+}, R^x_{j,-}\} + L^x_j r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}, \qquad \max\{R^v_{j,+}, R^v_{j,-}\} + L^v_j r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}$$

for every mesh index. This row supplies certificate preservation under one return.

This ledger is deliberately finite. Passing the seed-chart, coupled-corridor, solution-manifold compatibility, and returned-sample rows turns the domain-production burden into the self-map inclusion; the monodromy row identifies whether the returned-sample proof should use sensitivity control or boundary trapping. Failing any required row identifies the exact obstruction.

Certified topology row
After the finite closure audit supplies

$$P_\eta\big(\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}\big) \subseteq \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

precompactness is no longer a separate dynamical mystery: the returned histories already lie in the same certified

$$C^1$$

envelope. Continuity still requires one extra topological margin, namely strict transversality of the returned section.

Define the certified return-speed margin

$$u^{\mathrm{cert}}_{\mathrm{ret}} \equiv -\dot{\phi}_{\mathrm{cyc}}(0) - \frac{r_{\mathrm{cert}}}{4}$$

Because the mesh includes

$$\theta_N = 0$$

the returned-sample inequalities imply

$$\partial_\theta P_\eta(\phi)(0) \le \dot{\phi}_{\mathrm{cyc}}(0) + \frac{r_{\mathrm{cert}}}{4} = -u^{\mathrm{cert}}_{\mathrm{ret}}$$

Thus

$$u^{\mathrm{cert}}_{\mathrm{ret}} > 0$$

is the finite section-transversality check needed by the continuity proposition.

> **Proposition (Certified topology on the finite self-map domain).**
> Assume the finite-certificate invariant closure package, and assume in addition:
>
> 1. the certified return-speed margin satisfies

$$> u^{\mathrm{cert}}_{\mathrm{ret}} > 0; >$$

> 2. the certified branch-chart well-posedness proposition applies on the finite chart intervals covering the stored history and one-cycle continuation;
> 3. the active delayed roots persist on

> $\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$
>
> with the Jacobian floors, branch-count ceilings, and memory-depth bounds supplied by the certificate outside the certified fold-event atlas, and each fold layer in that atlas has a parity-preserving incoming-to-outgoing chart transition.
>
> Then
>
> $P_\eta : \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} \to \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$
>
> is continuous, and
>
> $P_\eta\big(\mathcal{K}^{\mathrm{cert}}_{x_*,\eta}\big)$
>
> is precompact in
>
> $C^1([-h,0]).$

Proof.
The finite-certificate invariant closure package gives

$$P_\eta(\phi) \in \mathcal{K}^{\mathrm{cert}}_{x_*,\eta} \subseteq \mathcal{C}_{x_*,\eta}$$

for every

$$\phi \in \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

Hence every returned history satisfies the position, speed, Lipschitz-velocity, and horizon bounds required by Proposition `Precompactness of the Return Image`; that proposition gives precompactness.

For continuity, certified branch-chart well-posedness gives continuous dependence of the controlled continuation on the initial history while the certificate keeps the same exterior charts, origin-layer charts, fold-event atlas, and Jacobian floors. The displayed return-speed estimate gives the uniform transverse return condition

$$\dot{x}(T(\phi);\phi) \le -u^{\mathrm{cert}}_{\mathrm{ret}} < 0$$

Therefore the continuity proposition for the return map applies with

$$\mathcal{K}_{x_*,\eta} = \mathcal{K}^{\mathrm{cert}}_{x_*,\eta}$$

and yields continuity of

$$P_\eta$$

on the certified domain.

The full Schauder-ready audit therefore has six rows:

1. the seed-chart row;
2. the coupled-corridor row;
3. the solution-manifold compatibility row;
4. the monodromy diagnostic row;
5. the returned-sample row;
6. and the topology row

$$u^{\mathrm{cert}}_{\mathrm{ret}} > 0$$

plus certified branch-chart well-posedness and a certified fold-event atlas on the controlled continuation.

Remaining blockers before Schauder

At this stage the remaining blockers are narrow and explicit:

No remaining blocker asks for an elementary closed-form orbit. The proof needs one instantiated candidate cycle

$$\phi_{\mathrm{cyc}}$$

defined against the dual-mollified absolute-time law, and finite certificate data proving that the same closed convex tame domain is self-mapping, continuous, and precompact under

$$P_\eta$$

The first explicit velocity-class packet has moved the obstruction from candidate absence to candidate falsification. A fixed cosine candidate supplies useful null-coordinate and fold-layer diagnostics, but it fails at the parent-complement part of the pre-ledger: after accepted simple-root windows and fold-layer diagnostics are removed, some parent complements still carry equality cores or non-strict null-coordinate overlap. The next admissible route is therefore a fresh fold-adapted collocation candidate, or an equivalent certified construction, whose pre-ledger closes before any seed-chart or branch-chart row begins.

There is now also a stricter sub-field-speed comparison branch. The held-release ODE segment and the exterior affine delayed-partner chart are action-generated baselines, not prescribed trajectories. They show that a normalized release from $x_0 > 1$ need not reach field speed during the held-source segment, and that the exterior delayed partner branch approaches $\dot{x} = -c_f$ only at the origin-layer limit. This does not prove a sub-field-speed breather, but it changes the proof burden: field-speed separators must be derived from the full dual-mollified dynamics or replaced by a certified sub-field return mechanism.

The negative-breather lesson is that even a formal expansion valid to all orders can miss a leakage channel outside the expansion scale. The collinear program therefore treats a small residual curve, a long-lived numerical trace, or a closed-looking ansatz as candidate evidence only. Promotion requires the existing certificate rows to close the leakage routes they control: the pre-ledger fold-layer budgets, coupled-corridor propagation, returned-sample preservation, and topology/self-map row on one certified domain. The same rule applies internally: partial diagnostics remain candidate evidence while parent-complement equality cores remain unresolved.

Before those five audit rows can be meaningful, the candidate must pass the named null-coordinate pre-ledger target from Closed-Form Collinear Breather Ansatz. Concretely, the proof must:

1. produce one candidate cycle

$$\phi_{\text{cyc}}$$

with a certificate mesh and either fold-adapted fractional basis data near field-speed separators or an interval-collocation representation with equivalent residual targets;
2. verify the `Null-Coordinate Causal Pre-Ledger` target for

$$|x(t) - x(s)| = c_f(t - s), \qquad s < t$$

using

$$u = c_f t - x, \qquad w = c_f t + x$$

to classify every ordered arc-pair block as empty, simple-root, or fold-layer. After accepted simple-root and fold-layer subblocks are removed, the remaining parent-complement strips must also be consumed by strict range separation, endpoint-excluded singleton contact under the declared boundary convention, exact fold-layer coverage, or another already accepted same-packet complement predicate. If this finite pre-ledger cannot be certified with strict gaps, derivative floors, fold-layer bounds, and consumed parent complements, the candidate or itinerary fails before the seed-cycle margin ledger is attempted;
3. verify the seed-cycle margin ledger

$$\nu_{\text{seed}} > 0, \qquad \gamma_{\text{gap}} > 0, \qquad \gamma_h > 0, \qquad \gamma_{\text{env}} > 0$$

then apply the quantitative radius criterion for

$$r_{\text{cert}}$$

to choose the sampled finite tame certificate for

$$\mathcal{K}^{\text{cert}}_{x_*,\eta}$$

4. verify the factorized coupled-corridor inequalities

$$C_{\text{in}}(\epsilon_c) > 0, \qquad P_{\text{out}} - S^{\text{out}}_{\text{ent}} - m_{\text{ent}} > 0, \qquad P_{\text{out}} - D_{\text{deep}}(\epsilon_c) - L_{\text{shell}}(\eta, \epsilon_c) > 0$$

then choose

$$g = \kappa\epsilon^2$$

above the displayed corridor threshold for the finite coupled-regime system in

$$(\eta, \epsilon_c, X_{\max}, U_{\max}, A_{\max}, T_{\max}, h, V_{\mathrm{ent}}^{\mathrm{out}}, a_{\mathrm{ent}}^{\mathrm{out}}, T_{\mathrm{ent}}^{\mathrm{out}}, \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}})$$

by producing one strict numeric or interval tuple

$$p_0$$

rather than treating local margins and envelope constants as independent;
5. compute the monodromy diagnostic for

$$DP_\eta(\phi_{\mathrm{cyc}})$$

on the section-anchored mesh. If the interval spectral enclosure has an unstable direction

$$|\lambda| > 1 + \delta_{\mathrm{mon}}$$

use the result to route the returned-sample proof to boundary trapping rather than residual-plus-sensitivity estimates.
6. derive returned-sample budgets. If the sample sensitivities

$$L_j^x, \qquad L_j^v$$

are too large to close the residual-plus-sensitivity route, use direct boundary trapping for

$$E_{j,\pm}^x, \qquad E_{j,\pm}^v$$

with strict sample slack. When sensitivities are small enough, the residual-plus-sensitivity inequalities

$$R_{j,\pm}^x + L_j^x r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}, \qquad R_{j,\pm}^v + L_j^v r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}$$

are sufficient. In either case, prove the finite checks that imply

$$P_\eta(\mathcal{K}_{x_*,\eta}^{\mathrm{cert}}) \subseteq \mathcal{K}_{x_*,\eta}^{\mathrm{cert}}$$

on that same domain.
7. verify the topology row:

$$u_{\mathrm{ret}}^{\mathrm{cert}} > 0$$

and certified branch-chart well-posedness for the dual-mollified vector field on the controlled continuation, including the origin-layer chart and certified fold-event atlas, so the certified topology proposition gives continuity and precompactness on

$$\mathcal{K}_{x_*,\eta}^{\mathrm{cert}}$$

Once the pre-ledger gate passes and the five audit rows are theorem-level, the finite-certificate invariant closure package supplies the self-map domain and the certified topology proposition supplies continuity and precompactness. The remaining Schauder step is then formally routine.

Schauder capstone

> **Conditional Theorem (Schauder Existence of a Dual-Mollified Collinear Breather).**
> Assume:
>
> 1. the theorem `Seed–to–Tame Full–Cycle Propagation`;
> 2. the finite-certificate invariant closure package, producing the nonempty closed convex tame self-map domain

$$> \mathcal{K}_{x_*,\eta}^{\mathrm{cert}}; >$$

> 3. and the certified topology proposition, giving continuity and precompactness of

$$> P_\eta >$$

> on that same certified domain.
>
> Then there exists

$$> \phi_\eta^* > \in > \mathcal{K}_{x_*,\eta}^{\mathrm{cert}} >$$

> such that

$$> P_\eta(\phi_\eta^*) = \phi_\eta^*. >$$

> The corresponding delayed trajectory is an exact bounded periodic origin-crossing two-body motion in the dual-mollified collinear model.

Proof.
`Seed-to-Tame Full-Cycle Propagation` supplies a nonempty tame class. The finite-certificate invariant closure package places that class inside a nonempty closed convex self-map domain

$$\mathcal{K}_{x_*,\eta}^{\mathrm{cert}}$$

Continuity and precompactness place the return image inside a compact subset of that same domain, while invariant-envelope closure prevents escape. Schauder therefore yields a fixed point of

$$P_\eta$$

on

$$\mathcal{K}_{x_*,\eta}^{\mathrm{cert}}$$

and by construction that fixed point is exactly a periodic returned history.
This capstone remains conditional on the finite closure audit and certified topology row. Without one nonempty closed convex tame self-map domain carrying propagation, continuity, precompactness, and invariance all at once, Schauder does not yet apply.

Seed history and tame-class nonemptiness

The remaining global nonvacuity issue is now easy to state. The invariant-envelope theorem is useful only if the section-side tame class is actually nonempty. The next theorem target is therefore to construct at least one explicit inbound history with controlled delayed geometry and then thicken it to a small nonempty tame neighborhood in the section topology.

The simplest seed is a strictly sub-field-speed affine inbound history on the right exterior branch. It is not meant to solve the full forward dynamics; its role is only to prove that the section-side tame constraints are simultaneously realizable.

> **Target Theorem (Seed History and Section-Tame Nonemptiness).**
> Fix

$$> x_* > 0, > \qquad > 0 < u_{\mathrm{seed}} < c_f, > \qquad > h \geq \frac{2x_*}{c_f - u_{\mathrm{seed}}}. >$$

> Then there exists an explicit inbound history

$$> \psi_{\mathrm{seed}} \in \Sigma_{x_*,\eta}^- \cap \mathcal{C}_{x_*,\eta} >$$

> such that:
>
> 1. the stored history lies in the position, speed, and acceleration envelope;
> 2. the stored partner root structure is finite and transversal;
> 3. there are no exact same-side self roots on the stored interval;
> 4. and a sufficiently small C^1 section neighborhood of

$$> \psi_{\mathrm{seed}} >$$

> remains inside a section-level tame subclass

$$> \mathcal{C}_{x_*,\eta}^{\mathrm{seed}} > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

This theorem is intentionally only a section-side nonemptiness statement. It does not yet say that the forward delayed flow preserves the same class for one full cycle. That stronger claim belongs to the later invariant-envelope theorem.

> **Proposition (Explicit affine inbound seed history).**
> Fix

$$> x_* > 0, > \qquad > 0 < u_{\mathrm{seed}} < c_f, > \qquad > h \geq \frac{2x_*}{c_f - u_{\mathrm{seed}}}. >$$

> Define

$$> \psi_{\mathrm{seed}}(\theta) > \equiv > x_* - u_{\mathrm{seed}}\theta, > \qquad > \theta \in [-h, 0]. >$$

> Then:
>
> 1. the section conditions hold:
>
> $$> \psi_{\text{seed}}(0) = x_*, > \qquad > \dot{\psi}_{\text{seed}}(0) = -u_{\text{seed}} < 0; >$$
>
> 2. the stored path is right exterior and monotone inbound:
>
> $$> x_* > \le > \psi_{\text{seed}}(\theta) > \le > x_* + u_{\text{seed}} h, > \qquad > \dot{\psi}_{\text{seed}}(\theta) = -u_{\text{seed}}, > \qquad > \ddot{\psi}_{\text{seed}}(\theta) = 0; >$$
>
> 3. there is exactly one partner root on the stored interval, located at
>
> $$> \theta_{p,\text{seed}} > = > -\frac{2x_*}{c_f - u_{\text{seed}}}, >$$
>
> and its Jacobian satisfies
>
> $$> J_{p,\text{seed}} > = > 1 - \frac{u_{\text{seed}}}{c_f} > 0; >$$
>
> 4. there are no exact same-side self roots on
>
> $$> [-h, 0). >$$
>
> Consequently, if
>
> $$> X_{\max} \ge x_* + u_{\text{seed}} h, > \qquad > U_{\max} \ge u_{\text{seed}}, > \qquad > A_{\max} > 0, > \qquad > \nu_{\text{seed}} \le 1 - \frac{u_{\text{seed}}}{c_f}, >$$
>
> then
>
> $$> \psi_{\text{seed}} \in \mathcal{C}_{x_*,\eta}, >$$
>
> and the stored-history transversality bounds hold with
>
> $$> |J_p| \ge \nu_{\text{seed}}, >$$
>
> while the self-root transversality condition is vacuous on the seed because there are no exact same-side self roots.

Proof.
The section anchoring and inbound sign are immediate from the definition of

$$\psi_{\text{seed}}$$

Since

$$\theta \in [-h, 0]$$

one has

$$\psi_{\text{seed}}(\theta) = x_* - u_{\text{seed}}\theta = x_* + u_{\text{seed}}|\theta|$$

so the stored path remains on the right exterior branch, decreases monotonically toward the section as

$$\theta \uparrow 0$$

and satisfies the displayed position, speed, and acceleration bounds.

For a partner root at the section time

$$\theta = 0$$

the delayed causal relation is

$$x_* + \psi_{\text{seed}}(\theta_p) = c_f(0 - \theta_p)$$

Writing

$$s = -\theta_p > 0$$

this becomes

$$x_* + \left(x_* + u_{\text{seed}} s\right) = c_f s$$

hence

$$2x_* = (c_f - u_{\text{seed}}) s$$

and therefore

$$s = \frac{2x_*}{c_f - u_{\text{seed}}}$$

The lower bound on

$$h$$

ensures that

$$\theta_{p,\text{seed}} = -s$$

lies inside

$$[-h, 0]$$

Since the seed velocity is constant,

$$J_{p,\text{seed}} = 1 + \frac{\dot{\psi}_{\text{seed}}(\theta_{p,\text{seed}})}{c_f} = 1 - \frac{u_{\text{seed}}}{c_f} > 0$$

Now consider exact same-side self roots on the stored interval. Such a root would satisfy

$$|\psi_{\text{seed}}(0) - \psi_{\text{seed}}(\theta_s)| = c_f(0 - \theta_s)$$

Again writing

$$s = -\theta_s > 0$$

the left-hand side equals

$$u_{\text{seed}} s$$

so the equation becomes

$$u_{\text{seed}} s = c_f s$$

Because

$$0 < u_{\text{seed}} < c_f$$

this has no solution for

$$s > 0$$

Hence there are no exact same-side self roots on

$$[-h, 0)$$

The final membership claim is then immediate from the displayed envelope inequalities.

> **Corollary (Nonempty section-level tame neighborhood).**
> Under the hypotheses of the proposition, there exists

$$> \varepsilon_{\text{seed}} > 0 >$$

> such that the set

$$> \mathcal{C}^{\text{seed}}_{x_*,\eta} > \equiv > \left\{ > \phi \in \mathcal{C}_{x_*,\eta} > \middle| > \phi(0) = x_*, > \quad > \dot{\phi}(0) \leq -\frac{u_{\text{seed}}}{2}, > \quad > \|\phi - \psi_{\text{seed}}\|_{C^1([-h,0])} \leq \varepsilon_{\text{seed}} > \right\} >$$

> is nonempty and consists of inbound section histories whose stored partner root persists uniquely and whose stored same-side exact self roots remain absent.

Proof sketch.
The set is nonempty because

$$\psi_{\mathrm{seed}} \in \mathcal{C}^{\mathrm{seed}}_{x_*,\eta}$$

for every

$$\varepsilon_{\mathrm{seed}} > 0$$

The seed has a strict sub-field-speed margin

$$\sigma_{\mathrm{seed}} \equiv c_f - u_{\mathrm{seed}} > 0$$

and a simple partner root with

$$J_{p,\mathrm{seed}} > 0$$

By continuity of the root equations and of the Jacobian factors under small C^1 perturbations of the stored history, these properties persist for all histories sufficiently close to

$$\psi_{\mathrm{seed}}$$

Likewise, the same-side self-root equation has a strict gap on the seed because

$$u_{\mathrm{seed}} < c_f$$

so exact same-side self roots cannot appear under a sufficiently small perturbation. Therefore a small enough neighborhood remains inside a section-level tame subclass.

This corollary is the first concrete nonvacuity statement for the theorem program. The remaining task is no longer to show that tame histories exist at all, but to propagate such a seed class through the full delayed cycle strongly enough that it becomes the nonempty class required by the invariant-envelope theorem.

Seed-to-tame propagation target

The seed construction resolves only the section-side nonvacuity issue. The next step is to promote a smaller neighborhood of seed histories to a genuinely nonempty tame class for the delayed flow itself. In other words, one wants to replace

$$\mathcal{C}^{\mathrm{seed}}_{x_*,\eta}$$

by a forward-propagated subclass on which the collapse, recapture, return, and root-control estimates all hold on one full cycle.

This is the precise bridge from the section-level seed construction to the invariant-envelope theorem.

> **Target Theorem (Seed-to-Tame Full-Cycle Propagation).**
> Assume the affine seed proposition and the nonempty section-level neighborhood corollary above. Then there exists a nonempty subclass

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} > \subseteq > \mathcal{C}^{\mathrm{seed}}_{x_*,\eta} > \subseteq > \mathcal{C}_{x_*,\eta} >$$

> such that:
>
> 1. every

$$> \psi \in \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} >$$

> admits a unique forward continuation through one full cycle;
> 2. the collapse-to-crossing control theorem applies uniformly on this class;
> 3. the explicit inner recapture regime and the unified trimmed-apocenter outer-turn criterion both apply uniformly on this class;
> 4. the turn-to-section return lemmas apply uniformly on this class;
> 5. and the returned history satisfies

$$> P_\eta(\psi) \in \mathcal{C}_{x_*,\eta}. >$$

> In particular,

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} \neq \varnothing, >$$

> the return map

$$> P_\eta >$$

> is well defined on a nonempty tame class, and the invariant-envelope theorem becomes nonvacuous.

This theorem is deliberately phrased as a propagation target rather than a proved proposition. The real remaining work is to show that the estimates already developed later in the note can be made uniform on a sufficiently small seed neighborhood rather than only along a single handpicked history.

Seed-propagation ladder

The intended proof order is:

1. **Local forward continuation from the seed neighborhood.**
   Show that a sufficiently small

$$C^1$$

neighborhood of

$$\psi_{\text{seed}}$$

evolves uniquely for at least one collapse phase while preserving the initial stored partner-root simplicity and same-side self-root exclusion.
2. **Seed-neighborhood collapse control.**
Prove that the collapse-to-crossing estimates can be made uniform on a smaller neighborhood

$$\mathcal{C}^{\text{seed,coll}}_{x_*,\eta} \subseteq \mathcal{C}^{\text{seed}}_{x_*,\eta}$$

3. **Seed-neighborhood realization of the inner regime.**
   Verify that the first crossing from this neighborhood lands uniformly in the Goldilocks window required by Proposition `Explicit short-window recapture regime`.
4. **Seed-neighborhood realization of the outer regime.**
   Verify that the same trajectories satisfy the hypotheses of the unified trimmed-apocenter outer-turn criterion on the final apocenter window.
5. **Returned-history reentry.**
   Show that the returned history segment lies back inside

$$\mathcal{C}_{x_*,\eta}$$

and, after shrinking once more if necessary, inside a forward-propagation subclass

$$\mathcal{C}^{\text{tame}}_{x_*,\eta}$$

The conceptual point is simple: the seed history does not need to solve the whole breather problem by itself. It only needs to provide one strict interior point of history space around which all the already-developed cycle estimates can be made uniform. Once such a neighborhood is propagated through one full cycle, the nonempty tame class required by the Schauder program is in hand.

> **Proposition (Local seed-neighborhood continuation with stored-root persistence).**
> Assume the affine seed proposition above, and strengthen the horizon choice slightly to

$$> h > \frac{2x_*}{c_f - u_{\text{seed}}}. >$$

> Then there exist constants

$$> 0 < \varepsilon_{\text{loc}} < \min\left\{\frac{u_{\text{seed}}}{2},\, c_f - u_{\text{seed}}\right\}, > \qquad > \nu_{\text{loc}} > 0, > \qquad > \tau_{\text{loc}} > 0, >$$

> and a nonempty subclass

$$> \mathcal{C}^{\text{seed,loc}}_{x_*,\eta} >\subseteq> \mathcal{C}^{\text{seed}}_{x_*,\eta} >$$

> such that for every

$$> \phi \in \mathcal{C}^{\text{seed,loc}}_{x_*,\eta} >$$

> one has:

> 1. **strict stored sub-field-speed bound:**

$$> |\dot{\phi}(\theta)| \le u_{\text{seed}} + \varepsilon_{\text{loc}} < c_f > \qquad > \text{for } \theta \in [-h, 0]; >$$

> 2. **absence of exact same-side self roots on the stored interval:**
> there is no

$$> \theta_s \in [-h, 0) >$$

> such that

$$> |\phi(0) - \phi(\theta_s)| = c_f(0 - \theta_s); >$$

> 3. **unique simple stored partner root:**
> there exists a unique

$$> \theta_p(\phi) \in [-h, 0) >$$

> satisfying

$$> \phi(0) + \phi(\theta_p) = c_f(0 - \theta_p), >$$

> and its Jacobian obeys

$$> J_p(\phi; \theta_p) \ge \nu_{\text{loc}} > 0; >$$

> 4. **short-time forward continuation:**
> if the dual-mollified vector field is locally Lipschitz on the stored-root branch determined above, then the history

$$> \phi >$$

> admits a unique forward continuation on

$$> [0, \tau_{\text{loc}}] >$$

> with continuous dependence on the initial history in the

$$> C^1([-h, 0]) >$$

> topology.
>
> In particular, the first item of the seed-propagation ladder holds on a nonempty neighborhood.

Proof sketch.
Because

$$h > \frac{2x_*}{c_f - u_{\text{seed}}}$$

there is a positive slack

$$\delta_h \equiv (c_f - u_{\text{seed}})h - 2x_* > 0$$

Choose

$$\varepsilon_{\text{loc}} > 0$$

small enough that

$$u_{\text{seed}} + \varepsilon_{\text{loc}} < c_f$$

and

$$\varepsilon_{\text{loc}} h < \delta_h$$

Let

$$\mathcal{C}^{\text{seed,loc}}_{x_*,\eta} \equiv \left\{ \phi \in \mathcal{C}^{\text{seed}}_{x_*,\eta} \mid \|\phi - \psi_{\text{seed}}\|_{C^1([-h,0])} \le \varepsilon_{\text{loc}} \right\}$$

This set is nonempty because it contains

$$\psi_{\mathrm{seed}}$$

For any

$$\phi \in \mathcal{C}^{\mathrm{seed,loc}}_{x_*,\eta}$$

the derivative bound gives

$$|\dot{\phi}(\theta)| \le u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}} < c_f$$

on

$$[-h, 0]$$

Now suppose a same-side self root

$$\theta_s < 0$$

existed. By the mean value theorem,

$$|\phi(0) - \phi(\theta_s)| \le (u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}})(0 - \theta_s) < c_f(0 - \theta_s)$$

contradicting the exact root equation. Hence no exact same-side self root exists on the stored interval.

For the partner root, define

$$F_\phi(\theta) \equiv \phi(0) + \phi(\theta) + c_f\theta$$

Then

$$F_\phi(0) = 2x_* > 0$$

while

$$F_\phi(-h) \le x_* + \big(x_* + u_{\mathrm{seed}}h + \varepsilon_{\mathrm{loc}}h\big) - c_f h = 2x_* - (c_f - u_{\mathrm{seed}} - \varepsilon_{\mathrm{loc}})h < 0$$

by the choice of

$$\varepsilon_{\mathrm{loc}}$$

Moreover,

$$F'_\phi(\theta) = \dot{\phi}(\theta) + c_f \ge c_f - (u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}}) > 0$$

so

$$F_\phi$$

is strictly increasing. Therefore it has a unique zero

$$\theta_p(\phi) \in [-h, 0)$$

At that root,

$$J_p(\phi;\theta_p) = 1 + \frac{\dot{\phi}(\theta_p)}{c_f} \ge 1 - \frac{u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}}}{c_f} \equiv \nu_{\mathrm{loc}} > 0$$

Finally, on this branch pattern the dual-mollified force law has one simple stored partner root and no exact same-side self roots on the initial history. Under the stated local Lipschitz hypothesis, standard local existence and continuous-dependence theory for functional differential equations yields a unique forward continuation on a short interval

$$[0, \tau_{\mathrm{loc}}]$$

This proves the proposition.

> **Proposition (Seed-neighborhood collapse control under a uniform inward bracket).**
> Let

$$> \mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta} > \subseteq > \mathcal{C}^{\mathrm{seed,loc}}_{x_*,\eta} >$$

> be a nonempty subclass. Assume there exist constants

$$> 0 < a_-^{\mathrm{seed}} \le a_+^{\mathrm{seed}}, > \qquad > \nu_{\mathrm{coll}} > 0, > \qquad > A_{\mathrm{coll}} > 0, >$$

> such that for every

$$> \psi \in \mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta} >$$

> the corresponding forward trajectory satisfies on its pre-crossing leg:
>
> 1. the two-sided inward acceleration bracket

$$> -a_+^{\mathrm{seed}} > \le > \ddot{x}(t;\psi) > \le > -a_-^{\mathrm{seed}} < 0; >$$

> 2. the acceleration ceiling

$$> |\ddot{x}(t;\psi)| \le A_{\mathrm{coll}}; >$$

> 3. and the active pre-crossing roots satisfy the uniform transversality bound

$$> |J_p| \ge \nu_{\mathrm{coll}}, > \qquad > |J_s| \ge \nu_{\mathrm{coll}}. >$$

> Then:
>
> 1. every

$$> \psi \in \mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta} >$$

> reaches the origin in finite time, with the uniform bound

$$> t_{\mathrm{cross}}(\psi) > \le > \sqrt{\frac{2x_*}{a_-^{\mathrm{seed}}}}; >$$

> 2. the pre-crossing tube bounds

$$> 0 \le x(t;\psi) \le X_{\mathrm{seed,max}}, > \qquad > |\dot{x}(t;\psi)| \le U_{\mathrm{seed,max}}, > \qquad > |\ddot{x}(t;\psi)| \le A_{\mathrm{coll}} >$$

> hold on the collapse leg for suitable class constants

$$> X_{\mathrm{seed,max}}, > \qquad > U_{\mathrm{seed,max}}; >$$

> 3. and if one chooses constants

$$> V^{\mathrm{seed}}_{\min}, > \qquad > V^{\mathrm{seed}}_{\max} >$$

> satisfying the uniform speed-window inequalities from Lemma 7 for every admissible section speed in

$$> \left[\frac{u_{\mathrm{seed}}}{2},\, u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}}\right], >$$

> then the collapse-to-crossing control theorem applies on

$$> \mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta}. >$$

Proof sketch.
For every

$$\psi \in \mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta}$$

the lower inward acceleration bound implies finite-time crossing by Lemma 6, yielding the displayed uniform bound on

$$t_{\mathrm{cross}}(\psi)$$

The two-sided acceleration bracket and the section-speed interval inherited from

$$\mathcal{C}^{\mathrm{seed,loc}}_{x_*,\eta}$$

allow Lemma 7 to be applied with

$$u_0 \in \left[\frac{u_{\mathrm{seed}}}{2},\, u_{\mathrm{seed}} + \varepsilon_{\mathrm{loc}}\right]$$

This produces a class-uniform crossing-speed window once

$$V_{\min}^{\mathrm{seed}}, \qquad V_{\max}^{\mathrm{seed}}$$

are chosen to dominate the resulting comparison bounds.

Finally, Lemma 8 upgrades the monotone inbound motion, the crossing-time bound, and the acceleration ceiling to the stated position-speed-acceleration tube bounds on the entire collapse leg. Together with the assumed Jacobian lower bound, these are exactly the ingredients required by the collapse-to-crossing theorem. Hence that theorem applies uniformly on

$$\mathcal{C}_{x_*,\eta}^{\mathrm{seed,coll}}$$

> **Proposition (Seed-neighborhood realization of the explicit inner recapture regime).**
> Let

$$> \mathcal{C}_{x_*,\eta}^{\mathrm{seed,in}} > \subseteq > \mathcal{C}_{x_*,\eta}^{\mathrm{seed,coll}} >$$

> be a nonempty subclass on which the collapse-to-crossing control theorem applies with uniform crossing-speed window

$$> V_{\min}^{\mathrm{seed}} > \leq > -\dot{x}(t_{\mathrm{cross}};\psi) > \leq > V_{\max}^{\mathrm{seed}} > \qquad > \text{for every } > \psi \in \mathcal{C}_{x_*,\eta}^{\mathrm{seed,in}}. >$$

> Assume further that the crossing histories issued from this class satisfy the admissible-crossing hypotheses with the same class constants entering Proposition `Explicit short-window recapture regime`, and that

$$> V_{\max}^{\mathrm{seed}} > < > \frac{\kappa\epsilon^2}{4\beta_{p,\max}\epsilon_c} > - > \frac{\overline{A}_s^{\rho}\,\epsilon_c}{2\beta_{p,\max}}, >$$

> together with

$$> \tau_\epsilon = \frac{\epsilon_c}{2\beta_{p,\max}} \leq \tau_1, > \qquad > \eta \leq \frac{\epsilon_c}{4c_f C_p}, > \qquad > \epsilon_c \leq \frac{\beta_{p,\max}^2}{c_f C_p}. >$$

> Then every first crossing launched from

$$> \mathcal{C}_{x_*,\eta}^{\mathrm{seed,in}} >$$

> lies in the explicit short-window recapture regime, and the corresponding post-crossing branch turns around on the class-uniform window

$$> [0,\tau_\epsilon]. >$$

Proof sketch.
By the collapse-to-crossing theorem, every

$$\psi \in \mathcal{C}_{x_*,\eta}^{\mathrm{seed,in}}$$

reaches a crossing history inside the admissible crossing subclass and with outgoing radial speed at most

$$V_{\max}^{\mathrm{seed}}$$

The displayed inequality is exactly the sufficient recapture condition from Proposition `Explicit short-window recapture regime`, with

$$V_{\max}$$

there replaced by the seed-neighborhood crossing-speed bound

$$V_{\max}^{\mathrm{seed}}$$

The three displayed small-window inequalities guarantee the same choice

$$\tau_\epsilon = \frac{\epsilon_c}{2\beta_{p,\max}}$$

is admissible. Therefore Proposition `Explicit short-window recapture regime` applies uniformly to every first crossing issued from

$$\mathcal{C}^{\mathrm{seed,in}}_{x_*,\eta}$$

yielding a class-uniform post-crossing turnaround on

$$[0,\tau_\epsilon]$$

This proposition closes the inner half of the seed-propagation program at the regime level: once the seed neighborhood is shrunk far enough that its collapse phase lands uniformly in the Goldilocks crossing window, the local post-crossing recapture mechanism becomes available without any additional pointwise tuning.

> **Proposition (Seed-neighborhood realization of the unified outer-turn regime).**
> Let

$$> \mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta} > \subseteq > \mathcal{C}^{\mathrm{seed,in}}_{x_*,\eta} >$$

> be a nonempty subclass such that the post-crossing recapture, return-half, and outer-branch delayed-geometry estimates developed later in the note hold uniformly on the corresponding trajectories. Assume in particular that for every

$$> \psi \in \mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta} >$$

> there is a trimmed apocenter window

$$> I_{\mathrm{deep}}(\psi) = [t_a(\psi) + \tau_{\mathrm{deep}},\ t_b(\psi)] >$$

> on which the unified trimmed-apocenter outer-turn criterion is applicable with the same class constants

$$> \underline{A}_p^{\mathrm{out}}, > \qquad > \tau_{\mathrm{deep}}, > \qquad > \sigma_{\mathrm{out}}, > \qquad > a_z^{\mathrm{out}}, > \qquad > a_{\mathrm{in,ref}}^{\mathrm{out}} > 0. >$$

> Assume moreover that the explicit inequalities

$$> z\big(t_{\mathrm{hinge}}^{\mathrm{out}}\big) > - > \frac{a_z^{\mathrm{out}}}{2} > \big(t_a + \tau_{\mathrm{deep}} - t_{\mathrm{hinge}}^{\mathrm{out}}\big)^2 > < 0, >$$

$$> \underline{A}_p^{\mathrm{out}} > - > \frac{\kappa\epsilon^2}{c_f^2\tau_{\mathrm{deep}}^2 + \epsilon_c^2} > - > \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{> \sigma_{\mathrm{out}}\epsilon_c^2} > \geq > a_{\mathrm{in,ref}}^{\mathrm{out}} > 0 >$$

> hold uniformly on that class.
>
> Then every trajectory issued from

$$> \mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta} >$$

> satisfies the outer-turn recapture mechanism uniformly: the trimmed-apocenter acceleration obeys

$$> \ddot{x}(t) \leq -a_{\mathrm{in,ref}}^{\mathrm{out}} < 0 > \qquad > \text{for } t \in I_{\mathrm{deep}}(\psi), >$$

> and, if

$$> |I_{\mathrm{deep}}(\psi)| > \geq > \frac{v_{\mathrm{deep}}}{a_{\mathrm{in,ref}}^{\mathrm{out}}}, >$$

> then a finite outer turn occurs on or just beyond the trimmed apocenter window for every member of the class.

Proof sketch.
By assumption, the same uniform constants entering the outer-turn layer apply to every trajectory launched from

$$\mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta}$$

The first displayed inequality is exactly the outbound-level exclusion condition from the

$$z$$

-descent layer, while the second displayed inequality is the refined trimmed-apocenter force margin. Therefore the unified trimmed-apocenter outer-turn criterion applies uniformly across the class.

It follows that every trajectory on the seed-out neighborhood has:

- outbound-level exclusion on the trimmed apocenter window,
- deep-past same-side root localization onto the pre-crossing inbound leg,

- the refined deep-past suppression bound,
- and the inward acceleration margin

$$\ddot{x}(t) \le -a^{\mathrm{out}}_{\mathrm{in,ref}} < 0$$

on

$$I_{\mathrm{deep}}(\psi)$$

The final turning claim is then exactly the conclusion of the unified trimmed-apocenter criterion once the window length dominates

$$\frac{v_{\mathrm{deep}}}{a^{\mathrm{out}}_{\mathrm{in,ref}}}$$

This proposition closes the outer half of the seed-propagation program at the regime level: once the seed neighborhood is small enough that the outer delayed geometry and trimmed-apocenter bounds are uniform, the outer-turn mechanism becomes class-uniform with no further history-by-history tuning.

> **Proposition (Returned-history reentry from uniform seed-cycle bounds).**
> Let

$$> \mathcal{C}^{\mathrm{seed,ret}}_{x_*,\eta} > \subseteq > \mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta} >$$

> be a nonempty subclass such that:
>
> 1. the collapse-to-crossing control theorem applies uniformly on this class;
> 2. the seed-neighborhood realization of the explicit inner recapture regime applies uniformly on this class;
> 3. the seed-neighborhood realization of the unified outer-turn regime applies uniformly on this class;
> 4. the turn-to-section return lemmas apply uniformly on this class with class constants

$$> X^{\mathrm{seed}}_{\mathrm{out,max}}, > \qquad > U^{\mathrm{seed}}_{\mathrm{sec,max}}, > \qquad > A^{\mathrm{seed}}_{\mathrm{cyc,max}}, > \qquad > T^{\mathrm{seed}}_{\mathrm{cyc,max}}; >$$

> 5. and the returned-history tameness estimates hold uniformly on this class.
>
> Assume moreover that the envelope parameters satisfy

$$> X_{\max} \ge \max\{x_*, X^{\mathrm{seed}}_{\mathrm{out,max}}\}, >$$

$$> U_{\max} \ge \max\{V^{\mathrm{seed}}_{\max}, U^{\mathrm{seed}}_{\mathrm{sec,max}}\}, >$$

$$> A_{\max} \ge A^{\mathrm{seed}}_{\mathrm{cyc,max}}, > \qquad > T_{\max} \ge T^{\mathrm{seed}}_{\mathrm{cyc,max}}, > \qquad > h \ge \frac{2X_{\max}}{c_f}. >$$

> Then

$$> P_\eta(\psi) \in \mathcal{C}_{x_*,\eta} > \qquad > \text{for every } > \psi \in \mathcal{C}^{\mathrm{seed,ret}}_{x_*,\eta}. >$$

> If, in addition, the same stored-history Jacobian, root-count, and local continuation bounds that define the seed-side propagation class persist on the returned segment, then after shrinking once more if necessary there exists a nonempty forward-propagation subclass

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} > \subseteq > \mathcal{C}^{\mathrm{seed,ret}}_{x_*,\eta} >$$

> such that

$$> P_\eta\big(\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}\big) > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

Proof sketch.
Items 1-3 provide the full dynamical cycle structure:

- finite-time first crossing with controlled speed,
- class-uniform inner turnaround after the first crossing,
- class-uniform outer turnaround on the trimmed apocenter window.

Item 4 then supplies the section-return consequences from the return-half layer:

$$0 \le x(t;\psi) \le X^{\mathrm{seed}}_{\mathrm{out,max}}, \qquad |\dot{x}(t;\psi)| \le U^{\mathrm{seed}}_{\mathrm{sec,max}}, \qquad |\ddot{x}(t;\psi)| \le A^{\mathrm{seed}}_{\mathrm{cyc,max}}$$

through the full cycle and up to the first inbound section return, together with the time bound

$$T(\psi) \le T^{\mathrm{seed}}_{\mathrm{cyc,max}}$$

The displayed envelope inequalities therefore imply that the returned history segment fits inside the convex envelope

$$\mathcal{C}_{x_*,\eta}$$

Hence

$$P_\eta(\psi) \in \mathcal{C}_{x_*,\eta}$$

for every

$$\psi \in \mathcal{C}^{\mathrm{seed,ret}}_{x_*,\eta}$$

If the returned segment also preserves the same stored-history root simplicity, Jacobian lower bounds, and local continuation control that defined the seed-side propagation class, then one may shrink the class once more to a nonempty subclass

$$\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}$$

on which those properties hold both before and after one full return. This gives exactly the forward-propagation tame class required by the invariant-envelope theorem.

This proposition closes the seed-propagation ladder at the nonvacuity level. The remaining logical step inside that ladder is to package the four seed-neighborhood propositions into a single nonempty tame-class theorem. The later Schauder route still requires the sampled tame-envelope certificate, coupled strict-slack arithmetic, and returned-sample preservation on the same domain.

> **Theorem (Nonempty tame class from seed propagation).**
> Assume:
>
> 1. the affine seed proposition and the nonempty section-level tame neighborhood corollary;
> 2. the proposition on local seed-neighborhood continuation with stored-root persistence;
> 3. the proposition on seed-neighborhood collapse control under a uniform inward bracket;
> 4. the proposition on seed-neighborhood realization of the explicit inner recapture regime;
> 5. the proposition on seed-neighborhood realization of the unified outer-turn regime;
> 6. and the proposition on returned-history reentry from uniform seed-cycle bounds.
>
> Then there exists a nonempty subclass

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} >\subseteq> \mathcal{C}_{x_*,\eta} >$$

> such that:
>
> 1. the return map

$$> P_\eta >$$

> is well defined on

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta}; >$$

> 2. the collapse-to-crossing, inner-recapture, outer-turn, and return-half bounds all apply uniformly on this class;
> 3. the returned histories satisfy

$$> P_\eta\big(\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}\big) >\subseteq> \mathcal{C}_{x_*,\eta}; >$$

> 4. and the invariant-envelope theorem is therefore nonvacuous on a genuine delayed history class.
>
> In particular, once the sampled certificate, coupled strict-slack inequalities, returned-sample preservation, continuity, and precompactness are verified on this same class, the Schauder route applies on a nonempty self-map domain.

Proof sketch.
The seed proposition and its neighborhood corollary provide a nonempty section-side class

$$\mathcal{C}^{\mathrm{seed}}_{x_*,\eta} \neq \varnothing$$

The local seed-neighborhood continuation proposition then produces a smaller nonempty subclass

$$\mathcal{C}^{\mathrm{seed,loc}}_{x_*,\eta}$$

on which the stored delayed geometry is simple and the forward flow is locally well defined. The collapse-control proposition shrinks again to a nonempty class

$$\mathcal{C}^{\mathrm{seed,coll}}_{x_*,\eta}$$

on which the collapse-to-crossing theorem applies uniformly.

The inner-regime proposition next yields a nonempty subclass

$$\mathcal{C}^{\mathrm{seed,in}}_{x_*,\eta}$$

whose first crossings lie uniformly in the explicit short-window recapture regime. The outer-regime proposition then yields a further nonempty subclass

$$\mathcal{C}^{\mathrm{seed,out}}_{x_*,\eta}$$

on which the trimmed-apocenter outer-turn mechanism applies uniformly. Finally, the returned-history reentry proposition produces a nonempty subclass

$$\mathcal{C}^{\mathrm{seed,ret}}_{x_*,\eta}$$

whose full-cycle images lie back in

$$\mathcal{C}_{x_*,\eta}$$

Choose

$$\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}$$

to be any nonempty forward-propagation subclass supplied by the last proposition. By construction, all cycle estimates invoked in the invariant-envelope synthesis hold uniformly on this class, and the return map is well defined there. The inclusion

$$P_\eta\big(\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}\big) \subseteq \mathcal{C}_{x_*,\eta}$$

is exactly the conclusion of the returned-history reentry step. Hence the invariant-envelope theorem is nonvacuous on a genuine delayed history class.

This theorem closes the seed-side nonvacuity gap in the global existence program. The note now contains:

- an explicit nonempty section-side seed,
- a propagation ladder from that seed to a nonempty tame class,
- explicit inner and outer recapture regimes,
- invariant-envelope closure on a certified closed convex history set, conditional on the sampled certificate and coupled strict-slack arithmetic,
- and the previously stated precompactness, continuity, and Schauder route.

The remaining work is therefore no longer to construct a nonempty delayed class. It is to verify the sampled tame-envelope certificate, verify the factorized coupled-corridor inequalities, and derive returned-sample budgets with strict sample slack, either through residual-plus-sensitivity control or direct boundary trapping, on that same class.

Collapse-to-crossing target

The next concrete theorem should attack Step 1 of the cycle ladder directly. Its role is to connect tame inbound section data to the already-audited local post-crossing recapture theorem.

> **Target Theorem (Collapse-to-Crossing Control).**
> Fix a dual-mollified tame inbound subclass

$$> \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

> Assume there exist class constants

$$> 0 < V_{\min} \le V_{\max}, > \qquad > A_{\max} > 0, > \qquad > \nu > 0, > \qquad > \tau_{\mathrm{cross,max}} > 0 >$$

> such that every trajectory launched from

$$> \psi \in \mathcal{C}^{\mathrm{tame}}_{x_*,\eta} >$$

> satisfies, on its inbound pre-crossing leg:
>
> 1. **finite-time crossing:**
>   there exists a first time

$$> t_{\mathrm{cross}}(\psi) \in (0, \tau_{\mathrm{cross,max}}] >$$

> with

$$> x(t_{\mathrm{cross}}(\psi); \psi) = 0; >$$

> 2. **crossing-speed window:**
>   the signed crossing speed obeys

$$> -V_{\max} \le \dot{x}(t_{\mathrm{cross}}(\psi); \psi) \le -V_{\min} < -c_f; >$$

> 3. **tube preservation before crossing:**

$$> 0 \le x(t; \psi) \le X_{\max}, > \qquad > |\dot{x}(t; \psi)| \le U_{\max}, > \qquad > |\ddot{x}(t; \psi)| \le A_{\max} >$$

> for

$$> 0 \le t \le t_{\mathrm{cross}}(\psi); >$$

> 4. **pre-crossing transversality:**
>   all active roots on the pre-crossing leg satisfy the same Jacobian lower bound

$$> |J_p| \ge \nu, > \qquad > |J_s| \ge \nu. >$$

> Then the translated crossing history belongs to a uniform admissible crossing subclass of the type used by the local origin-crossing recapture theorem. In particular, the post-crossing local recapture theorem applies immediately after the crossing.

This theorem is the missing hinge between the inbound section map and the local post-crossing analysis. It says: if tame inbound histories reach the origin in finite time with a controlled super-field-speed crossing and without losing the tube bounds, then the entire local recapture machine developed earlier becomes available automatically.

Collapse-to-crossing lemma ladder

The intended proof order for the collapse phase is:

1. **Inbound partner-dominance lemma.**
   Produce a class-uniform lower bound on inward partner acceleration on the pre-crossing leg.
2. **Finite-time crossing lemma.**
   Use the inward acceleration and inbound section sign to show that

$$x(t)$$

reaches zero in bounded time.
3. **Crossing-speed bounds.**
Estimate the speed gain accumulated before the crossing and show the resulting crossing speed lies inside

$$[V_{\min}, V_{\max}]$$

4. **Pre-crossing tube preservation.**
   Verify that position, speed, and acceleration remain inside the tame envelope up to

$$t_{\mathrm{cross}}$$

5. **Crossing-history admissibility.**
   Show that the translated history at

$$t_{\mathrm{cross}}$$

satisfies the sorting-map, Jacobian, and branch-count hypotheses needed by the local origin-crossing theorem.

Among these, the most delicate step is not finite-time arrival itself, but the quantitative crossing-speed window. The local post-crossing theorem needs the crossing to land in the Goldilocks regime:

- fast enough that the sorting map stays on the descending side and the caustic remains behind the trajectory,
- but not so fast that the integrated post-crossing partner impulse can no longer erase the outward radial speed.

So the collapse-to-crossing theorem is not merely a reachability statement. It is a controlled entry theorem into the local recapture regime.

Lemma 5: Inbound partner-dominance lower bound.
Assume the pre-crossing leg of a tame inbound trajectory satisfies:

- right exterior inbound geometry,

$$0 \le x(t) \le X_{\max}, \qquad \dot{x}(t) \le 0$$

- at least one inward exterior active partner branch for each

$$t \in [0, t_{\text{cross}}]$$

with

$$x(t) + x(t_p) > 0, \qquad 0 \le x(t_p) \le X_{\max}$$

- the speed bound

$$|\dot{x}(t)| \le U_{\max}$$

- and the partner Jacobian transversality bound

$$|J_p(t; t_p)| \ge \nu$$

on every active partner root.

Then the partner contribution to the inward acceleration obeys the class-uniform lower bound

$$A_p(t) \ge \underline{A}_p^{\text{in}} \equiv \frac{\kappa \epsilon^2}{(4X_{\max}^2 + \epsilon_c^2)\left(1 + \frac{U_{\max}}{c_f}\right)}$$

Equivalently, the partner acceleration satisfies

$$a_p(t) = -A_p(t) \le -\underline{A}_p^{\text{in}} < 0$$

Proof.
Along the retained inward exterior partner channel, the delayed source remains on the opposite side of the current right-hand particle, so each retained partner contribution points inward and has the form

$$a_p(t) = -A_p(t)$$

For any retained active partner root

$$t_p < t$$

the delayed separation is

$$r_p(t; t_p) = x(t) + x(t_p)$$

Because both the current and delayed positions lie in the tame position envelope,

$$0 \le x(t) \le X_{\max}, \qquad 0 \le x(t_p) \le X_{\max}$$

one has

$$r_p(t; t_p) \le 2X_{\max}$$

Hence the dual-mollified amplitude denominator satisfies

$$r_p(t; t_p)^2 + \epsilon_c^2 \le 4X_{\max}^2 + \epsilon_c^2$$

On the same branch the 1D partner line-of-action sign is

$$\hat{r}_p = +1$$

so

$$J_p(t;t_p) = 1 + \frac{\dot{x}(t_p)}{c_f}$$

Using the speed bound gives the crude upper estimate

$$|J_p(t;t_p)| \le 1 + \frac{U_{\max}}{c_f}$$

Therefore each retained active partner branch contributes at least

$$\kappa\epsilon^2 \frac{1}{(r_p(t;t_p)^2 + \epsilon_c^2)\,|J_p(t;t_p)|} \ge \frac{\kappa\epsilon^2}{(4X_{\max}^2 + \epsilon_c^2)\left(1 + \frac{U_{\max}}{c_f}\right)}$$

Since at least one inward exterior partner branch is active, summing over the retained active partner roots yields

$$A_p(t) \ge \underline{A}_p^{\text{in}}$$

which proves the lemma.

Lemma 6: Finite-time crossing under a net inward acceleration floor.
Assume the pre-crossing leg starts from the inbound section

$$x(0) = x_* > 0, \qquad \dot{x}(0) \le 0$$

and suppose there exists a constant

$$a_{\text{in}} > 0$$

such that the full pre-crossing acceleration obeys

$$\ddot{x}(t) \le -a_{\text{in}} \qquad \text{for } 0 \le t \le t_{\text{cross}}$$

Then the trajectory reaches the origin in finite time, with

$$t_{\text{cross}} \le \sqrt{\frac{2x_*}{a_{\text{in}}}}$$

In particular, a sufficient realization is

$$A_s^{\text{in}}(t) - A_s^{\text{out}}(t) \le \theta\, \underline{A}_p^{\text{in}} \qquad \text{for } 0 \le t \le t_{\text{cross}}$$

for some

$$0 \le \theta < 1$$

since then

$$\ddot{x}(t) = -A_p(t) - A_s^{\text{out}}(t) + A_s^{\text{in}}(t) \le -(1-\theta)\underline{A}_p^{\text{in}} \equiv -a_{\text{in}}$$

Proof.
Integrating the acceleration bound once gives

$$\dot{x}(t) \le \dot{x}(0) - a_{\text{in}}t \le -a_{\text{in}}t$$

because

$$\dot{x}(0) \le 0$$

Integrating again from

$$x(0) = x_*$$

yields

$$x(t) \le x_* + \dot{x}(0)t - \frac{a_{\text{in}}}{2}t^2 \le x_* - \frac{a_{\text{in}}}{2}t^2$$

Therefore

$$x(t) \le 0$$

whenever

$$t \ge \sqrt{\frac{2x_*}{a_{\text{in}}}}$$

By continuity of the trajectory, there is a first crossing time

$$t_{\text{cross}} \in \left(0, \sqrt{\frac{2x_*}{a_{\text{in}}}}\right]$$

such that

$$x(t_{\text{cross}}) = 0$$

This proves the lemma.

Lemma 7: Crossing-speed bounds under two-sided acceleration control.
Assume the pre-crossing leg starts from

$$x(0) = x_* > 0, \qquad \dot{x}(0) = -u_0, \qquad u_0 \ge 0$$

and suppose there exist positive constants

$$0 < a_- \le a_+$$

such that

$$-a_+ \le \ddot{x}(t) \le -a_- \qquad \text{for } 0 \le t \le t_{\text{cross}}$$

Define the quadratic comparison roots

$$\tau_\pm \equiv \frac{\sqrt{u_0^2 + 2a_\pm x_*} - u_0}{a_\pm}$$

Then the crossing time satisfies

$$\tau_+ \le t_{\text{cross}} \le \tau_-$$

and the crossing speed obeys

$$u_0 + a_- \tau_+ \le -\dot{x}(t_{\text{cross}}) \le u_0 + a_+ \tau_-$$

In particular, if the class constants satisfy

$$V_{\min} \le u_0 + a_- \tau_+, \qquad u_0 + a_+ \tau_- \le V_{\max}$$

then the crossing lands in the Goldilocks speed window

$$V_{\min} \le -\dot{x}(t_{\text{cross}}) \le V_{\max}$$

Proof.
Integrating the two-sided acceleration bound gives

$$-u_0 - a_+ t \le \dot{x}(t) \le -u_0 - a_- t$$

Integrating again yields the quadratic comparison bounds

$$x_* - u_0 t - \frac{a_+}{2} t^2 \le x(t) \le x_* - u_0 t - \frac{a_-}{2} t^2$$

Let

$$q_\pm(t) \equiv x_* - u_0 t - \frac{a_\pm}{2} t^2$$

Each $q_\pm$ has a unique positive root, namely

$$\tau_\pm = \frac{\sqrt{u_0^2 + 2a_\pm x_*} - u_0}{a_\pm}$$

Because

$$x(t) \le q_-(t)$$

the crossing must occur no later than the first time the upper comparison reaches zero:

$$t_{\text{cross}} \le \tau_-$$

Likewise,

$$x(t) \ge q_+(t)$$

so the trajectory cannot cross before the lower comparison reaches zero:

$$t_{\text{cross}} \ge \tau_+$$

Hence

$$\tau_+ \le t_{\text{cross}} \le \tau_-$$

Evaluating the velocity bounds at the crossing time gives

$$-\dot{x}(t_{\text{cross}}) \ge u_0 + a_- t_{\text{cross}} \ge u_0 + a_- \tau_+$$

and

$$-\dot{x}(t_{\text{cross}}) \le u_0 + a_+ t_{\text{cross}} \le u_0 + a_+ \tau_-$$

This proves the claimed crossing-speed bracket.

Lemma 8: Pre-crossing tube preservation from monotonicity and bounded acceleration.
Assume the pre-crossing leg starts from the inbound section

$$x(0) = x_*, \qquad \dot{x}(0) = -u_0, \qquad 0 \le u_0 \le U_{\text{in}}$$

with

$$0 < x_* \le X_{\max}$$

Assume moreover that on

$$[0, t_{\text{cross}}]$$

the trajectory satisfies:

- inward monotonicity,

$$\dot{x}(t) \le 0$$

- bounded crossing time,

$$t_{\text{cross}} \le \tau_{\text{cross,max}}$$

- and a uniform acceleration bound,

$$|\ddot{x}(t)| \le A_{\max}$$

If

$$U_{\text{in}} + A_{\max}\tau_{\text{cross,max}} \le U_{\max}$$

then the full pre-crossing tube bounds hold:

$$0 \le x(t) \le X_{\max}, \qquad |\dot{x}(t)| \le U_{\max}, \qquad |\ddot{x}(t)| \le A_{\max} \qquad \text{for } 0 \le t \le t_{\text{cross}}$$

Proof.
Because

$$\dot{x}(t) \leq 0 \qquad \text{for } 0 \leq t \leq t_{\text{cross}}$$

the position is nonincreasing on the pre-crossing leg. Since the first crossing occurs at

$$x(t_{\text{cross}}) = 0$$

one has

$$0 \leq x(t) \leq x(0) = x_* \leq X_{\max} \qquad \text{for } 0 \leq t \leq t_{\text{cross}}$$

For the velocity, the acceleration bound gives

$$|\dot{x}(t) - \dot{x}(0)| \leq \int_0^t |\ddot{x}(s)|\, ds \leq A_{\max} t \leq A_{\max} \tau_{\text{cross,max}}$$

Therefore

$$|\dot{x}(t)| \leq |\dot{x}(0)| + A_{\max}\tau_{\text{cross,max}} \leq U_{\text{in}} + A_{\max}\tau_{\text{cross,max}} \leq U_{\max}$$

The acceleration bound is already part of the hypotheses, so the full tube estimate follows.

This lemma does not by itself control Jacobian transversality or branch-count growth. It isolates the easier kinematic part of tube preservation: once monotone inbound motion, bounded crossing time, and bounded acceleration are known, the position-speed tube closes automatically up to the crossing.

Lemma 9: Crossing-history admissibility.
Let

$$\psi \in \mathcal{C}^{\text{tame}}_{x_*,\eta}$$

be an inbound history for which the pre-crossing leg satisfies Lemmas 6-8. Let

$$t_{\text{cross}} = t_{\text{cross}}(\psi)$$

denote the first crossing time and define the translated crossing history

$$\phi_{\text{cross}}(\theta) \equiv x(t_{\text{cross}} + \theta; \psi), \qquad \theta \in [-h, 0]$$

Assume, in addition, that the pre-crossing collapse provides:

- a crossing-speed window

$$V_{\min} \leq -\dot{x}(t_{\text{cross}}) \leq V_{\max}, \qquad V_{\min} > c_f$$

- a stored-past sorting-map geometry with class-uniform data

$$t_{\text{zero}} < t_{\text{hinge}} < 0, \qquad \delta_w(\phi_{\text{cross}}; \gamma_w) \geq \delta_{w,\min}$$

- sub-field-speed source transversality on the pre-hinge portion of the translated history,

$$\dot{\phi}_{\text{cross}}(\theta) \geq -c_f + \nu \qquad \text{for } \theta \in [-h, t_{\text{zero}}]$$

- and the same class-uniform acceleration and root-count bounds used in the definition of

$$\mathcal{K}^{\text{cross}}_{\eta,\epsilon_c}$$

If

$$\eta < \frac{\delta_{w,\min}}{2}, \qquad h \geq \frac{2X_{\max}}{c_f}$$

then

$$\phi_{\text{cross}} \in \mathcal{K}^{\text{cross}}_{\eta,\epsilon_c}$$

In particular, the local origin-crossing recapture theorem applies to the translated crossing history.

Proof.
By construction of the translated segment,

$$\phi_{\text{cross}}(0) = x(t_{\text{cross}}) = 0$$

The crossing-speed hypothesis gives

$$\dot{\phi}_{\text{cross}}(0) = \dot{x}(t_{\text{cross}}) \in [-V_{\max}, -V_{\min}]$$

with

$$V_{\min} > c_f$$

which is exactly the origin-crossing speed requirement of the admissible crossing subclass.

The stored-past sorting-map assumptions are likewise phrased directly on the translated history

$$\phi_{\text{cross}}$$

They therefore supply the required times

$$t_{\text{zero}} < t_{\text{hinge}} < 0$$

the interior compact-subinterval gap

$$\delta_w(\phi_{\text{cross}}; \gamma_w) \geq \delta_{w,\min}$$

and the sub-field-speed source transversality bound

$$\dot{\phi}_{\text{cross}}(\theta) \geq -c_f + \nu \qquad \text{for } \theta \in [-h, t_{\text{zero}}]$$

Because

$$\eta < \frac{\delta_{w,\min}}{2}$$

the shell-width condition required in the local post-crossing theorem is also satisfied.

The acceleration and root-count hypotheses are inherited by assumption from the pre-crossing tame tube and the translated-history bounds. Finally, the horizon choice

$$h \geq \frac{2X_{\max}}{c_f}$$

ensures that all causal delays compatible with the position envelope fit inside the stored history window.

Thus every defining condition of

$$\mathcal{K}^{\text{cross}}_{\eta,\epsilon_c}$$

holds for

$$\phi_{\text{cross}}$$

which proves the lemma.

This lemma isolates the exact last handoff in the proof architecture. The collapse phase does not itself prove local recapture; it only has to deliver the trajectory into the admissible crossing subclass where the already-established post-crossing theorem takes over.

Pre-crossing caustic-transit target

The collapse-to-crossing ladder now has its kinematic part in place. The remaining hard issue is delayed geometry, but it must be framed correctly. Because the inbound speed rises from a sub-field-speed regime to a crossing speed strictly larger than c_f, the trajectory must pass through the hinge

$$\dot{x} = -c_f$$

At that hinge, the self-hit sorting map necessarily creates a self root, and the corresponding self Jacobian reaches

$$J_s = 0$$

at the instant of birth. So the correct theorem target is not global self-root transversality on the whole pre-crossing leg. The correct target is a **controlled caustic transit**:

- the partner branch stays safely away from the caustic,
- the self branch is born at the hinge,
- the resulting inward impulse remains bounded in the dual-mollified model,
- and the self Jacobian recovers to a strictly positive lower bound before the origin crossing.

> **Target Theorem (Pre-crossing Caustic Transit and Recovery).**
> Fix a dual-mollified tame inbound subclass

$$> \mathcal{C}^{\text{tame}}_{x_*,\eta} > \subseteq > \mathcal{C}_{x_*,\eta}. >$$

> Assume the pre-crossing leg from every

$$> \psi \in \mathcal{C}^{\text{tame}}_{x_*,\eta} >$$

> satisfies the kinematic hypotheses of Lemmas 5-8. Suppose moreover that there exist class constants

$$> \nu_p > 0, > \qquad > \nu_s > 0, > \qquad > N_{p,\max} \geq 1, > \qquad > N_{s,\max} \geq 1, > \qquad > \Delta V_{\text{cau},\max} < \infty, > \qquad > \delta_{w,\min} > 0 >$$

> such that on the entire pre-crossing leg:
>
> 1. **partner-root safety:** the active partner branch persists continuously and remains transversal,

$$> |J_p| \geq \nu_p; >$$

> 2. **hinge birth of the self branch:** exactly one principal self root is born when

$$> \dot{x} = -c_f, >$$

> and no uncontrolled branch proliferation occurs before the crossing;
> 3. **bounded caustic impulse:** the dual-mollified inward velocity gain contributed during the self-root birth and immediate caustic transit is bounded by

$$> \Delta V_{\text{cau},\max}; >$$

> 4. **post-hinge Jacobian recovery:** by the time of the origin crossing, the active self branch has receded into the sub-field-speed past strongly enough that

$$> |J_s| \geq \nu_s; >$$

> 5. **sorting-gap inheritance:** the translated crossing history satisfies

$$> \delta_w(\phi_{\text{cross}}; \gamma_w) \geq \delta_{w,\min}. >$$

> Then the collapse phase preserves the full delayed geometry needed by Lemma 9, and the translated crossing history lies in the admissible crossing subclass

$$> \mathcal{K}^{\text{cross}}_{\eta,\epsilon_c}. >$$

This is the genuine delayed-geometry bottleneck behind the collapse phase. Lemmas 5-8 reduce the kinematic part of the infall to ordinary differential inequalities; the theorem above is what must control the compulsory self-root birth and show that it helps the collapse without destroying the Goldilocks crossing window.

Pre-crossing propagation ladder

The intended proof order for this delayed-geometry step is:

1. **Partner-root safety lemma.**
   Show that the active partner branch persists continuously along the inbound leg and never reaches a caustic before the crossing.
2. **Hinge-birth lemma.**
   Prove that exactly one principal self root is born when the trajectory passes through

$$\dot{x} = -c_f$$

3. **Caustic-transit impulse bound.**
   Show that the dual-mollified self-root birth contributes only a bounded inward velocity kick

$$\Delta V_{\mathrm{cau}} \leq \Delta V_{\mathrm{cau,max}}$$

and therefore does not destroy the Goldilocks crossing-speed upper bound.
4. **Root-count bound.**
Show that the total number of active branches on the inbound leg remains bounded by class constants

$$N_{p,\max}, \quad N_{s,\max}$$

5. **Sorting-gap inheritance and Jacobian recovery.**
   Prove that by the time of the origin crossing the active self root has moved far enough into the sub-field-speed past that

$$|J_s| \geq \nu_s$$

and the translated crossing history inherits the compact-subinterval sorting gap needed by the local post-crossing theorem.

The first item is a partner-branch regularity statement. The second and third items explicitly embrace the self-root caustic instead of assuming it away. The fifth is the exact handoff needed to pass from the inbound collapse theorem to the local origin-crossing recapture theorem.

Lemma 10: Partner-root safety on the inbound leg.
Assume the pre-crossing leg satisfies:

- right exterior inbound geometry

$$x(t) \geq 0, \qquad \dot{x}(t) \leq 0$$

- a unique hinge time

$$t_{\mathrm{hinge}} \in (0, t_{\mathrm{cross}})$$

with

$$\dot{x}(t_{\mathrm{hinge}}) = -c_f$$

- and strict post-hinge super-field-speed infall

$$\dot{x}(t) < -c_f \qquad \text{for } t \in (t_{\mathrm{hinge}}, t_{\mathrm{cross}}]$$

Define

$$w(t) \equiv x(t) + c_f t, \qquad y(t) \equiv c_f t - x(t)$$

Then the active partner root on the inbound leg is selected by

$$w(t_p) = y(t), \qquad t_p < t$$

Moreover:

1. the partner branch persists continuously on

$$[0, t_{\mathrm{cross}}]$$

2. it remains strictly on the ascending side of the sorting map,

$$t_p(t) < t_{\mathrm{hinge}}$$

3. and therefore the partner Jacobian stays strictly positive:

$$J_p(t; t_p) = \frac{\dot{w}(t_p)}{c_f} > 0$$

Proof.
On the inbound leg,

$$\dot{y}(t) = c_f - \dot{x}(t) \geq c_f > 0$$

so $y(t)$ is strictly increasing. Also,

$$\dot{w}(t) = \dot{x}(t) + c_f$$

which is positive before the hinge, zero at the hinge, and negative after the hinge. Hence

$$w(t)$$

has a strict maximum at

$$t = t_{\mathrm{hinge}}$$

It therefore suffices to show that

$$y(t) < w(t_{\mathrm{hinge}}) \qquad \text{for every } t \in [0, t_{\mathrm{cross}}]$$

Since y is increasing, it is enough to check this at the crossing time. Using

$$x(t_{\mathrm{cross}}) = 0$$

gives

$$y(t_{\mathrm{cross}}) = c_f t_{\mathrm{cross}}$$

On the other hand,

$$w(t_{\mathrm{hinge}}) = x(t_{\mathrm{hinge}}) + c_f t_{\mathrm{hinge}}$$

By the mean value theorem and the strict post-hinge inequality

$$\dot{x} < -c_f \qquad \text{on } (t_{\mathrm{hinge}}, t_{\mathrm{cross}}]$$

one has

$$x(t_{\mathrm{cross}}) - x(t_{\mathrm{hinge}}) < -c_f\,(t_{\mathrm{cross}} - t_{\mathrm{hinge}})$$

Since

$$x(t_{\mathrm{cross}}) = 0$$

this rearranges to

$$x(t_{\mathrm{hinge}}) > c_f\,(t_{\mathrm{cross}} - t_{\mathrm{hinge}})$$

hence

$$w(t_{\mathrm{hinge}}) = x(t_{\mathrm{hinge}}) + c_f t_{\mathrm{hinge}} > c_f t_{\mathrm{cross}} = y(t_{\mathrm{cross}})$$

Therefore

$$y(t) < w(t_{\mathrm{hinge}})$$

for all pre-crossing times, so the partner branch never reaches the hinge maximum.

Because

$$w$$

is strictly increasing on the ascending side, there is a unique solution

$$t_p(t) < t_{\mathrm{hinge}}$$

to

$$w(t_p) = y(t)$$

for each

$$t \in [0, t_{\mathrm{cross}}]$$

This gives continuous persistence of the partner branch. Finally, on that ascending side,

$$\dot{w}(t_p) > 0$$

so

$$J_p(t;t_p) = \frac{\dot{w}(t_p)}{c_f} > 0$$

Thus the partner branch remains safe from the caustic throughout the infall.

Lemma 11: Hinge birth and uniqueness of the principal self root.
Assume the pre-crossing leg satisfies a strict inward acceleration floor

$$\ddot{x}(t) \le -a_- < 0 \qquad \text{for } 0 \le t \le t_{\text{cross}}$$

and define

$$w(t) \equiv x(t) + c_f t$$

Let

$$t_{\text{hinge}}$$

be the unique time at which

$$\dot{x}(t_{\text{hinge}}) = -c_f$$

Then:

1. w is strictly concave on the pre-crossing interval,
2. w has a unique global maximum at

$$t = t_{\text{hinge}}$$

3. for each

$$t \in (t_{\text{hinge}}, t_{\text{cross}}]$$

there exists a unique self root

$$t_s(t) < t_{\text{hinge}}$$

satisfying

$$w(t_s) = w(t)$$

4. and this self branch is born at the hinge with

$$\lim_{t \downarrow t_{\text{hinge}}} t_s(t) = t_{\text{hinge}}, \qquad J_s(t;t_s) \to 0^+$$

Proof.
Differentiate twice:

$$\dot{w}(t) = \dot{x}(t) + c_f, \qquad \ddot{w}(t) = \ddot{x}(t)$$

By hypothesis,

$$\ddot{w}(t) \le -a_- < 0$$

so

$$w$$

is strictly concave on the full pre-crossing interval. Hence

$$\dot{w}$$

is strictly decreasing and can vanish at most once. Since

$$\dot{w}(t_{\text{hinge}}) = \dot{x}(t_{\text{hinge}}) + c_f = 0$$

the hinge is the unique critical point of

$$w$$

and therefore its unique global maximum.

For

$$t < t_{\mathrm{hinge}}$$

the function

$$w$$

is strictly increasing, so no earlier time can satisfy

$$w(t_s) = w(t)$$

with

$$t_s < t$$

For

$$t > t_{\mathrm{hinge}}$$

strict concavity implies that

$$w(t) < w(t_{\mathrm{hinge}})$$

and because the ascending branch is strictly increasing up to the hinge, there is a unique

$$t_s(t) < t_{\mathrm{hinge}}$$

such that

$$w(t_s) = w(t)$$

This is the unique principal self root on the pre-crossing leg.

As

$$t \downarrow t_{\mathrm{hinge}}$$

continuity and uniqueness force

$$t_s(t) \uparrow t_{\mathrm{hinge}}$$

On the relevant outer-memory branch,

$$J_s(t; t_s) = 1 + \frac{\dot{x}(t_s)}{c_f} = \frac{\dot{w}(t_s)}{c_f}$$

Since

$$t_s(t) < t_{\mathrm{hinge}}$$

lies on the ascending side,

$$\dot{w}(t_s) > 0$$

so

$$J_s(t; t_s) > 0$$

But as

$$t_s(t) \uparrow t_{\mathrm{hinge}}$$

one has

$$\dot{w}(t_s) \downarrow \dot{w}(t_{\text{hinge}}) = 0$$

hence

$$J_s(t; t_s) \to 0^+$$

So the self branch is born exactly at the hinge and is unique.

Lemma 12: Bounded caustic-transit impulse in the dual-mollified model.
Fix a hinge-centered time window

$$I_{\text{cau}} \equiv [t_{\text{hinge}} - \tau_{\text{cau}},\, t_{\text{hinge}} + \tau_{\text{cau}}]$$

on which the dual-mollified self interaction is evaluated through the regularized time-integral representation with:

- shell mollifier

$$\delta_\eta$$

bounded by

$$\|\delta_\eta\|_\infty < \infty$$

- core mollifier

$$\epsilon_c > 0$$

- and memory horizon

$$h > 0$$

Assume the self integral on this window is taken only over the stored history

$$t_0 \in [t-h, t]$$

Then the total inward velocity kick contributed by the self branch across the hinge window is finite and obeys the crude bound

$$\Delta V_{\text{cau}} \le \frac{2\kappa\epsilon^2\, h\, \tau_{\text{cau}} \|\delta_\eta\|_\infty}{\epsilon_c^2} \equiv \Delta V_{\text{cau,max}}$$

In particular, the self-root birth at

$$J_s = 0$$

does not produce an infinite velocity jump in the dual-mollified model.

Proof.
On the hinge window, evaluate the self contribution in the regularized integral form rather than the branch-sum form. By construction of the dual mollification, the self acceleration satisfies the absolute bound

$$|a_s(t)| \le \kappa\epsilon^2 \int_{t-h}^{t} \frac{\delta_\eta(\cdots)}{|x(t) - x(t_0)|^2 + \epsilon_c^2}\, dt_0$$

Because

$$|x(t) - x(t_0)|^2 + \epsilon_c^2 \ge \epsilon_c^2$$

and

$$\delta_\eta(\cdots) \le \|\delta_\eta\|_\infty$$

one obtains

$$|a_s(t)| \le \kappa\epsilon^2 \int_{t-h}^{t} \frac{\|\delta_\eta\|_\infty}{\epsilon_c^2}\, dt_0 = \frac{\kappa\epsilon^2\, h\, \|\delta_\eta\|_\infty}{\epsilon_c^2}$$

Integrating over the hinge window gives

$$\Delta V_{\text{cau}} \le \int_{I_{\text{cau}}} |a_s(t)|\,dt \le \frac{\kappa\epsilon^2\,h\,\|\delta_\eta\|_\infty}{\epsilon_c^2} \cdot |I_{\text{cau}}|$$

Since

$$|I_{\text{cau}}| = 2\tau_{\text{cau}}$$

this yields

$$\Delta V_{\text{cau}} \le \frac{2\kappa\epsilon^2\,h\,\tau_{\text{cau}}\|\delta_\eta\|_\infty}{\epsilon_c^2} \equiv \Delta V_{\text{cau,max}}$$

Thus the caustic transit contributes a finite inward impulse in the dual-mollified model.

Lemma 13: Post-hinge Jacobian recovery and sorting-gap inheritance.
Assume the pre-crossing leg satisfies Lemmas 10-12, and let

$$t_{\text{cross}}$$

denote the first origin crossing. Define

$$w(t) \equiv x(t) + c_f t$$

on the unshifted inbound leg, and let

$$t_{\text{zero}} < t_{\text{hinge}}$$

be the unique time satisfying

$$w(t_{\text{zero}}) = w(t_{\text{cross}})$$

Assume moreover that on the ascending side of the sorting map one has the lower derivative bound

$$\dot{w}(t) \ge \nu_s > 0 \qquad \text{for } t \in [t_{\text{zero}}, t_{\text{hinge}} - \gamma_w]$$

for some

$$\gamma_w > 0$$

Then:

1. the active self root at the crossing is exactly

$$t_s(t_{\text{cross}}) = t_{\text{zero}}$$

2. the recovered self Jacobian at the crossing satisfies

$$J_s(t_{\text{cross}}; t_{\text{zero}}) = \frac{\dot{w}(t_{\text{zero}})}{c_f} \ge \frac{\nu_s}{c_f}$$

3. and the translated crossing history

$$\phi_{\text{cross}}(\theta) = x(t_{\text{cross}} + \theta)$$

inherits a compact-subinterval sorting gap:
for every

$$0 < \gamma < \min\{t_{\text{hinge}} - t_{\text{zero}}, -t_{\text{hinge}}\}$$

the translated sorting function

$$\widetilde{w}(\theta) \equiv \phi_{\text{cross}}(\theta) + c_f\theta = w(t_{\text{cross}} + \theta) - w(t_{\text{cross}})$$

satisfies

$$\widetilde{w}(\theta) > 0 \qquad \text{for } \theta \in (t_{\text{zero}} - t_{\text{cross}}, 0)$$

and therefore

$$\delta_w(\phi_{\text{cross}};\gamma) > 0$$

Proof.
By Lemma 11, for each

$$t \in (t_{\text{hinge}}, t_{\text{cross}}]$$

there exists a unique principal self root

$$t_s(t) < t_{\text{hinge}}$$

such that

$$w(t_s) = w(t)$$

Evaluating this at the crossing time and using the defining property of

$$t_{\text{zero}}$$

shows

$$t_s(t_{\text{cross}}) = t_{\text{zero}}$$

On the relevant outer-memory branch,

$$J_s(t_{\text{cross}}; t_{\text{zero}}) = 1 + \frac{\dot{x}(t_{\text{zero}})}{c_f} = \frac{\dot{w}(t_{\text{zero}})}{c_f}$$

The assumed lower derivative bound on the ascending side therefore yields

$$J_s(t_{\text{cross}}; t_{\text{zero}}) \geq \frac{\nu_s}{c_f} > 0$$

which is the desired post-hinge Jacobian recovery.

For the sorting-gap inheritance, define

$$\widetilde{w}(\theta) = w(t_{\text{cross}} + \theta) - w(t_{\text{cross}})$$

Then

$$\widetilde{w}(0) = 0$$

and

$$\widetilde{w}(t_{\text{zero}} - t_{\text{cross}}) = 0$$

Because

$$w(t) < w(t_{\text{hinge}})$$

for all

$$t \neq t_{\text{hinge}}$$

and because the level

$$w(t_{\text{cross}})$$

intersects the strictly concave graph of

$$w$$

exactly at

$$t_{\text{zero}} \quad \text{and} \quad t_{\text{cross}}$$

it follows that

$$\widetilde{w}(\theta) > 0$$

for every

$$\theta \in (t_{\text{zero}} - t_{\text{cross}}, 0)$$

Restricting to any compact subinterval away from the two zeros, continuity yields a positive minimum, which is precisely the required compact-subinterval sorting gap

$$\delta_w(\phi_{\text{cross}}; \gamma) > 0$$

This proves the lemma.

Turn-to-section return target

Once the local post-crossing theorem has produced a turning point, the remaining analytic burden is to close the excursion back to the inbound section. This is the return-half analogue of the collapse-to-crossing theorem.

> **Target Theorem (Turn-to-Section Return).**
> Fix a dual-mollified tame crossing subclass

$$> \mathcal{K}^{\text{cross}}_{\eta,\epsilon_c} >$$

> and suppose the local origin-crossing recapture theorem produces, for every admissible crossing history, a turning time

$$> t_{\text{turn}} \le \tau_{\text{env}} >$$

> with

$$> \dot{\rho}(t_{\text{turn}}) = 0. >$$

> Assume there exist class constants

$$> X_{\max}, > \qquad > U_{\max}, > \qquad > A_{\max}, > \qquad > T_{\text{ret},\max} > 0 >$$

> such that for every post-turn branch:
>
> 1. **inward return after the turn:**
>   the trajectory re-enters toward the origin after

$$> t_{\text{turn}}, >$$

> crosses the center a second time, reaches the reflected section state

$$> x = x_*, > \qquad > \dot{x} > 0, >$$

> on the right exterior branch, and after one further outer turn returns to the section

$$> x = x_* >$$

> as an inbound branch;
> 2. **bounded excursion on the return half:**

$$> 0 \le x(t) \le X_{\max} > \qquad > \text{for } t_{\text{turn}} \le t \le T(\psi); >$$

> 3. **bounded speed and acceleration on the return half:**

$$> |\dot{x}(t)| \le U_{\max}, > \qquad > |\ddot{x}(t)| \le A_{\max} > \qquad > \text{for } t_{\text{turn}} \le t \le T(\psi); >$$

> 4. **bounded return time:**
>   the first inbound section return satisfies

$$> 0 < T(\psi) - t_{\text{turn}} \le T_{\text{ret},\max}; >$$

> 5. **bounded inbound section speed:**

$$> -\dot{x}(T(\psi)) \le U_{\max}; >$$

> 6. **returned-history control:**
>   the translated segment

$$> x_{T(\psi)} >$$

> satisfies the same acceleration, Jacobian, and branch-count bounds required by the tame envelope.
>
> Then the post-turn branch closes the full cycle back to the inbound section, and the return map

$$> P_\eta >$$

> is well defined on the corresponding tame class with

$$> P_\eta(\psi) \in \mathcal{C}_{x_*,\eta}. >$$

This theorem is the last missing dynamical segment of the cycle. The collapse-to-crossing theorem feeds the local recapture theorem; the turn-to-section return theorem feeds the invariant-envelope and Schauder steps.

Turn-to-section return ladder

The intended proof order for the return half is:

1. **Post-turn inward-drive lemma.**
   Show that after the turning time the net delayed force drives the trajectory back toward the origin strongly enough to prevent outward re-escape.
2. **Second-crossing lemma.**
   Prove that the trajectory crosses the origin a second time in finite time after the turn.
3. **Reflected-section lemma.**
   Show that after the second crossing, the trajectory reaches

$$x = x_*$$

on the right exterior branch with

$$\dot{x} > 0$$

4. **Outer-turn closure lemma.**
   Show that after one further outer turn on the right branch, the trajectory returns to

$$x = x_*$$

again on an inbound branch.
5. **Return-speed bound.**
Estimate the inbound speed at the section and show

$$-\dot{x}(T(\psi)) \le U_{\max}$$

6. **Returned-history tameness.**
   Prove that the translated return segment inherits the tame acceleration, Jacobian, and branch-count bounds.

The first, fourth, and fifth items are the real analytic bottlenecks on the return half. The middle two are reachability statements once the sign of the post-turn drive is controlled.

Lemma 14: Post-turn inward-drive lemma.
Let

$$t_{\text{turn}}$$

be a turning time produced by the local origin-crossing recapture theorem, so that

$$\rho(t_{\text{turn}}) = \rho_{\max} > 0, \qquad \dot{\rho}(t_{\text{turn}}) = 0$$

Assume there exists a post-turn window

$$[t_{\text{turn}},\, t_{\text{turn}} + \tau_{\text{ret}}]$$

and a constant

$$a_{\mathrm{ret}} > 0$$

such that on that window the radial acceleration satisfies

$$\ddot{\rho}(t) \leq -a_{\mathrm{ret}}$$

Then:

1. the trajectory cannot re-escape outward on that window,
2. the radial speed becomes strictly inward immediately after the turn,

$$\dot{\rho}(t) \leq -a_{\mathrm{ret}}(t - t_{\mathrm{turn}}) \qquad \text{for } t \in [t_{\mathrm{turn}},\, t_{\mathrm{turn}} + \tau_{\mathrm{ret}}]$$

3. and the radius decreases monotonically there, with

$$\rho(t) \leq \rho_{\max} - \frac{a_{\mathrm{ret}}}{2}(t - t_{\mathrm{turn}})^2$$

In particular, a sufficient realization is

$$A_p^{\rho}(t) - A_s^{\rho}(t) \geq a_{\mathrm{ret}} > 0 \qquad \text{for } t \in [t_{\mathrm{turn}},\, t_{\mathrm{turn}} + \tau_{\mathrm{ret}}]$$

because then

$$\ddot{\rho}(t) \leq -a_{\mathrm{ret}}$$

Proof.
Integrating the radial acceleration bound from the turning time gives

$$\dot{\rho}(t) = \dot{\rho}(t_{\mathrm{turn}}) + \int_{t_{\mathrm{turn}}}^{t} \ddot{\rho}(s)\, ds \leq 0 - a_{\mathrm{ret}}(t - t_{\mathrm{turn}})$$

which proves the velocity estimate and shows that

$$\dot{\rho}(t) < 0 \qquad \text{for } t > t_{\mathrm{turn}}$$

Thus the trajectory moves strictly inward immediately after the turn and cannot re-escape outward on the stated window.

Integrating once more yields

$$\rho(t) = \rho(t_{\mathrm{turn}}) + \int_{t_{\mathrm{turn}}}^{t} \dot{\rho}(s)\, ds \leq \rho_{\max} - \frac{a_{\mathrm{ret}}}{2}(t - t_{\mathrm{turn}})^2$$

which proves the monotone decrease of the radius on the post-turn window.

Lemma 15: Finite-time second crossing after the turn.
Assume the hypotheses of Lemma 14 and suppose, in addition, that the return window is long enough to satisfy

$$\tau_{\mathrm{ret}} \geq \sqrt{\frac{2\rho_{\max}}{a_{\mathrm{ret}}}}$$

Then the trajectory reaches the center again in finite time: there exists

$$t_{\mathrm{cross}}^{(2)} \in \left(t_{\mathrm{turn}},\, t_{\mathrm{turn}} + \sqrt{\frac{2\rho_{\max}}{a_{\mathrm{ret}}}}\right]$$

such that

$$\rho\left(t_{\mathrm{cross}}^{(2)}\right) = 0$$

Equivalently, in signed coordinates the trajectory crosses the origin a second time by that time.

Proof.
Lemma 14 gives the comparison bound

$$\rho(t) \leq \rho_{\max} - \frac{a_{\mathrm{ret}}}{2}(t - t_{\mathrm{turn}})^2$$

for

$$t \in [t_{\text{turn}}, t_{\text{turn}} + \tau_{\text{ret}}]$$

Therefore

$$\rho(t) \le 0$$

whenever

$$t - t_{\text{turn}} \ge \sqrt{\frac{2\rho_{\max}}{a_{\text{ret}}}}$$

Because the assumed window length satisfies

$$\tau_{\text{ret}} \ge \sqrt{\frac{2\rho_{\max}}{a_{\text{ret}}}}$$

the comparison reaches zero before the end of the return window. Since

$$\rho(t_{\text{turn}}) = \rho_{\max} > 0$$

and

$$\rho$$

is continuous, there exists a first time

$$t^{(2)}_{\text{cross}} \in \left(t_{\text{turn}}, t_{\text{turn}} + \sqrt{\frac{2\rho_{\max}}{a_{\text{ret}}}}\right]$$

for which

$$\rho\left(t^{(2)}_{\text{cross}}\right) = 0$$

This proves the lemma.

Lemma 16: Return to the reflected section state after the second crossing.
Assume the hypotheses of Lemma 15 and let

$$t^{(2)}_{\text{cross}}$$

denote the second origin crossing. Assume, in addition, that there exists a post-second-crossing window

$$[t^{(2)}_{\text{cross}}, t^{(2)}_{\text{cross}} + \tau_*]$$

on which:

- the trajectory lies on the right exterior branch,

$$x(t) \ge 0$$

- the motion is outward,

$$\dot{x}(t) \ge v_* > 0$$

- and the position remains bounded above by the global excursion envelope,

$$x(t) \le X_{\max}$$

If

$$\tau_* \ge \frac{x_*}{v_*}$$

then there exists a first time

$$t_* \in \left[t^{(2)}_{\text{cross}}, t^{(2)}_{\text{cross}} + \frac{x_*}{v_*}\right]$$

such that

$$x(t_*) = x_*, \qquad \dot{x}(t_*) \geq v_* > 0$$

Equivalently, by reflection symmetry of the two-body state, the trajectory has returned to the reflected section state corresponding to the inbound section at radius

$$x_*$$

Proof.
For

$$t \in [t_{\text{cross}}^{(2)}, t_{\text{cross}}^{(2)} + \tau_*]$$

the lower speed bound gives

$$x(t) = x(t_{\text{cross}}^{(2)}) + \int_{t_{\text{cross}}^{(2)}}^{t} \dot{x}(s)\, ds \geq v_*(t - t_{\text{cross}}^{(2)})$$

because

$$x(t_{\text{cross}}^{(2)}) = 0$$

Hence

$$x(t) \geq x_*$$

whenever

$$t - t_{\text{cross}}^{(2)} \geq \frac{x_*}{v_*}$$

Since

$$\tau_* \geq \frac{x_*}{v_*}$$

the trajectory reaches radius

$$x_*$$

within the stated window. Continuity of

$$x$$

then gives a first time

$$t_* \in \left[t_{\text{cross}}^{(2)}, t_{\text{cross}}^{(2)} + \frac{x_*}{v_*}\right]$$

such that

$$x(t_*) = x_*$$

The outward speed bound on the window implies

$$\dot{x}(t_*) \geq v_* > 0$$

Thus the trajectory reaches the reflected section state in finite time.

For the full return-map program this is the natural intermediate object: literal signed return to

$$x = x_*$$

with

$$\dot{x} < 0$$

requires one further outer-turn control step, whereas return to the reflected section state is the immediate consequence of the second crossing plus outward continuation on the right branch.

Lemma 17: Outer-turn closure from the reflected section state.
Assume the hypotheses of Lemma 16 and let

$$t_*$$

denote the reflected-section time, so that

$$x(t_*) = x_*, \qquad \dot{x}(t_*) \geq v_* > 0$$

Assume, in addition, that there exists a later outer turning time

$$t_{\text{turn}}^{\text{out}} > t_*$$

with

$$x\big(t_{\text{turn}}^{\text{out}}\big) = X_{\text{out}} \geq x_*, \qquad \dot{x}\big(t_{\text{turn}}^{\text{out}}\big) = 0$$

and a post-turn window

$$\big[t_{\text{turn}}^{\text{out}},\, t_{\text{turn}}^{\text{out}} + \tau_{\text{in}}\big]$$

on which

$$\ddot{x}(t) \leq -a_{\text{in}}^{\text{out}} < 0$$

If

$$\tau_{\text{in}} \geq \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_{\text{in}}^{\text{out}}}}$$

then there exists a first return time

$$T(\psi) \in \left[t_{\text{turn}}^{\text{out}},\, t_{\text{turn}}^{\text{out}} + \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_{\text{in}}^{\text{out}}}}\right]$$

such that

$$x(T(\psi)) = x_*, \qquad \dot{x}(T(\psi)) < 0$$

Proof.
Integrating the acceleration bound from the outer turning time gives

$$\dot{x}(t) = \dot{x}\big(t_{\text{turn}}^{\text{out}}\big) + \int_{t_{\text{turn}}^{\text{out}}}^{t} \ddot{x}(s)\, ds \leq -a_{\text{in}}^{\text{out}}(t - t_{\text{turn}}^{\text{out}})$$

for

$$t \in \big[t_{\text{turn}}^{\text{out}},\, t_{\text{turn}}^{\text{out}} + \tau_{\text{in}}\big]$$

because

$$\dot{x}\big(t_{\text{turn}}^{\text{out}}\big) = 0$$

Hence

$$\dot{x}(t) < 0$$

for all

$$t > t_{\text{turn}}^{\text{out}}$$

in the window, so the trajectory moves strictly inward on the right branch after the outer turn.

Integrating once more yields

$$x(t) \leq X_{\text{out}} - \frac{a_{\text{in}}^{\text{out}}}{2}\big(t - t_{\text{turn}}^{\text{out}}\big)^2$$

Therefore

$$x(t) \le x_*$$

whenever

$$t - t_{\text{turn}}^{\text{out}} \ge \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_{\text{in}}^{\text{out}}}}$$

By the assumed lower bound on

$$\tau_{\text{in}}$$

the comparison reaches

$$x_*$$

before the end of the window. Since

$$x\big(t_{\text{turn}}^{\text{out}}\big) = X_{\text{out}} \ge x_*$$

and

$$x$$

is continuous, there exists a first time

$$T(\psi) \in \left[t_{\text{turn}}^{\text{out}},\, t_{\text{turn}}^{\text{out}} + \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_{\text{in}}^{\text{out}}}}\right]$$

for which

$$x(T(\psi)) = x_*$$

The strict inward velocity bound implies

$$\dot{x}(T(\psi)) < 0$$

Thus the trajectory returns to the inbound section in finite time.

Lemma 18: Inbound section-speed bound after the outer turn.
Assume the hypotheses of Lemma 17 and, in addition, that on the post-turn window

$$\big[t_{\text{turn}}^{\text{out}},\, T(\psi)\big]$$

the acceleration satisfies the two-sided bound

$$-a_+^{\text{out}} \le \ddot{x}(t) \le -a_-^{\text{out}} < 0, \qquad 0 < a_-^{\text{out}} \le a_+^{\text{out}}$$

Then the inbound section speed satisfies

$$0 < -\dot{x}(T(\psi)) \le a_+^{\text{out}} \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_-^{\text{out}}}}$$

In particular, a sufficient condition for the tame return-speed bound is

$$a_+^{\text{out}} \sqrt{\frac{2(X_{\text{out}} - x_*)}{a_-^{\text{out}}}} \le U_{\max}$$

Proof.
Integrating the upper acceleration bound from the outer turning time to the inbound section return gives

$$\dot{x}(T(\psi)) = \dot{x}\big(t_{\text{turn}}^{\text{out}}\big) + \int_{t_{\text{turn}}^{\text{out}}}^{T(\psi)} \ddot{x}(s)\, ds \ge -a_+^{\text{out}} \big(T(\psi) - t_{\text{turn}}^{\text{out}}\big)$$

because

$$\dot{x}\big(t_{\mathrm{turn}}^{\mathrm{out}}\big) = 0$$

Since Lemma 17 already gives

$$\dot{x}(T(\psi)) < 0$$

this implies

$$0 < -\dot{x}(T(\psi)) \le a_{+}^{\mathrm{out}}\big(T(\psi) - t_{\mathrm{turn}}^{\mathrm{out}}\big)$$

It remains to bound the elapsed time. By the lower acceleration floor,

$$x(t) \le X_{\mathrm{out}} - \frac{a_{-}^{\mathrm{out}}}{2}\big(t - t_{\mathrm{turn}}^{\mathrm{out}}\big)^2$$

on

$$\big[t_{\mathrm{turn}}^{\mathrm{out}},\, T(\psi)\big]$$

Evaluating at

$$t = T(\psi)$$

and using

$$x(T(\psi)) = x_*$$

yields

$$T(\psi) - t_{\mathrm{turn}}^{\mathrm{out}} \le \sqrt{\frac{2(X_{\mathrm{out}} - x_*)}{a_{-}^{\mathrm{out}}}}$$

Substituting this into the previous speed bound proves

$$0 < -\dot{x}(T(\psi)) \le a_{+}^{\mathrm{out}}\sqrt{\frac{2(X_{\mathrm{out}} - x_*)}{a_{-}^{\mathrm{out}}}}$$

The stated sufficient condition for

$$-\dot{x}(T(\psi)) \le U_{\max}$$

is immediate.

Lemma 19: Returned-history tameness from final-window bounds.
Let

$$T(\psi)$$

be an inbound section return time produced by Lemma 17, and define the translated return history

$$x_{T(\psi)}(\theta) = x\big(T(\psi) + \theta\big), \qquad \theta \in [-h, 0]$$

Assume that on the final window

$$[T(\psi) - h,\, T(\psi)]$$

the trajectory satisfies:

- the section anchoring and sign conditions

$$x(T(\psi)) = x_*, \qquad \dot{x}(T(\psi)) < 0$$

- the envelope bounds

$$0 \le x(t) \le X_{\max}, \qquad |\dot{x}(t)| \le U_{\max}, \qquad |\ddot{x}(t)| \le A_{\max}$$

- and the same Jacobian and active-root count bounds that define the tame return class.

Then

$$x_{T(\psi)}$$

lies in the tame return envelope. In particular, if those final-window bounds are exactly the defining bounds of

$$\mathcal{C}^{\mathrm{tame}}_{x_*,\eta}$$

then

$$P_\eta(\psi) = x_{T(\psi)} \in \mathcal{C}^{\mathrm{tame}}_{x_*,\eta}$$

Proof.
For

$$\theta \in [-h, 0]$$

the translated history satisfies

$$x_{T(\psi)}(\theta) = x\bigl(T(\psi) + \theta\bigr)$$

so every point of the history segment is sampled from the final window

$$[T(\psi) - h,\, T(\psi)]$$

Therefore the pointwise bounds on

$$x, \qquad \dot{x}, \qquad \ddot{x}$$

transfer directly to

$$x_{T(\psi)}, \qquad \dot{x}_{T(\psi)}, \qquad \ddot{x}_{T(\psi)}$$

The section anchoring conditions at

$$\theta = 0$$

follow from

$$x(T(\psi)) = x_*, \qquad \dot{x}(T(\psi)) < 0$$

Likewise, because the Jacobian and active-root count bounds are assumed uniformly on the same final window, they transfer directly to the translated segment.

Hence the translated history satisfies the defining bounds of the tame return class, which proves the lemma.

Outer-turn recapture target

The remaining major dynamical gap on the return half is no longer kinematic. Lemmas 17 and 18 show that, once an outer turning point exists with a post-turn inward acceleration floor, the literal inbound section return follows by comparison geometry. The unresolved question is therefore whether the delayed forces actually create such an outer turn on the right exterior branch.

> **Target Theorem (Outer-Turn Recapture).**
> Fix a dual-mollified tame return class and suppose the return-half branch has already reached the reflected section state

$$> x = x_*, > \qquad > \dot{x} > 0 >$$

> on the right exterior branch. Assume there exist class constants

$$> X_{\mathrm{out,max}}, > \qquad > a^{\mathrm{out}}_{\mathrm{in}} > 0, > \qquad > a^{\mathrm{out}}_{+} > 0 >$$

> such that on the subsequent outer branch:
>
> 1. **bounded outward excursion:**
> the trajectory remains in

$$> x_* \le x(t) \le X_{\mathrm{out,max}} >$$

> until the first outer turn;
> 2. **outer-turn existence:**
> there exists a first time

$$> t_{\rm turn}^{\rm out} >$$

> with

$$> \dot x(t_{\rm turn}^{\rm out}) = 0, > \qquad > x(t_{\rm turn}^{\rm out}) = X_{\rm out} \in [x_*, X_{\rm out,max}]; >$$

> 3. **post-turn inward-force margin:**
> on a window after

$$> t_{\rm turn}^{\rm out}, >$$

> the signed acceleration satisfies

$$> -a_+^{\rm out} >\leq> \ddot x(t) >\leq> -a_{\rm in}^{\rm out} ><> 0; >$$

> 4. **final-window tame bounds:**
> the trajectory on the last

$$> h >$$

> units before the section return satisfies the same position, speed, acceleration, Jacobian, and root-count bounds used in Lemma 19.
>
> Then the outer branch closes back to the literal inbound section by Lemmas 17–19, and the remaining task is reduced to proving the force bounds that realize items 1–3.

This theorem isolates the outer-branch analogue of the inner recapture problem: near the apocenter one must show that delayed partner attraction plus favorable path-history geometry dominate the outward self-drive strongly enough to force one more turn.

Outer-turn recapture ladder

The intended proof order for the outer branch is:

1. **Outer-branch partner lower bound.**
   Derive a class-uniform lower bound for the inward partner contribution on the right exterior branch.
2. **Outer-branch self-drive upper bound.**
   Bound the outward self contribution there, using the longer partner distance and sub-field-speed Jacobian dilation on the delayed branches.
3. **Outer-force margin theorem.**
   Prove a quantitative inequality of the form

$$A_p(t) - A_s(t) \geq a_{\rm in}^{\rm out} > 0$$

on an apocenter window.
4. **Outer-turn existence theorem.**
Integrate the force margin to show that the outward branch stops at a finite radius

$$X_{\rm out} \leq X_{\rm out,max}$$

and develops a true outer turn.
5. **Post-turn window theorem.**
Show that the same force margin, or a weaker two-sided acceleration bracket, persists long enough after the turn to trigger Lemmas 17 and 18.

The third and fourth items are the main analytic bottlenecks. Once a robust outer-force margin is available, the remaining return-to-section estimates are already in place.

Lemma 20: Outer-branch partner lower bound.
Assume the post-second-crossing outer branch satisfies:

- right exterior outbound geometry,

$$x_* \leq x(t) \leq X_{\rm out,max}, \qquad \dot x(t) \geq 0$$

- at least one retained active partner branch for each

$$t \in [t_*, t_{\rm turn}^{\rm out}]$$

- the speed bound

$$|\dot{x}(t)| \le U_{\max}$$

- the partner roots retained in this lower bound remain inward exterior roots with

$$x(t) + x(t_p) > 0, \qquad 0 \le x(t_p) \le X_{\mathrm{out,max}}$$

- and the partner Jacobian upper bound

$$|J_p(t;t_p)| \le J_{p,\max}^{\mathrm{out}}$$

on every retained active partner root.
The speed bound permits the conservative choice

$$J_{p,\max}^{\mathrm{out}} = 1 + \frac{U_{\max}}{c_f}$$

Then the partner contribution to the inward acceleration obeys the class-uniform lower bound

$$A_p(t) \ge \underline{A}_p^{\mathrm{out}} \equiv \frac{\kappa \epsilon^2}{\left(4X_{\mathrm{out,max}}^2 + \epsilon_c^2\right) J_{p,\max}^{\mathrm{out}}}$$

Equivalently, the partner acceleration satisfies

$$a_p(t) = -A_p(t) \le -\underline{A}_p^{\mathrm{out}} < 0$$

on the outer branch.

Proof.
Along the retained inward exterior partner channel, the delayed source remains on the opposite side of the current right-hand particle, so each retained contribution points inward and therefore contributes with signed acceleration

$$a_p(t) = -A_p(t)$$

For any active partner root

$$t_p < t$$

the delayed partner separation satisfies

$$r_p(t;t_p) = x(t) + x(t_p)$$

Because both the current and delayed positions remain within the outer excursion envelope,

$$0 \le x(t) \le X_{\mathrm{out,max}}, \qquad 0 \le x(t_p) \le X_{\mathrm{out,max}}$$

we obtain

$$0 < r_p(t;t_p) \le 2X_{\mathrm{out,max}}$$

Hence the core-mollified denominator obeys

$$r_p(t;t_p)^2 + \epsilon_c^2 \le 4X_{\mathrm{out,max}}^2 + \epsilon_c^2$$

Each retained active partner contribution therefore has magnitude at least

$$\frac{\kappa \epsilon^2}{\left(r_p(t;t_p)^2 + \epsilon_c^2\right) |J_p(t;t_p)|} \ge \frac{\kappa \epsilon^2}{\left(4X_{\mathrm{out,max}}^2 + \epsilon_c^2\right) J_{p,\max}^{\mathrm{out}}}$$

Summing over the retained active partner branches and retaining only one branch yields

$$A_p(t) \ge \underline{A}_p^{\mathrm{out}}$$

which proves the lemma.

Lemma 21: Conditional outer-branch self-drive upper bound.
Assume that on the outer branch

$$[t_*, t_{\mathrm{turn}}^{\mathrm{out}}]$$

the active self branches satisfy:

- a root-count bound

$$N_s(t) \le N^{\mathrm{out}}_{s,\max}$$

- a self-Jacobian transversality bound

$$|J_s(t;t_s)| \ge \nu^{\mathrm{out}}_s > 0$$

on every active self root,

- and a delayed self-separation lower bound

$$r_s(t;t_s) \ge r^{\mathrm{out}}_{s,\min} > 0$$

on every active self root.

Then the outward self contribution obeys the class-uniform upper bound

$$A_s(t) \le \overline{A}^{\mathrm{out}}_s \equiv N^{\mathrm{out}}_{s,\max} \frac{\kappa \epsilon^2}{\left((r^{\mathrm{out}}_{s,\min})^2 + \epsilon_c^2\right) \nu^{\mathrm{out}}_s}$$

Proof.
For each active self root

$$t_s < t$$

the contribution to the outward self-drive has magnitude bounded by

$$\frac{\kappa \epsilon^2}{\left(r_s(t;t_s)^2 + \epsilon_c^2\right) |J_s(t;t_s)|}$$

Using the assumed lower bounds on

$$r_s(t;t_s) \qquad \text{and} \qquad |J_s(t;t_s)|$$

gives the branchwise estimate

$$\frac{\kappa \epsilon^2}{\left(r_s(t;t_s)^2 + \epsilon_c^2\right) |J_s(t;t_s)|} \le \frac{\kappa \epsilon^2}{\left((r^{\mathrm{out}}_{s,\min})^2 + \epsilon_c^2\right) \nu^{\mathrm{out}}_s}$$

Summing over at most

$$N^{\mathrm{out}}_{s,\max}$$

active self branches yields

$$A_s(t) \le \overline{A}^{\mathrm{out}}_s$$

which proves the lemma.

Lemma 22: Outer-force margin on the apocenter window.
Assume that on an outer-branch window

$$[t_*,\, t_* + \tau_{\mathrm{apo}}]$$

the signed dynamics can be written in the form

$$\ddot{x}(t) \le -A_p(t) + A_s(t)$$

where

$$A_p(t)$$

is the inward partner contribution and

$$A_s(t)$$

is the total outward delayed self contribution on that window. If

$$A_p(t) \geq \underline{A}_p^{\mathrm{out}} \qquad \text{and} \qquad A_s(t) \leq \overline{A}_s^{\mathrm{out}}$$

there with

$$\underline{A}_p^{\mathrm{out}} - \overline{A}_s^{\mathrm{out}} \geq a_{\mathrm{in}}^{\mathrm{out}} > 0$$

then

$$\ddot{x}(t) \leq -a_{\mathrm{in}}^{\mathrm{out}} < 0$$

for every

$$t \in [t_*,\, t_* + \tau_{\mathrm{apo}}]$$

In particular, Lemmas 20 and 21 reduce the outer-turn force margin to the parameter inequality

$$\frac{\kappa \epsilon^2}{\left(4X_{\mathrm{out,max}}^2 + \epsilon_c^2\right) J_{p,\mathrm{max}}^{\mathrm{out}}} - N_{s,\mathrm{max}}^{\mathrm{out}} \frac{\kappa \epsilon^2}{\left((r_{s,\mathrm{min}}^{\mathrm{out}})^2 + \epsilon_c^2\right) \nu_s^{\mathrm{out}}} \geq a_{\mathrm{in}}^{\mathrm{out}} > 0$$

Proof.
By hypothesis,

$$\ddot{x}(t) \leq -A_p(t) + A_s(t)$$

Using the lower bound for the inward partner term and the upper bound for the outward self term yields

$$\ddot{x}(t) \leq -\underline{A}_p^{\mathrm{out}} + \overline{A}_s^{\mathrm{out}} \leq -a_{\mathrm{in}}^{\mathrm{out}} < 0$$

which proves the claim.

Lemma 23: Finite-radius outer turn under an apocenter force margin.
Assume the hypotheses of Lemma 22 and suppose, in addition, that at the reflected section time

$$t_*$$

the trajectory satisfies

$$x(t_*) = x_*, \qquad \dot{x}(t_*) = v_* > 0$$

If the apocenter window is long enough to satisfy

$$\tau_{\mathrm{apo}} \geq \frac{v_*}{a_{\mathrm{in}}^{\mathrm{out}}}$$

then there exists a first outer turning time

$$t_{\mathrm{turn}}^{\mathrm{out}} \in \left[t_*,\, t_* + \frac{v_*}{a_{\mathrm{in}}^{\mathrm{out}}}\right]$$

such that

$$\dot{x}\left(t_{\mathrm{turn}}^{\mathrm{out}}\right) = 0$$

Moreover, the turning radius obeys the explicit bound

$$X_{\mathrm{out}} = x\left(t_{\mathrm{turn}}^{\mathrm{out}}\right) \leq x_* + \frac{v_*^2}{2a_{\mathrm{in}}^{\mathrm{out}}}$$

In particular, a sufficient condition for the outer-turn radius envelope is

$$x_* + \frac{v_*^2}{2a_{\mathrm{in}}^{\mathrm{out}}} \leq X_{\mathrm{out,max}}$$

Proof.
Lemma 22 gives the uniform acceleration bound

$$\ddot{x}(t) \leq -a_{\mathrm{in}}^{\mathrm{out}}$$

on

$$[t_*,\, t_* + \tau_{\text{apo}}]$$

Integrating from

$$t_*$$

to any later time

$$t$$

in that window yields

$$\dot{x}(t) = \dot{x}(t_*) + \int_{t_*}^{t} \ddot{x}(s)\, ds \le v_* - a_{\text{in}}^{\text{out}}(t - t_*)$$

Therefore

$$\dot{x}(t) \le 0$$

whenever

$$t - t_* \ge \frac{v_*}{a_{\text{in}}^{\text{out}}}$$

Because

$$\tau_{\text{apo}} \ge \frac{v_*}{a_{\text{in}}^{\text{out}}}$$

the comparison velocity reaches zero before the end of the apocenter window. Since

$$\dot{x}(t_*) = v_* > 0$$

and

$$\dot{x}$$

is continuous, there exists a first time

$$t_{\text{turn}}^{\text{out}} \in \left[t_*,\, t_* + \frac{v_*}{a_{\text{in}}^{\text{out}}}\right]$$

for which

$$\dot{x}\big(t_{\text{turn}}^{\text{out}}\big) = 0$$

Integrating the velocity estimate once more gives

$$x(t) \le x_* + v_*(t - t_*) - \frac{a_{\text{in}}^{\text{out}}}{2}(t - t_*)^2$$

Evaluating at

$$t = t_{\text{turn}}^{\text{out}}$$

and using

$$t_{\text{turn}}^{\text{out}} - t_* \le \frac{v_*}{a_{\text{in}}^{\text{out}}}$$

yields

$$X_{\text{out}} \le x_* + \frac{v_*^2}{2a_{\text{in}}^{\text{out}}}$$

which proves the radius bound.

Lemma 24: Post-turn acceleration bracket after the outer turn.
Assume the hypotheses of Lemma 23 and let

$$t_{\mathrm{turn}}^{\mathrm{out}}$$

be the first outer turning time, with

$$x\big(t_{\mathrm{turn}}^{\mathrm{out}}\big) = X_{\mathrm{out}}, \qquad \dot{x}\big(t_{\mathrm{turn}}^{\mathrm{out}}\big) = 0$$

Assume, in addition, that there exists a post-turn window

$$\big[t_{\mathrm{turn}}^{\mathrm{out}},\, t_{\mathrm{turn}}^{\mathrm{out}} + \tau_{\mathrm{in}}\big]$$

on which the delayed force contributions satisfy

$$A_p(t) \ge \underline{A}_{p,\mathrm{post}}^{\mathrm{out}}, \qquad A_s(t) \le \overline{A}_{s,\mathrm{post}}^{\mathrm{out}}$$

with

$$\underline{A}_{p,\mathrm{post}}^{\mathrm{out}} - \overline{A}_{s,\mathrm{post}}^{\mathrm{out}} \ge a_-^{\mathrm{out}} > 0$$

and also admit a class-uniform upper acceleration bound

$$|\ddot{x}(t)| \le a_+^{\mathrm{out}}$$

Then on that post-turn window one has the two-sided acceleration bracket

$$-a_+^{\mathrm{out}} \le \ddot{x}(t) \le -a_-^{\mathrm{out}} < 0$$

Consequently, if

$$\tau_{\mathrm{in}} \ge \sqrt{\frac{2(X_{\mathrm{out}} - x_*)}{a_-^{\mathrm{out}}}}$$

then the hypotheses of Lemmas 17 and 18 hold with

$$a_{\mathrm{in}}^{\mathrm{out}} = a_-^{\mathrm{out}}$$

Proof.
On the post-turn window the signed equation has the form

$$\ddot{x}(t) \le -A_p(t) + A_s(t)$$

Using the assumed lower bound for the inward partner term and the upper bound for the outward self term yields

$$\ddot{x}(t) \le -\underline{A}_{p,\mathrm{post}}^{\mathrm{out}} + \overline{A}_{s,\mathrm{post}}^{\mathrm{out}} \le -a_-^{\mathrm{out}} < 0$$

The assumed absolute acceleration bound gives

$$\ddot{x}(t) \ge -a_+^{\mathrm{out}}$$

so the stated two-sided bracket follows.

If

$$\tau_{\mathrm{in}} \ge \sqrt{\frac{2(X_{\mathrm{out}} - x_*)}{a_-^{\mathrm{out}}}}$$

then Lemma 17 applies with

$$a_{\mathrm{in}}^{\mathrm{out}} = a_-^{\mathrm{out}}$$

and Lemma 18 applies with the pair

$$(a_-^{\mathrm{out}}, a_+^{\mathrm{out}})$$

This is exactly the required post-turn handoff.

Outer-branch delayed-geometry target

The outer-turn force-margin lemmas are now in place, but Lemma 21 is still conditional on delayed self geometry. The remaining task is to prove that on the right exterior outbound branch the active self roots stay both sparse and noncaustic long enough to make the outer-force margin genuine rather than assumed.

> **Target Theorem (Outer-Branch Self-Root Separation and Transversality).**
> Fix a tame outer-branch class on the right exterior outbound interval
>
> $$> [t_*, t_{\mathrm{turn}}^{\mathrm{out}}]. >$$
>
> Suppose the branch stays within the excursion tube
>
> $$> x_* \le x(t) \le X_{\mathrm{out,max}}, > \qquad > 0 \le \dot{x}(t) \le U_{\max}, >$$
>
> and that its same-side delayed self interactions are organized by the outer sorting map
>
> $$> z(t) \equiv x(t) - c_f t. >$$
>
> Assume there exist class constants
>
> $$> r_{s,\min}^{\mathrm{out}} > 0, > \qquad > \nu_s^{\mathrm{out}} > 0, > \qquad > N_{s,\max}^{\mathrm{out}} \in \mathbb{N} >$$
>
> such that on the apocenter window:
>
> 1. active self roots satisfy a uniform delayed-separation lower bound
>
> $$> r_s(t; t_s) \ge r_{s,\min}^{\mathrm{out}}, >$$
>
> 2. active self roots stay on a noncaustic side of the outer sorting map with
>
> $$> |J_s(t; t_s)| \ge \nu_s^{\mathrm{out}}, >$$
>
> 3. and the number of active self branches obeys
>
> $$> N_s(t) \le N_{s,\max}^{\mathrm{out}}. >$$
>
> Then Lemma 21 applies, and the outer-force margin reduces to the explicit parameter inequality of Lemma 22.

This theorem is the outer-branch analogue of the earlier pre-crossing and post-crossing delayed-geometry steps: the partner floor is comparatively easy, while the decisive issue is keeping the self branches away from both short-distance concentration and Jacobian collapse.

Outer-branch delayed-geometry ladder

The intended proof order is:

1. **Outer sorting-map lemma.**
   Identify the correct same-side sorting map on the right exterior outbound branch and show that active self roots are organized by its level sets.
2. **Delayed-separation lemma.**
   Prove that the active outer self roots cannot approach the current point closer than a class-uniform radius

$$r_{s,\min}^{\mathrm{out}} > 0$$

on the apocenter window.
3. **Outer self-transversality lemma.**
Show that the active self roots stay on a noncaustic side of the sorting map, giving

$$|J_s| \ge \nu_s^{\mathrm{out}} > 0$$

4. **Outer root-count lemma.**
   Prove that the number of active same-side self branches remains bounded by

$$N_{s,\max}^{\mathrm{out}}$$

5. **Self-drive upper-bound corollary.**
   Feed the preceding three items into Lemma 21.

The second and third items are the real bottlenecks. Once they are available, the outer-turn recapture theorem becomes a direct comparison argument.

Lemma 25: Outer sorting-map identity on the right exterior outbound branch.
Assume the trajectory lies on the right exterior outbound branch,

$$x(t) \geq 0, \qquad \dot{x}(t) \geq 0$$

and consider same-side self roots

$$t_s < t$$

for which the delayed self-hit condition is

$$x(t) - x(t_s) = c_f(t - t_s)$$

Define the outer sorting map

$$z(t) \equiv x(t) - c_f t$$

Then every such active self root is selected by the level-set identity

$$z(t_s) = z(t)$$

Consequently, the same-side outer self branches on the right exterior outbound leg are organized by level sets of

$$z$$

Proof.
The same-side delayed self-hit condition is

$$x(t) - x(t_s) = c_f(t - t_s)$$

Rearranging gives

$$x(t) - c_f t = x(t_s) - c_f t_s$$

which is exactly

$$z(t) = z(t_s)$$

Thus every active same-side self root on the right exterior outbound branch is a level-set root of

$$z$$

which proves the lemma.

Lemma 26: Exact same-side self-root exclusion on a strictly sub-field-speed outer window.
Assume there exists an outer-branch window

$$[t_a, t_b] \subseteq [t_*, \infty)$$

on which the outbound speed stays strictly below field speed:

$$0 \leq \dot{x}(t) \leq c_f - \sigma_{\text{out}} < c_f \qquad \text{for } t \in [t_a, t_b]$$

with some

$$\sigma_{\text{out}} > 0$$

Then the outer sorting map

$$z(t) = x(t) - c_f t$$

is strictly decreasing on that window. Consequently, there are no exact same-side self roots

$$t_s < t$$

with both

$$t_s, t \in [t_a, t_b]$$

and

$$x(t) - x(t_s) = c_f(t - t_s)$$

Proof.
On the stated window one has

$$\dot{z}(t) = \dot{x}(t) - c_f \le -\sigma_{\text{out}} < 0$$

so

$$z$$

is strictly decreasing on

$$[t_a, t_b]$$

If there existed an exact same-side self root pair

$$t_s < t$$

with both times in that window, Lemma 25 would give

$$z(t_s) = z(t)$$

But strict monotonicity of

$$z$$

implies

$$z(t_s) > z(t)$$

whenever

$$t_s < t$$

which is impossible. Therefore no such exact same-side self root exists on the strictly sub-field-speed outer window.

This shows that on a strictly sub-field-speed apocenter window the exact delayed self geometry is maximally favorable: same-side outer self roots are absent. The remaining issue for the dual-mollified model is then not exact root multiplicity, but control of the shell-smeared near-diagonal contribution.

Lemma 27: Shell-tail bound on a strictly sub-field-speed outer window.
Assume the hypotheses of Lemma 26 on a window

$$[t_a, t_b] \subseteq [t_*, \infty)$$

and assume that the same-side outer self contribution is evaluated in the dual-mollified integral form with:

- shell mollifier

$$\delta_\eta$$

supported where its argument lies in

$$[-\eta, \eta]$$

- essential bound

$$\|\delta_\eta\|_\infty < \infty$$

- core mollifier

$$\epsilon_c > 0$$

- and memory horizon

$$h > 0$$

For each fixed

$$t \in [t_a, t_b]$$

let the **local** same-side shell contribution be integrated only over delayed times

$$t_0 \in [t_a, t]$$

for which the outer sorting-map mismatch

$$z(t_0) - z(t)$$

lies in the shell support. Then the local same-side outer self contribution obeys the pointwise bound

$$A^{\text{out}}_{s,\text{shell,loc}}(t) \le \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\text{out}}\,\epsilon_c^2}$$

This lemma controls only the same-window shell leakage. Same-side contributions from

$$t_0 < t_a$$

are deep-past channels and must be excluded or bounded separately by the later deep-past suppression package.

In particular, on a strictly sub-field-speed apocenter window the local outer self term coming from same-side shell leakage is uniformly bounded by an

$$\mathcal{O}\left(\frac{\eta}{\sigma_{\text{out}}\epsilon_c^2}\right)$$

quantity, even though the exact same-side root set is empty.

Proof.
Fix

$$t \in [t_a, t_b]$$

By Lemma 26,

$$z(t) = x(t) - c_f t$$

is strictly decreasing with derivative bounded above by

$$\dot{z}(t) \le -\sigma_{\text{out}} < 0$$

Hence for any delayed time

$$t_0 < t$$

in the same window one has

$$z(t_0) - z(t) \ge \sigma_{\text{out}}(t - t_0)$$

Therefore, if

$$|z(t_0) - z(t)| \le \eta$$

then necessarily

$$0 \le t - t_0 \le \frac{\eta}{\sigma_{\text{out}}}$$

So the set of delayed times inside the shell support has measure at most

$$\frac{\eta}{\sigma_{\text{out}}} \le \frac{2\eta}{\sigma_{\text{out}}}$$

Evaluating the local same-side self term in integral form and using

$$|x(t) - x(t_0)|^2 + \epsilon_c^2 \geq \epsilon_c^2$$

gives

$$A^{\text{out}}_{s,\text{shell,loc}}(t) \leq \kappa\epsilon^2 \int_{t_a}^{t} \frac{\delta_\eta(\cdots)}{|x(t) - x(t_0)|^2 + \epsilon_c^2}\, dt_0 \leq \frac{\kappa\epsilon^2\, \|\delta_\eta\|_\infty}{\epsilon_c^2} \cdot |\text{supp}_t(\delta_\eta)|$$

Using the support-measure bound yields

$$A^{\text{out}}_{s,\text{shell,loc}}(t) \leq \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{\sigma_{\text{out}}\, \epsilon_c^2}$$

which proves the lemma.

Corollary 28: Sub-field-speed outer-force margin from partner floor versus local shell tail.
Assume there exists an apocenter window

$$[t_a, t_b] \subseteq [t_*, \infty)$$

on which the branch has not yet turned and:

- the outer branch is strictly sub-field-speed,

$$0 \leq \dot{x}(t) \leq c_f - \sigma_{\text{out}} < c_f$$

- the partner lower bound of Lemma 20 holds with

$$A_p(t) \geq \underline{A}_p^{\text{out}}$$

- the only local same-window outward self contribution on that window is the same-side shell tail estimated in Lemma 27,
- and deep-past outward self channels are absent or have already been bounded by zero on this local-only corollary.

If

$$\underline{A}_p^{\text{out}} - \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{\sigma_{\text{out}}\, \epsilon_c^2} \geq a^{\text{out}}_{\text{in,shell}} > 0$$

then on that window one has the unconditional inward acceleration bound

$$\ddot{x}(t) \leq -a^{\text{out}}_{\text{in,shell}} < 0$$

In particular, on a strictly sub-field-speed apocenter window with no remaining deep-past outward self channel, the outer-force margin reduces to a direct parameter race between the partner floor and the shell-mollified same-window self leakage.

Proof.
Lemma 20 gives

$$A_p(t) \geq \underline{A}_p^{\text{out}}$$

By Lemma 26, there are no exact same-side self roots with both times on the stated window, and Lemma 27 therefore bounds the surviving same-window shell contribution by

$$A_s^{\text{out}}(t) = A^{\text{out}}_{s,\text{shell,loc}}(t) \leq \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{\sigma_{\text{out}}\, \epsilon_c^2}$$

Using the signed dynamics

$$\ddot{x}(t) \leq -A_p(t) + A_s^{\text{out}}(t)$$

yields

$$\ddot{x}(t) \leq -\underline{A}_p^{\text{out}} + \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{\sigma_{\text{out}}\, \epsilon_c^2} \leq -a^{\text{out}}_{\text{in,shell}} < 0$$

which proves the corollary.

Lemma 29: Coarse speed-decay entry into a strict sub-field-speed apocenter window.
Fix a desired strict sub-field-speed gap

$$\sigma_{\text{out}} > 0$$

and write

$$v_{\text{sub}}^{\text{out}} \equiv c_f - \sigma_{\text{out}}$$

Assume there is an outbound outer-entry interval

$$I_{\text{ent}} \equiv [t_0, t_1]$$

on which the branch has not yet been shown to turn, but the following non-circular data are available:

- the trajectory is on the right exterior outbound branch as long as no turn has occurred,

$$x_* \le x(t) \le X_{\text{out,max}}, \qquad \dot{x}(t) \ge 0$$

- the entry interval carries a coarse inward braking margin at the sub-field-speed boundary and above it:

$$\ddot{x}(t) \le -a_{\text{ent}}^{\text{out}} < 0 \qquad \text{whenever } t \in I_{\text{ent}} \text{ and } \dot{x}(t) \ge v_{\text{sub}}^{\text{out}}$$

- the interval is long enough for entry plus a retained sub-field-speed window of length

$$\tau_{\text{sub}}^{\text{out}} > 0$$

$$t_1 - t_0 \ge \frac{\left(\dot{x}(t_0) - v_{\text{sub}}^{\text{out}}\right)_+}{a_{\text{ent}}^{\text{out}}} + \tau_{\text{sub}}^{\text{out}}$$

Then one of the following alternatives holds:

1. a finite outer turn occurs before the retained sub-field-speed window is exhausted; or
2. there exists an entry time

$$t_a \in \left[t_0,\, t_0 + \frac{\left(\dot{x}(t_0) - v_{\text{sub}}^{\text{out}}\right)_+}{a_{\text{ent}}^{\text{out}}}\right]$$

such that the branch remains strictly sub-field-speed and outbound on

$$I_{\text{sub}} \equiv [t_a, t_a + \tau_{\text{sub}}^{\text{out}}]$$

namely

$$0 \le \dot{x}(t) \le c_f - \sigma_{\text{out}} \qquad \text{for every } t \in I_{\text{sub}}$$

The coarse margin can be certified without using the sub-field-speed sorting argument. For example, it is enough to have on

$$I_{\text{ent}}$$

a partner floor and a coarse total outward ceiling satisfying

$$\underline{A}_p^{\text{out}} - \overline{A}_{s,\text{ent}}^{\text{out}} \ge a_{\text{ent}}^{\text{out}} > 0$$

where

$$\overline{A}_{s,\text{ent}}^{\text{out}}$$

includes all outward self, fold, shell, and deep-past channels on the entry interval. This ceiling is deliberately coarse: it is not allowed to use Lemma 26 or Lemma 27, because those lemmas are consequences of the sub-field-speed window produced here.

Proof.
Let

$$v(t) \equiv \dot{x}(t)$$

If

$$v(t_0) \le v_{\text{sub}}^{\text{out}}$$

set

$$t_a = t_0$$

Otherwise, as long as

$$v(t) \geq v_{\mathrm{sub}}^{\mathrm{out}}$$

and no turn has occurred, the coarse margin gives

$$v'(t) = \ddot{x}(t) \leq -a_{\mathrm{ent}}^{\mathrm{out}}$$

Integrating from

$$t_0$$

shows that

$$v(t) \leq v(t_0) - a_{\mathrm{ent}}^{\mathrm{out}}(t - t_0)$$

throughout the portion of the interval where

$$v \geq v_{\mathrm{sub}}^{\mathrm{out}}$$

Hence either the velocity reaches zero first, giving a finite outer turn, or it reaches

$$v_{\mathrm{sub}}^{\mathrm{out}}$$

no later than

$$t_0 + \frac{\left(v(t_0) - v_{\mathrm{sub}}^{\mathrm{out}}\right)_+}{a_{\mathrm{ent}}^{\mathrm{out}}}$$

Call the first such time

$$t_a$$

It remains to show that the trajectory cannot immediately exit back above

$$v_{\mathrm{sub}}^{\mathrm{out}}$$

before the retained window is exhausted. Suppose instead that, after entry and before any turn, there is a first time

$$t_{\mathrm{exit}} > t_a$$

at which

$$v(t_{\mathrm{exit}}) = v_{\mathrm{sub}}^{\mathrm{out}}$$

and the velocity is about to cross from

$$v \leq v_{\mathrm{sub}}^{\mathrm{out}}$$

to

$$v > v_{\mathrm{sub}}^{\mathrm{out}}$$

At this boundary point the same coarse margin applies, so

$$v'(t_{\mathrm{exit}}) \leq -a_{\mathrm{ent}}^{\mathrm{out}} < 0$$

which is incompatible with an upward first exit. Therefore the sub-field-speed inequality is forward invariant on the retained part of

$$I_{\mathrm{ent}}$$

until a turn occurs.

The length hypothesis ensures that

$$[t_a, t_a + \tau_{\mathrm{sub}}^{\mathrm{out}}] \subseteq I_{\mathrm{ent}}$$

If no turn occurs on that retained interval, then the outbound condition supplies

$$v(t) \geq 0$$

and the forward-invariance argument supplies

$$v(t) \leq v_{\mathrm{sub}}^{\mathrm{out}} = c_f - \sigma_{\mathrm{out}}$$

This is exactly the claimed strict sub-field-speed apocenter window. The final displayed partner-floor condition implies the coarse acceleration hypothesis directly from the signed dynamics

$$\ddot{x}(t) \leq -A_p(t) + A_s^{\mathrm{out}}(t)$$

using

$$A_p(t) \geq \underline{A}_p^{\mathrm{out}}, \qquad A_s^{\mathrm{out}}(t) \leq \overline{A}_{s,\mathrm{ent}}^{\mathrm{out}}$$

Corollary 29.1: Strict sub-field-speed apocenter window.
Assume there exists an apocenter window

$$I_{\mathrm{sub}} \equiv [t_a, t_b] \subseteq [t_*, \infty)$$

on which the branch has not yet turned and satisfies

$$0 \leq \dot{x}(t) \leq c_f - \sigma_{\mathrm{out}} < c_f \qquad \text{for every } t \in I_{\mathrm{sub}}$$

Then the hypotheses of Lemmas 26 and 27 hold on

$$I_{\mathrm{sub}}$$

This corollary is intentionally separated from the entry mechanism. Lemma 29 supplies

$$I_{\mathrm{sub}}$$

under the coarse speed-decay hypotheses; if that lemma instead reaches the first alternative, then the outer turn has already occurred and the local sub-field-speed criterion is not needed for existence.

Proof.
The displayed speed bound is exactly the strict sub-field-speed hypothesis used by Lemma 26. Lemma 27 then applies to the local shell-smeared contribution on the same window.

Proposition: Explicit sub-field-speed apocenter recapture regime.
Assume the outer branch reaches an apocenter window

$$I_{\mathrm{sub}} \equiv [t_a, t_b]$$

before any known outer turn, and on that window:

- the branch remains outbound,

$$0 \leq \dot{x}(t)$$

- the branch is strictly sub-field-speed,

$$\dot{x}(t) \leq c_f - \sigma_{\mathrm{out}} < c_f$$

- the partner lower bound of Lemma 20 holds with

$$A_p(t) \geq \underline{A}_p^{\mathrm{out}}$$

- and deep-past outward self channels are absent or already bounded by zero, so the only outward self contribution on this local criterion is the same-window shell leakage of Lemma 27.

If, in addition, the parameter inequality

$$\frac{\kappa\epsilon^2}{\left(4X_{\mathrm{out,max}}^2 + \epsilon_c^2\right) J_{p,\mathrm{max}}^{\mathrm{out}}} - \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\, \epsilon_c^2} \geq a_{\mathrm{in,shell}}^{\mathrm{out}} > 0$$

holds, then on

$$I_{\mathrm{sub}}$$

one has the inward acceleration bound

$$\ddot{x}(t) \le -a^{\mathrm{out}}_{\mathrm{in,shell}} < 0$$

In particular, if

$$t_b - t_a \ge \frac{\dot{x}(t_a)}{a^{\mathrm{out}}_{\mathrm{in,shell}}}$$

then a finite outer turn occurs on

$$\left[t_a,\, t_a + \frac{\dot{x}(t_a)}{a^{\mathrm{out}}_{\mathrm{in,shell}}}\right]$$

with radius bound

$$X_{\mathrm{out}} \le x(t_a) + \frac{\dot{x}(t_a)^2}{2a^{\mathrm{out}}_{\mathrm{in,shell}}}$$

Proof.
Corollary 29.1 activates Lemmas 26 and 27 on

$$I_{\mathrm{sub}}$$

so the same-window outer self contribution is reduced to the shell-tail bound

$$A^{\mathrm{out}}_{s,\mathrm{shell,loc}}(t) \le \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\,\epsilon_c^2}$$

Combining this with the partner lower bound from Lemma 20 gives exactly the hypothesis of Corollary 28, hence

$$\ddot{x}(t) \le -a^{\mathrm{out}}_{\mathrm{in,shell}} < 0$$

on

$$I_{\mathrm{sub}}$$

Integrating from

$$t_a$$

to

$$t \in I_{\mathrm{sub}}$$

gives

$$\dot{x}(t) \le \dot{x}(t_a) - a^{\mathrm{out}}_{\mathrm{in,shell}}(t - t_a)$$

If the displayed window-length condition holds, continuity of

$$\dot{x}$$

forces a first zero of the velocity inside the stated interval. Integrating the same comparison once more gives the radius bound.

Deep-past outer self suppression target

The outer-turn program is now reduced to one explicit remaining issue. On the final sub-field-speed apocenter window, the local same-side self roots are annihilated by the monotonicity of

$$z(t) = x(t) - c_f t$$

so the local outward self-drive is only the shell tail bounded in Lemma 27. The remaining possible outward self contributions are therefore the roots that come from much earlier times

$$t_s < t_a$$

outside the local sub-field-speed window but still satisfy

$$z(t_s) = z(t)$$

> **Target Theorem (Deep-Past Outer Self Suppression).**
> Fix a final sub-field-speed apocenter window

$$> [t_a, t_b] \subseteq [t_*, \infty) >$$

> on which

$$> 0 \le \dot{x}(t) \le c_f - \sigma_{\text{out}} < c_f. >$$

> Assume that every outward-driving same-side self root

$$> t_s < t_a >$$

> satisfying

$$> z(t_s) = z(t) >$$

> obeys:
>
> 1. a macroscopic delayed-separation lower bound

$$> r_s(t; t_s) \ge R_{\text{deep}}^{\text{out}} > 0, >$$

> 2. a deep-past transversality bound

$$> |J_s(t; t_s)| \ge \nu_{s,\text{deep}}^{\text{out}} > 0, >$$

> 3. and a deep-past root-count bound

$$> N_{s,\text{deep}}^{\text{out}}(t) \le N_{s,\text{deep,max}}^{\text{out}}. >$$

> Then the total outward self contribution from deep-past roots satisfies

$$> A_{s,\text{deep}}^{\text{out}}(t) > \le > \overline{A}_{s,\text{deep}}^{\text{out}} > \equiv > N_{s,\text{deep,max}}^{\text{out}} > \frac{\kappa \epsilon^2}{> \left((R_{\text{deep}}^{\text{out}})^2 + \epsilon_c^2 \right) \nu_{s,\text{deep}}^{\text{out}}}. >$$

> Consequently, if

$$> \underline{A}_p^{\text{out}} > - > \overline{A}_{s,\text{deep}}^{\text{out}} > - > \frac{2\kappa \epsilon^2 \, \eta \, \|\delta_\eta\|_\infty}{> \sigma_{\text{out}} \, \epsilon_c^2} > \ge > a_{\text{in,full}}^{\text{out}} > 0, >$$

> then the full outward self-drive on the apocenter window is dominated and the outer-force margin becomes unconditional there.

This is the final missing outer-branch analogue of the post-crossing self-drive bound: local same-side roots are eliminated by the sub-field-speed sorting geometry, while the deep-past roots must be shown harmless by distance and Jacobian dilution.

Deep-past suppression ladder

The intended proof order is:

1. **Deep-past separation lemma.**
   Show that any same-side outer root with

$$t_s < t_a$$

must satisfy a macroscopic delay gap and hence a macroscopic spatial separation

$$r_s(t; t_s) \ge R_{\text{deep}}^{\text{out}}$$

2. **Deep-past transversality lemma.**
   Prove that the emitting velocities at those earlier times stay away from the outer caustic side, giving

$$|J_s| \ge \nu_{s,\text{deep}}^{\text{out}}$$

3. **Deep-past root-count lemma.**
   Bound the number of such roots by a class constant.
4. **Deep-past suppression corollary.**
   Combine the three bounds into the explicit amplitude estimate above.

The first two items are the real bottlenecks. Once deep-past roots are diluted by distance and Jacobian control, the outer-turn proposition becomes a direct explicit parameter race.

Lemma 30: Deep-past separation on a trimmed apocenter window.
Assume the hypotheses of Lemma 26 on a final sub-field-speed apocenter window

$$[t_a, t_b]$$

and fix a trimming parameter

$$0 < \tau_{\text{deep}} \le t_b - t_a$$

Let

$$I_{\text{deep}} \equiv [t_a + \tau_{\text{deep}},\, t_b]$$

If

$$t \in I_{\text{deep}}$$

and

$$t_s < t_a$$

is a same-side outward-driving self root satisfying

$$z(t_s) = z(t)$$

then:

1. the delayed time gap is uniformly bounded below,

$$t - t_s \ge \tau_{\text{deep}}$$

2. and the causal self separation is therefore macroscopic,

$$r_s(t; t_s) = c_f(t - t_s) \ge c_f \tau_{\text{deep}}$$

In particular, on the trimmed subwindow

$$I_{\text{deep}}$$

one may take

$$R^{\text{out}}_{\text{deep}} = c_f \tau_{\text{deep}}$$

Proof.
Because

$$t \in [t_a + \tau_{\text{deep}},\, t_b]$$

and

$$t_s < t_a$$

one immediately has

$$t - t_s > (t_a + \tau_{\text{deep}}) - t_a = \tau_{\text{deep}}$$

hence in particular

$$t - t_s \ge \tau_{\text{deep}}$$

For an outward-driving same-side self root on the right exterior outbound branch, the causal relation is

$$x(t) - x(t_s) = c_f(t - t_s)$$

Therefore

$$r_s(t; t_s) = c_f(t - t_s) \geq c_f \tau_{\text{deep}}$$

which proves the lemma.

Lemma 31: Deep-past transversality from a sub-field-speed source region.
Assume there exists a source interval

$$I_{\text{src}}^{\text{deep}} \subseteq (-\infty, t_a]$$

on which the emitting velocities satisfy the strict sub-field-speed bound

$$0 \leq \dot{x}(\theta) \leq c_f - \nu_{\text{deep}} < c_f \qquad \text{for every } \theta \in I_{\text{src}}^{\text{deep}}$$

with some

$$\nu_{\text{deep}} > 0$$

Let

$$t \in I_{\text{deep}}$$

and let

$$t_s \in I_{\text{src}}^{\text{deep}}$$

be a same-side outward-driving self root satisfying

$$z(t_s) = z(t)$$

Then the self Jacobian at the emitting time obeys

$$J_s(t; t_s) = 1 - \frac{\dot{x}(t_s)}{c_f} \geq \frac{\nu_{\text{deep}}}{c_f} > 0$$

In particular, on such deep-past roots one may take

$$\nu_{s,\text{deep}}^{\text{out}} = \frac{\nu_{\text{deep}}}{c_f}$$

Proof.
For a same-side outward-driving self root on the right exterior outbound branch one has

$$x(t) - x(t_s) = c_f(t - t_s), \qquad x(t) > x(t_s)$$

so the line-of-action sign is

$$\hat{r}_s(t; t_s) = +1$$

Therefore the self Jacobian reduces to

$$J_s(t; t_s) = 1 - \frac{\dot{x}(t_s)}{c_f}$$

Because

$$t_s \in I_{\text{src}}^{\text{deep}}$$

and the source interval is strictly sub-field-speed, we have

$$\dot{x}(t_s) \leq c_f - \nu_{\text{deep}}$$

Substituting gives

$$J_s(t;t_s) \ge 1 - \frac{c_f - \nu_{\mathrm{deep}}}{c_f} = \frac{\nu_{\mathrm{deep}}}{c_f} > 0$$

which proves the lemma.

Corollary 32: Deep-past amplitude suppression on a trimmed apocenter window.
Assume:

- the hypotheses of Lemma 30 on the trimmed apocenter window

$$I_{\mathrm{deep}} = [t_a + \tau_{\mathrm{deep}},\, t_b]$$

- the hypotheses of Lemma 31 with a deep-past sub-field-speed source interval

$$I_{\mathrm{src}}^{\mathrm{deep}} \subseteq (-\infty, t_a]$$

- and a deep-past root-count bound

$$N_{s,\mathrm{deep}}^{\mathrm{out}}(t) \le N_{s,\mathrm{deep,max}}^{\mathrm{out}}$$

for

$$t \in I_{\mathrm{deep}}$$

Then the total outward self contribution from deep-past same-side roots satisfies

$$A_{s,\mathrm{deep}}^{\mathrm{out}}(t) \le \overline{A}_{s,\mathrm{deep}}^{\mathrm{out}} \equiv N_{s,\mathrm{deep,max}}^{\mathrm{out}} \frac{\kappa\epsilon^2}{\left(c_f^2\tau_{\mathrm{deep}}^2 + \epsilon_c^2\right)\left(\nu_{\mathrm{deep}}/c_f\right)}$$

for every

$$t \in I_{\mathrm{deep}}$$

In particular, on the trimmed apocenter window the full outward self-drive is bounded by

$$A_s^{\mathrm{out}}(t) \le \overline{A}_{s,\mathrm{deep}}^{\mathrm{out}} + \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\,\epsilon_c^2}$$

provided the only remaining local same-side contribution is the shell tail of Lemma 27.

Proof.
Fix

$$t \in I_{\mathrm{deep}}$$

and let

$$t_s < t_a$$

be any outward-driving same-side deep-past root with

$$z(t_s) = z(t)$$

Lemma 30 gives the macroscopic separation bound

$$r_s(t;t_s) \ge c_f\tau_{\mathrm{deep}}$$

Lemma 31 gives the transversality bound

$$|J_s(t;t_s)| \ge \frac{\nu_{\mathrm{deep}}}{c_f}$$

Therefore each deep-past branch contributes at most

$$\frac{\kappa\epsilon^2}{\left(r_s(t;t_s)^2 + \epsilon_c^2\right)|J_s(t;t_s)|} \le \frac{\kappa\epsilon^2}{\left(c_f^2\tau_{\mathrm{deep}}^2 + \epsilon_c^2\right)\left(\nu_{\mathrm{deep}}/c_f\right)}$$

Summing over at most

$$N^{\mathrm{out}}_{s,\mathrm{deep,max}}$$

deep-past roots yields

$$A^{\mathrm{out}}_{s,\mathrm{deep}}(t) \le \overline{A}^{\mathrm{out}}_{s,\mathrm{deep}}$$

which proves the first claim.

If, in addition, the only local same-side outward contribution is the shell tail on the final sub-field-speed window, Lemma 27 supplies the second term, and the stated total bound follows by addition.

Corollary 33: Full trimmed-apocenter outer-force margin.
Assume on the trimmed apocenter window

$$I_{\mathrm{deep}} = [t_a + \tau_{\mathrm{deep}},\, t_b]$$

that:

- the partner lower bound of Lemma 20 holds,

$$A_p(t) \ge \underline{A}^{\mathrm{out}}_p$$

- the same-side local self contribution is only the shell tail controlled by Lemma 27,
- and the deep-past outward self contribution satisfies the suppression estimate of Corollary 32.

If

$$\underline{A}^{\mathrm{out}}_p - \overline{A}^{\mathrm{out}}_{s,\mathrm{deep}} - \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\,\epsilon_c^2} \ge a^{\mathrm{out}}_{\mathrm{in,trim}} > 0$$

then on

$$I_{\mathrm{deep}}$$

the full outward self-drive is dominated and one has the unconditional inward acceleration bound

$$\ddot{x}(t) \le -a^{\mathrm{out}}_{\mathrm{in,trim}} < 0$$

In particular, if

$$|I_{\mathrm{deep}}| \ge \frac{v_{\mathrm{deep}}}{a^{\mathrm{out}}_{\mathrm{in,trim}}}$$

where

$$v_{\mathrm{deep}} \equiv \sup_{t\in I_{\mathrm{deep}}} \dot{x}(t)$$

then the same comparison argument as in Lemma 23 forces a finite outer turn inside or immediately after the trimmed window.

Proof.
By Lemma 20,

$$A_p(t) \ge \underline{A}^{\mathrm{out}}_p$$

By Corollary 32,

$$A^{\mathrm{out}}_s(t) \le \overline{A}^{\mathrm{out}}_{s,\mathrm{deep}} + \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\,\epsilon_c^2}$$

Therefore the signed dynamics satisfy

$$\ddot{x}(t) \le -A_p(t) + A^{\mathrm{out}}_s(t) \le -\underline{A}^{\mathrm{out}}_p + \overline{A}^{\mathrm{out}}_{s,\mathrm{deep}} + \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\mathrm{out}}\,\epsilon_c^2}$$

The assumed parameter inequality gives

$$\ddot{x}(t) \le -a^{\mathrm{out}}_{\mathrm{in,trim}} < 0$$

which proves the first claim.

If the trimmed window length dominates

$$\frac{v_{\mathrm{deep}}}{a^{\mathrm{out}}_{\mathrm{in,trim}}}$$

then integrating the acceleration comparison exactly as in Lemma 23 forces the outward velocity to hit zero in finite time. This yields a finite outer turn on or just beyond the trimmed apocenter interval.

Lemma 34: Deep-past source localization by outer-level exclusion.
Assume the first origin crossing occurs at

$$t = 0, \qquad x(0) = 0$$

and let

$$I_{\mathrm{deep}} = [t_a + \tau_{\mathrm{deep}},\, t_b] \subseteq [t_*, t^{\mathrm{out}}_{\mathrm{turn}}]$$

be a trimmed apocenter window on the later right exterior outbound branch. Assume moreover that the outer sorting levels on the trimmed window lie strictly below the entire earlier outbound range:

$$\sup_{t \in I_{\mathrm{deep}}} z(t) < \inf_{0 \le s \le t_a} z(s)$$

If

$$t \in I_{\mathrm{deep}}$$

and

$$t_s < t_a$$

satisfies

$$z(t_s) = z(t)$$

then necessarily

$$t_s < 0$$

In particular, every deep-past same-side root on the trimmed apocenter window is forced onto the pre-crossing leg.

Proof.
Fix

$$t \in I_{\mathrm{deep}}$$

and suppose for contradiction that

$$0 \le t_s \le t_a$$

Then by the assumed outbound-level exclusion one has

$$z(t) \le \sup_{r \in I_{\mathrm{deep}}} z(r) < \inf_{0 \le s \le t_a} z(s) \le z(t_s)$$

which contradicts

$$z(t_s) = z(t)$$

Therefore

$$t_s < 0$$

as claimed.

Lemma 35: Deep-past root uniqueness and automatic transversality on the pre-crossing inbound leg.
Assume the hypotheses of Lemma 34, and assume the pre-crossing source interval

$$[-h, 0]$$

satisfies

$$\dot{x}(s) < 0 \qquad \text{for } s \in [-h, 0]$$

If

$$t \in I_{\text{deep}}$$

and

$$t_s < 0$$

is a same-side outward-driving self root with

$$z(t_s) = z(t)$$

then:

1. the source root is unique on

$$[-h, 0]$$

2. the self Jacobian satisfies the automatic lower bound

$$J_s(t; t_s) = 1 - \frac{\dot{x}(t_s)}{c_f} > 1$$

3. and hence one may take

$$N^{\text{out}}_{s,\text{deep,max}} \le 1, \qquad \nu^{\text{out}}_{s,\text{deep}} \ge 1$$

on the trimmed apocenter window.

Proof.
On the pre-crossing inbound leg one has

$$\dot{z}(s) = \dot{x}(s) - c_f < -c_f < 0 \qquad \text{for } s \in [-h, 0]$$

Therefore

$$z$$

is strictly decreasing on

$$[-h, 0]$$

Hence the level equation

$$z(s) = z(t)$$

can have at most one solution

$$s \in [-h, 0]$$

which proves uniqueness of the deep-past source root on that interval.

For a same-side outward-driving self root on the right exterior outbound branch one has

$$\hat{r}_s(t; t_s) = +1$$

so

$$J_s(t; t_s) = 1 - \frac{\dot{x}(t_s)}{c_f}$$

Since

$$\dot{x}(t_s) < 0$$

it follows immediately that

$$J_s(t;t_s) > 1$$

Thus

$$|J_s(t;t_s)| \geq 1$$

and the stated bounds

$$N^{\text{out}}_{s,\text{deep,max}} \leq 1, \qquad \nu^{\text{out}}_{s,\text{deep}} \geq 1$$

follow.

Corollary 36: Refined deep-past suppression from outbound-level exclusion.
Assume:

- the hypotheses of Lemma 30 on the trimmed apocenter window

$$I_{\text{deep}} = [t_a + \tau_{\text{deep}},\, t_b]$$

- the outbound-level exclusion hypothesis of Lemma 34,

$$\sup_{t \in I_{\text{deep}}} z(t) < \inf_{0 \leq s \leq t_a} z(s)$$

- and the pre-crossing inbound monotonicity hypothesis of Lemma 35,

$$\dot{x}(s) < 0 \qquad \text{for } s \in [-h, 0]$$

Then every deep-past same-side outward-driving root on

$$I_{\text{deep}}$$

lies on the pre-crossing inbound leg, is unique, and satisfies

$$|J_s(t;t_s)| \geq 1$$

Consequently,

$$A^{\text{out}}_{s,\text{deep}}(t) \leq \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2} \qquad \text{for every } t \in I_{\text{deep}}$$

Proof.
By Lemma 34, any deep-past same-side root with

$$z(t_s) = z(t)$$

must satisfy

$$t_s < 0$$

Lemma 35 then shows that on the pre-crossing inbound leg such a root is unique and obeys

$$|J_s(t;t_s)| \geq 1$$

Lemma 30 gives the separation bound

$$r_s(t;t_s) \geq c_f \tau_{\text{deep}}$$

Therefore the single deep-past branch contributes at most

$$\frac{\kappa\epsilon^2}{\left(r_s(t;t_s)^2 + \epsilon_c^2\right) |J_s(t;t_s)|} \leq \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2}$$

which proves the claim.

Corollary 37: Refined trimmed-apocenter outer-force margin.
Assume on the trimmed apocenter window

$$I_{\text{deep}} = [t_a + \tau_{\text{deep}},\, t_b]$$

that:

- the partner lower bound of Lemma 20 holds,

$$A_p(t) \geq \underline{A}_p^{\text{out}}$$

- the same-side local self contribution is only the shell tail controlled by Lemma 27,
- the hypotheses of Corollary 36 hold, so the deep-past same-side contribution satisfies

$$A_{s,\text{deep}}^{\text{out}}(t) \leq \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2}$$

- and there are no additional outward-driving self branches on

$$I_{\text{deep}}$$

beyond those two channels.

If

$$\underline{A}_p^{\text{out}} - \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2} - \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\text{out}}\,\epsilon_c^2} \geq a_{\text{in,ref}}^{\text{out}} > 0$$

then on

$$I_{\text{deep}}$$

one has the unconditional inward acceleration bound

$$\ddot{x}(t) \leq -a_{\text{in,ref}}^{\text{out}} < 0$$

In particular, if

$$|I_{\text{deep}}| \geq \frac{v_{\text{deep}}}{a_{\text{in,ref}}^{\text{out}}}$$

then the same comparison argument as in Lemma 23 forces a finite outer turn on or just beyond the trimmed apocenter window.

Proof.
By Lemma 20,

$$A_p(t) \geq \underline{A}_p^{\text{out}}$$

By Corollary 36,

$$A_{s,\text{deep}}^{\text{out}}(t) \leq \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2}$$

By Lemma 27, the local same-side shell leakage satisfies

$$A_{s,\text{shell,loc}}^{\text{out}}(t) \leq \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\text{out}}\,\epsilon_c^2}$$

Under the stated hypothesis that these exhaust the outward-driving self channels on

$$I_{\text{deep}}$$

the full outward self contribution is bounded by the sum of those two terms. Therefore

$$\ddot{x}(t) \leq -A_p(t) + A_s^{\text{out}}(t) \leq -\underline{A}_p^{\text{out}} + \frac{\kappa\epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2} + \frac{2\kappa\epsilon^2\,\eta\,\|\delta_\eta\|_\infty}{\sigma_{\text{out}}\,\epsilon_c^2}$$

The assumed parameter inequality gives

$$\ddot{x}(t) \leq -a_{\text{in,ref}}^{\text{out}} < 0$$

which proves the first claim.

If

$$|I_{\rm deep}| \ge \frac{v_{\rm deep}}{a^{\rm out}_{\rm in,ref}}$$

then integrating the acceleration comparison exactly as in Lemma 23 forces the outward velocity to hit zero in finite time, yielding a finite outer turn on or just beyond the trimmed apocenter interval.

z-map descent target

The outer-turn and deep-past layers are now reduced to one remaining geometric hypothesis:

$$\sup_{t\in I_{\rm deep}} z(t) < \inf_{0\le s\le t_a} z(s), \qquad z(t) = x(t) - c_f t$$

This is an exclusion statement saying that the late apocenter levels of

$$z$$

have descended below the entire earlier outbound range. Once this holds, the deep-past roots are forced onto the pre-crossing inbound leg by Lemma 34, and the refined outer-force margin becomes fully explicit.

> **Target Theorem (Outbound-Level Exclusion by z-Descent).**
> Assume the right exterior outbound branch starts at the first origin crossing with

$$> x(0) = 0, > \qquad > \dot{x}(0) = V_0 > c_f, >$$

> and later develops an outer hinge time

$$> t^{\rm out}_{\rm hinge} >$$

> defined by

$$> \dot{x}\big(t^{\rm out}_{\rm hinge}\big) = c_f. >$$

> Assume further that on the post-hinge outbound branch there is a sub-field-speed deceleration window on which

$$> \ddot{x}(t) \le -a^{\rm out}_z < 0. >$$

> If the resulting descent of

$$> z(t) = x(t) - c_f t >$$

> is large enough that, on a trimmed apocenter window

$$> I_{\rm deep} = [t_a + \tau_{\rm deep},\, t_b], >$$

> one has

$$> \sup_{t\in I_{\rm deep}} z(t) ><> \inf_{0\le s\le t_a} z(s), >$$

> then Lemma 34 applies and the deep-past same-side roots are forced onto the pre-crossing inbound leg.

This theorem isolates the final missing global shape statement for the outer sorting map. The earlier sections now reduce the outer-turn problem to proving enough descent of

$$z$$

after the outer hinge.

z-map descent ladder

The intended proof order is:

1. **Outer-hinge lemma.**
   Show that the outbound branch has a first time

$$t^{\rm out}_{\rm hinge}$$

with

$$\dot{x} = c_f$$

so

$$\dot{z} = 0$$

2. **Post-hinge monotonicity lemma.**
   Prove that once

$$\dot{x} < c_f$$

the sorting map

$$z(t) = x(t) - c_f t$$

is strictly decreasing.
3. **Quadratic descent lemma.**
Use the post-hinge acceleration floor to obtain an explicit estimate

$$z(t) \le z\big(t_{\mathrm{hinge}}^{\mathrm{out}}\big) - \frac{a_z^{\mathrm{out}}}{2}\big(t - t_{\mathrm{hinge}}^{\mathrm{out}}\big)^2$$

4. **Outbound-level exclusion corollary.**
   Compare this late-time upper bound with the earlier outbound range

$$[0, t_a]$$

to verify the hypothesis of Lemma 34.

The third and fourth items are the real bottlenecks. Once the descent estimate pushes the late

$$z$$

levels below the earlier outbound range, the deep-past topology is fully controlled.

Lemma 38: Outer hinge and z-monotonicity on the outbound branch.
Assume the right exterior outbound branch satisfies

$$\dot{x}(0) = V_0 > c_f$$

and later reaches a first outer turn at time

$$t_{\mathrm{turn}}^{\mathrm{out}}$$

with

$$\dot{x}\big(t_{\mathrm{turn}}^{\mathrm{out}}\big) = 0$$

Then there exists a first outer hinge time

$$t_{\mathrm{hinge}}^{\mathrm{out}} \in (0, t_{\mathrm{turn}}^{\mathrm{out}})$$

such that

$$\dot{x}\big(t_{\mathrm{hinge}}^{\mathrm{out}}\big) = c_f$$

Moreover, for

$$z(t) = x(t) - c_f t$$

one has

$$\dot{z}(t) = \dot{x}(t) - c_f$$

so

$$\dot{z}(t) > 0$$

for

$$0 \le t < t_{\text{hinge}}^{\text{out}}$$

and

$$\dot{z}(t) \le 0$$

for

$$t_{\text{hinge}}^{\text{out}} \le t \le t_{\text{turn}}^{\text{out}}$$

Proof.
The velocity

$$\dot{x}$$

is continuous on the outbound branch. At the crossing,

$$\dot{x}(0) = V_0 > c_f$$

while at the outer turn,

$$\dot{x}\big(t_{\text{turn}}^{\text{out}}\big) = 0 < c_f$$

By the intermediate value theorem there exists at least one time

$$t \in (0, t_{\text{turn}}^{\text{out}})$$

for which

$$\dot{x}(t) = c_f$$

Define

$$t_{\text{hinge}}^{\text{out}}$$

to be the first such time. Then

$$\dot{x}(t) > c_f \qquad \text{for } 0 \le t < t_{\text{hinge}}^{\text{out}}$$

and by definition

$$\dot{x}\big(t_{\text{hinge}}^{\text{out}}\big) = c_f$$

Therefore

$$\dot{z}(t) = \dot{x}(t) - c_f > 0$$

before the hinge, and

$$\dot{z}\big(t_{\text{hinge}}^{\text{out}}\big) = 0$$

If in addition the post-hinge branch remains sub-field-speed, then

$$\dot{z}(t) \le 0$$

there. This proves the stated monotonicity.

Lemma 39: Quadratic post-hinge z-descent.
Assume there exists a post-hinge interval

$$[t_{\text{hinge}}^{\text{out}},\, t_c] \subseteq [t_{\text{hinge}}^{\text{out}},\, t_{\text{turn}}^{\text{out}}]$$

on which

$$\ddot{x}(t) \le -a_z^{\text{out}} < 0$$

Then for every

$$t \in [t_{\text{hinge}}^{\text{out}},\, t_c]$$

one has

$$\dot{z}(t) \le -a_z^{\mathrm{out}}\left(t - t_{\mathrm{hinge}}^{\mathrm{out}}\right)$$

and

$$z(t) \le z\left(t_{\mathrm{hinge}}^{\mathrm{out}}\right) - \frac{a_z^{\mathrm{out}}}{2}\left(t - t_{\mathrm{hinge}}^{\mathrm{out}}\right)^2$$

Proof.
Since

$$z(t) = x(t) - c_f t$$

one has

$$\ddot{z}(t) = \ddot{x}(t)$$

On the stated interval this gives

$$\ddot{z}(t) \le -a_z^{\mathrm{out}} < 0$$

At the outer hinge,

$$\dot{z}\left(t_{\mathrm{hinge}}^{\mathrm{out}}\right) = \dot{x}\left(t_{\mathrm{hinge}}^{\mathrm{out}}\right) - c_f = 0$$

Integrating the acceleration bound from

$$t_{\mathrm{hinge}}^{\mathrm{out}}$$

to

$$t$$

yields

$$\dot{z}(t) \le -a_z^{\mathrm{out}}\left(t - t_{\mathrm{hinge}}^{\mathrm{out}}\right)$$

which is the first claim. Integrating once more gives

$$z(t) \le z\left(t_{\mathrm{hinge}}^{\mathrm{out}}\right) - \frac{a_z^{\mathrm{out}}}{2}\left(t - t_{\mathrm{hinge}}^{\mathrm{out}}\right)^2$$

which proves the quadratic descent estimate.

Corollary 40: Outbound-level exclusion from explicit z-descent.
Assume the hypotheses of Lemma 39 and let

$$I_{\mathrm{deep}} = [t_a + \tau_{\mathrm{deep}},\, t_b] \subseteq [t_{\mathrm{hinge}}^{\mathrm{out}},\, t_c]$$

be a trimmed apocenter window on the post-hinge branch. Define the earlier outbound floor

$$m_{\mathrm{out}}^{\mathrm{early}} \equiv \inf_{0 \le s \le t_a} z(s)$$

If

$$z\left(t_{\mathrm{hinge}}^{\mathrm{out}}\right) - \frac{a_z^{\mathrm{out}}}{2}\left(t_a + \tau_{\mathrm{deep}} - t_{\mathrm{hinge}}^{\mathrm{out}}\right)^2 < m_{\mathrm{out}}^{\mathrm{early}}$$

then the outbound-level exclusion hypothesis of Lemma 34 holds:

$$\sup_{t \in I_{\mathrm{deep}}} z(t) < \inf_{0 \le s \le t_a} z(s)$$

Proof.
Because

$$I_{\mathrm{deep}} \subseteq [t_{\mathrm{hinge}}^{\mathrm{out}},\, t_c]$$

and Lemma 39 gives

$$\dot{z}(t) \le -a_z^{\text{out}}\left(t - t_{\text{hinge}}^{\text{out}}\right) \le 0$$

on that interval, the function

$$z$$

is nonincreasing there. Therefore its supremum on

$$I_{\text{deep}}$$

is attained at the left endpoint:

$$\sup_{t \in I_{\text{deep}}} z(t) = z(t_a + \tau_{\text{deep}})$$

Applying Lemma 39 at

$$t = t_a + \tau_{\text{deep}}$$

yields

$$z(t_a + \tau_{\text{deep}}) \le z\left(t_{\text{hinge}}^{\text{out}}\right) - \frac{a_z^{\text{out}}}{2}\left(t_a + \tau_{\text{deep}} - t_{\text{hinge}}^{\text{out}}\right)^2$$

If the right-hand side is strictly smaller than

$$m_{\text{out}}^{\text{early}} = \inf_{0 \le s \le t_a} z(s)$$

then

$$\sup_{t \in I_{\text{deep}}} z(t) < \inf_{0 \le s \le t_a} z(s)$$

which is exactly the required outbound-level exclusion.

Remark (Simplified earlier-outbound floor).
Under the hypotheses of Lemma 38, the function

$$z(t) = x(t) - c_f t$$

is strictly increasing on

$$[0, t_{\text{hinge}}^{\text{out}})$$

and nonincreasing on

$$[t_{\text{hinge}}^{\text{out}}, t_a]$$

Therefore the earlier outbound floor satisfies

$$m_{\text{out}}^{\text{early}} = \inf_{0 \le s \le t_a} z(s) = \min\{z(0), z(t_a)\} = \min\{0, z(t_a)\}$$

because

$$z(0) = x(0) - c_f \cdot 0 = 0$$

In particular, a sufficient condition for outbound-level exclusion is simply

$$z(t_a + \tau_{\text{deep}}) < 0$$

or more conservatively, the explicit descent inequality

$$z\left(t_{\text{hinge}}^{\text{out}}\right) - \frac{a_z^{\text{out}}}{2}\left(t_a + \tau_{\text{deep}} - t_{\text{hinge}}^{\text{out}}\right)^2 < 0$$

> **Proposition (Unified trimmed-apocenter outer-turn criterion).**
> Assume:

> 1. the partner lower bound of Lemma 20 holds on a trimmed apocenter window

$$> I_{\text{deep}} = [t_a + \tau_{\text{deep}},\, t_b], >$$

> 2. the same-side local self contribution on that window is only the shell tail controlled by Lemma 27,
> 3. the pre-crossing inbound source interval satisfies

$$> \dot{x}(s) < 0 > \qquad > \text{for } s \in [-h, 0], >$$

> 4. the post-hinge branch satisfies the quadratic descent estimate of Lemma 39,
> 5. and the following two explicit inequalities hold:

$$> z\big(t^{\text{out}}_{\text{hinge}}\big) > - > \frac{a^{\text{out}}_z}{2} > \big(t_a + \tau_{\text{deep}} - t^{\text{out}}_{\text{hinge}}\big)^2 >< 0, >$$

$$> \underline{A}^{\text{out}}_p > - > \frac{\kappa \epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2} > - > \frac{2\kappa\epsilon^2\, \eta\, \|\delta_\eta\|_\infty}{> \sigma_{\text{out}}\, \epsilon_c^2} >\geq> a^{\text{out}}_{\text{in,ref}} > 0. >$$

> Then:
>
> 1. outbound-level exclusion holds on

$$> I_{\text{deep}}, >$$

> 2. every deep-past same-side outward-driving root is forced onto the pre-crossing inbound leg and is unique there,
> 3. the trimmed-apocenter acceleration satisfies

$$> \ddot{x}(t) \leq -a^{\text{out}}_{\text{in,ref}} < 0 > \qquad > \text{for } t \in I_{\text{deep}}, >$$

> 4. and if

$$> |I_{\text{deep}}| >\geq> \frac{v_{\text{deep}}}{a^{\text{out}}_{\text{in,ref}}}, >$$

> then a finite outer turn occurs on or just beyond the trimmed apocenter window.

Proof.
The first displayed inequality and Corollary 40 imply the outbound-level exclusion hypothesis of Lemma 34. Lemma 35 then forces any deep-past same-side outward-driving root onto the pre-crossing inbound leg, where it is unique and satisfies

$$|J_s| \geq 1$$

Therefore Corollary 36 yields the deep-past bound

$$A^{\text{out}}_{s,\text{deep}}(t) \leq \frac{\kappa \epsilon^2}{c_f^2 \tau_{\text{deep}}^2 + \epsilon_c^2}$$

Combining that with the shell-tail bound of Lemma 27 and the partner floor of Lemma 20 gives exactly the second displayed inequality, so Corollary 37 applies and yields

$$\ddot{x}(t) \leq -a^{\text{out}}_{\text{in,ref}} < 0 \qquad \text{on } I_{\text{deep}}$$

If the trimmed window length dominates

$$\frac{v_{\text{deep}}}{a^{\text{out}}_{\text{in,ref}}}$$

then the same comparison argument as in Lemma 23 forces the outward velocity to hit zero in finite time, yielding a finite outer turn on or just beyond

$$I_{\text{deep}}$$

Equal-amplitude cycling

The current delayed geometry does not naturally point to a continuous family of equal-amplitude cycles. In a purely causal delayed system, the more plausible generic picture is:

- net delayed braking at large excursion,
- net delayed pumping at small excursion,

- and an isolated balance point where the two effects cancel over one cycle.

If that picture is correct, the relevant mathematical object is an isolated fixed point of P_η, possibly attracting, rather than a one-parameter conservative orbit family. Equal-amplitude cycling is therefore plausible only if some stronger cycle-balance or exact history-functional structure is present; otherwise one should expect amplitude drift to be the generic behavior away from the fixed point.

Residual Scope Boundaries

The scaffold is now coherent enough to freeze as a proof program, but the following scope boundaries still need to stay explicit.

- **Origin singularity.** The shell regularization δ_η does not by itself remove the divergence of the amplitude factor $1/r^2$ at the origin crossing. For the current braking-dominance theorem target, an explicit core mollifier of the denominator should be treated as required rather than optional, for example by replacing r^{-2} with $(r^2+\epsilon_c^2)^{-1}$ or an equivalent short-distance regularization.
- **State-space labeling.** The theorem program is safest in true signed coordinates $x \in \mathbb{R}$, with recapture phrased in the radial variable $\rho = |x|$. Any language suggesting a rebound on the same $x > 0$ branch before the origin should be treated as provisional shorthand rather than as a derived dynamical fact.
- **Physical plausibility boundary.** In the collinear geometry the self term is not a centrifugal barrier. On the physically relevant post-crossing outbound branch it tends to reinforce the current radial motion. So the only plausible recapture mechanism in this model is that delayed partner attraction eventually dominates that outward self-drive on the outer leg. If the outer-turn theorem target fails, then the collinear breather should be read as a failed stabilization test rather than as an almost-closed proof.
- **Apocenter-entry window.** Lemma 29 now supplies the strict sub-field-speed window from a coarse entry-brake margin, or else reaches the outer turn before that window is needed. The global proof still has to include the coarse entry-brake ceiling inside the coupled parameter regime rather than smuggling it in through the local z-map argument.
- **Past-velocity transversality.** The Jacobians J_p and J_s depend on emission-time velocities, not current velocity. Turning through $\dot{x}=0$ at the present time does not by itself preserve transversality, so the lower bounds on $|J|$ must be checked against the delayed high-speed part of the history.
- **Partner-root inequality, not equality.** As the trajectory brakes after the crossing, the true partner distance can only become smaller than the leading linear prediction, which strengthens the partner force. So the partner-root estimate should be used as an upper bound on $r_p(t)$ and therefore a lower bound on $A_p^\rho(t)$, not as an exact identity on the nonlinear window.
- **Inner rebound region.** The theorem program still packages the actual near-center reversal into the admissible history class. That is acceptable for the current reduced problem, but it means the hardest local dynamics near the inner rebound is not yet derived from first principles here.
- **Root multiplicity control.** The branch sums defining A_p, A_s^{out}, and A_s^{in} are only tame if the number of active roots stays controlled. The regularized model softens each branch contribution, but it does not by itself prevent root proliferation from defeating the envelope bounds.
- **Candidate-packet falsification.** A rejected candidate packet may preserve useful diagnostics, such as strict subrows, fold normal forms, or range gaps, but those diagnostics do not promote the packet into a branch chart. Once a pre-ledger leaves a positive-width parent-complement overlap, a residual equality core, or an uncertified endpoint-scale gap, the same packet cannot feed the corridor, monodromy, returned-sample, topology, or Schauder rows.
- **Compactness is conditional.** The added acceleration bound is the right first step toward precompactness in C^1, but a later fixed-point theorem will still need the exact topology and continuity properties of the return map to be verified rather than assumed.
- **Continuity through the crossing.** The theorem uses a history class in which velocity is continuous through $t=0$, but the dual-mollified acceleration can still develop a very sharp gradient near the origin. Any Banach-space formulation must therefore keep enough Lipschitz-velocity, or weak acceleration, control near the boundary of the history interval that the delayed integrals remain well behaved at the crossing.

Capstone Statement

The existence capstone of the manuscript is the Schauder theorem target above. In compressed form, the final 1D statement is:

> **Theorem Target (Dual-Mollified Collinear Breather).**
> For some nonempty parameter regime
>
> $$> (\kappa, \epsilon, c_f, \eta, h, x_*) >$$
>
> and some closed convex tame envelope
>
> $$> \mathcal{K}_{x_*,\eta} \subseteq \Sigma^-_{x_*,\eta}, >$$
>
> the return map P_η has a fixed point
>
> $$> \phi^*_\eta \in \mathcal{K}_{x_*,\eta}, > \qquad > P_\eta(\phi^*_\eta) = \phi^*_\eta. >$$
>
> The corresponding trajectory is a bounded periodic two-body motion in which:

> 1. partner attraction drives the inward phase,
> 2. a post-crossing outward self-hit drive is eventually overcome strongly enough for radial recapture,
> 3. the motion returns to the same inbound section data after one full cycle.

The stability version is stronger:

> **Further Target (Stable Breather).**
> The Fréchet derivative $DP_\eta(\phi_\eta^*)$ has spectral radius < 1 on the section modulo time-shift symmetry, so the fixed point attracts nearby admissible histories.

This is the first clean theorem target for a self-hit-assisted bounded-recapture mechanism. It avoids the 2D circular tangential obstruction and does not require the full nested shell braid architecture.

Why This Reduced Problem Comes First

This model should be attacked before the full circular MCB or full nested shell braid for three reasons.

1. No tangential obstruction

The circular binary has a tangential no-go problem. The 1D model has no tangential channel at all. That removes the main obstruction already visible in the planar circular analysis.

2. Exact scalar Jacobians

In 1D,

$$\hat{r} \in \{-1, +1\}$$

so the delay-map Jacobians reduce to explicit scalar factors

$$J_p = 1 + \frac{\dot{x}(t_0)\hat{r}_p}{c_f}, \qquad J_s = 1 - \frac{\dot{x}(t_0)\hat{r}_s}{c_f}$$

This makes the branch geometry much easier to track analytically.

3. Direct test of the self-hit mechanism

If the collinear breather does not exist even after regularization, then the claim that delayed self-interaction can participate in a bounded binary-recapture mechanism is badly weakened. If the target theorem is eventually closed, then the theory would gain its first rigorous bounded delayed attractor.

What Counts as Success or Failure

Success

This reduced problem succeeds if it supports a proof program for:

- local well-posedness of the regularized 1D dynamics,
- well-defined first-return times on a nontrivial section,
- existence of a fixed point of P_η,
- and, ideally, local stability of that fixed point.

Failure

This reduced problem fails as a stabilization test if:

- the return map is not well defined on any robust section,
- all trajectories escape or collapse instead of returning,
- the self-hit branches do not produce reversal strongly enough to create recurrence,
- or the $\eta \to 0^+$ limit destroys every regularized bounded orbit.

Appendix: Why a Closed-Form Solution Is Unlikely

The following boxed aside is heuristic rather than theorem-level. Its purpose is not to prove a no-closed-form theorem, but to explain why the fixed-point and envelope route is mathematically more realistic than a search for an explicit formula

$$x(t) = f(t, X_0, V_0)$$

Here “closed-form solution” means an elementary formula for the orbit. It does not mean that the evolution law itself is unavailable. The dual-mollified absolute-time integral law is already an exact certified-law target; branch sums are local reductions on simple-root charts.

> **Heuristic aside.**

The symmetries of the 1D line make a closed formula tempting. One might hope for an expression

$$> x(t) = f(t, X_0, V_0) >$$

that captures the entire orbit from elementary initial data.

The delayed system does not support that expectation. It is a dynamical object with infinite-dimensional memory, state-dependent delays, and changing root topology, so the natural proof target is an invariant history-space return map rather than a global elementary solution.

1. The phase space is infinite-dimensional.
In ordinary Newtonian mechanics, the state is a point

$$> (X_0, V_0) >$$

in a finite-dimensional phase space. But the delayed master equation of $\mathbb{AAA}$ is non-Markovian. To compute the acceleration at

$$> t = 0^+, >$$

it is not enough to know only

$$> X_0 > \qquad > \text{and} > \qquad > V_0. >$$

One must know the stored path history

$$> \phi(\theta), > \qquad > \theta \in [-h, 0], >$$

because the active causal roots depend on how the particle arrived at the present state. The genuine initial datum is therefore a function, not a point.

2. The delays are state-dependent and implicit.
Even a linear constant-delay equation already resists elementary closed forms. Here the delay times are not fixed constants at all; they are roots of the implicit equations

$$> |x(t) \pm x(t_s)| = c_f(t - t_s). >$$

The timeline is being solved for at the same moment as the trajectory. The equation is not merely nonlinear; it is continually rewriting its own delayed arguments through the unknown path history.

3. The caustic changes the root topology.
At the hinge

$$> \dot{x} = -c_f, >$$

a new self-hit branch is born. The number of active roots changes with the motion itself. Whatever one chooses to call a “closed form,” it should not be expected to glide effortlessly across a dynamics in which the active branch structure changes as the trajectory passes through a causal fold.

4. The shadow of the three-body problem still hangs over the room.
Even instantaneous inverse-square dynamics already taught us that explicit formulas are not to be expected in generic nonlinear few-body problems. Here the 1D breather may look like a two-body problem, but the delayed self-interaction makes it behave like a path-history problem with an effectively infinite braid of past images. One should not expect such a system to become simpler merely because it lives on a line.

The silver lining.
This is why the present strategy is mathematically appropriate. It replaces the search for a global closed-form solution with a proof target that is stronger for the purpose of the chapter:

- existence of the delayed orbit,
- uniqueness once the history is fixed,
- boundedness inside an invariant envelope,
- and a fixed point of the return map.

In other words, we do not need a formula for

$$> x(t) >$$

valid for arbitrary data. The proof needs one candidate cycle

$$> \phi_{\mathrm{cyc}}, >$$

> and a finite certificate proving that the return map is continuous, precompact, and self-mapping on one closed convex tame domain.
>
> If one ever seeks formulas again, the natural place is not the global initial-value problem but the periodic orbit itself: after a fixed point

$$> \phi^*_\eta >$$

> is established, one might try an asymptotic or Fourier-type representation of that specific limit cycle. But that would be a local description of the attractor, not a universal closed form for arbitrary initial data.

Related Chapters

- master-equation.md
- binary-dynamics.md
- causal-action-functional.md
- energy.md
- dyadic-resonance-lock.md

Closed-Form Collinear Breather Ansatz

This note starts a parallel ansatz program for the 1D collinear breather. It does not replace the fixed-point proof architecture in collinear-breather.md. Its purpose is to generate certificate data for that proof program. A closed-form or closed-by-quadrature orbit is useful only insofar as it produces a candidate cycle, a branch chart, a mesh, and return residuals with strict audit slack.

This program is optional for the existence proof. The proof does not need an elementary closed-form orbit; it needs one candidate certified cycle and a finite certificate for the return map on a closed convex tame domain.

The external breather literature supplies useful terminology pressure but not a mechanism that can be imported into this proof. In this chapter, `breather` means a bounded delayed return-map fixed point in the collinear $\mathbb{AAA}$ reduction. Standard nonlinear-wave breathers are comparison objects; they do not replace the causal-root ledger, fold-layer integrals, returned-history residuals, or Schauder-domain audit needed here.

Negative breather results sharpen the same rule. In a nonintegrable wave equation, a formal expansion can be valid to all orders while the true dynamics still leak energy and fail to contain an exact localized periodic solution. The $\mathbb{AAA}$ consequence is not to import that radiation mechanism; it is to refuse promotion from formal closure alone. A candidate history remains approximate until fold-layer budgets, returned-history residuals, and the closed convex self-map audit are all certified on the same packet.

The integrable and near-integrable nonlinear Schrodinger catalogs strengthen the terminology boundary. Their coherent profiles, Darboux constructions, rogue-wave limits, and degenerate-breather limits depend on equation classes and conservation structures not present in the delayed architrino law. They are useful only as a checklist for native certification: the ansatz must declare which variables are certificate coordinates, which limit or degeneration is being taken, and which separator or fold layer remains bounded in the causal-root ledger. A limit that exists only in the external equation is not an ansatz transfer.

Perturbation nonpersistence results give the same refusal in another form. If a candidate survives only because an exact integrable symmetry or cancellation is kept intact, it is not a proof route for this certificate. The collinear program must show survival under the dual-mollified delayed law itself, with leakage channels closed by certificate rows rather than by analogy to a special wave equation.

The same discipline applies to construction methods. A numerical enclosure, validated quadrature orbit, or interval-collocation solve is equivalent to a closed-form ansatz only if it produces the same finite candidate packet: period, section/symmetry chart, representation coefficients, mesh, residual targets, causal pre-ledger inputs, and branch-chart inputs on one certified domain.

The state-dependent-delay periodic-orbit literature gives the methodological reason for this rule. A periodic boundary-value problem can be reduced locally to algebraic root finding only when the finite vector is tied to a projection from histories and a reconstruction back into the history space. For this collinear certificate, a residual vector is therefore not just a numerical fit: it must record the projection/reconstruction convention, the local neighborhood where the reduction is meant to hold, and the regularity assumptions that make the returned history meaningful.

Collocation adds a second discipline. The piecewise polynomial is a candidate representation of a periodic boundary-value problem, not a proof object by itself. A collocation packet must state the subinterval partition, polynomial degree, collocation nodes, period normalization, and section anchoring used to remove time-translation symmetry. Mesh-node superconvergence, meaning extra accuracy at selected nodes, is not assumed as a global bound; separator and origin layers need interval residual bounds on cells, not only small residuals at mesh points.

Continuation and finite auxiliary ODE constructions are useful only at the candidate-source level. A continued branch point or auxiliary-system orbit must reconstruct to the declared signed history, period, mesh, separator layers, and causal-root ledger before it can feed the certificate. Nonuniform transition-layer behavior near separators must be bounded by interval cell estimates, not inferred from small residuals at isolated nodes.

The accepted output of this note is therefore a certificate packet

$$\mathfrak{C}_{\mathrm{ans}} = \left(\phi_{\mathrm{cyc}}, T, \mathcal{B}_{\mathrm{act}}, \{\theta_j\}_{j=0}^{N}, \{R_j^x, R_j^v\}_{j=0}^{N}\right)$$

where

$$\phi_{\mathrm{cyc}}$$

is the candidate cycle,

$$T$$

is its proposed period,

$$\mathcal{B}_{\mathrm{act}}$$

is the finite active branch list with inactive complements, and the sampled residuals feed the finite audit in collinear-breather.md.

The guiding suspicion is:

- below field speed, the active causal roots are tame and the force may reduce to a small number of effective $1/r$ phase-space curves, with conservative potential curves only as a certified special case;
- at field speed, the sorting maps become marginal and the orbit passes through a metastable separator;
- above field speed, the active branch structure changes and must be matched by explicit crossing laws rather than by one smooth formula.

If a closed-form collinear breather exists, it is likely not one elementary expression on the whole line. The more plausible object is a piecewise analytic orbit whose pieces are joined by causal matching conditions at the field-speed separators and at origin crossings.

Status

This is an ansatz document, not a theorem. It records the first closed-form search path and the algebraic tests needed before it can feed the finite certificate program in collinear-breather.md.

The target object is a candidate history

$$\phi_{\mathrm{cyc}}$$

because the finite Schauder audit now needs an instantiated center history, a mesh, and certificate data. A closed-form ansatz is useful exactly if it can produce that

$$\phi_{\mathrm{cyc}}$$

without first solving the return-map fixed point abstractly. A numerical enclosure, validated quadrature orbit, or other certified construction would serve the same proof role if it supplies the same certificate rows.

The governing law for that certification is the dual-mollified absolute-time integral law from collinear-breather.md. Branch-sum formulas inside this note are working reductions on finite simple-root charts, not replacements for the integral law through separator layers or causal folds.

The first explicit velocity-class packet has sharpened this status without proving a breather. A fixed cosine candidate fails at the parent-complement part of the null-coordinate pre-ledger: after the accepted simple-root windows and fold-layer diagnostics are removed, residual equality cores remain in the parent complements. Those diagnostics are useful, but they do not authorize branch-chart construction. The next candidate source must therefore be a fresh fold-adapted collocation packet, or an equivalent certified construction, whose null-coordinate pre-ledger passes before any active branch chart is built.

Variables and Speed Classes

Work in the same reflection-symmetric 1D reduction as the main note:

$$x_1(t) = -x(t), \qquad x_2(t) = x(t)$$

with field speed

$$c_f > 0$$

For this ansatz it is useful to introduce the radial speed

$$u(t) \equiv |\dot{x}(t)|$$

and the field-speed shorthand

$$v_f \equiv c_f$$

The three speed classes are:

1. **sub-field branch**

$$u(t) < v_f$$

2. **field-speed separator**

$$u(t) = v_f$$

3. **super-field branch**

$$u(t) > v_f$$

The signed sorting maps from the main proof remain the natural branch variables:

$$w(t) = x(t) + c_f t, \qquad z(t) = x(t) - c_f t$$

On an outbound right branch,

$$\dot{z}(t) = \dot{x}(t) - c_f$$

Thus

$$\dot{x} < c_f, \qquad \dot{x} = c_f, \qquad \dot{x} > c_f$$

mean, respectively, that

$$z$$

is decreasing, stationary, or increasing. The separator

$$\dot{x} = c_f$$

is therefore not merely a speed value; it is a causal sorting transition.

Constant-Velocity Causal Algebra

The simplest way to see where a closed form might come from is to freeze the velocity locally. Suppose

$$x(t_0) \approx x(t) - v(t - t_0), \qquad v = \dot{x}(t)$$

For a partner hit, the causal equation is

$$x(t) + x(t_0) = c_f(t - t_0)$$

Writing

$$\tau = t - t_0$$

gives

$$2x = (c_f + v)\tau, \qquad \tau_p = \frac{2x}{c_f + v}$$

The causal partner distance is therefore

$$r_p = c_f \tau_p = \frac{2c_f x}{c_f + v}$$

and the branch Jacobian is

$$J_p = 1 + \frac{v}{c_f}$$

Ignoring the short-distance core for a moment, the partner force scale becomes

$$A_p \sim \frac{\kappa\epsilon^2}{r_p^2 J_p} = \frac{\kappa\epsilon^2(c_f + v)}{4c_f x^2}$$

With

$$\beta \equiv \frac{v}{c_f}, \qquad g \equiv \kappa\epsilon^2$$

this reads

$$A_p \sim \frac{g(1 + \beta)}{4x^2}$$

This is not, by itself, a conservative potential curve. The Jacobian factor makes the affine partner force velocity dependent, so the local model is a Lienard-type phase equation. On the bare affine partner chart,

$$\ddot{x} = -\frac{g}{4x^2}\left(1 + \frac{\dot{x}}{c_f}\right)$$

Writing

$$v(x) = \dot{x}, \qquad \ddot{x} = v\frac{dv}{dx}$$

gives the separable phase equation

$$\frac{v}{1 + v/c_f}\, dv = -\frac{g}{4x^2}\, dx$$

Hence the exact affine partner invariant is

$$c_f v - c_f^2 \ln|c_f + v| = \frac{g}{4x} + C_{\mathcal{R}}$$

This implicit phase-space curve replaces the naive energy curve on the affine partner chart. The logarithmic term also exposes a useful topology check: in the unsoftened affine partner model, reaching

$$v = -c_f$$

requires

$$x \to 0$$

Thus the inbound field-speed separator and the origin-crossing layer are tightly coupled in the bare model. The dual core scale

$$\epsilon_c$$

and shell width

$$\eta$$

soften this coincidence, but the certificate should still treat the separator and origin layer as coupled events unless interval data prove a strict separation.

The exact core-mollified version replaces

$$x^2$$

by the corresponding branch distance square plus

$$\epsilon_c^2$$

When a conservative approximation is separately certified, the candidate potential curves should use

$$R_{\epsilon_c}(r) \equiv \sqrt{r^2 + \epsilon_c^2}$$

rather than a bare

$$|r|$$

Sub-field-speed partner-only benchmark

The sub-field comparison case must be generated from the force law, not prescribed as a future path. On the exterior affine partner chart above, fix initial data

$$x(0) = x_0 > 0, \qquad \dot{x}(0) = c_f \beta_0, \qquad -1 < \beta_0 \le 0$$

and evolve by

$$\ddot{x} = -\frac{g}{4x^2}\left(1 + \frac{\dot{x}}{c_f}\right)$$

The held-release preparation supplies one concrete source of such initial data. If the pre-release source is held at $-x_0$ and

$$y(t) \equiv x(t) + x_0$$

then the held-source segment has

$$y(\theta) = 2x_0 \cos^2\theta, \qquad x(\theta) = x_0 \cos(2\theta), \qquad \dot{x}(\theta) = -\sqrt{\frac{g}{x_0}}\tan\theta$$

The first moving-partner wake reaches the receiver at the unique angle satisfying

$$\cos^2\theta_* = \rho\left(\theta_* + \sin\theta_* \cos\theta_*\right), \qquad \rho \equiv c_f\sqrt{\frac{x_0}{g}}$$

For $x_0 = 1.25$, $g = 1$, and $c_f = 1$, this gives

$$x_* \approx 0.8707972823389274, \qquad \beta_* \approx -0.37820836925058077$$

Thus the exterior Lambert branch can be initialized from a finite sub-field-speed handoff rather than from the rejected exact field-speed head-on prehistory.

With

$$\alpha = \frac{g}{4c_f^2}$$

the exact phase invariant is

$$\beta - \beta_0 - \ln\left(\frac{1+\beta}{1+\beta_0}\right) = \alpha\left(\frac{1}{x} - \frac{1}{x_0}\right), \qquad \beta = \frac{\dot{x}}{c_f}$$

Equivalently, if

$$S(x) = \beta_0 - \ln(1+\beta_0) + \alpha\left(\frac{1}{x} - \frac{1}{x_0}\right)$$

then the two analytic velocity branches are

$$\beta_k(x) = -1 - \mathrm{W}_k\left(-e^{-(S(x)+1)}\right)$$

The inbound sub-field branch is

$$\beta_{\mathrm{in}}(x) = -1 - \mathrm{W}_0\left(-e^{-(S(x)+1)}\right)$$

and satisfies $-1 < \beta_{\mathrm{in}}(x) < 0$ for every $x > 0$ on the exterior chart. The outbound branch uses the other real Lambert branch when the same invariant is continued away from the core layer. The branch time is recovered by

$$t - t_0 = \int_x^{x_0} \frac{d\xi}{-c_f\,\beta_{\mathrm{in}}(\xi)}$$

This gives a controlled analytic baseline for a sub-field-speed breather search. The exterior partner branch does not reach

$$|\dot{x}| = c_f$$

at any finite $x > 0$; the logarithm diverges as $\beta \to -1^+$. Therefore a finite-radius field-speed separator is not produced by this action-generated partner chart. It must come from a core-layer effect, finite shell width, nonaffine path history, a self-image contribution, or a different certified branch chart.

The same branch also supplies an exact self-root exclusion test in the sharp-shell limit. If a candidate history satisfies

$$|\dot{x}(t)| \le c_f - \sigma \qquad \text{on a stored interval}$$

for some $\sigma > 0$, then for all $s < t$ in that interval,

$$|x(t) - x(s)| \le (c_f - \sigma)(t - s) < c_f(t - s)$$

Thus the exact same-side self-hit equation has no nontrivial solution there. For finite shell width η, the possible self contribution is confined to the near-diagonal collar

$$0 < t - s \le \frac{\eta}{\sigma}$$

and must be bounded from the dual-mollified integral law rather than inserted as an exact simple-root branch. This separates the analytic sub-field test from the field-speed fold program: the test asks whether partner attraction plus the finite-width self-collar can close a return without ever producing a true field-speed separator.

Signed partner branch table

The local affine partner calculation should now be kept as a table of certified branch data. Work on an exterior chart

$$x(t) = \sigma q(t), \qquad q(t) > 0, \qquad \sigma \in \{-1, +1\}$$

with radial velocity

$$u_r(t) \equiv \dot{q}(t)$$

On a locally affine same-exterior window,

$$q(s) \approx q(t) - u_r(t)(t - s)$$

the partner root has

$$\tau_p = t - s = \frac{2q}{c_f + u_r}, \qquad r_p = c_f \tau_p, \qquad \hat{r}_p = \sigma, \qquad J_p = 1 + \frac{u_r}{c_f}$$

The signed partner acceleration in the

$$x$$

coordinate points as

$$\operatorname{sgn}(a_p) = -\sigma$$

that is, inward toward the origin.

Arc chart	Radial assumptions	τ_p	$\hat{r}_p$	J_p	Partner sign in x	Validity conditions
inbound exterior	$q > 0,\, u_r < 0$	$\dfrac{2q}{c_f + u_r}$	σ	$1 + \dfrac{u_r}{c_f}$	$-\sigma$	$c_f + u_r \geq \nu c_f$, no origin crossing inside the affine window
field-speed hinge	$u_r = -c_f$	singular	σ before the fold	0	fold-controlled	branch-sum form invalid; use the dual-mollified fold integral
origin-crossing layer	$q \lesssim \epsilon_c$ or σ changes	not a single affine root	changes by layer	chart-dependent	core-controlled	use the absolute-time integral law, not one exterior branch table
outbound exterior	$q > 0,\, u_r > 0$	$\dfrac{2q}{c_f + u_r}$	σ	$1 + \dfrac{u_r}{c_f} > 1$	$-\sigma$	same exterior chart and certified active root
apocenter sub-field	$q > 0,\, \lvert u_r\rvert < c_f,$ $u_r \to 0$	$\dfrac{2q}{c_f + u_r}$	σ	near 1	$-\sigma$	strict sub-field margin and active-root separation on the apocenter window

This table is only the partner column of the certificate packet. The self-image columns must be produced separately because their source and receiver are the same labeled path and their active roots can change at field-speed separators.

Why the Field-Speed Separator Matters

For same-side self hits on an affine segment,

$$|x(t) - x(t_0)| = \|\mathbf{v}\|\tau$$

The exact causal-isochron equation is

$$\|\mathbf{v}\|\tau = c_f\tau$$

For

$$\tau > 0$$

this is possible only when

$$\|\mathbf{v}\| = c_f$$

Therefore a perfectly affine segment has no same-side exact self root away from the field-speed separator. Self branches appear because the real trajectory is not globally affine: acceleration, origin crossing, and later return geometry let a present point meet older path-history images.

This suggests a closed-form strategy:

1. solve sub-field and super-field segments as certified phase-space arcs, using Lienard quadrature where the causal Jacobian remains velocity dependent;
2. treat the field-speed separator as the event where causal images are born, die, or switch branch labels;
3. impose matching laws at those separator events.

The separator is metastable in the sense that small perturbations decide whether the sorting map keeps descending, stalls, or reverses. In the dual-mollified model the separator should become a thin transition layer rather than an infinite impulse.

Separator normal form and fold scaling

For certificate purposes, the field-speed separator is a codimension-one event surface in the reduced phase data together with an active branch label:

$$\Sigma_{\mathcal{B}} = \{(x, \mathbf{v}, \mathcal{B}) : \|\mathbf{v}\| = c_f\}$$

Here

$$\mathcal{B}$$

is part of the state description, because crossing

$$\Sigma_{\mathcal{B}}$$

can create, annihilate, or relabel path-history roots even when

$$(x, v)$$

remains continuous.

Near a separator event, the dual-mollified vector field should be treated as a regularized perturbation of the bare branch-sum field. The shell width

$$\eta$$

and core radius

$$\epsilon_c$$

are then small but fixed certificate parameters, not limiting symbols to be discarded before the impulse budget is computed.

Let

$$g(t, s; \lambda) = 0$$

denote one signed causal-root defect on a local chart, with

$$\lambda$$

the transverse separator coordinate. A generic branch-topology change has the fold normal form

$$g(t, s; \lambda) = a(s - s_\Sigma)^2 + b\lambda + O(|s - s_\Sigma|^3 + |\lambda||s - s_\Sigma| + \lambda^2), \qquad ab \neq 0$$

Thus the active-root change is a saddle-node of branch labels: two simple roots are born or annihilated as the sign of

$$b\lambda/a$$

changes. In the dual-mollified chart, the shell support

$$|g| \lesssim \eta$$

gives the fold-root thickness

$$|s - s_\Sigma| = O(\eta^{1/2})$$

Under a transverse passage through the fold coordinate, the unresolved fold layer has the same

$$O(\eta^{1/2})$$

clock-time scale after reparametrizing by the local fold coordinate. If a concrete chart uses a different clock normalization, the certificate must record the interval enclosure directly.

Consequently the fold impulse ceiling is not a free assertion. It must be supplied by an interval bound of the form

$$|\Delta v_\Sigma| \leq I^{\mathrm{fold}}_{\eta,\epsilon_c} \leq C_\Sigma \eta^{1/2} A_{\Sigma,\eta,\epsilon_c}$$

where

$$A_{\Sigma,\eta,\epsilon_c}$$

is an interval upper bound for the dual-mollified acceleration on the certified fold tube and

$$C_\Sigma$$

is the corresponding transversality constant. The certificate may use a sharper direct quadrature bound, but it must expose the normal-form constants and the resulting finite slack.

Piecewise Chart Ansatz

Let

$$\mathcal{R} \in \{<, =, >\}$$

denote a speed class relative to

$$v_f$$

On each open region away from the separator, first fix a branch chart

$$\mathcal{I}_{\mathcal{R}}$$

containing the active partner and self-image data. On that chart the delayed force should first be written as a phase-space law

$$v\frac{dv}{dx} = F_{\mathcal{R}}(x, v; \mathcal{I}_{\mathcal{R}}), \qquad v = \dot{x}$$

with the path-history data in

$$\mathcal{I}_{\mathcal{R}}$$

held fixed by the certificate.

The affine partner calculation gives the model row

$$v\frac{dv}{dx} = -\frac{g}{4x^2}\left(1 + \frac{v}{c_f}\right)$$

with exact implicit quadrature

$$c_f v - c_f^2 \ln|c_f + v| = \frac{g}{4x} + C_{\mathcal{R}}$$

More generally, if the certified branch chart yields a separable Lienard row

$$\frac{v}{Q_{\mathcal{R}}(v)}\, dv = P_{\mathcal{R}}(x)\, dx$$

the quadrature invariant is

$$\int^{v} \frac{\zeta}{Q_{\mathcal{R}}(\zeta)}\, d\zeta - \int^{x} P_{\mathcal{R}}(\xi)\, d\xi = C_{\mathcal{R}}$$

This is the preferred closed-form object for branch charts with velocity-dependent causal Jacobians.

A conservative potential is allowed only as a special certified reduction. The required condition is

$$F_{\mathcal{R}}(x, v; \mathcal{I}_{\mathcal{R}}) = -\partial_x U_{\mathcal{R}}(x; \mathcal{I}_{\mathcal{R}})$$

with no residual

$$v$$

dependence after the active image data are fixed. If that identity is proved, the arc may use the energy equation

$$\frac{1}{2}\dot{x}^2 + U_{\mathcal{R}}(x; \mathcal{I}_{\mathcal{R}}) = E_{\mathcal{R}}$$

Absent that proof, the chart must use the Lienard phase invariant, direct interval quadrature, or collocation residuals for the dual-mollified absolute-time law.

Thus the ansatz search reduces to two questions:

1. Can the active image distances

$$r_p(x), \qquad r_{s,m}(x)$$

be expressed and certified on each fixed branch chart?
2. Do the separator impulse laws and returned-history residuals close after one full cycle?

Minimal Four-Arc Breather Skeleton

A first closed-form skeleton should use four arcs:

1. **Inbound sub-field arc**

$$x = x_*, \qquad \dot{x} = -u_*, \qquad 0 < u_* < c_f$$

This arc falls toward the origin under partner attraction and controlled self-image terms.

2. **Origin-crossing layer**
The signed coordinate changes branch. The dual core scale

$$\epsilon_c$$

regularizes the near-origin amplitude, and the shell width

$$\eta$$

regularizes the causal-isochron selection.

3. **Outbound super-field or near-field-speed arc**
The right branch moves outward. If

$$\dot{x} > c_f$$

the sorting map

$$z = x - c_f t$$

reverses monotonicity and the active image list must be updated.

4. **Apocenter sub-field recapture arc**
The branch enters

$$0 \leq \dot{x} < c_f$$

before turning. On this arc the partner term should dominate the outward self-image terms, producing the final turn and return to

$$x = x_*, \qquad \dot{x} < 0$$

Velocity-class itinerary ledger

The four arc names above are a compressed return graph, not yet a complete velocity-class itinerary. Define

$$\mathfrak{v}(t) \in \{\mathsf{S}_{\text{sub}}, \mathsf{S}_{\text{sep}}, \mathsf{S}_{\text{sup}}\}$$

by

$$\mathsf{S}_{\text{sub}}: \ |\dot{x}| < c_f, \qquad \mathsf{S}_{\text{sep}}: \ |\dot{x}| = c_f, \qquad \mathsf{S}_{\text{sup}}: \ |\dot{x}| > c_f$$

A full origin-crossing breather may pass through more separator events than the compressed four-arc naming suggests. The current self-image table below assumes the simple compressed itinerary in which the apocenter recapture remains sub-field after the outer separator. Before using that table as a certificate input, the ansatz packet must specify the actual itinerary.

Two admissible itinerary templates are:

$$\mathsf{S}_{\text{sub}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sup}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sub}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sup}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sub}}$$

the doubled four-arc itinerary, and

$$\mathsf{S}_{\text{sub}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sup}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sub}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sep}} \to \mathsf{S}_{\text{sub}}$$

a glancing apocenter itinerary in which the path touches the separator without entering another super-field arc. These templates have different self-image tables. The certificate generator should therefore key every branch table by the chosen itinerary

$$\mathcal{K}$$

and its ordered interval list

$$I_1(\mathcal{K}), \ldots, I_m(\mathcal{K})$$

For the first certificate attempt, use the doubled four-arc itinerary. It is the generic transverse choice: every field-speed separator is treated as a simple fold event, while the glancing itinerary is reserved as a fallback if the generic branch enumeration fails or the corridor arithmetic forces a degenerate outer turn.

The periodicity condition is not merely

$$x(T) = x(0)$$

It is the returned-history condition

$$P_\eta(\phi) = \phi$$

For the closed-form ansatz, the finite approximation is to require equality on the sampled certificate mesh:

$$P_\eta(\phi)(\theta_j) = \phi(\theta_j), \qquad \partial_\theta P_\eta(\phi)(\theta_j) = \dot{\phi}(\theta_j), \qquad 0 \le j \le N$$

Itinerary-Keyed Self-Image Enumeration

The decisive algebraic test is not the partner root. It is the same-path self-root equation

$$|x(t) - x(s)| = c_f(t-s), \qquad s < t$$

across the four-arc skeleton.

For the compressed four-arc itinerary, let the candidate cycle be partitioned into four time intervals:

$$I_1 = \text{inbound sub-field}, \qquad I_2 = \text{origin-crossing layer}$$

$$I_3 = \text{outbound super-field or near-field-speed}, \qquad I_4 = \text{apocenter sub-field recapture}$$

For any richer itinerary

$$\mathcal{K}$$

replace this list by

$$I_1(\mathcal{K}), \ldots, I_m(\mathcal{K})$$

and fill the same table over all ordered interval pairs. The sixteen-row table below is therefore not the universal branch table; it is the compressed four-arc case.

For each ordered pair

$$(\alpha, \beta) \in \{1,2,3,4\}^2$$

with

$$t \in I_\alpha, \qquad s \in I_\beta, \qquad s < t$$

solve the two signed defects

$$g^{\pm}_{\alpha\beta}(t,s) \equiv \pm\big(x_\alpha(t) - x_\beta(s)\big) - c_f(t-s) = 0$$

subject to the sign consistency condition

$$\pm\big(x_\alpha(t) - x_\beta(s)\big) > 0$$

On an affine pair of arcs,

$$x_\alpha(t) = a_\alpha + v_\alpha t, \qquad x_\beta(s) = a_\beta + v_\beta s$$

write the orientation sign as

$$\chi \in \{-1, +1\}$$

The signed self-image defect is

$$g^{\chi}_{\alpha\beta}(t,s) = \chi\big(x_\alpha(t) - x_\beta(s)\big) - c_f(t-s)$$

If the source-side denominator has a certified floor

$$|c_f - \chi v_\beta| \ge \nu_{\alpha\beta} c_f > 0$$

then the affine root is explicit:

$$s^{\chi}_{\alpha\beta}(t) = \frac{(c_f - \chi v_\alpha)t - \chi(a_\alpha - a_\beta)}{c_f - \chi v_\beta}$$

The source Jacobian on that row is

$$J^{\chi}_{\alpha\beta} = 1 - \frac{\chi v_\beta}{c_f} = \frac{c_f - \chi v_\beta}{c_f}$$

Thus every affine self-image row reduces to interval validation of the following predicates:

$$t \in I_\alpha, \qquad s^{\chi}_{\alpha\beta}(t) \in I_\beta, \qquad s^{\chi}_{\alpha\beta}(t) < t$$

$$0 < t - s^{\chi}_{\alpha\beta}(t) \le h, \qquad \chi\bigl(x_\alpha(t) - x_\beta(s^{\chi}_{\alpha\beta}(t))\bigr) > 0, \qquad \left|J^{\chi}_{\alpha\beta}\right| \ge \nu_{\alpha\beta}$$

If the denominator loses its floor, the row is not a simple affine branch; it is a separator or fold row and must be certified by the dual-mollified fold normal form rather than by the branch-sum formula.

Null-coordinate causal pre-ledger

Before running interval root validation, reduce the search by a 1D Minkowski diagnostic. Use null coordinates

$$u(t) = c_f t - x(t), \qquad w(t) = c_f t + x(t)$$

The self-image equation

$$|x(t) - x(s)| = c_f(t - s), \qquad s < t$$

splits into two exact ledgers:

$$x(t) > x(s) \quad \Longleftrightarrow \quad u(t) = u(s)$$

and

$$x(t) < x(s) \quad \Longleftrightarrow \quad w(t) = w(s)$$

Geometrically, this is just the intersection of the path with the past-directed causal cone from

$$(x(t), c_f t)$$

Computationally, it means that each ordered arc pair

$$(I_\alpha, I_\beta)$$

should be preclassified by interval ranges of

$$u(I_\alpha), \quad u(I_\beta), \quad w(I_\alpha), \quad w(I_\beta)$$

If the relevant null-coordinate ranges are disjoint, that block of the self-image table is empty before any root solve. If the ranges overlap on monotone subarcs, the root count is the number of interval-certified level crossings, and the sign of

$$\hat{r}_s$$

is already known from whether the

$$u$$

or

$$w$$

ledger is active.

This also fixes the Jacobian sign test in a coordinate-free way:

$$J_u = \frac{du/ds}{c_f} = 1 - \frac{\dot{x}(s)}{c_f}, \qquad J_w = \frac{dw/ds}{c_f} = 1 + \frac{\dot{x}(s)}{c_f}$$

The interval validator should therefore start from a causal pre-ledger with three outcomes for each block:

1. null-coordinate ranges disjoint, so the block is certified empty;
2. ranges overlap with monotone source and receiver subarcs, so the root count and sign are bounded before solving;
3. a separator or turning interval is present, so the block must be split or sent to the fold-layer certificate.

> **Target Theorem (Null-Coordinate Causal Pre-Ledger).**
> Fix a proposed velocity-class itinerary

$$> \mathcal{K} >$$

> with ordered arc partition

$$> I_1(\mathcal{K}), \ldots, I_m(\mathcal{K}) >$$

> and a compact certificate tube around a candidate history. Suppose the interval enclosures for

$$> u = c_f t - x, > \qquad > w = c_f t + x >$$

> split every ordered receiver-source block

$$> (I_\alpha, I_\beta) >$$

> into finitely many subblocks, each of which is either range-disjoint, monotone with a positive derivative floor, or contained in a certified separator/fold layer. Then the self-image equation

$$> |x(t) - x(s)| = c_f(t - s), > \qquad > s < t, >$$

> admits a finite causal pre-ledger

$$> \mathcal{L}_{\mathcal{K}} >$$

> assigning each subblock one of three certified statuses:
> empty, simple-root, or fold-layer. Empty subblocks contain no self-image roots. Simple-root subblocks carry interval enclosures for the root count, root sign, source Jacobian floor, memory-depth range, and contribution sign. Fold-layer subblocks are excluded from branch-sum reduction until the dual-mollified fold certificate supplies a parity-preserving incoming-to-outgoing transition.
>
> The finite partition must also consume the parent-complement strips left after accepted simple-root and fold-layer subblocks have been removed. A parent-complement strip

$$> B >$$

> is accepted only if it has strict null-coordinate range separation, endpoint-excluded singleton contact under the declared boundary convention, exact certified fold-layer coverage, or another already accepted same-packet complement predicate. Positive-width null-coordinate overlap, a residual equality core, or an uncertified endpoint-scale gap rejects the candidate before branch-chart certification.
>
> Completing this theorem target is the first seed-chart gate. If

$$> \mathcal{L}_{\mathcal{K}} >$$

> cannot be made finite with strict empty-block gaps, monotone-block floors, and fold-layer bounds, the chosen itinerary or candidate history fails before quadrature, collocation residuals, or coupled-corridor arithmetic become relevant.

Proof route. Range-disjoint blocks are empty by direct interval separation of the relevant null coordinate. On monotone subblocks, the one-dimensional inverse function theorem and interval endpoint tests give finite level crossings, root enclosures, and the corresponding

$$J_u \quad \text{or} \quad J_w$$

floor. Separator and turning blocks are not forced into simple-root charts; they are routed to the fold normal form and must preserve

$$\Delta N \in 2\mathbb{Z}, \qquad \Delta D = 0$$

before the pre-ledger can feed the active branch chart.

For every root branch, record

$$\hat{r}_s = \operatorname{sgn}(x_\alpha(t) - x_\beta(s)), \qquad J_s = 1 - \frac{\dot{x}_\beta(s)\hat{r}_s}{c_f}$$

the interval of existence, and the contribution sign in the reduced equation. Also record the signed degree contribution

$$D_{\alpha\beta} = \sum_{g^{\pm}_{\alpha\beta}(t,s)=0} \operatorname{sgn} J_s$$

with the sum taken over certified root branches on that interval pair. On a simple-root chart with a positive Jacobian floor, this degree equals the unsigned root count. Near separators it is the invariant that survives the fold.

Separator fold rows and the excluded diagonal

The first local repair to the affine self-image table is to keep the diagonal exclusion and the fold layer in the same calculation. Let

$$y \in \{u, w\}$$

be the active null coordinate near a separator source time

$$s_\Sigma$$

and assume a nondegenerate local maximum

$$y'(s_\Sigma) = 0, \qquad y''(s_\Sigma) = -\alpha, \qquad \alpha > 0$$

For a receiver level

$$y(t) = y(s_\Sigma) - \lambda, \qquad \lambda > 0$$

the source-side fold equation has the normal form

$$y(s) - y(t) = \lambda - \frac{\alpha}{2}(s - s_\Sigma)^2 + O(|s - s_\Sigma|^3)$$

Hence the two local source branches are

$$s_\pm(t) = s_\Sigma \pm \sqrt{\frac{2\lambda}{\alpha}} + O(\lambda)$$

Their null-coordinate Jacobians are

$$J_y(s_\pm) = \frac{y'(s_\pm)}{c_f} = \mp \frac{\sqrt{2\alpha\lambda}}{c_f} + O(\lambda)$$

so the two branches carry opposite signed degree and the fold preserves

$$\Delta D = 0$$

The memory-depth tests are

$$0 < t - s_\Sigma + \sqrt{\frac{2\lambda}{\alpha}} + O(\lambda) \le h$$

for

$$s_-$$

and

$$0 < t - s_\Sigma - \sqrt{\frac{2\lambda}{\alpha}} + O(\lambda) \le h$$

for

$$s_+$$

When the receiver is still on the same outgoing source arc, the

$$s_+$$

branch may coincide with the excluded diagonal

$$s = t$$

to leading order. That branch is not an accepted simple-root contribution, but it is still part of the separator fold layer. It becomes a nontrivial branch only after the receiver leaves the outgoing source arc and the memory-depth inequality becomes strict.

Applied to the simplified doubled four-arc affine check, this repairs the apparent odd branch birth at the first and third separators. At

$$\Sigma_1$$

the active fold is the

$$w$$

ledger. The pre-fold branch has positive degree and matches the nontrivial

$$w$$

roots that continue through the adjacent source copies; the post-fold branch has negative degree and is initially diagonal-carried before becoming the second nontrivial

$$w$$

root on the later receiver block. At

$$\Sigma_3$$

the same calculation holds in the

$$u$$

ledger. Thus a one-root affine row immediately after a separator is not by itself a parity violation. It is a separator fold row whose missing opposite-degree partner is carried by the excluded diagonal until it emerges into a later ordered block.

This calculation gives a concrete obstruction to using a piecewise-affine table as a complete certificate: the affine row can identify the visible simple-root branch, but it cannot certify the separator unless the fold-layer chart records the hidden diagonal-carried partner, its opposite Jacobian sign, and its memory-depth exit into a nontrivial source interval.

The enumeration deliverable is the following table, filled with exact formulas or interval-validated enclosures:

Receiver arc I_α	Source arc I_β	Root count N	Signed degree D	Root formula or enclosure	$\hat{r}_s$	J_s floor	Contribution sign	Separator jumps	Certificate status
I_1	I_1	target	target	target	target	target	target	target	open
I_1	I_2	target	target	target	target	target	target	target	open
I_1	I_3	target	target	target	target	target	target	target	open
I_1	I_4	target	target	target	target	target	target	target	open
I_2	I_1	target	target	target	target	target	target	target	open
I_2	I_2	target	target	target	target	target	target	target	open
I_2	I_3	target	target	target	target	target	target	target	open
I_2	I_4	target	target	target	target	target	target	target	open
I_3	I_1	target	target	target	target	target	target	target	open
I_3	I_2	target	target	target	target	target	target	target	open
I_3	I_3	target	target	target	target	target	target	target	open
I_3	I_4	target	target	target	target	target	target	target	open
I_4	I_1	target	target	target	target	target	target	target	open
I_4	I_2	target	target	target	target	target	target	target	open
I_4	I_3	target	target	target	target	target	target	target	open
I_4	I_4	target	target	target	target	target	target	target	open

The parity check is imported from Proposition 3 in master-equation.md: generic folds create or annihilate one root pair, so

$$\Delta N \in 2\mathbb{Z}, \qquad \Delta D = 0$$

On a closed cycle the branch ledger must return to itself, hence

$$\sum_{\Sigma} \Delta N = 0, \qquad \sum_{\Sigma} \Delta D = 0$$

with every local unsigned jump even. This is a discrete consistency test on the ansatz. A candidate branch list that fails it should be rejected before any quadrature or collocation residual is computed.

Causal-Root Ledger and Action Bookkeeping

The enumeration table is also the bridge to the discrete-step language in energy.md. Let

$$N_{\alpha\beta}$$

denote the unsigned self-root count in an itinerary-keyed row, and let

$$M_{\alpha\beta}$$

denote the analogous partner-root channel count supplied by the partner branch table. The pair

$$(N_{\alpha\beta}, M_{\alpha\beta})$$

is the local causal-root ledger for that arc pair.

On a fixed simple-root chart with fixed

$$(N_{\alpha\beta}, M_{\alpha\beta}, D_{\alpha\beta})$$

the motion is still continuous and any energy or phase quadrature is ordinary continuous bookkeeping. No separate energy atom is inserted. A discrete action step enters only when a separator or fold changes the admissible integer ledger. In the raw self-root table, a generic fold changes the unsigned root count by an even jump,

$$\Delta N \in 2\mathbb{Z}$$

while preserving

$$\Delta D = 0$$

When that root pair is grouped as one newly active channel for action-angle bookkeeping, the same event is recorded as one channel update. This is the sense in which an h-like transaction can correspond to

$$N \to N+1 \qquad \text{or} \qquad M \to M+1$$

in the grouped causal-root ledger, without treating energy itself as discontinuous at the substrate level.

Thus any claimed h-like or $2h$-like energy step must be backed by three certificate facts: the branch-list update across the separator, the parity law for the underlying simple roots, and returned-history closure of the full cycle. This is the precise route by which continuous delayed geometry can produce discrete effective action bookkeeping.

If this table closes to a finite branch list with strict separation, memory-depth, and Jacobian floors, the ansatz can feed the finite certificate audit. If the self images do not close algebraically into a finite list, the next certificate generator should be a piecewise fractionally augmented Chebyshev or cubic

$$C^1$$

collocation history

$$\phi_{\mathrm{cyc}}$$

with interval validation of the finite active branches and the returned-history residuals. The accepted output is strict residual slack, not a compact symbolic formula.

Separator Matching Laws

At every separator time

$$t_\Sigma$$

where

$$|\dot{x}(t_\Sigma)| = c_f$$

the matching law must come from the dual-mollified fold calculation rather than from an assumed conservative energy jump. Choose a fold layer

$$[t_\Sigma - \Delta, t_\Sigma + \Delta]$$

on which the simple-root branch-sum chart is replaced by the dual-mollified integral law. The separator impulse is

$$\Delta v_\Sigma = \int_{t_\Sigma-\Delta}^{t_\Sigma+\Delta} a^{\mathrm{fold}}_{\eta,\epsilon_c}(t)\,dt$$

The ansatz must impose four matching conditions:

1. **position continuity**

$$x(t_\Sigma^-) = x(t_\Sigma^+)$$

2. **controlled velocity increment across the fold layer**

$$\dot{x}(t_\Sigma+\Delta) - \dot{x}(t_\Sigma-\Delta) = \Delta v_\Sigma$$

3. **branch-list update**

$$\mathcal{I}_{\mathcal{R}^-} \longrightarrow \mathcal{I}_{\mathcal{R}^+}$$

4. **certificate budget update**

$$|\Delta v_\Sigma| \le I^{\mathrm{fold}}_{\eta,\epsilon_c}$$

where

$$I^{\mathrm{fold}}_{\eta,\epsilon_c}$$

is the finite caustic-transit impulse ceiling imported from the proof scaffold.

The normal-form section above makes this ceiling an auditable number. For each separator, the ansatz packet must report the local fold coefficients

$$a, \qquad b$$

the transversality constant

$$C_\Sigma$$

the shell and core parameters

$$(\eta, \epsilon_c)$$

and either the bound

$$I^{\mathrm{fold}}_{\eta,\epsilon_c} \le C_\Sigma \eta^{1/2} A_{\Sigma,\eta,\epsilon_c}$$

or a sharper interval quadrature bound over the certified fold layer. The matching law is usable only after this finite impulse estimate has strict slack against the adjacent arc budgets.

This formulation keeps the separator tied to the same estimates used in collinear-breather.md. Energy constants on the adjacent arcs may still be useful bookkeeping devices, but they are not the primitive matching data at

$$|\dot{x}| = c_f$$

Fold-Adapted Fractional Basis

Pure polynomial splines are not the preferred certificate basis near a field-speed separator. The fold normal form produces square-root source-time scaling in the simple-root reduction. In the bare fold model this gives a local hierarchy of the form

$$\Delta v(\tau) \sim |\tau|^{1/2}, \qquad \Delta x(\tau) \sim |\tau|^{3/2}, \qquad \tau = t - t_\Sigma$$

The dual mollifiers make the actual certificate function smooth at fixed

$$(\eta, \epsilon_c)$$

but the unsoftened fold asymptotic remains the right shape for reducing residuals and avoiding artificial derivative ringing.

Near every certified separator, use a fractionally augmented local basis

$$\phi_{\mathrm{local}}(\tau) = a_0 + a_1\tau + a_{3/2}|\tau|^{3/2} + a_2\tau^2 + a_{5/2}|\tau|^{5/2} + \cdots$$

optionally multiplied by a compact blending function that hands off to the ordinary polynomial or Chebyshev basis outside the fold layer. The coefficients

$$a_{3/2}, \qquad a_{5/2}, \ldots$$

are not aesthetic parameters; they encode the known separator singularity budget. The interval report should record which separator layers use the fractional basis, the layer radii, and the residual improvement against the velocity sample budget

$$R_j^v + L_j^v r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}$$

Away from separators, ordinary Chebyshev, cubic, or other validated bases remain acceptable. The required standard is not polynomial purity; it is strict interval slack in the returned-history residuals and the branch-chart margins.

The parent-complement obstruction gives the fresh collocation packet a concrete construction test, not merely another rejection condition. Let

$$C(\mathbf{a}) = 0$$

denote the structural constraints of a candidate packet: section anchoring, symmetry, separator equations, C^1 matching, fold nondegeneracy, origin placement, and neutral-coordinate fixing. For each unresolved parent complement

$$C_m = R_m \times S_m$$

choose a signed null-coordinate gap

$$\delta_m(\mathbf{a})$$

that is positive exactly when the receiver and source ranges are strictly separated. A useful collocation basis must admit a tangent direction

$$DC(\mathbf{a}_0)\xi = 0, \qquad D\delta_m(\mathbf{a}_0)\xi > 0$$

for all unresolved complements at the provisional packet

$$\mathbf{a}_0$$

Then a nearby structural candidate opens those gaps to first order, while already strict margins persist for sufficiently small deformation. This is the mathematical reason the next packet must change the null-coordinate geometry itself; refining the rejected cosine mesh cannot remove fixed-history equality collars.

What Would Count as a Successful Closed-Form Candidate

A closed-form candidate is successful only as a certificate generator. It is not a separate proof route.

A candidate ansatz packet must produce:

1. a history

$$\phi_{\mathrm{cyc}} \in C^1([-h, 0])$$

2. a period

$$T > 0$$

3. a finite active branch list

$$\mathcal{B}_{\mathrm{act}}$$

on every arc, together with inactive branch complements;
4. an itinerary ledger

$$\mathcal{K}$$

and an itinerary-keyed self-image table with root counts, signed degrees, grouped channel counts, and separator parity jumps;
5. a symmetry chart, either apocenter-even in

$$q$$

or origin-crossing-odd in

$$x$$

together with the paired branch-label rule;
6. a neutral-coordinate audit identifying every continuous freedom that leaves the same physical certificate unchanged. At minimum this includes the removed time-shift freedom, any declared reflection or relabeling symmetry, and any ansatz parameter whose first variation is tangent to the candidate branch rather than transverse to it. In finite form, if

$$\alpha^a$$

are ansatz coordinates and

$$Z_a(\theta) \equiv \frac{\partial \phi_{\mathrm{cyc}}(\theta;\alpha)}{\partial \alpha^a}$$

then the certificate must classify each

$$Z_a$$

as section-fixed, symmetry-neutral, or genuinely deforming before monodromy or residual rows are interpreted;
7. a null-coordinate causal pre-ledger in

$$u = c_f t - x, \qquad w = c_f t + x$$

marking empty, candidate nonempty, and fold-split self-image blocks before interval root solving;
8. a certificate mesh

$$\{\theta_j\}_{j=0}^{N} \subset [-h,0]$$

9. algebraic, Lienard phase quadrature, fractionally augmented Chebyshev or cubic

$$C^1$$

or other interval-validated formulas for each arc;
10. separator impulse laws at every

$$|\dot{x}| = c_f$$

event, including the fold normal-form constants and finite impulse bounds;
11. a bifurcation-parameter sweep over

$$(\eta, \epsilon_c, V_{\max})$$

or a justified lower-dimensional slice, identifying the region where the itinerary is admissible, the required roots exist, inactive-root gaps are positive, and fold impulses are finite;
12. returned-history residuals

$$R_j^x, \qquad R_j^v$$

small enough to feed the finite certificate audit in collinear-breather.md.

The last item is essential. A visually plausible orbit is not enough. The ansatz must produce the certificate data:

$$\nu_{\mathrm{seed}}, \quad \gamma_{\mathrm{gap}}, \quad \gamma_h, \quad \gamma_{\mathrm{env}}$$

the factorized corridor coefficients,
and the returned-sample residuals or boundary budgets.

Seed-chart pre-ledger acceptance rule

The first machine-checkable gate is the null-coordinate pre-ledger, not the returned residual. For every ordered receiver-source block

$$(I_\alpha, I_\beta)$$

define the range gaps

$$\Delta^u_{\alpha\beta} = \operatorname{dist}\big(u(I_\alpha), u(I_\beta)\big), \qquad \Delta^w_{\alpha\beta} = \operatorname{dist}\big(w(I_\alpha), w(I_\beta)\big)$$

The row is empty when the relevant gap is strictly positive. It is a simple-root row only when the corresponding source-side derivative floor is positive:

$$\inf_{s\in I_\beta}\left|1-\frac{\dot{x}(s)}{c_f}\right|>0 \qquad \text{for the } u \text{ ledger}$$

or

$$\inf_{s\in I_\beta}\left|1+\frac{\dot{x}(s)}{c_f}\right|>0 \qquad \text{for the } w \text{ ledger}$$

Rows that satisfy neither test must be split or routed to a fold-layer certificate. A candidate

$$\phi_{\text{cyc}}$$

does not advance to branch-chart certification while any ordered block remains unresolved.

The same acceptance rule applies to parent complements. After accepted simple-root and fold-layer subrows are removed from a parent block, every leftover parent-complement strip must be accepted by strict null-coordinate range separation, endpoint-excluded singleton contact under the declared boundary convention, exact fold-layer coverage, or another already accepted same-packet complement predicate. Any positive-width overlap, residual equality core, or uncertified endpoint-scale gap rejects the packet before branch-chart work.

At a separator row, a single visible simple root adjacent to the fold is not enough to pass the pre-ledger. The fold-layer certificate must also account for any opposite-degree branch that is temporarily carried by the excluded diagonal

$$s=t$$

and prove either its continued diagonal exclusion or its later strict memory-depth entry as a nontrivial source branch.

This rule makes the pre-ledger a genuine falsification gate. A failed row rejects the candidate history, the chosen split, or the itinerary before corridor arithmetic, monodromy, or returned-sample preservation is attempted. A passed pre-ledger still does not prove the breather; it only permits construction of the active branch chart with inactive complements, Jacobian floors, memory-depth ranges, and contribution signs on the same sampled domain.

First Working Guess

Closed-by-quadrature is only one possible certificate generator. A two-parameter family is generally too small unless the cycle symmetry is built into the parametrization: the compressed skeleton has arc-junction conditions, separator impulse conditions, branch-list updates, and a returned-history residual.

The first analytic guess should therefore be at least a three-parameter family:

$$\phi_{\text{cyc}}(\theta;u_*,X_*,C_>)$$

where

$$X_*=x_*, \qquad 0<u_*<c_f$$

and

$$C_>$$

is the phase-curve or shape parameter for the super-field arc. It should not be interpreted as a conservative energy unless the fixed branch chart has separately passed the potential-reduction test. In a collocation version,

$$C_>$$

is replaced by the analogous independent shape coefficient for the inner super-field segment.

On a fixed affine partner chart, the default quadrature arc is generated by the Lienard phase invariant

$$c_f v - c_f^2\ln|c_f+v| = \frac{g}{4x}+C_{\mathcal{R}}, \qquad v=\dot{x}$$

When self-image terms are included, this invariant is replaced by the corresponding certified phase quadrature or by interval collocation of the dual-mollified absolute-time law. A potential curve of the form

$$\frac{1}{2}\dot{x}^2+U_{\mathcal{R}}(x)=E_{\mathcal{R}}$$

is admissible only on a fixed branch chart where the delayed force has already been shown to be an exact

$$-\partial_x U_{\mathcal{R}}$$

derivative along that chart.

The more certificate-friendly parallel guess is a piecewise fractionally augmented Chebyshev or cubic

$$C^1$$

history with unknown coefficients

$$\phi_{\mathrm{cyc}}(\theta; \mathbf{a})$$

chosen by collocation against the dual-mollified absolute-time law. In that version, the active branch list and returned residuals are interval-validated directly rather than inferred from symbolic quadrature.

The first reduction should impose cycle-reversal symmetry rather than leave periodicity to unrestricted shooting. One convenient phase choice places the apocenter at

$$\theta = 0$$

and imposes the radial condition

$$q(-\theta) = q(\theta), \qquad \dot{q}(0) = 0$$

Equivalently, a signed-coordinate chart centered on an origin crossing may impose the odd sheet condition

$$x(-\theta) = -x(\theta)$$

The certificate must state which symmetry chart is used and how the branch labels pair under the symmetry. When the paired branch ledger and regularization preserve this cycle-reversal symmetry, the net-work integral cancels by parity:

$$\oint F_{\mathrm{net}}\, dx = 0$$

If the causal-delay branch data do not pair in this way, the failure appears as a returned-history residual rather than as an adjustable energy defect. This is why the symmetry constraint belongs in the ansatz, not as a post-hoc interpretation of a numerically closed orbit.

The parameters

$$u_*, \qquad X_*, \qquad C_>$$

are then chosen so that the returned section state satisfies

$$x(T) = x_*, \qquad \dot{x}(T) = -u_*$$

the outer and inner separator impulses match the adjacent arcs, and the sampled history residuals are minimized. In the intended symmetric case,

$$C_>$$

is determined by separator matching from the apocenter side while

$$u_*$$

is determined by the outer separator and returned section condition.

In the strict closed-form version, the residuals vanish:

$$R^x_{j,\pm} = 0, \qquad R^v_{j,\pm} = 0$$

In a certificate version, they only need to satisfy

$$R^x_{j,\pm} + L^x_j r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}, \qquad R^v_{j,\pm} + L^v_j r_{\mathrm{cert}} < \frac{r_{\mathrm{cert}}}{4}$$

Immediate Derivation Tasks

1. Use the action-generated sub-field test case as the first analytic baseline: compare the held-source energy segment, the Lambert-W exterior partner branch, and the finite-width self-collar before accepting any field-speed separator as dynamically produced.
2. Complete the signed partner branch table for affine and fixed-chart arcs, including the core-mollified force coefficient and validity margins.
3. Compute the separator normal-form constants and fold-layer impulse bounds for every proposed

$$|\dot{x}| = c_f$$

event.
4. Use the doubled four-arc itinerary as the first admissible velocity-class itinerary

$$\mathcal{K}$$

and key the arc partition to that itinerary rather than assuming the compressed four-arc graph by default.
5. Choose the symmetry chart: apocenter-even in

$$q$$

or origin-crossing-odd in

$$x$$

and record the paired branch-label rule.
6. Build and discharge the theorem target `Null-Coordinate Causal Pre-Ledger` in

$$u = c_f t - x, \qquad w = c_f t + x$$

producing the finite ledger

$$\mathcal{L}_{\mathcal{K}}$$

with certified empty blocks, simple-root blocks, and separator/fold blocks.
6. Use

$$\mathcal{L}_{\mathcal{K}}$$

to fill the itinerary-keyed self-image enumeration table for

$$|x(t) - x(s)| = c_f(t - s)$$

on every ordered arc pair

$$(I_\alpha, I_\beta)$$

7. Add the parity ledger

$$\Delta N \in 2\mathbb{Z}, \qquad \Delta D = 0$$

at every generic fold and verify that the closed-cycle sums vanish.
8. Record the grouped causal-root ledger

$$(N, M)$$

used for action bookkeeping, distinguishing it from the raw simple-root counts whenever fold pairs are grouped into one active channel.
9. If the pre-ledger or self-image table fails to close with strict finite margins, reject the current itinerary/candidate packet before attempting quadrature or collocation residuals. A fixed candidate whose parent-complement ranges retain positive-width overlap is not rescued by mesh refinement alone.
10. If the current candidate fails at that gate, instantiate a fresh fold-adapted piecewise collocation candidate, with the same-packet null-coordinate pre-ledger as its first acceptance row.
11. If the self-image table closes, convert it into

$$\mathcal{B}_{\mathrm{act}}$$

inactive branch complements, Jacobian floors, separation margins, and memory-depth bounds.
12. If the self-image table closes topologically but does not close algebraically, build a piecewise fractionally augmented Chebyshev or cubic

$$C^1$$

collocation history

$$\phi_{\mathrm{cyc}}$$

and certify the finite active branches numerically by interval validation.
13. Sweep

$$(\eta, \epsilon_c, V_{\max})$$

or a justified lower-dimensional slice to locate the itinerary-admissible parameter region before attempting the full corridor certificate.
14. Build the first certificate packet

$$\mathfrak{C}_{\mathrm{ans}}$$

and compute its returned section residuals.
15. If the residuals have strict slack, compute the finite certificate data and test the five audit rows in collinear-breather.md.

Provisional Assessment

The ansatz is plausible because the partner force on a locally affine branch already collapses to a velocity-weighted inverse-square law with an exact implicit phase invariant. On a fixed branch chart, that is the sort of structure that can generate a natural

$$1/r$$

phase-space curve; a conservative potential curve is only a special certified reduction.

The hard part is the same-side self-image term. If the self images collapse to a finite branch list across the field-speed separator, a closed-form or closed-by-quadrature certificate packet is credible. If they do not close algebraically, the next route is still productive: use a spline or collocation

$$\phi_{\mathrm{cyc}}$$

and certify the finite active branches numerically.

The first doubled-itinerary affine check has a sharper conclusion: the apparent odd simple-root births at the first and third separators are separator fold rows with one opposite-degree branch carried by the excluded diagonal, not completed branch-chart rows. The first fixed candidate also shows that parent-complement equality cores can remain after useful subrow and fold diagnostics are extracted. The next concrete calculation is therefore not another branch chart on that fixed candidate, but a fresh fold-adapted collocation packet whose null-coordinate pre-ledger consumes every ordered row before any branch-chart, residual, or corridor work begins. The accepted output is the finite audit packet, not an elegant formula.

Planar Bridge Closure

Purpose

This chapter isolates the first higher-dimensional closure problem that can move the dynamics stack forward in a decisive way. The exact delayed law is already stated in Master Equation, and the branch-topology machinery is already formalized in Causal Action Functional. What is still missing is a theorem-backed bridge showing that a genuinely planar delayed system admits a controlled section class, local branch regularity, bounded caustic transit, a genuine radial turnaround, and a return map that closes on a controlled envelope.

The planar bridge is the first regime where the proof architecture must leave the line while still retaining enough symmetry to remain mathematically tractable. If this bridge closes, it becomes the substrate basis for planar lock, terminal aligned modes, and the horizon-facing chirality questions developed in Horizon Chirality and Planar Spin. If it fails, the failure should identify the exact geometric obstruction rather than leaving the whole closure program underdetermined.

Position in the Dynamics Stack

The present chapter sits between four existing layers:

1. the exact delayed equations in Master Equation,
2. the topological branch formalism in Causal Action Functional,
3. the reduced return-map architecture in Collinear Breather,
4. the higher-dimensional program statement in Master-Equation Breather Program.

The role of this chapter is narrower than the full breather program. It does not attempt immediate many-body closure. It focuses on the first planar binary regime in which line-order arguments fail, tangential escape becomes real, and branch topology must be controlled in tandem with radial recapture rather than in a separate later chapter.

Reduced Planar Bridge Regime

Work in the reflection-symmetric planar two-body subclass

$$\mathbf{x}_1(t) = -\mathbf{r}(t), \qquad \mathbf{x}_2(t) = \mathbf{r}(t), \qquad \mathbf{r}(t) \in \Pi \cong \mathbb{R}^2, \qquad q_1 = -\epsilon, \qquad q_2 = +\epsilon$$

Write

$$\rho(t) \equiv \|\mathbf{r}(t)\|, \qquad \hat{\mathbf{e}}_r(t) \equiv \frac{\mathbf{r}(t)}{\rho(t)}, \qquad \hat{\mathbf{e}}_\theta(t) \equiv R_{\pi/2}\hat{\mathbf{e}}_r(t)$$

and decompose

$$\dot{\mathbf{r}}(t) = u_r(t)\hat{\mathbf{e}}_r(t) + u_\theta(t)\hat{\mathbf{e}}_\theta(t)$$

This is the smallest regime that still contains all the new burdens that matter:

- branch sorting is no longer inherited from a line order,
- recapture must control both radial and tangential escape,
- rotational symmetry must be reduced without destroying a convex return domain,
- and self-hit geometry must remain tame across a full excursion.

Rotational Gauge and Return Section

The natural section should remove rigid planar rotation locally and fix only one genuine return constraint. The reduced gauge choice is

$$\mathbf{r}(0) = \rho_* \mathbf{e}_1, \qquad \mathbf{e}_2 \cdot \dot{\mathbf{r}}(0) > 0$$

so that the section-defining equality is

$$\rho(0) = \rho_*$$

This choice serves three purposes:

- it removes time-shift and rigid-rotation redundancy cleanly enough for a return map,
- it keeps the section codimension one in the reduced history space,
- and it avoids building the proof on a non-affine section whose geometry is hard to close under convexity arguments.

The first local target is stricter than mere section definition: histories in the seed packet should satisfy a quantitative transversality condition

$$\dot{\rho}(0) \le -u_r < 0$$

so that the first return time is not born tangent to the section. Without such a margin, the gauge-reset map for the returned history need not depend continuously on the return event.

Local Cone Control and Branch Regularity

The first branch theorem should be local, not global. The correct opening package is a directional-cone result showing that the post-section planar velocity remains inside a strict admissible angular sector for a controlled time window. This is the planar replacement for 1D line ordering.

The target statement is of the following form:

> On an admissible planar seed packet, the delayed chords and instantaneous velocities remain inside a finite directional atlas on a short post-section interval, with quantitative angular separation from the Jacobian-null directions.

The right conceptual bridge to Causal Action Functional is the causal locus picture. In the regular regime, branch labels remain locally constant and can change only when

$$F(t,t') = 0, \qquad \nabla F(t,t') = 0$$

So the opening burden is not yet a whole-cycle branch census. It is to prove enough local transversality that the planar delayed geometry stays away from the singular directions long enough to support a finite branch atlas on an initial excursion slab.

This local cone control is the first point at which the planar program can either gain traction or expose a real obstruction.

Delay-Adapted Angle Control

The Jacobian

$$J = 1 - \frac{\mathbf{v} \cdot \hat{\mathbf{r}}}{c_f}$$

depends on the angle between the instantaneous velocity and the delayed chord. Planar closure therefore needs a delay-adapted moving frame that tracks this angle directly rather than only through coarse Cartesian bounds.

The working geometric data are:

- the radial/tangential decomposition relative to $(\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta)$,
- the angular offset between $\mathbf{v}(t)$ and each active delayed chord,
- and a finite sector atlas controlling how those offsets evolve.

The immediate theorem target is a finite-time cone-transversality estimate implying

$$J \geq \nu > 0$$

on a controlled interval before the first fold tube is entered.

Bounded Caustic Transit

The planar bridge should not assume that the full excursion avoids every Jacobian-null event. The hinge at which self-hit branches are born is part of the mechanism and must be crossed in a controlled way.

The right target is therefore a bounded fold-transit theorem:

> When an admissible history enters a sufficiently small tubular neighborhood of a planar fold where $F = 0$ and $\nabla F = 0$, the dual-mollified delayed impulse remains finite and the outgoing history stays inside an explicitly controlled post-fold sector.

This is the first place where the topological criterion from Causal Action Functional must be combined with quantitative delayed dynamics rather than cited abstractly.

Radial Turnaround Versus Centrifugal Leakage

A planar breather is fundamentally a radial turnaround problem. The main escape channel is not an abstract vector norm; it is the centrifugal barrier generated by tangential motion. The correct recapture target is therefore a strict radial majorization of the form

$$\ddot{\rho} = a_r^{\text{partner}} + a_r^{\text{self}} + \rho\dot{\vartheta}^2 \leq -\mathfrak{M}_{\text{in}} < 0$$

on the inbound leg, with sign conventions chosen consistently with the delayed force decomposition.

The proof burden splits into two parts:

- partner attraction and delayed memory must produce enough inward radial impulse,
- tangential forcing must remain bounded strongly enough that the centrifugal term does not outrun radial braking before the turn.

This makes tangential control subordinate but essential. Tangential dynamics do not supply a separate closure theorem; they must be bounded tightly enough to prevent centrifugal leakage from destroying the radial return.

Radial leakage-budget target

The external breather comparison adds one useful discipline: a formal oscillatory ansatz is not enough when a nonintegrable system can leak energy or drift out of the putative bound state. In the planar bridge, the corresponding failure channel is tangential leakage. The first radial-turnaround theorem should therefore be written as an integrated budget, not only as a pointwise slogan.

Nonpersistence and degenerate-breather comparisons make this burden stricter. A planar bridge is not allowed to inherit stability from a line or from an integrable breather family. The theorem must show that tangential leakage, gauge-reset discontinuity, and separator or fold transit remain bounded after the symmetry-breaking perturbation is present. If the required cancellation exists only in a comparison equation, the planar bridge fails rather than being rescued by analogy.

Rigidity comparisons sharpen this into a transport criterion. A collinear fixed point is not planar evidence unless the planar packet proves its own branch chart, continuous gauge reset, bounded fold/separator transit, and radial leakage budget with strict margins. If those rows close only because the line removes tangential escape or because an external integrable equation supplies cancellations, the correct conclusion is non-transport rather than inherited stability.

Let

$$I_{\text{turn}} = [t_a, t_b]$$

be the first candidate outward-to-inward turnaround window in the reduced planar history, with

$$\dot{\rho}(t_a) > 0$$

Write the net radial acceleration from the delayed master equation as

$$a_r(t) = \mathbf{a}(t) \cdot \hat{\mathbf{e}}_r(t)$$

so that

$$\ddot{\rho}(t) = a_r(t) + \frac{u_\theta^2(t)}{\rho(t)}$$

Define the inward delayed budget and tangential leakage budget by

$$B_{\text{in}} \equiv \int_{t_a}^{t_b} [-a_r(t)]_+ \, dt, \qquad B_\theta \equiv \int_{t_a}^{t_b} \frac{u_\theta^2(t)}{\rho(t)} \, dt$$

Let

$$E_{\text{fold}}, \qquad E_{\text{branch}}, \qquad E_{\text{gauge}}$$

denote certified upper bounds for unresolved fold-layer impulse, delayed-branch classification error, and section/gauge-reset error on the same controlled window. The first useful planar recapture certificate is the strict inequality

$$B_{\text{in}} - B_\theta - E_{\text{fold}} - E_{\text{branch}} - E_{\text{gauge}} \geq \dot{\rho}(t_a) + \gamma_{\text{turn}}, \qquad \gamma_{\text{turn}} > 0$$

This implies

$$\dot{\rho}(t_b) \leq -\gamma_{\text{turn}}$$

under the certified error budget. If the inequality cannot be made strict on any admissible seed packet, the planar bridge fails for a precise reason: centrifugal leakage and uncertified branch/fold uncertainty outrun radial recapture before the return map can close.

This budget also fixes what the later gauge-continuity row must provide. The return event must satisfy a transverse crossing margin

$$|\dot{\rho}(T_{\text{ret}})| \geq \nu_{\text{ret}} > 0$$

and the compensating rotation angle must have a bounded sensitivity on the same history box. Otherwise the gauge-reset map can lose continuity even if the radial budget itself turns the orbit around.

Tame-Envelope and Gauge Closure

The eventual return theorem needs a closed domain on which a fixed-point or continuation argument can act. The desired envelope should control at least:

- section radius and admissible radial excursion,
- tangential speed and accumulated angle,
- minimum branch separation,
- distance from Jacobian-null caustics,
- and history norms strong enough to pass compactness and continuity steps.

The closure target is:

> The one-cycle return map sends a convex tame envelope of reduced planar histories into itself.

The gauge-reset operator must be included in that statement. After one excursion, the returned history must be rotated back into the section gauge. That step is continuous only if the return event is quantitatively transverse; otherwise the return time and the compensating rotation angle need not vary continuously with the incoming history.

The planar bridge should therefore treat the phase, rotation, and section-time variables as collective coordinates rather than as ordinary stability directions. If

$$\alpha = (t_0, \psi, \rho_*, \ldots)$$

records the finite chart data of a candidate reduced cycle, then the tangent rows

$$Z_a(\theta) = \partial_{\alpha^a} \mathbf{r}_{\mathrm{cyc}}(\theta; \alpha)$$

must be classified before the return spectrum is interpreted. The rotation and time rows are removed by gauge and section choices; any remaining geometric row must either close by a holonomy residual or enter the transverse stability certificate. This finite zero-mode ledger is the clean way to state what the planar proof means by "the same cycle" after one excursion.

This is the exact higher-dimensional replacement for the collinear tame-class closure. If it succeeds, the abstract fixed-point capstone from Master-Equation Breather Program becomes actionable rather than aspirational.

Failure Alternatives

This chapter is useful even if the bridge does not close. There are only a few meaningful failure modes:

1. no seed packet with quantitative section transversality can be maintained;
2. local cone control fails before the first useful excursion slab is complete;
3. fold transit produces unbounded delayed impulse or uncontrolled branch proliferation;
4. centrifugal leakage outruns radial recapture before turnaround;
5. the reduced return map loses continuity under gauge reset.

Each failure would be informative. It would tell us whether the theory needs:

- a more restrictive planar regime,
- a different section choice,
- a different regularity class,
- or a revision of the stabilization claims made in the binary and nested shell braid chapters.

Immediate Theorem Program

The next sequence should be short and disciplined.

1. Define the reduced planar history space, seed packet, and quantitative section transversality.
2. Prove local sectorized cone control and short-time branch regularity on the first excursion slab.
3. Prove a bounded caustic-transit theorem for the first planar fold tube.
4. Prove the radial leakage-budget inequality in which inward delayed forcing beats centrifugal leakage, fold uncertainty, branch uncertainty, and gauge-reset error with a strict

$$\gamma_{\mathrm{turn}} > 0$$

5. Assemble these ingredients into a tame-envelope return theorem with continuous gauge reset.

That order matters. Without a transverse seed packet, the return map is not well-defined. Without local cone control, the branch atlas is not stable enough to transport. Without bounded fold transit, the self-hit mechanism is not mathematically usable. Without a radial-turnaround inequality, no planar breather can exist.

Why This Is the Best Queued Planar Target

After the collinear certificate either passes or produces a precise obstruction that planar geometry is meant to resolve, this chapter is the top higher-dimensional bottleneck because it is upstream of several attractive but softer narratives.

- It is upstream of the terminal aligned-mode story in Mapping the Planck Scale to the Nested Shell Braid Geometry.
- It is upstream of the planar-lock and branch-selection story in Horizon Chirality and Planar Spin.
- It is upstream of any reliable effective reduction in Effective Lagrangian and Gauge Symmetries.

If planar bridge closure fails, those higher-level chapters must become more conditional. If it succeeds, they gain a much firmer substrate basis.

Interfaces to Other Chapters

- Master Equation: exact delayed law, root equations, and Jacobian structure.
- Causal Action Functional: branch labels, coarea reduction, and the Jacobian-null bifurcation criterion.
- Collinear Breather: reduced return-map architecture and tame-envelope philosophy.
- Master-Equation Breather Program: global roadmap that this chapter now instantiates in the first planar regime.
- Nested Shell Braid Dynamics: higher-dimensional geometric target that eventually inherits the planar bridge machinery.

- Horizon Chirality and Planar Spin: downstream interpretation of planar branch selection once the planar bridge is mathematically under control.